# Assessment of walleye pollock in the eastern Bering Sea

James Ianelli, Sophia Wassermann, Abigail McCarthy, Carey McGilliard, Elizabeth Siddon, Kalei Shotwell, Grant Adams, Nathan Lauffenburger

Alaska Fisheries Science Center, NOAA Fisheries

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Tools assisted data processing, code execution, figure preparation and report writing under author direction. The authors retain responsibility for the scientific content, analyses, interpretations and conclusions.

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# 1 Executive summary

The last full EBS pollock assessment was completed in 2024 (Ianelli et al. 2024). A catch report was prepared in 2025 (Alaska Fisheries Science Center, National Marine Fisheries Service 2025). Changes in this assessment are described relative to 2024.

This executive summary presents Model 26.0 (Rceattle), the base model for this assessment and its Tier 3 harvest recommendations. Model 23.0 (ADMB), accepted in 2024, provides the comparison with the previous assessment.

## 1.1 Changes in assessment inputs

Catch, fishery age compositions, survey indices and age compositions, and weight-at-age schedules have been updated since 2024. Section 2.1.1 describes the years, sources and data treatments.

## 1.2 Changes in assessment methods

Model 26.0 implements the age-structured assessment in Rceattle and uses weight-at-age schedules estimated by a separate random-effects model. Section 2.1.2 describes these changes from Model 23.0.

## 1.3 Summary of results

Estimated total biomass in 2026 is 12.162 million t (approximate 95% log-scale Wald confidence interval 9.534–15.515 million t). Female spawning biomass (SSB) is 3.057 million t (2.489–3.754 million t, using the same interval method). The largest fishing mortality rate among ages is 0.3 per year.

Scenario 2, the expected-catch projection scenario, assumes catches of 1.375 million t in each of 2027 and 2028, equal to the specified 2027 annual TAC. The projections use mean 2023–2025 fishery selectivity, fixed biological inputs and recruitment equal to the fitted 1978–2023 mean (Section 7).

The advice table compares the 2025–2026 recommendations from the 2024 assessment with this assessment's 2027–2028 recommendations. Projected female spawning biomass and age 3+ biomass use Scenario 2 with the assumed catches above.

The final specifications for 2026 and 2027, issued in the absence of a 2025 stock assessment, specified the Bering Sea subarea pollock OFL at 2,496,000 t and ABC at 2,036,000 t; specified annual TAC is 1,375,000 t. These are the adopted specifications, separate from the recommendations from this assessment.

The stock is above MSST in 2026 and is not approaching an overfished condition (Section 9). All risk categories are Level 1, supporting maximum ABC (Section 8).

| Model 26.0 advice from revised fits | | | | |
|---|---|---|---|---|
| Quantity | As estimated or specified in 2024 for: | | Primary baseline Model 26.0 for: | |
| | 2025 | 2026 | 2027 | 2028 |
| Natural mortality (annual rate, ages 3+) | 0.300 | 0.300 | 0.300 | 0.300 |
| Tier and sub-tier | 3a | 3a | 3a | 3a |
| Projected total biomass (ages 3+, t) | 8,526,000 | 8,075,000 | 11,270,000 | 12,367,000 |
| Projected female spawning biomass (t) | 3,118,000 | 3,342,000 | 3,198,000 | 3,394,000 |
| B100% (t) | 5,902,000 | 5,902,000 | 6,158,000 | 6,158,000 |
| B40% (t) | 2,361,000 | 2,361,000 | 2,463,000 | 2,463,000 |
| B35% (t) | 2,066,000 | 2,066,000 | 2,155,000 | 2,155,000 |
| F_OFL (annual rate) | 0.513 | 0.513 | 0.658 | 0.658 |
| Maximum F_ABC (annual rate) | 0.394 | 0.394 | 0.513 | 0.513 |
| Recommended F_ABC (annual rate) | 0.394 | 0.394 | 0.513 | 0.513 |
| OFL (t) | 2,957,000 | 2,496,000 | 2,328,000 | 2,422,000 |
| Maximum ABC (t) | 2,417,000 | 2,036,000 | 1,896,000 | 1,968,000 |
| Recommended ABC (t) | 2,417,000 | 2,036,000 | 1,896,000 | 1,968,000 |
| Historical source: December 2024 SAFE, Executive Summary, Tier 3 version, p. 4; B40% is printed in section 7.2, p. 35. | | | | |
| 2027–2028 projections assume catch of 1,375,000 t in each year, mean 2023–2025 selectivity and fixed biological inputs. Total biomass covers ages 3+ at the start of the year. F_OFL and maximum F_ABC are the fully selected F35% and F40% rates. Biomass and catch are rounded to the nearest 1,000 t. | | | | |
| Spawning biomass exceeds B40% in both years, placing the stock in Tier 3a. All risk categories are Level 1, supporting the maximum ABC. Recruitment is fixed at the fitted 1978–2023 mean; these projections show point estimates. Adopted catch specifications are reported separately. | | | | |

| Model 26.0 stock-status summary. The 2025–2026 columns report status for 2025 and 2026; historical determinations are from the 2024 assessment. | | | | |
|---|---|---|---|---|
| Status | As determined in 2024 for: | | As determined this year for: | |
| | 2023 | 2024 | 2025 | 2026 |
| Overfishing | No | n/a | No | n/a |
| Overfished | n/a | No | n/a | No |
| Approaching overfished | n/a | No | n/a | No |
| Historical determinations are from the December 2024 EBS pollock SAFE, Executive Summary, p. 4. | | | | |
| Assessment conclusions are separate from final NMFS determinations. Updated 2025 catch (1,326,215.61 t) is below its 2,957,000-t OFL. For Model 26.0, 2026 spawning biomass (3.057 million t) and Alternative 7 biomass in 2028 (3.123 million t) both exceed B35% (2.155 million t), supporting the two 2026 "No" entries. Section 9 describes the status tests. Each two-year column pair reports overfishing for the earlier catch year and stock status for the later assessment year; cells outside those year-specific determinations are marked n/a. NMFS issues the final status determinations. | | | | |

## 1.4 SSC and Plan Team requests

The September 2026 Plan Team requested the following work:

Present Model 26.0 for management use and Model 23.0 for continuity; both are compared in this assessment.

Examine age-1 BTS selectivity and the influence of BTS variability on retrospective patterns. Additional selectivity and data-influence tests are outside the scope of this assessment.

Extend the simulation self-test and examine retrospective patterns in time-varying selectivity. The completed simulation experiment and its limitations are described in Section 6.3.1.

Present retrospective results for Models 23.0 and 26.0 side by side. Matched runs were completed and compared; both models showed similar spawning-biomass retrospective patterns (Appendix A).

Document age-composition weighting and evaluate residual patterns. Selected sample sizes, one-step-ahead residuals and Francis comparisons are presented in the assessment.

Continue development of the Ecosystem and Socioeconomic Profile; we developed a draft using AI to compile and coordinate information from disparate sources (Appendix B).

Earlier SSC requests address model development, uncertainty, survey treatment and composition weighting. The individual requests and responses are presented in Appendix A.

# 2 Introduction

## 2.1 General

Walleye pollock (*Gadus chalcogrammus*; hereafter referred to as pollock) are broadly distributed throughout the North Pacific with the largest concentrations found in the eastern Bering Sea (EBS). Also known as Alaska pollock, this species continues to play important roles ecologically and economically.

### 2.1.1 Changes in assessment inputs

The 2026 assessment updates the accepted December 2024 assessment; the 2025 document was a catch report. Updated inputs include annual catches from 1991 onward, fishery age compositions for 2024–2025, BTS indices and age compositions for 2025–2026, the 2026 ATS index and provisional age composition, and AVO indices through 2026. Historical BTS estimates were also revised. Spatiotemporal survey estimation continues the approach adopted in 2024; the current biomass-index and abundance-at-age products use sdmTMB (Anderson et al. 2025) and tinyVAST (Thorson et al. 2025), respectively.

The 2024–2025 fishery age compositions use year-specific FT-NIRS-to-traditional-microscopic-age conversions, with likelihood sample sizes of 578 and 530. The provisional 2026 ATS age composition applies the BTS age–length key to ATS lengths; FT-NIRS ages from the ATS samples are expected in 2027.

Fishery weight-at-age retains the fixed 1964–1990 schedule and annual observation means for 1991–2025. A separate random-effects model fitted to fishery and BTS weights through 2025 supplies 2026 predictions and 2027–2028 forecasts; its fit omits the 2026 BTS observations. Population weights use BTS means through 2026, including revised historical estimates. Spawning weights at ages 3–15+ use fitted 2024–2025 A-season weights and a 2026 projection, retaining the previous ages 1–2 schedule. Section 4 and Appendix G and Appendix H describe the data treatments and their uncertainty.

### 2.1.2 Changes in assessment methods

The main assessment-method change is the transition from the ADMB implementation of Model 23.0 to the Rceattle implementation of Model 26.0. Weight-at-age is estimated in a separate random-effects model and supplied as fixed schedules, whereas Model 23.0 estimates cohort and year effects within the assessment. Model 26.0 uncertainty therefore conditions on the estimated weights. Natural mortality and catchability assumptions retain the previous treatment. The configuration comparisons, diagnostics and sensitivities evaluate the effects of the data updates and implementation change (Section 5, Section 6 and Appendix I).

## 2.2 Review of Life History

In the EBS, pollock generally spawn during March–May and in relatively localized regions during specific periods (Bailey 2000). Generally spawning begins nearshore north of Unimak Island in March and April and later near the Pribilof Islands (Bacheler et al. (2010)). Females spawn in batches with up to 10 batches of eggs per female per year (during the peak spawning period). Eggs and larvae of EBS pollock are planktonic for a period of about 90 days and appear to be sensitive to environmental conditions. These conditions likely affect their dispersal into favorable areas (for subsequent separation from predators) and also affect general food requirements for over-wintering survival (Gann et al. (2015), Heintz et al. (2013), Hunt Jr. et al. (2011), Ciannelli et al. (2004)). Duffy-Anderson et al. (2016) provide a review of the early life history of EBS pollock.

Throughout their range juvenile pollock feed on a variety of planktonic crustaceans, including calanoid copepods and euphausiids. In the EBS shelf region, one-year-old pollock are found throughout the water column, but also commonly occur in the NMFS bottom trawl survey. Ages 2 and 3 year old pollock are rarely caught in summer bottom trawl survey gear and are more common in the midwater zone as detected by mid-water acoustic trawl surveys. Younger pollock are generally found in the more northern parts of the survey area and appear to move to the southeast as they age (Buckley et al. (2009)). Euphausiids, principally *Thysanoessa inermis* and *T. raschii*, are among the most important prey items for pollock in the Bering Sea (Livingston (1991); Lang et al. (2000); Brodeur et al. (2002); Ciannelli et al. (2004); Lang et al. (2005)). Pollock diets become more piscivorous with age, and cannibalism has been commonly observed in this region. However, Buckley et al. (2015) showed that spatial patterns of pollock foraging, including cannibalism, vary with the size of the foraging pollock. For example, pollock in the northern shelf region between the 100 and 200 m isobaths (closest to the shelf break) tend to be more piscivorous than pollock in shallow nearshore areas.

## 2.3 Stock structure

Stock structure for EBS pollock was evaluated in Ianelli et al. (2015). In that review past work on genetics (e.g., Bailey et al. (1999), Canino et al. (2005)) provided insight on genetic differentiation. The investigation also compared synchrony in year-classes and growth patterns by region. Pollock samples from areas including Zhemchug Canyon, Japan, Prince William Sound, Bogoslof, Shelikof, and the northern Bering Sea were processed and results presented in Ianelli et al. (2021). Genetics research at the AFSC (Ianelli et al. 2023) indicated that genetic groups mostly aligned with current stock-management delineations. The Aleutian Islands and Bogoslof stocks showed subtle differentiation from the Gulf of Alaska stocks.

# 3 Fishery and management history

Directed foreign fishing began in 1964. US vessels entered the fishery in 1980, and it became fully domestic by 1988. The domestic observer program formally began in 1990; later monitoring developments include electronic monitoring on participating catcher vessels and biological sampling during offloading (Ianelli et al. 2024). The directed fishery primarily uses pelagic trawl gear, with inshore, catcher/processor and mothership sectors. The historical catch and adopted harvest-limit series is continued through 2026 in Table 1.

Updated reported catches are 1,311,261 t in 2024 (2025 catch report) and 1,326,193 t in 2025 (National Marine Fisheries Service Alaska Region 2026b). The catch table reports two 2026 values: reported catch of 1,297,206 t through October 17 (National Marine Fisheries Service Alaska Region 2026a) and the annual catch estimate of 1,312,504.58 t used in this assessment.

For 2026, adopted ABC and OFL are 2,036,000 and 2,496,000 t. The initial Bering Sea subarea TAC of 1,375,000 t became 1,389,500 t after inseason reallocations; the August allocation table specifies 139,400 t for the Community Development Quota (CDQ) directed fishing allowance and 46,000 t incidental catch allowance. CDQ directed catch and the non-Bogoslof incidental account are each included once in the reported Bering Sea subarea total; the latter already includes incidental CDQ catch. Bogoslof and Aleutian Islands accounts remain separate (National Marine Fisheries Service 2026a, 2026c).

Continuing 2026 measures allocate 45% of Bering Sea subarea directed allowances to the A season (January 20–June 10) and 55% to the B season (June 10–November 1). The non-CDQ directed allowance is divided 50%, 40% and 10% among the inshore, catcher/processor and mothership sectors. The Steller sea lion conservation area limit before noon April 1 is 28% of each sector's annual directed allowance (National Marine Fisheries Service 2026a). These regulatory seasons differ from the week-ending month groups used in several historical fishery displays. An additional 2026 spatial action closed the Winter Herring Savings Area (58–60° N, 172–175° W) to directed pollock trawling, including CDQ, from noon September 30, 2026 to noon March 1, 2027 (National Marine Fisheries Service 2026b).

Management has combined area restrictions, allocations, monitoring and incentives to avoid prohibited species. Selected historical milestones are summarized in Table 2; Council recommendations, voluntary programs and regulatory implementation retain their separate dates. The voluntary rolling-hotspot programs began for chum salmon in 2001 and Chinook salmon in 2002. The Council selected Amendment 84 in 2005, followed by exempted fishing in 2006 and implementing regulations in 2007 (National Marine Fisheries Service 2007, 2023).

Amendment 91's rule became effective on September 29, 2010, and its Chinook bycatch program operated in the 2011 fishery. It combined a 60,000-fish overall prohibited species catch (PSC) limit with incentive plan agreements (IPAs) and sector shares of a 47,591-fish performance standard. A sector retains the higher allocation if it exceeds its share of the standard in at most two of seven consecutive years. Exceeding the standard in three of those years triggers the lower allocation in all subsequent years. Amendment 110 followed Council final action in April 2015 and a final rule effective July 11, 2016. It strengthened Chinook and chum avoidance requirements and linked the following year's Chinook limits to the State of Alaska's index for the Kuskokwim, Unalakleet and Upper Yukon systems. At an index of 250,000 fish or less, the PSC limit and performance standard fall to **45,000 and 33,318 fish**, respectively. The index first informed the 2017 fishing year. The pollock fishery uses the **45% / 55% pollock seasonal allocations** described above (National Marine Fisheries Service 2010, 2016).

Salmon accounting and genetic stock identification serve different purposes. The Bering Sea pollock program requires census counts of salmon in catches, with observer verification at at-sea processors or during offloading. Participating catcher vessels may use compliance electronic monitoring (EM), while observers at shoreside processors collect biological samples. EM verifies handling and discard compliance. Genetic origin is determined from biological samples. Representative biological samples, rather than every counted fish, are genetically analyzed to estimate stock composition. These are fishery-dependent records, separate from the assessment's fishery-independent survey indices (National Marine Fisheries Service 2010). Compliance monitoring and shoreside sampling are described by NOAA Fisheries (National Marine Fisheries Service Alaska Regional Office 2025). The genetic analysis of the 2015 Chinook salmon bycatch illustrates this subsampling approach (Guthrie et al. 2017).

## 3.1 Fishery conditions

The A-season fishery targets pre-spawning pollock and supports roe production. Fishing has historically concentrated north and west of Unimak Island and along the shelf toward the Pribilof Islands, with access affected by ice, weather and fish distribution. The summer–fall B-season fishery follows changes in fish availability across the shelf. Salmon-avoidance measures and vessel choices also influence where fishing occurs. The assessment authors report good fishing conditions in the 2026 A season; later in the B season fishing moved farther north and pollock of desirable sizes were less available.

Catch by area and season is summarized in Table 3. The week-ending extract records 1,267,543 tons through October 17, 2026: 604,041 tons in the A-season grouping and 663,501 tons in the B-season grouping. These seasonal groupings follow the reporting weeks described in the caption.

The annual catch-accounting extract summarizes retained and discarded catch in Table 4. The 2026 total is 1,312,505 tons, including 8,997 tons discarded (0.7%). The annual and seasonal extracts have different coverage and update schedules.

Nominal pollock catch per tow-hour is generally higher in the A season than in the B season, with A-season peaks in 2019 and 2023 (Figure 1). Species-specific patterns differ: A-season Chinook rates peak in 2006–2007, whereas the chum salmon series has its largest B-season rates in 2005 and 2017. Herring and sablefish each have a pronounced peak in 2020, in the A and B seasons respectively. These are nominal rates for available pollock-dominated EBS hauls, calculated from summed catch and valid tow-hours with panel-specific units.

Estimated catch numbers for both sexes decline to a trough in 2009, rise again during the 2010s and reach their largest annual totals in 2021 (Figure 2). B-season numbers generally exceed A-season numbers within each sex, and female annual catch numbers decline from 1,075.8 million fish in 2023 to 916.7 million in 2025, while male catch numbers decline from 1,218.9 million to 952.6 million. Catch numbers are expressed in thousands of fish.

The cumulative monthly depiction shows whether A-season fishing activity has shifted toward earlier or later months within the year. Cumulative catch through April stays near 40% of annual specified TAC over much of 2000–2017 and near 44% from 2018 onward (Figure 3). In 2026 the cumulative shares are 6.69%, 23.14%, 39.42% and 43.72% at the end of January, February, March and April, respectively. Monthly totals are based on weekly catch reports and the specified annual TAC before quota transfers. These observations cover the first four months of the year; the A season continues through June 10.

As expected, A-season roe production exceeds B-season production throughout the 2000–2026 series (Figure 4). A-season production peaks in 2006 and again in 2019, declines through 2021 and rises from 2022 to 2026. The latest A-season value is 19,092.9 t of roe. The 2025 B-season total is 1,795.8 t; the 2026 total of 406.4 t covers June–October and represents a partial-season estimate. The series combines offshore Bering Sea and inshore Bering Sea and Aleutian Islands (BSAI) records in the original January–May and June–December groups.

The broader BSAI fishery's catch, vessel participation and market context are summarized in the ESP socioeconomic tables. Those statistics retain their BSAI, national-trade and global-catch scopes and their latest available observation years.

### 3.1.1 Recent spatial catch patterns

The seasonal maps compare 2024–2026 using the available catch-location records (Figure 5; Figure 6). A-season maps cover January 20–June 9. B-season maps use June 10–September 18 in every year, matching the 2026 data cutoff. Vertical bar heights show recorded catch tons. The rotated, elevated view separates neighboring columns. Each season uses a fixed vertical scale across its three years, with a 10,000-t reference column in each panel. The vertical scales differ between seasons.

A-season fishing remained concentrated on the southeastern shelf, with more catch extending westward in 2025. The share of available observed catch west of 170°W increased from 15% in 2024 to 39% in 2025, then returned to 15% in 2026. The catch-weighted mean latitude changed from 56.07°N to 56.43°N and 56.19°N, respectively. The 2026 pattern therefore shifted back toward the southeastern grounds relative to 2025.

The B-season distribution extended farther north and west in 2025–2026 than in 2024. The share of recorded catch north of 58°N increased from 54% in 2024 to 75% in 2025 and 79% in 2026. The share west of 170°W rose from 76% to about 92% in both later years. Catch-weighted mean latitude was 57.66°N in 2024, 58.56°N in 2025 and 58.52°N in 2026. Thus, the recent northward shift persisted in 2026, while the catch-weighted mean longitude moved eastward relative to 2025, from 174.65°W to 173.26°W.

These summaries describe the available observed catch, including catcher-processors, motherships and observed catcher vessels. The extract omits electronic-monitoring catcher-vessel records, so changes in fleet coverage can also affect the mapped patterns. The maps show catch concentration; the updated fleet-dispersion index below measures the separation among observed tows.

### 3.1.2 Fleet dispersion

Fleet dispersion describes how widely separated fishing locations are within a day (Figure 7). Following the 2024 assessment, we calculate the great-circle distance between each unique pair of tows made on the same day, separately for catcher-processors and observed catcher vessels. Distance sums and pair counts are then pooled by year and season. Days with more tow pairs contribute more to the seasonal average. Larger values indicate a more dispersed pattern of observed fishing locations; smaller values indicate greater concentration.

A-season dispersion increased from 47.8 nautical miles in 2024 to 68.8 in 2025, then declined to 36.5 in 2026. This change is consistent with the 2025 westward extension and the more concentrated southeastern pattern in 2026 shown in the catch maps. For B-season records through September 18 in each year, dispersion declined from 78.5 nautical miles in 2024 to 67.8 in 2025 and 56.5 in 2026. Thus, the observed B-season fishing locations became more concentrated even as much of the catch remained farther north than in 2024. A shift in fishing location and a change in spatial spread describe different aspects of fleet behavior.

The figure retains the historical day-of-year boundary at day 161, which approximates the seasonal transition. Dashed lines provide a common September 18 cutoff to support comparisons with the partial 2026 B season. The full series uses consistent duration calculations and vessel-type selections. Catcher-vessel coverage is incomplete because electronic-monitoring records are absent. Changes in the mix of observed vessels, bycatch avoidance and fishing conditions can influence this index, so it should be interpreted together with catch distribution and catch rates.

### 3.1.3 Seasonal changes in fish size

Following the approach described in Section 3.1.1 of the 2024 assessment, we summarize the mean weight of pollock within each source record and weight that record by its pollock catch. Mean fish weight is the recorded pollock weight divided by the corresponding fish count. Grouping these record means into weight bins shows how the sizes encountered by the fishery change over time (Figure 8).

The dominant A-season mean-weight bin increased from 550 g in 2023 to 600 g in 2024–2025 and 700 g in 2026. For the B season, the dominant bin increased from 450 g in 2023 to 550 g in 2024 and 700 g in 2025, then shifted to 450 g in the available 2026 records. This recent contrast is consistent with the reported difficulty finding larger pollock during the 2026 B season. Changes in these distributions can reflect growth, recruitment, the areas fished and the mixture of vessels represented in the observations.

Weekly displays show how size patterns develop within the A-season and B-season windows (Figure 9; Figure 10). The source extract extends through September 18, 2026, so the B-season results describe the available portion of the season. Catcher-vessel observations exclude electronic-monitoring records and represent only part of the shoreside fleet. Annual panels use a source-week boundary at week 24; the weekly panels cover weeks 2–15 and 25–40. These plotting windows differ from the regulatory season dates. Ridge heights are scaled to reveal distribution shape; catch totals should be compared using the underlying bin totals.

### 3.2 Bycatch

Pacific ocean perch (3,995 t) and Pacific cod (3,315 t) were the largest bycatch categories in the pollock fishery in 2025, followed by flathead sole (1,813 t). Preliminary 2026 estimates through October 24 were 2,412 t of Pacific ocean perch, 6,549 t of Pacific cod, and 3,367 t of flathead sole. See Table 6.

Pollock catch in identified non-pollock fisheries declined from 46,996 t in 2024 to 29,263 t in 2025. Rock sole, yellowfin sole, and Pacific cod fisheries accounted for 10,776 t, 8,710 t, and 6,427 t, respectively, in 2025. The preliminary 2026 total through October 31 was 34,729 t; the different cutoffs and partial-year coverage qualify comparisons with the other bycatch tables. See Table 7.

Prohibited-species bycatch is summarized in Table 8. Chinook catches increased from 8,054 fish in 2024 to 19,825 in 2025, and non-Chinook catches increased from 35,152 to 151,457 fish. Halibut mortality rose from 53 to 66 t. Preliminary 2026 catches through October 24 were 15,153 Chinook, 149,468 non-Chinook, and 56 t of halibut mortality.

Jellyfish dominated the selected nontarget-species catch by weight. Estimated jellyfish catch declined from 4,402 t in 2024 to 1,769 t in 2025. Miscellaneous fish contributed 99 t in 2025, with smaller catches in each of the other displayed groups. Preliminary jellyfish catch through October 24, 2026 was 4,418 t. These catch estimates describe fishery encounters and should be interpreted with fishing effort, distribution, and reporting coverage. See Table 9.

## 4 Data

Fishery catch data begin in 1964, whereas the BTS began in 1982 and the ATS in 1994 for the purposes of this assessment (see the table below). Available fishery age compositions end in 2025, one year before the requested 2026 catch and survey updates.

| Source | Type | Years |
|---|---|---|
| Fishery | Catch biomass | 1964–2025; 2026 year to date and projected annual input |
| Fishery | Catch age composition | 1964–2025 |
| Fishery | Historical trawl CPUE | 1965–1976 |
| Bottom-trawl survey | Biomass index and age-specific proportions | 1982–2019, 2021–2026 |
| Acoustic-trawl survey | Biomass index and age-specific proportions | 1994, 1996, 1997, 1999, 2000, 2002, 2004, 2006–2010, 2012, 2014, 2016, 2018, 2020, 2022, 2024, 2026 |
| Acoustic vessels of opportunity (AVO) | Relative backscatter index | 2006–2019, 2021–2026 |

## 4.1 Fishery data

### 4.1.1 Catch

Annual catch rose from about 0.175 million t in 1964 to the series maximum of 1.875 million t in 1972 (Figure 11). Catch subsequently fluctuated, fell to 0.811 and 0.810 million t in 2009 and 2010, respectively, and recovered to 1.326 million t in 2025. The 2026 records total 1.268 million t through October 17 and represent a partial year. Records before 1991 provide annual totals; later records distinguish the A-season and northwest and southeast B-season catches.

The annual assessment catch includes retained and discarded commercial pollock across gears and targets in the Bering Sea, with CDQ counted once and the separately reported Bogoslof and Aleutian Islands catches excluded. The catch update uses the catch estimates for 1991–2026, with pre-1991 model inputs retained. The revised totals are 1,311,260.74 t in 2024, 1,326,215.61 t in 2025 and 1,312,504.58 t in 2026. The 2026 estimate is the annual input selected for this assessment. Explicit Bogoslof (518) and outside-stock area codes are excluded. The 1998 records with blank area codes remain included under their Bering Sea classification. Noncommercial removals remain separately documented in Appendix C.

### 4.1.2 Age compositions

Fishery catch-at-age estimates for 1964–2025 are summarized in Table 5. Ages 15 and older are pooled as 15+. Where complete annual catch numbers are unavailable, we calculate numbers at age from the assessment proportions, annual fishery weight-at-age and catch biomass.

We estimate fishery age compositions from biological samples collected by at-sea observers and during shore-side offloading. The sampling records include fish measured for length frequencies, paired weight–length measurements,

and age estimates (Table 10; Table 11; Table 12). The processing keeps recent unsexed fish separate rather than assigning them to males or females. Within each sex and sampling stratum, we expand length frequencies to the associated catch biomass using mean weight at length. Age–length keys allocate the expanded numbers to ages; we then sum the numbers across strata and sexes and normalize them to give annual proportions (Kimura 1989; Barbeaux et al. 2005; Ianelli et al. 2024). Earlier assessments provide the pre-1991 age compositions. For the 2024–2025 update, the processing strata are January–May across Bering Sea areas, followed by June–December split into northwest (NMFS reporting areas 520 and above) and southeast (areas below 520). The 2024 assessment describes January–June and July–December groups; the current sampling groups use the calendar periods defined above. The new length and age records cover all 12 months.

The linked bootstrap represents variation among sampling units and among length and age observations within those units. Its composition precision estimates inform the selected likelihood sample sizes: 443, 578 and 530 for 2023, 2024 and 2025, respectively. They represent effective information for fitting the annual proportions, rather than numbers of fish measured or aged (Section 4.3.1). The model fits annual sex-combined proportions at ages 1–14 and 15+. Fishery age observations end in 2025.

The 2024–2025 age-estimate data are from pelagic trawl fisheries. For 2025, FT-NIRS predictions from model a00-d3 are expressed on an independent traditional microscopic-ageing (TMA) basis using 668 paired estimates (Appendix G). Within each FT-NIRS age, the paired observations define the proportions assigned to each tester age. These probabilities redistribute the catch-expanded numbers among ages while preserving their total. Source age 13 lacks paired tester observations; its conditional probabilities are the midpoint of the age-12 and age-14 rows. The 2025 source and tester ages range from 2 to 15, and the assessment fits ages 1–14 and 15+. The likelihood uses the selected input sample size of 530. Uncertainty in the conversion key and the age-13 interpolation is additional to the sampling uncertainty represented by that input weight.

The 2024 fishery age composition is expressed on the independent TMA basis using 693 paired reader–tester observations. For source age 13, the key pools TMA-minus-FT-NIRS age offsets from the same year's source ages 9–12 (34 pairs) and adds those offsets to age 13. Empirical key rows retain their observed probabilities.

### 4.1.3 Weight-at-age

Mean weight-at-age connects the assessment's estimates of fish numbers to catch biomass and harvest projections. Seasonal and spatial differences in fish condition and fishery distribution affect these means. Fishery sampling is therefore organized by season and area, with stratum estimates combined using the corresponding catch information. Completed-year observations provide the basis for estimating weights for the observation year and forecasting the weights needed for future catches.

For recent fishery samples, Fourier-transform near-infrared spectroscopy (FT-NIR) of otoliths provides predicted ages. Year-specific paired FT-NIR and traditional morphological ageing (TMA) observations align these predictions with the TMA age basis. Estimated abundance and biomass in each FT-NIR age bin are redistributed using the same conditional tester-age key. Mean weight at a target age is then converted biomass divided by converted abundance. For the 15+ group, biomass and abundance are pooled before taking their ratio. The conversion retains each source bin's mean weight when allocating it among tester ages. Its reported uncertainty conditions on the selected key; the supported ages and approved tail-allocation rules are documented in Appendix G.

The weight-model approach described in Section 14.5 of the 2024 assessment (Ianelli et al. 2024) links growth within cohorts to annual changes in condition. A smooth mean age pattern defines the typical weight increments. A cohort's weight in the previous year provides the starting point for the next age, while annual random effects allow variation in starting weight and growth. Observed mean weights and their variances inform this model, separating sampling

precision from variability among years. The fitted model supplies predictions for missing observations and forecasts for future years. Model 26.0 uses the resulting weight schedules as fixed biological inputs.

The annual fishery weights in Table 28 retain the historical observational inputs through 2025, with fitted values for missing 2025 ages 13 and 14. The 2026 predictions and 2027–2028 forecasts use the standalone WtAgeRE fit. Ages 1–2 retain the 2024 inputs, the 1964–1990 schedule remains fixed, and the terminal age is 15+.

Cohort growth and annual changes in condition link observed weights to the schedules used for catch and spawning projections. Observed fishery and BTS weights and the selected fishery-scale estimates for 2014–2028 are shown in Figure 23. The forecast follows the 2026 age-3 cohort from 0.368 kg to 0.482 kg at age 4 in 2027 and 0.601 kg at age 5 in 2028. The 2026 biological weight schedules are compared with the selected 2027 catch and spawning projection weights in Figure 35.

## 4.2 Survey data

### 4.2.1 Bottom-trawl survey

The AFSC bottom-trawl survey provides indices of pollock abundance and population age structure, complementing the acoustic-trawl survey's sampling of the midwater component. Historical and recent sampling effort is summarized in Table 13. Annual surveys began in 1979, with standardized gear and methods from 1982 (Ianelli et al. 2024). The 2025 and 2026 observations extend the 42 survey years available for the 2024 assessment to 44 observed years during 1982–2026. The survey was absent in 2020; that year remains a gap in the observation series. The biomass estimates and assessment CVs are listed in Table 14. Availability to bottom-trawl gear varies with age and vertical distribution. Ages 2–3 are often relatively scarce in the catches because juveniles tend to occupy the water column (Monnahan et al. 2021; O'Leary et al. 2022).

Bottom temperature appears to influence the distribution of pollock available to the bottom-trawl survey. The AKsf survey-mapping example (Ianelli 2026) places annual CPUE distributions alongside the 0, 2 and 4 °C bottom-temperature contours for 2018, 2019 and 2021–2026 (Figure 12). High CPUE is concentrated along parts of the western and southern shelf, with several dense patches near the 2 °C contour in 2022, 2024 and 2026. The more extensive warm-water conditions in 2018–2019 contrast with the broader areas of cold bottom water in later panels, while the locations and extent of the pollock concentrations also change.

These patterns suggest that temperature helps structure near-bottom habitat and survey availability, with concentrations often near the margins of colder shelf water. Depth, prey, age composition and vertical movements can also contribute to the pattern. The maps show spatial associations between survey distributions and temperature. Other environmental and biological factors may also contribute. They use locally interpolated, density-dependent-corrected CPUE within survey coverage. The example's correction version and adoption status are unspecified, so its CPUE surface serves as a descriptive illustration alongside the assessment indices.

The standard EBS survey expanded northwest in 1987. Sampling in the separate NBS region has varied among years, while pollock distribution has shifted between regions. Model-based estimates integrate station observations, including zero catches, across the EBS and NBS and account for spatial and temporal variation (O'Leary et al. 2020). The biomass index uses sdmTMB and abundance at age uses tinyVAST, replacing the earlier VAST implementation. The complete selected abundance-at-age series incorporates historical revisions as well as the recent observations

(Table 15). The regional coverage, model specifications, annual covariance and remaining production-source qualifications are described in Appendix H. Estimates for unsampled years are model predictions.

The BTS length distributions show pronounced changes in the sizes represented in the survey (Figure 13). Peaks among small fish and their progression toward larger lengths illustrate cohort growth, while the relative contribution of small and large fish varies among years. These distributions complement the age compositions by showing changes in size directly.

The BTS age compositions show changes in the ages represented in the survey and the progression of cohorts through time (Figure 14). Highlighted bars follow the 1992, 2012 and 2018 year classes through successive ages. The 2012 year class progresses from age 1 in 2013 to age 10 in 2022, while the 2018 year class progresses from age 1 in 2019 to age 8 in 2026. The blank 2020 row marks the gap when the survey was absent.

The age compositions combined traditional age readings and FT-NIRS estimates. We converted the 2026 FT-NIRS estimates to traditional ages using 307 screened paired readings (Appendix G). The conversion preserved total survey abundance and pooled ages 15 and older into the 15+ group.

Spatiotemporal estimation of BTS weight-at-age, adopted in 2022, replaces the earlier conversion of mean length to weight (Indivero et al. 2023). The full BTS-derived weight schedules used for population biomass are presented in Table 16; the compact biological-input summary retains the recent fishery and spawning schedules and fixed biological parameters. We applied the screened 2026 age key consistently to BTS weight-at-age means and their delta-method standard errors, retaining the joint covariance treatment. We used the same key for the provisional ATS abundance and biomass-derived weights, including the weights used to predict the 2026 AVO index. We refitted the standalone random-effects weight-at-age model with the revised BTS means and standard errors under the existing equations, likelihood and source-covariance treatment, preserving fishery observations and the 2025 conversion key. These uncertainty calculations exclude empirical age-key uncertainty. The conversion, weight calculations and assessment use are described in Appendix G.

The missing 2020 survey and subsequent 2021 observation affect interpretation of the time series. Retrospective comparisons describe changes in population estimates as observations accumulate (Section 6.1.5).

### 4.2.2 Acoustic-trawl survey

The 2026 ATS ages are provisional: we apply the BTS age–length key (ALK) to ATS population length-frequency estimates. FT-NIRS ages from the 2026 ATS samples will be available in 2027.

Acoustic-trawl surveys (ATS) are typically conducted every other year and estimate the off-bottom component of the pollock stock, complementing the annual BTS index of near-bottom pollock. Historical and acoustic-trawl sampling effort is summarized in Table 17. The 2026 collections retain their regional scope, while unavailable direct age counts remain NA.

Estimated pollock biomass for the EBS shelf has averaged about 3.202 million t since 1994, when the revised time series begins (Table 18). The revised estimates include the water column down to 0.5 m above the seafloor; the historical midwater index extended to 3 m above the seafloor. Low recruitment during the relatively warm early 2000s was subsequently reflected in lower survey biomass during the cold period of 2006–2012 (Honkalehto and McCarthy 2015). With growth of the strong 2012 year class, biomass exceeded 4 million t in 2014 and 2016, another warm period. The 2026 regional estimates included the Steller sea lion conservation area (SCA) and the Northern Extension (Table 19). The SCA is a traditionally reported subset of the survey used to evaluate pollock abundance in areas considered sensitive to Steller sea lions.

### Historical abundance at age

The historical series showed strong pulses of age-1 fish in 1997, 2007, 2009, 2014 and 2024 (Table 20). The provisional 2026 estimates were largest at ages 2 and 3: 3.73 and 3.96 billion fish, respectively, compared with 2.03 billion age-1 fish.

The ATS age compositions show pulses of young fish and the progression of cohorts through successive surveys (Figure 15). The highlighted 2018 year class appears at age 4 in 2022, age 6 in 2024 and age 8 in 2026. Ages 2 and 3 dominate the provisional 2026 composition. The 2026 estimates use the BTS age–length key applied to ATS population length frequencies; FT-NIRS ages from the 2026 ATS samples are expected in 2027. Blank years represent gaps in the age-composition series.

### 2026 survey design and results

The 2026 ATS covered the EBS shelf from approximately 60 m depth to the shelf break between 30 May and 4 August. Core-area transects were spaced at 20 nmi; selected transects extended into the Northern Extension at 40 nmi spacing. Survey timing was similar to 2022, with a longer duration in 2026 because of special projects. The area east of 170° W was surveyed from 30 May to 25 June, and the area west of 170° W from 26 June to 4 August.

The 2026 core-area estimate was 12.7 billion fish and 3.682 million t of biomass (Table 21). Biomass increased by 28% relative to the 2024 estimate of 2.875 million t (11.4 billion fish). The 2026 biomass estimate was 4% below the 2022 estimate of 3.834 million t (9.67 billion fish) and 16% above the 1994–2024 survey mean of 3.176 million t. These are survey estimates; assessment intake and adoption remain separate decisions.

Provisional 2026 ATS age compositions over supported lengths were estimated by applying the FT-NIRS-based BTS age–length key to ATS abundance-at-length (Appendix G). Midwater pollock were concentrated between the 100 m contour and the shelf break, particularly northwest of Unimak Pass and between Zhemchug and Pervenets canyons. Length distributions suggested that presumed age-1 fish (<20 cm fork length) formed dense schools on the northwestern shelf, with smaller concentrations east of the Pribilof Islands and near Unimak Pass. Presumed age-3+ fish accounted for most biomass, with 74% of their biomass west of 170° W.

### Provisional abundance at age from the 2026 BTS keys

We estimated provisional ages by matching ATS lengths to the 2026 BTS FT-NIRS age–length key, then converting those ages to traditional ages using paired readings. Ages 2 and 3 dominated the resulting distribution (Table 22; Figure 16). Fish beyond the key's length range represented less than 0.001% of core-area abundance. The technical appendix describes the keys, calibration and allocation assumptions.

### Biomass and mean weight-at-age

We applied the same age keys to biomass and divided biomass by abundance to estimate mean weights at age. The resulting mean weights are presented in Table 23 and Figure 17.

### Survey uncertainty and assessment treatment

We estimated ATS biomass sampling error with the established one-dimensional geostatistical method, which accounted for spatial patterns along survey transects (Petitgas 1993; Walline 2007; Williamson and Traynor 1996). The 2026 sampling error was 3.4% of estimated biomass, compared with a historical average of 4.4%. Complete core-area coverage at 20 nmi transect spacing likely contributed to the lower error.

For assessment use, we increased the survey errors to allow for additional uncertainty, including target strength and trawl selectivity. We scaled the assessment coefficients of variation to average 20% over 1994–2026 and assigned a value of 45% to 2000 (Table 18).

#### Spatial and historical patterns

The ATS index used in the assessment covered the water column down to 0.5 m above the seafloor, while the historical heatmap showed biomass from near the surface to 3 m above the bottom (Figure 18). Biomass was low during 2008–2012, especially in 2008–2009, and increased in 2014–2016. The maps also showed shifts in the strength and location of concentrations along the western and southern shelf. The Northern Extension contained an additional 0.078 million t beyond the 3.682 million t core-area estimate (Table 19).

**Biomass time series and observation errors**

The series contained 20 survey estimates between 1994 and 2026. We plotted approximate 95% sampling intervals around each biomass estimate using the survey standard errors (Figure 19; Table 18). Gaps in the annual series marked years without a survey estimate. These intervals described survey sampling variability; the assessment used larger errors to allow for additional uncertainty.

Biomass varied strongly along the survey tracks. Concentrations were highest on parts of the western and southern shelf and generally lower in the far north (Figure 20). The map included the Northern Extension and showed GMT contours of mean biomass per sampling interval (tons per EDSU). The map uses approximately 10-nautical-mile cell means, with the smoothed surface displayed within 30 km of sampled cell centers.

### 4.2.3 Acoustic vessel-of-opportunity index

The acoustic vessel-of-opportunity (AVO) index provides a relative-abundance observation for age-1 and older midwater pollock within the U.S. EEZ (Honkalehto et al. 2011; Lauffenburger et al. 2025). The 2026 observations were collected at 38 kHz from 29 May to 5 August aboard the summer bottom-trawl charter vessels. The index covers the water column from 3 m above the seabed to 16 m below the surface.

**Annual index**

Relative AVO backscatter is lowest at 0.299 in 2009 and highest at 1.55 in 2015 (Table 25; Figure 21). It is 0.863 in 2024, 1.07 in 2025 and 1.12 in 2026, with sampling CVs of 2.72%, 5.26% and 4.55%, respectively. The 2025 and 2026 sampling confidence intervals overlap. The display series covers 2009–2026, excluding 2020, when no bottom-trawl survey took place. Each annual area-integrated estimate is divided by the mean of the 17 estimates to give a mean-one series. The table also retains total pollock backscatter, in millions of square metres, with sampling errors and scaled confidence intervals. This is an acoustic backscattering cross section, rather than geographic area (Lauffenburger et al. 2025).

For the nine paired survey years in the 2009–2026 display period, AVO total pollock backscatter and revised ATS biomass have a squared Pearson correlation of 0.89. This association is descriptive: AVO covers 3 m above the seabed to 16 m below the surface, while the revised ATS index extends from 0.5 m above the seabed to 16 m below the surface. The vertical domains differ. The 2026 AVO report gives 0.90 using an ATS 2026 estimate of 3.62 million t; the comparison here uses the revised ATS estimate of 3.682 million t (AFSC-RACE MACE 2026).

**Spatial distribution and sampling coverage**

Strong backscatter occurs along western EBS tracks in the recent maps, with weaker returns interspersed along central and southeastern tracks (Figure 22). The annual panels use a common map extent and color scale while showing differences in track coverage.

The longer series also contains alternating strong and weak returns along the sampled shelf (Figure 22). The year 2020 is omitted because AVO observations are unavailable. Survey coverage and processing vary among years. Inferences about population redistribution require comparable sampling and integration across years.

The number of observed survey intervals ranges from 5,557 in 2009 to 12,229 in 2024 (Table 26). Recorded-zero shares fall from 56.3% in 2009 to 7.8% in 2015, vary subsequently, and are 14.9%, 5.0% and 14.2% in 2024–2026, respectively.

The maps summarize observed pollock backscatter in approximately 20-nautical-mile cells, including recorded zeros. GMT gridding and contours show the mean backscatter in $m^2$ $nmi^{-2}$ on a common color scale. Surfaces extend up to 45 km from sampled cell centers; more distant areas remain masked. The annual AVO index integrates backscatter

over the survey area and is reported in million $m^2$. Survey coverage and processing vary among years, so comparisons of spatial patterns depend on comparable sampling.

### Assessment use and observation uncertainty

The AVO assessment index is 2.01 million $m^2$ in 2024, 2.50 million $m^2$ in 2025 and 2.60 million $m^2$ in 2026 (Table 27). Its 20 observations cover 2006–2026, excluding 2020. The assessment uses the newly processed 2011 and 2013 estimates and their sampling errors (AFSC-RACE MACE 2026). The 2006–2008 estimates retain the legacy rescaling described in the 2024 assessment (Ianelli et al. 2024).

Survey sampling errors describe one-dimensional geostatistical variability. The assessment applies additional observation-error inflation to an arithmetic mean CV of 23% across its 20 observations. The two newly processed years use the existing modern-year error inflation rule, followed by common recalibration of all 20 CVs to that mean. Assessment errors differ from sampling errors and from the ATS mean assessment CV of 20%.

For the Normal index likelihood, arithmetic observation SEs equal the retained assessment index multiplied by its selected CV. For the display series, let ItI_t denote the original index, I¯\bar I its 17-year mean and Rt=It/I¯R_t=I_t/\bar I. The same constant rescales its index, SEs and confidence limits, leaving the CV unchanged:

SE(Rt)=RtCVt,[Lt,Ut]=It±1.96ItCVtI¯. \mathrm{SE}(R_t)=R_t\,\mathrm{CV}_t,\qquad [L_t,U_t]=\frac{I_t\pm1.96\,I_t\mathrm{CV}_t}{\bar I}.

The source 95% interval is represented by the half-width 1.96ItCVt1.96 I_t\mathrm{CV}_t, checked for every observed year. The displayed intervals describe survey sampling uncertainty with normalization treated as a change of units; assessment error inflation is a separate model input.

## 4.3 Other data

Model 26.0 fitted the 12 historical CPUE observations from 1965–1976 using fishery selectivity and fishery weights at age. We retained the observations and errors from the reference fit through 2024.

#### *Fixed biological assumptions*

We retained the 2024 biological assumptions (Ianelli et al. 2024). Natural mortality was 0.9 $yr^{-1}$ at age 1, 0.45 $yr^{-1}$ at age 2 and 0.3 $yr^{-1}$ at ages 3–15+. Maturity at ages 1–15+ was 0, .008, .289, .641, .842, .901, .947, .963, .970, 1, 1, 1, 1, 1, 1. The model calculated April spawning biomass one quarter-year after the start of the year and applied a female fraction of 0.5 once to sex-combined abundance. We used a 15-by-15 identity matrix for ageing error, matching the archived ADMB configuration. This likelihood assumption treated ages as exact after data preparation, which used FT-NIRS age conversion and age–length keys.

All weight schedules covered ages 1–15+ for the combined sexes in kg per fish. The model used fishery weights to calculate catch biomass, BTS weights to calculate population biomass and A-season weights to calculate spawning biomass.

#### *Population and spawning weights*

Population weights use BTS observations through 2026. Three historical values retain their previous treatment; details are in the technical supplement. For spawning biomass, we replaced the older weights at ages 3–15+ with A-season fitted values for 2024–2025 and projections for 2026. Within the common-year comparison, this update changed the 13 values at ages 3–15+ in 2024. We retained baseline weights at ages 1–2; age-2 maturity was .008.

#### *ATS and AVO weights*

We retained each historical schedule through 2024 and carried its weights forward to 2025. After BTS-ALK and conditional TMA preparation, we calculated provisional 2026 ATS weights by dividing biomass by the corresponding abundance. For age 15+, we pooled biomass and abundance before division. Appendix G documented the supported lengths and unallocated tail. We used the 2026 ATS vector as a proxy for AVO weights, although the historical schedules differed.

Section 4.1.3 described the rationale for changing weight estimation. Weight-schedule uncertainty remained separate from the assessment's conditional parameter uncertainty. Native forecasts repeated the 2026 weight vectors.

#### 4.3.1 Age-composition input sample sizes

The full-data Model 23.0 fit uses fishery composition N=443, 578 and 530 for 2023, 2024 and 2025, respectively (Table 24). These are the rounded continuity-scaled harmonic estimates from the corrected sampler bootstrap, applied in the completed full-data Model 23.0 fit at the author's direction. The bootstrap preserves linked sampling units within fishery strata and uses 10,000 replicates per year. ATS 2026 uses N=94, and BTS retains the selected 2026 N, shown as 131. Fishery N is fixed at 10 in each year from 1964–1977. The three composition-likelihood multipliers are 1.

Model 26.0 fits age compositions with a multinomial model. The baseline retains likelihood sample sizes from the bridge workbook. These likelihood sample sizes differ from counts of fish sampled and from diagnostic effective sample sizes. For every fishery or survey year, the final assessment must provide the raw sampling information, starting sample size, adjustment method, and final sample size used in the likelihood. The 2020 biomass estimate remains a separate fitted survey observation.

One-step-ahead (OSA) residuals compare an observation with its conditional prediction after preceding age categories have been considered. The OSA and Francis comparisons document the selected weights and observed-data residual patterns in Section 6.1.6.

# 5 Analytic approach

## 5.1 Main and comparison models

Model 26.0 (Rceattle) is the adopted model for Tier 3 calculations and specifications. Models 23.0 and 23.2 provide comparisons with the previous assessment and the transition in model structure. Other sensitivities separate the early Foreign fishery (1964–1990) from the later domestic fishery (1991–2026) and vary the penalty on declining selectivity. Model 26.1.2 is the Early-late fishery split sensitivity. The fits and retrospective comparisons hold the process standard deviations fixed at the values estimated by its marginal model. Model 26.0 uncertainty and retrospective diagnostics are described below.

#### 5.1.1 Model 23.0 and impact of new data

As in past years, we incrementally introduced new data relative to the previous assessment while retaining the Model 23.0 formulation. We began with total catch through 2026, incorporating the revised catch estimates for 1991–2026 (2026: 1,312,504.58 t) and extending the model endpoint and parameter and projection horizons to accommodate the added years. At this first step, fishery age compositions and survey observations retained their historical coverage. We then updated weight-at-age and A-season spawning weights, followed by fishery catch-at-age data and their information weights.

The survey updates followed in sequence: the bottom-trawl survey (BTS), the acoustic-trawl survey (ATS), and the acoustic vessels of opportunity (AVO) index. BTS retained the 2020 survey gap and annual information weights, including the real-valued 2026 input. ATS incorporated the selected age-1 rule and provisional 2026 age information, and AVO incorporated its annual observations and assessment-error recalibration. Historical revisions and uncertainty recalibration were included with each survey's update (Section 4.2.2; Section 4.2.3).

Each step includes all preceding updates, so its estimated effect depends on the order in which the data were added. The resulting 2026 estimates are summarized in Table 29, and the cumulative annual point estimates are shown in Figure 24. These comparisons use matched definitions of population-weight biomass, female spawning biomass, age-1 recruitment and maximum realized fishing mortality across ages. Model equations are given in the model-equations appendix.

The saved 2027 spawning-biomass and young-abundance projections from the cumulative input-update analyses are compared in Figure 25. These diagnostic projections retain the earlier 2026 catch estimate of 1,310,000 t; the 2026

estimates above use the revised catch inputs. The display starts with Catch + 2026 endpoint, followed by + Fishery age data and the subsequent survey updates. The intervening weight-at-age and A-season spawning-weight updates remain incorporated in the fishery-age and later fits. Abundance at ages 1–3 is measured at the start of 2027, before removals; spawning biomass includes mortality to spawning time. Projection settings are detailed in the technical supplement. Management projections retain their separate catch, selectivity, weight and recruitment assumptions (Section 7).

The input sequence and configuration comparison answer different questions. The former retains the Model 23.0 formulation while adding observations and updating schedules. The latter compares formulations on matched observations and biology (Table 30).

Approximate distributions compare the change in spawning biomass from 2026 to 2027 (Figure 26) and the historical change in total abundance at ages 1–4 from 2025 to 2026 (Figure 27). They show the same selected cumulative sequence, followed by Models 23.2 and 26.0. Their input and projection settings are documented in the technical supplement. Each distribution retains its matching fitted-parameter covariance between years and, for abundance, among ages. The normal approximations are conditional on the model and biological and projection assumptions.

Among the additions in the saved 2027 projection comparison, ATS produced the largest increase in female spawning biomass: from 2.685 to 2.975 million t, an increase of about 290,000 t (10.8%) relative to the preceding BTS step. AVO increased projected spawning biomass a further 61,000 t (2.0%), to 3.036 million t. The effects on estimated 2026 spawning biomass differed: fishery age data produced the largest increase, about 213,000 t (8.4%); ATS reduced it by about 73,000 t (2.6%), and AVO increased it by about 47,000 t (1.7%). These are successive changes within the specified sequence, with each survey contrast including its historical revisions and error calibration.

### 5.1.2 Model 23.2 fitted to the 2026 assessment data

Model 23.2 is a structural bridge between the ADMB continuity model and the Rceattle configuration. Its annual spawning-biomass and recruitment trajectories retain their documented scope. The established zero-variance management quantities are excluded from management comparisons. Current configuration effects use verified saved population outputs and their stated units. The comparison of Models 23.0 and 23.2 includes matching local asymptotic uncertainty ranges (Figure 28).

### 5.1.3 Model 26.0 fitted to the 2026 assessment data

Model 26.0 uses one fishery and fixed weight-at-age schedules. The updated base model fit retains its native working-likelihood, parameter bounds, selectivity forms, priors and fixed biological assumptions. We check outputs against the written inputs and saved native quantities. Population-weight biomass, female spawning biomass, age-1 recruitment and maximum fishing mortality retain distinct definitions. Their annual estimates and matching local asymptotic uncertainty are compared with Model 23.2 in Figure 29.

#### 5.1.4 Model 26.1: Foreign and Modern domestic fisheries

The split-fishery alternatives distinguish Foreign fishing in 1964–1990 from Modern domestic fishing in 1991–2026. Model 26.1.0 supplies the fitted split-fishery comparison; Model 26.1.1 reduces the declining-selectivity penalty, and the zero-penalty case removes it. Model 26.1.2 extends that formulation by estimating the Modern annual selectivity-process SD while integrating over annual selectivity effects. The Foreign process SD remains fixed. Local uncertainty uses the verified marginal information for this alternative; its observation diagnostics condition on the fitted annual effects.

# 6 Results

## 6.1 Model 26.0 assessment

Model 26.0 provides the main estimates of stock size and the basis for the Tier 3 catch limits presented here. Results from the other model configurations, including Model 26.1.2, show how alternative assumptions affect the assessment. Additional analyses describe uncertainty in Model 26.0 estimates and how those estimates change when recent years of data are removed. Comparisons with past assessments use the data and catch advice reported in each assessment year.

#### 6.1.1 Population estimates and uncertainty

The Model 26.0 base model estimates 2026 total biomass at **12.162 million t**, female spawning biomass at **3.057 million t**, age-1 recruitment at **36.3 billion fish**, and maximum fishing mortality across ages at **0.299 per year** (Table 31).

The local asymptotic 95% ranges in the tables use the revised fit's native covariance. The uncertainty figure is shown in Figure 30. Annual estimates and ranges appear in Table 32.

Regularized-posterior intervals use all 5,000 existing draws, using the final reported summaries (Table 33). The 2026 female spawning-biomass median is **2.946 million t**, with a central 95% credible interval of **2.450–3.584 million t**. Total-biomass median is **11.462 million t**, with interval **9.215–14.719 million t**. Age-1 recruitment median is **31.1518 billion fish**, with interval **12.2880–91.0616 billion fish**. These are the final reported MCMC summaries; production point estimates and asymptotic intervals retain their separate roles.

Numbers-at-age track cohorts through the population (Figure 31). This shows periods of how recruitment and changes in fishing mortality by age affect abundance relative to their mean abundance estimates (and uncertainty) propagates through time. Actual point estimates are reported in Table 34, while the 2026 abundance-at-age estimates (and asymptotic uncertainties) are shown in Table 35.

**Comparison with the 2024 assessment**

The 2018 year class remains a large recruitment point estimate (Figure 32). Its age-1 abundance in 2019 is **87.8 billion fish** in the Model 26.0 base model, compared with **81.1 billion** in the accepted 2024 assessment, a revision of **8.3%**. The revision reflects both updated observations and changes in model structure.

The estimate for the same 2024 female spawning biomass changes from **3.389 million t** in the saved 2024 assessment to **3.526 million t** in Model 26.0, a revision of **4.0%**. Within the trajectory, spawning biomass changes from that 2024 estimate to **3.057 million t** in 2026 (**-13.3%**). Total biomass changes from **10.767 million t** to **12.162 million t** over the same years (**13.0%**). These within-trajectory changes differ from shared-year assessment revisions.

The recruitment comparison uses calendar age-1 year and billion-fish units (Figure 32). Model 26.1.2 results are included to examine how recruitment estimates for the early period change relative to the 2024 assessment. Historical values are point estimates. Current error bars use each model's local asymptotic covariance: the base-model covariance for Model 26.0 and the marginal information for Model 26.1.2. Recent years without a historical estimate are identified. These intervals describe uncertainty around the fitted estimates.

The two assessments project different spawning-biomass paths through 2028 (Figure 33). The comparison retains each assessment's Scenario 2 catch assumptions; the current uncertainty band propagates local assessment-parameter uncertainty.

**Recruitment per unit of spawning biomass**

The log recruits-per-spawner diagnostic from the 2024 assessment is updated using the Model 26.0 estimates (Figure 34). For each year of hatching, age-1 recruitment in the following year is divided by female spawning biomass in the year of hatching. Taking the natural logarithm puts proportional changes on a comparable scale: higher values indicate more recruits per unit of spawning biomass. The series shows substantial variation among year classes. The 1978, 2008 and 2018 year classes have particularly high recruits-per-spawner values. Following the strong 2018 year class, the ratio declined through the 2022 year class; estimates for 2023–2025 are higher but remain less certain. The spawning-biomass panel generally shows lower ratios at higher biomass, with considerable variation among years at similar biomass. This pattern can reflect density dependence, environmental conditions and estimation uncertainty. Spawning biomass appears in both the horizontal axis and the ratio, which can contribute to the apparent relationship. Model 26.0 uses mean recruitment rather than an estimated Ricker curve. The figure provides a descriptive check on productivity using point estimates of the recruitment-to-biomass ratio.

Biological schedules and selectivity retain their age, sex and year definitions (Figure 35; Table 36; Table 37). Fishing mortality is also shown by age (Figure 36). Whole-year spawning exploitation compares paired spawning-biomass calculations from the base model production fit (Figure 37). For the interval from spawning in 2026 to spawning in 2027, its production-point estimate is **13.7%**: **3.326 million t** with fishing versus **3.853 million t** without fishing. The branches share prior fishing history and their spawning-date starting population, with 1.375 million t annual catch assumed in 2027. The separate MCMC posterior median for this exploitation fraction is **14.5%**, with a central 95% credible interval of **11.2–18.4%**. These paired-state calculations carry the revised fishery-weight schedule into 2027 and retain the fixed 1.375 million t catch assumption. The production-fit point remains separate from that posterior interval.

### 6.1.2 Estimated model parameters

The parameter-family summary identifies fitted parameter groups and their asymptotic uncertainty (Table 38). Complete estimates and the largest parameter correlations are retained in the detailed estimates appendix. The summary likelihood contributions are in Table 39.

### 6.1.3 Fits to the observations

The fitted indices follow the main rises and falls in the survey series, including the low ATS biomass in 2006–2009 and its subsequent increase (Figure 38). Annual departures remain. The BTS residuals include negative departures in 2006

and 2009 and positive departures in 2014–2015, each outside the plotted 95% reference range. The AVO observation in 2015 is also higher than fitted, whereas all ATS biomass residuals fall within that range (Figure 39; Table 40). Thus, the broad survey trends are represented more closely than some individual annual observations. BTS residuals account for covariance among survey years.

The fishery selectivity curves show substantial changes in the ages most vulnerable to fishing, with dome-shaped patterns in many years and flatter patterns in others (Figure 40). BTS selectivity generally increases with age and approaches a plateau, while the ATS curve, shared with AVO, is relatively flat at older ages (Figure 41; Figure 42). Recruitment varies strongly among years: the 2018 year class, entering at age 1 in 2019, has the largest positive departure from the fitted recruitment level.

The mean-age comparisons show that the model follows most of the observed changes in survey age composition and closely tracks fishery mean age in the domestic period. The clearest sustained fishery discrepancy occurs in the early years, when predicted mean ages are higher than observed (Figure 43). The observed bars show plus or minus two standard errors of mean age; they describe sampling uncertainty, rather than uncertainty in the fitted population. Fishery age observations end in 2025. The annual composition panels show how these mean ages arise from the proportions at each age and allow individual cohorts to be followed as they age (Figure 44; Figure 45; Figure 46; Figure 47).

Aggregate observed and fitted age proportions agree closely, especially for the fishery, but the residual panels retain patterns by age and year (Figure 48; Stewart and Monnahan (2025); Monnahan et al. (2026)). For example, BTS residuals include extended runs of the same sign at several ages, and the early fishery residuals also show structured departures. Averaging across years can obscure these differences. The OSA residual standard deviations are below one for all three sources (about 0.76 for the fishery, 0.86 for BTS and 0.64 for ATS), and the QQ plots are generally narrower than the standard-normal reference, although some BTS tail residuals are large. Close aggregate fits therefore need to be considered alongside the annual patterns. The afscOSA v0.0.1 displays retain the native Model 26.0 OSAs and selected likelihood sample sizes. They describe conditional observation fit under the working-score approximation; parameter uncertainty is outside their scope.

#### 6.1.4 Profile likelihood diagnostics

A profile examines how strongly the observations and model assumptions constrain ATS survey catchability in Model 26.0. Catchability scales the survey index relative to the population available to the survey. We fixed this parameter at **17 values from 0.45 to 0.65**, including the freely estimated value of **0.51694**, and re-estimated all other active parameters. The grid uses eight equal intervals on each side of the estimated value: approximately 0.00837 below it and 0.01663 above it. Data, bounds, likelihood terms, and penalties match the Model 26.0 fit, including the updated 2011 and 2013 AVO observations (Figure 49).

The best fit remains at catchability **0.51694**. Lower or higher values increase the negative log likelihood (NLL), indicating a poorer overall fit after the other parameters have adjusted. The conventional change of 1.92 NLL units is crossed between **0.45000 and 0.45837** on the lower side and between **0.58347 and 0.60010** on the upper side. Linear interpolation between these points gives approximate crossings of **0.4576 and 0.5838**. These are diagnostic bounds from the sampled profile; the objective includes model penalties, and the crossings have been interpolated rather than directly fitted.

The component breakdown shows how the model balances the different data sources. At catchability 0.45, the total NLL increases by **2.48** units. Index likelihoods contribute 2.36 units, while the combined composition fit improves by 0.43 units. The ATS index and compositions fit less well, while BTS compositions fit better. At catchability 0.65, the total increase is **6.80** units. Compositions contribute 4.50 units, largely from the BTS, and selectivity penalties contribute 1.93 units. The combined index term changes by only 0.14 units because improved BTS index fit partly offsets poorer ATS index fit. Thus, the preferred catchability reflects agreement across surveys, fishery observations, and selectivity assumptions.

All 17 fits have maximum absolute gradients below $10^{-4}$ and positive-definite Hessians. Seven points required additional numerical refinement or successful optimizer confirmation. Weighted NLL components sum to the total objective at every point. Scientific evaluation should also consider sensitivity to composition weighting and selectivity assumptions.

The final ATS-q posterior median is **0.5093**, with central 95% credible interval **0.4514–0.5755**. The profile minimum is **0.51694**, with approximate profile limits **0.4576–0.5838**. Figure 50 compares the negative log ATS-q marginal density with the optimized profile NLL. The kernel density integrates to one, and each displayed curve is centered at its own minimum over ATS q. The marginal integrates over the other parameters and includes posterior regularization; the profile optimizes those parameters under the native penalized objective. Panel B fits weighted quadratic curves directly to individual-draw native NLL components and compares them with the profile components. All 5,000 draws contribute; binning is excluded. MCMC curves are centered at the ATS-q marginal mode and displayed over the central 99% of sampled q. These conditional mean curves have a different interpretation from the negative log marginal density, which has no unique additive decomposition into the native likelihood terms. The smoothers summarize the broad component trends; tail coverage is limited by the available draws.

The paired MCMC draws show the relationships among ATS q, summed abundance at ages 2–5 in 2026, age-1 abundance in 2019 (the 2018 year class), and projected 2027 female spawning biomass (Figure 51). Diagonal panels show their marginal densities. All four quantities are paired within each saved draw.

Projected 2027 female spawning biomass is most strongly associated with the summed abundance of ages 2–5 in 2026 among the quantities shown ($r$=0.84). This contrasts sharply with the 2024 assessment, in which projected 2025 spawning biomass depended heavily on the estimated magnitude of the 2018 year class (Figure 53 of the 2024 SAFE). The 2018 year-class dependence remains in the projection, as shown by its positive correlation with 2027 spawning biomass ($r$=0.36), but is less pronounced than the dependence on the younger fish now approaching or contributing to spawning biomass.

At each fixed ATS catchability, we also project 2027 female spawning biomass and its conditional 95% confidence interval (Table 41). The projection uses 1,375 kt catch in 2027, mean 2023–2025 fishery selectivity, April 1 spawning, the weight schedules, and the age-15+ group. These conditional profile calculations retain the biological schedules of the fitted assessment.

These intervals propagate fitted-parameter uncertainty using the inverse fixed-q penalized Hessian and the delta method on log spawning biomass. They condition on each ATS q and the stated projection assumptions. They exclude uncertainty in ATS q, future recruitment process variation, and weight-at-age schedules. Thus, they describe uncertainty in the estimated 2027 SSB at each profile point.

#### 6.1.5 Retrospective patterns

Across the full nine peels ending in 2017–2025, Model 26.0 female spawning-biomass Mohn's rho is +**0.121** (Table 42). Age-1 recruitment rho is –**0.0985**, and maximum fishing-mortality rho is –**0.0636**. Each statistic averages the relative revision between a peel's terminal estimate and the full-2026 estimate for that same calendar year. The annual revisions are reported in Table 43.

The paired retrospective comparison shows female spawning biomass, age-1 recruitment and maximum fishing mortality for Model 26.0 and the Early-late fishery split Model 26.1.2 sensitivity (Figure 52). Each colored curve removes one through nine terminal years and is compared with its own analysis's full 2026 reference. Model 26.0's full nine-peel statistics are reported above.

The conditional Model 26.1.2 comparison has nine-peel spawning-biomass rho of +**0.0735**, recruitment rho of −**0.133**, and maximum fishing-mortality rho of +**0.00226**. Both configurations show positive average spawning-biomass revisions and negative average recruitment revisions, while their average fishing-mortality revisions differ. These full-window summaries describe the overall direction of retrospective change across all nine endpoints.

The Model 26.0 simulated-data and posterior-predictive retrospective experiments provide separate conditional comparisons (Section 6.3.1).

**Historical retrospectives**

Successive assessments provide different estimates of biomass as observations accumulate and methods change. Following Figure 63 of the 2024 SAFE, Figure 53 compares beginning-of-year age-3+ biomass from earlier assessments with the Model 26.0 series. The 2024 comparison uses the published values in the official SAFE Table 27. There was no assessment fit in 2025.

For this historical comparison, age-3+ biomass is the sum of beginning-of-year numbers at ages 3–15+ multiplied by the corresponding fishery weights at age in each year. This definition differs from total population-weight biomass reported elsewhere in the assessment.

The 2026 assessment estimates age-3+ biomass in 2023 at 11.983 million t, compared with 9.695 million t in the 2024 assessment, an increase of about 24%. The corresponding estimates for 2024 are 10.185 and 9.415 million t, a difference of about 8.2%. The Model 26.0 trajectory declines from 14.438 million t in 2022 to 8.774 million t in 2025, then rises to 13.286 million t in 2026. Earlier assessment trajectories show that both the magnitude and timing of biomass changes have been revised over time.

These historical comparisons combine the effects of new observations, revised biological inputs and changes in model structure. The retrospective peels above instead remove recent years within a specified model framework. The historical figure is therefore a record of assessment revisions, with each series retaining its original assumptions and available years.

### 6.1.6 Age-composition weighting comparison

The age-composition diagnostics suggest relatively minor changes to the weights used in the 2026 assessment (Figure 54; Figure 55). Given the small suggested adjustments, the 2023–2025 fishery age compositions therefore retain input information weights of 443, 578 and 530. These weights determine the influence of the compositions in model fitting. The data section reports the raw numbers of fish sampled.

Composition weighting can be evaluated more fully in future assessments, including alternatives such as the Dirichlet-multinomial distribution, which allows greater variation among age samples than the multinomial model. Such comparisons would help assess whether estimating this additional variation within the model improves the treatment of age-composition data and changes the resulting stock estimates.

## 6.2 Model 26.1 split-fishery alternatives

Models 26.1.0 and 26.1.1 separate the Foreign fishery through 1990 from the Modern domestic fishery beginning in 1991. Their annual selectivity standard deviations are fixed at 0.5, while the declining-age penalty is 12.5 in Model 26.1.0 and 3.125 in Model 26.1.1. The zero-penalty case provides a further comparison. The Model 26.1.2 sensitivity shown here retains the 3.125 penalty and fixes the Foreign and Modern annual process standard deviations at 0.1235 and 0.582, respectively. The remaining parameters and annual selectivity effects are estimated conditionally on these values.

The likelihood comparison separates data-likelihood contributions, penalty terms and the marginal fitting criterion (Table 44). The projections use the revised ordinary fits and the Early-late fishery split Model 26.1.2 comparison. The alternatives retain matched observations, likelihood weights and projection assumptions (Table 45).

#### 6.2.1 Selectivity

Separating the Foreign (1964–1990) and Modern domestic (1991–2026) fisheries changes the fitted selectivity most visibly near the boundary between the two periods. Over much of the later period, Models 26.0 and 26.1.0 retain similar annual age patterns, including the peaks and declines at older ages (Figure 56). The split allows the Foreign curve to differ more freely from the Modern curve, while Model 26.0 represents the history as a single fishery.

Both models estimate a shift from 1990 to 1991 toward lower relative selectivity at younger ages and higher relative selectivity at ages 11 and older. The shift is larger in Model 26.1.0: at age 5, the change is about −0.69 on the native mean-one scale, compared with −0.31 in Model 26.0; at ages 12 and older, the increases are about 0.55 and 0.24, respectively (Figure 57). This comparison shows how the split changes the estimated transition. The fitted curves alone establish neither an abrupt change in fishing practices nor an improvement in the population estimates; the observation fits and retrospective results provide the complementary checks below.

The full-history curves are divided by each year's maximum so that their shapes can be compared. The boundary-year display instead retains native mean-one normalization, and its differences refer to that scale. These are fitted point curves. The Model 26.1.2 curves used in subsequent comparisons retain the fitted selectivity states from its fixed-SD conditional fit and remain distinct from the Model 26.1.0 comparison here.

#### 6.2.2 Fishery age-composition fits

We compared Models 26.0 and 26.1.2 using the same 62 annual fishery age compositions and likelihood weights (Figure 58; Figure 59). Model 26.0 fitted one fishery across the full record; Model 26.1.2 separated the Foreign fishery through 1990 from the Modern domestic fishery beginning in 1991. We displayed both models over those same periods to compare their fits directly. Both models reproduced the dominant age groups in the combined compositions, while the annual plots revealed differences that the combined summaries obscured.

Model 26.1.2 fitted the Modern domestic observations more closely. It reduced the sample-weighted root-mean-square difference in age proportions from **0.693 to 0.590 percentage points** and the root-mean-square error in annual mean age from **0.123 to 0.0963 years**. Both models predicted a younger mean age than the observations showed in 1991, but Model 26.1.2 reduced that discrepancy from **0.510 to 0.339 years**. The smaller differences across ages and years showed that the split-fishery configuration improved the modern fit rather than simply matching its combined age distribution.

Model 26.0 fitted the Foreign-period observations more closely. Model 26.1.2 increased the sample-weighted age-proportion error from **2.67 to 3.43 percentage points** and the annual mean-age error from **0.655 to 0.696 years**. Both models predicted older mean ages than the early observations showed, particularly in 1964–1965 and 1970. The age-by-year panels showed persistent overprediction at several older ages and underprediction at younger ages during much of the early record. Separating the fisheries therefore improved the modern fit while leaving larger discrepancies in the Foreign period.

## 6.3 Further model evaluations

#### 6.3.1 Observation-error retrospective self-test

In response to Plan Team and SSC requests to investigate retrospective patterns, we developed a new simulation-based retrospective experiment for this assessment (Figure 60). Its distinctive feature is a paired design that examines revisions both at the fitted parameter estimates and across plausible parameter states from the regularized posterior. One set of ten simulated datasets uses the fitted parameters; a second set uses ten posterior parameter draws. Each dataset has its own full 2026 fit and nine optimization peels ending in 2017–2025. Revisions compare each peel's terminal estimate with its own full fit for the same calendar year. This approach extends the usual retrospective analysis by examining how

observation variation and uncertainty in the generating parameters influence the revisions, providing additional context for interpreting the assessment's retrospective pattern.

Across all nine peels ending in 2017–2025, Model 26.0 assessment Mohn's rho for female spawning biomass is **+0.121**, compared with median values of **−0.0117** for datasets generated at the fitted parameters and **−0.0574** for datasets generated from posterior draws. The assessment value exceeds every simulated rho in both sets: the fixed-parameter range is **−0.115 to +0.0335**, and the posterior-draw range is **−0.0971 to +0.116**. Thus, the positive average spawning-biomass revision in the assessment is stronger than that reproduced by these simulated datasets.

For age-1 recruitment, the assessment's nine-peel rho is **−0.0985**, within both simulated ranges. Median recruitment rho is **−0.00751** in the fixed-parameter set and **+0.0830** in the posterior-draw set. The contrast between spawning biomass and recruitment shows that the retrospective pattern depends on the population quantity considered.

The paired design explores how observation variation and uncertainty in generating parameters contribute to the full nine-peel pattern. Ten datasets per branch provide an exploratory comparison. The technical supplement describes the fixed process history, data-generation approximations and other conditioning assumptions. These experiments assess biomass and recruitment revisions. Confidence bands describing retrospective ABC or OFL patterns are beyond the scope of this work but would be interesting to evaluate.

### 6.3.2 Posterior diagnostics and predictive checks

The regularized posterior explores parameter uncertainty separately from the production base-model estimates (Figure 61; Figure 62; Figure 63). Four chains each retained 1,250 draws after 1,000 warmup iterations. Combined parameter diagnostics give maximum R-hat 1.024 and minimum bulk ESS 242, with zero divergences and zero maximum-treedepth events. The reported population quantities mix more consistently: maximum R-hat is 1.0013 and minimum bulk ESS is 3,914. MCMC standard errors of the 2026 spawning-biomass and biomass means are about 4,110 t and 20,600 t; their first-half versus second-half mean shifts are less than 0.03 posterior standard deviations. These diagnostics describe the saved chains and retain the recorded parameter-level qualifications.

**Final posterior summaries**

The posterior mean female spawning biomass in 2026 is 2.965 million t, with a central 95% credible interval of 2.450–3.584 million t (Table 46). These summaries use all 5,000 retained MCMC draws under the regularizing priors and fixed biological inputs. Full-precision results are downloadable as CSV files containing terminal stock and selectivity summaries (CSV), annual population intervals (CSV), summaries of all 1,266 parameters (CSV), and spawning-exploitation intervals (CSV).

Predictive checks ask whether the fitted model can generate data that resemble the observations (Figure 64; Figure 65). We selected 1,000 MCMC draws spanning the four chains. Each draw supplies a possible set of model parameters, from which a new set of observations is simulated. The shaded ranges therefore include both parameter uncertainty and the sampling variation assumed by the model. Points outside these ranges, or repeated patterns of differences, help identify features the model struggles to reproduce. The ranges apply to each age or year individually; some points outside them are expected even when the assumptions are appropriate.

For the indices, simulated observations use the same error assumptions used in fitting, including the relationships among BTS errors. The overall BTS mismatch measure is as large as the observed measure in only 0.2% of the simulations, indicating that this feature of the BTS data is rarely reproduced. Normal error assumptions also allow some simulated indices to be negative.

For age compositions, simulated fish are randomly assigned to age groups using each draw's predicted proportions and the sample size used in fitting, rounded to a whole number. The simulations approximate the fitting method, which adds a small constant (0.001) to age proportions. The figure compares the latest three actual samples for each source with these simulated age proportions. Age conversions, weights and other biological inputs are held fixed. These checks assess how well the model reproduces observations; the retrospective analyses address how estimates change as additional years of data become available.

# 7 Projections and harvest recommendations

## 7.1 Proposed Tier 3 configuration and reference points

The proposed fishery selectivity is the mean of the 2023–2025 annual curves, the latest three years with fishery age observations. Each age is averaged before scaling the resulting curve to a maximum of one. The separately inferred 2026 curve is distinguished from directly age-informed years (the projection-input appendix).

The deterministic recruitment level uses the arithmetic mean of age-1 entries in 1978–2023, corresponding to birth cohorts 1977–2022. The less precise 2024–2026 entries are excluded. Every entry in the advice tables equals this selected historical mean. The separate projection figures add future recruitment variation to illustrate a range of possible paths. Reference points and fishing rates are reported in Table 47. The Model 26.1.2 column uses the revised Early-late fishery split comparison under the same projection assumptions.

The Tier 3 reference points are spawning-potential quantities. The ABC rule reduces fishing mortality below B40% and sets it to zero below B20%. Projections start from each fitted population and use the stated catch scenarios and biological inputs. The author-rate fraction is one, subject to the risk evaluation and final review.

The previous-year OFL-equivalent fishing mortality is reported in Section 10.

## 7.2 Standard harvest scenarios

The proposed projections retain the standard harvest scenarios and their common biology (Table 48). The expected-catch scenario assumes 1.375 million t in each of 2027 and 2028; the 2028 catch is a planning assumption. Short-term outputs are reported in Table 49.

The catch and female spawning-biomass panels compare Alternative 2 with recent-mean-F Alternative 3 (Figure 66; Figure 67). Black lines show the means of 1,000 future-recruitment simulations, ribbons show their 5th–95th percentiles, and ten colored trajectories illustrate individual outcomes. Each color follows the same recruitment realization across both alternatives and figures. Recruitment follows the established inverse-Gaussian method, using the arithmetic and harmonic means of the fitted 1978–2023 age-1 entries. Initial age-1 abundance in 2027 equals the historical arithmetic mean; each subsequent entry varies among simulations. The widening bands show how future recruitment variation accumulates as new cohorts enter the fishery and spawning population. Alternative 2 catch paths coincide in 2027 and 2028 because those catches are prescribed.

The projection tables retain all seven scenarios with recruitment fixed at the historical arithmetic mean (Table 50; Table 51; Table 52). The historical phase plot places the fitted 1981–2026 trajectory and deterministic Alternative 2 projections for 2027–2028 on common axes (Figure 68). The horizontal axis is female spawning biomass divided by B35%; the vertical axis is maximum fishing mortality at age divided by F35%. Both denominators are the current primary projection reference points (B35% = 2.155 million t; F35% = 0.658 per year), held constant across the trajectory. This common benchmark shows changes in the fitted stock and fishing mortality through time. The 2024 assessment used annually varying reference points; the historical positions here therefore also reflect the choice of a common current benchmark. F35% is a spawning-per-recruit proxy for FMSY, and B35% is the corresponding spawning-biomass reference point. Solid lines mark ratios of one, and the dashed vertical line marks half B35%. Two-digit year labels trace the trajectory, with the two projection years boxed and shaded yellow. The fitted 2026 position and both projected positions are above B35% and below F35%.

### 7.2.1 Decision table

As in past assessments, we present estimated consequences and probabilities for a range of future catches and the subsequent fishing effort implied by each choice (Table 53; Table 54). The decision table compares eight possible 2027 catches, from 10 to 2,540 kt, using Model 26.0 with the updated AVO data. The indicator definitions identify the quantities, comparison benchmarks and years used in the calculations. Together, the tables show how catch choices affect fishing mortality, future spawning biomass and age structure, and how likely each choice is to meet the specified benchmarks.

Each catch column represents a different fishing strategy. Fishing mortality is calculated to achieve that catch in 2027, then held constant through 2031. Catches in subsequent years therefore change with the projected abundance and age composition of the stock. The effort indicator gives the probability that mean age-specific fishing mortality in 2027 exceeds its 2026 value. Interpreting it as relative effort assumes that catchability and the age pattern of fishery selectivity remain fixed. It provides a comparison among strategies rather than a forecast of vessel numbers, fishing days or operating costs.

Readers can compare columns to examine the trade-off between the initial catch and subsequent stock conditions. The probability rows summarize the chance of crossing the stated biomass, fishing-mortality or age-structure benchmarks. These are conditional comparisons that support catch decisions alongside the standard harvest projections and risk table.

Entries are approximate chances out of 100. They propagate uncertainty in the fitted population, selectivity, estimated mean recruitment and comparison benchmarks. Future recruitment is held at the estimated 1978–2023 mean, and biological schedules are fixed. These probabilities describe uncertainty in the assessment estimates under each catch choice. Additional year-to-year recruitment variation and uncertainty in weights and maturity are outside these calculations. Values displayed as 0 or 100 reflect rounding.

The 1,270-kt column is retained for comparison with the earlier decision-table analysis. The expected-catch scenario elsewhere in this section assumes 1,375 kt in 2027 and 2028. The decision tables provide a broader comparison of catch choices; they use local asymptotic uncertainty rather than MCMC draws.

### 7.2.2 Reference point sensitivity to fishery selectivity specification

Fishery selectivity determines how fishing mortality is distributed across ages and therefore affects both the reference fishing rates and the catches obtained at those rates. The central projection uses the age-specific arithmetic mean of the raw 2023–2025 fishery selectivity curves, rescaled so that its maximum is one. This recent three-year average smooths annual variation while retaining the age pattern from the latest years with fishery age observations. Averaging precedes rescaling because normalizing each annual curve first would give a different average. Figure 69 introduces this recent-average specification and contrasts it with individual annual estimates. The inferred 2026 curve is shown separately because fishery age observations end in 2025. It provides a terminal-year comparison, while the management projections retain the 2023–2025 average.

The recent curves show that changes in fishery recruitment involve particular ages rather than a uniform shift toward younger fish. The first ascending half-maximum crossing was near age 6 in 2021–2022, compared with approximately 4.5, 5.0 and 5.2 in 2023–2025. Relative selectivity at age 5 was especially high in 2023 and then declined in 2024–2025. Selectivity at ages 1–3 remained low and varied independently. These differences matter because a curve that exposes age-5 fish more strongly can produce a different catch and spawning impact from one that concentrates fishing at older ages, even when both have the same maximum selectivity.

The full annual comparisons show which parts of each curve differ from the projection assumption (Figure 70). The repeated 2023–2025 reference makes shifts in the ascending limb, peak and older-age tail visible across 2007–2026. For example, the 2019 curve peaks near age 7 and declines substantially at older ages, whereas several earlier curves retain high selectivity across the older ages. Thus, the annual panels provide the age-specific explanation for the management differences in the next figure. The older curves represent alternative fishing patterns for sensitivity analysis. Their relevance as forecasts depends on evidence about current and future fishery conditions.

Annual selectivity alternatives give both higher and lower catch ceilings than the recent-average baseline (Figure 71).

Of the twenty curves, 14 produce a higher maximum ABC in 2027 and 12 do so in 2028. Changes range from -8.2% to +12.0% in 2027 and from -5.9% to +8.4% in 2028. These are conditional sensitivities under common population, recruitment, biological and catch assumptions.

The lowest annual-curve maximum ABC occurs with the 2018 curve in 2027 and the 2019 curve in 2028. The 2025 curve changes the ceilings by +1.5% and -4.4%, respectively. The age pattern affects immediate catches and the ages surviving into the following year. F40% and F35% are recalculated for each curve; the assumed catch remains 1.375 million t in each year.

The younger-age summaries explain why a single description such as "younger selection" is insufficient (Figure 72). They separate exposure of ages 1–3, selectivity at ages 4 and 5, the first ascending half-maximum age, and changes across the full curve. For example, from 2022 to 2023 the half-maximum age falls from about 6.0 to 4.5 years, while the curve-weighted mean age changes only from 10.09 to 10.00 years. Conversely, from 2023 to 2025 the half-maximum age rises while the curve-weighted mean age falls. The ascending limb and older-age tail therefore move in different directions. This figure connects the detailed annual panels to the reference-point results: the age at which fish begin entering the fishery and the relative exposure of older fish carry different information. Its mean-age measure weights selectivity equally across model ages 1–15, with age 15 representing the plus group; population abundance and catch composition are excluded from that shape index.

The reference-rate comparison makes the dependence on age selection explicit (Figure 73). Under fixed biology, each curve has its own fully selected F35% rate, the fishing-mortality scalar that reduces lifetime spawning output per recruit to 35% of the unfished level. Later exposure can allow more spawning before capture and therefore support a higher fully selected rate at the same spawning-per-recruit target. However, the full curve, including its older-age tail, determines this response. The high 2019 F35% illustrates how a reference scalar can rise sharply under a particular age pattern. Such an increase describes the rate permitted under that pattern; it provides no independent evidence of increased stock productivity or a larger sustainable catch.

Two distinctions are essential when interpreting an FMSY proxy against selected age. First, F35% is a spawning-per-recruit proxy for FMSY, whereas FMSY itself depends on the equilibrium yield and recruitment relationship. The historical FMSY comparison and this F35% diagnostic therefore represent different reference quantities. Second, fully selected F is a scalar: fishing mortality at each age equals that scalar multiplied by selectivity at that age. Comparisons among years require the same selectivity normalization and the corresponding age-specific mortality pattern. The scatterplot uses a selectivity-weighted age index over ages 1–10, whereas its spawning-per-recruit calculation includes ages 1–14 and 15+, fixed natural mortality, maturity and the selected projection spawning weights, and quarter-year spawning timing. Its age index also differs from the ages 1–15 index in the preceding figure and from the abundance-weighted age of selected fish. It is a compact descriptor of curve shape, and a common mean selected age can accompany different reference rates. The complete selectivity curve and resulting catch and spawning effects provide the appropriate basis for interpreting the FMSY proxy and comparing harvest specifications.

# 8 Risk table and ABC recommendation

The author risk evaluation retains all four categories at **Level 1 – Normal** (Table 55). The ecosystem and fishery observations are summarized below. Working recommendations remain subject to author scientific evaluation and Plan Team, SSC and Council review. The Level 1 evaluation supports no reduction from maximum permissible ABC.

Table 55: 2026 risk-category scores and supporting summaries. Assessment-related and fishery-informed stock evaluations are Level 1 – Normal; population dynamics and ecosystem evaluations draw on the synthesis by Elizabeth Siddon and Kalei Shotwell.

| Considerations | | | |
|---|---|---|---|
| Assessment-related | Population Dynamics | Ecosystem | Fishery-informed Stock |
| Level 1 – Normal | Level 1 – Normal | Level 1 – Normal | Level 1 – Normal |
| Model developments improved fitting, with minor implications for stock status and projected ABC/OFL. | Egg and larval abundance was low; later prey conditions were favorable and cannibalism was low. | Cool conditions, abundant prey and improving adult condition indicated favorable ecosystem conditions. | A-season fishing appeared good; later B-season fishing was farther north than normal, with limited desirable sizes. |

## 8.1 Assessment-related considerations

Model 26.0 (Rceattle) is the adopted assessment model and main proposal for Tier 3 calculations and specifications. The alternative models examine how the assessment responds to changes in its structure and treatment of annual selectivity variation. Comparisons with Models 23.0 and 23.2 connect the current assessment to the previous formulation and help distinguish the effects of updated observations from changes in model structure. The split-fishery alternatives test whether separate selectivity patterns for the Foreign fishery in 1964–1990 and the Modern domestic fishery from 1991 onward better represent the fishing history. Further comparisons vary the constraint on declining selectivity and the treatment of year-to-year changes in selectivity.

Model 26.1.2 provides the principal sensitivity to these choices. Its underlying marginal formulation estimates the variability of Modern fishery selectivity while accounting for annual selectivity effects. The conditional comparison presented here fixes the Foreign and Modern process standard deviations at their selected full-data values and estimates the remaining parameters and annual selectivity effects. This configuration fits the Modern fishery age compositions more closely than Model 26.0, while larger discrepancies remain in the Foreign period. The result identifies a potential benefit of separating the fisheries, alongside a trade-off in how the model represents the earlier observations.

The Model 26.1.2 retrospective analysis asks whether estimates change as the most recent observations are successively removed. Each shortened-data fit holds the process standard deviations fixed and is compared with a full-data fit using the same conditional estimation method. This matching reference makes the retrospective differences interpretable within the fixed-SD configuration. The diagnostic therefore describes stability when the assumed magnitude of annual selectivity variation is held constant. Re-estimating that variation in each shortened dataset could change the retrospective pattern, and uncertainty in the process standard deviations remains outside this comparison. Improved fits to the Modern observations consequently provide one part of the model evaluation; broader retrospective reliability requires evidence across the relevant estimation choices. Model 26.0 remains the basis for the Tier 3 specifications, with Model 26.1.2 used to assess sensitivity to fishery structure and selectivity assumptions.

## 8.2 Population dynamics risk summary

Population dynamics risk is **Level 1 – Normal**. Cold spring conditions coincided with low egg and larval abundance; temperature-dependent timing complicates interpretation of these early-life indices. Southeastern Bering Sea (SEBS) age-0 abundance was also low. Above-average late-summer large copepod abundance, lipid-rich copepods and euphausiids, and above-average SEBS age-0 energy density indicate favorable feeding and potential survival through the first winter. Northern Bering Sea (NBS) age-0 energy density was below average, while adult-pollock cannibalism was low overall; age-1 specificity remains unverified. The regional contrasts and supporting evidence are described in Section 11.1 within Section 11.

## 8.3 Environmental and ecosystem risk summary

The environmental and ecosystem synthesis supports a **Level 1 – Normal** ecosystem score. Following the warm period of 2014–2021 and near-average thermal conditions in 2022–2025, the EBS shelf shifted to cool conditions in 2026. The expanded cold pool, abundant lipid-rich prey and improving adult condition provide favorable context for juvenile and adult pollock. Increased pollock biomass may increase competition; prey and condition indicators suggest adequate food, while the effects of increasing sablefish abundance remain a research question. Section 11 presents the detailed distribution, environmental, prey, competitor and predator evidence, including conditions for later juveniles and adults in Section 11.2. These category scores support the recommendation for no reduction from maximum permissible ABC.

## 8.4 Fishery-informed Stock Considerations

Fishing conditions differed between the two seasons. The assessment author's observations indicate that A-season fishing in 2026 appeared good. Later in the B-season, fishing occurred farther north than normal, and pollock of desirable sizes were limited. The late-season observations therefore qualify the favorable A-season picture and identify changes in where fishing occurred and the sizes available to the fishery.

We retain the **Level 1 – Normal** score for fishery-informed stock considerations. This score reflects the author's qualitative assessment of fishing conditions across the year. The observations provide context for interpreting fishery performance; their implications for stock-wide abundance remain uncertain. A standardized comparison of fishing locations, effort and size composition would help distinguish changes in fishery availability from changes in overall stock abundance.

## 8.5 Ecosystem and Socioeconomic Profile development

Development of the EBS pollock Ecosystem and Socioeconomic Profile (ESP) continues alongside the assessment (Shotwell et al. 2026). The archived September 2026 development report supplies exploratory background, and the AI-assisted draft appears as Appendix B. The 2026 Eastern Bering Sea Ecosystem Status Report describes an external Dynamic Structural Equation Modeling (DSEM) analysis that identified **sea ice extent**, **sea surface temperature (SST)** and **summer large copepods** as indicators with statistically supported relationships to age-1 recruitment (Siddon 2026). The exploratory Rceattle DSEM pilot in the appendix uses late-summer SST and cold-pool extent as two environmental proxies. Its scope supports a separate assessment of those proxies. Evaluating all three external relationships within Rceattle requires additional testing.

The first application of DSEM has been completed as a draft in the ESP draft. Development and implementation of an integrated version within Rceattle continue from the work presented in the September 2026 development document (Ianelli et al. 2026). We see the Ecosystem Status Report (ESR), the ESP, and integration of ecosystem information within the full assessment model as logical applications of ecosystem-based methods for providing informed catch advice to the Council.

Since all four categories are Level 1 – Normal, we recommend no reduction from maximum permissible ABC.

# 9 Status determination

Under the Magnuson–Stevens Fishery Conservation and Management Act, stock status is evaluated for overfishing, an overfished condition, and the risk of approaching an overfished condition.

For Tier 3, B35% is the BMSY proxy. MSST is evaluated using the biomass and projection tests below. Alternative 6 fishes at the OFL rate in every projected year. Alternative 7 fishes at the maximum-ABC rate in 2027–2028 and the OFL rate from 2029 onward.

Overfishing in 2025. Catch of 1,326,215.61 t was below the specified Bering Sea subarea OFL of 2,957,000 t. Thus, the stock was not subjected to overfishing in 2025.

Overfished in 2026. Biomass below half B35% indicates an overfished condition; biomass above B35% indicates the stock is above MSST. Between those levels, status depends on Alternative 6 biomass in 2036: below B35% indicates an overfished condition; otherwise, the stock is above MSST. Current biomass is 3.057 million t for Model 26.0 and 3.065 million t for the Early-late fishery split Model 26.1.2 sensitivity, above their respective B35% values of 2.155 and 2.177 million t. Both therefore indicate the stock is not overfished; the 2036 test is unnecessary.

Approaching an overfished condition. Under Alternative 7, 2028 biomass below half B35% indicates an approaching overfished condition; biomass above B35% indicates it is not approaching that condition. If that biomass falls between half B35% and B35%, use Alternative 7 biomass in 2038 for the long-term test. Biomass below B35% in that long-term test indicates an approaching overfished condition. Projected 2028 biomass is 3.123 million t for Model 26.0 and 3.179 million t for Model 26.1.2, both above B35%. Neither model indicates an approaching overfished condition; the 2038 test is unnecessary.

These comparisons use deterministic historical-mean recruitment. Assessment conclusions and adopted catch specifications remain separate from final NMFS status determinations.

Status comparison (female spawning biomass in tons)

| Quantity | Year / basis | Model 26.0 | Model 26.1.2 |
|---|---|---|---|
| Estimated female spawning biomass | 2026 | 3,057,000 | 3,065,000 |
| BMSY proxy: B35% | Reference point | 2,155,000 | 2,177,000 |
| Lower status boundary: half B35% | Reference point | 1,078,000 | 1,088,000 |
| Alternative 6 female spawning biomass | 2036 | 2,263,000 | 2,288,000 |
| Alternative 7 female spawning biomass | 2028 | 3,123,000 | 3,179,000 |
| Alternative 7 female spawning biomass | 2038 | 2,263,000 | 2,288,000 |
| Maximum ABC: expected-catch scenario | 2027 | 1,896,000 | 1,855,000 |
| OFL: expected-catch scenario | 2027 | 2,328,000 | 2,269,000 |
| Maximum ABC: expected-catch scenario | 2028 | 1,968,000 | 1,958,000 |
| OFL: expected-catch scenario | 2028 | 2,422,000 | 2,399,000 |

B35% is the Tier 3 BMSY proxy. The 2036 Alternative 6 and 2038 Alternative 7 values are shown for completeness; their conditional tests are unnecessary here. ABC/OFL rows use the expected-catch scenario with 1,375,000 t in 2027 and 2028. Model 26.1.2 is the sensitivity comparison; adopted specifications remain separate.

# 10 F limit

Following section 4.15 of the assessment guidelines, F limit is reverse-engineered as the annual fishing mortality that would have produced the previous year's OFL using the recommended model's estimated population for that year. For this 2026 assessment, the calculation therefore uses the adopted **2025 Bering Sea pollock OFL of 2.957 million t** (National Marine Fisheries Service (2025)) and the Model 26.0 beginning-of-year 2025 numbers at age, fishery weights, natural mortality and realized selectivity.

Solving the annual Baranov catch equation gives **F limit = 0.787 per year** on the maximum-F-across-ages scale used in this report, with the maximum at age 6. The equivalent scalar on Rceattle's native mean-normalized selectivity scale is **0.439 per year**. Both scalings give the same fishing mortality at age and reproduce the adopted 2025 OFL. This calculation holds the fitted population and biological inputs fixed while changing fishing mortality.

The calculation finds the value of $F_{limit}$ satisfying

$$C_{2025}^{OFL} = \sum_{a=1}^{15+} N_{a,2025} w_{a,2025}^{fishery} \frac{F_{limit}^{*} s_{a,2025}}{M_{a,2025} + F_{limit}^{*} s_{a,2025}} [1 - \exp\{-(M_{a,2025} + F_{limit} s_{a,2025}^{*})\}] \cdot (1)$$

Here $N$ is beginning-of-year abundance, $w^{fishery}$ is fishery weight-at-age, $M$ is annual natural mortality, and $s^{*}$ is the fitted selectivity curve rescaled to a maximum of one. Thus $F_{limit}$ is the maximum annual fishing mortality across ages. The sum includes ages 1–15+. Abundance in millions of fish and weight in kg give catch in thousand tons.

# 11 Ecosystem considerations

A draft Ecosystem and Socioeconomic Profile (ESP) is presented in Appendix B. It brings together indicators of pollock habitat, food-web conditions and fishery performance to provide context for the assessment. The ESP complements the population model by organizing evidence about processes that can affect year-class strength, fish growth and the distribution of fish available to surveys and the fishery. This broader perspective helps interpret changes in abundance and age composition alongside the environmental conditions experienced by successive cohorts.

The draft 2026 Eastern Bering Sea Ecosystem Status Report provides the environmental and food-web evidence summarized here (Siddon 2026). In 2026, cool spring conditions and an extensive cold pool coincided with low egg, larval and southeastern Bering Sea age-0 abundance. At the same time, favorable zooplankton abundance and lipid content, together with above-average southeastern Bering Sea age-0 energy density, indicated good feeding conditions for surviving juveniles. These contrasting signals emphasize the importance of timing, region and life stage: low early abundance and favorable later feeding conditions can occur within the same year. Their implications for recruitment depend on survival through the subsequent winter and later juvenile stages.

The following subsections connect these observations to pollock population processes. The first considers environmental conditions, prey, competitors and predators affecting eggs, larvae and early juveniles. The second addresses the distribution, feeding conditions and condition of later juveniles and adults, including how temperature and vertical habitat use can affect survey availability. Together, these indicators help distinguish changes in population abundance from changes in where fish occur and how effectively surveys sample them. The synthesis informs interpretation of the assessment and the risk-category evaluation summarized in Section 8; the individual indicators provide complementary evidence rather than a direct forecast of recruitment or an automatic adjustment to catch advice.

## 11.1 Ecosystem impacts of EBS pollock population dynamics

**Environmental Processes**

A very weak winter Aleutian Low in 2025/2026 brought cool Arctic air across the EBS (Lemagie and Wang 2026). **Sea ice extent** remained above average for six weeks during March–April 2026 (Thoman 2026). The resulting 2°C cold pool was approximately 150% larger than in 2025 and the fifth-largest during 1982–2026; the 0°C cold pool was the largest since 2013 (Rohan and Barnett 2026). **Sea surface temperatures** in the SEBS were colder than in 2025, especially during January–April, and remained below average for most of the year (Lemagie and Callahan 2026). In May 2026, spring winds favored Ekman transport off the shelf, influencing larval dispersal trajectories (Hennon 2026).

**Prey**

Large diatom anomalies were above average during 2023–2025 (Ostle and Batten 2026), and spring 2026 chlorophyll-a concentrations were average to high across the NBS and SEBS (Gann et al. 2026). These records suggest favorable prey-production conditions in their respective periods. Increased surface phosphate and a deeper mixed layer in 2024 imply wind mixing, larger phytoplankton cells, shorter food chains and more efficient energy transfer to juvenile fish (Gann 2026).

The spring Rapid Zooplankton Assessment indicated adequate small copepod abundance, with **above-average large copepod abundance by late summer** (Kimmel et al. 2026). Euphausiid abundance increased from 2025 (Levine et al. 2026; Spear et al. 2026). Copepod and euphausiid lipid content in the SEBS were both above average in 2026, indicating good prey quality for pelagic juveniles (Pinger et al. 2026).

Spring larval abundance and egg numbers were low, as was the larval condition ratio. Such observations are typical of cold spring conditions over the shelf, and temperature-dependent seasonal timing complicates their interpretation (Siddon 2026). SEBS age-0 abundance was low (Andrews, Domke, Murphy, Spear, et al. 2026), and age-0 energy density was above average (Page et al. 2026). The suggested increase in age-1 abundance in this juvenile-survey comparison remains unverified. Length–weight residuals for small pollock (100–250 mm) in the EBS survey improved to just below the long-term mean (Prohaska and Rohan 2026). In the NBS, age-0 abundance was average (Andrews, Domke, Murphy, Spear, et al. 2026) and energy density was below average (Page et al. 2026).

**Competitors**

Pelagic forager biomass increased from 2025 to 2026 to above its long-term mean, largely driven by a 58% increase in pollock biomass (Siddon 2026, 7–9). Above-average zooplankton quantity and quality moderate concerns about competition for prey among young pollock. SEBS jellyfish biomass remained average (Buser 2026) to below average (Domke et al. 2026). NBS jellyfish biomass continued its slow decline since 2023 and remained just above the time-series mean (Domke et al. 2026).

Pacific herring and capelin biomass were below average in both the NBS and SEBS (Andrews, Domke, Murphy, Dimond, et al. 2026). Bristol Bay sockeye salmon continued to decline from peak 2021/2022 levels (Cunningham et al. 2026). Strong 2020 and 2022 sablefish year classes contributed to a growing, aging population that could compete for pelagic juvenile prey; direct evidence for that competition remains unavailable (Siddon 2026).

**Predators**

Adult pollock diets consisted primarily of copepods and euphausiids, with a low overall proportion of cannibalism in 2026 (Aydin 2026). CNRG estimates show that age-1 predation mortality declined from 2025, while mortality due to cannibalism remained slightly above its long-term mean. The biomass of pollock consumed by conspecifics was approximately average and lower than in 2024 and 2025 (Holsman et al. 2026). Adult Bristol Bay sockeye salmon, which prey on juvenile pollock, decreased from 2025 (Cunningham et al. 2026).

Northern fur seal pup weights at St. Paul Island exceeded the average by more than one standard deviation, and maternal trip durations were the shortest in the time series. These indicators suggest high foraging success near the island. Age-0 pollock and herring had the highest frequency of occurrence in the September 2025 fur-seal diet samples; analysis of September 2026 samples was ongoing in the draft (McHuron et al. 2026). The apex predator guild increased from 2025, driven by a 17% increase in Pacific cod, and remained below its long-term mean (Siddon 2026, 7–9). This guild level suggests low predation pressure on juvenile pollock (Siddon 2026).

## 11.2 Ecosystem considerations for later juveniles and adults

**Distribution**

Pollock biomass estimated by the standard bottom-trawl survey increased approximately 50% from 2025 to 2026, alongside a pronounced spatial pattern associated with the extensive cold pool. Adult pollock occurred primarily outside bottom waters colder than 2°C, with concentrations along the northwestern outer shelf (Siddon 2026). The cold pool therefore delineated the near-bottom habitat occupied by much of the adult population during the survey. This pattern is relevant to interpreting the biomass increase because survey catches reflect both the number and size of fish present and their availability to bottom-trawl gear. The higher survey estimate and the concentration of adults along the cold-pool margin should be considered together when comparing years. The spatial association supports a role for temperature in shaping survey availability, while the contribution of redistribution to the biomass increase remains unresolved.

**Environmental Processes**

Bottom temperatures in 2026 were comparable to the historical cool years 1986, 1992, 2006 and 2010 (Siddon 2026). The large cold pool provided further evidence of cool bottom conditions (Rohan and Barnett 2026). **Below-average SSTs** characterized the SEBS, particularly during spring (Lemagie and Callahan 2026). Adult pollock condition was above its long-term mean (Prohaska and Rohan 2026). These temperature and condition patterns describe concurrent conditions; their causal relationship remains an interpretation.

**Prey**

Adult pollock diets continued to be dominated by **copepods** and euphausiids (Aydin 2026). Summer and late-summer euphausiid abundance increased from 2025 to 2026 (Levine et al. 2026; Spear et al. 2026), and above-average SEBS copepod and euphausiid lipid content provided lipid-rich prey (Pinger et al. 2026). Cannibalism formed a low proportion of adult pollock diets (Aydin 2026). CNRG estimates separately indicate slightly above-average mortality of age-1 pollock from cannibalism, with approximately average biomass consumed by conspecifics (Holsman et al. 2026).

Morphometric condition, measured as length–weight residuals, for large adult pollock (>250 mm) in the EBS survey increased from 2023 to 2026 to above the long-term mean. The NBS condition series extends through 2025, when large-pollock condition remained positive and continued the increasing trend observed since 2021 (Prohaska and Rohan 2026).

**Competitors**

The increase in pelagic forager biomass from 2025 to 2026 was primarily driven by pollock (Siddon 2026, 7–9), suggesting a potential increase in competition within the pollock population. Above-average prey quantity, prey quality and pollock length–weight residuals indicate adequate food resources. Increasing sablefish abundance represents potential competition for pelagic prey, with direct evidence still pending (Siddon 2026).

**Predators**

The apex predator guild, driven by Pacific cod, increased from 2025 and remained below its long-term mean (Siddon 2026, 7–9). This pattern suggests low predation pressure on pollock (Siddon 2026). These associations provide context for the stock; independent evidence is needed to establish the underlying mechanisms. The Plan Team requested further examination of ecosystem contributions to retrospective behavior alongside the direct data-influence tests.

# 12 Data gaps and research priorities

The 2026 assessment compares the separately fitted configurations and the Model 26.1.2 marginal-estimation sensitivity using their matching observations and conditional observation diagnostics. Model 26.0 posterior uncertainty, predictive observation checks and nine retrospective fits are complete. The Model 26.1.2 retrospective diagnostic uses fixed full-model process SDs and a conditional full 2026 reference. The two Model 26.0 simulation-retrospective experiments compare conditional biomass and recruitment revisions with the observed-data results (Section 6.3.1). Each experiment uses ten simulated datasets to examine biomass and recruitment revisions.

Additional age-1 BTS selectivity and BTS data-influence tests are outside the scope of this assessment (PT-02 and PT-05). The comparisons describe differences in retrospective patterns. The weight-at-age comparison in Section 4.1.3 retains the distinction between estimated schedules and uncertainty carried into population and projection outputs.

A high research priority is to refine the statistical treatment of length compositions and age observations from the fishery, bottom-trawl survey (BTS), and acoustic-trawl survey (ATS), together with paired Fourier-transform near-infrared spectroscopy (FT-NIR) and traditional morphological ageing (TMA) observations. A more statistically rigorous framework should account for sampling structure, ageing error, uncertainty in conversions between ageing methods, and dependence among observations derived from the same samples. This work should guide composition weighting and the integration of length and age information, with the resulting methods and uncertainty treatment carried into future model specifications.

The treatment of early fishery data should also be refined relative to more recent observations. Research should evaluate changes in sampling coverage, age and length information, and fishing practices across periods, and use those differences to inform selectivity assumptions and observation weights. Comparisons should assess how alternative treatments affect recruitment, biomass trends, and retrospective behavior.

Further development of random-effects models should address temporal variation in selectivity and other population or observation processes. Priorities include identifying which sources of variability the data can resolve, evaluating correlation across ages and years, and estimating process variation within a coherent framework. Simulation tests, retrospective analyses, and predictive checks should evaluate model performance and the propagation of uncertainty into stock status and catch projections.

The Ecosystem and Socioeconomic Profile (ESP) provides a basis for fuller implementation and use of dynamic structural equation models (DSEM). This work should develop and test hypothesized links among environmental conditions, prey, pollock condition, and recruitment, while accounting for observation uncertainty, missing years, and time lags. Evaluation should establish the stability and predictive value of these relationships and how ESP results can inform assessment interpretation, recruitment scenarios, and future model specifications.

Additional research questions include AVO uncertainty, the 2018 recruitment profile, and stock boundaries and connectivity.

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# Tables

**Table 1:** Catch and adopted harvest limits for EBS pollock, 1964–2026 (round-weight metric tons). Catch from 1991 onward uses the catch estimates; the highlighted 2026 value is the selected annual model input, with the extract coverage cutoff unspecified.

| EBS catch and adopted harvest limits | | | | |
|---|---|---|---|---|
| Year | OFL (t) | ABC (t) | TAC (t) | Catch (t) |
| 1964 | — | — | — | 174,792 |
| 1965 | — | — | — | 230,551 |
| 1966 | — | — | — | 261,678 |
| 1967 | — | — | — | 550,362 |
| 1968 | — | — | — | 702,181 |
| 1969 | — | — | — | 862,789 |
| 1970 | — | — | — | 1,256,565 |
| 1971 | — | — | — | 1,743,763 |
| 1972 | — | — | — | 1,874,534 |
| 1973 | — | — | — | 1,758,919 |
| 1974 | — | — | — | 1,588,390 |
| 1975 | — | — | — | 1,356,736 |
| 1976 | — | — | — | 1,177,822 |
| 1977 | — | 950,000 | 950,000 | 978,370 |
| 1978 | — | 950,000 | 950,000 | 979,431 |
| 1979 | — | 1,100,000 | 950,000 | 935,714 |
| 1980 | — | 1,300,000 | 1,000,000 | 958,280 |
| 1981 | — | 1,300,000 | 1,000,000 | 973,502 |
| 1982 | — | 1,300,000 | 1,000,000 | 955,964 |
| 1983 | — | 1,300,000 | 1,000,000 | 981,450 |
| 1984 | — | 1,300,000 | 1,200,000 | 1,092,055 |
| 1985 | — | 1,300,000 | 1,200,000 | 1,139,676 |
| 1986 | — | 1,300,000 | 1,200,000 | 1,141,993 |
| 1987 | — | 1,300,000 | 1,200,000 | 859,416 |
| 1988 | — | 1,500,000 | 1,300,000 | 1,228,721 |
| 1989 | — | 1,340,000 | 1,340,000 | 1,229,600 |
| 1990 | — | 1,450,000 | 1,280,000 | 1,455,193 |
| 1991 | — | 1,676,000 | 1,300,000 | 1,195,644 |
| 1992 | 1,770,000 | 1,490,000 | 1,300,000 | 1,390,299 |
| 1993 | 1,340,000 | 1,340,000 | 1,300,000 | 1,326,602 |
| 1994 | 1,590,000 | 1,330,000 | 1,330,000 | 1,329,352 |
| 1995 | 1,500,000 | 1,250,000 | 1,250,000 | 1,264,247 |
| 1996 | 1,460,000 | 1,190,000 | 1,190,000 | 1,192,781 |
| 1997 | 1,980,000 | 1,130,000 | 1,130,000 | 1,124,433 |
| 1998 | 2,060,000 | 1,110,000 | 1,110,000 | 1,102,159 |
| 1999 | 1,720,000 | 992,000 | 992,000 | 989,680 |
| 2000 | 1,680,000 | 1,139,000 | 1,139,000 | 1,132,710 |
| 2001 | 3,536,000 | 1,842,000 | 1,400,000 | 1,387,197 |
| 2002 | 3,530,000 | 2,110,000 | 1,485,000 | 1,480,774 |
| 2003 | 3,530,000 | 2,330,000 | 1,491,760 | 1,490,779 |
| 2004 | 2,740,000 | 2,560,000 | 1,492,000 | 1,480,552 |
| 2005 | 2,100,000 | 1,960,000 | 1,478,500 | 1,483,022 |
| 2006 | 2,090,000 | 1,930,000 | 1,485,000 | 1,488,031 |

| EBS catch and adopted harvest limits | | | | |
|---|---|---|---|---|
| Year | OFL (t) | ABC (t) | TAC (t) | Catch (t) |
| 2007 | 1,640,000 | 1,394,000 | 1,394,000 | 1,354,502 |
| 2008 | 1,440,000 | 1,000,000 | 1,000,000 | 990,578 |
| 2009 | 977,000 | 815,000 | 815,000 | 810,784 |
| 2010 | 918,000 | 813,000 | 813,000 | 810,186 |
| 2011 | 2,450,000 | 1,270,000 | 1,252,000 | 1,199,031 |
| 2012 | 2,474,000 | 1,220,000 | 1,200,000 | 1,205,214 |
| 2013 | 2,550,000 | 1,375,000 | 1,247,000 | 1,270,760 |
| 2014 | 2,795,000 | 1,369,000 | 1,267,000 | 1,297,417 |
| 2015 | 3,330,000 | 1,637,000 | 1,310,000 | 1,321,574 |
| 2016 | 3,910,000 | 2,090,000 | 1,340,000 | 1,352,681 |
| 2017 | 3,640,000 | 2,800,000 | 1,345,000 | 1,359,181 |
| 2018 | 4,797,000 | 2,592,000 | 1,364,341 | 1,379,288 |
| 2019 | 3,914,000 | 2,163,000 | 1,397,000 | 1,409,346 |
| 2020 | 4,085,000 | 2,043,000 | 1,425,000 | 1,367,236 |
| 2021 | 2,594,000 | 1,626,000 | 1,375,000 | 1,376,258 |
| 2022 | 1,469,000 | 1,111,000 | 1,111,000 | 1,105,419 |
| 2023 | 3,381,000 | 1,910,000 | 1,300,000 | 1,310,592 |
| 2024 | 3,162,000 | 2,313,000 | 1,313,580 | 1,311,261 |
| 2025 | 2,957,000 | 2,417,000 | 1,389,000 | 1,326,216 |
| **2026** | 2,496,000 | 2,036,000 | 1,389,500 | **1,312,505** |

Catch for 1991–2026 uses the revised annual estimates. Model inputs use the same totals at full precision, converted to thousand tons. Pre-1991 catch and adopted harvest limits retain their previous sources. Dashes mean unavailable.

All supplied Bering Sea pollock retained/discarded, gear and sector records are summed once. Explicit Bogoslof (518) and outside-stock area codes are excluded. The 23 records with blank reporting-area codes in 1998 retain their Bering Sea classification and contribute 83,077.58 t. The year-by-year reconciliation is in the catch-revision table.

The 2026 CSV total is 1,312,504.580845496 t and is used as the annual input under the author's instruction. The CSV supplies no within-year catch cutoff; the author has confirmed this CSV as the correct annual input version. The previous 1,310,000-t forecast and October 17 catch-to-date are archived in the revision record. Annual source totals.

**Table 2:** Selected management and monitoring milestones affecting the EBS pollock fishery, 1977–2017. Dates distinguish Council recommendations, enactment, voluntary programs and regulatory implementation. BSAI- or GOA-wide scope is stated where applicable; the table is selective rather than a complete regulatory history.

| Historical pollock management and monitoring measures | | | |
|---|---|---|---|
| Year | Action stage | Measure and scope | Sources |
| 1977 | Implemented | Preliminary BSAI fishery management plan (FMP), including closure areas. | SAFE 2024; History 1997 |
| 1982 | Implemented | BSAI groundfish FMP implemented; Chinook salmon bycatch limits established for foreign trawlers. | SAFE 2024; History 1997 |
| 1984 | Implemented | BSAI groundfish optimum yield capped at 2 million t; Chinook salmon bycatch limits for foreign trawling reduced. | SAFE 2024; History 1997 |
| 1990 | Implemented | Domestic observer program and associated data reporting began. | SAFE 2024; NOAA history |
| 1992 | Implemented | Pollock allocations under the Western Alaska Community Development Quota (CDQ) program began. | SAFE 2024 |
| 1993–1994 | Donation program | Prohibited Species Donation Program established in 1993; voluntary salmon donations in 1994 are reported in the accepted assessment history. | Council overview; SAFE 2024 |
| 1994 | Council recommendation | Council recommended minimum codend mesh/design standards; an implemented NMFS mesh rule is not established by the reviewed sources. | History 1997; Gear review |
| 1994 | Reporting | NMFS began publishing individual-vessel bycatch rates online. | SAFE 2024; History 1997 |
| 1994–1995 | Implemented | Emergency Chum Salmon Savings Area in 1994; permanent chum savings-area provisions under Amendment 35 in 1995, alongside the historical Chinook closure/trigger regime. | 2023 notice history; SAFE 2024 |
| 1998 | Implemented | Amendment 49 improved retention/utilization requirements for pollock and Pacific cod took effect January 3. | Amendment 49 |
| 1998 | Enacted | American Fisheries Act (AFA) enacted, establishing the framework for sector allocations and fishery cooperatives. | SAFE 2024; AFA rule |
| 1999 | Implemented | AFA provisions implemented for catcher/processors. | SAFE 2024; AFA rule |
| 1999 | Emergency rule | Temporal/spatial pollock dispersal measures and trawl exclusion zones around selected Steller sea lion rookeries and haulouts in BSAI and GOA. | 1999 emergency rule |
| 2000 | Implemented | AFA provisions implemented for the remaining catcher-vessel and mothership sectors. | SAFE 2024; AFA rule |
| 2001 | Voluntary | Pollock industry began the chum salmon rolling-hotspot program. | 2023 notice history; SAFE 2024 |
| 2002 | Voluntary | Pollock industry began the Chinook salmon rolling-hotspot program. | 2023 notice history; SAFE 2024 |
| 2005 | Council action | Council selected Amendment 84 in October to provide a salmon bycatch intercooperative agreement (ICA) alternative to fixed savings-area closures. | Amendment 84 |
| 2006 | Exempted fishing | The B-season exempted fishing permit tested the ICA/rolling-hotspot approach before general regulatory implementation. | Amendment 84 |
| 2007 | Implemented | Amendment 84 regulations, effective November 28, exempted pollock vessels participating in an approved salmon bycatch ICA from specified savings-area closures. | Amendment 84 |
| 2010–2011 | Rule / program | Amendment 91 final rule effective September 29, 2010; Chinook PSC limits, incentive plan agreements (IPAs), performance standards and enhanced monitoring operated in the 2011 fishery. | Amendment 91 |
| 2015 | Council action | Council took final action on Amendment 110 in April, recommending linked Chinook/chum avoidance measures and abundance-based Chinook limits. | Amendment 110 |

| Historical pollock management and monitoring measures | | | |
|---|---|---|---|
| Year | Action stage | Measure and scope | Sources |
| 2016 | Implemented | Amendment 110 final rule published June 10, effective July 11. It strengthened IPAs and changed pollock seasonal allocations to 45% A season / 55% B season. | Amendment 110 |
| 2017 | First index year | First fishing year informed by the Amendment 110 three-system abundance index supplied in 2016. Low-abundance limits apply when the index is at or below 250,000 Chinook salmon. | Amendment 110 |

Historical entries were checked against accepted 2024 SAFE Table 4 (p. 63) and the linked primary sources. The 1994 codend item is a Council recommendation; the reviewed sources do not establish NMFS implementation. The 2005 Amendment 84 action preceded exempted fishing and 2007 regulations. Council action and rule effective dates are distinct.

The 45% / 55% allocation in the 2016 rule concerns pollock, distinct from Chinook PSC apportionments. Continuing and dated 2026 measures are described in Fishery and management history. Age-estimation and survey developments are documented in their respective report sections; they are not listed here as regulatory implementations.

Supplied CSV; supplied Markdown; row-by-row verification and corrections. Original source records remain unchanged.

**Table 3:** EBS pollock catch by reporting area and season, 1991–2026 (tons). Southeast (SE) comprises NMFS areas 520–539; Northwest (NW) comprises areas 500–517 and 519. Area 518 (Bogoslof) is excluded. A and B group records with week-ending dates before June 1 and from June 1 onward, respectively. The seasonal extract extends through October 17, 2026. Only the 1998 records included catch without an area; this catch is allocated between NW and SE in proportion to reported catch within each season. Totals include retained and discarded catch; values are rounded to whole tons.

| Year | NW A | NW B | SE A | SE B | NW | SE | A | B | Total |
|---|---|---|---|---|---|---|---|---|---|
| 1991 | 297,202 | 356,353 | 12,191 | 529,917 | 653,555 | 542,109 | 309,393 | 886,271 | 1,195,664 |
| 1992 | 454,947 | 375,612 | 101,597 | 458,143 | 830,559 | 559,741 | 556,544 | 833,755 | 1,390,299 |
| 1993 | 475,143 | 619,287 | 93,013 | 139,160 | 1,094,429 | 232,173 | 568,156 | 758,446 | 1,326,602 |
| 1994 | 586,105 | 566,470 | 11,026 | 165,751 | 1,152,575 | 176,777 | 597,131 | 732,221 | 1,329,352 |
| 1995 | 600,499 | 571,807 | 3,631 | 88,310 | 1,172,306 | 91,941 | 604,129 | 660,117 | 1,264,247 |
| 1996 | 549,297 | 537,545 | 1,804 | 104,135 | 1,086,843 | 105,939 | 551,101 | 641,680 | 1,192,781 |
| 1997 | 557,106 | 262,784 | 2,363 | 302,180 | 819,889 | 304,544 | 559,469 | 564,964 | 1,124,433 |
| 1998 | 515,420 | 443,514 | 1,632 | 141,593 | 958,934 | 143,226 | 517,052 | 585,107 | 1,102,159 |
| 1999 | 414,030 | 368,953 | 4,139 | 202,558 | 782,983 | 206,698 | 418,169 | 571,511 | 989,680 |
| 2000 | 439,023 | 400,154 | 19,363 | 274,170 | 839,177 | 293,532 | 458,386 | 674,324 | 1,132,710 |
| 2001 | 444,565 | 517,412 | 108,013 | 317,207 | 961,977 | 425,220 | 552,578 | 834,619 | 1,387,197 |
| 2002 | 561,061 | 599,272 | 36,693 | 283,749 | 1,160,334 | 320,442 | 597,754 | 883,022 | 1,480,776 |
| 2003 | 446,152 | 487,039 | 151,465 | 406,123 | 933,191 | 557,588 | 597,617 | 893,162 | 1,490,779 |
| 2004 | 556,469 | 533,538 | 50,661 | 339,883 | 1,090,008 | 390,544 | 607,130 | 873,422 | 1,480,552 |
| 2005 | 476,970 | 325,184 | 120,320 | 560,548 | 802,154 | 680,868 | 597,290 | 885,732 | 1,483,022 |
| 2006 | 585,507 | 241,700 | 18,789 | 642,034 | 827,207 | 660,824 | 604,297 | 883,734 | 1,488,031 |
| 2007 | 505,306 | 222,943 | 62,504 | 563,748 | 728,249 | 626,253 | 567,811 | 786,691 | 1,354,502 |
| 2008 | 358,936 | 123,762 | 44,801 | 463,079 | 482,698 | 507,880 | 403,738 | 586,840 | 990,578 |
| 2009 | 247,944 | 110,307 | 81,575 | 370,957 | 358,252 | 452,532 | 329,520 | 481,264 | 810,784 |
| 2010 | 124,670 | 130,443 | 199,011 | 356,061 | 255,114 | 555,072 | 323,681 | 486,505 | 810,186 |
| 2011 | 401,755 | 346,128 | 102,252 | 348,897 | 747,882 | 451,149 | 504,007 | 695,024 | 1,199,031 |
| 2012 | 380,373 | 238,497 | 104,926 | 481,418 | 618,870 | 586,344 | 485,299 | 719,915 | 1,205,214 |
| 2013 | 415,014 | 280,652 | 94,763 | 480,331 | 695,667 | 575,094 | 509,777 | 760,983 | 1,270,760 |
| 2014 | 469,973 | 388,267 | 50,465 | 388,712 | 858,240 | 439,177 | 520,438 | 776,979 | 1,297,417 |
| 2015 | 267,095 | 429,153 | 258,574 | 366,752 | 696,248 | 625,326 | 525,669 | 795,905 | 1,321,574 |
| 2016 | 458,044 | 709,028 | 81,531 | 104,078 | 1,167,072 | 185,609 | 539,576 | 813,106 | 1,352,681 |
| 2017 | 544,028 | 633,993 | 37,730 | 143,430 | 1,178,021 | 181,159 | 581,758 | 777,423 | 1,359,181 |
| 2018 | 598,533 | 450,160 | 3,842 | 326,753 | 1,048,692 | 330,595 | 602,375 | 776,913 | 1,379,288 |
| 2019 | 614,949 | 487,217 | 2,649 | 304,532 | 1,102,166 | 307,181 | 617,598 | 791,749 | 1,409,346 |
| 2020 | 560,438 | 299,978 | 85,717 | 421,102 | 860,416 | 506,820 | 646,156 | 721,080 | 1,367,236 |
| 2021 | 554,774 | 469,234 | 56,779 | 295,470 | 1,024,009 | 352,249 | 611,554 | 764,704 | 1,376,258 |
| 2022 | 487,216 | 411,573 | 9,018 | 197,610 | 898,790 | 206,629 | 496,235 | 609,184 | 1,105,419 |
| 2023 | 568,455 | 319,251 | 11,335 | 411,552 | 887,705 | 422,887 | 579,789 | 730,803 | 1,310,592 |
| 2024 | 530,902 | 347,432 | 55,087 | 377,840 | 878,334 | 432,927 | 585,989 | 725,272 | 1,311,261 |
| 2025 | 445,959 | 270,015 | 157,475 | 452,767 | 715,974 | 610,242 | 603,434 | 722,782 | 1,326,216 |
| 2026 | 528,440 | 205,467 | 75,601 | 458,034 | 733,908 | 533,635 | 604,041 | 663,501 | 1,267,543 |

**Table 4:** Retained and discarded EBS pollock catch by reporting area, 1991–2026 (tons), from the annual catch-accounting extract. Southeast (SE) comprises NMFS areas 520–539; Northwest (NW) comprises areas 500–517 and 519. Area 518 (Bogoslof) is excluded. Discarded percentage equals discarded catch divided by total catch. Values are rounded to whole tons; 2026 estimates are preliminary.

| Year | NW Discarded | SE Discarded | NW Retained | SE Retained | Total | Discarded % |
|---|---|---|---|---|---|---|
| 1991 | 66,792 | 48,257 | 586,763 | 493,852 | 1,195,664 | 9.6 |
| 1992 | 71,194 | 57,578 | 759,365 | 502,163 | 1,390,299 | 9.3 |
| 1993 | 83,986 | 26,100 | 1,010,443 | 206,073 | 1,326,602 | 8.3 |
| 1994 | 88,098 | 16,084 | 1,064,476 | 160,693 | 1,329,352 | 7.8 |
| 1995 | 87,492 | 9,715 | 1,084,813 | 82,226 | 1,264,247 | 7.7 |
| 1996 | 71,368 | 4,838 | 1,015,474 | 101,100 | 1,192,781 | 6.4 |
| 1997 | 71,032 | 22,557 | 748,858 | 281,986 | 1,124,433 | 8.3 |
| 1998 | 14,291 | 1,581 | 944,511 | 141,777 | 1,102,159 | 1.4 |
| 1999 | 26,912 | 1,912 | 756,071 | 204,786 | 989,680 | 2.9 |
| 2000 | 19,678 | 1,942 | 819,499 | 291,591 | 1,132,710 | 1.9 |
| 2001 | 14,874 | 2,450 | 947,103 | 422,770 | 1,387,197 | 1.2 |
| 2002 | 19,430 | 1,441 | 1,140,904 | 319,002 | 1,480,776 | 1.4 |
| 2003 | 13,795 | 2,959 | 919,397 | 554,629 | 1,490,779 | 1.1 |
| 2004 | 20,380 | 2,781 | 1,069,628 | 387,763 | 1,480,552 | 1.6 |
| 2005 | 14,838 | 2,586 | 787,316 | 678,282 | 1,483,022 | 1.2 |
| 2006 | 11,877 | 3,677 | 815,330 | 657,147 | 1,488,031 | 1.0 |
| 2007 | 12,334 | 3,769 | 715,915 | 622,484 | 1,354,502 | 1.2 |
| 2008 | 5,968 | 1,643 | 476,730 | 506,237 | 990,578 | 0.8 |
| 2009 | 4,014 | 1,936 | 354,238 | 450,596 | 810,784 | 0.7 |
| 2010 | 2,490 | 1,270 | 252,623 | 553,802 | 810,186 | 0.5 |
| 2011 | 3,444 | 1,376 | 744,438 | 449,773 | 1,199,031 | 0.4 |
| 2012 | 4,080 | 1,190 | 614,791 | 585,154 | 1,205,214 | 0.4 |
| 2013 | 4,084 | 1,225 | 691,582 | 573,869 | 1,270,760 | 0.4 |
| 2014 | 12,556 | 1,786 | 845,684 | 437,391 | 1,297,417 | 1.1 |
| 2015 | 7,055 | 2,418 | 689,193 | 622,907 | 1,321,574 | 0.7 |
| 2016 | 8,124 | 1,036 | 1,158,948 | 184,574 | 1,352,681 | 0.7 |
| 2017 | 6,848 | 1,356 | 1,171,173 | 179,803 | 1,359,181 | 0.6 |
| 2018 | 9,170 | 2,005 | 1,039,523 | 328,590 | 1,379,288 | 0.8 |
| 2019 | 7,126 | 1,979 | 1,095,040 | 305,202 | 1,409,346 | 0.6 |
| 2020 | 9,363 | 2,450 | 851,053 | 504,370 | 1,367,236 | 0.9 |
| 2021 | 12,379 | 1,534 | 1,011,629 | 350,715 | 1,376,258 | 1.0 |
| 2022 | 9,898 | 3,538 | 888,891 | 203,091 | 1,105,419 | 1.2 |
| 2023 | 10,219 | 4,685 | 877,486 | 418,201 | 1,310,592 | 1.1 |
| 2024 | 12,935 | 4,131 | 865,399 | 428,796 | 1,311,261 | 1.3 |
| 2025 | 6,355 | 899 | 709,618 | 609,343 | 1,326,216 | 0.5 |
| 2026 | 5,815 | 3,182 | 769,440 | 534,068 | 1,312,505 | 0.7 |

**Table 5:** EBS pollock fishery catch-at-age estimates, 1964–2025 (million fish). Ages 15 and older are pooled as 15+. Catch numbers are calculated from assessment proportions, annual fishery weights at age and catch biomass where complete annual numbers are unavailable. Age-specific values are rounded to 0.1 million fish; All ages totals are rounded to the nearest million fish.

| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ | All ages |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1964 | 8.3 | 34.8 | 54.6 | 63.8 | 31.5 | 88.5 | 39.8 | 8.3 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 330 |
| 1965 | 7.2 | 7.8 | 103.3 | 278.4 | 68.2 | 17.0 | 16.4 | 7.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 506 |
| 1966 | 16.9 | 100.2 | 34.2 | 250.2 | 157.8 | 14.4 | 15.7 | 6.1 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 596 |
| 1967 | 14.1 | 164.0 | 859.9 | 130.6 | 238.2 | 56.6 | 17.1 | 8.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1,489 |
| 1968 | 57.3 | 310.9 | 244.1 | 815.5 | 186.9 | 53.2 | 63.7 | 9.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1,741 |
| 1969 | 33.6 | 182.6 | 1051.1 | 722.5 | 248.2 | 5.3 | 27.8 | 17.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 2,289 |
| 1970 | 730.1 | 1109.2 | 1743.2 | 823.0 | 159.5 | 51.9 | 25.9 | 14.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 4,658 |
| 1971 | 766.6 | 229.0 | 1278.9 | 858.5 | 1272.7 | 123.6 | 81.2 | 22.8 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 4,633 |
| 1972 | 155.0 | 747.6 | 1724.5 | 1381.2 | 578.7 | 227.0 | 86.2 | 31.2 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 4,931 |
| 1973 | 36.3 | 327.2 | 396.1 | 861.1 | 659.1 | 450.3 | 250.5 | 181.5 | 81.4 | 14.7 | 1.8 | 0.5 | 0.1 | 0.5 | 0.0 | 3,261 |
| 1974 | 117.7 | 2558.9 | 945.7 | 269.0 | 361.3 | 195.1 | 124.9 | 105.3 | 103.3 | 43.5 | 17.7 | 4.6 | 4.6 | 0.8 | 0.7 | 4,853 |
| 1975 | 34.3 | 602.7 | 2774.6 | 213.3 | 56.7 | 106.6 | 102.1 | 55.6 | 32.7 | 23.0 | 5.7 | 2.3 | 0.4 | 0.5 | 0.0 | 4,010 |
| 1976 | 0.6 | 357.3 | 1351.8 | 909.8 | 89.7 | 41.4 | 57.3 | 66.5 | 51.7 | 25.5 | 11.8 | 8.0 | 2.1 | 0.0 | 0.0 | 2,973 |
| 1977 | 104.3 | 685.8 | 916.8 | 664.3 | 238.5 | 53.0 | 19.5 | 28.1 | 22.5 | 18.0 | 9.3 | 4.6 | 3.4 | 0.7 | 0.2 | 2,769 |
| 1978 | 29.4 | 455.2 | 752.9 | 638.6 | 368.2 | 99.7 | 28.8 | 17.3 | 19.7 | 15.8 | 11.1 | 4.3 | 1.8 | 0.5 | 0.2 | 2,444 |
| 1979 | 89.6 | 479.6 | 635.7 | 371.0 | 346.6 | 190.3 | 49.7 | 22.7 | 31.7 | 24.3 | 15.6 | 7.0 | 2.6 | 0.5 | 0.5 | 2,267 |
| 1980 | 9.0 | 424.0 | 754.9 | 406.7 | 231.2 | 193.5 | 76.8 | 34.5 | 19.9 | 21.9 | 23.3 | 14.6 | 7.1 | 2.3 | 1.1 | 2,221 |
| 1981 | 0.6 | 75.8 | 1062.5 | 669.3 | 238.2 | 107.9 | 54.2 | 31.1 | 16.9 | 9.8 | 7.9 | 4.8 | 1.6 | 0.7 | 0.3 | 2,282 |
| 1982 | 4.5 | 24.0 | 152.6 | 1108.4 | 399.3 | 98.1 | 34.0 | 34.1 | 20.4 | 8.6 | 5.1 | 3.0 | 1.8 | 0.7 | 0.3 | 1,895 |
| 1983 | 5.2 | 121.3 | 161.4 | 320.2 | 835.8 | 223.3 | 42.4 | 25.3 | 20.3 | 11.4 | 7.8 | 5.0 | 3.6 | 1.8 | 0.4 | 1,785 |
| 1984 | 1.9 | 41.8 | 80.8 | 392.7 | 448.3 | 596.3 | 122.0 | 32.4 | 22.9 | 14.3 | 6.5 | 2.3 | 2.7 | 1.5 | 1.8 | 1,768 |
| 1985 | 2.4 | 50.9 | 351.3 | 112.2 | 337.0 | 296.3 | 408.4 | 103.6 | 33.7 | 23.8 | 22.9 | 9.8 | 8.7 | 3.7 | 4.7 | 1,770 |
| 1986 | 2.8 | 78.5 | 84.2 | 683.0 | 195.4 | 345.0 | 202.5 | 195.5 | 54.5 | 13.9 | 3.0 | 2.4 | 0.2 | 1.1 | 0.0 | 1,862 |
| 1987 | 0.0 | 16.3 | 91.9 | 63.9 | 340.5 | 114.4 | 100.9 | 74.7 | 203.6 | 44.5 | 31.9 | 17.7 | 23.8 | 5.0 | 6.6 | 1,136 |
| 1988 | 0.0 | 9.7 | 410.9 | 381.6 | 228.2 | 492.7 | 203.4 | 95.0 | 35.5 | 87.6 | 16.5 | 9.2 | 3.4 | 5.0 | 5.6 | 1,984 |
| 1989 | 0.0 | 4.3 | 49.7 | 134.4 | 406.8 | 150.4 | 516.1 | 86.9 | 93.6 | 29.2 | 116.4 | 9.8 | 3.6 | 2.3 | 5.3 | 1,609 |
| 1990 | 1.4 | 36.1 | 62.2 | 239.8 | 219.3 | 522.1 | 141.2 | 402.6 | 71.8 | 111.4 | 9.9 | 65.6 | 9.3 | 5.1 | 10.1 | 1,908 |
| 1991 | 0.7 | 112.9 | 43.8 | 90.3 | 153.9 | 184.4 | 497.5 | 78.1 | 288.0 | 28.9 | 144.1 | 18.7 | 91.2 | 22.8 | 53.3 | 1,808 |
| 1992 | 1.1 | 84.7 | 675.8 | 130.1 | 79.6 | 108.7 | 133.8 | 253.6 | 102.3 | 147.0 | 58.0 | 46.3 | 13.4 | 43.8 | 34.5 | 1,913 |
| 1993 | 0.1 | 7.4 | 260.5 | 1146.7 | 103.0 | 66.2 | 66.4 | 56.5 | 86.2 | 21.2 | 32.7 | 12.4 | 13.5 | 6.8 | 16.1 | 1,895 |
| 1994 | 0.7 | 30.2 | 55.1 | 361.2 | 1059.7 | 175.7 | 53.5 | 19.1 | 13.1 | 20.2 | 9.7 | 9.4 | 7.6 | 4.0 | 8.3 | 1,827 |
| 1995 | 0.0 | 0.5 | 72.9 | 146.8 | 395.5 | 761.1 | 136.3 | 34.6 | 12.3 | 7.5 | 17.5 | 5.0 | 5.8 | 1.7 | 8.9 | 1,606 |
| 1996 | 0.0 | 21.7 | 48.1 | 71.8 | 161.0 | 361.9 | 481.7 | 184.6 | 33.6 | 13.4 | 7.9 | 8.8 | 4.3 | 5.8 | 5.3 | 1,410 |
| 1997 | 1.0 | 77.7 | 40.3 | 119.0 | 455.1 | 289.0 | 256.4 | 198.6 | 64.0 | 13.3 | 6.0 | 4.6 | 2.9 | 4.8 | 9.1 | 1,542 |
| 1998 | 0.3 | 42.0 | 84.5 | 70.5 | 153.4 | 702.8 | 199.6 | 131.7 | 110.7 | 27.8 | 6.1 | 5.6 | 2.6 | 2.5 | 4.5 | 1,545 |
| 1999 | 0.2 | 10.3 | 298.7 | 225.0 | 103.0 | 157.0 | 469.7 | 131.0 | 56.5 | 33.2 | 4.0 | 2.2 | 0.9 | 0.5 | 2.1 | 1,494 |
| 2000 | 0.0 | 16.1 | 82.4 | 428.5 | 346.5 | 106.7 | 168.4 | 357.7 | 84.9 | 29.7 | 22.0 | 5.2 | 1.4 | 0.6 | 1.0 | 1,651 |
| 2001 | 0.0 | 3.2 | 42.8 | 154.5 | 581.0 | 415.0 | 137.2 | 129.0 | 157.2 | 57.9 | 33.6 | 16.2 | 5.5 | 3.1 | 1.9 | 1,738 |
| 2002 | 0.8 | 47.0 | 108.1 | 217.8 | 287.6 | 606.3 | 268.0 | 98.5 | 85.9 | 93.9 | 34.6 | 14.4 | 11.0 | 3.0 | 1.8 | 1,879 |
| 2003 | 0.0 | 14.5 | 411.8 | 324.1 | 360.4 | 301.5 | 337.6 | 158.6 | 49.4 | 39.3 | 35.7 | 22.9 | 6.6 | 3.6 | 3.2 | 2,069 |
| 2004 | 0.0 | 0.5 | 89.6 | 831.1 | 480.6 | 236.8 | 169.3 | 156.3 | 65.0 | 16.1 | 17.1 | 25.3 | 9.4 | 5.9 | 6.9 | 2,110 |
| 2005 | 0.0 | 4.8 | 52.2 | 392.9 | 863.8 | 484.6 | 159.5 | 68.1 | 66.6 | 30.1 | 10.0 | 9.1 | 3.2 | 2.5 | 3.4 | 2,151 |
| 2006 | 0.0 | 9.9 | 84.2 | 295.8 | 619.6 | 597.6 | 278.6 | 107.3 | 48.0 | 38.4 | 17.7 | 8.2 | 8.3 | 5.3 | 7.2 | 2,126 |
| 2007 | 1.7 | 15.7 | 59.2 | 139.2 | 389.4 | 512.0 | 300.8 | 137.1 | 47.6 | 27.5 | 21.9 | 8.9 | 6.5 | 3.4 | 10.8 | 1,682 |
| 2008 | 0.0 | 25.2 | 58.8 | 79.2 | 147.0 | 309.7 | 242.3 | 148.7 | 84.3 | 22.2 | 17.6 | 14.4 | 8.6 | 2.8 | 12.7 | 1,173 |
| 2009 | 0.0 | 1.3 | 175.5 | 200.6 | 82.6 | 114.4 | 124.3 | 104.3 | 66.6 | 40.2 | 23.5 | 7.6 | 7.5 | 2.7 | 8.7 | 960 |

| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ | All ages |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2010 | 1.1 | 26.5 | 31.8 | 559.4 | 220.5 | 54.7 | 43.0 | 57.7 | 51.8 | 31.8 | 16.0 | 8.6 | 6.0 | 4.4 | 5.1 | 1,118 |
| 2011 | 0.4 | 10.4 | 193.3 | 115.4 | 809.2 | 284.7 | 63.6 | 37.5 | 38.5 | 41.4 | 25.8 | 12.5 | 1.8 | 4.0 | 4.3 | 1,643 |
| 2012 | 0.0 | 22.3 | 116.7 | 946.7 | 172.8 | 432.6 | 141.6 | 36.7 | 17.4 | 14.6 | 15.9 | 13.5 | 7.4 | 6.1 | 3.3 | 1,948 |
| 2013 | 1.8 | 1.0 | 64.0 | 342.4 | 955.9 | 194.4 | 156.5 | 70.0 | 20.7 | 12.8 | 12.8 | 10.9 | 7.8 | 4.8 | 5.7 | 1,861 |
| 2014 | 0.0 | 39.2 | 31.0 | 167.8 | 399.3 | 751.8 | 210.2 | 86.2 | 29.8 | 9.0 | 4.5 | 4.5 | 4.6 | 2.8 | 6.0 | 1,747 |
| 2015 | 0.0 | 15.5 | 632.4 | 196.4 | 228.6 | 384.3 | 510.1 | 88.8 | 42.1 | 17.7 | 2.9 | 2.1 | 3.1 | 2.7 | 1.2 | 2,128 |
| 2016 | 0.0 | 0.5 | 90.6 | 1390.3 | 159.9 | 174.5 | 174.8 | 224.7 | 34.1 | 13.8 | 8.0 | 0.5 | 1.2 | 0.7 | 1.0 | 2,274 |
| 2017 | 0.0 | 2.2 | 28.2 | 549.1 | 899.0 | 215.5 | 147.6 | 122.1 | 97.3 | 21.8 | 7.2 | 5.6 | 0.5 | 0.2 | 0.1 | 2,096 |
| 2018 | 0.0 | 1.3 | 13.8 | 115.0 | 1216.1 | 506.6 | 104.8 | 82.0 | 60.6 | 25.9 | 4.3 | 1.1 | 0.4 | 0.6 | 0.5 | 2,133 |
| 2019 | 0.7 | 10.9 | 12.3 | 18.4 | 157.6 | 916.8 | 422.5 | 93.2 | 52.1 | 53.0 | 10.1 | 2.9 | 0.8 | 0.0 | 0.0 | 1,751 |
| 2020 | 3.5 | 247.7 | 85.5 | 101.1 | 135.2 | 551.1 | 597.8 | 124.0 | 52.8 | 38.6 | 27.3 | 6.8 | 1.8 | 1.1 | 0.0 | 1,974 |
| 2021 | 0.0 | 111.5 | 1297.0 | 144.2 | 110.1 | 107.2 | 309.6 | 296.0 | 72.1 | 26.5 | 16.2 | 8.5 | 2.1 | 0.0 | 0.4 | 2,501 |
| 2022 | 0.0 | 62.3 | 174.5 | 1035.4 | 183.7 | 70.4 | 79.5 | 141.9 | 81.1 | 16.3 | 11.3 | 4.1 | 6.0 | 1.0 | 0.0 | 1,867 |
| 2023 | 0.0 | 25.2 | 86.6 | 263.0 | 1473.1 | 227.2 | 29.4 | 45.4 | 63.6 | 42.6 | 18.8 | 5.4 | 8.6 | 2.6 | 0.4 | 2,292 |
| 2024 | 0.7 | 2.9 | 104.8 | 131.5 | 362.9 | 1277.6 | 113.7 | 22.3 | 32.9 | 22.8 | 25.3 | 10.2 | 3.4 | 1.9 | 0.1 | 2,113 |
| 2025 | 0.0 | 28.8 | 50.7 | 100.3 | 129.5 | 526.2 | 852.1 | 92.5 | 29.3 | 26.0 | 8.0 | 19.7 | 0.0 | 0.0 | 6.2 | 1,869 |

**Table 6:**Bycatch estimates (t) of other species caught in the Bering Sea directed pollock fishery, 1991-2026.

| Year | P.cod | Rock.Sole | Flathead | Arrowtooth | POP | Yellowfin | Sablefish | Shark | Atka.Mkrl | Skates | Other |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1991 | 24,310 | 5,121 | NA | 5,719 | 418 | 418 | 10 | NA | 24 | NA | 10,973 |
| 1992 | 24,005 | 7,234 | 3 | 4,311 | 173 | 892 | 8 | NA | 261 | NA | 15,552 |
| 1993 | 20,931 | 8,714 | NA | 1,223 | 283 | 1,102 | 2 | NA | 40 | NA | 8,127 |
| 1994 | 14,409 | 3,010 | NA | 2,011 | 171 | 1,208 | 2 | NA | 44 | NA | 4,435 |
| 1995 | 19,777 | 2,180 | 2,175 | 1,177 | 142 | 675 | 12 | NA | 211 | NA | 1,897 |
| 1996 | 15,174 | 2,042 | 3,208 | 1,844 | 303 | 1,798 | 7 | NA | 384 | NA | 3,718 |
| 1997 | 8,263 | 1,523 | 2,350 | 985 | 428 | 606 | 2 | NA | 83 | NA | 3,187 |
| 1998 | 6,255 | 770 | 2,048 | 1,713 | 617 | 1,745 | 2 | NA | 10 | NA | 2,449 |
| 1999 | 3,220 | 1,059 | 1,885 | 273 | 121 | 350 | 7 | NA | 158 | NA | 1,740 |
| 2000 | 3,433 | 2,688 | 2,510 | 979 | 21 | 1,466 | 12 | NA | 2 | NA | 1,859 |
| 2001 | 3,880 | 1,673 | 2,199 | 530 | 574 | 594 | 21 | NA | 41 | NA | 3,226 |
| 2002 | 5,887 | 1,886 | 1,845 | 607 | 543 | 768 | 34 | NA | 221 | NA | 2,775 |
| 2003 | 5,968 | 1,419 | 1,501 | 618 | 935 | 210 | 48 | NA | 762 | NA | 1,992 |
| 2004 | 6,437 | 2,554 | 2,104 | 557 | 394 | 841 | 17 | NA | 1,053 | NA | 2,649 |
| 2005 | 7,413 | 1,125 | 2,352 | 651 | 652 | 63 | 11 | NA | 678 | NA | 2,759 |
| 2006 | 7,291 | 1,361 | 2,862 | 1,089 | 736 | 256 | 9 | NA | 789 | NA | 4,094 |
| 2007 | 5,630 | 510 | 4,226 | 2,795 | 625 | 86 | 12 | NA | 315 | NA | 3,731 |
| 2008 | 6,971 | 2,150 | 4,315 | 1,715 | 336 | 552 | 5 | NA | 15 | NA | 5,217 |
| 2009 | 7,875 | 7,591 | 4,666 | 2,202 | 114 | 271 | 3 | NA | 25 | NA | 4,969 |
| 2010 | 6,908 | 2,240 | 4,357 | 1,454 | 231 | 1,057 | 2 | NA | 55 | NA | 2,877 |
| 2011 | 10,043 | 8,481 | 4,886 | 1,602 | 658 | 1,083 | 1 | 66 | 900 | 2,352 | 1,156 |
| 2012 | 10,066 | 6,702 | 3,968 | 749 | 706 | 1,517 | 1 | 55 | 263 | 2,018 | 1,483 |
| 2013 | 8,955 | 6,379 | 3,146 | 966 | 611 | 2,097 | 0 | 43 | 70 | 1,750 | 746 |
| 2014 | 5,242 | 4,360 | 2,554 | 758 | 1,301 | 1,954 | 1 | 75 | 117 | 813 | 2,526 |
| 2015 | 8,303 | 1,709 | 2,260 | 403 | 2,525 | 863 | 0 | 52 | 202 | 824 | 2,875 |
| 2016 | 4,980 | 1,095 | 1,554 | 291 | 3,191 | 870 | 18 | 60 | 68 | 423 | 1,716 |
| 2017 | 6,151 | 1,708 | 908 | 204 | 4,438 | 578 | 96 | 91 | 61 | 448 | 2,120 |
| 2018 | 4,271 | 1,077 | 974 | 271 | 3,804 | 743 | 396 | 62 | 543 | 508 | 2,048 |
| 2019 | 6,213 | 1,117 | 1,087 | 422 | 7,972 | 444 | 1,237 | 101 | 367 | 509 | 790 |
| 2020 | 9,184 | 866 | 1,971 | 688 | 5,970 | 1,211 | 3,453 | 132 | 569 | 827 | 1,465 |
| 2021 | 9,104 | 830 | 1,531 | 413 | 2,468 | 754 | 1,105 | 173 | 545 | 912 | 566 |
| 2022 | 3,785 | 677 | 948 | 242 | 1,445 | 887 | 321 | 55 | 157 | 561 | 357 |
| 2023 | 3,821 | 549 | 844 | 245 | 1,345 | 750 | 487 | 271 | 40 | 400 | 458 |
| 2024 | 2,923 | 888 | 961 | 332 | 1,811 | 523 | 103 | 130 | 45 | 371 | 518 |
| 2025 | 3,315 | 557 | 1,813 | 293 | 3,995 | 231 | 51 | 155 | 12 | 438 | 873 |
| 2026 | 6,549 | 1,586 | 3,367 | 489 | 2,412 | 566 | 125 | 194 | 13 | 823 | 636 |

Source: supplied AKFIN / NMFS Alaska Regional Office FMP catch export. Estimates sum retained and discarded catch across both pollock target categories and all reported gears in the BS subarea. Pollock, salmon, halibut, Non TAC Species, and records without a species group are excluded. Other combines remaining named source groups, including other flatfish and rockfish, Greenland turbot, Alaska plaice, squid, octopus, sculpins, and historical Other / Other Species groups; definitions follow the source export rather than a fixed modern FMP list. NA means no records for that category and year, rather than a measured zero. Source group names and the complete column map are available with the annual CSV. Values are summed before rounding; units are metric tons. The 2026 estimates are preliminary through the week ending October 24. Download annual estimates (CSV) · Column definitions (CSV).

**Table 7:**Pollock bycatch (t) in identified non-pollock directed Bering Sea fisheries, 1991-2026.

| Year | Rock.sole | Other.flats | Other.spp | P.cod | Yellowfin.sole | Total |
|---|---|---|---|---|---|---|
| 1991 | 9,711 | 7,992 | 756 | 10,695 | NA | 29,155 |
| 1992 | 9,825 | 1,372 | 568 | 20,778 | 13,101 | 45,644 |
| 1993 | 18,583 | 2,581 | 524 | 31,299 | 15,253 | 68,240 |
| 1994 | 15,784 | 6,770 | 89 | 26,594 | 33,200 | 82,438 |
| 1995 | 7,766 | 5,211 | 58 | 25,691 | 27,041 | 65,768 |
| 1996 | 7,699 | 5,457 | 205 | 22,382 | 22,255 | 57,997 |
| 1997 | 9,123 | 3,480 | 13 | 33,658 | 24,101 | 70,375 |
| 1998 | 3,960 | 3,011 | 61 | 10,469 | 15,339 | 32,840 |
| 1999 | 5,207 | 4,772 | 258 | 21,131 | 8,701 | 40,070 |
| 2000 | 5,481 | 7,068 | 43 | 14,509 | 13,425 | 40,526 |
| 2001 | 4,578 | 4,740 | 488 | 11,571 | 16,503 | 37,879 |
| 2002 | 9,942 | 2,220 | 52 | 15,256 | 14,490 | 41,960 |
| 2003 | 4,925 | 3,672 | 261 | 15,926 | 11,579 | 36,362 |
| 2004 | 8,976 | 6,396 | 194 | 18,650 | 10,384 | 44,600 |
| 2005 | 7,235 | 5,058 | 202 | 14,110 | 10,313 | 36,917 |
| 2006 | 6,986 | 3,826 | 143 | 15,168 | 5,967 | 32,091 |
| 2007 | 3,245 | 4,353 | 276 | 20,320 | 4,021 | 32,215 |
| 2008 | 4,931 | 4,822 | 17 | 9,534 | 9,828 | 29,132 |
| 2009 | 6,172 | 3,505 | 14 | 7,876 | 7,037 | 24,603 |
| 2010 | 6,074 | 3,317 | 86 | 6,576 | 5,179 | 21,232 |
| 2011 | 6,931 | 2,302 | 301 | 8,981 | 8,674 | 27,189 |
| 2012 | 6,704 | 1,751 | 413 | 8,377 | 11,198 | 28,443 |
| 2013 | 7,327 | 4,048 | 238 | 9,801 | 20,172 | 41,586 |
| 2014 | 11,259 | 6,404 | 203 | 11,502 | 24,713 | 54,080 |
| 2015 | 9,387 | 4,994 | 430 | 9,062 | 21,282 | 45,154 |
| 2016 | 11,850 | 3,687 | 450 | 9,071 | 22,306 | 47,365 |
| 2017 | 5,616 | 3,612 | 508 | 8,319 | 23,414 | 41,471 |
| 2018 | 5,182 | 3,525 | 957 | 8,008 | 28,235 | 45,908 |
| 2019 | 3,176 | 7,972 | 1,367 | 7,594 | 23,153 | 43,262 |
| 2020 | 6,401 | 2,373 | 728 | 5,512 | 31,651 | 46,666 |
| 2021 | 2,399 | 5,098 | 1,415 | 4,316 | 24,845 | 38,073 |
| 2022 | 2,977 | 5,505 | 2,627 | 6,260 | 26,515 | 43,884 |
| 2023 | 11,048 | 4,586 | 3,076 | 7,182 | 22,348 | 48,239 |
| 2024 | 15,126 | 4,356 | 1,929 | 6,690 | 18,895 | 46,996 |
| 2025 | 10,776 | 2,070 | 1,279 | 6,427 | 8,710 | 29,263 |
| 2026 | 13,847 | 7,040 | 1,345 | 6,599 | 5,898 | 34,729 |

Source: supplied AKFIN / NMFS Alaska Regional Office FMP catch export, BS subarea. Estimates include retained and discarded pollock catch and all reported gears. Pollock-directed trips, blank trip targets, and No retained catch targets are excluded. Other.flats combines flathead sole, arrowtooth flounder, other flatfish, Alaska plaice, Kamchatka flounder, and Greenland turbot targets. Other.spp combines the remaining identified targets: rockfish, sablefish, Atka mackerel, Other Species, and halibut. Rock.sole is the source Rock Sole - BSAI target; it is not restricted to a separately identified northern rock sole target. NA means no records for that target group and year. Total sums the included unrounded values; displayed columns may differ from Total by rounding. Units are metric tons. The 2026 estimates are preliminary through the week ending October 31. The complete target map and exclusion totals accompany the annual CSV. Download annual estimates (CSV) · Column definitions (CSV).

**Table 8:**Bycatch estimates of prohibited species in the BSAI directed pollock fishery, 1991–2026.

| Year | Bairdi | Chinook | Halib | Halib.M | Herring | NonChin | Snow Crab | Red King | Blue King | Golden King |
|---|---|---|---|---|---|---|---|---|---|---|
| **1991** | 1,398,107 | 39,054 | 2,156 | NA | 3,159 | 28,709 | 4,380,023 | 17,777 | NA | NA |
| **1992** | 1,500,765 | 33,672 | 2,220 | NA | 647 | 40,187 | 4,569,662 | 43,874 | NA | NA |
| **1993** | 1,649,103 | 36,619 | 1,326 | NA | 527 | 241,980 | 738,259 | 58,140 | NA | NA |
| **1994** | 371,214 | 31,890 | 963 | 689 | 1,627 | 92,011 | 811,734 | 42,361 | NA | NA |
| **1995** | 153,993 | 13,403 | 492 | 397 | 905 | 17,755 | 206,651 | 4,644 | NA | NA |
| **1996** | 89,416 | 55,472 | 382 | 321 | 1,242 | 77,174 | 63,398 | 5,934 | NA | NA |
| **1997** | 17,046 | 44,320 | 257 | 200 | 1,135 | 65,415 | 216,152 | 137 | NA | NA |
| **1998** | 57,071 | 51,245 | 353 | 278 | 801 | 60,677 | 123,758 | 14,287 | NA | NA |
| **1999** | 7,549 | 11,978 | 154 | 125 | 800 | 45,169 | 20,386 | 92 | NA | NA |
| **2000** | 1,517 | 4,961 | 110 | 91 | 483 | 58,568 | 6,580 | NA | NA | NA |
| **2001** | 5,146 | 33,444 | 243 | 200 | 225 | 57,004 | 5,709 | 106 | NA | NA |
| **2002** | 2,172 | 34,495 | 165 | 138 | 109 | 80,641 | 2,908 | 20 | NA | NA |
| **2003** | 771 | 45,661 | 89 | 74 | 967 | 188,339 | 750 | 52 | 9 | 0 |
| **2004** | 1,139 | 51,751 | 96 | 92 | 1,096 | 442,148 | 712 | 27 | 4 | 2 |
| **2005** | 631 | 68,126 | 120 | 112 | 594 | 703,080 | 2,260 | 0 | 0 | 1 |
| **2006** | 1,345 | 82,653 | 132 | 122 | 434 | 305,793 | 2,796 | 289 | 0 | 3 |
| **2007** | 1,497 | 121,956 | 312 | 290 | 352 | 86,380 | 3,165 | 7 | 0 | 3 |
| **2008** | 9,179 | 21,242 | 373 | 336 | 128 | 15,133 | 9,327 | 676 | 8 | 33 |
| **2009** | 6,204 | 12,731 | 541 | 457 | 65 | 45,960 | 7,392 | 1,137 | 20 | 0 |
| **2010** | 12,846 | 9,835 | 335 | 267 | 349 | 13,734 | 10,291 | 1,124 | 35 | 0 |
| **2011** | 10,986 | 25,499 | 460 | 378 | 377 | 193,754 | 6,584 | 577 | 26 | 0 |
| **2012** | 5,621 | 11,343 | 463 | 388 | 2,353 | 22,297 | 6,250 | 344 | 0 | 0 |
| **2013** | 12,427 | 13,091 | 334 | 272 | 959 | 125,525 | 8,704 | 508 | 34 | 107 |
| **2014** | 12,522 | 15,135 | 239 | 200 | 160 | 219,837 | 19,455 | 368 | 0 | 148 |
| **2015** | 8,872 | 18,329 | 152 | 130 | 1,487 | 237,803 | 8,339 | 0 | 0 | 0 |
| **2016** | 2,295 | 22,204 | 121 | 103 | 1,431 | 343,208 | 1,166 | 440 | 0 | 106 |
| **2017** | 7,270 | 30,079 | 97 | 89 | 964 | 467,750 | 3,407 | 203 | 0 | 67 |
| **2018** | 2,249 | 13,726 | 75 | 62 | 474 | 295,818 | 5,143 | 565 | 0 | 53 |
| **2019** | 3,146 | 25,038 | 134 | 113 | 1,102 | 348,631 | 6,228 | 453 | 99 | 445 |
| **2020** | 10,749 | 32,204 | 128 | 102 | 3,861 | 343,625 | 40,005 | 479 | 1 | 522 |
| **2021** | 8,417 | 13,852 | 145 | 131 | 1,709 | 546,472 | 4,668 | 52 | 0 | 115 |
| **2022** | 4,758 | 6,415 | 170 | 156 | 1,708 | 242,375 | 1,952 | 311 | 59 | 88 |
| **2023** | 11,997 | 11,874 | 88 | 70 | 3,087 | 112,512 | 4,100 | 54 | 0 | 132 |
| **2024** | 10,120 | 8,054 | 81 | 53 | 1,281 | 35,152 | 10,553 | 230 | 1 | 4 |
| **2025** | 4,822 | 19,825 | 79 | 66 | 3,030 | 151,457 | 14,424 | 1,018 | 1 | 2 |
| **2026** | 3,336 | 15,153 | 64 | 56 | 3,354 | 149,468 | 4,353 | 148 | 0 | 5 |

Source: author-provided AKFIN / NMFS Alaska Regional Office export, received from AKR October 27, 2026 and loaded November 7, 2026 (earlier source dates remain in some historical records). Both pollock target categories and all reported gears are included. Herring, halibut catch (Halib), and halibut mortality (Halib.M) are metric tons; other columns are estimated numbers of individuals. Values are summed before rounding. NA denotes unavailable estimates; zero denotes a reported zero. Sums use available values when individual records are missing; halibut mortality is unavailable for 1991–1993 and has some missing records in 1998–2003. The 2026 values are preliminary, through the week ending October 24. Other King Crab is outside the species columns of the example table. Download annual PSC estimates (CSV) · Column definitions (CSV).

**Table 9:**Estimated catch (t) of selected nontarget-species groups in the BSAI directed pollock fishery, 2003-2026.

| Year | Jellies | MiscFish | SeaStar | Osmerid | Grenadier | Eelpouts | Sea.pen | Anemone | Snails | Other |
|---|---|---|---|---|---|---|---|---|---|---|
| 2003 | 5,644 | 101 | 89 | 10 | 21 | 7 | 1 | 0 | 1 | 2 |
| 2004 | 6,590 | 90 | 7 | 22 | 14 | 1 | 1 | 0 | 1 | 1 |
| 2005 | 5,198 | 158 | 10 | 13 | 14 | 1 | 2 | 0 | 7 | 3 |
| 2006 | 2,718 | 154 | 11 | 102 | 16 | 21 | 2 | 1 | 0 | 14 |
| 2007 | 2,403 | 203 | 5 | 140 | 28 | 119 | 4 | 0 | 0 | 13 |
| 2008 | 4,185 | 121 | 20 | 4 | 28 | 9 | 1 | 1 | 2 | 9 |
| 2009 | 8,118 | 135 | 10 | 6 | 5 | 4 | 3 | 1 | 1 | 3 |
| 2010 | 2,518 | 150 | 12 | 1 | 2 | 1 | 3 | 2 | 1 | 10 |
| 2011 | 8,233 | 277 | 27 | 2 | 2 | 1 | 3 | 2 | 1 | 8 |
| 2012 | 3,519 | 143 | 7 | 2 | 2 | 1 | 3 | 1 | 1 | 4 |
| 2013 | 5,294 | 121 | 15 | 1 | 2 | 2 | 2 | 2 | 1 | 10 |
| 2014 | 12,767 | 45 | 30 | 2 | 10 | 8 | 3 | 2 | 2 | 9 |
| 2015 | 4,951 | 90 | 42 | 24 | 4 | 11 | 2 | 2 | 1 | 5 |
| 2016 | 2,203 | 75 | 55 | 5 | 4 | 23 | 1 | 1 | 0 | 4 |
| 2017 | 6,153 | 48 | 12 | 3 | 3 | 18 | 1 | 1 | 1 | 2 |
| 2018 | 8,252 | 52 | 25 | 1 | 9 | 4 | 1 | 1 | 0 | 4 |
| 2019 | 3,889 | 73 | 51 | 1 | 9 | 2 | 1 | 0 | 0 | 6 |
| 2020 | 3,150 | 94 | 61 | 1 | 43 | 6 | 1 | 6 | 1 | 26 |
| 2021 | 7,831 | 35 | 20 | 0 | 55 | 1 | 2 | 3 | 1 | 3 |
| 2022 | 7,609 | 23 | 185 | 1 | 11 | 0 | 1 | 1 | 1 | 11 |
| 2023 | 7,088 | 38 | 28 | 0 | 86 | 1 | 2 | 3 | 1 | 6 |
| 2024 | 4,402 | 43 | 16 | 3 | 18 | 3 | 1 | 4 | 0 | 3 |
| 2025 | 1,769 | 99 | 8 | 4 | 6 | 14 | 2 | 1 | 1 | 3 |
| 2026 | 4,418 | 68 | 5 | 3 | 1 | 1 | 2 | 0 | 1 | 2 |

Source: supplied AKFIN / NMFS Alaska Regional Office nontarget catch-accounting export. BSAI records (BS and AI) are included; GOA and inside-water records are excluded. Both pollock target categories, all reported gears, and retained and discarded records are combined using Nontarget Estimate (mt), not the species-count field. Squid, sculpins, and seabirds are outside the selected groups of this table and are excluded. MiscFish is the source Misc fish group. Osmerid combines capelin, eulachon, other osmerids, family Osmeridae smelts, and surf smelt; Grenadier combines all named grenadier groups. Other contains the remaining selected fish and invertebrate groups; the complete mapping accompanies the annual CSV. NA means no records for a group-year; a displayed zero may be a positive estimate below 0.5 t. Values are summed before rounding to whole metric tons. The 2026 estimates are preliminary through the week ending October 24. The November 2024 summary approximately doubled the 2019-2022 Jellies, MiscFish, SeaStar, Osmerid, Grenadier, and Eelpouts columns relative to September. Sea pen, Anemone, Snails, and Other changed non-uniformly, with cross-column numerical matches indicating changed alignment or allocation. These are comparisons of historical headers. The current table uses explicit source species names and both catch dispositions, with no scaling adjustment. The upstream preparation step remains unconfirmed. View the jellyfish source-history and filter audit. Download annual estimates (CSV) · Column definitions (CSV).

**Table 10:** Numbers of EBS pollock measured for fishery length frequencies, by recorded sex and processing stratum, 1977–2025.

| Year | A | | B southeast | | B northwest | | Total |
|---|---|---|---|---|---|---|---|
| | Male | Female | Male | Female | Male | Female | |
| 1977 | 26,411 | 25,923 | 4,301 | 4,511 | 29,075 | 31,219 | 121,440 |
| 1978 | 25,110 | 31,653 | 9,829 | 9,524 | 46,349 | 46,072 | 168,537 |
| 1979 | 59,782 | 62,512 | 3,461 | 3,113 | 62,298 | 61,402 | 252,568 |
| 1980 | 42,726 | 42,577 | 3,380 | 3,464 | 47,030 | 49,037 | 188,214 |
| 1981 | 64,718 | 57,936 | 2,401 | 2,147 | 53,161 | 53,570 | 233,933 |
| 1982 | 74,172 | 70,073 | 16,265 | 14,885 | 181,606 | 163,272 | 520,273 |
| 1983 | 94,118 | 90,778 | 16,604 | 16,826 | 193,031 | 174,589 | 585,946 |
| 1984 | 158,329 | 161,876 | 106,654 | 105,234 | 243,877 | 217,362 | 993,332 |
| 1985 | 119,384 | 109,230 | 96,684 | 97,841 | 284,850 | 256,091 | 964,080 |
| 1986 | 186,505 | 189,497 | 135,444 | 123,413 | 164,546 | 131,322 | 930,727 |
| 1987 | 373,163 | 399,072 | 14,170 | 21,162 | 24,038 | 22,117 | 853,722 |
| 1988 | NA | NA | NA | NA | NA | NA | NA |
| 1989 | NA | NA | NA | NA | NA | NA | NA |
| 1990 | NA | NA | NA | NA | NA | NA | NA |
| 1991 | 155,876 | 143,625 | 148,385 | 132,539 | 123,516 | 122,241 | 826,182 |
| 1992 | 152,566 | 148,024 | 150,829 | 152,003 | 93,073 | 94,701 | 791,196 |
| 1993 | 136,211 | 126,635 | 145,280 | 137,384 | 24,797 | 26,057 | 596,364 |
| 1994 | 138,179 | 146,067 | 154,311 | 148,497 | 26,431 | 26,380 | 639,865 |
| 1995 | 128,719 | 125,847 | 175,115 | 150,323 | 16,142 | 16,327 | 612,473 |
| 1996 | 147,992 | 139,905 | 193,493 | 149,814 | 18,101 | 18,288 | 667,593 |
| 1997 | 123,454 | 102,619 | 114,846 | 106,001 | 58,492 | 51,498 | 556,910 |
| 1998 | 135,136 | 109,119 | 205,282 | 174,676 | 31,968 | 39,475 | 695,656 |
| 1999 | 36,035 | 32,407 | 38,229 | 35,084 | 16,258 | 18,321 | 176,334 |
| 2000 | 64,430 | 58,030 | 63,746 | 41,027 | 40,839 | 39,105 | 307,177 |
| 2001 | 79,190 | 75,491 | 54,037 | 51,179 | 44,232 | 45,766 | 349,895 |
| 2002 | 71,502 | 69,467 | 65,299 | 64,243 | 37,661 | 39,285 | 347,457 |
| 2003 | 74,902 | 77,533 | 49,307 | 52,899 | 51,764 | 53,435 | 359,840 |
| 2004 | 75,208 | 75,811 | 63,146 | 61,957 | 47,261 | 44,220 | 367,603 |
| 2005 | 75,784 | 68,665 | 43,271 | 33,917 | 68,831 | 63,022 | 353,490 |
| 2006 | 72,543 | 63,349 | 35,378 | 27,939 | 77,620 | 67,219 | 344,048 |
| 2007 | 66,533 | 63,969 | 38,104 | 29,558 | 76,755 | 70,504 | 345,423 |
| 2008 | 51,303 | 46,296 | 23,467 | 20,462 | 64,126 | 60,678 | 266,332 |
| 2009 | 43,476 | 41,540 | 17,343 | 16,148 | 45,418 | 47,926 | 211,851 |
| 2010 | 41,019 | 39,495 | 20,577 | 19,194 | 40,914 | 40,449 | 201,648 |
| 2011 | 62,295 | 58,481 | 65,057 | 60,208 | 48,055 | 50,927 | 345,023 |
| 2012 | 57,946 | 53,557 | 46,942 | 45,024 | 53,243 | 49,968 | 306,680 |
| 2013 | 62,148 | 51,984 | 44,582 | 37,307 | 49,649 | 49,161 | 294,831 |
| 2014 | 58,066 | 55,954 | 51,743 | 46,568 | 46,067 | 46,642 | 305,040 |
| 2015 | 56,419 | 55,646 | 43,601 | 46,853 | 41,183 | 45,117 | 288,819 |
| 2016 | 58,915 | 57,478 | 69,654 | 72,973 | 9,015 | 10,264 | 278,299 |
| 2017 | 64,693 | 55,965 | 65,982 | 70,285 | 14,125 | 15,871 | 286,921 |
| 2018 | 64,628 | 57,156 | 49,653 | 56,243 | 32,796 | 35,811 | 296,287 |
| 2019 | 64,665 | 49,191 | 54,927 | 59,416 | 27,753 | 34,955 | 290,907 |
| 2020 | 65,609 | 60,018 | 47,791 | 53,161 | 48,459 | 53,985 | 329,023 |
| 2021 | 75,561 | 59,580 | 53,205 | 61,208 | 27,641 | 30,833 | 308,028 |
| 2022 | 114,918 | 93,686 | 86,598 | 94,900 | 33,622 | 36,012 | 459,736 |
| 2023 | 73,284 | 54,056 | 38,537 | 42,259 | 38,222 | 37,101 | 283,459 |
| 2024 | 72,356 | 60,933 | 47,300 | 53,442 | 33,511 | 33,693 | 301,397 |
| 2025 | 78,854 | 65,020 | 39,796 | 42,845 | 43,374 | 48,084 | 319,026 |

**Table 11:** Numbers of EBS pollock with paired fish-weight and length measurements, by recorded sex and processing stratum, 1977–2025.

| Year | A | | B southeast | | B northwest | | Total |
|---|---|---|---|---|---|---|---|
| | Male | Female | Male | Female | Male | Female | |
| 1977 | 1,222 | 1,338 | 137 | 166 | 1,461 | 1,664 | 5,988 |
| 1978 | 1,991 | 2,686 | 409 | 516 | 2,200 | 2,623 | 10,425 |
| 1979 | 2,709 | 3,151 | 152 | 209 | 1,469 | 1,566 | 9,256 |
| 1980 | 1,849 | 2,156 | 99 | 144 | 612 | 681 | 5,541 |
| 1981 | 1,821 | 2,045 | 51 | 52 | 1,623 | 1,810 | 7,402 |
| 1982 | 2,030 | 2,208 | 181 | 176 | 2,852 | 3,043 | 10,490 |
| 1983 | 1,199 | 1,200 | 144 | 122 | 3,268 | 3,447 | 9,380 |
| 1984 | 980 | 1,046 | 117 | 136 | 1,273 | 1,378 | 4,930 |
| 1985 | 520 | 499 | 46 | 55 | 426 | 488 | 2,034 |
| 1986 | 689 | 794 | 518 | 501 | 286 | 286 | 3,074 |
| 1987 | 1,351 | 1,466 | 25 | 33 | 72 | 63 | 3,010 |
| 1988 | NA | NA | NA | NA | NA | NA | NA |
| 1989 | NA | NA | NA | NA | NA | NA | NA |
| 1990 | NA | NA | NA | NA | NA | NA | NA |
| 1991 | 2,893 | 2,791 | 1,209 | 1,116 | 2,536 | 2,408 | 12,953 |
| 1992 | 1,605 | 1,537 | 556 | 600 | 2,003 | 1,940 | 8,241 |
| 1993 | 1,278 | 1,205 | 451 | 437 | 1,412 | 1,459 | 6,242 |
| 1994 | 1,638 | 1,553 | 174 | 166 | 1,591 | 1,584 | 6,706 |
| 1995 | 1,258 | 1,220 | 232 | 223 | 1,352 | 1,331 | 5,616 |
| 1996 | 2,165 | 2,117 | NA | NA | 1,393 | 1,421 | 7,096 |
| 1997 | 629 | 630 | 552 | 536 | 674 | 620 | 3,641 |
| 1998 | 1,958 | 1,865 | 357 | 335 | 936 | 982 | 6,433 |
| 1999 | 4,813 | 5,337 | 3,767 | 3,546 | 7,182 | 7,954 | 32,599 |
| 2000 | 11,346 | 12,457 | 7,736 | 7,991 | 7,800 | 12,463 | 59,793 |
| 2001 | 14,411 | 14,965 | 9,064 | 8,803 | 10,460 | 10,871 | 68,574 |
| 2002 | 13,564 | 14,098 | 7,648 | 7,213 | 13,004 | 12,988 | 68,515 |
| 2003 | 15,535 | 14,857 | 10,272 | 10,031 | 10,111 | 9,437 | 70,243 |
| 2004 | 7,924 | 7,742 | 4,318 | 4,617 | 6,868 | 6,850 | 38,319 |
| 2005 | 7,039 | 7,428 | 6,426 | 6,947 | 4,114 | 5,139 | 37,093 |
| 2006 | 6,566 | 7,381 | 6,442 | 7,406 | 3,045 | 4,006 | 34,846 |
| 2007 | 6,640 | 6,695 | 7,081 | 7,798 | 3,202 | 4,305 | 35,721 |
| 2008 | 4,501 | 4,865 | 5,855 | 6,264 | 2,236 | 2,624 | 26,345 |
| 2009 | 4,033 | 4,382 | 4,655 | 4,511 | 1,723 | 1,934 | 21,238 |
| 2010 | 4,258 | 4,536 | 3,883 | 4,125 | 2,012 | 2,261 | 21,075 |
| 2011 | 5,845 | 6,388 | 4,954 | 4,647 | 5,929 | 6,456 | 34,219 |
| 2012 | 5,494 | 5,979 | 4,923 | 5,346 | 4,507 | 4,774 | 31,023 |
| 2013 | 5,689 | 6,525 | 4,844 | 4,920 | 3,599 | 4,313 | 29,890 |
| 2014 | 5,675 | 5,871 | 4,785 | 4,652 | 4,753 | 5,180 | 30,916 |
| 2015 | 5,310 | 5,323 | 4,648 | 4,194 | 4,365 | 4,064 | 27,904 |
| 2016 | 5,312 | 5,725 | 1,077 | 909 | 6,872 | 6,635 | 26,530 |
| 2017 | 5,238 | 6,047 | 1,586 | 1,343 | 6,575 | 6,254 | 27,043 |
| 2018 | 5,583 | 6,174 | 3,430 | 3,172 | 5,506 | 4,850 | 28,715 |
| 2019 | 4,513 | 6,086 | 3,594 | 2,953 | 5,809 | 5,499 | 28,454 |
| 2020 | 6,116 | 6,846 | 5,325 | 4,815 | 5,376 | 4,900 | 33,378 |
| 2021 | 5,852 | 7,368 | 6,247 | 5,468 | 2,886 | 2,698 | 30,519 |
| 2022 | 4,890 | 5,847 | 1,705 | 1,566 | 4,899 | 4,364 | 23,271 |
| 2023 | 10,676 | 14,487 | 6,171 | 6,173 | 5,911 | 5,436 | 48,854 |
| 2024 | 7,253 | 6,276 | 5,027 | 5,679 | 3,298 | 3,421 | 30,958 |
| 2025 | 7,391 | 6,333 | 4,257 | 4,597 | 4,379 | 4,922 | 31,927 |

**Table 12:** Numbers of EBS pollock contributing age estimates, by recorded sex and processing stratum, 1977–2025. The 2024–2025 entries are released FT-NIRS age predictions.

| Year | A | | B southeast | | B northwest | | Total |
|---|---|---|---|---|---|---|---|
| | Male | Female | Male | Female | Male | Female | |
| 1977 | 1,229 | 1,344 | 137 | 166 | 1,415 | 1,613 | 5,904 |
| 1978 | 1,992 | 2,686 | 407 | 514 | 2,188 | 2,611 | 10,398 |
| 1979 | 2,647 | 3,088 | 152 | 209 | 1,464 | 1,561 | 9,121 |
| 1980 | 1,854 | 2,158 | 93 | 138 | 606 | 675 | 5,524 |
| 1981 | 1,819 | 2,042 | 51 | 52 | 1,620 | 1,807 | 7,391 |
| 1982 | 2,030 | 2,210 | 181 | 176 | 2,865 | 3,062 | 10,524 |
| 1983 | 1,200 | 1,200 | 144 | 122 | 3,249 | 3,420 | 9,335 |
| 1984 | 980 | 1,046 | 117 | 136 | 1,272 | 1,379 | 4,930 |
| 1985 | 520 | 499 | 46 | 55 | 426 | 488 | 2,034 |
| 1986 | 689 | 794 | 518 | 501 | 286 | 286 | 3,074 |
| 1987 | 1,351 | 1,466 | 25 | 33 | 72 | 63 | 3,010 |
| 1988 | NA | NA | NA | NA | NA | NA | NA |
| 1989 | NA | NA | NA | NA | NA | NA | NA |
| 1990 | NA | NA | NA | NA | NA | NA | NA |
| 1991 | 439 | 431 | 367 | 349 | 263 | 289 | 2,138 |
| 1992 | 399 | 396 | 178 | 180 | 391 | 375 | 1,919 |
| 1993 | 476 | 445 | 122 | 124 | 496 | 507 | 2,170 |
| 1994 | 201 | 200 | 142 | 132 | 574 | 571 | 1,820 |
| 1995 | 313 | 299 | 131 | 123 | 420 | 439 | 1,725 |
| 1996 | 465 | 479 | NA | NA | 436 | 443 | 1,823 |
| 1997 | 437 | 434 | 343 | 341 | 313 | 286 | 2,154 |
| 1998 | 663 | 595 | 237 | 222 | 311 | 316 | 2,344 |
| 1999 | 506 | 541 | 298 | 308 | 748 | 750 | 3,151 |
| 2000 | 629 | 667 | 293 | 254 | 596 | 847 | 3,286 |
| 2001 | 563 | 603 | 205 | 178 | 697 | 736 | 2,982 |
| 2002 | 672 | 663 | 247 | 202 | 890 | 839 | 3,513 |
| 2003 | 653 | 588 | 274 | 262 | 701 | 671 | 3,149 |
| 2004 | 547 | 561 | 221 | 245 | 698 | 600 | 2,872 |
| 2005 | 599 | 617 | 420 | 422 | 490 | 614 | 3,162 |
| 2006 | 528 | 609 | 507 | 568 | 367 | 459 | 3,038 |
| 2007 | 627 | 642 | 552 | 568 | 485 | 594 | 3,468 |
| 2008 | 513 | 497 | 538 | 650 | 342 | 368 | 2,908 |
| 2009 | 404 | 484 | 440 | 432 | 240 | 299 | 2,299 |
| 2010 | 545 | 624 | 413 | 466 | 418 | 505 | 2,971 |
| 2011 | 581 | 808 | 404 | 396 | 582 | 660 | 3,431 |
| 2012 | 517 | 571 | 485 | 579 | 480 | 533 | 3,165 |
| 2013 | 666 | 703 | 525 | 568 | 401 | 518 | 3,381 |
| 2014 | 609 | 629 | 413 | 407 | 475 | 553 | 3,086 |
| 2015 | 653 | 642 | 511 | 493 | 508 | 513 | 3,320 |
| 2016 | 488 | 599 | 157 | 125 | 929 | 969 | 3,267 |
| 2017 | 604 | 778 | 179 | 163 | 777 | 753 | 3,254 |
| 2018 | 569 | 662 | 366 | 358 | 621 | 591 | 3,167 |
| 2019 | 552 | 778 | 387 | 332 | 558 | 531 | 3,138 |
| 2020 | 757 | 899 | 405 | 420 | 450 | 408 | 3,339 |
| 2021 | 760 | 910 | 588 | 542 | 270 | 256 | 3,326 |
| 2022 | 609 | 777 | 209 | 212 | 616 | 563 | 2,986 |
| 2023 | 452 | 633 | 350 | 313 | 399 | 342 | 2,489 |
| 2024 | 700 | 616 | 493 | 534 | 291 | 350 | 2,985 |
| 2025 | 697 | 562 | 401 | 433 | 439 | 456 | 2,992 |

**Table 13:** Pollock sampling effort from the NMFS bottom-trawl survey, 1982–2026 (counts). Historical values are as reported; recent haul counts use the survey records and recent ages are released EBS assignments, including FT-NIRS ages.

| Year | Hauls | Lengths | Aged / age assigned |
|---|---|---|---|
| 1982 | 329 | 40,001 | 1,611 |
| 1983 | 354 | 78,033 | 1,931 |
| 1984 | 355 | 40,530 | 1,806 |
| 1985 | 434 | 48,642 | 1,913 |
| 1986 | 354 | 41,101 | 1,344 |
| 1987 | 356 | 40,144 | 1,607 |
| 1988 | 373 | 40,408 | 1,173 |
| 1989 | 373 | 38,926 | 1,227 |
| 1990 | 371 | 34,814 | 1,257 |
| 1991 | 371 | 43,406 | 1,083 |
| 1992 | 356 | 34,024 | 1,263 |
| 1993 | 375 | 43,278 | 1,385 |
| 1994 | 375 | 38,901 | 1,141 |
| 1995 | 376 | 25,673 | 1,156 |
| 1996 | 375 | 40,789 | 1,387 |
| 1997 | 376 | 35,536 | 1,193 |
| 1998 | 375 | 37,673 | 1,261 |
| 1999 | 373 | 32,532 | 1,385 |
| 2000 | 372 | 41,762 | 1,545 |
| 2001 | 375 | 47,335 | 1,641 |
| 2002 | 375 | 43,361 | 1,695 |
| 2003 | 376 | 46,480 | 1,638 |
| 2004 | 375 | 44,102 | 1,660 |
| 2005 | 373 | 35,976 | 1,676 |
| 2006 | 376 | 39,211 | 1,573 |
| 2007 | 376 | 29,679 | 1,484 |
| 2008 | 375 | 24,635 | 1,251 |
| 2009 | 376 | 24,819 | 1,342 |
| 2010 | 517 | 23,142 | 1,385 |
| 2011 | 376 | 36,227 | 1,734 |
| 2012 | 376 | 35,782 | 1,785 |
| 2013 | 376 | 35,908 | 1,847 |
| 2014 | 376 | 43,042 | 2,099 |
| 2015 | 376 | 54,241 | 2,320 |
| 2016 | 376 | 50,857 | 1,766 |
| 2017 | 519 | 47,873 | 1,623 |
| 2018 | 376 | 48,673 | 1,486 |
| 2019 | 520 | 42,382 | 1,519 |
| 2020 | NA | NA | NA |
| 2021 | 520 | 53,545 | 2,050 |
| 2022 | 520 | 36,687 | 1,980 |
| 2023 | 492 | 38,618 | 1,677 |
| 2024 | 350 | 36,137 | 1,526 |
| 2025 | 487 | 30,862 | 1,619 |
| 2026 | 350 | 31,296 | 1,582 |

NA denotes an unavailable count; the survey was absent in 2020. The 2025 haul count includes 350 EBS and 137 NBS records; the 2026 count includes 350 EBS records. Historical coverage varies among years.

The 2025 and 2026 length counts use the EBS production summaries. Recent length and age counts cover EBS; haul counts include NBS where stated.

Recent age assignments are distinct from traditional tester-pair counts; historical age counts retain their published definitions.

**Table 14:** Bottom-trawl survey (BTS) pollock biomass for the combined eastern and northern Bering Sea, 1982–2026, with assessment standard errors (SE) and coefficients of variation (CV).

| Year | Biomass (million t) | SE (thousand t) | CV (%) |
|---|---|---|---|
| 1982 | 4.064 | 732 | 18.0% |
| 1983 | 10.457 | 1,174 | 11.2% |
| 1984 | 6.777 | 977 | 14.4% |
| 1985 | 7.794 | 1,008 | 12.9% |
| 1986 | 7.459 | 1,078 | 14.5% |
| 1987 | 8.131 | 1,399 | 17.2% |
| 1988 | 12.000 | 1,896 | 15.8% |
| 1989 | 10.554 | 1,435 | 13.6% |
| 1990 | 11.449 | 2,352 | 20.5% |
| 1991 | 7.472 | 1,427 | 19.1% |
| 1992 | 6.961 | 1,121 | 16.1% |
| 1993 | 8.275 | 1,060 | 12.8% |
| 1994 | 7.385 | 1,159 | 15.7% |
| 1995 | 7.631 | 2,335 | 30.6% |
| 1996 | 4.179 | 551 | 13.2% |
| 1997 | 4.842 | 890 | 18.4% |
| 1998 | 3.665 | 565 | 15.4% |
| 1999 | 5.732 | 1,020 | 17.8% |
| 2000 | 7.778 | 1,350 | 17.4% |
| 2001 | 6.271 | 705 | 11.2% |
| 2002 | 7.117 | 971 | 13.6% |
| 2003 | 12.148 | 4,671 | 38.5% |
| 2004 | 6.063 | 739 | 12.2% |
| 2005 | 6.867 | 952 | 13.9% |
| 2006 | 4.181 | 513 | 12.3% |
| 2007 | 6.511 | 992 | 15.2% |
| 2008 | 4.271 | 633 | 14.8% |
| 2009 | 2.860 | 482 | 16.8% |
| 2010 | 5.174 | 839 | 16.2% |
| 2011 | 4.524 | 520 | 11.5% |
| 2012 | 5.063 | 681 | 13.5% |
| 2013 | 6.739 | 790 | 11.7% |
| 2014 | 11.838 | 1,266 | 10.7% |
| 2015 | 11.145 | 982 | 8.8% |
| 2016 | 8.825 | 1,006 | 11.4% |
| 2017 | 8.781 | 709 | 8.1% |
| 2018 | 5.119 | 813 | 15.9% |
| 2019 | 9.026 | 967 | 10.7% |
| 2021 | 4.706 | 338 | 7.2% |
| 2022 | 6.348 | 649 | 10.2% |
| 2023 | 4.754 | 429 | 9.0% |
| 2024 | 8.354 | 1,196 | 14.3% |
| 2025 | 5.801 | 792 | 13.7% |
| 2026 | 9.034 | 956 | 10.6% |

CV is the annual standard error divided by biomass. These are the uncertainties used in the assessment, which also accounts for correlations among years. The survey was absent in 2020.

**Table 15:** Combined eastern and northern Bering Sea model-based pollock abundance at age used for the assessment, 1982–2026 (million fish). Ages 15 and older are pooled as 15+.

| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1982 | 1220.6 | 2780.1 | 3221.8 | 4394.2 | 1481.2 | 196.2 | 126.3 | 60.7 | 37.7 | 23.7 | 14.8 | 8.8 | 3.3 | 1.4 | 1.4 | 13572.2 |
| 1983 | 4548.2 | 1080.4 | 2880.1 | 4538.0 | 9238.2 | 2681.1 | 458.1 | 210.7 | 97.6 | 87.7 | 62.5 | 22.9 | 8.3 | 7.7 | 3.6 | 25925.1 |
| 1984 | 494.0 | 404.5 | 803.7 | 2218.2 | 2460.1 | 5226.9 | 929.5 | 201.8 | 92.6 | 31.1 | 21.3 | 8.0 | 4.9 | 6.1 | 3.1 | 12905.9 |
| 1985 | 5028.8 | 787.5 | 3999.5 | 1409.8 | 4629.4 | 2808.1 | 1856.2 | 353.5 | 86.3 | 73.3 | 25.6 | 8.0 | 8.6 | 0.7 | 0.7 | 21075.9 |
| 1986 | 3061.3 | 918.2 | 793.9 | 3276.1 | 1686.3 | 2542.1 | 2103.4 | 1790.2 | 534.1 | 83.7 | 37.4 | 16.5 | 1.5 | 4.6 | 1.6 | 16850.8 |
| 1987 | 495.7 | 706.6 | 1227.4 | 956.9 | 5641.8 | 1477.3 | 1356.5 | 522.5 | 1628.6 | 256.8 | 79.2 | 30.0 | 5.9 | 2.9 | 2.6 | 14390.8 |
| 1988 | 1588.1 | 769.9 | 2287.4 | 4488.9 | 1959.7 | 6093.4 | 1659.9 | 1198.1 | 666.8 | 1611.9 | 155.7 | 91.5 | 20.6 | 24.6 | 14.8 | 22631.4 |
| 1989 | 944.5 | 576.1 | 979.3 | 2967.0 | 6523.0 | 1245.0 | 4451.9 | 628.4 | 703.5 | 263.0 | 825.0 | 141.4 | 126.8 | 59.2 | 95.4 | 20529.5 |
| 1990 | 2285.0 | 272.5 | 107.0 | 1089.6 | 2485.0 | 8094.0 | 1492.5 | 3554.8 | 337.0 | 587.7 | 89.3 | 832.7 | 71.8 | 52.6 | 73.2 | 21424.7 |
| 1991 | 3250.4 | 947.9 | 368.5 | 134.1 | 952.9 | 836.1 | 2526.4 | 881.9 | 1708.2 | 430.0 | 576.7 | 117.6 | 346.1 | 52.7 | 60.4 | 13189.9 |
| 1992 | 1617.8 | 418.5 | 2737.7 | 428.0 | 520.3 | 836.6 | 731.6 | 1010.7 | 438.8 | 810.1 | 281.6 | 355.1 | 153.0 | 121.0 | 97.3 | 10558.3 |
| 1993 | 2776.9 | 444.9 | 1162.3 | 5158.9 | 1074.2 | 819.5 | 409.8 | 538.4 | 681.7 | 416.6 | 359.3 | 261.2 | 209.2 | 115.1 | 140.3 | 14568.1 |
| 1994 | 1722.8 | 695.5 | 525.9 | 2019.7 | 5779.3 | 950.8 | 229.2 | 178.9 | 196.0 | 361.9 | 221.7 | 308.3 | 118.8 | 117.1 | 198.7 | 13624.6 |
| 1995 | 1631.4 | 204.8 | 456.2 | 1978.6 | 2724.3 | 4580.8 | 1714.5 | 388.3 | 229.3 | 147.8 | 272.0 | 112.0 | 206.2 | 84.3 | 134.9 | 14865.5 |
| 1996 | 2000.9 | 437.8 | 198.4 | 408.0 | 1127.1 | 1494.9 | 1288.5 | 423.6 | 102.1 | 111.4 | 74.7 | 142.5 | 48.6 | 87.4 | 128.8 | 8074.7 |
| 1997 | 2338.7 | 398.5 | 216.5 | 265.8 | 3005.0 | 1397.2 | 817.3 | 1038.9 | 178.6 | 90.3 | 66.3 | 71.4 | 124.0 | 40.3 | 152.6 | 10201.3 |
| 1998 | 902.8 | 708.6 | 371.2 | 262.0 | 548.3 | 3044.7 | 782.6 | 476.6 | 364.3 | 94.0 | 41.5 | 14.1 | 32.9 | 35.1 | 86.7 | 7765.4 |
| 1999 | 1000.6 | 1041.4 | 1090.9 | 1116.1 | 634.3 | 1180.5 | 2883.0 | 751.1 | 362.3 | 337.0 | 126.5 | 54.0 | 22.4 | 33.6 | 123.5 | 10757.3 |
| 2000 | 1053.0 | 376.4 | 703.0 | 2187.0 | 2206.2 | 1225.8 | 970.7 | 2977.1 | 1110.5 | 573.6 | 246.2 | 158.1 | 50.3 | 23.0 | 103.9 | 13964.7 |
| 2001 | 1985.9 | 1224.0 | 712.7 | 713.1 | 1687.0 | 1656.8 | 688.4 | 335.0 | 987.1 | 704.8 | 269.8 | 212.6 | 82.1 | 30.2 | 87.2 | 11376.7 |
| 2002 | 1319.8 | 393.9 | 794.7 | 1211.9 | 1379.5 | 1789.1 | 892.4 | 430.4 | 575.4 | 1076.3 | 538.2 | 249.8 | 149.1 | 48.0 | 57.8 | 10906.2 |
| 2003 | 565.2 | 170.1 | 1149.5 | 1885.8 | 2127.9 | 1800.1 | 2248.6 | 1153.8 | 477.1 | 626.8 | 1226.1 | 562.9 | 236.1 | 98.8 | 109.5 | 14438.2 |
| 2004 | 401.3 | 266.2 | 191.1 | 1614.3 | 1597.9 | 1188.0 | 665.2 | 702.3 | 346.8 | 219.4 | 220.1 | 406.2 | 181.4 | 48.1 | 40.8 | 8089.1 |
| 2005 | 397.3 | 153.0 | 251.1 | 1214.6 | 3364.3 | 2117.6 | 1068.1 | 485.3 | 369.1 | 269.1 | 75.0 | 156.8 | 261.1 | 108.4 | 123.4 | 10414.1 |
| 2006 | 822.5 | 76.0 | 113.3 | 525.4 | 1405.2 | 1644.7 | 972.2 | 442.0 | 245.2 | 204.2 | 98.3 | 61.0 | 87.9 | 119.3 | 123.5 | 6940.6 |
| 2007 | 2730.3 | 65.0 | 224.4 | 719.4 | 2148.5 | 2411.1 | 1628.2 | 1147.3 | 449.4 | 195.6 | 180.2 | 154.0 | 71.7 | 88.1 | 178.4 | 12391.4 |
| 2008 | 501.3 | 122.5 | 141.0 | 269.9 | 777.1 | 1573.6 | 1152.4 | 734.2 | 437.9 | 158.4 | 129.9 | 97.2 | 44.2 | 24.6 | 159.1 | 6323.3 |
| 2009 | 863.4 | 210.1 | 435.1 | 497.8 | 325.1 | 488.6 | 690.0 | 521.6 | 371.6 | 171.0 | 107.8 | 35.1 | 35.2 | 18.3 | 80.1 | 4850.6 |
| 2010 | 480.0 | 136.6 | 315.5 | 4277.0 | 1722.7 | 479.2 | 379.3 | 389.4 | 391.1 | 268.4 | 227.6 | 82.8 | 49.9 | 29.5 | 68.1 | 9296.8 |
| 2011 | 1331.7 | 136.3 | 359.0 | 431.6 | 2165.5 | 1048.1 | 294.4 | 160.1 | 244.1 | 240.6 | 205.5 | 158.0 | 69.4 | 33.1 | 98.2 | 6975.5 |
| 2012 | 1341.7 | 211.1 | 485.1 | 4164.0 | 951.6 | 1431.6 | 456.1 | 171.2 | 115.0 | 158.5 | 127.7 | 110.9 | 96.5 | 34.9 | 67.6 | 9923.3 |
| 2013 | 1257.5 | 134.9 | 273.7 | 1127.2 | 5422.6 | 1228.5 | 750.4 | 262.1 | 88.5 | 79.4 | 105.0 | 79.8 | 77.2 | 44.9 | 63.6 | 10995.3 |
| 2014 | 2531.9 | 477.4 | 279.9 | 497.3 | 2312.4 | 7741.2 | 3820.1 | 797.9 | 435.3 | 157.6 | 62.1 | 89.1 | 93.0 | 46.9 | 141.4 | 19483.5 |
| 2015 | 1460.1 | 947.2 | 2835.6 | 767.6 | 1528.6 | 2738.5 | 5150.1 | 1522.0 | 385.3 | 192.1 | 27.0 | 25.2 | 49.9 | 30.5 | 70.4 | 17730.0 |
| 2016 | 782.9 | 507.3 | 744.5 | 4155.6 | 1720.4 | 1196.9 | 1767.8 | 2703.3 | 547.2 | 224.0 | 79.7 | 17.8 | 18.7 | 6.8 | 14.1 | 14486.9 |
| 2017 | 789.3 | 312.7 | 526.1 | 2619.5 | 3497.6 | 1602.5 | 1236.2 | 1161.9 | 1488.5 | 425.9 | 163.7 | 55.1 | 8.4 | 4.8 | 16.2 | 13908.4 |
| 2018 | 1179.6 | 474.1 | 251.5 | 376.6 | 2531.9 | 1556.8 | 541.1 | 393.9 | 405.0 | 333.0 | 93.2 | 14.6 | 1.9 | 0.3 | 5.8 | 8159.1 |
| 2019 | 2685.3 | 712.4 | 584.7 | 619.6 | 1763.6 | 6308.4 | 2520.5 | 597.1 | 431.0 | 254.9 | 133.9 | 60.8 | 24.1 | 6.9 | 2.7 | 16705.9 |
| 2020 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 2021 | 1027.6 | 827.5 | 1953.4 | 765.3 | 703.6 | 508.1 | 1291.4 | 1479.8 | 391.6 | 152.5 | 127.5 | 82.6 | 18.2 | 12.9 | 4.4 | 9346.2 |
| 2022 | 821.4 | 348.4 | 595.5 | 4025.4 | 1888.9 | 733.5 | 815.5 | 1143.0 | 855.9 | 271.4 | 116.2 | 96.7 | 37.6 | 21.3 | 2.2 | 11773.0 |
| 2023 | 1152.8 | 388.7 | 341.0 | 630.2 | 2710.5 | 804.7 | 436.7 | 424.2 | 702.6 | 503.9 | 126.4 | 68.4 | 38.1 | 11.7 | 10.6 | 8350.8 |
| 2024 | 1683.9 | 957.6 | 893.4 | 610.1 | 1944.1 | 6404.9 | 1426.1 | 447.7 | 424.6 | 389.1 | 275.1 | 107.7 | 33.5 | 14.6 | 5.4 | 15617.7 |
| 2025 | 1534.2 | 344.2 | 318.5 | 504.0 | 491.9 | 1366.7 | 3303.4 | 827.7 | 327.6 | 208.0 | 299.8 | 162.0 | 81.4 | 50.4 | 26.5 | 9846.2 |
| 2026 | 3466.8 | 608.7 | 1637.4 | 948.4 | 1034.1 | 1390.7 | 2587.6 | 2390.8 | 738.5 | 253.8 | 442.7 | 369.9 | 114.1 | 11.4 | 69.8 | 16064.7 |
| Mean | 1615.3 | 549.5 | 921.3 | 1692.5 | 2315.4 | 2225.9 | 1444.8 | 863.9 | 495.2 | 321.1 | 205.4 | 142.5 | 80.1 | 41.2 | 69.3 | 12983.4 |

The full current revised series is shown; historical values differ from the earlier published VAST series. Northern survey coverage varies among years. These survey-expansion estimates are distinct from model population numbers at age.

The 2025 and 2026 FT-NIRS products use their year-specific traditional-age keys; the 2026 key retains 307 pairs after excluding 22 flagged pairs. Exact traditional ages 1–18 are converted before 15+ pooling, conserving total abundance. Empirical-key uncertainty is excluded. NA in 2020 denotes no survey observation; the mean uses the 44 observed-year estimates. Values are rounded to 0.1 million fish.

**Table 16:** Annual BTS-derived population weight-at-age inputs used in Model 26.0, 1982–2026 (kg per fish). Ages 15 and older are pooled as 15+.

| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1982 | 0.027 | 0.076 | 0.180 | 0.369 | 0.550 | 0.862 | 1.151 | 1.326 | 1.463 | 1.648 | 1.790 | 1.778 | 2.142 | 1.770 | 2.633 |
| 1983 | 0.017 | 0.091 | 0.179 | 0.341 | 0.491 | 0.691 | 0.988 | 1.304 | 1.494 | 1.539 | 1.634 | 1.600 | 1.807 | 1.555 | 1.961 |
| 1984 | 0.017 | 0.062 | 0.187 | 0.293 | 0.465 | 0.605 | 0.955 | 1.375 | 1.520 | 1.710 | 1.957 | 1.829 | 1.880 | 1.675 | 2.091 |
| 1985 | 0.017 | 0.070 | 0.183 | 0.333 | 0.485 | 0.753 | 0.941 | 1.238 | 1.667 | 1.648 | 1.779 | 2.062 | 2.135 | 1.817 | 2.505 |
| 1986 | 0.014 | 0.066 | 0.137 | 0.356 | 0.433 | 0.657 | 0.829 | 1.025 | 1.270 | 1.635 | 1.714 | 1.952 | 2.051 | 2.212 | [*] 2.191 |
| 1987 | 0.028 | 0.105 | 0.239 | 0.291 | 0.424 | 0.555 | 0.705 | 0.861 | 0.993 | 1.224 | 1.506 | 1.539 | 1.931 | 2.215 | 2.339 |
| 1988 | 0.019 | 0.079 | 0.246 | 0.331 | 0.444 | 0.527 | 0.661 | 0.883 | 0.910 | 1.101 | 1.366 | 1.390 | 1.872 | 1.169 | 1.715 |
| 1989 | 0.022 | 0.085 | 0.166 | 0.293 | 0.384 | 0.509 | 0.710 | 0.802 | 1.088 | 1.072 | 1.201 | 1.268 | 1.303 | 1.290 | 1.400 |
| 1990 | 0.019 | 0.101 | 0.145 | 0.337 | 0.441 | 0.536 | 0.633 | 0.741 | 0.916 | 1.134 | 1.256 | 1.313 | 1.393 | 1.498 | 1.692 |
| 1991 | 0.025 | 0.109 | 0.202 | 0.339 | 0.549 | 0.629 | 0.714 | 0.809 | 0.986 | 1.108 | 1.364 | 1.400 | 1.552 | 1.824 | 1.930 |
| 1992 | 0.018 | 0.096 | 0.270 | 0.381 | 0.510 | 0.575 | 0.774 | 0.916 | 0.969 | 1.083 | 1.262 | 1.455 | 1.601 | 1.662 | 1.630 |
| 1993 | 0.015 | 0.087 | 0.271 | 0.470 | 0.524 | 0.646 | 0.757 | 0.895 | 1.101 | 1.112 | 1.318 | 1.406 | 1.561 | 1.673 | 1.786 |
| 1994 | 0.019 | 0.083 | 0.209 | 0.388 | 0.559 | 0.675 | 0.719 | 1.053 | 1.239 | 1.199 | 1.322 | 1.413 | 1.525 | 1.680 | 1.758 |
| 1995 | 0.016 | 0.057 | 0.139 | 0.339 | 0.485 | 0.657 | 0.693 | 0.912 | 1.079 | 1.328 | 1.314 | 1.373 | 1.501 | 1.481 | 1.747 |
| 1996 | 0.021 | 0.077 | 0.137 | 0.297 | 0.500 | 0.632 | 0.797 | 0.903 | 1.043 | 1.176 | 1.386 | 1.511 | 1.680 | 1.682 | 1.789 |
| 1997 | 0.018 | 0.078 | 0.193 | 0.299 | 0.380 | 0.623 | 0.754 | 0.899 | 1.122 | 1.185 | 1.319 | 1.529 | 1.583 | 1.645 | 1.764 |
| 1998 | 0.018 | 0.070 | 0.185 | 0.323 | 0.460 | 0.524 | 0.801 | 0.895 | 1.012 | 1.082 | 1.286 | 1.349 | 1.485 | 1.886 | 1.886 |
| 1999 | 0.016 | 0.078 | 0.219 | 0.347 | 0.407 | 0.561 | 0.700 | 0.897 | 1.087 | 1.152 | 1.292 | 1.323 | 1.754 | 1.756 | 1.982 |
| 2000 | 0.014 | 0.079 | 0.236 | 0.400 | 0.477 | 0.562 | 0.704 | 0.786 | 0.946 | 1.164 | 1.364 | 1.476 | 1.720 | 1.752 | 2.001 |
| 2001 | 0.018 | 0.078 | 0.172 | 0.359 | 0.530 | 0.669 | 0.797 | 0.881 | 0.996 | 1.118 | 1.337 | 1.499 | 1.589 | 1.794 | 1.880 |
| 2002 | 0.017 | 0.094 | 0.244 | 0.389 | 0.521 | 0.674 | 0.760 | 0.935 | 1.033 | 1.121 | 1.127 | 1.401 | 1.563 | 1.741 | 1.934 |
| 2003 | 0.027 | 0.113 | 0.279 | 0.461 | 0.621 | 0.735 | 0.815 | 0.968 | 1.045 | 1.201 | 1.250 | 1.371 | 1.548 | 1.550 | 1.744 |
| 2004 | 0.023 | 0.107 | 0.275 | 0.421 | 0.567 | 0.689 | 0.819 | 0.928 | 1.041 | 1.152 | 1.210 | 1.304 | 1.456 | 1.748 | 1.746 |
| 2005 | 0.022 | 0.088 | 0.217 | 0.395 | 0.510 | 0.630 | 0.809 | 0.942 | 1.020 | 1.108 | 1.361 | 1.314 | 1.299 | 1.480 | 1.599 |
| 2006 | 0.011 | 0.087 | 0.170 | 0.421 | 0.598 | 0.696 | 0.815 | 0.894 | 1.042 | 1.164 | 1.275 | 1.390 | 1.467 | 1.434 | 1.489 |
| 2007 | 0.014 | 0.102 | 0.263 | 0.497 | 0.637 | 0.802 | 0.907 | 1.039 | 1.146 | 1.406 | 1.372 | 1.411 | 1.723 | 1.636 | 1.689 |
| 2008 | 0.014 | 0.056 | 0.221 | 0.455 | 0.585 | 0.717 | 0.846 | 0.997 | 1.058 | 1.237 | 1.537 | 1.509 | 1.726 | 1.850 | 1.894 |
| 2009 | 0.012 | 0.074 | 0.224 | 0.499 | 0.675 | 0.791 | 0.999 | 1.135 | 1.174 | 1.481 | 1.568 | 1.759 | 2.049 | 2.277 | 2.360 |
| 2010 | 0.020 | 0.077 | 0.249 | 0.510 | 0.675 | 0.875 | 1.094 | 1.188 | 1.245 | 1.352 | 1.453 | 1.651 | 1.876 | 2.213 | 2.443 |
| 2011 | 0.017 | 0.085 | 0.209 | 0.499 | 0.663 | 0.773 | 0.938 | 1.118 | 1.247 | 1.296 | 1.455 | 1.530 | 1.638 | 1.727 | 2.015 |
| 2012 | 0.015 | 0.075 | 0.195 | 0.420 | 0.606 | 0.784 | 0.933 | 1.137 | 1.333 | 1.401 | 1.432 | 1.529 | 1.558 | 1.845 | 2.015 |
| 2013 | 0.018 | 0.079 | 0.213 | 0.501 | 0.582 | 0.744 | 0.993 | 1.166 | 1.354 | 1.409 | 1.542 | 1.582 | 1.710 | 1.860 | 1.923 |
| 2014 | 0.019 | 0.109 | 0.257 | 0.438 | 0.571 | 0.716 | 0.791 | 1.106 | 1.231 | 1.423 | 1.676 | 1.606 | 1.672 | 1.847 | 1.903 |
| 2015 | 0.020 | 0.090 | 0.327 | 0.445 | 0.588 | 0.686 | 0.784 | 0.905 | 1.166 | 1.229 | 1.496 | 1.550 | 1.432 | 1.492 | 1.750 |
| 2016 | 0.023 | 0.080 | 0.279 | 0.531 | 0.595 | 0.697 | 0.779 | 0.828 | 0.904 | 1.004 | 1.137 | 1.554 | 1.351 | 1.536 | 1.421 |
| 2017 | 0.021 | 0.092 | 0.231 | 0.472 | 0.625 | 0.673 | 0.755 | 0.814 | 0.892 | 0.919 | 1.057 | 0.993 | 1.289 | 1.123 | 1.485 |
| 2018 | 0.023 | 0.092 | 0.215 | 0.421 | 0.591 | 0.684 | 0.777 | 0.816 | 0.896 | 0.933 | 1.036 | 0.944 | 1.410 | [*] 1.720 | 1.540 |
| 2019 | 0.022 | 0.105 | 0.286 | 0.487 | 0.634 | 0.731 | 0.822 | 0.925 | 0.937 | 1.035 | 0.965 | 1.066 | 1.038 | 1.063 | [*] 1.910 |
| 2020 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 2021 | 0.022 | 0.101 | 0.245 | 0.442 | 0.585 | 0.713 | 0.803 | 0.857 | 0.977 | 1.118 | 1.040 | 1.117 | 1.298 | 0.999 | 1.361 |
| 2022 | 0.018 | 0.112 | 0.270 | 0.447 | 0.590 | 0.697 | 0.771 | 0.873 | 0.969 | 1.034 | 1.254 | 1.251 | 1.324 | 1.205 | 1.812 |
| 2023 | 0.015 | 0.106 | 0.283 | 0.416 | 0.536 | 0.657 | 0.812 | 0.955 | 0.968 | 1.013 | 1.179 | 1.200 | 1.414 | 1.759 | 1.487 |
| 2024 | 0.017 | 0.090 | 0.309 | 0.454 | 0.560 | 0.607 | 0.693 | 0.923 | 0.997 | 1.020 | 1.126 | 1.137 | 1.259 | 1.406 | 1.754 |
| 2025 | 0.020 | 0.079 | 0.226 | 0.526 | 0.587 | 0.676 | 0.734 | 0.812 | 0.903 | 1.087 | 1.066 | 1.150 | 1.258 | 1.399 | 1.547 |
| 2026 | 0.020 | 0.066 | 0.397 | 0.544 | 0.640 | 0.725 | 0.809 | 0.874 | 0.949 | 1.027 | 1.048 | 1.192 | 1.416 | 1.449 | 1.360 |
| Mean | 0.019 | 0.086 | 0.227 | 0.409 | 0.542 | 0.673 | 0.815 | 0.967 | 1.101 | 1.218 | 1.351 | 1.431 | 1.587 | 1.645 | 1.838 |

[*] The selected value is retained. Its current treatment has not been reconciled with an older preparation record identifying these historical cells for filling; the technical supplement preserves that provenance qualification.

The table reproduces the selected schedules, including historical revisions, rather than labeling every cell as an unmodified survey measurement. NA in 2020 marks absence of a survey observation; the model retains its separate annual schedule. The mean uses the 44 displayed annual schedules. Values are rounded to 0.001 kg.

**Table 17:** Sampling effort and biological collections from eastern Bering Sea acoustic-trawl surveys, 1979–2026 (counts). E and W denote collections east and west of 170°W within the US EEZ; US is the reported US subtotal and RU is the Russian-region count. The 2024 US aged-fish count is 2,260 (McCarthy et al. 2026, p. 15).

| Year | Hauls | | | | Length records | | | | Otoliths | | | | Number aged | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | E | W | US | RU | E | W | US | RU | E | W | US | RU | E | W | US | RU |
| 1979 | NA | NA | 25 | NA | NA | NA | 7722 | NA | NA | NA | 0 | NA | NA | NA | 2610 | NA |
| 1982 | 13 | 31 | 48 | NA | 1725 | 6689 | 8687 | NA | 840 | 2324 | 3164 | NA | 783 | 1958 | 2741 | NA |
| 1985 | NA | NA | 73 | NA | NA | NA | 19872 | NA | NA | NA | 2739 | NA | NA | NA | 2739 | NA |
| 1988 | NA | NA | 25 | NA | NA | NA | 6619 | NA | NA | NA | 1471 | NA | NA | NA | 1471 | NA |
| 1991 | NA | NA | 62 | NA | NA | NA | 16343 | NA | NA | NA | 2062 | NA | NA | NA | 1663 | NA |
| 1994 | 25 | 51 | 76 | 19 | 4553 | 21011 | 25564 | 8930 | 1560 | 3694 | 4966 | 1270 | 612 | 932 | 1770 | 455 |
| 1996 | 15 | 42 | 57 | NA | 3551 | 13273 | 16824 | NA | 669 | 1280 | 1949 | NA | 815 | 1111 | 1926 | NA |
| 1997 | 25 | 61 | 86 | NA | 6493 | 23043 | 29536 | NA | 966 | 2669 | 3635 | NA | 936 | 1349 | 2285 | NA |
| 1999 | 41 | 77 | 118 | NA | 13841 | 28521 | 42362 | NA | 1945 | 3001 | 4946 | NA | 946 | 1500 | 2446 | NA |
| 2000 | 29 | 95 | 124 | NA | 7721 | 36008 | 43729 | NA | 850 | 2609 | 3459 | NA | 850 | 1403 | 2253 | NA |
| 2002 | 47 | 79 | 126 | NA | 14601 | 25633 | 40234 | NA | 1424 | 1883 | 3307 | NA | 1000 | 1200 | 2200 | NA |
| 2004 | 33 | 57 | 90 | 15 | 8896 | 18262 | 27158 | 5893 | 1167 | 2002 | 3169 | 461 | 798 | 1192 | 2351 | 461 |
| 2006 | 27 | 56 | 83 | NA | 4939 | 19326 | 24265 | NA | 822 | 1871 | 2693 | NA | 822 | 1870 | 2692 | NA |
| 2007 | 23 | 46 | 69 | 4 | 5492 | 14863 | 20355 | 1407 | 871 | 1961 | 2832 | 319 | 823 | 1737 | 2560 | 315 |
| 2008 | 9 | 53 | 62 | 6 | 2394 | 15354 | 17748 | 1754 | 341 | 1698 | 2039 | 177 | 338 | 1381 | 1719 | 176 |
| 2009 | 13 | 33 | 46 | 3 | 1576 | 9257 | 10833 | 282 | 308 | 1210 | 1518 | 54 | 306 | 1205 | 1511 | 54 |
| 2010 | 11 | 48 | 59 | 9 | 2432 | 20263 | 22695 | 3502 | 653 | 1868 | 2521 | 381 | 652 | 1598 | 2250 | 379 |
| 2012 | 17 | 60 | 77 | 14 | 4422 | 23929 | 28351 | 5620 | 650 | 2045 | 2695 | 418 | 646 | 1483 | 2129 | 416 |
| 2014 | 52 | 87 | 139 | 3 | 28857 | 8645 | 37502 | 747 | 1739 | 849 | 2588 | 72 | 845 | 1735 | 2580 | 72 |
| 2016 | 37 | 71 | 108 | NA | 10912 | 24134 | 35046 | NA | 880 | 1514 | 2394 | NA | 876 | 1513 | 2388 | NA |
| 2018 | 36 | 64 | 100 | NA | 11031 | 21173 | 32204 | NA | 1105 | 1750 | 2855 | NA | 1100 | 1743 | 2843 | NA |
| 2020 | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA | NA |
| 2022 | 26 | 36 | 62 | NA | 4954 | 11994 | 16948 | NA | 671 | 1691 | 2362 | NA | 668 | 1688 | 2356 | NA |
| 2024 | 14 | 41 | 55 | NA | 4535 | 14472 | 19007 | NA | 725 | 1561 | 2286 | NA | NA | NA | 2260 | NA |
| 2026 | 36 | 62 | 99 | NA | 11916 | 23031 | 35128 | NA | 1051 | 1571 | 2642 | NA | NA | NA | NA | NA |

**Table 18:** Midwater pollock biomass by area from summer ATS surveys of the U.S. EEZ Bering Sea shelf, 1994–2026 (million tons; near surface to 0.5 m above the seafloor). The core shelf excludes the Northern Extension. Assessment CVs average 20%, with 45% assigned to 2000; SE equals biomass × CV. The 2020 survey used uncrewed sailing vessels; 2026 estimates are preliminary.

| Year | Date | Area ($nmi^2$) | Biomass (million t) | | | | Rel. error | Assessment CV | Assessment SE (kt) |
|---|---|---|---|---|---|---|---|---|---|
| | | | SCA | E170–SCA | W170 | 0.5 m total | | | |
| 1994 | 9 Jul - 7 Sept | 78,251 | 0.382 | 0.652 | 2.594 | 3.628 | 0.040 | 17.0% | 618.4 |
| 1996 | 20 Jul - 30 Aug | 93,810 | 0.275 | 0.492 | 2.185 | 2.952 | 0.032 | 13.6% | 402.5 |
| 1997 | 17 Jul - 5 Sept | 102,770 | 0.272 | 0.703 | 2.470 | 3.446 | 0.029 | 12.4% | 425.8 |
| 1999 | 12 Jun - 29 Jul | 103,670 | 0.383 | 0.757 | 3.060 | 4.200 | 0.044 | 18.8% | 787.5 |
| 2000 | 12 Jun - 1 Aug | 106,140 | 0.458 | 0.715 | 2.441 | 3.615 | 0.028 | 45.0% | 1,626.6 |
| 2002 | 9 Jun - 21 Jul | 99,526 | 0.762 | 0.938 | 2.601 | 4.301 | 0.027 | 11.5% | 494.9 |
| 2004 | 4 Jun -29 Jul | 99,659 | 0.550 | 0.918 | 2.528 | 3.996 | 0.031 | 13.2% | 527.9 |
| 2006 | 3 Jun -25 Jul | 89,550 | 0.147 | 0.340 | 1.387 | 1.874 | 0.033 | 14.1% | 263.6 |
| 2007 | 2 Jun -30 Jul | 92,944 | 0.136 | 0.245 | 1.902 | 2.283 | 0.038 | 16.2% | 369.7 |
| 2008 | 2 Jun -31 Jul | 95,374 | 0.122 | 0.087 | 1.196 | 1.404 | 0.056 | 23.9% | 335.2 |
| 2009 | 9 Jun -7 Aug | 91,414 | 0.156 | 0.058 | 1.117 | 1.331 | 0.069 | 29.4% | 391.5 |
| 2010 | 5 Jun -7 Aug | 92,849 | 0.098 | 0.193 | 2.345 | 2.636 | 0.054 | 23.0% | 606.6 |
| 2012 | 7 Jun -10 Aug | 96,852 | 0.196 | 0.319 | 1.764 | 2.279 | 0.034 | 14.5% | 330.3 |
| 2014 | 12 Jun -13 Aug | 94,361 | 0.571 | 1.458 | 2.714 | 4.743 | 0.034 | 14.5% | 687.2 |
| 2016 | 12 Jun -17 Aug | 100,674 | 0.542 | 1.268 | 3.028 | 4.838 | 0.019 | 8.1% | 391.7 |
| 2018 | 12 Jun -22 Aug | 92,283 | 0.234 | 0.510 | 1.753 | 2.497 | 0.039 | 16.6% | 415.1 |
| 2020 | 4 Jul - 20 Aug | 102,320 | 0.398 | 0.531 | 2.688 | 3.617 | 0.096 | 40.9% | 1,479.8 |
| 2022 | 1 Jun -5 Aug | 103,942 | 0.538 | 0.823 | 2.473 | 3.834 | 0.068 | 29.0% | 1,111.1 |
| 2024 | 5 Jun - 22 Jul | 100,365 | 0.497 | 0.593 | 1.785 | 2.875 | 0.056 | 23.9% | 686.1 |
| 2026 | 28 May - 12 Aug | 95,091 | 0.239 | 0.706 | 2.738 | 3.682 | 0.034 | 14.5% | 533.6 |

**Table 19:** Biomass and abundance summed across the 2026 survey intervals, including the below-3 m analysis and northern extension. The three core-shelf regions sum to the historical EBS estimate; the Northern Extension is additional.

| Region | EDSU records | Biomass (t) | Abundance (million fish) |
|---|---|---|---|
| SCA | 1,079 | 238,999.6 | 720.50 |
| East of 170 | 3,232 | 705,748.5 | 1,396.55 |
| West of 170 | 5,137 | 2,737,718.7 | 10,545.58 |
| EBS Northern Extension | 1,476 | 78,311.1 | 104.29 |

**Table 20:** ATS estimates of EBS pollock abundance at age, 1979–2026 (million fish). Ages 10 and older are pooled as 10+. The 2026 estimates use the BTS age–length key applied to ATS length distributions and remain provisional; ages from the 2026 ATS samples are expected in 2027. Mean and median summarize surveys with age compositions during 1994–2026; 1979–1991 provides historical context.

| Year | Age | | | | | | | | | | Age 2+ | Total at age |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10+ | | |
| 1979 | 69,110 | 41,132 | 3,884 | 413 | 534 | 128 | 30 | 4 | 28 | 161 | 46,314 | 115,424 |
| 1982 | 108 | 3,401 | 4,108 | 7,637 | 1,790 | 283 | 141 | 178 | 90 | 177 | 17,805 | 17,913 |
| 1985 | 2,076 | 929 | 8,149 | 898 | 2,186 | 1,510 | 1,127 | 130 | 21 | 15 | 14,965 | 17,041 |
| 1988 | 11 | 1,112 | 3,586 | 3,864 | 739 | 1,882 | 403 | 151 | 130 | 414 | 12,280 | 12,292 |
| 1991 | 639 | 5,942 | 967 | 215 | 224 | 133 | 120 | 39 | 37 | 53 | 7,730 | 8,369 |
| 1994 | 1,140 | 4,969 | 1,424 | 1,819 | 2,252 | 389 | 109 | 96 | 56 | 221 | 11,335 | 12,475 |
| 1996 | 1,800 | 567 | 552 | 2,741 | 915 | 634 | 585 | 142 | 39 | 165 | 6,338 | 8,139 |
| 1997 | 13,227 | 2,881 | 440 | 536 | 2,330 | 546 | 313 | 290 | 75 | 220 | 7,633 | 20,860 |
| 1999 | 607 | 1,780 | 3,717 | 1,810 | 652 | 398 | 1,548 | 526 | 180 | 249 | 10,859 | 11,466 |
| 2000 | 460 | 1,322 | 1,230 | 2,588 | 1,012 | 327 | 308 | 950 | 278 | 252 | 8,266 | 8,726 |
| 2002 | 796 | 4,944 | 3,385 | 1,295 | 661 | 935 | 538 | 140 | 162 | 493 | 12,554 | 13,351 |
| 2004 | 83 | 313 | 1,217 | 3,123 | 1,634 | 567 | 288 | 283 | 121 | 265 | 7,811 | 7,894 |
| 2006 | 525 | 217 | 291 | 654 | 783 | 659 | 390 | 145 | 75 | 171 | 3,386 | 3,910 |
| 2007 | 5,775 | 1,041 | 345 | 478 | 794 | 729 | 407 | 241 | 98 | 135 | 4,267 | 10,042 |
| 2008 | 71 | 2,915 | 1,047 | 166 | 161 | 288 | 235 | 136 | 102 | 120 | 5,169 | 5,240 |
| 2009 | 5,197 | 816 | 1,734 | 281 | 77 | 94 | 129 | 111 | 77 | 114 | 3,433 | 8,630 |
| 2010 | 2,568 | 6,404 | 984 | 2,295 | 446 | 73 | 33 | 37 | 38 | 91 | 10,400 | 12,968 |
| 2012 | 177 | 1,989 | 1,693 | 2,710 | 280 | 367 | 113 | 36 | 25 | 103 | 7,315 | 7,492 |
| 2014 | 4,751 | 8,655 | 969 | 1,161 | 1,119 | 1,770 | 740 | 170 | 79 | 99 | 14,762 | 19,513 |
| 2016 | 174 | 1,038 | 4,496 | 4,476 | 715 | 348 | 392 | 420 | 96 | 64 | 12,046 | 12,220 |
| 2018 | 450 | 517 | 249 | 621 | 2,268 | 944 | 198 | 112 | 107 | 104 | 5,120 | 5,570 |
| 2022 | 142 | 332 | 975 | 6,578 | 819 | 211 | 133 | 239 | 166 | 79 | 9,533 | 9,674 |
| 2024 | 5,075 | 1,146 | 681 | 379 | 820 | 2,485 | 450 | 110 | 86 | 127 | 6,284 | 11,359 |
| 2026 | 2,026 | 3,729 | 3,958 | 375 | 319 | 403 | 714 | 656 | 198 | 285 | 10,637 | 12,663 |
| Mean 1994–2026 | 2,371 | 2,399 | 1,547 | 1,794 | 950 | 640 | 401 | 255 | 108 | 177 | 8,271 | 10,642 |
| Median 1994–2026 | 796 | 1,322 | 1,047 | 1,295 | 794 | 403 | 313 | 145 | 96 | 135 | 7,811 | 10,042 |

**Table 21:** Recent core-area ATS pollock biomass and abundance estimates. The 2026 estimates are preliminary.

| Recent core-area ATS estimates | | |
|---|---|---|
| Survey year | Biomass (million t) | Abundance (billion fish) |
| 2022 | 3.8 | 9.67 |
| 2024 | 2.87 | 11.4 |
| 2026 (preliminary) | 3.68 | 12.7 |

**Table 22:** Provisional core-area 2026 ATS abundance by age, in million fish, restricted to supported lengths. FT-NIRS source age 15 retains its released label; the corresponding TMA row pools target ages 15–18. The upper-length tail remains unallocated.

| Age (FT-NIRS / TMA) | FT-NIRS abundance | TMA abundance |
|---:|---:|---:|
| 1 | 2,194.690 | 2,025.867 |
| 2 | 5,340.635 | 3,729.246 |
| 3 | 1,978.834 | 3,958.120 |
| 4 | 597.222 | 374.975 |
| 5 | 338.742 | 318.742 |
| 6 | 299.224 | 403.011 |
| 7 | 619.494 | 713.961 |
| 8 | 786.844 | 656.327 |
| 9 | 343.438 | 197.732 |
| 10 | 99.819 | 63.788 |
| 11 | 45.953 | 106.936 |
| 12 | 12.907 | 80.216 |
| 13 | 2.903 | 19.546 |
| 14 | 0.836 | 1.844 |
| 15 / 15+ | 0.980 | 12.210 |

**Table 23:** Provisional core-area 2026 ATS TMA biomass, abundance and mean weight-at-age over supported lengths. The 15+ row sums exact ages 15–18 before dividing biomass by abundance. Unsupported upper-length biomass and fish are excluded from these age-specific estimates and retained separately.

| TMA age | Biomass (t) | Abundance (million fish) | Mean weight (kg/fish) |
|---|---|---|---|
| 1 | 89,550.280 | 2,025.867 | 0.0442 |
| 2 | 525,587.105 | 3,729.246 | 0.1409 |
| 3 | 949,648.466 | 3,958.120 | 0.2399 |
| 4 | 182,528.304 | 374.975 | 0.4868 |
| 5 | 185,878.529 | 318.742 | 0.5832 |
| 6 | 263,257.538 | 403.011 | 0.6532 |
| 7 | 525,152.651 | 713.961 | 0.7355 |
| 8 | 518,643.709 | 656.327 | 0.7902 |
| 9 | 168,006.366 | 197.732 | 0.8497 |
| 10 | 57,043.013 | 63.788 | 0.8943 |
| 11 | 96,454.798 | 106.936 | 0.9020 |
| 12 | 81,491.146 | 80.216 | 1.0159 |
| 13 | 22,896.782 | 19.546 | 1.1714 |
| 14 | 2,181.579 | 1.844 | 1.1832 |
| 15+ | 13,821.450 | 12.210 | 1.1320 |

**Table 24:** Assumed age-composition likelihood sample sizes used in the completed full-data Model 23.0 fit, 1964–2026. Fishery inputs for 2023–2025 are the applied rounded values 443, 578 and 530; other values are rounded for display.

| Year | Fishery | BTS | ATS |
|---|---|---|---|
| 1964–1977 | 10 | | |
| 1978 | 39 | | |
| 1979 | 39 | | |
| 1980 | 39 | | |
| 1981 | 39 | | |
| 1982 | 39 | 153 | |
| 1983 | 39 | 134 | |
| 1984 | 39 | 61 | |
| 1985 | 39 | 76 | |
| 1986 | 39 | 153 | |
| 1987 | 39 | 60 | |
| 1988 | 39 | 81 | |
| 1989 | 39 | 55 | |
| 1990 | 39 | 48 | |
| 1991 | 260 | 79 | |
| 1992 | 228 | 86 | |
| 1993 | 344 | 92 | |
| 1994 | 286 | 91 | 43 |
| 1995 | 257 | 103 | |
| 1996 | 189 | 227 | 32 |
| 1997 | 319 | 77 | 49 |
| 1998 | 355 | 64 | |
| 1999 | 476 | 97 | 67 |
| 2000 | 482 | 88 | 70 |
| 2001 | 328 | 178 | |
| 2002 | 487 | 187 | 72 |
| 2003 | 441 | 130 | |
| 2004 | 391 | 195 | 51 |
| 2005 | 494 | 188 | |
| 2006 | 506 | 321 | 47 |
| 2007 | 500 | 184 | 39 |
| 2008 | 524 | 238 | 35 |
| 2009 | 420 | 148 | 26 |
| 2010 | 549 | 107 | 34 |
| 2011 | 728 | 172 | |
| 2012 | 607 | 108 | 44 |
| 2013 | 753 | 64 | |
| 2014 | 606 | 117 | 79 |
| 2015 | 821 | 104 | |
| 2016 | 704 | 124 | 61 |
| 2017 | 607 | 241 | |
| 2018 | 668 | 156 | 50 |
| 2019 | 701 | 120 | |
| 2020 | 565 | | 1 |
| 2021 | 818 | 150 | |
| 2022 | 641 | 181 | 52 |
| 2023 | 443 | 135 | |
| 2024 | 578 | 63 | 79 |
| 2025 | 530 | 111 | |
| 2026 | | 131 | 94 |
| Mean | 503 | 129 | 51 |

**Table 25:** Revised mean-one AVO display index for 17 observations during 2009–2026, excluding 2020, including newly processed 2011 and 2013 values. Sampling relative errors, standard errors and scaled source 95% confidence intervals are retained. The blank 2020 row preserves calendar-year coverage.

| Year | Index (million $m^2$) | Relative error (%) | SE (million $m^2$) | Relative index | SE (relative) | 95% CI (relative) |
|---|---|---|---|---|---|---|
| 2009 | 0.6951 | 12.22% | 0.0849 | 0.2990 | 0.0365 | 0.2274–0.3707 |
| 2010 | 1.9218 | 6.56% | 0.1261 | 0.8268 | 0.0542 | 0.7205–0.9331 |
| 2011 | 1.5679 | 4.51% | 0.0707 | 0.6745 | 0.0304 | 0.6149–0.7342 |
| 2012 | 1.5206 | 5.32% | 0.0810 | 0.6542 | 0.0348 | 0.5859–0.7225 |
| 2013 | 2.2347 | 4.64% | 0.1037 | 0.9614 | 0.0446 | 0.8740–1.0488 |
| 2014 | 3.0769 | 4.11% | 0.1263 | 1.3238 | 0.0543 | 1.2172–1.4303 |
| 2015 | 3.5927 | 2.46% | 0.0884 | 1.5457 | 0.0380 | 1.4711–1.6202 |
| 2016 | 2.8319 | 2.91% | 0.0825 | 1.2184 | 0.0355 | 1.1488–1.2879 |
| 2017 | 2.2630 | 3.05% | 0.0691 | 0.9736 | 0.0297 | 0.9153–1.0319 |
| 2018 | 2.0841 | 2.66% | 0.0555 | 0.8966 | 0.0239 | 0.8498–0.9434 |
| 2019 | 2.8292 | 4.89% | 0.1383 | 1.2172 | 0.0595 | 1.1006–1.3338 |
| 2020 | | | | | | |
| 2021 | 2.4101 | 4.79% | 0.1154 | 1.0369 | 0.0496 | 0.9396–1.1342 |
| 2022 | 2.9031 | 3.41% | 0.0990 | 1.2490 | 0.0426 | 1.1655–1.3325 |
| 2023 | 2.4810 | 2.94% | 0.0730 | 1.0674 | 0.0314 | 1.0058–1.1289 |
| 2024 | 2.0070 | 2.72% | 0.0546 | 0.8635 | 0.0235 | 0.8174–0.9095 |
| 2025 | 2.4954 | 5.26% | 0.1313 | 1.0736 | 0.0565 | 0.9628–1.1843 |
| 2026 | 2.5997 | 4.55% | 0.1182 | 1.1185 | 0.0508 | 1.0188–1.2181 |

Source: NOAA AFSC MACE AVO workbook, 2009–2026. The 2020 observation is unavailable, so its data cells are blank.

SE = index × CV. Relative index and SE share the mean-1 scaling; relative error is unchanged. Source 95% intervals are scaled by the same constant.

**Table 26:** Counts of AVO sampling intervals during 2009–2026, including positive intervals and recorded zeros; these counts describe survey coverage. There are no AVO observations in 2020.

| Year | Survey intervals | Positive intervals | Recorded zeros |
|---|---|---|---|
| 2009 | 5,557 | 2,430 | 56.3% |
| 2010 | 5,699 | 2,584 | 54.7% |
| 2011 | 7,829 | 5,171 | 34.0% |
| 2012 | 6,740 | 4,750 | 29.5% |
| 2013 | 8,021 | 4,725 | 41.1% |
| 2014 | 10,032 | 8,315 | 17.1% |
| 2015 | 10,524 | 9,700 | 7.8% |
| 2016 | 8,603 | 7,912 | 8.0% |
| 2017 | 8,798 | 6,833 | 22.3% |
| 2018 | 8,880 | 8,044 | 9.4% |
| 2019 | 8,836 | 7,301 | 17.4% |
| 2020 | — | — | — |
| 2021 | 6,939 | 6,038 | 13.0% |
| 2022 | 9,350 | 8,502 | 9.1% |
| 2023 | 8,740 | 7,435 | 14.9% |
| 2024 | 12,229 | 10,406 | 14.9% |
| 2025 | 7,320 | 6,952 | 5.0% |
| 2026 | 8,699 | 7,462 | 14.2% |

**Table 27:** ATS biomass (million tons) and AVO backscatter (million m²), 2006–2026. AVO indices and sampling relative errors for 2009–2026 match Table 25. Assessment CVs from the saved Model 26.0 inputs average 23% across 20 observations; SE equals index × CV. The 2020 USV survey provides ATS biomass; AVO has no observation that year.

| Year | ATS biomass (million t) | AVO index (million $m^2$) | Sampling RE | Assessment CV | Assessment SE (million $m^2$) |
|---|---|---|---|---|---|
| 2006 | 1.8741 | 1.7415 | 0.0510 | 27.64% | 0.4814 |
| 2007 | 2.2827 | 2.0019 | 0.0865 | 53.90% | 1.0790 |
| 2008 | 1.4045 | 0.9915 | 0.0643 | 19.84% | 0.1968 |
| 2009 | 1.3314 | 0.6951 | 0.1222 | 56.89% | 0.3955 |
| 2010 | 2.6357 | 1.9218 | 0.0656 | 30.55% | 0.5872 |
| 2011 | — | 1.5679 | 0.0451 | 21.01% | 0.3294 |
| 2012 | 2.2794 | 1.5206 | 0.0532 | 24.80% | 0.3771 |
| 2013 | — | 2.2347 | 0.0464 | 21.60% | 0.4827 |
| 2014 | 4.7428 | 3.0769 | 0.0411 | 19.12% | 0.5882 |
| 2015 | — | 3.5927 | 0.0246 | 11.46% | 0.4118 |
| 2016 | 4.8377 | 2.8319 | 0.0291 | 13.57% | 0.3843 |
| 2017 | — | 2.2630 | 0.0305 | 14.22% | 0.3218 |
| 2018 | 2.4973 | 2.0841 | 0.0266 | 12.40% | 0.2585 |
| 2019 | — | 2.8292 | 0.0489 | 22.77% | 0.6441 |
| 2020 | 3.6170 | — | — | — | — |
| 2021 | — | 2.4101 | 0.0479 | 22.29% | 0.5372 |
| 2022 | 3.8340 | 2.9031 | 0.0341 | 15.88% | 0.4610 |
| 2023 | — | 2.4810 | 0.0294 | 13.70% | 0.3400 |
| 2024 | 2.8750 | 2.0070 | 0.0272 | 12.67% | 0.2543 |
| 2025 | — | 2.4954 | 0.0526 | 24.51% | 0.6117 |
| 2026 | 3.6825 | 2.5997 | 0.0455 | 21.17% | 0.5504 |
| Mean | 2.9149 | 2.2125 | 0.0486 | 23.00% | 0.4646 |

**Table 28:** Annual fishery mean weight-at-age (kg), with observational inputs through 2025 and standalone WtAgeRE predictions and forecasts for 2026–2028. Observational inputs extend through 2025; the 2026 row is predicted. Age 15 is a 15+ group. The fixed early schedule and mean/CV definitions are described below.

| Year / basis | Age (years) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ |
| 1964–1990 | 0.007 | 0.170 | 0.303 | 0.447 | 0.589 | 0.722 | 0.840 | 0.942 | 1.029 | 1.102 | 1.163 | 1.212 | 1.253 | 1.286 | 1.312 |
| 1991 | 0.007 | 0.150 | 0.285 | 0.483 | 0.604 | 0.730 | 0.844 | 0.882 | 1.020 | 1.123 | 1.135 | 1.237 | 1.236 | 1.296 | 1.247 |
| 1992 | 0.007 | 0.179 | 0.397 | 0.465 | 0.651 | 0.714 | 0.819 | 0.986 | 1.030 | 1.200 | 1.237 | 1.269 | 1.193 | 1.357 | 1.431 |
| 1993 | 0.007 | 0.331 | 0.495 | 0.612 | 0.652 | 0.775 | 0.934 | 1.062 | 1.198 | 1.240 | 1.423 | 1.540 | 1.576 | 1.609 | 1.508 |
| 1994 | 0.007 | 0.233 | 0.400 | 0.652 | 0.732 | 0.746 | 0.727 | 1.070 | 1.380 | 1.325 | 1.335 | 1.409 | 1.397 | 1.278 | 1.370 |
| 1995 | 0.007 | 0.153 | 0.386 | 0.505 | 0.729 | 0.843 | 0.847 | 0.970 | 1.232 | 1.296 | 1.401 | 1.402 | 1.392 | 1.095 | 1.306 |
| 1996 | 0.007 | 0.293 | 0.336 | 0.445 | 0.684 | 0.797 | 0.948 | 0.956 | 1.025 | 1.100 | 1.418 | 1.489 | 1.521 | 1.702 | 1.602 |
| 1997 | 0.007 | 0.187 | 0.327 | 0.477 | 0.559 | 0.748 | 0.889 | 1.074 | 1.095 | 1.236 | 1.287 | 1.400 | 1.561 | 1.363 | 1.338 |
| 1998 | 0.007 | 0.191 | 0.369 | 0.589 | 0.618 | 0.622 | 0.780 | 1.040 | 1.169 | 1.276 | 1.316 | 1.428 | 1.448 | 1.437 | 1.528 |
| 1999 | 0.007 | 0.188 | 0.404 | 0.507 | 0.643 | 0.702 | 0.729 | 0.894 | 1.038 | 1.253 | 1.224 | 1.422 | 0.995 | 0.616 | 1.239 |
| 2000 | 0.007 | 0.218 | 0.353 | 0.526 | 0.630 | 0.732 | 0.780 | 0.807 | 0.968 | 1.015 | 1.253 | 1.286 | 1.108 | 1.084 | 1.359 |
| 2001 | 0.006 | 0.227 | 0.329 | 0.505 | 0.668 | 0.786 | 0.964 | 0.986 | 1.061 | 1.133 | 1.320 | 1.411 | 1.568 | 1.472 | 1.495 |
| 2002 | 0.007 | 0.231 | 0.385 | 0.510 | 0.667 | 0.799 | 0.911 | 1.026 | 1.113 | 1.102 | 1.284 | 1.442 | 1.579 | 1.290 | 1.568 |
| 2003 | 0.006 | 0.276 | 0.489 | 0.549 | 0.652 | 0.769 | 0.863 | 0.953 | 1.086 | 1.202 | 1.212 | 1.194 | 1.374 | 1.355 | 1.709 |
| 2004 | 0.007 | 0.135 | 0.408 | 0.584 | 0.641 | 0.760 | 0.888 | 0.924 | 1.036 | 1.176 | 1.127 | 1.167 | 1.309 | 1.254 | 1.185 |
| 2005 | 0.007 | 0.283 | 0.351 | 0.508 | 0.641 | 0.742 | 0.880 | 0.960 | 1.062 | 1.074 | 1.216 | 1.268 | 1.217 | 1.076 | 1.342 |
| 2006 | 0.007 | 0.174 | 0.306 | 0.448 | 0.606 | 0.755 | 0.858 | 0.959 | 1.060 | 1.117 | 1.190 | 1.218 | 1.280 | 1.384 | 1.417 |
| 2007 | 0.007 | 0.155 | 0.349 | 0.507 | 0.642 | 0.783 | 0.961 | 1.100 | 1.192 | 1.266 | 1.327 | 1.488 | 1.444 | 1.729 | 1.512 |
| 2008 | 0.007 | 0.208 | 0.328 | 0.519 | 0.653 | 0.774 | 0.900 | 1.054 | 1.117 | 1.289 | 1.452 | 1.528 | 1.560 | 1.874 | 1.645 |
| 2009 | 0.007 | 0.136 | 0.340 | 0.525 | 0.705 | 0.879 | 0.999 | 1.130 | 1.398 | 1.479 | 1.558 | 1.576 | 1.807 | 2.026 | 2.222 |
| 2010 | 0.050 | 0.175 | 0.381 | 0.490 | 0.668 | 0.909 | 1.114 | 1.277 | 1.374 | 1.586 | 1.679 | 1.923 | 1.948 | 2.077 | 2.271 |
| 2011 | 0.031 | 0.205 | 0.290 | 0.508 | 0.666 | 0.809 | 0.971 | 1.224 | 1.342 | 1.513 | 1.582 | 1.623 | 2.080 | 1.707 | 2.242 |
| 2012 | 0.029 | 0.142 | 0.271 | 0.409 | 0.643 | 0.824 | 0.974 | 1.170 | 1.303 | 1.509 | 1.599 | 1.637 | 1.680 | 2.031 | 2.062 |
| 2013 | 0.095 | 0.144 | 0.290 | 0.442 | 0.564 | 0.781 | 1.130 | 1.281 | 1.440 | 1.685 | 1.827 | 1.786 | 1.934 | 2.159 | 2.182 |
| 2014 | 0.014 | 0.193 | 0.319 | 0.454 | 0.617 | 0.751 | 0.894 | 1.156 | 1.307 | 1.386 | 1.669 | 1.773 | 1.704 | 1.623 | 2.215 |
| 2015 | 0.025 | 0.181 | 0.404 | 0.462 | 0.571 | 0.690 | 0.786 | 0.887 | 1.141 | 1.195 | 1.315 | 1.671 | 1.389 | 1.559 | 2.600 |
| 2016 | 0.025 | 0.181 | 0.409 | 0.531 | 0.557 | 0.646 | 0.732 | 0.800 | 0.941 | 1.043 | 1.178 | 0.788 | 0.911 | 1.684 | 1.429 |
| 2017 | 0.025 | 0.191 | 0.408 | 0.499 | 0.650 | 0.694 | 0.752 | 0.827 | 0.894 | 0.911 | 1.028 | 0.961 | 0.312 | 0.701 | 0.688 |
| 2018 | 0.025 | 0.186 | 0.377 | 0.467 | 0.573 | 0.734 | 0.809 | 0.853 | 0.906 | 1.039 | 0.936 | 1.110 | 0.568 | 0.435 | 1.130 |
| 2019 | 0.025 | 0.186 | 0.422 | 0.565 | 0.643 | 0.759 | 0.878 | 0.962 | 1.007 | 1.065 | 1.035 | 1.182 | 0.754 | 0.698 | 0.769 |
| 2020 | 0.025 | 0.186 | 0.387 | 0.522 | 0.632 | 0.716 | 0.799 | 0.955 | 1.006 | 1.040 | 1.189 | 1.072 | 1.208 | 0.961 | 0.769 |
| 2021 | 0.025 | 0.186 | 0.393 | 0.480 | 0.574 | 0.690 | 0.757 | 0.841 | 1.011 | 1.130 | 1.160 | 1.269 | 1.215 | 0.936 | 0.408 |
| 2022 | 0.025 | 0.186 | 0.440 | 0.506 | 0.574 | 0.724 | 0.837 | 0.883 | 0.984 | 0.972 | 1.271 | 1.044 | 1.235 | 0.910 | 0.770 |
| 2023 | 0.025 | 0.186 | 0.494 | 0.483 | 0.524 | 0.602 | 0.812 | 0.923 | 0.957 | 1.041 | 1.095 | 1.294 | 1.331 | 1.218 | 1.131 |
| 2024 | 0.025 | 0.186 | 0.435 | 0.481 | 0.560 | 0.617 | 0.696 | 0.825 | 1.016 | 1.017 | 1.078 | 1.251 | 1.308 | 1.328 | 1.559 |
| 2025 | 0.025 | 0.186 | 0.314 | 0.556 | 0.621 | 0.693 | 0.730 | 0.855 | 0.966 | 1.140 | 1.194 | 1.082 | 1.314 | 1.343 | 1.268 |
| 2026 predicted | 0.025 | 0.186 | 0.368 | 0.440 | 0.677 | 0.734 | 0.813 | 0.864 | 0.953 | 1.075 | 1.194 | 1.232 | 1.307 | 1.420 | 1.448 |
| 2027 forecast | | | 0.368 | 0.482 | 0.560 | 0.800 | 0.858 | 0.936 | 0.985 | 1.069 | 1.187 | 1.300 | 1.333 | 1.402 | 1.508 |
| 2028 forecast | | | 0.368 | 0.482 | 0.601 | 0.683 | 0.924 | 0.980 | 1.056 | 1.101 | 1.181 | 1.293 | 1.400 | 1.427 | 1.491 |
| Mean | 0.018 | 0.196 | 0.373 | 0.508 | 0.629 | 0.745 | 0.863 | 0.987 | 1.111 | 1.205 | 1.301 | 1.361 | 1.356 | 1.362 | 1.472 |
| CV (%) | 97.9% | 22.7% | 15.6% | 10.0% | 7.8% | 9.1% | 12.1% | 13.2% | 13.6% | 15.1% | 15.5% | 18.1% | 26.8% | 31.0% | 33.7% |

| Year / basis | Age (years) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ |

1964–1990 uses the identical fixed schedule in every year. Annual rows through 2025 retain the actual main-data inputs: observed means at ages 3–15+ where available, with model-predicted fills for 2025 ages 13 and 14. The 2026 ages 3–15+ are the selected standalone WtAgeRE predictions.

Ages 1–2 in 2025 and 2026 retain the 2024 input values. Forecast rows 2027–2028 have blank ages 1–2 because that annual model estimates ages 3–15+ only. Blank cells indicate unavailable estimates. Future rows use the selected standalone WtAgeRE fishery forecasts. Projections hold the 2027 forecast fixed for catch and spawning weights. Detailed source records are in the technical appendix.

Mean and CV use the 35 main-input schedules from 1991–2025, including the identified fills and younger-age values retained from the baseline. The fixed early row and all 2026–2028 predictions/forecasts are excluded. CV = sample SD / mean, expressed as percent; it describes variation in the weight schedules among years.

**Table 29:** Effects of cumulative input updates within Model 23.0, expressed as 2026 estimates. Biomass uses the same reference population weight schedule in every fit, and fishing mortality is the maximum realized rate across ages. Each row includes preceding updates. ATS and AVO contrasts combine new observations, historical revisions and the stated uncertainty calibrations; the fits estimate their combined effect.

| Input updates within Model 23.0 | | | | |
|---|---|---|---|---|
| Cumulative input update | Biomass (million t) | Spawning biomass (million t) | Age-1 fish (billion) | Fishing mortality (per year) |
| Catch + endpoint | 9.030 | 2.623 | 18.525 | 0.348 |
| + Fishery weights 2024 | 8.856 | 2.542 | 18.541 | 0.348 |
| + Fishery weights 2026 | 8.857 | 2.533 | 18.541 | 0.348 |
| + Fishery age data | 9.323 | 2.746 | 18.523 | 0.341 |
| + BTS | 9.966 | 2.834 | 24.345 | 0.335 |
| + ATS | 11.043 | 2.761 | 31.731 | 0.345 |
| + AVO | 11.227 | 2.808 | 31.948 | 0.341 |

**Table 30:** Deterministic 2026 population estimates with the selected catch inputs. Model 26.1.2 is the Early-late fishery split comparison; its marginal sampling estimator is excluded.

| 2026 estimates after the catch revision | | | | |
|---|---|---|---|---|
| Model | Biomass (million t) | Spawning biomass (million t) | Age-1 fish (billion) | Maximum F (per year) |
| Model 23.2 | 12.267 | 3.031 | 37.532 | 0.304 |
| Model 23.0 | 11.227 | 2.808 | 31.948 | 0.341 |
| Model 26.0 | 12.162 | 3.057 | 36.347 | 0.299 |
| Model 26.1.0 | 12.153 | 3.053 | 36.334 | 0.300 |
| Model 26.1.1 | 12.198 | 3.064 | 36.826 | 0.316 |
| Model 26.1 decline penalty 0 | 12.351 | 3.110 | 37.608 | 0.338 |
| Model 26.1.2 Early-late fishery split | 12.235 | 3.065 | 36.518 | 0.315 |

ADMB biomass uses the same population weights as Model 26.0. Model 26.1.2 is a deterministic native joint mode conditional on fixed Foreign and Modern SDs; its sampling-based marginal estimator was excluded by author instruction. This table presents point estimates; scientific model adoption remains separate.

**Table 31:** Model 26.0 estimated population quantities in 2026. Ranges are approximate 95% confidence intervals under the selected model and fixed biological schedules. Fishing mortality is the largest realized rate across ages in 2026.

| Estimated 2026 population | | | | | |
|---|---|---|---|---|---|
| Quantity | Estimate | SE | 95% lower | 95% upper | Units |
| Total biomass | 12.162 | 1.511 | 9.534 | 15.515 | Million tons |
| Spawning biomass | 3.057 | 0.320 | 2.489 | 3.754 | Million tons |
| Age-1 recruitment | 36.347 | 19.395 | 12.772 | 103.440 | Billion fish |
| Fishing mortality: most exposed age | 0.299 | 0.067 | 0.194 | 0.463 | Per year |

**Table 32:** Model 26.0 annual estimates and approximate 95% confidence intervals, 1964–2026. Each cell gives the estimate followed by its 95% interval. Biomass and spawning biomass are million tons, recruitment is billion age-1 fish and fishing mortality is the maximum realized rate across ages per year. Recruitment is labeled by the calendar year fish enter age 1.

| Annual population estimates and 95% intervals | | | |
|---|---|---|---|
| Year Biomass (million t) | Spawning biomass (million t) | Age-1 fish (billion) | Fishing mortality (per year) |
| 1964 4.827 (3.351–6.953) | 1.348 (0.909–1.998) | 5.727 (2.316–14.165) | 0.077 (0.043–0.135) |
| 1965 5.201 (3.701–7.308) | 1.376 (0.939–2.016) | 23.494 (13.757–40.121) | 0.085 (0.049–0.146) |
| 1966 5.420 (3.893–7.545) | 1.398 (0.946–2.067) | 15.880 (8.083–31.199) | 0.090 (0.053–0.153) |
| 1967 6.159 (4.520–8.392) | 1.508 (1.027–2.215) | 27.213 (16.078–46.059) | 0.161 (0.091–0.286) |
| 1968 6.803 (5.019–9.223) | 1.674 (1.131–2.476) | 22.441 (12.605–39.953) | 0.178 (0.107–0.297) |
| 1969 7.387 (5.544–9.843) | 1.895 (1.292–2.781) | 26.281 (15.584–44.321) | 0.179 (0.104–0.308) |
| 1970 8.087 (6.171–10.599) | 2.006 (1.388–2.899) | 23.326 (13.486–40.346) | 0.265 (0.126–0.556) |
| 1971 8.012 (6.127–10.478) | 2.117 (1.476–3.036) | 13.629 (6.794–27.342) | 0.386 (0.178–0.839) |
| 1972 7.273 (5.512–9.596) | 1.978 (1.384–2.825) | 10.896 (5.305–22.380) | 0.426 (0.196–0.929) |
| 1973 6.521 (4.990–8.522) | 1.697 (1.159–2.484) | 26.743 (18.276–39.133) | 0.471 (0.221–1.005) |
| 1974 5.485 (4.193–7.175) | 1.293 (0.834–2.007) | 19.281 (13.001–28.596) | 0.499 (0.303–0.821) |
| 1975 4.711 (3.663–6.059) | 1.200 (0.784–1.835) | 16.451 (11.522–23.489) | 0.467 (0.327–0.668) |
| 1976 4.304 (3.468–5.342) | 1.121 (0.794–1.581) | 12.944 (9.258–18.097) | 0.423 (0.278–0.643) |
| 1977 4.036 (3.351–4.860) | 1.305 (0.984–1.732) | 14.023 (10.506–18.716) | 0.377 (0.171–0.830) |
| 1978 4.311 (3.700–5.023) | 1.158 (0.892–1.504) | 26.513 (21.903–32.094) | 0.436 (0.197–0.968) |
| 1979 5.530 (4.931–6.200) | 1.068 (0.839–1.359) | 62.439 (54.916–70.993) | 0.462 (0.220–0.969) |
| 1980 6.294 (5.684–6.969) | 1.268 (1.044–1.541) | 29.662 (24.950–35.264) | 0.424 (0.263–0.684) |
| 1981 7.831 (7.139–8.589) | 1.934 (1.693–2.210) | 34.243 (29.110–40.280) | 0.308 (0.201–0.472) |
| 1982 9.443 (8.615–10.352) | 2.680 (2.387–3.008) | 19.180 (15.520–23.704) | 0.201 (0.139–0.291) |
| 1983 10.311 (9.464–11.233) | 3.594 (3.219–4.013) | 52.001 (46.089–58.672) | 0.174 (0.113–0.267) |
| 1984 10.272 (9.441–11.176) | 3.732 (3.350–4.158) | 14.229 (11.444–17.692) | 0.172 (0.130–0.227) |
| 1985 12.273 (11.351–13.271) | 4.262 (3.876–4.686) | 32.543 (28.408–37.280) | 0.181 (0.132–0.249) |
| 1986 11.664 (10.861–12.526) | 4.145 (3.805–4.516) | 12.743 (10.478–15.497) | 0.168 (0.115–0.246) |
| 1987 10.839 (10.212–11.504) | 4.157 (3.857–4.480) | 7.323 (5.855–9.158) | 0.129 (0.086–0.194) |
| 1988 9.780 (9.224–10.370) | 3.909 (3.657–4.179) | 5.712 (4.552–7.168) | 0.181 (0.122–0.268) |
| 1989 8.195 (7.706–8.714) | 3.382 (3.165–3.613) | 11.908 (10.240–13.848) | 0.209 (0.159–0.275) |
| 1990 7.731 (7.305–8.183) | 2.870 (2.674–3.080) | 50.462 (46.960–54.226) | 0.292 (0.214–0.396) |
| 1991 8.525 (8.086–8.988) | 2.320 (2.144–2.511) | 27.363 (24.774–30.224) | 0.406 (0.289–0.569) |
| 1992 9.146 (8.688–9.628) | 2.202 (2.049–2.366) | 23.766 (21.304–26.512) | 0.484 (0.330–0.711) |
| 1993 10.389 (9.875–10.931) | 2.878 (2.704–3.063) | 46.285 (42.750–50.113) | 0.304 (0.211–0.438) |
| 1994 10.238 (9.733–10.769) | 3.341 (3.143–3.551) | 15.931 (14.075–18.031) | 0.233 (0.157–0.346) |
| 1995 8.867 (8.408–9.350) | 3.572 (3.367–3.790) | 11.171 (9.736–12.818) | 0.296 (0.252–0.348) |
| 1996 9.339 (8.859–9.845) | 3.509 (3.307–3.723) | 22.654 (20.657–24.844) | 0.376 (0.308–0.458) |
| 1997 8.875 (8.422–9.352) | 3.412 (3.210–3.626) | 30.307 (28.043–32.753) | 0.339 (0.229–0.504) |
| 1998 8.748 (8.302–9.217) | 2.901 (2.726–3.087) | 14.951 (13.463–16.604) | 0.275 (0.199–0.380) |
| 1999 8.937 (8.494–9.403) | 3.020 (2.847–3.204) | 16.029 (14.509–17.708) | 0.213 (0.180–0.251) |
| 2000 9.070 (8.635–9.528) | 2.903 (2.741–3.075) | 24.585 (22.686–26.642) | 0.272 (0.224–0.330) |
| 2001 9.067 (8.643–9.512) | 2.939 (2.776–3.113) | 34.415 (32.234–36.744) | 0.291 (0.239–0.354) |
| 2002 9.528 (9.109–9.966) | 2.728 (2.577–2.889) | 22.938 (21.267–24.742) | 0.374 (0.318–0.440) |
| 2003 10.763 (10.308–11.238) | 2.748 (2.605–2.899) | 14.341 (13.115–15.681) | 0.415 (0.338–0.510) |
| 2004 9.630 (9.214–10.065) | 3.032 (2.882–3.189) | 6.525 (5.751–7.404) | 0.361 (0.279–0.467) |
| 2005 7.764 (7.409–8.136) | 2.663 (2.529–2.804) | 4.624 (3.988–5.362) | 0.335 (0.282–0.399) |
| 2006 6.805 (6.471–7.156) | 2.431 (2.302–2.567) | 11.539 (10.424–12.774) | 0.406 (0.348–0.474) |
| 2007 6.328 (6.007–6.666) | 1.994 (1.876–2.119) | 24.986 (23.081–27.048) | 0.432 (0.370–0.505) |
| 2008 4.826 (4.568–5.099) | 1.528 (1.427–1.636) | 13.553 (12.109–15.170) | 0.443 (0.372–0.528) |

| Annual population estimates and 95% intervals | | | |
|---|---|---|---|
| Year Biomass (million t) | Spawning biomass (million t) | Age-1 fish (billion) | Fishing mortality (per year) |
| 2009 5.899 (5.593–6.221) | 1.617 (1.512–1.730) | 49.521 (46.316–52.948) | 0.381 (0.256–0.569) |
| 2010 7.701 (7.321–8.101) | 1.654 (1.549–1.767) | 23.135 (21.102–25.363) | 0.317 (0.237–0.425) |
| 2011 8.485 (8.068–8.924) | 2.029 (1.907–2.158) | 14.211 (12.693–15.910) | 0.433 (0.356–0.526) |
| 2012 8.811 (8.359–9.287) | 2.475 (2.331–2.628) | 12.310 (10.960–13.827) | 0.429 (0.266–0.691) |
| 2013 9.756 (9.251–10.287) | 2.824 (2.655–3.003) | 43.285 (40.546–46.208) | 0.428 (0.253–0.723) |
| 2014 10.381 (9.869–10.919) | 2.782 (2.606–2.969) | 49.480 (46.207–52.984) | 0.403 (0.226–0.719) |
| 2015 11.568 (11.013–12.151) | 2.597 (2.436–2.768) | 21.258 (19.132–23.620) | 0.449 (0.245–0.823) |
| 2016 12.325 (11.722–12.960) | 2.855 (2.690–3.030) | 8.625 (7.265–10.239) | 0.312 (0.244–0.400) |
| 2017 11.425 (10.826–12.056) | 3.300 (3.110–3.502) | 7.890 (6.545–9.511) | 0.295 (0.211–0.414) |
| 2018 9.899 (9.330–10.502) | 3.152 (2.957–3.361) | 16.823 (14.472–19.557) | 0.276 (0.232–0.329) |
| 2019 10.741 (10.071–11.457) | 3.067 (2.851–3.299) | 87.831 (78.998–97.651) | 0.438 (0.360–0.532) |
| 2020 11.795 (10.940–12.717) | 2.364 (2.171–2.575) | 27.572 (23.257–32.688) | 0.419 (0.353–0.497) |
| 2021 11.818 (10.823–12.905) | 2.619 (2.394–2.866) | 12.958 (9.881–16.992) | 0.416 (0.332–0.523) |
| 2022 13.137 (11.866–14.545) | 3.742 (3.383–4.139) | 12.683 (9.030–17.815) | 0.291 (0.220–0.385) |
| 2023 11.901 (10.613–13.345) | 3.779 (3.369–4.238) | 11.616 (7.811–17.274) | 0.232 (0.155–0.348) |
| 2024 10.767 (9.397–12.337) | 3.526 (3.076–4.041) | 39.254 (25.030–61.560) | 0.243 (0.201–0.294) |
| 2025 10.134 (8.563–11.993) | 3.228 (2.732–3.814) | 27.976 (16.327–47.937) | 0.293 (0.225–0.380) |
| 2026 12.162 (9.534–15.515) | 3.057 (2.489–3.754) | 36.347 (12.772–103.440) | 0.299 (0.194–0.463) |

**Table 33:** Final 2026 population summaries from the regularized Model 26.0 posterior. All 5,000 saved draws contribute. Intervals are central 95% credible intervals. Priors, parameter support and supplied biology remain fixed. Production-fit estimates and projections retain their separate roles.

| Posterior population uncertainty | | | | |
|---|---|---|---|---|
| Quantity and units | Mean | Median | 2.5% | 97.5% |
| SSB 2026 (million t) | 2.9645 | 2.9465 | 2.4497 | 3.5844 |
| Biomass 2026 (million t) | 11.5860 | 11.4616 | 9.2151 | 14.7192 |
| Recruitment 2026 (billion fish) | 36.4680 | 31.1518 | 12.2880 | 91.0616 |

**Table 34:** Model 26.0 start-of-year numbers at age, 1964–2026, in billion fish. Age 15+ includes all older fish. Values are rounded to 0.001 billion fish; full precision is retained in the supporting tables.

| Numbers at age (billion fish) | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ |
| 1964 | 5.727 | 4.350 | 2.577 | 0.877 | 0.514 | 0.633 | 0.390 | 0.247 | 0.198 | 0.173 | 0.149 | 0.127 | 0.106 | 0.088 | 0.159 |
| 1965 | 23.494 | 2.328 | 2.763 | 1.872 | 0.612 | 0.356 | 0.434 | 0.269 | 0.171 | 0.137 | 0.120 | 0.104 | 0.089 | 0.074 | 0.172 |
| 1966 | 15.880 | 9.549 | 1.476 | 1.987 | 1.274 | 0.417 | 0.243 | 0.297 | 0.184 | 0.117 | 0.094 | 0.083 | 0.072 | 0.061 | 0.170 |
| 1967 | 27.213 | 6.454 | 6.040 | 1.053 | 1.345 | 0.864 | 0.283 | 0.165 | 0.202 | 0.125 | 0.080 | 0.064 | 0.057 | 0.049 | 0.158 |
| 1968 | 22.441 | 11.052 | 4.038 | 4.077 | 0.664 | 0.850 | 0.548 | 0.180 | 0.105 | 0.129 | 0.080 | 0.051 | 0.041 | 0.036 | 0.132 |
| 1969 | 26.281 | 9.109 | 6.877 | 2.693 | 2.528 | 0.414 | 0.533 | 0.345 | 0.113 | 0.066 | 0.081 | 0.050 | 0.032 | 0.026 | 0.106 |
| 1970 | 23.326 | 10.660 | 5.638 | 4.445 | 1.667 | 1.574 | 0.259 | 0.334 | 0.216 | 0.071 | 0.041 | 0.051 | 0.031 | 0.020 | 0.082 |
| 1971 | 13.629 | 9.439 | 6.459 | 3.435 | 2.654 | 0.984 | 0.930 | 0.152 | 0.193 | 0.124 | 0.041 | 0.024 | 0.029 | 0.018 | 0.058 |
| 1972 | 10.896 | 5.500 | 5.584 | 3.709 | 1.888 | 1.418 | 0.529 | 0.500 | 0.079 | 0.100 | 0.064 | 0.021 | 0.012 | 0.015 | 0.038 |
| 1973 | 26.743 | 4.398 | 3.158 | 2.995 | 1.895 | 0.965 | 0.726 | 0.271 | 0.249 | 0.039 | 0.049 | 0.031 | 0.010 | 0.006 | 0.026 |
| 1974 | 19.281 | 10.796 | 2.452 | 1.583 | 1.418 | 0.896 | 0.457 | 0.345 | 0.126 | 0.116 | 0.018 | 0.023 | 0.014 | 0.005 | 0.014 |
| 1975 | 16.451 | 7.788 | 5.813 | 1.103 | 0.714 | 0.641 | 0.407 | 0.208 | 0.155 | 0.057 | 0.052 | 0.008 | 0.010 | 0.007 | 0.009 |
| 1976 | 12.944 | 6.658 | 4.425 | 2.700 | 0.526 | 0.347 | 0.314 | 0.200 | 0.102 | 0.077 | 0.028 | 0.026 | 0.004 | 0.005 | 0.007 |
| 1977 | 14.023 | 5.244 | 3.863 | 2.220 | 1.311 | 0.261 | 0.174 | 0.159 | 0.102 | 0.052 | 0.039 | 0.014 | 0.013 | 0.002 | 0.006 |
| 1978 | 26.513 | 5.685 | 3.073 | 2.134 | 1.152 | 0.671 | 0.135 | 0.090 | 0.083 | 0.053 | 0.027 | 0.020 | 0.007 | 0.007 | 0.004 |
| 1979 | 62.439 | 10.753 | 3.356 | 1.697 | 1.095 | 0.568 | 0.333 | 0.067 | 0.045 | 0.041 | 0.025 | 0.013 | 0.009 | 0.003 | 0.005 |
| 1980 | 29.662 | 25.334 | 6.512 | 1.953 | 0.894 | 0.526 | 0.271 | 0.160 | 0.032 | 0.021 | 0.019 | 0.012 | 0.006 | 0.004 | 0.004 |
| 1981 | 34.243 | 12.042 | 15.722 | 4.120 | 1.061 | 0.440 | 0.255 | 0.132 | 0.078 | 0.016 | 0.010 | 0.009 | 0.006 | 0.003 | 0.004 |
| 1982 | 19.180 | 13.910 | 7.571 | 10.735 | 2.474 | 0.577 | 0.240 | 0.140 | 0.073 | 0.043 | 0.009 | 0.006 | 0.005 | 0.003 | 0.004 |
| 1983 | 52.001 | 7.794 | 8.800 | 5.406 | 7.005 | 1.499 | 0.350 | 0.146 | 0.085 | 0.044 | 0.026 | 0.005 | 0.003 | 0.003 | 0.004 |
| 1984 | 14.229 | 21.135 | 4.935 | 6.345 | 3.678 | 4.454 | 0.933 | 0.219 | 0.091 | 0.053 | 0.028 | 0.017 | 0.003 | 0.002 | 0.005 |
| 1985 | 32.543 | 5.783 | 13.399 | 3.558 | 4.364 | 2.346 | 2.778 | 0.583 | 0.137 | 0.057 | 0.033 | 0.017 | 0.010 | 0.002 | 0.004 |
| 1986 | 12.743 | 13.228 | 3.666 | 9.626 | 2.464 | 2.876 | 1.450 | 1.721 | 0.363 | 0.086 | 0.036 | 0.021 | 0.011 | 0.007 | 0.004 |
| 1987 | 7.323 | 5.180 | 8.390 | 2.635 | 6.620 | 1.641 | 1.821 | 0.908 | 1.088 | 0.229 | 0.054 | 0.023 | 0.013 | 0.007 | 0.007 |
| 1988 | 5.712 | 2.977 | 3.291 | 6.063 | 1.859 | 4.529 | 1.089 | 1.207 | 0.591 | 0.709 | 0.150 | 0.036 | 0.015 | 0.009 | 0.009 |
| 1989 | 11.908 | 2.322 | 1.889 | 2.308 | 4.180 | 1.215 | 2.892 | 0.674 | 0.753 | 0.369 | 0.446 | 0.095 | 0.023 | 0.009 | 0.011 |
| 1990 | 50.462 | 4.841 | 1.474 | 1.348 | 1.573 | 2.721 | 0.768 | 1.739 | 0.412 | 0.467 | 0.229 | 0.278 | 0.059 | 0.014 | 0.013 |
| 1991 | 27.363 | 20.513 | 3.069 | 1.051 | 0.883 | 0.939 | 1.582 | 0.426 | 0.962 | 0.232 | 0.268 | 0.128 | 0.158 | 0.034 | 0.015 |
| 1992 | 23.766 | 11.123 | 13.004 | 2.212 | 0.709 | 0.545 | 0.550 | 0.835 | 0.237 | 0.517 | 0.130 | 0.133 | 0.065 | 0.080 | 0.025 |
| 1993 | 46.285 | 9.660 | 7.040 | 9.058 | 1.484 | 0.440 | 0.299 | 0.266 | 0.386 | 0.109 | 0.241 | 0.059 | 0.061 | 0.030 | 0.048 |
| 1994 | 15.931 | 18.816 | 6.141 | 5.016 | 5.788 | 0.975 | 0.266 | 0.164 | 0.145 | 0.212 | 0.061 | 0.136 | 0.033 | 0.034 | 0.043 |
| 1995 | 11.171 | 6.476 | 11.970 | 4.488 | 3.422 | 3.424 | 0.588 | 0.156 | 0.097 | 0.087 | 0.127 | 0.037 | 0.082 | 0.020 | 0.046 |
| 1996 | 22.654 | 4.542 | 4.119 | 8.785 | 3.202 | 2.200 | 1.886 | 0.337 | 0.090 | 0.057 | 0.052 | 0.077 | 0.022 | 0.050 | 0.040 |
| 1997 | 30.307 | 9.210 | 2.882 | 3.010 | 6.386 | 2.202 | 1.297 | 0.960 | 0.172 | 0.049 | 0.032 | 0.030 | 0.045 | 0.013 | 0.053 |
| 1998 | 14.951 | 12.321 | 5.830 | 2.098 | 2.142 | 4.358 | 1.377 | 0.729 | 0.524 | 0.091 | 0.027 | 0.018 | 0.017 | 0.026 | 0.037 |
| 1999 | 16.029 | 6.078 | 7.822 | 4.233 | 1.487 | 1.457 | 2.659 | 0.843 | 0.421 | 0.295 | 0.052 | 0.016 | 0.011 | 0.010 | 0.037 |
| 2000 | 24.585 | 6.517 | 3.865 | 5.567 | 2.949 | 1.011 | 0.946 | 1.592 | 0.514 | 0.259 | 0.182 | 0.032 | 0.010 | 0.007 | 0.029 |
| 2001 | 34.415 | 9.995 | 4.144 | 2.792 | 3.765 | 1.891 | 0.648 | 0.560 | 0.899 | 0.307 | 0.159 | 0.112 | 0.020 | 0.006 | 0.022 |
| 2002 | 22.938 | 13.991 | 6.359 | 3.017 | 1.923 | 2.277 | 1.047 | 0.364 | 0.318 | 0.524 | 0.184 | 0.095 | 0.068 | 0.012 | 0.017 |
| 2003 | 14.341 | 9.325 | 8.893 | 4.605 | 2.049 | 1.174 | 1.160 | 0.540 | 0.192 | 0.175 | 0.301 | 0.108 | 0.057 | 0.040 | 0.017 |
| 2004 | 6.525 | 5.830 | 5.931 | 6.270 | 3.117 | 1.213 | 0.615 | 0.567 | 0.271 | 0.102 | 0.098 | 0.176 | 0.063 | 0.033 | 0.034 |
| 2005 | 4.624 | 2.653 | 3.710 | 4.296 | 3.941 | 1.884 | 0.684 | 0.318 | 0.295 | 0.147 | 0.058 | 0.058 | 0.105 | 0.038 | 0.040 |
| 2006 | 11.539 | 1.880 | 1.688 | 2.690 | 2.846 | 2.166 | 0.998 | 0.370 | 0.176 | 0.165 | 0.085 | 0.034 | 0.035 | 0.064 | 0.047 |
| 2007 | 24.986 | 4.691 | 1.195 | 1.193 | 1.747 | 1.593 | 1.069 | 0.507 | 0.191 | 0.093 | 0.089 | 0.047 | 0.020 | 0.020 | 0.063 |
| 2008 | 13.553 | 10.158 | 2.982 | 0.845 | 0.768 | 0.978 | 0.766 | 0.526 | 0.262 | 0.101 | 0.050 | 0.048 | 0.026 | 0.011 | 0.045 |
| 2009 | 49.521 | 5.510 | 6.459 | 2.150 | 0.555 | 0.441 | 0.465 | 0.369 | 0.260 | 0.132 | 0.053 | 0.026 | 0.025 | 0.013 | 0.029 |

| Numbers at age (billion fish) | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15+ |
| 2010 | 23.135 | 20.133 | 3.507 | 4.643 | 1.417 | 0.335 | 0.226 | 0.238 | 0.190 | 0.135 | 0.068 | 0.027 | 0.013 | 0.013 | 0.022 |
| 2011 | 14.211 | 9.405 | 12.817 | 2.564 | 2.970 | 0.863 | 0.197 | 0.127 | 0.128 | 0.103 | 0.074 | 0.037 | 0.014 | 0.007 | 0.019 |
| 2012 | 12.310 | 5.777 | 5.985 | 9.326 | 1.780 | 1.497 | 0.415 | 0.096 | 0.062 | 0.063 | 0.050 | 0.036 | 0.019 | 0.007 | 0.013 |
| 2013 | 43.285 | 5.005 | 3.675 | 4.338 | 6.114 | 1.155 | 0.723 | 0.201 | 0.047 | 0.031 | 0.031 | 0.024 | 0.018 | 0.009 | 0.010 |
| 2014 | 49.480 | 17.597 | 3.185 | 2.667 | 2.916 | 3.703 | 0.691 | 0.385 | 0.098 | 0.023 | 0.015 | 0.015 | 0.012 | 0.008 | 0.009 |
| 2015 | 21.258 | 20.116 | 11.197 | 2.315 | 1.823 | 1.817 | 2.084 | 0.356 | 0.201 | 0.051 | 0.011 | 0.007 | 0.008 | 0.006 | 0.009 |
| 2016 | 8.625 | 8.642 | 12.808 | 7.777 | 1.536 | 1.146 | 1.015 | 1.080 | 0.185 | 0.105 | 0.025 | 0.005 | 0.004 | 0.004 | 0.007 |
| 2017 | 7.890 | 3.507 | 5.505 | 9.386 | 4.577 | 0.972 | 0.694 | 0.596 | 0.586 | 0.103 | 0.059 | 0.014 | 0.003 | 0.002 | 0.006 |
| 2018 | 16.823 | 3.208 | 2.233 | 4.045 | 6.465 | 2.573 | 0.548 | 0.391 | 0.334 | 0.323 | 0.058 | 0.033 | 0.008 | 0.002 | 0.004 |
| 2019 | 87.831 | 6.840 | 2.042 | 1.641 | 2.896 | 3.781 | 1.446 | 0.314 | 0.224 | 0.194 | 0.191 | 0.034 | 0.020 | 0.005 | 0.004 |
| 2020 | 27.572 | 35.707 | 4.349 | 1.497 | 1.185 | 1.992 | 2.008 | 0.691 | 0.164 | 0.123 | 0.109 | 0.113 | 0.020 | 0.012 | 0.005 |
| 2021 | 12.958 | 11.209 | 22.578 | 3.140 | 1.035 | 0.768 | 1.044 | 0.979 | 0.355 | 0.082 | 0.062 | 0.056 | 0.060 | 0.011 | 0.009 |
| 2022 | 12.683 | 5.268 | 7.061 | 15.644 | 2.187 | 0.668 | 0.471 | 0.534 | 0.478 | 0.181 | 0.042 | 0.033 | 0.030 | 0.032 | 0.011 |
| 2023 | 11.616 | 5.156 | 3.322 | 5.072 | 10.705 | 1.455 | 0.427 | 0.283 | 0.296 | 0.276 | 0.107 | 0.024 | 0.019 | 0.018 | 0.025 |
| 2024 | 39.254 | 4.722 | 3.268 | 2.384 | 3.526 | 6.678 | 0.891 | 0.267 | 0.171 | 0.174 | 0.164 | 0.063 | 0.014 | 0.011 | 0.025 |
| 2025 | 27.976 | 15.959 | 3.002 | 2.342 | 1.659 | 2.308 | 3.879 | 0.546 | 0.166 | 0.104 | 0.106 | 0.100 | 0.038 | 0.009 | 0.022 |
| 2026 | 36.347 | 11.374 | 10.149 | 2.172 | 1.644 | 1.112 | 1.276 | 2.157 | 0.320 | 0.098 | 0.062 | 0.064 | 0.060 | 0.023 | 0.019 |

**Table 35:** Model 26.0 start-of-2026 numbers at age and uncertainty, in billion fish. Ranges are approximate 95% confidence intervals, accounting for covariance in the fitted model. Age 15+ includes all older fish.

| 2026 numbers at age and uncertainty | | | | | |
|---|---|---|---|---|---|
| Age | Estimate | SE | CV | 95% lower | 95% upper |
| 1 | 36.347 | 19.395 | 53.4% | 12.772 | 103.440 |
| 2 | 11.374 | 3.125 | 27.5% | 6.638 | 19.489 |
| 3 | 10.149 | 2.335 | 23.0% | 6.466 | 15.932 |
| 4 | 2.172 | 0.449 | 20.7% | 1.448 | 3.259 |
| 5 | 1.644 | 0.309 | 18.8% | 1.138 | 2.376 |
| 6 | 1.112 | 0.184 | 16.5% | 0.804 | 1.537 |
| 7 | 1.276 | 0.176 | 13.8% | 0.973 | 1.672 |
| 8 | 2.157 | 0.259 | 12.0% | 1.704 | 2.730 |
| 9 | 0.320 | 0.045 | 13.9% | 0.243 | 0.420 |
| 10 | 0.098 | 0.016 | 16.6% | 0.071 | 0.136 |
| 11 | 0.062 | 0.011 | 18.1% | 0.043 | 0.088 |
| 12 | 0.064 | 0.012 | 18.0% | 0.045 | 0.091 |
| 13 | 0.060 | 0.012 | 19.7% | 0.041 | 0.089 |
| 14 | 0.023 | 0.006 | 25.1% | 0.014 | 0.038 |
| 15+ | 0.019 | 0.006 | 31.3% | 0.010 | 0.034 |

**Table 36:** Model 26.0 biological schedules and fishery weight predictions at ages 1–15+, with weights for 2024–2026. Mortality is an annual instantaneous rate; maturity and female fraction are proportions; weights are kg per fish. Population weights are the BTS schedule. Fishery 2026 values show the current predictions; projections use fixed 2027 fishery forecasts for catch and spawning weights.

| Biological schedules and recent weights | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Age | Natural mortality | Mature | Female | Population weights (kg) | | | Fishery weights (kg) | | | Spawning weights (kg) | | |
| | | | | 2024 | 2025 | 2026 | 2024 | 2025 | 2026 | 2024 | 2025 | 2026 |
| 1 | 0.900 | 0.000 | 0.500 | 0.017 | 0.020 | 0.020 | 0.025 | 0.025 | 0.025 | 0.085 | 0.085 | 0.085 |
| 2 | 0.450 | 0.008 | 0.500 | 0.090 | 0.079 | 0.066 | 0.186 | 0.186 | 0.186 | 0.196 | 0.196 | 0.196 |
| 3 | 0.300 | 0.289 | 0.500 | 0.309 | 0.226 | 0.397 | 0.435 | 0.314 | 0.368 | 0.387 | 0.415 | 0.317 |
| 4 | 0.300 | 0.641 | 0.500 | 0.454 | 0.526 | 0.544 | 0.481 | 0.556 | 0.440 | 0.452 | 0.523 | 0.541 |
| 5 | 0.300 | 0.842 | 0.500 | 0.560 | 0.587 | 0.640 | 0.560 | 0.621 | 0.677 | 0.521 | 0.596 | 0.657 |
| 6 | 0.300 | 0.901 | 0.500 | 0.607 | 0.676 | 0.725 | 0.617 | 0.693 | 0.734 | 0.633 | 0.668 | 0.732 |
| 7 | 0.300 | 0.947 | 0.500 | 0.693 | 0.734 | 0.809 | 0.696 | 0.730 | 0.813 | 0.750 | 0.779 | 0.804 |
| 8 | 0.300 | 0.963 | 0.500 | 0.923 | 0.812 | 0.874 | 0.825 | 0.855 | 0.864 | 0.879 | 0.893 | 0.911 |
| 9 | 0.300 | 0.970 | 0.500 | 0.997 | 0.903 | 0.949 | 1.016 | 0.966 | 0.953 | 0.987 | 1.016 | 1.019 |
| 10 | 0.300 | 1.000 | 0.500 | 1.020 | 1.087 | 1.027 | 1.017 | 1.140 | 1.075 | 1.055 | 1.116 | 1.135 |
| 11 | 0.300 | 1.000 | 0.500 | 1.126 | 1.066 | 1.048 | 1.078 | 1.194 | 1.194 | 1.127 | 1.175 | 1.227 |
| 12 | 0.300 | 1.000 | 0.500 | 1.137 | 1.150 | 1.192 | 1.251 | 1.082 | 1.232 | 1.245 | 1.239 | 1.279 |
| 13 | 0.300 | 1.000 | 0.500 | 1.259 | 1.258 | 1.416 | 1.308 | 1.314 | 1.307 | 1.294 | 1.347 | 1.333 |
| 14 | 0.300 | 1.000 | 0.500 | 1.406 | 1.399 | 1.449 | 1.328 | 1.343 | 1.420 | 1.336 | 1.388 | 1.434 |
| 15+ | 0.300 | 1.000 | 0.500 | 1.754 | 1.547 | 1.360 | 1.559 | 1.268 | 1.448 | 1.391 | 1.421 | 1.466 |

**Table 37:** Model 26.0 selectivity at age in 2026. Cells give the estimate and approximate 95% confidence interval. ATS and AVO share the same estimated curve. Relative selectivity can exceed one because the existing curve convention uses a different normalization than division by its maximum.

| 2026 selectivity and uncertainty | | | |
|---|---|---|---|
| Age | Fishery | BTS | ATS and AVO |
| 1 | 0.000 (0.000–0.003) | 0.054 (0.019–0.155) | 0.000 (0.000–0.000) |
| 2 | 0.016 (0.006–0.045) | 0.026 (0.012–0.055) | 0.776 (0.516–1.169) |
| 3 | 0.145 (0.055–0.382) | 0.066 (0.039–0.112) | 0.789 (0.535–1.163) |
| 4 | 0.337 (0.133–0.852) | 0.156 (0.099–0.244) | 0.709 (0.483–1.039) |
| 5 | 0.646 (0.261–1.594) | 0.326 (0.207–0.513) | 0.779 (0.541–1.122) |
| 6 | 1.589 (1.018–2.481) | 0.559 (0.374–0.836) | 0.974 (0.690–1.376) |
| 7 | 1.556 (1.076–2.249) | 0.769 (0.583–1.013) | 1.141 (0.831–1.567) |
| 8 | 1.536 (1.098–2.148) | 0.897 (0.770–1.046) | 1.125 (0.992–1.275) |
| 9 | 1.429 (1.042–1.960) | 0.958 (0.889–1.033) | 1.125 (0.992–1.275) |
| 10 | 1.354 (0.999–1.837) | 0.984 (0.950–1.018) | 1.125 (0.992–1.275) |
| 11 | 1.295 (0.963–1.743) | 0.994 (0.979–1.009) | 1.125 (0.992–1.275) |
| 12 | 1.274 (0.937–1.733) | 0.998 (0.991–1.004) | 1.125 (0.992–1.275) |
| 13 | 1.274 (0.937–1.733) | 0.999 (0.996–1.002) | 1.125 (0.992–1.275) |
| 14 | 1.274 (0.937–1.733) | 1.000 (0.998–1.001) | 1.125 (0.992–1.275) |
| 15+ | 1.274 (0.937–1.733) | 1.000 (0.999–1.000) | 1.125 (0.992–1.275) |

**Table 38:** Estimated parameter groups and their uncertainty in Model 26.0. Standard errors use each group's estimation scale; differences in scales make direct comparisons among groups inappropriate.

| Estimated parameter groups | | | | | |
|---|---|---|---|---|---|
| Parameter group | Count | Minimum estimate | Maximum estimate | Median SE | Maximum SE |
| BTS age-1 deviation | 44 | −0.401 | 0.359 | 0.138 | 0.211 |
| BTS age-1 log selectivity | 1 | −3.048 | −3.048 | 0.307 | 0.307 |
| BTS midpoint deviation | 44 | −0.351 | 0.192 | 0.099 | 0.123 |
| BTS slope deviation | 44 | 0.246 | 1.589 | 0.234 | 0.398 |
| Initial-age deviation | 14 | −1.480 | −0.395 | 0.651 | 0.803 |
| Log catchability | 3 | −8.225 | −0.248 | 0.071 | 0.185 |
| Log fishing scale | 63 | −2.939 | −0.821 | 0.134 | 0.233 |
| Log logistic slope | 1 | −0.838 | −0.838 | 0.196 | 0.196 |
| Logistic midpoint | 1 | 6.981 | 6.981 | 0.641 | 0.641 |
| Recruitment deviation | 63 | −1.234 | 1.710 | 0.142 | 0.538 |
| Recruitment level | 1 | 9.673 | 9.673 | 0.121 | 0.121 |
| Selectivity increment | 987 | −1.459 | 1.287 | 0.280 | 0.597 |

**Table 39:** Contributions to the Model 26.0 fit, grouped by data source or stabilizing assumption. Rounded rows may differ slightly from their sum. Negative values arise from likelihood constants. Contributions across different data and model structures use different scales and assumptions.

| Contributions to the fit | |
|---|---|
| Contribution | Value |
| Catch data | −127.771 |
| Composition data | 421.207 |
| Index data | 160.956 |
| Initial abundance deviates | 19.860 |
| Non-parametric selectivity | 18.325 |
| Recruitment deviates | 71.193 |
| Selectivity deviates | 188.221 |

**Table 40:** Observed-index residual summaries for Model 26.0. Residuals measure observed-minus-predicted differences relative to the selected observation errors. The retained 2020 ATS placeholders are excluded.

| Model 26.0 index residuals | | | |
|---|---|---|---|
| Series | Observations | Mean residual | Residual SD |
| ATS | 20 | −0.10 | 1.04 |
| ATS age 1 | 19 | 0.06 | 1.13 |
| AVO | 20 | 0.12 | 0.97 |
| BTS | 44 | 0.09 | 1.19 |
| Historical CPUE | 12 | 0.05 | 0.53 |

BTS residuals account for the covariance among survey years. A spread near 1 agrees with the selected observation-error scale. These are fitted-model diagnostics.

**Table 41:** Model 26.0 profile likelihood points and conditional 2027 female spawning biomass (thousand tons, kt), with approximate 95% confidence intervals. ATS catchability is fixed separately at each point; all other active parameters are re-estimated. The projection assumes 1,375 kt catch in 2027 and the mean 2023–2025 fishery selectivity curve.

| 2027 female spawning biomass at fixed ATS catchability | | | | |
|---|---|---|---|---|
| ATS q | ΔNLL | 2027 SSB (kt) | Lower 95% (kt) | Upper 95% (kt) |
| 0.45000 | 2.477 | 3,567.3 | 2,737.3 | 4,649.0 |
| 0.45837 | 1.866 | 3,533.9 | 2,709.7 | 4,608.9 |
| 0.46674 | 1.349 | 3,501.6 | 2,682.8 | 4,570.3 |
| 0.47510 | 0.922 | 3,470.2 | 2,656.8 | 4,532.6 |
| 0.48347 | 0.581 | 3,439.7 | 2,631.5 | 4,496.0 |
| 0.49184 | 0.322 | 3,410.1 | 2,607.0 | 4,460.5 |
| 0.50021 | 0.141 | 3,381.2 | 2,583.6 | 4,425.0 |
| 0.50858 | 0.035 | 3,353.0 | 2,560.3 | 4,391.0 |
| 0.51694 | 0.000 | 3,325.5 | 2,538.0 | 4,357.3 |
| 0.53358 | 0.130 | 3,272.7 | 2,494.8 | 4,293.2 |
| 0.55021 | 0.505 | 3,222.5 | 2,453.8 | 4,232.0 |
| 0.56684 | 1.102 | 3,174.7 | 2,414.2 | 4,174.7 |
| 0.58347 | 1.901 | 3,129.3 | 2,376.6 | 4,120.4 |
| 0.60010 | 2.886 | 3,086.1 | 2,340.4 | 4,069.5 |
| 0.61674 | 4.039 | 3,045.0 | 2,306.3 | 4,020.4 |
| 0.63337 | 5.347 | 3,005.8 | 2,273.8 | 3,973.5 |
| 0.65000 | 6.797 | 2,968.3 | 2,242.7 | 3,928.7 |

Intervals use the fixed-q penalized Hessian and the delta method on log SSB. They propagate fitted-parameter uncertainty, including fitted abundance and selectivity, conditional on the stated catch, biological schedules and ATS q. They exclude uncertainty in ATS q, future recruitment process variation and weight-at-age schedules. Spawning occurs on April 1; age 15+ is retained.

**Table 42:** Model 26.0 Mohn's rho against the full-2026 fit. Rho is the mean proportional terminal revision.

| Model 26.0 retrospective Mohn's rho | | | |
|---|---|---|---|
| Quantity | 2024–2025 (2 peels) | 2023–2025 (3 peels) | 2017–2025 (9 peels) |
| Maximum fishing mortality | 0.04411 | 0.15391 | −0.06364 |
| Age-1 recruitment | −0.26348 | 0.04013 | −0.09848 |
| Total biomass | −0.10792 | −0.10774 | 0.00006 |
| Female spawning biomass | −0.07636 | −0.09393 | 0.12096 |

**Table 43:** Model 26.0 retrospective terminal estimates and revisions: All rows use the full-2026 reference. Each row omits all observations after its terminal year, using the same data-lag rules. Relative revisions are 100 × (peel estimate / full-2026-fit estimate for the same calendar year − 1). SSB is female spawning biomass; F is the maximum age-specific fishing mortality. These are point-estimate comparisons; the peel comparisons cover population estimates only.

| Retrospective revisions in Model 26.0 | | | | | | |
|---|---|---|---|---|---|---|
| Terminal year | F revision (%) | Recruitment revision (%) | Biomass revision (%) | SSB revision (%) | Peel SSB (million t) | Full-fit SSB (million t) |
| 2025 | −9.7 | −30.8 | −11.0 | −4.5 | 3.08 | 3.23 |
| 2024 | 18.5 | −21.9 | −10.5 | −10.8 | 3.15 | 3.53 |
| 2023 | 37.4 | 64.7 | −10.7 | −12.9 | 3.29 | 3.78 |
| 2022 | 7.6 | −35.0 | 1.6 | 2.3 | 3.83 | 3.74 |
| 2021 | 34.7 | 44.6 | −9.8 | −13.8 | 2.26 | 2.62 |
| 2020 | −35.6 | −59.0 | −11.0 | 40.7 | 3.33 | 2.36 |
| 2019 | −53.4 | −79.4 | 0.8 | 28.2 | 3.93 | 3.07 |
| 2018 | −17.2 | −21.3 | 14.3 | 26.9 | 4.00 | 3.15 |
| 2017 | −39.5 | 49.4 | 36.5 | 52.8 | 5.04 | 3.30 |

**Table 44:** Native fitting objectives with the selected catch inputs, including the Early-late fishery split comparison. Model-specific penalties and normalizations preclude direct likelihood ranking.

| model | objective | method |
|---|---|---|
| 26_0 | 751.989578 | Established deterministic penalized fit |
| 26_1_0 | 746.780971 | Established deterministic penalized fit |
| 26_1_1 | 726.234542 | Established deterministic penalized fit |
| 26_1_decline_zero | 706.054843 | Established deterministic penalized fit |
| 26_1_2_conditional | 731.019469 | Early-late fishery split; Foreign 0.1235; Modern fixed at prior marginal estimate |

Objectives have model-specific penalties and normalizations. The Early-late fishery split comparison supplies its native joint objective. These values support run verification; direct AIC or likelihood-ratio ranking is excluded.

**Table 45:** Maximum ABC and OFL for the four separately fitted Rceattle configurations and the Early-late fishery split Model 26.1.2 comparison under matched Scenario 2 assumptions, in million tons. Model 26.0 (Rceattle) is the adopted assessment model and main proposal for Tier 3 calculations and specifications; Model 26.1.2 is the sensitivity analysis. Adopted catch specifications remain separate.

| Maximum ABC and OFL under matched Scenario 2 catches | | | | |
|---|---|---|---|---|
| Model | 2027 | | 2028 | |
| | Maximum ABC (difference) | OFL | Maximum ABC (difference) | OFL |
| Model 26.0 | 1.896 (+0.00%) | 2.328 | 1.968 (+0.00%) | 2.422 |
| Model 26.1.0 | 1.895 (-0.09%) | 2.326 | 1.966 (-0.09%) | 2.420 |
| Model 26.1.1 | 1.863 (-1.75%) | 2.281 | 1.953 (-0.74%) | 2.396 |
| 26.1 decline penalty = 0 | 1.867 (-1.57%) | 2.278 | 1.970 (+0.09%) | 2.408 |
| 26.1.2 Early-late fishery split | 1.855 (-2.19%) | 2.269 | 1.958 (-0.51%) | 2.399 |

Million tons; differences are relative to Model 26.0 in the same year. Recruitment is fixed at each model's historical arithmetic mean. The 26.1.2 row is the Early-late fishery split comparator. Simulation, retrospective and MCMC analyses are excluded. Adopted specifications remain unchanged.

**Table 46:** Final Model 26.0 posterior stock and selectivity summaries from 5,000 saved draws. SD is posterior standard deviation; intervals are central 95% credible intervals. R-hat and chain ESS describe the saved chains.

| Posterior stock and selectivity summaries | | | | | |
|---|---|---|---|---|---|
| Quantity and units | Mean | Posterior SD | Median | 2.5% | 97.5% |
| SSB 2026 (million t) | 2.9645 | 0.2928 | 2.9465 | 2.4497 | 3.5844 |
| Biomass 2026 (million t) | 11.5860 | 1.4111 | 11.4616 | 9.2151 | 14.7192 |
| Recruitment 2026 (billion fish) | 36.4680 | 21.2479 | 31.1518 | 12.2880 | 91.0616 |
| Ages1-3 2026 (billion fish) | 56.0356 | 22.4635 | 50.9012 | 28.5376 | 113.5991 |
| BTS selectivity age 1 | 0.0702 | 0.0354 | 0.0635 | 0.0207 | 0.1578 |
| BTS selectivity age 2 | 0.0330 | 0.0120 | 0.0314 | 0.0152 | 0.0605 |
| BTS selectivity age 5 | 0.3631 | 0.0795 | 0.3541 | 0.2332 | 0.5394 |
| BTS selectivity age 10 | 0.9772 | 0.0215 | 0.9839 | 0.9172 | 0.9985 |
| BTS selectivity age 15 | 0.9995 | 0.0009 | 0.9999 | 0.9969 | 1.0000 |

**Table 47:** Reference points for revised Model 26.0 and the Early-late fishery split Model 26.1.2 comparison. Biomass is female spawning biomass in million tons; F is the annual rate on the most selected age. Each column uses that model's primary projection selectivity and fixed biological schedules.

| Quantity | Model 26.0 | 26.1.2 Early-late fishery split | Units |
|---|---|---|---|
| B100% | 6.158 | 6.220 | Million tons |
| B40% | 2.463 | 2.488 | Million tons |
| B35% | 2.155 | 2.177 | Million tons |
| F40% | 0.513 | 0.524 | Per year |
| F35% | 0.658 | 0.666 | Per year |

Revised fits; model-specific 1978–2023 arithmetic mean recruitment. These SPR reference points condition on fixed biology and the stated selectivity window.

**Table 48:** Projections from the revised fits. Harvest scenarios under the Model 26.0 base fit. Each scenario retains its own rule; specified catches apply only to Scenario 2.

| Scenario | Rule |
|---|---|
| 1 | Fishing at maximum permissible ABC each year. |
| 2 | Expected catch of 1,375 kt in 2027 and 2028, followed by maximum permissible ABC; author F fraction 1. |
| 3 | Mean annual maximum fishery F over 2021–2025, held constant. |
| 4 | One-half the maximum permissible ABC fishing mortality, without fixed-catch conditioning. |
| 5 | No fishing. |
| 6 | Fishing at the OFL threshold each year. |
| 7 | Maximum ABC fishing in 2027 and 2028, followed by OFL fishing from 2029. |

**Table 49:** Projections from the revised fits. Short-term Tier 3 calculations under Scenario 2. Catch, spawning biomass, maximum ABC and OFL are thousand tons; F is the annual rate on the most selected age. The 2028 catch is a planning assumption.

| Year | Assumed catch | Spawning biomass | Maximum ABC | OFL | F |
|---|---|---|---|---|---|
| 2027 | 1,375 | 3,198 | 1,896 | 2,328 | 0.354 |
| 2028 | 1,375 | 3,394 | 1,968 | 2,422 | 0.341 |

**Table 50:** Projections from the revised fits. Thirteen-year catch table for the proposed harvest scenarios, in thousand tons.

| Year | Catch (kt) by harvest scenario | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1: Maximum ABC | 2: Expected catch | 3: Recent F | 4: Half FABC | 5: No fishing | 6: OFL | 7: ABC, then OFL |
| 2026 | 1,313 | 1,313 | 1,313 | 1,313 | 1,313 | 1,313 | 1,313 |
| 2027 | 1,896 | 1,375 | 1,166 | 1,026 | 0 | 2,328 | 1,896 |
| 2028 | 1,828 | 1,375 | 1,240 | 1,112 | 0 | 2,114 | 1,828 |
| 2029 | 1,997 | 2,268 | 1,460 | 1,328 | 0 | 2,211 | 2,449 |
| 2030 | 1,900 | 2,060 | 1,468 | 1,351 | 0 | 2,047 | 2,175 |
| 2031 | 1,845 | 1,933 | 1,482 | 1,375 | 0 | 1,940 | 2,015 |
| 2032 | 1,675 | 1,720 | 1,384 | 1,294 | 0 | 1,662 | 1,700 |
| 2033 | 1,602 | 1,625 | 1,334 | 1,251 | 0 | 1,603 | 1,612 |
| 2034 | 1,569 | 1,580 | 1,307 | 1,228 | 0 | 1,594 | 1,596 |
| 2035 | 1,550 | 1,556 | 1,289 | 1,212 | 0 | 1,594 | 1,594 |
| 2036 | 1,541 | 1,544 | 1,278 | 1,202 | 0 | 1,595 | 1,595 |
| 2037 | 1,536 | 1,538 | 1,271 | 1,195 | 0 | 1,596 | 1,595 |
| 2038 | 1,534 | 1,535 | 1,267 | 1,191 | 0 | 1,596 | 1,596 |

**Table 51:** Projections from the revised fits. Thirteen-year female spawning-biomass table for the proposed harvest scenarios, in thousand tons.

| Year | SSB (kt) by harvest scenario | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1: Maximum ABC | 2: Expected catch | 3: Recent F | 4: Half FABC | 5: No fishing | 6: OFL | 7: ABC, then OFL |
| 2026 | 3,057 | 3,057 | 3,057 | 3,057 | 3,057 | 3,057 | 3,057 |
| 2027 | 3,127 | 3,198 | 3,225 | 3,243 | 3,366 | 3,064 | 3,127 |
| 2028 | 3,123 | 3,394 | 3,496 | 3,569 | 4,123 | 2,911 | 3,123 |
| 2029 | 3,181 | 3,528 | 3,774 | 3,898 | 4,919 | 2,871 | 3,115 |
| 2030 | 2,987 | 3,188 | 3,722 | 3,885 | 5,348 | 2,632 | 2,760 |
| 2031 | 2,815 | 2,925 | 3,632 | 3,823 | 5,664 | 2,446 | 2,508 |
| 2032 | 2,641 | 2,697 | 3,475 | 3,678 | 5,790 | 2,302 | 2,325 |
| 2033 | 2,555 | 2,585 | 3,382 | 3,590 | 5,904 | 2,269 | 2,275 |
| 2034 | 2,510 | 2,525 | 3,321 | 3,530 | 5,981 | 2,263 | 2,263 |
| 2035 | 2,488 | 2,495 | 3,286 | 3,494 | 6,051 | 2,263 | 2,263 |
| 2036 | 2,476 | 2,480 | 3,262 | 3,469 | 6,091 | 2,263 | 2,263 |
| 2037 | 2,470 | 2,472 | 3,247 | 3,452 | 6,119 | 2,263 | 2,263 |
| 2038 | 2,466 | 2,467 | 3,237 | 3,441 | 6,137 | 2,263 | 2,263 |

**Table 52:** Thirteen-year fishing-mortality table for the proposed harvest scenarios. F is the annual rate on the most selected age.

| Year | F ($yr^{-1}$) by harvest scenario | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1: Maximum ABC | 2: Expected catch | 3: Recent F | 4: Half FABC | 5: No fishing | 6: OFL | 7: ABC, then OFL |
| 2026 | 0.299 | 0.299 | 0.299 | 0.299 | 0.299 | 0.299 | 0.299 |
| 2027 | 0.513 | 0.354 | 0.295 | 0.257 | 0.000 | 0.658 | 0.513 |
| 2028 | 0.513 | 0.341 | 0.295 | 0.257 | 0.000 | 0.658 | 0.513 |
| 2029 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.658 | 0.658 |
| 2030 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.658 | 0.658 |
| 2031 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.653 | 0.658 |
| 2032 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.612 | 0.619 |
| 2033 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.603 | 0.605 |
| 2034 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.601 | 0.601 |
| 2035 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.601 | 0.601 |
| 2036 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.601 | 0.601 |
| 2037 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.601 | 0.601 |
| 2038 | 0.513 | 0.513 | 0.295 | 0.257 | 0.000 | 0.601 | 0.601 |

**Table 53:** Definitions of the nine Model 26.0 catch-decision indicators. All benchmarks are calculated from the Model 26.0 fit.

| Probability event | Definition | Purpose |
|---|---|---|
| $P(F_{2027} > F_{35\%})$ | 2027 fishing mortality exceeds the rate giving 35% of unfished spawning potential per recruit. | Compare the catch scenario with the Tier 3 reference rate. |
| $P(B_{2028} <$ mean B$)$ | 2028 female spawning biomass at spawning is below mean estimated SSB over 1978–2025. | Compare the following year's stock with its historical mean. |
| $P(B_{2031} <$ mean B$)$ | 2031 female spawning biomass at spawning is below mean estimated SSB over 1978–2025. | Compare medium-term stock condition with its historical mean. |
| $P(B_{2031} < B_{2027})$ | 2031 female spawning biomass at spawning is below the first projected-year value under the same scenario. | Describe medium-term change relative to 2027. |
| $P(B_{2029} < B_{20\%})$ | 2029 female spawning biomass at spawning is below 20% of the model-specific unfished reference biomass. | Describe the historical low-biomass benchmark used in this decision table; Model 26.0 uses mean-recruitment $B_{100}$ and ADMB uses stock–recruit Bzero. |
| $P(Y_{2031} > Y_{\text{hist}})$ | The 2031 abundance ratio of ages 1–5 to ages 6–15+ exceeds the corresponding ratio pooled over 1964–2026. | Describe relative abundance of young fish. The historical threshold pools abundance across years, rather than averaging annual proportions. |
| $P(D_{2028} < D_{1994})$ | The effective number of mature-biomass age classes in 2028 is below its estimated 1994 value. | Describe near-term age diversity relative to a historical low-diversity year. |
| $P(D_{2031} < D_{1994})$ | The effective number of mature-biomass age classes in 2031 is below its estimated 1994 value. | Describe medium-term age diversity relative to 1994. |
| $P(E_{2027} > E_{2026})$ | Mean age-specific fishing mortality in 2027 exceeds mean age-specific fishing mortality in 2026. | Provide an F-based relative-effort proxy. Observed vessel effort, catchability changes and costs are outside this indicator. |

B denotes female spawning biomass at spawning. Y denotes the young-to-old abundance ratio. D is exp(−sum(p × log(p))), with p the beginning-year mature-biomass age shares. E is the mean-age F proxy. Model-specific biological schedules and reference points are retained.

**Table 54:** Model 26.0 probabilities under eight alternative 2027 catches (kt), expressed as chances out of 100. Fishing mortality is set to achieve each 2027 catch and then held constant through 2031.

| Probability event | 2027 catch (kt) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 10.0 | 317.5 | 635.0 | 952.5 | 1270.0 | 1587.5 | 1905.0 | 2540.0 |
| $P(F_{2027} > F_{35\%})$ | 0 | 0 | 0 | 0 | 0 | 0 | 4 | 73 |
| $P(B_{2028} <$ mean B$)$ | 2 | 3 | 6 | 10 | 16 | 23 | 32 | 51 |
| $P(B_{2031} <$ mean B$)$ | 0 | 0 | 2 | 6 | 15 | 30 | 52 | 89 |
| $P(B_{2031} < B_{2027})$ | 0 | 1 | 3 | 10 | 27 | 54 | 80 | 99 |
| $P(B_{2029} < B_{20\%})$ | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2 |
| $P(Y_{2031} > Y_{\text{hist}})$ | 0 | 0 | 0 | 2 | 11 | 27 | 44 | 69 |
| $P(D_{2028} < D_{1994})$ | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 5 |
| $P(D_{2031} < D_{1994})$ | 0 | 0 | 0 | 1 | 2 | 4 | 10 | 41 |
| $P(E_{2027} > E_{2026})$ | 0 | 0 | 0 | 5 | 79 | 99 | 100 | 100 |

Nine supported events. Assessment covariance is propagated through the catch solution, projections and estimated benchmarks. Future recruitment equals the estimated 1978–2023 mean; independent future recruitment innovations and weight-estimation uncertainty are excluded.

Entries are approximate chances out of 100, rounded to whole numbers. A displayed 0 or 100 reflects rounding.

# Figures

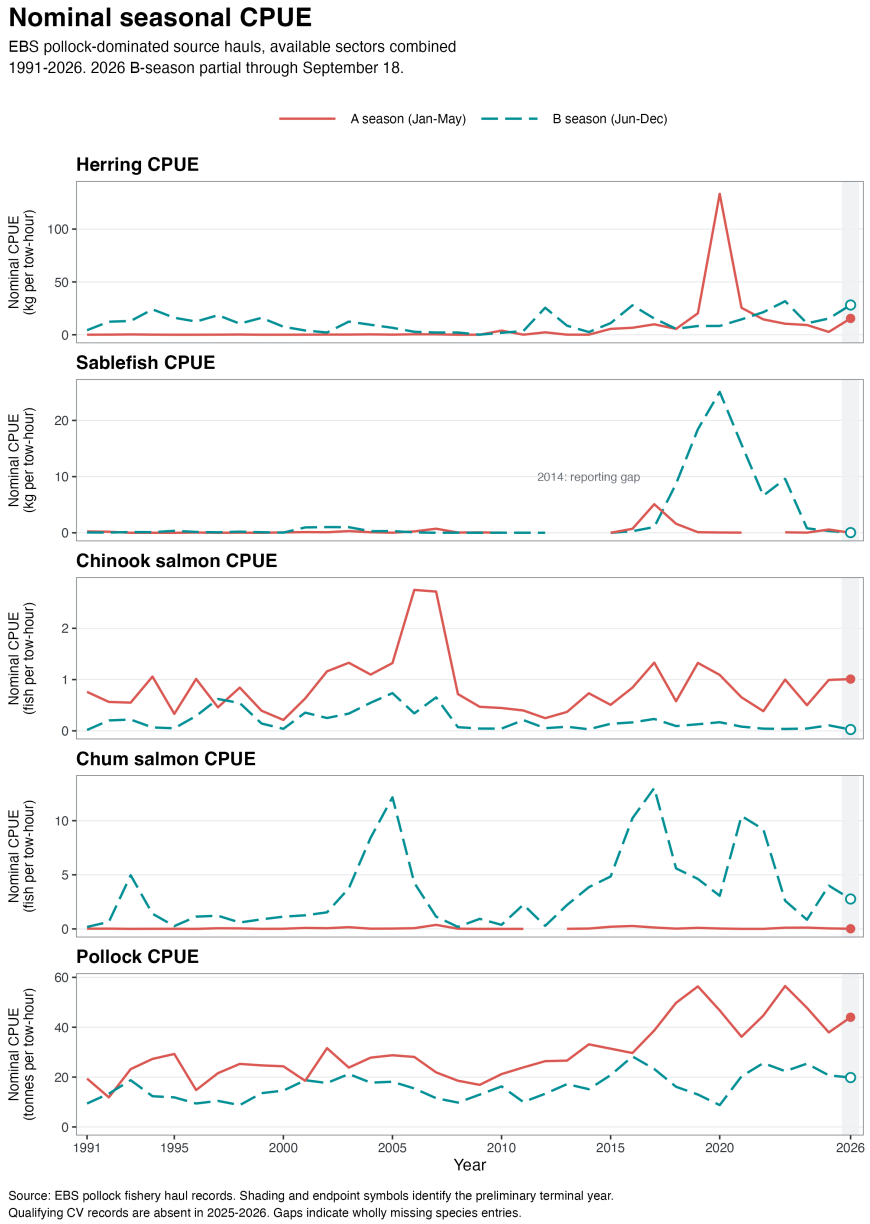



Figure 1: Nominal seasonal catch per tow-hour for herring, sablefish, Chinook salmon, chum salmon, and pollock in available EBS pollock-dominated haul records, 1991-2026, with source vessel types 1-3 combined. Each rate is the sum of available extrapolated catch divided by the sum of positive valid haul hours in the same year and season; the calculation uses the ratio of totals. Weight rates are kg per tow-hour for herring and sablefish and tons per tow-hour for pollock. Salmon rates are extrapolated fish per tow-hour. A season is January-May and B season is June-December, retaining the original figure's calendar split. Selection requires recorded NMFS area 500-539, positive official total catch, and pollock kg / official total catch in tons >800 (a >80% pollock proxy). 2026 values are preliminary; B-season records extend through September 18, 2026 and are partial. Historical years use their full available seasons. Blank species entries are omitted from available catch sums, retained in the effort denominator, and counted in the plotting table. Entirely unreported species-season combinations are gaps, including sablefish in 2014. The supplied file has incomplete CV/EM coverage; no CV records pass the target filter in 2025-2026. Rates describe the available source records; fleet-wide catch accounting uses a separate dataset.

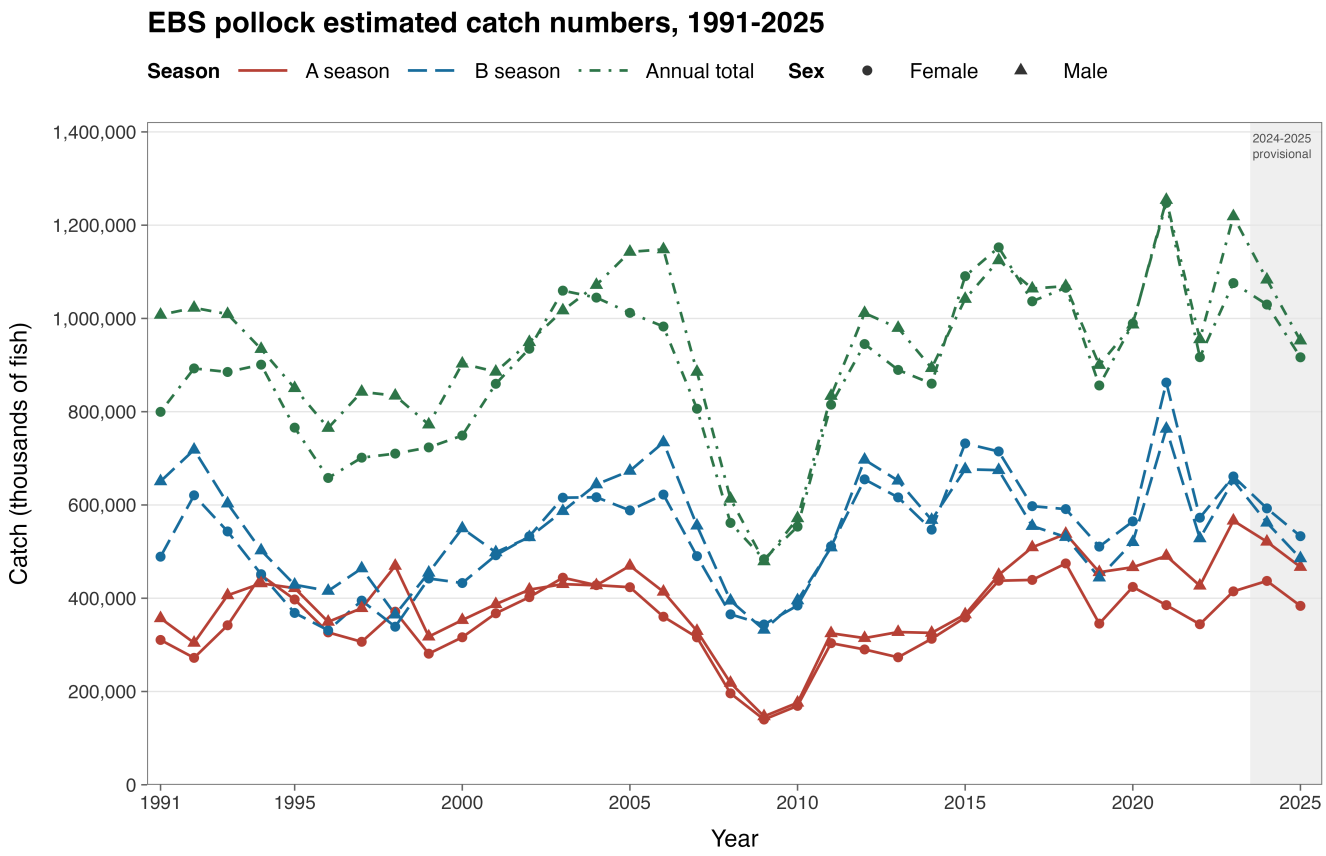



Figure 2: Estimated EBS pollock catch numbers (thousands of fish) by sex for the A season (January-May), B season (June- December), and annual total, 1991-2025. Circles denote females and triangles denote males. B-season estimates combine the northwest and southeast strata, and annual totals equal A+B. Shading marks provisional 2024-2025 estimates on the original sampler age basis.

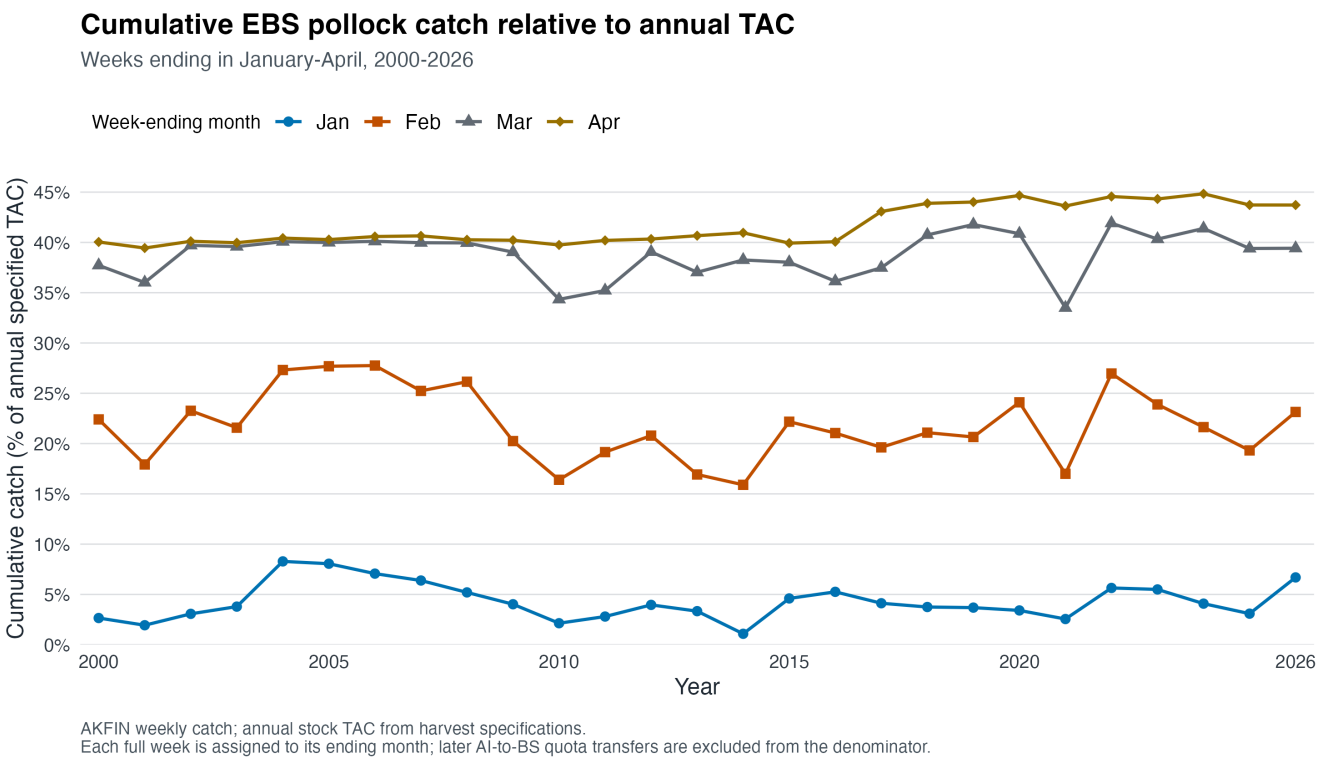



Figure 3: Cumulative EBS pollock catch as a proportion of annual specified TAC for weeks ending in January, February, March, and April, 2000–2026. Catch includes retained and discarded pollock from all gears and targets in the Bering Sea subarea, excluding Bogoslof reporting area 518. Each full reporting week is assigned to its week-ending month. The denominator is the annual stock TAC from harvest specifications, matching the original figure convention, before later AI-to-BS quota transfers. Amendment 110 shifted five percentage points of the directed fishery seasonal allocation from B to A to provide greater flexibility to avoid Chinook salmon bycatch; 2017 was the first A season after the rule took effect.

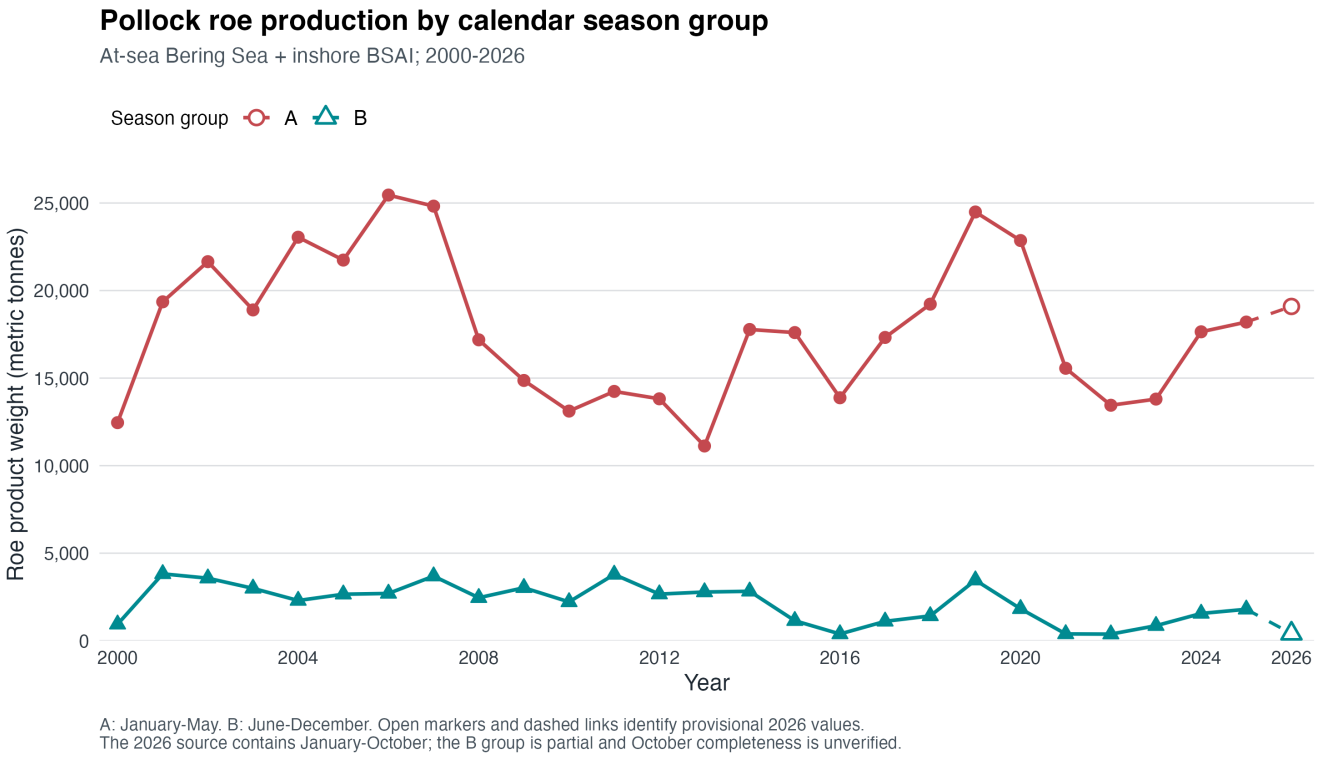



Figure 4: Pollock roe production in the supplied EBS source series by calendar season group, A (January–May) and B (June– December), 2000–2026. Production combines at-sea records labeled Bering Sea (BS) and inshore records labeled Bering Sea and Aleutian Islands (BSAI). Values are roe product metric tons. Open markers and dashed links identify provisional 2026 values. The 2026 export contains January–October; the B group is partial and October completeness is unverified. Calendar groups follow the original figure definition; applying the June 10 regulatory season boundary requires daily production records.

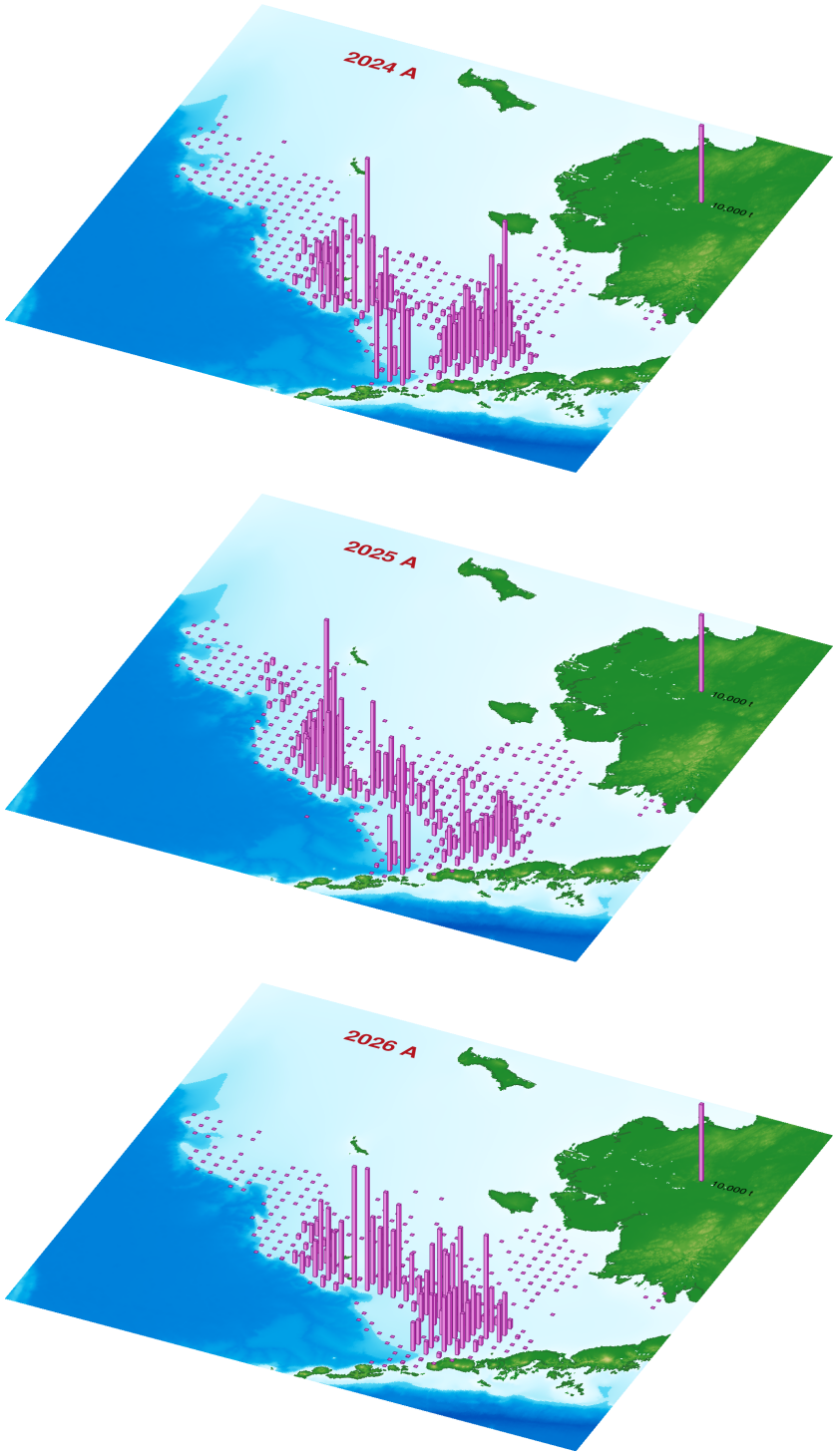



Figure 5: Spatial distribution of observed pollock catch in the A season, 2024–2026, using January 20–June 9 in each year. Vertical bar height represents recorded catch tons within 0.5-degree longitude by 0.25-degree latitude cells· the height scale is fixed across years within each season, with separate A- and B-season scales and a 10,000-t reference column in each panel. Data are from inseas26.csv and include available catcher-processor, mothership and observed catcher-vessel records in NMFS areas below 540. Electronic-monitoring catcher-vessel records are absent. These are observed-record catch totals rather than fleet-wide expanded catches. The B-season comparison is restricted to the September 18 cutoff in all three years. Rendered with GMT using the AKsf coastline and the local bottom_topo.grd bathymetry grid. Blue shades show ocean depth and green shades show land elevation. Mercator projection viewed at azimuth 155 degrees and elevation 35 degrees, with increased column heights; vertical bars represent catch rather than seafloor height. Updated in the style of Figures 4 and 7 of the 2024 assessment.

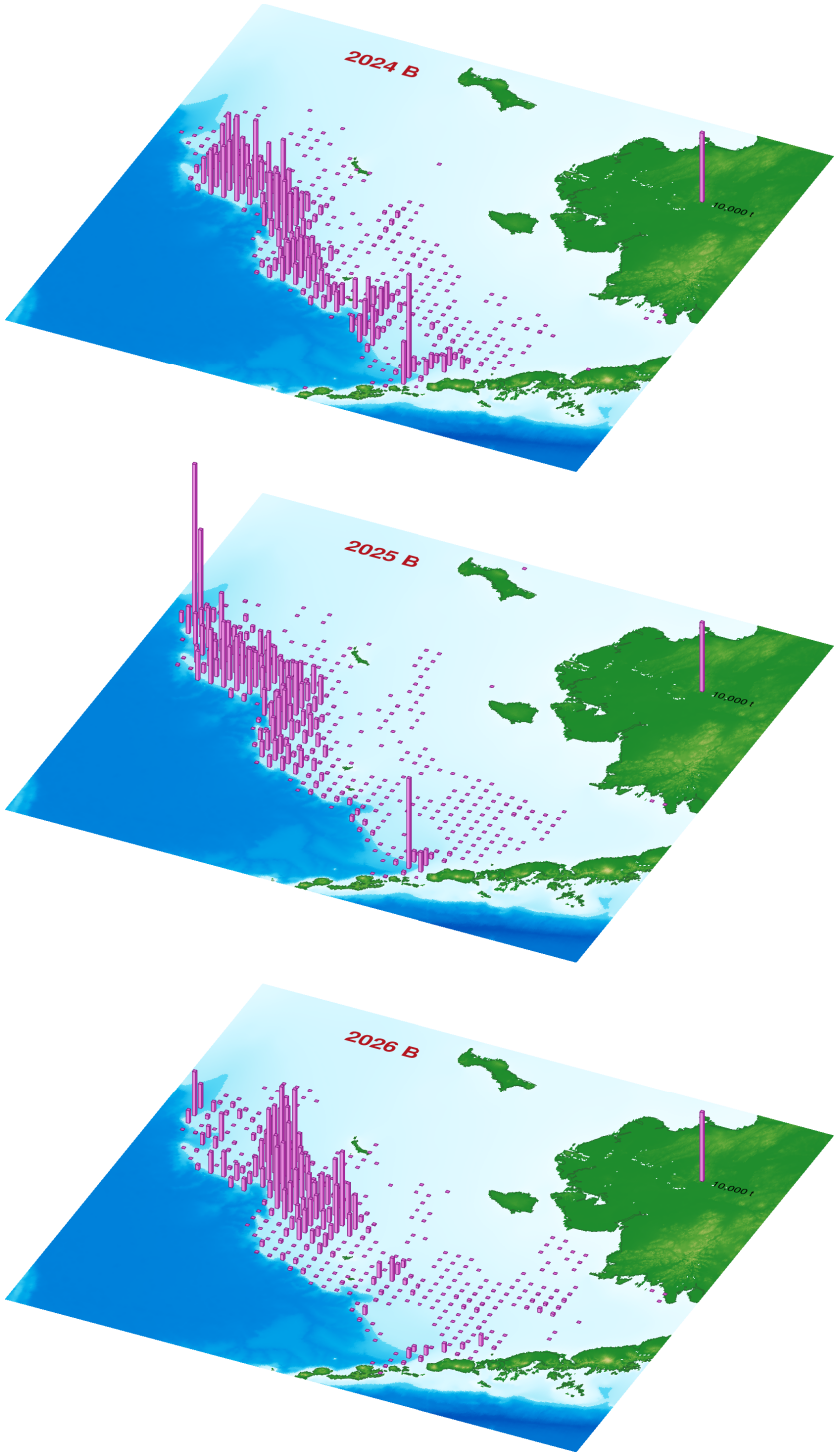



Figure 6: Spatial distribution of observed pollock catch in the B season, 2024–2026, using June 10–September 18 in each year. Vertical bar height represents recorded catch tons within 0.5-degree longitude by 0.25-degree latitude cells· the height scale is fixed across years within each season, with separate A- and B-season scales and a 10,000-t reference column in each panel. Data are from inseas26.csv and include available catcher-processor, mothership and observed catcher-vessel records in NMFS areas below 540. Electronic-monitoring catcher-vessel records are absent. These are observed-record catch totals rather than fleet-wide expanded catches. The B-season comparison is restricted to the September 18 cutoff in all three years. Rendered with GMT using the AKsf coastline and the local bottom_topo.grd bathymetry grid. Blue shades show ocean depth and green shades show land elevation. Mercator projection viewed at azimuth 155 degrees and elevation 35 degrees, with increased column heights; vertical bars represent catch rather than seafloor height. Updated in the style of Figures 4 and 7 of the 2024 assessment.

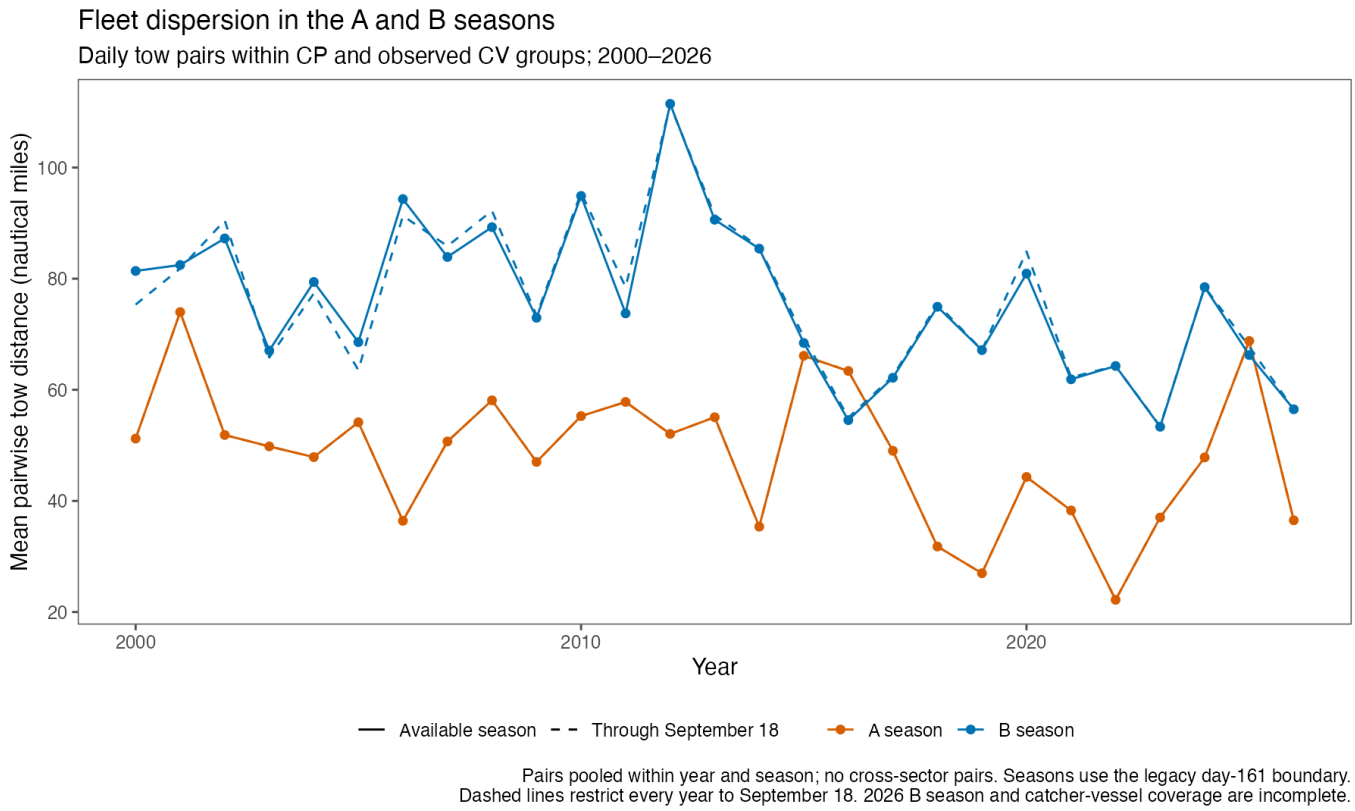



Figure 7: Fleet dispersion in the A and B seasons, 2000–2026, recalculated from inseas26.csv using the 2024 assessment's daily pairwise-distance approach. Unique tow pairs are formed within each day and vessel group (catcher-processors and observed catcher vessels); seasonal means pool distance sums and pair counts across both groups. No cross-group pairs are included. Distances are great-circle nautical miles. The legacy boundary assigns day of year 161 and earlier to A and later days to B. Solid lines use available seasonal records; dashed lines restrict every year to September 18. The 2026 B season is partial. Records require more than 80% pollock by catch weight, valid positions, EBS reporting areas 500–539, and tow duration between zero and 40 hours. Duration is calculated from timestamps with recorded minutes as fallback. Vessel-type codes replace the unavailable historical interactive vessel lists; electronic-monitoring catcher-vessel records are absent.

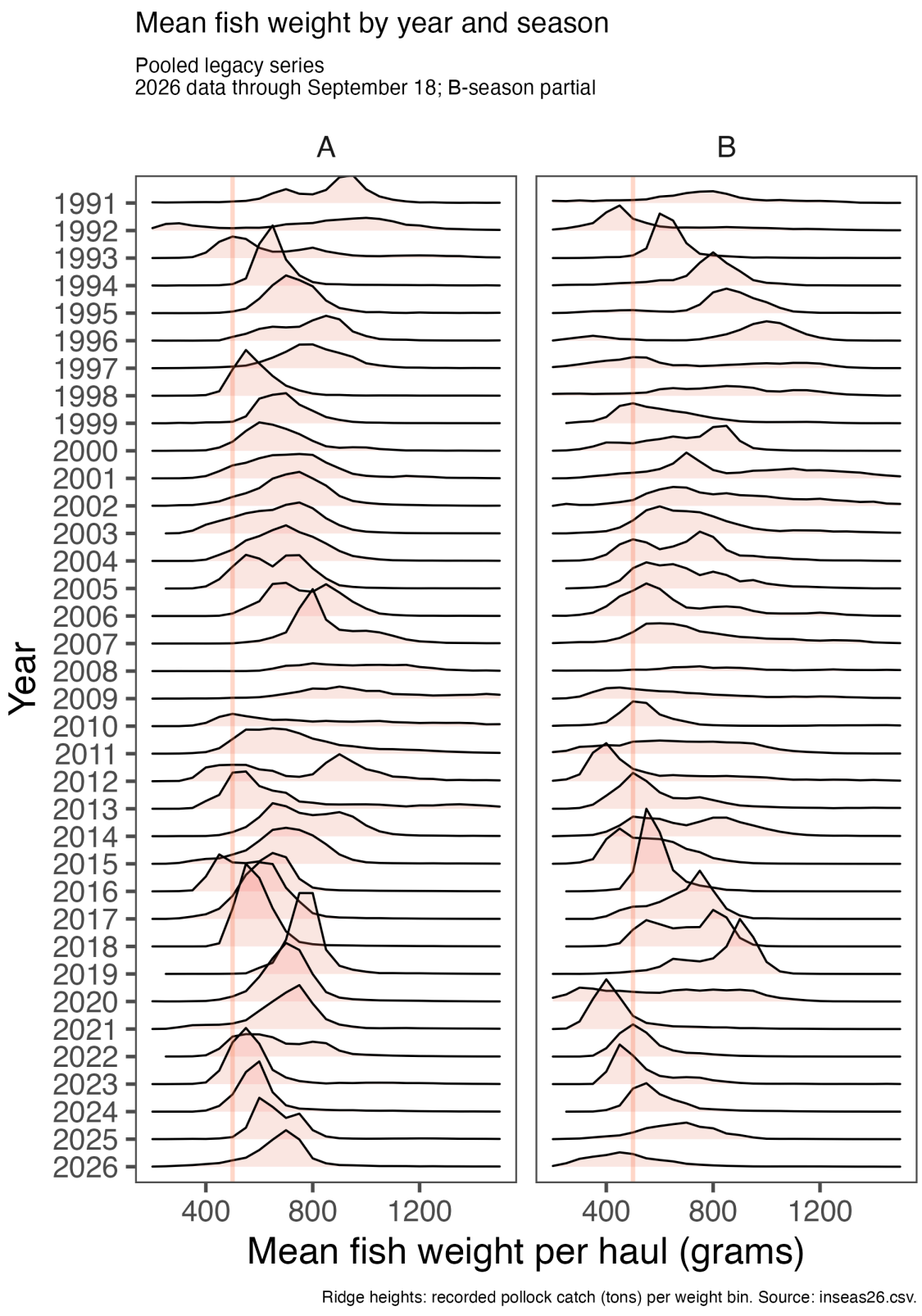



Figure 8: Catch-weighted distributions of mean pollock weight per source record for A and B seasons, 1991–2026. Catch tons are grouped in 50-g bins; the vertical line marks 500 g. A is source week 24 or earlier and B is later than week 24. The extract ends September 18, 2026, so B-season coverage is partial. Available catcher-vessel records exclude electronic-monitoring records. Ridges describe the distribution of record means, with heights scaled for display.

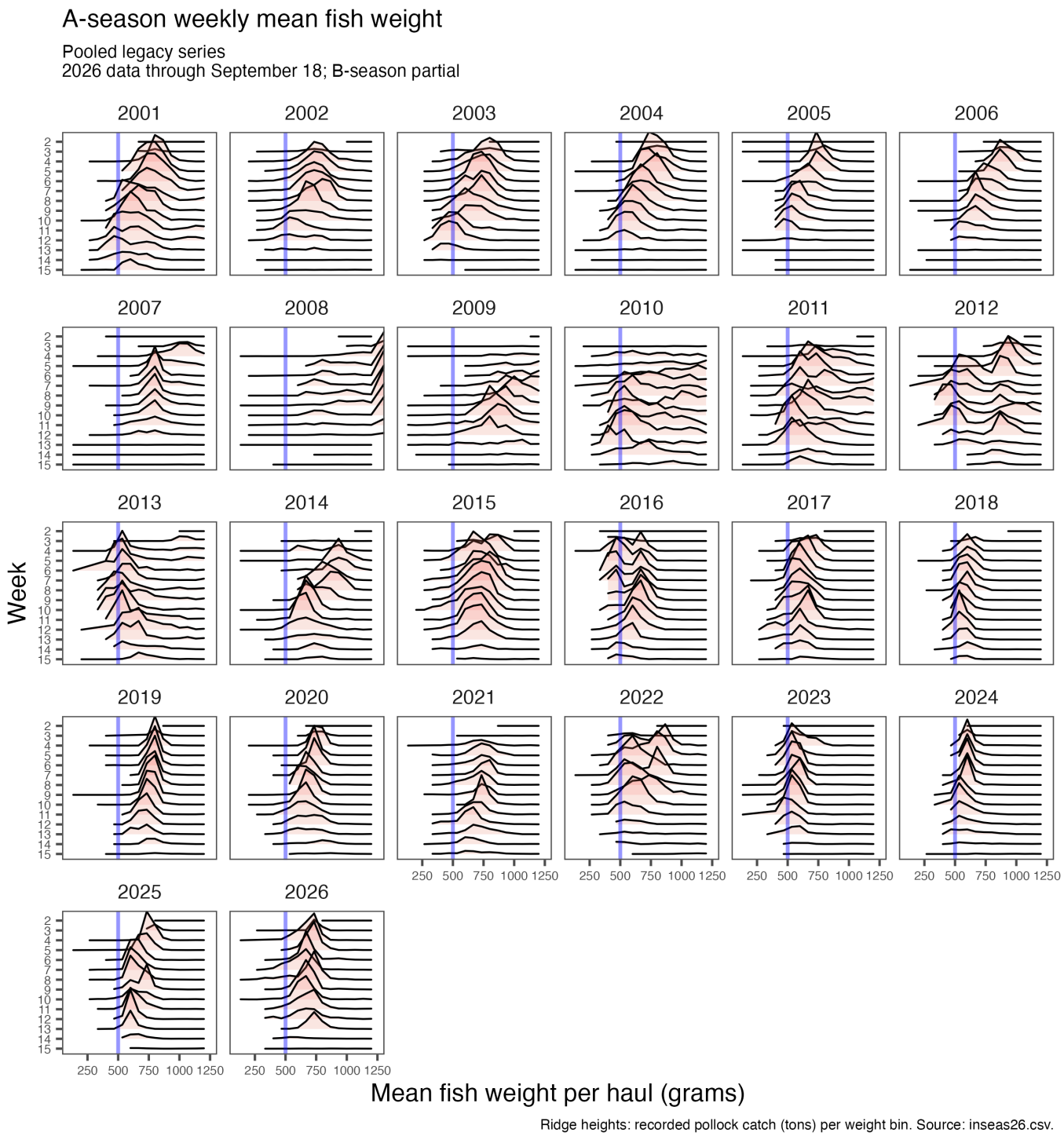



Figure 9: Weekly A-season-window distributions of mean pollock weight, 2001–2026, for source weeks 2–15. Catch tons are grouped in approximately 66.7-g bins; the vertical line marks 500 g. Source weeks are seven-day periods beginning January 1. The window differs from the regulatory A season. Available catcher-vessel coverage is incomplete.

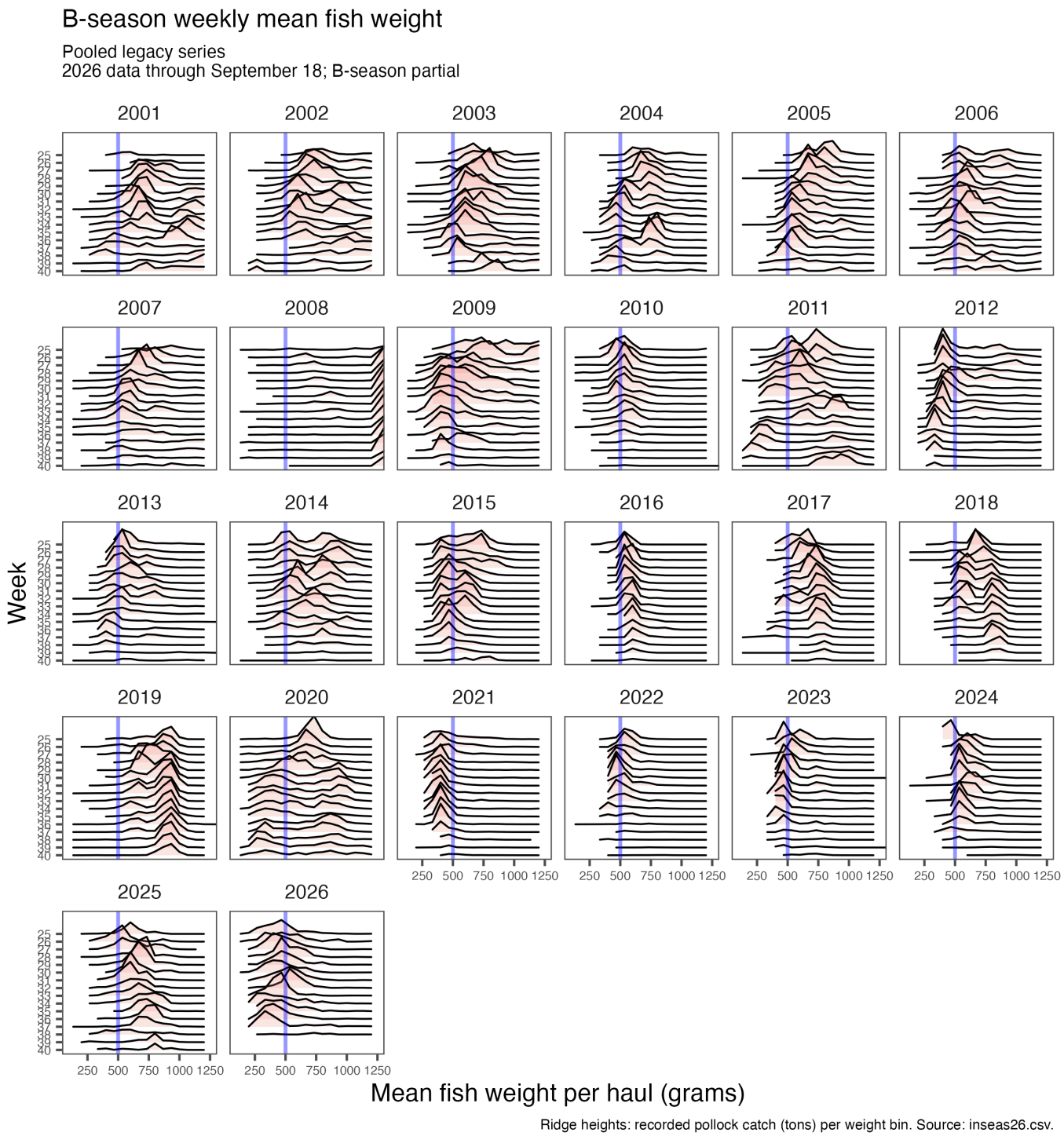



Figure 10: Weekly B-season-window distributions of mean pollock weight, 2001–2026, for source weeks 25–40. Catch tons are grouped in approximately 66.7-g bins; the vertical line marks 500 g. The 2026 extract ends September 18 and therefore covers only part of this window. Available catcher-vessel coverage is incomplete.

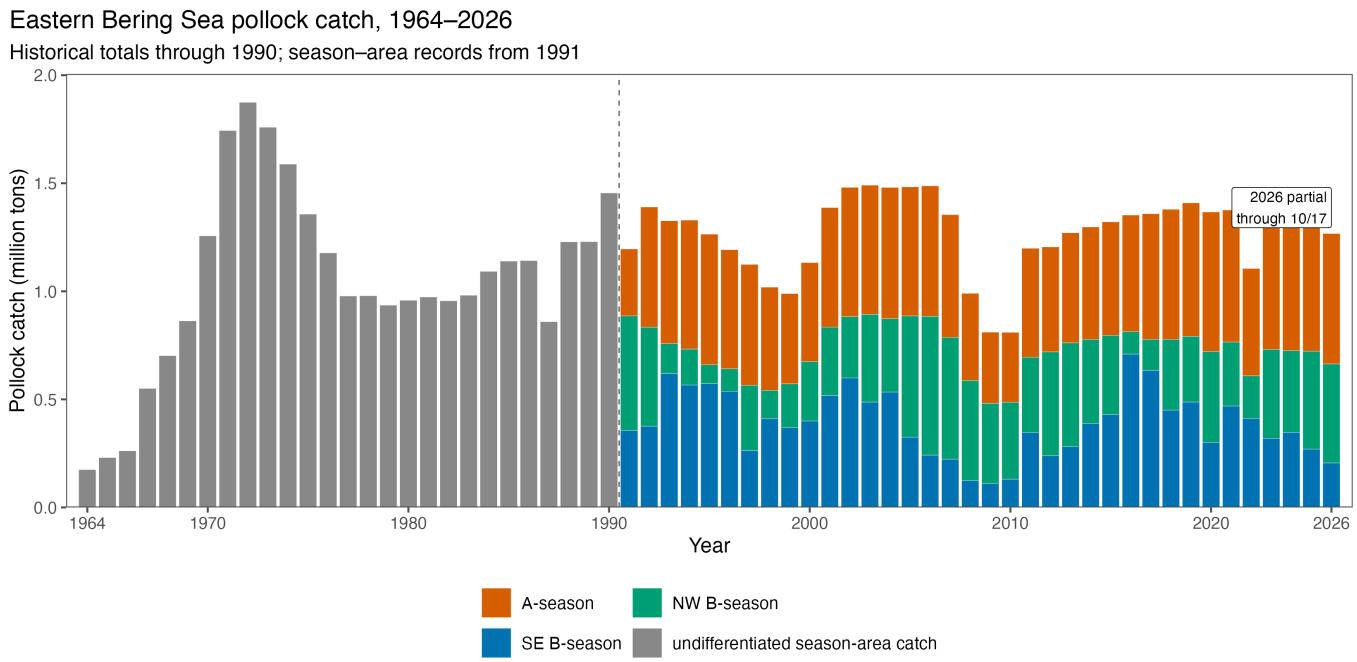



Figure 11: Eastern Bering Sea pollock catch, 1964–2026, in million tons. Catches for 1964–1990 are undifferentiated season-area catch; from 1991 the A-season combines northwest and southeast records, with separate NW and SE B-season catches. The source recipe classifies reported week-ending dates before June 1 as A and all remaining dates as B (June–December), including small November–December catches. NW comprises NMFS areas 520–529; SE comprises 500–517 and 519. Annual bars avoid interpolating splits at the 1990/1991 boundary. The 2026 bar includes 1.268 million tons of recorded catch through October 17 and is incomplete; it excludes the annual completion forecast.

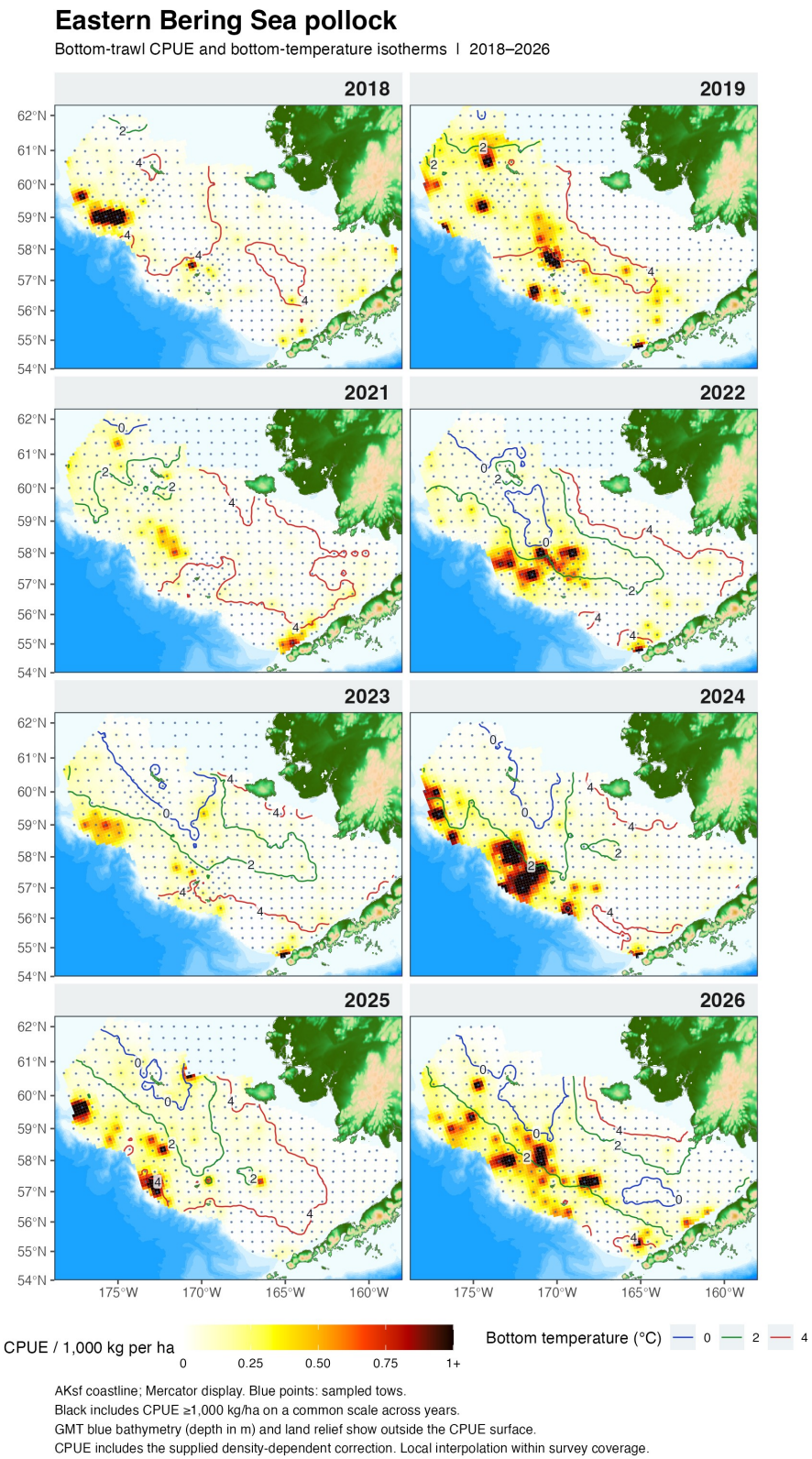



Figure 12: Bottom-trawl pollock CPUE and bottom-temperature contours in the eastern Bering Sea, 2018, 2019 and 2021–2026, reproduced from the AKsf pollock-mapping example (Ianelli 2026), snapshot dated 30 September 2026. Blue points mark sampled tows; contours are labeled 0, 2 and 4 °C. The common CPUE color scale saturates at 1,000 kg/ha. Surfaces are descriptive local interpolations within survey coverage, with no spatial prediction intervals. The supplied CPUE includes a density-dependent correction whose version and adoption status are unspecified in the example. The 2020 survey is absent.

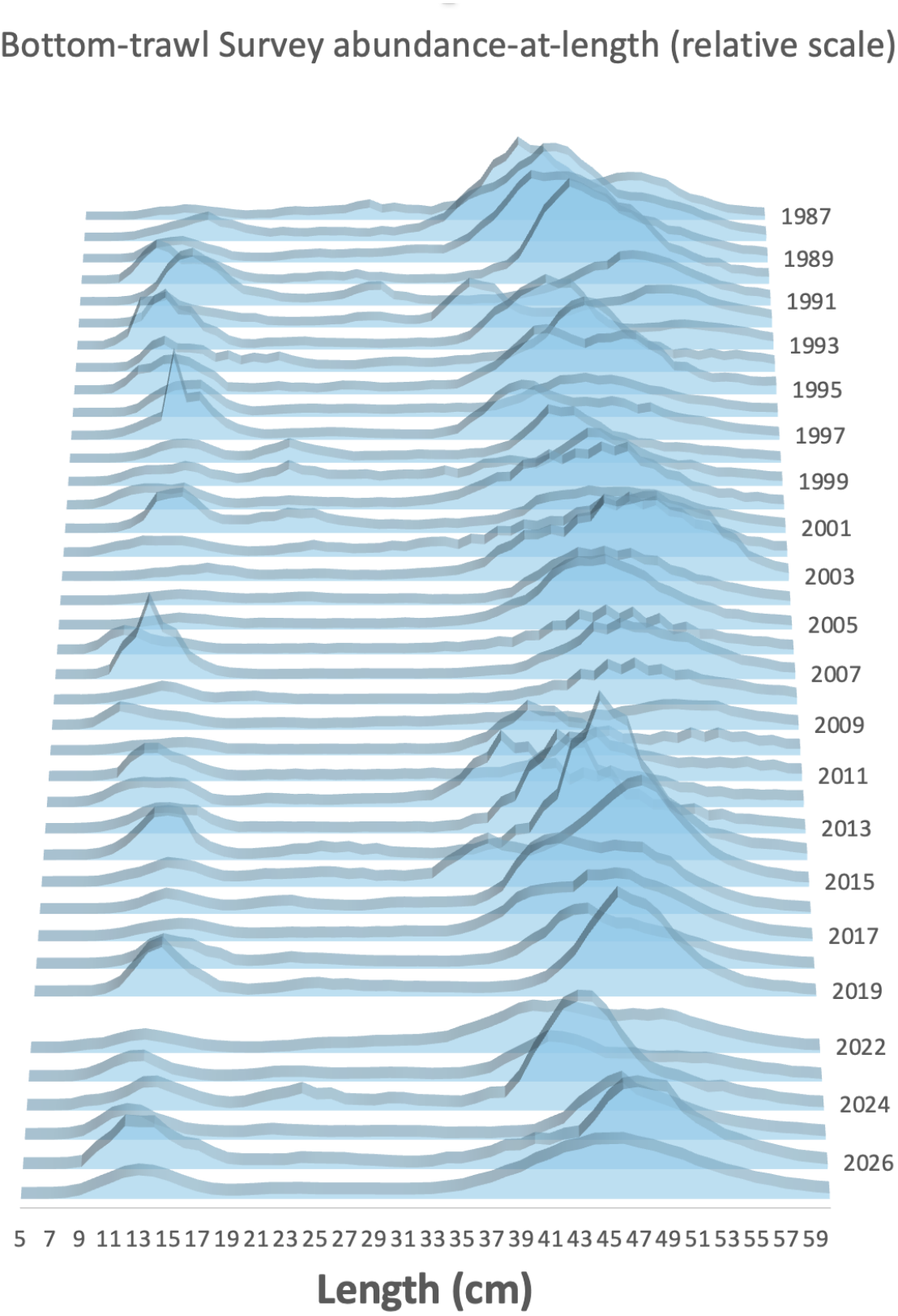



Figure 13: BTS abundance at length on a relative scale, 1987–2026. Annual length distributions are arranged from top to bottom, with length in centimetres on the horizontal axis. The gap after 2019 marks the unsampled 2020 survey year.

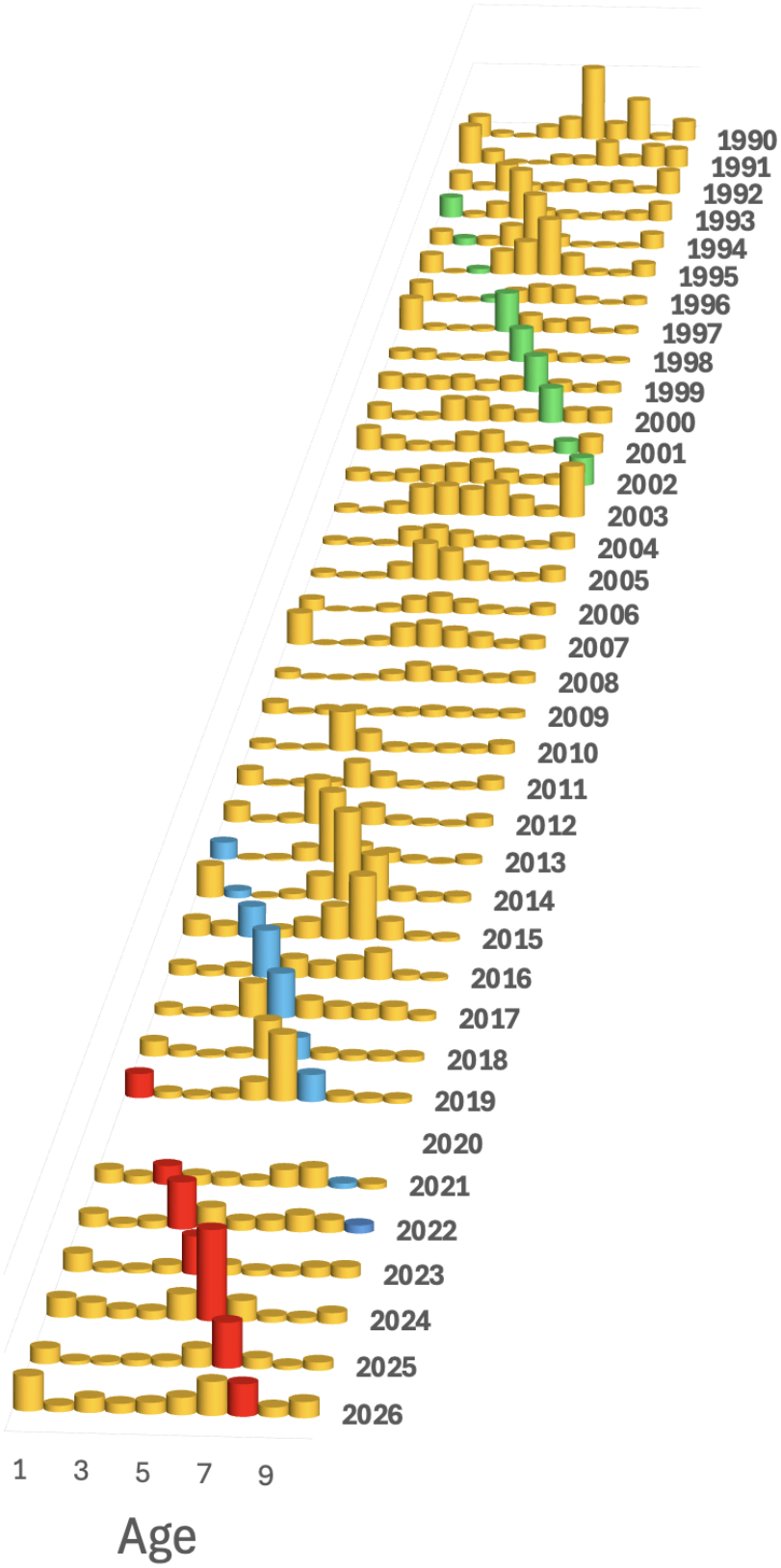



Figure 14: BTS age composition by survey year, 1990–2026. Ages increase from left to right and years run from top to bottom. Highlighted bars trace the 1992 (green), 2012 (blue) and 2018 (red) year classes as they advance through successive ages. The blank 2020 row marks the unsampled survey year.

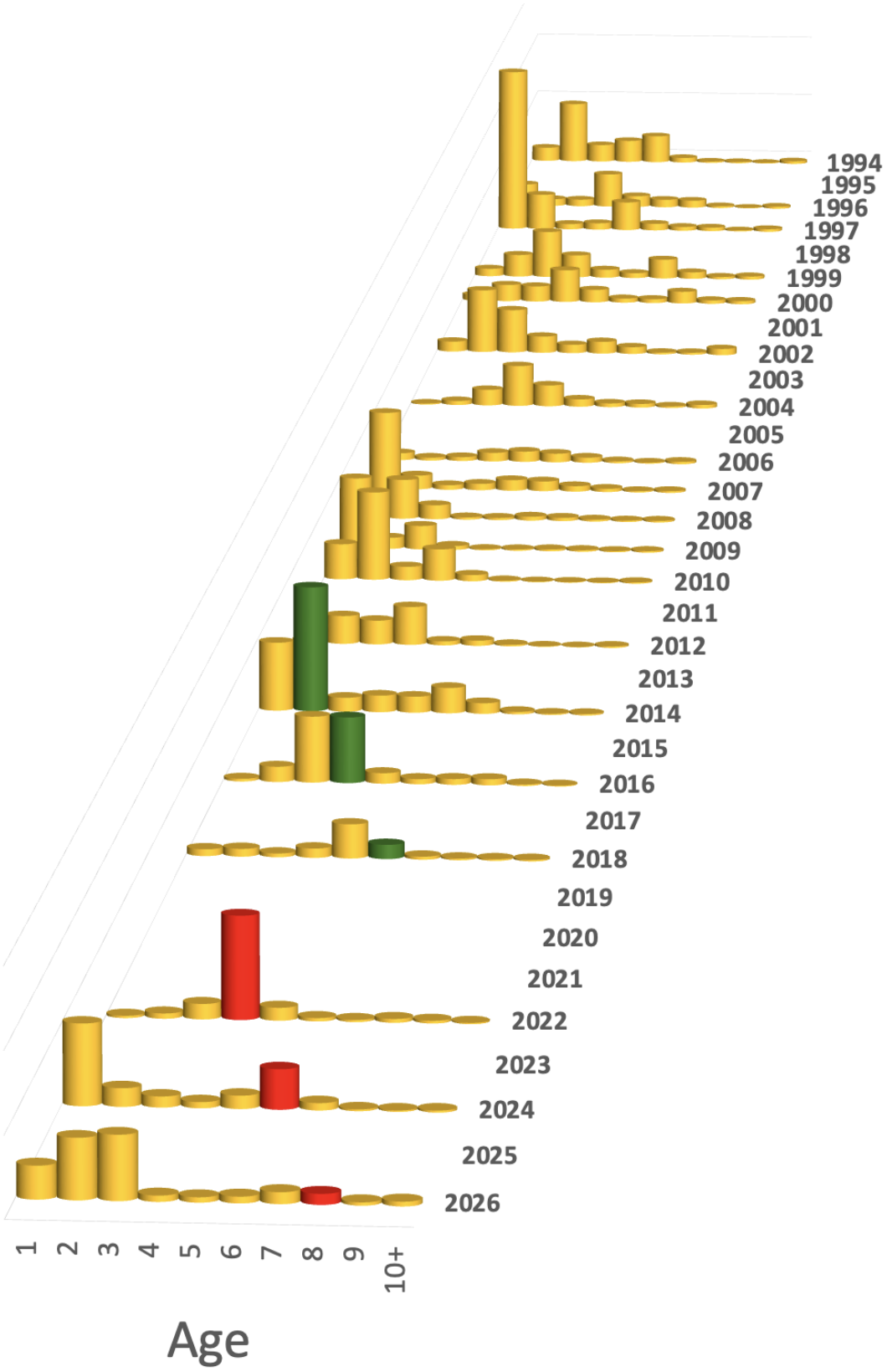



Figure 15: ATS age composition by survey year, 1994–2026. Ages increase from left to right, with ages 10 and older pooled as 10+. Blank rows indicate years lacking age-composition observations. Red bars follow the 2018 year class. The 2026 ages are provisional estimates derived by applying the BTS age–length key to ATS population length frequencies.

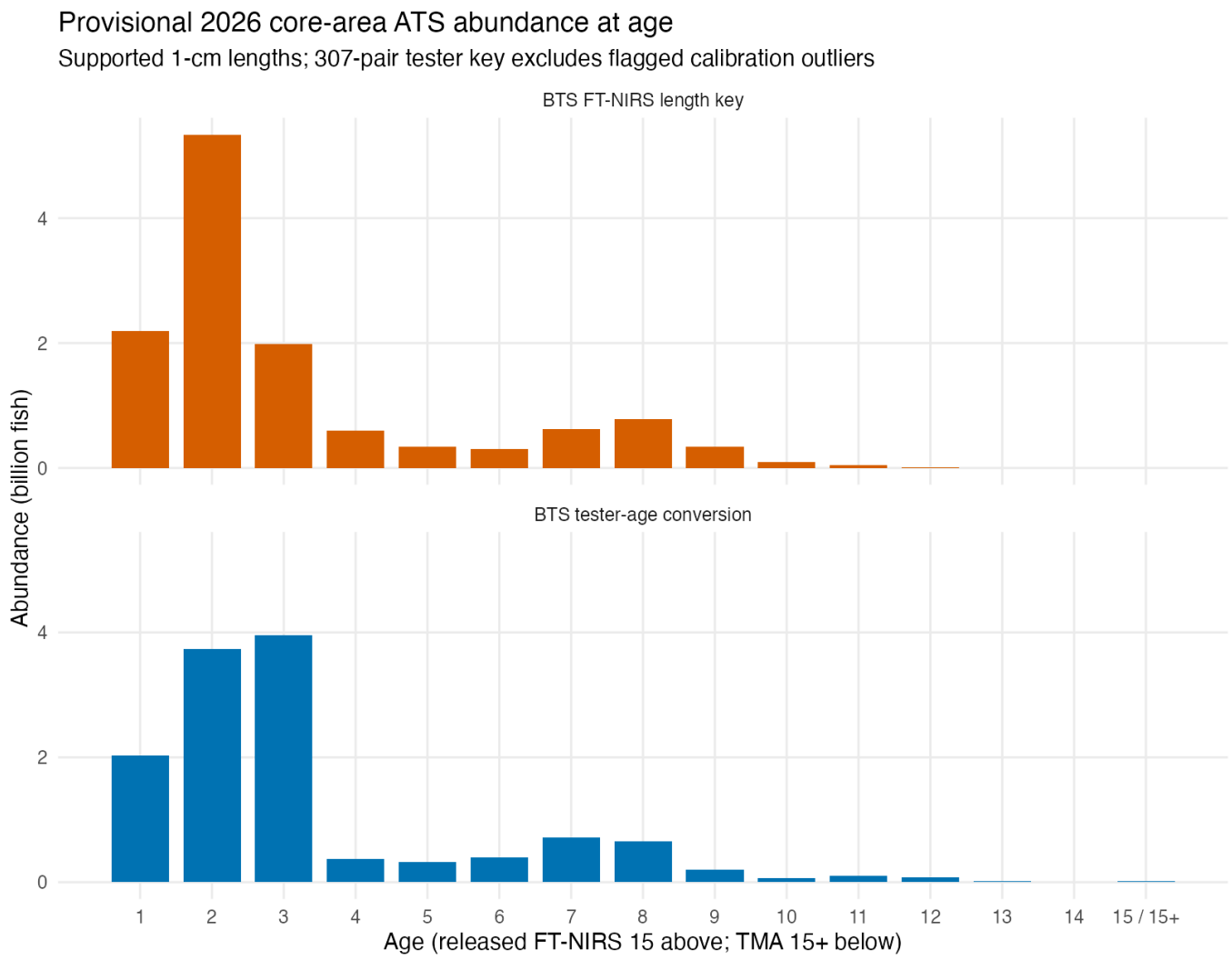



Figure 16: Provisional core-area 2026 ATS abundance at age after applying a BTS FT-NIRS length key (upper panel) and the screened 307-pair BTS conditional tester-age key (lower panel). Bars show billion fish over supported lengths only, with no uncertainty intervals. Released FT-NIRS age 15 is unpooled; the TMA terminal bar is 15+.

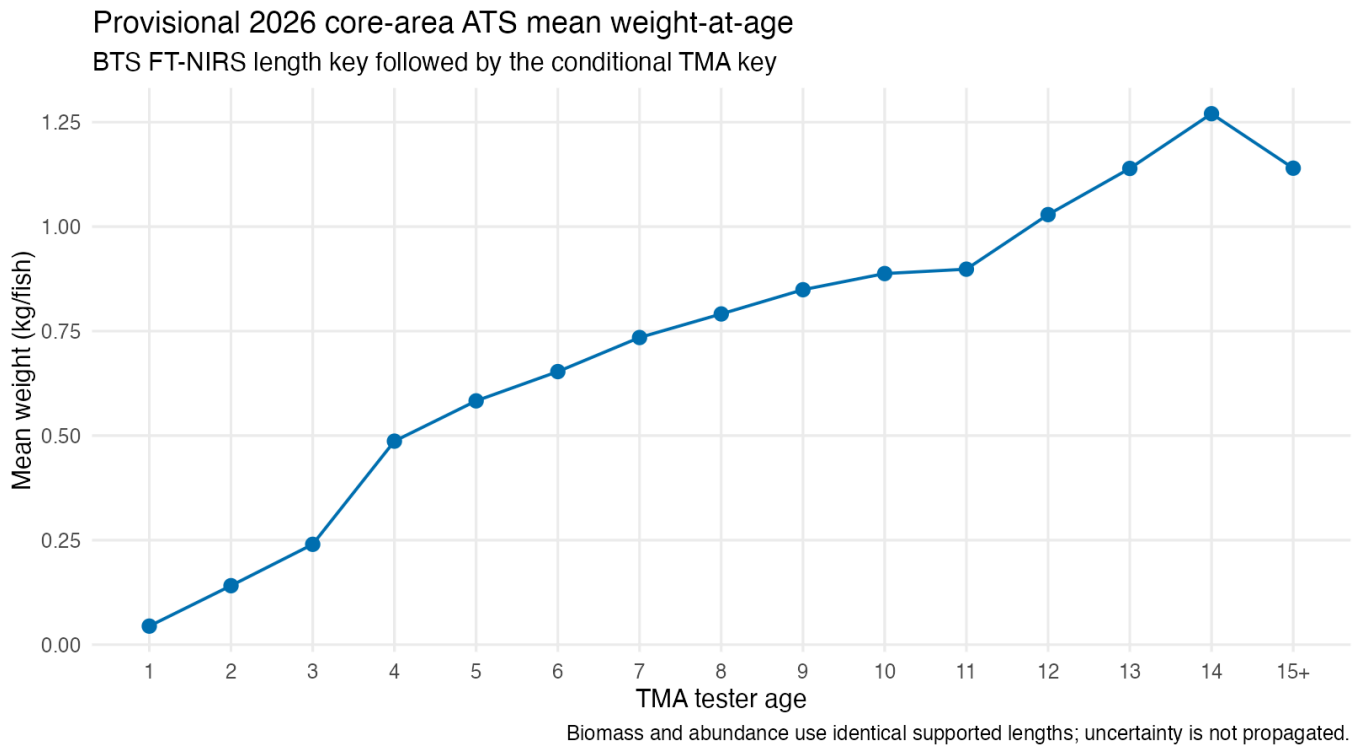



Figure 17: Provisional core-area 2026 ATS mean weight at TMA tester age, obtained by dividing biomass after the unchanged BTS length-key transfer and screened 307-pair tester-key transfer by the matching abundance estimates. The 15+ point uses pooled biomass and numbers for exact ages 15–18. These point estimates cover supported lengths only.

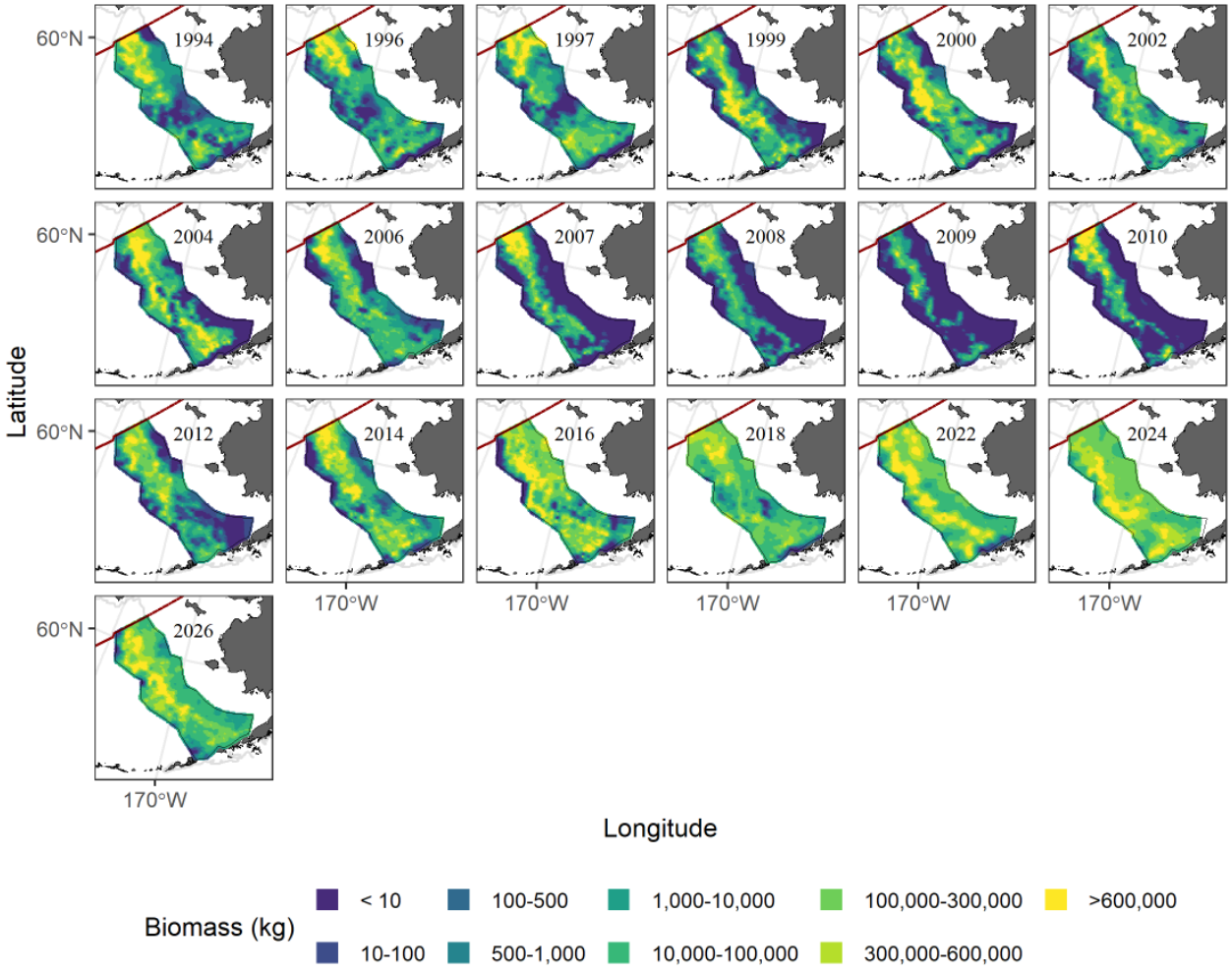



Figure 18: Historical pollock biomass between near surface and 3.0 m above the bottom in the summer eastern Bering Sea acoustic- trawl surveys of the U.S. EEZ, 1994–2026. Biomass between 3.0 m and the seafloor is excluded. The 2026 panel is preliminary; the legend gives biomass in kg.

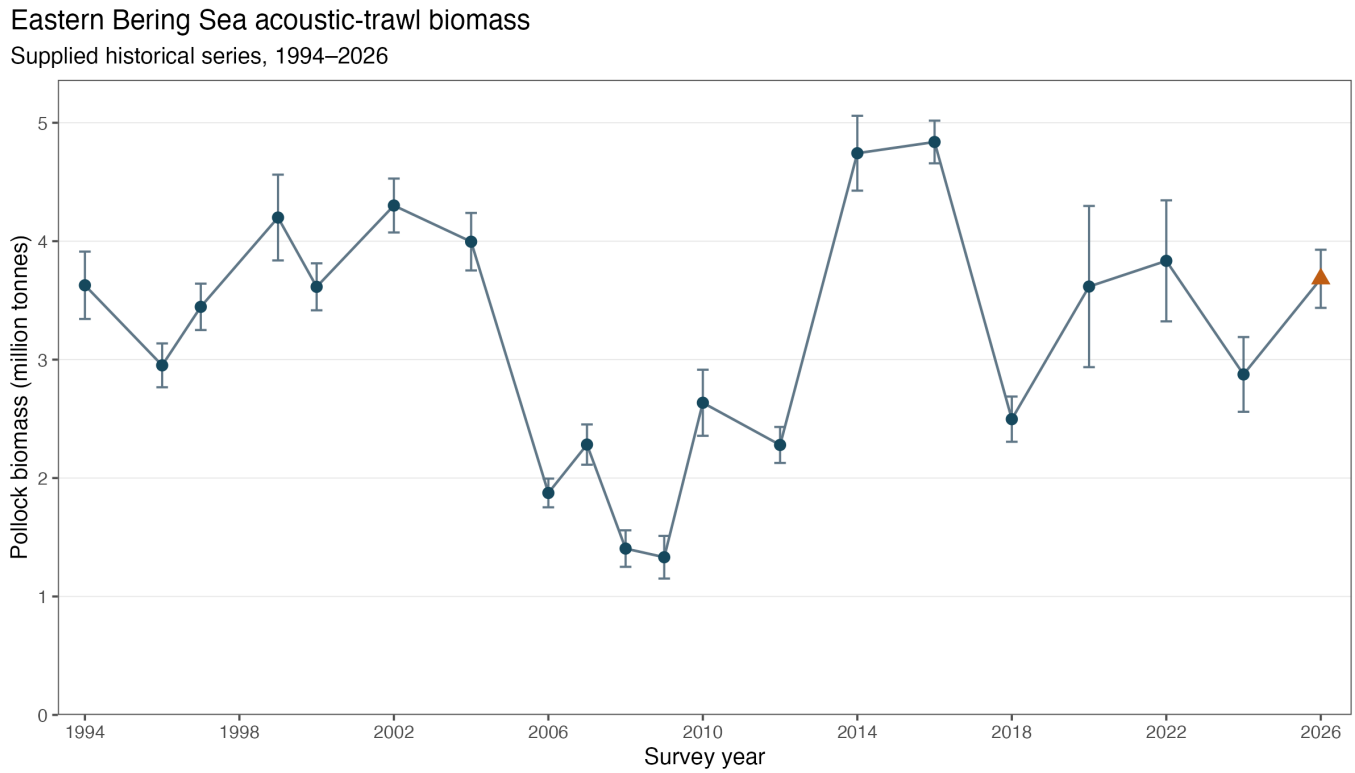



Figure 19: ATS core-shelf biomass from the historical series. Bars show approximate normal 95% intervals using survey relative errors. The line connects observed survey estimates; missing calendar years have no observation. The orange triangle marks preliminary 2026.

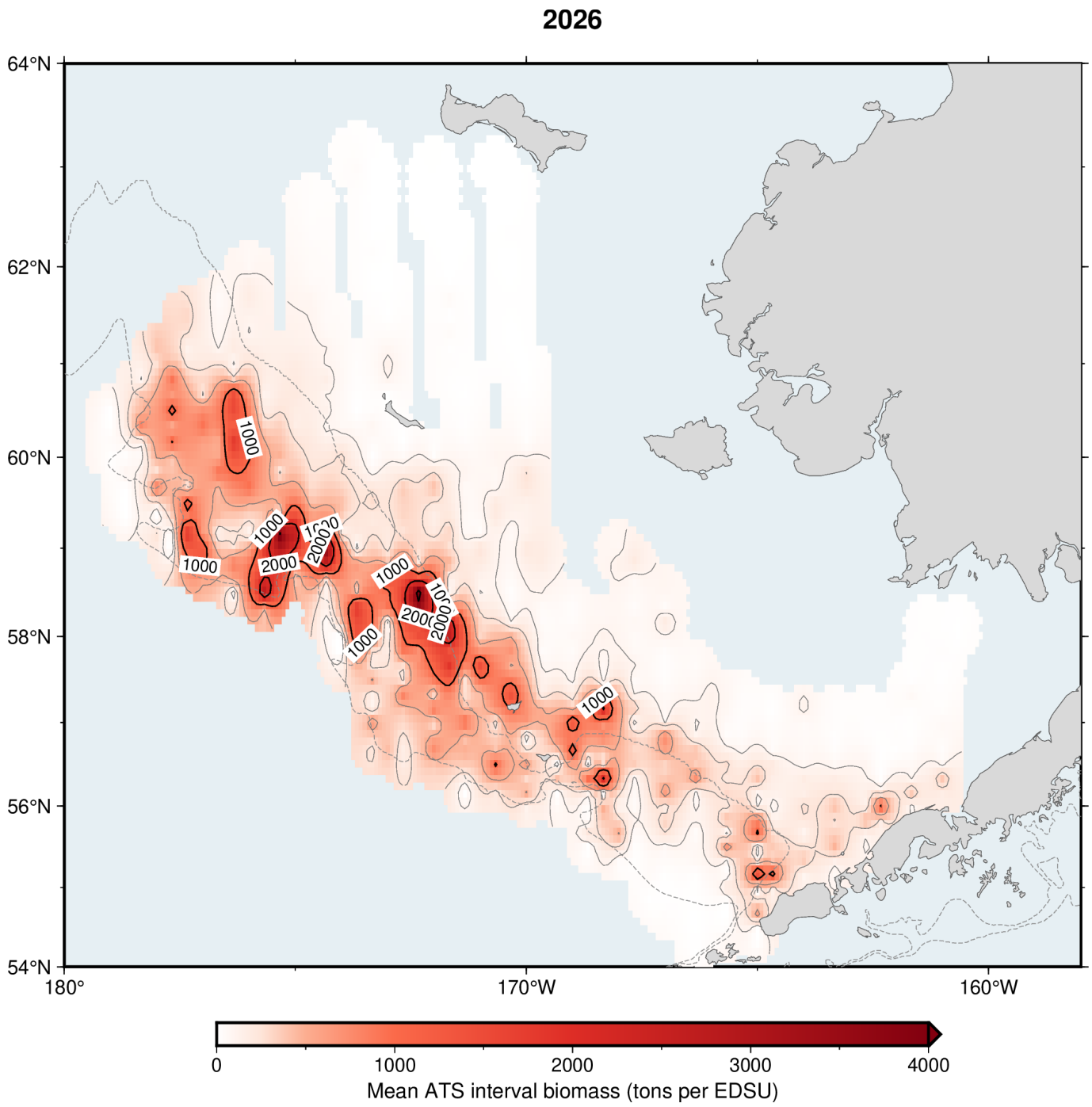



Figure 20: Preliminary 2026 ATS pollock distribution, including the Northern Extension. Colors and GMT contours show gridded mean interval biomass (tons per EDSU), using all 10,924 observations. Approximately 10-nautical-mile cell means are smoothed and displayed within 30 km of sampled cell centers. Dashed gray lines mark 100 and 200 m depths. Coastline: AKsf.

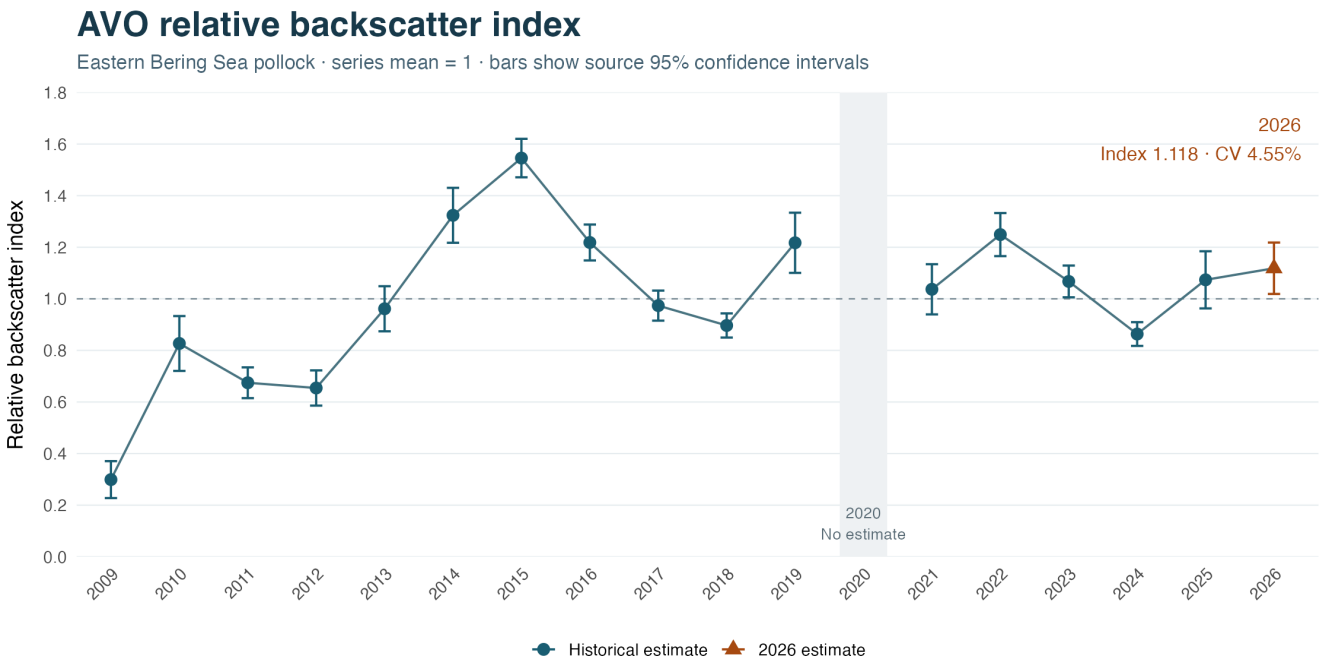



Figure 21: Annual AVO relative backscatter index for 2009–2026, with the source 95% confidence intervals rescaled by the same mean. The dashed line marks a relative index of 1. The line has a gap at 2020; the orange triangle identifies 2026.

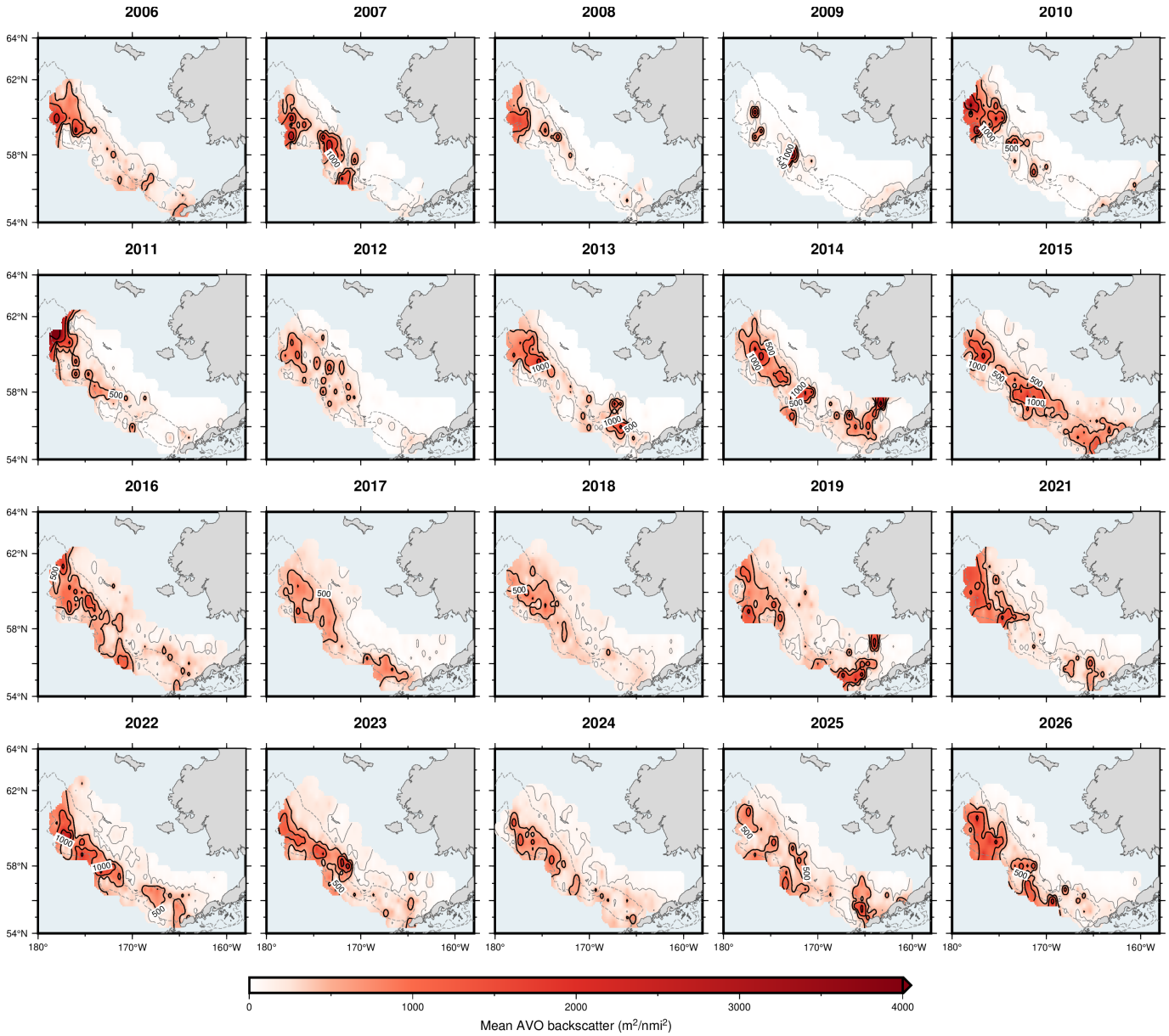



Figure 22: AVO pollock backscatter, 2006–2026 ($m^2/nmi^2$), mapped with common GMT gridding and contour levels. Twenty survey years are shown; AVO observations are unavailable for 2020. Cell means include observed zeros and use an approximately 20-nautical-mile grid. Surfaces are displayed within 45 km of sampled cell centers. Dashed gray lines mark 100 and 200 m depths. Coastline: AKsf. The 2026 observations are preliminary.

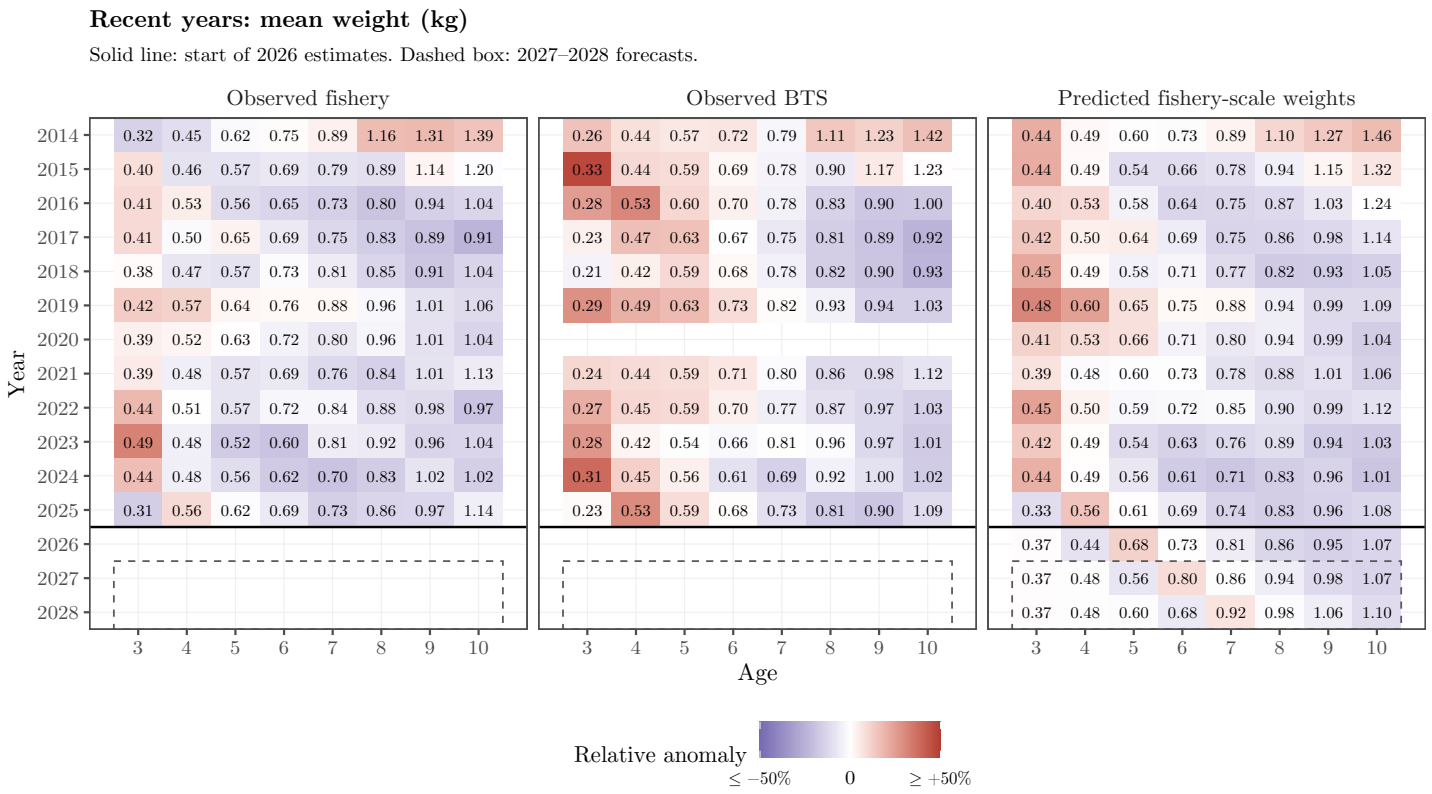



**Figure 23:** Observed and predicted weight-at-age, 2014–2028, after omitting 2026 BTS weight observations from the standalone random-effects fit. Cell labels give mean weight in kg per fish for ages 3–10. Fishery and BTS observations end in 2025; the right-hand panel includes 2026 estimates without weight observations and 2027–2028 forecasts. The solid line marks 2026; dashed boxes mark forecast years. Colours show proportional deviations from each age's mean through 2025 within its own panel, saturating at ±50%. Blank cells indicate absent observations, including the 2020 BTS gap. This figure describes the selected fit; it provides no direct comparison with the fit that included 2026 BTS weights.

Figure data: tables/waa-omit-bts2026-figure.csv

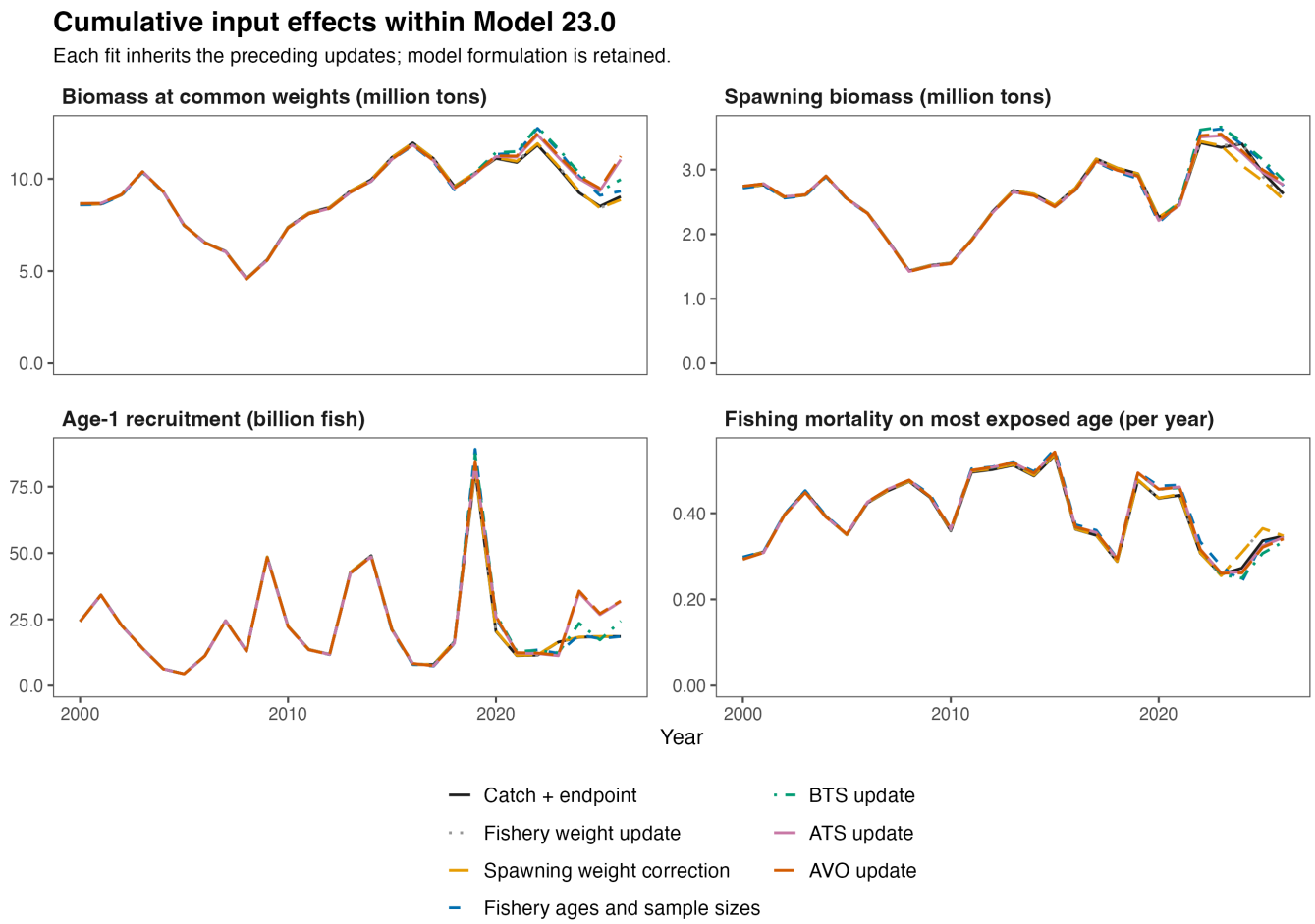



Figure 24: Cumulative input-update comparisons within Model 23.0, 2000–2026. Each fit includes the preceding updates. Biomass uses the same population-weight schedule in every comparison, and fishing mortality is the maximum realized rate across ages. ATS and AVO updates combine observations, historical revisions and the stated error calibrations. Lines show point estimates. Historical stages retain their actual source identities.

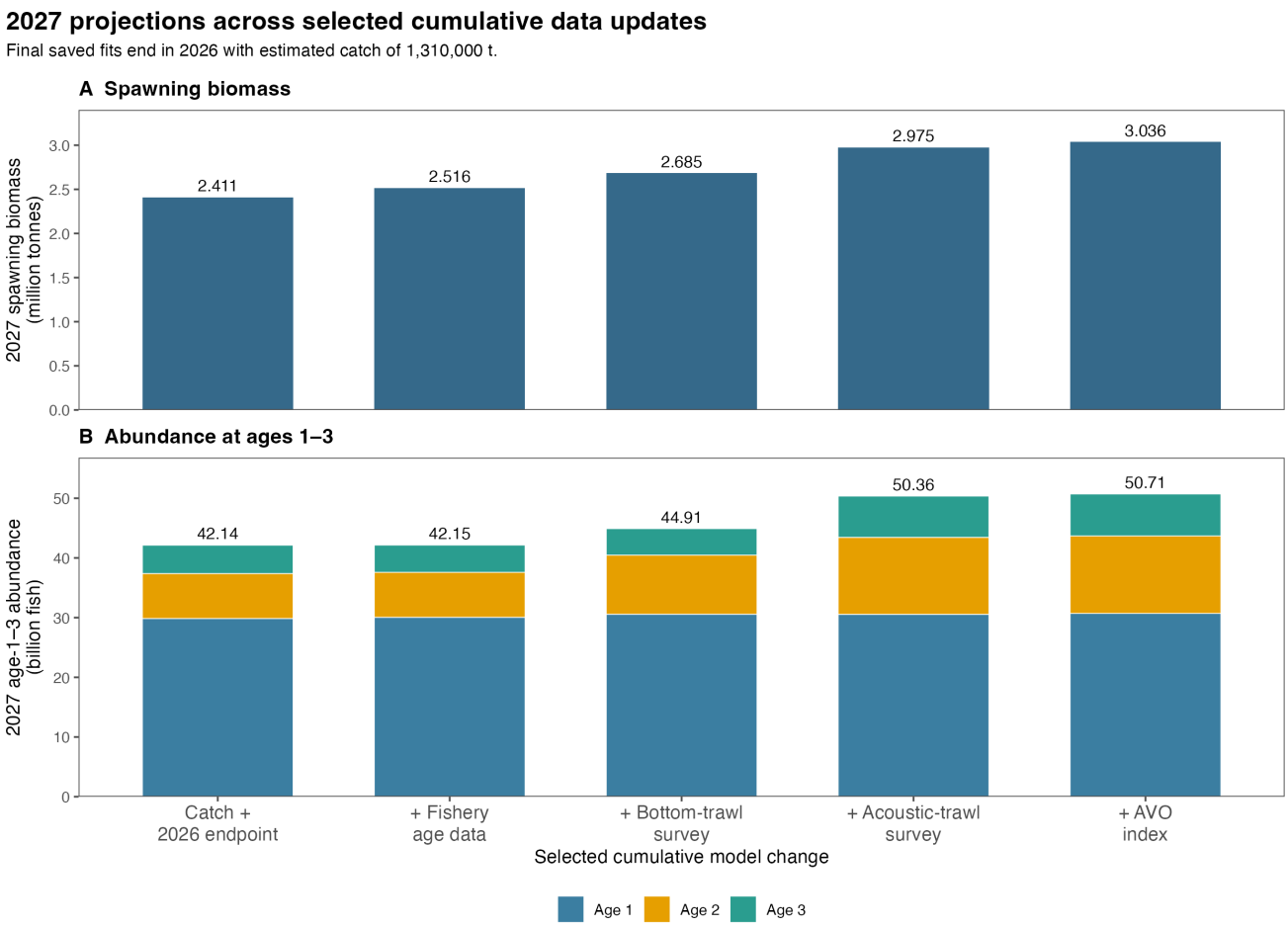



Figure 25: Projected 2027 spawning biomass and combined abundance at ages 1–3 across five selected cumulative Model 23.0 input-update stages. The final saved fits use estimated annual 2026 catch of 1,310,000 t. The displayed stages are cumulative; the omitted weight-at-age and A-season spawning-weight updates remain included in the fishery-age and later stages. Lower-panel columns stack ages 1, 2 and 3 at the start of 2027, before removals; spawning biomass includes mortality to spawning time. Saved diagnostic projection conventions are retained and documented in the technical supplement.

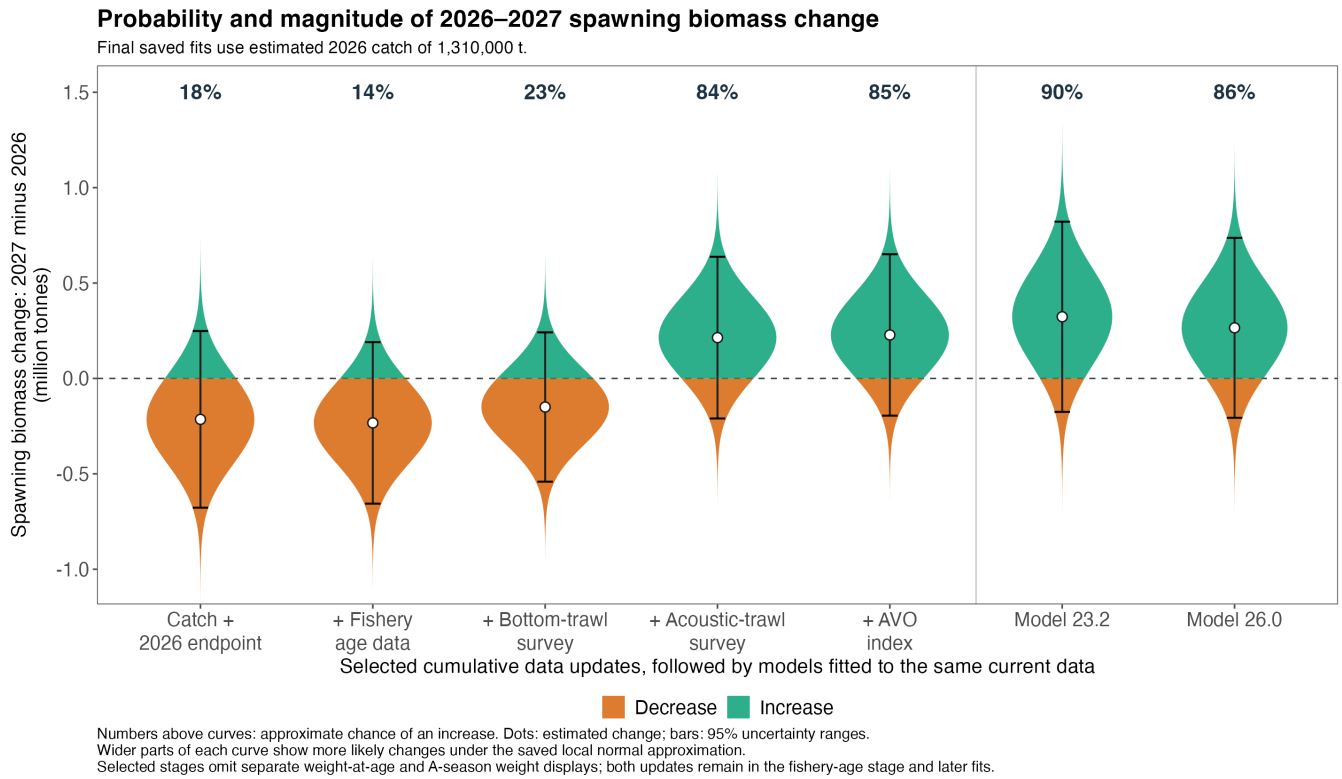



Figure 26: Approximate distributions of spawning-biomass change from 2026 to 2027 across five selected cumulative input-update stages, followed by Models 23.2 and 26.0 fitted to the same assessment data. The final saved fits use estimated annual 2026 catch of 1,310,000 t. The displayed stages are cumulative; the omitted weight-at-age and A-season spawning-weight updates remain included in the fishery-age and later stages. Green area represents an increase and orange area a decrease. Dots show estimated changes, bars show 95% local asymptotic ranges, and labels give approximate chances of an increase rounded to whole percentages. Matching fitted covariance and saved diagnostic projection conventions are retained; methods are documented in the technical supplement.

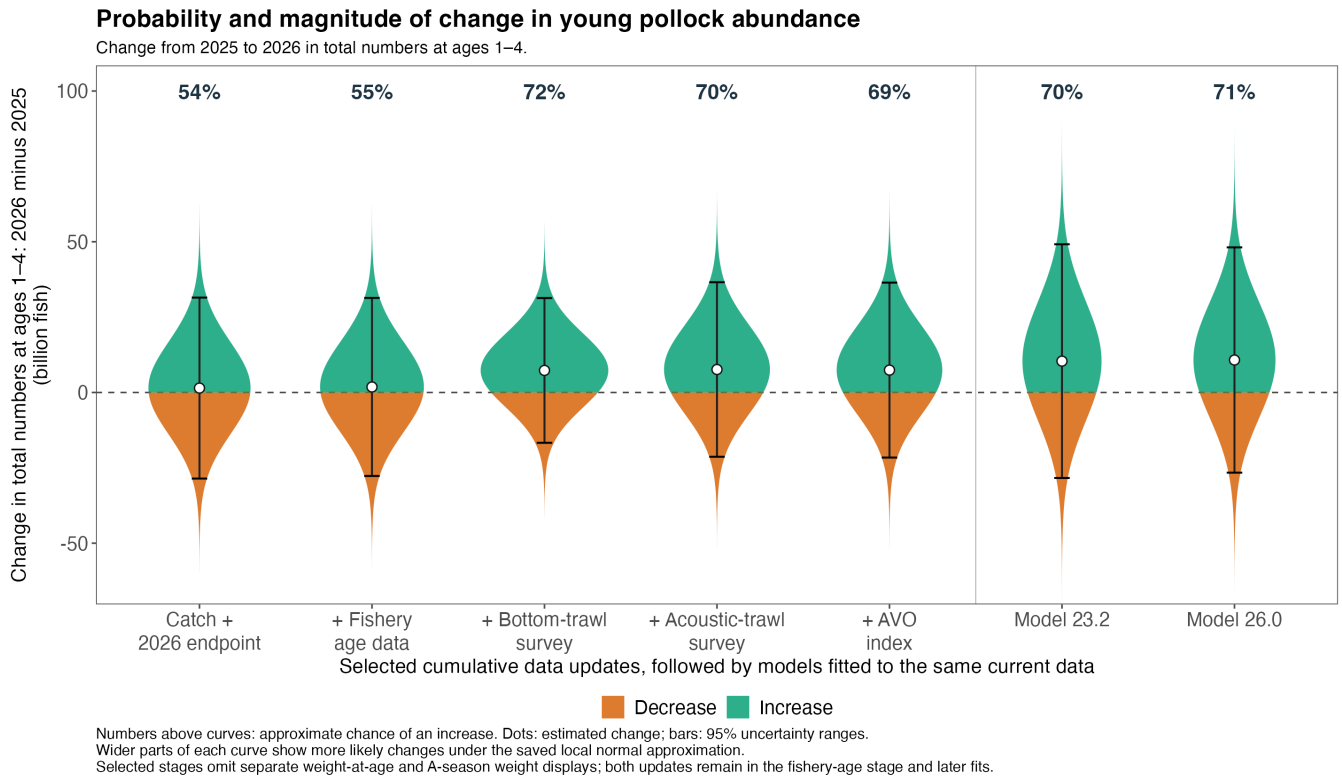



Figure 27: Approximate distributions of change from 2025 to 2026 in total numbers at ages 1–4. Five selected cumulative input- update stages are followed by Models 23.2 and 26.0 fitted to the same assessment data. The displayed stages are cumulative; the omitted weight-at-age and A-season spawning-weight updates remain included in the fishery-age and later stages. Annual totals sum ages 1–4 at the start of the year and retain both cross-age and cross-year fitted covariance. Green area represents an increase and orange area a decrease. Dots show estimates, bars show 95% uncertainty ranges and labels give approximate chances of an increase. This historical contrast retains its saved values.

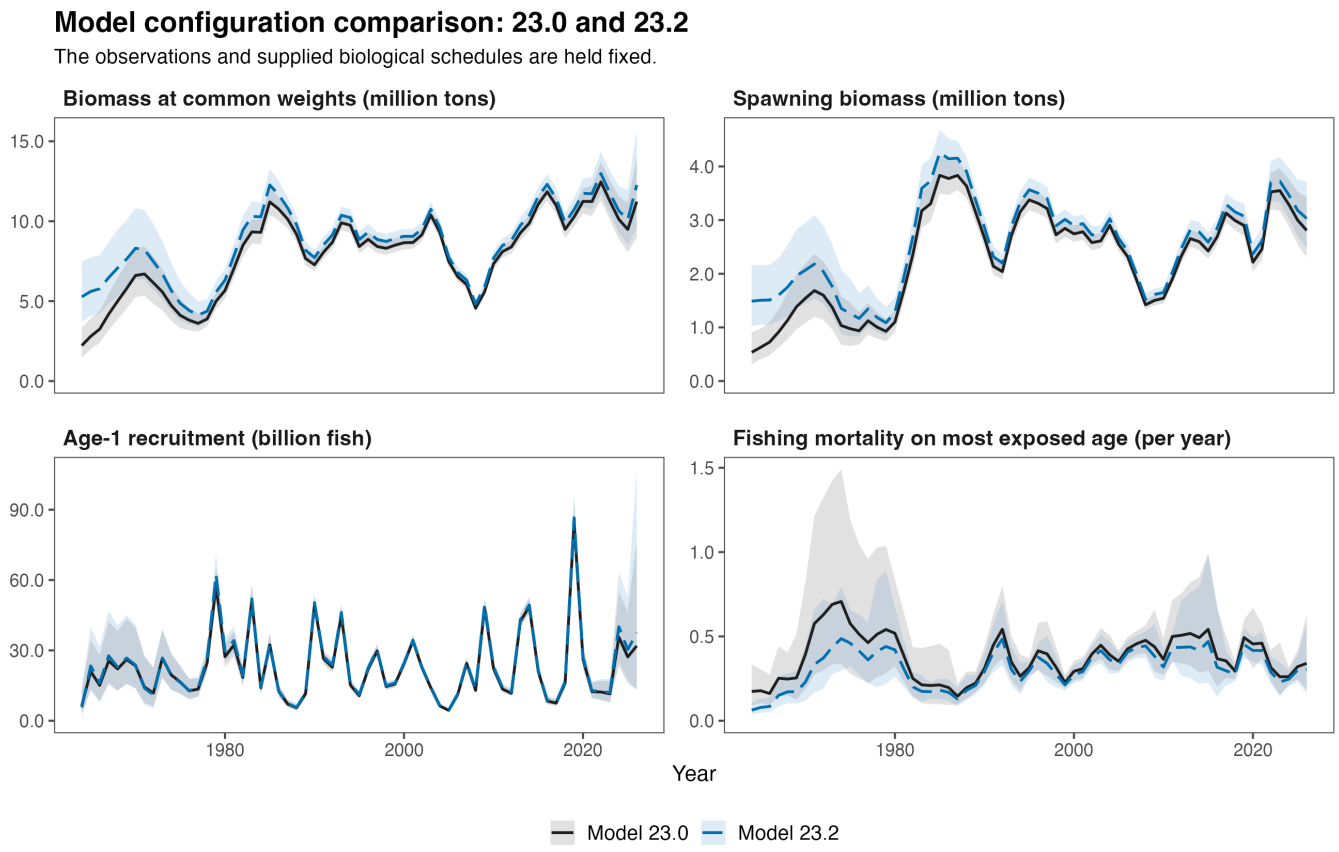



Figure 28: Models 23.0 and 23.2 fitted to the same observations and supplied biological schedules. Lines show annual estimates and shading gives approximate 95% local asymptotic uncertainty ranges. Panels show biomass at common population weights, spawning biomass, age-1 recruitment and maximum realized fishing mortality across ages.

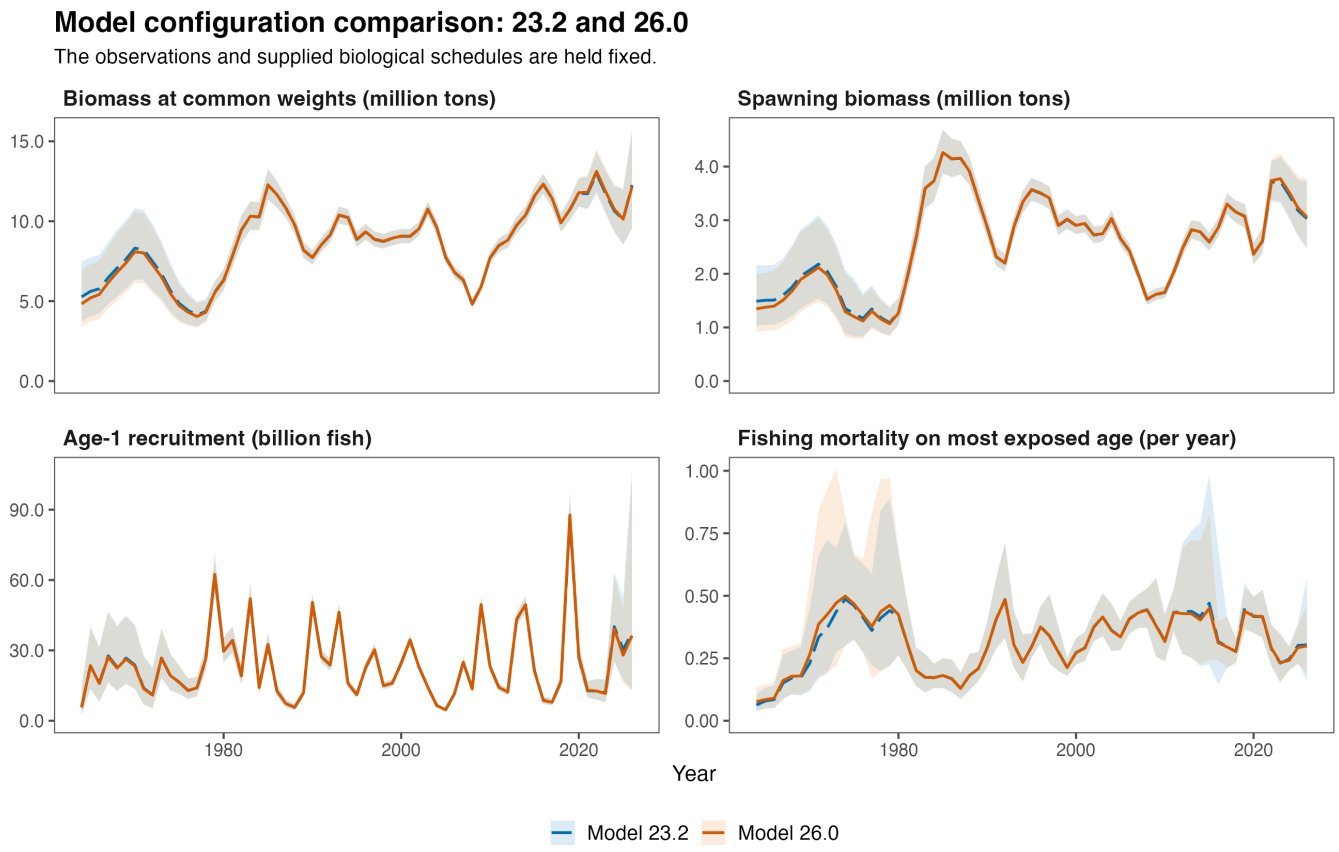



Figure 29: Models 23.2 and 26.0 fitted to the same observations and supplied biological schedules. Lines show annual estimates and shading gives approximate 95% local asymptotic uncertainty ranges. Panels show biomass at common population weights, spawning biomass, age-1 recruitment and maximum realized fishing mortality across ages. These estimation ranges are conditional on the stated model and supplied inputs.

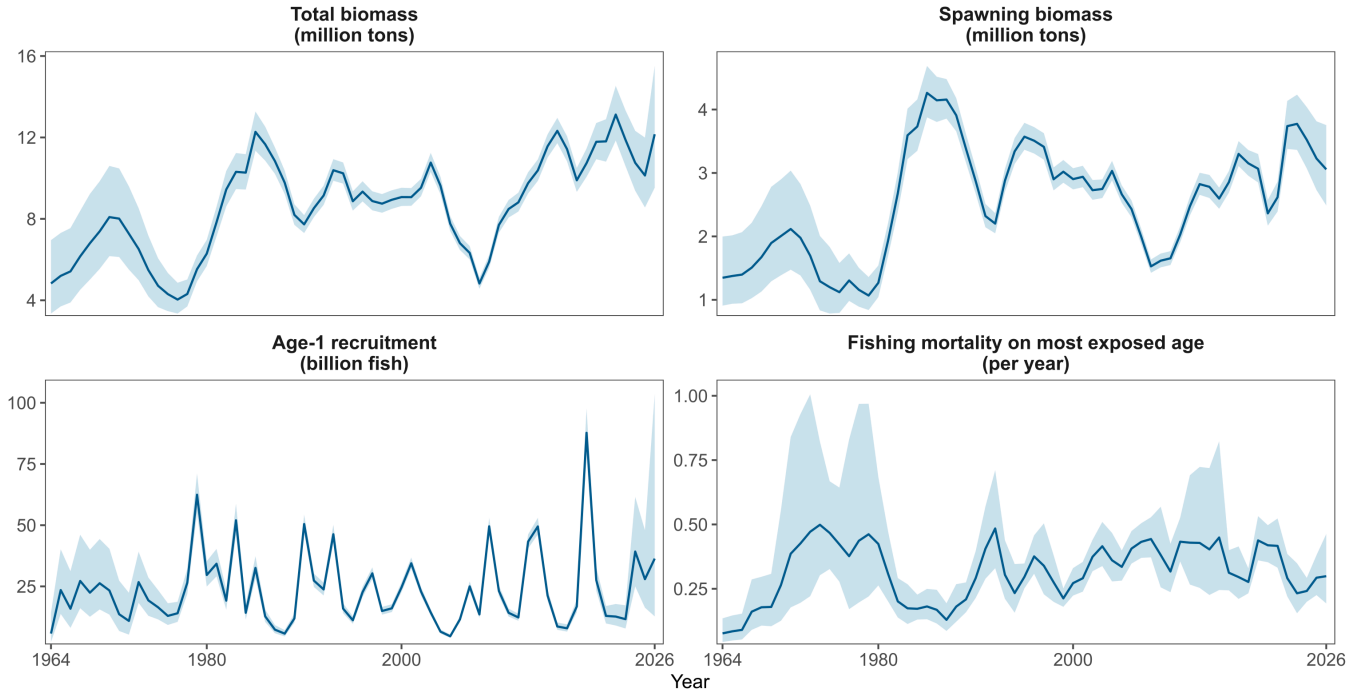
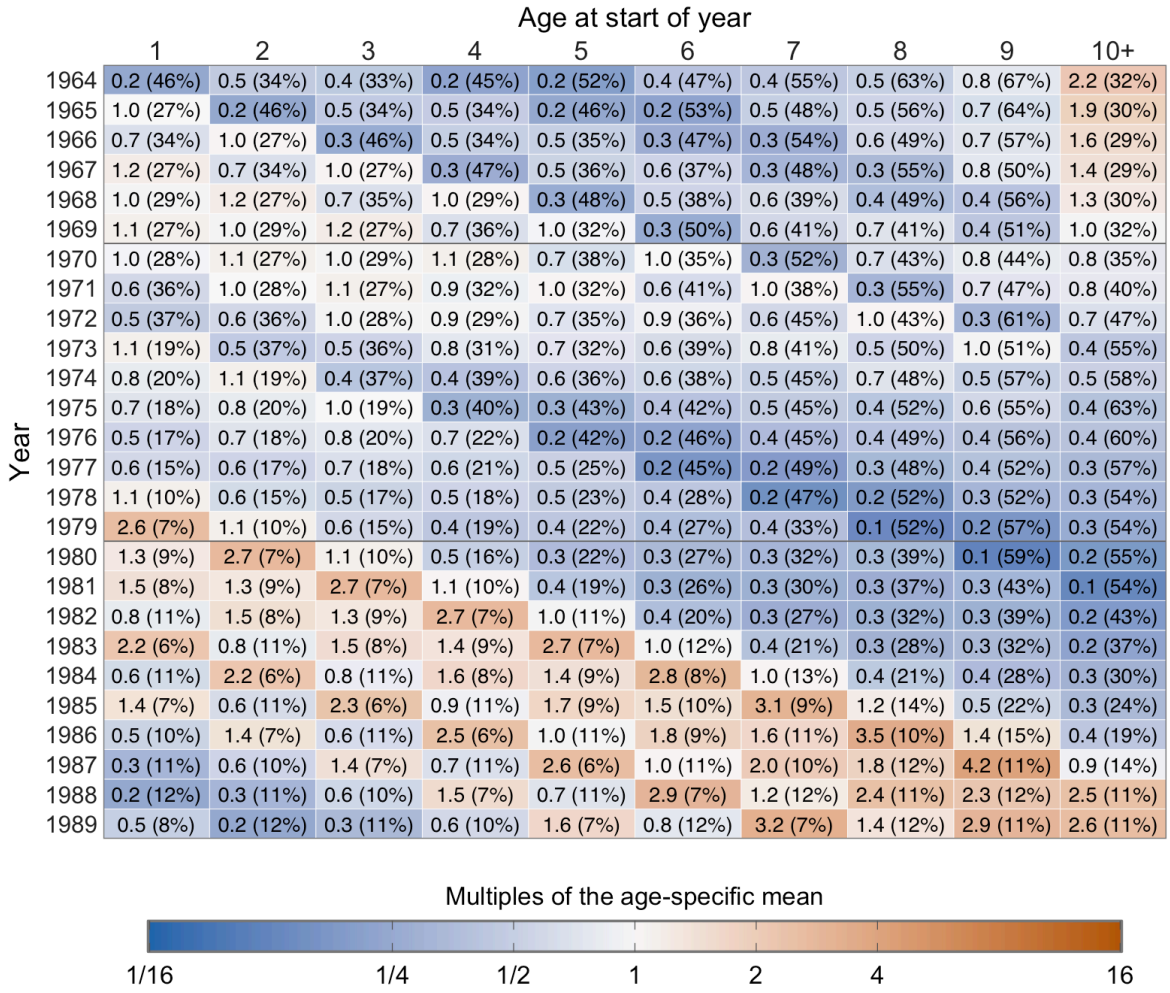



Figure 30: Model 26.0 annual total biomass, female spawning biomass, age-1 recruitment and fishing mortality through 2026, with approximate 95% asymptotic estimation ranges. Total biomass uses population weights; spawning biomass uses mature female spawning weights and survival to spawning. Recruitment is labeled by the year fish enter age 1; fishing mortality is the maximum realized rate across ages. Supplied biological schedules are fixed.

## Model 26.0 numbers at age

1964-1989 | abundance relative to each age's 1964-2026 mean
Cells: abundance / age-specific mean (CV%); CV = asymptotic SE as a percentage of the estimate




Colour denominators: each age's arithmetic mean over 1964-2026 (all 63 years).
Labels: multiples to 1 decimal; CV to whole %. 1 = age mean; 2 = twice the mean.
10+ sums ages 10-14 and 15+ once; its SE retains all cross-age covariance.

Figure 31: Model 26.0 start-of-year numbers at age, 1964–1989, followed by 1990–2026 as the same figure continued. The color legend shows multiples of the age-specific mean. Cells show abundance as multiples of each age-group mean, rounded to one decimal , with asymptotic CV (%) in parentheses. Each age-group mean uses the full 1964–2026 period. Ages 10–14 and 15+ are pooled once into 10+; its SE uses the summed saved Jacobian and full matching parameter covariance. Both pages use identical scales, denominators and cell sizing.

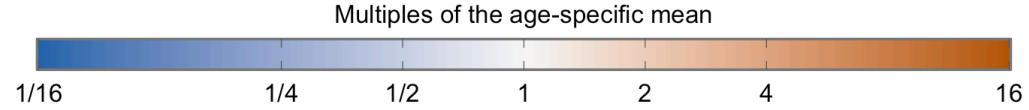
Figure 31 (continued): 1990–2026. Age groups, full-period denominators, scales and row sizing match the preceding page.

## Model 26.0 numbers at age (continued)

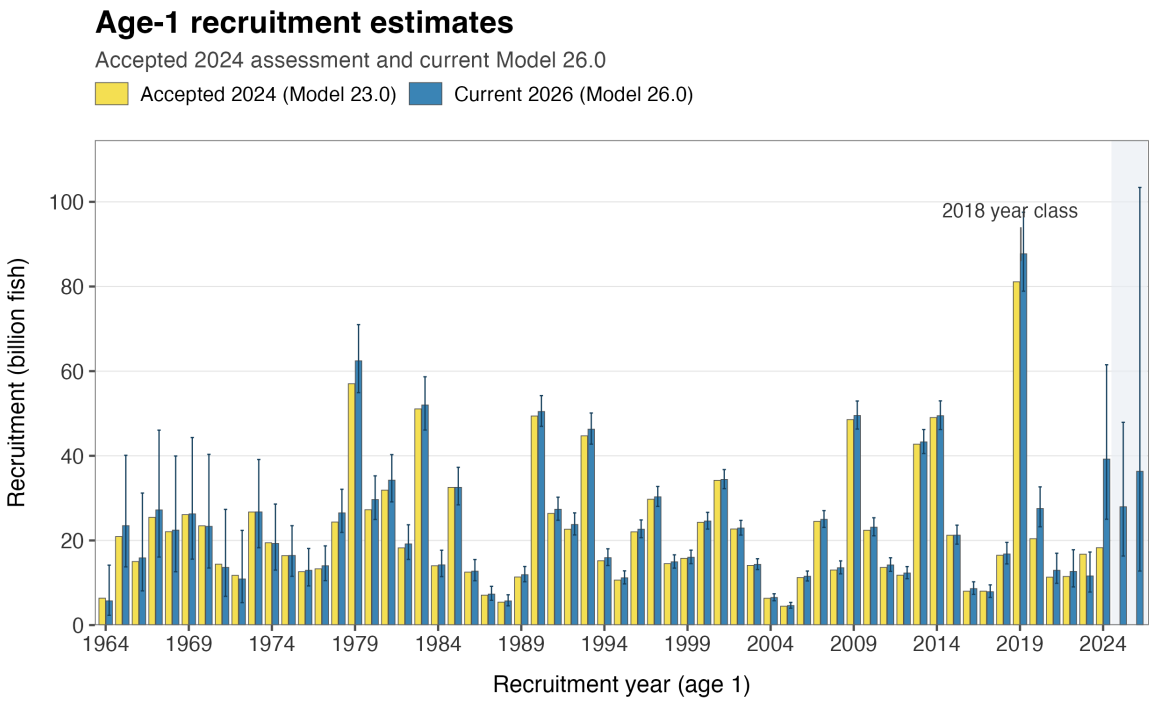
1990-2026 | abundance relative to each age's 1964-2026 mean
Cells: abundance / age-specific mean (CV%); CV = asymptotic SE as a percentage of the estimate

| Year | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10+ |
|---|---|---|---|---|---|---|---|---|---|---|
| 1990 | 2.1 (4%) | 0.5 (8%) | 0.3 (12%) | 0.3 (11%) | 0.6 (10%) | 1.7 (7%) | 0.9 (12%) | 3.5 (8%) | 1.6 (13%) | 2.8 (9%) |
| 1991 | 1.2 (5%) | 2.2 (4%) | 0.5 (8%) | 0.3 (12%) | 0.3 (11%) | 0.6 (10%) | 1.8 (7%) | 0.9 (12%) | 3.7 (8%) | 2.2 (9%) |
| 1992 | 1.0 (6%) | 1.2 (5%) | 2.2 (4%) | 0.6 (8%) | 0.3 (12%) | 0.3 (13%) | 0.6 (12%) | 1.7 (9%) | 0.9 (14%) | 2.5 (9%) |
| 1993 | 2.0 (4%) | 1.0 (6%) | 1.2 (5%) | 2.3 (4%) | 0.6 (8%) | 0.3 (13%) | 0.3 (15%) | 0.5 (15%) | 1.5 (13%) | 1.5 (11%) |
| 1994 | 0.7 (6%) | 2.0 (4%) | 1.1 (6%) | 1.3 (5%) | 2.3 (4%) | 0.6 (9%) | 0.3 (14%) | 0.3 (17%) | 0.6 (17%) | 1.4 (11%) |
| 1995 | 0.5 (7%) | 0.7 (6%) | 2.1 (4%) | 1.1 (6%) | 1.3 (6%) | 2.2 (5%) | 0.7 (9%) | 0.3 (15%) | 0.4 (18%) | 1.1 (12%) |
| 1996 | 1.0 (5%) | 0.5 (7%) | 0.7 (6%) | 2.2 (4%) | 1.3 (6%) | 1.4 (6%) | 2.1 (6%) | 0.7 (10%) | 0.3 (16%) | 0.8 (12%) |
| 1997 | 1.3 (4%) | 1.0 (5%) | 0.5 (7%) | 0.8 (6%) | 2.5 (4%) | 1.4 (6%) | 1.4 (7%) | 1.9 (8%) | 0.7 (12%) | 0.6 (12%) |
| 1998 | 0.6 (5%) | 1.3 (4%) | 1.0 (5%) | 0.5 (7%) | 0.8 (6%) | 2.8 (4%) | 1.5 (7%) | 1.5 (9%) | 2.0 (10%) | 0.6 (12%) |
| 1999 | 0.7 (5%) | 0.6 (5%) | 1.3 (4%) | 1.1 (5%) | 0.6 (7%) | 0.9 (7%) | 3.0 (5%) | 1.7 (7%) | 1.6 (10%) | 1.1 (11%) |
| 2000 | 1.0 (4%) | 0.7 (5%) | 0.7 (5%) | 1.4 (4%) | 1.2 (5%) | 0.6 (8%) | 1.1 (8%) | 3.2 (6%) | 2.0 (8%) | 1.4 (10%) |
| 2001 | 1.5 (3%) | 1.1 (4%) | 0.7 (5%) | 0.7 (5%) | 1.5 (4%) | 1.2 (6%) | 0.7 (8%) | 1.1 (9%) | 3.4 (7%) | 1.7 (8%) |
| 2002 | 1.0 (4%) | 1.5 (3%) | 1.1 (4%) | 0.8 (5%) | 0.8 (6%) | 1.5 (5%) | 1.2 (7%) | 0.7 (9%) | 1.2 (10%) | 2.4 (7%) |
| 2003 | 0.6 (5%) | 1.0 (4%) | 1.5 (3%) | 1.2 (4%) | 0.8 (5%) | 0.7 (6%) | 1.3 (6%) | 1.1 (8%) | 0.7 (11%) | 1.9 (7%) |
| 2004 | 0.3 (6%) | 0.6 (5%) | 1.0 (4%) | 1.6 (3%) | 1.2 (4%) | 0.8 (6%) | 0.7 (8%) | 1.1 (8%) | 1.0 (10%) | 1.4 (8%) |
| 2005 | 0.2 (8%) | 0.3 (6%) | 0.6 (5%) | 1.1 (4%) | 1.5 (4%) | 1.2 (5%) | 0.8 (7%) | 0.6 (9%) | 1.1 (9%) | 1.2 (8%) |
| 2006 | 0.5 (5%) | 0.2 (8%) | 0.3 (6%) | 0.7 (5%) | 1.1 (4%) | 1.4 (4%) | 1.1 (6%) | 0.7 (8%) | 0.7 (10%) | 1.2 (8%) |
| 2007 | 1.1 (4%) | 0.5 (5%) | 0.2 (8%) | 0.3 (7%) | 0.7 (5%) | 1.0 (5%) | 1.2 (5%) | 1.0 (7%) | 0.7 (9%) | 0.9 (8%) |
| 2008 | 0.6 (6%) | 1.1 (4%) | 0.5 (5%) | 0.2 (8%) | 0.3 (7%) | 0.6 (6%) | 0.9 (6%) | 1.1 (7%) | 1.0 (8%) | 0.8 (9%) |
| 2009 | 2.1 (3%) | 0.6 (6%) | 1.1 (4%) | 0.5 (5%) | 0.2 (8%) | 0.3 (8%) | 0.5 (7%) | 0.7 (8%) | 1.0 (9%) | 0.7 (9%) |
| 2010 | 1.0 (5%) | 2.1 (3%) | 0.6 (6%) | 1.2 (4%) | 0.6 (6%) | 0.2 (9%) | 0.3 (11%) | 0.5 (10%) | 0.7 (9%) | 0.7 (9%) |
| 2011 | 0.6 (6%) | 1.0 (5%) | 2.2 (3%) | 0.7 (6%) | 1.2 (5%) | 0.6 (7%) | 0.2 (11%) | 0.3 (13%) | 0.5 (12%) | 0.7 (10%) |
| 2012 | 0.5 (6%) | 0.6 (6%) | 1.0 (5%) | 2.4 (3%) | 0.7 (6%) | 1.0 (6%) | 0.5 (9%) | 0.2 (13%) | 0.2 (16%) | 0.5 (12%) |
| 2013 | 1.8 (3%) | 0.5 (6%) | 0.6 (6%) | 1.1 (5%) | 2.4 (4%) | 0.7 (7%) | 0.8 (8%) | 0.4 (11%) | 0.2 (17%) | 0.3 (15%) |
| 2014 | 2.1 (3%) | 1.9 (3%) | 0.5 (6%) | 0.7 (6%) | 1.1 (5%) | 2.4 (4%) | 0.8 (8%) | 0.8 (11%) | 0.4 (16%) | 0.2 (19%) |
| 2015 | 0.9 (5%) | 2.1 (3%) | 1.9 (3%) | 0.6 (6%) | 0.7 (6%) | 1.2 (6%) | 2.3 (5%) | 0.7 (10%) | 0.8 (14%) | 0.2 (18%) |
| 2016 | 0.4 (9%) | 0.9 (5%) | 2.2 (3%) | 2.0 (4%) | 0.6 (6%) | 0.7 (7%) | 1.1 (7%) | 2.2 (7%) | 0.7 (12%) | 0.4 (17%) |
| 2017 | 0.3 (10%) | 0.4 (9%) | 0.9 (5%) | 2.4 (4%) | 1.8 (4%) | 0.6 (7%) | 0.8 (8%) | 1.2 (9%) | 2.2 (9%) | 0.5 (14%) |
| 2018 | 0.7 (8%) | 0.3 (10%) | 0.4 (9%) | 1.0 (5%) | 2.5 (4%) | 1.6 (5%) | 0.6 (8%) | 0.8 (9%) | 1.3 (10%) | 1.1 (12%) |
| 2019 | 3.7 (5%) | 0.7 (8%) | 0.4 (10%) | 0.4 (9%) | 1.1 (6%) | 2.4 (4%) | 1.6 (6%) | 0.6 (9%) | 0.9 (11%) | 1.2 (11%) |
| 2020 | 1.2 (9%) | 3.8 (5%) | 0.7 (8%) | 0.4 (10%) | 0.5 (9%) | 1.3 (6%) | 2.2 (6%) | 1.4 (9%) | 0.6 (11%) | 1.0 (11%) |
| 2021 | 0.5 (14%) | 1.2 (9%) | 3.9 (5%) | 0.8 (8%) | 0.4 (10%) | 0.5 (10%) | 1.2 (7%) | 2.0 (8%) | 1.4 (11%) | 0.8 (14%) |
| 2022 | 0.5 (17%) | 0.6 (14%) | 1.2 (9%) | 4.0 (6%) | 0.9 (8%) | 0.4 (11%) | 0.5 (11%) | 1.1 (10%) | 1.8 (11%) | 0.9 (14%) |
| 2023 | 0.5 (20%) | 0.5 (17%) | 0.6 (14%) | 1.3 (9%) | 4.2 (6%) | 0.9 (9%) | 0.5 (13%) | 0.6 (13%) | 1.1 (12%) | 1.3 (12%) |
| 2024 | 1.7 (23%) | 0.5 (20%) | 0.6 (17%) | 0.6 (14%) | 1.4 (10%) | 4.3 (7%) | 1.0 (10%) | 0.5 (14%) | 0.7 (15%) | 1.2 (13%) |
| 2025 | 1.2 (27%) | 1.7 (23%) | 0.5 (20%) | 0.6 (18%) | 0.6 (15%) | 1.5 (11%) | 4.3 (9%) | 1.1 (12%) | 0.6 (15%) | 1.0 (13%) |
| 2026 | 1.5 (53%) | 1.2 (27%) | 1.7 (23%) | 0.6 (21%) | 0.6 (19%) | 0.7 (17%) | 1.4 (14%) | 4.4 (12%) | 1.2 (14%) | 0.9 (14%) |

Age at start of year




Colour denominators: each age's arithmetic mean over 1964-2026 (all 63 years).
Labels: multiples to 1 decimal; CV to whole %. 1 = age mean; 2 = twice the mean.
10+ sums ages 10-14 and 15+ once; its SE retains all cross-age covariance.



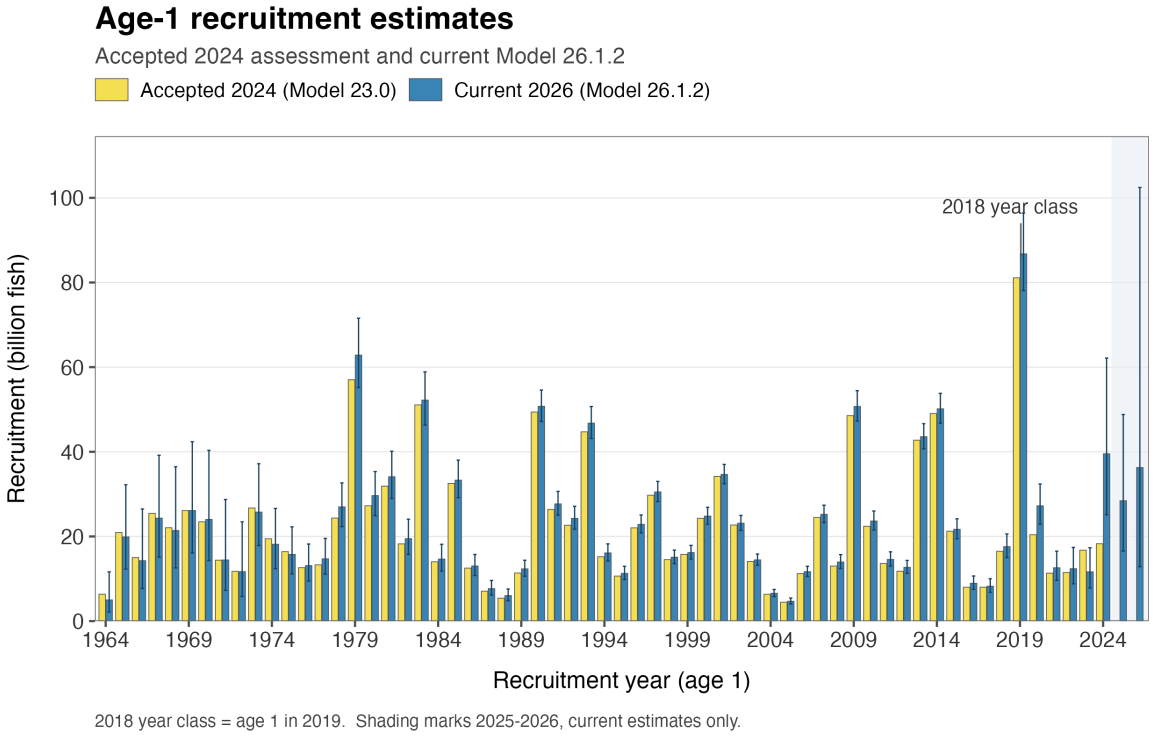




Figure 32: Age-1 recruitment from the accepted 2024 assessment, Model 23.0, compared with Models 26.0 (panel A) and 26.1.2 (panel B). Bars show point estimates for both sexes in billions of fish. Historical estimates are repeated unchanged without intervals. Whiskers show local 95% log-scale intervals: the native asymptotic covariance for Model 26.0 and the marginal fit's Louis observed information for Model 26.1.2, conditional on its integration samples and normalizer grid. Calendar year denotes age 1; the 2018 year class enters at age 1 in 2019.

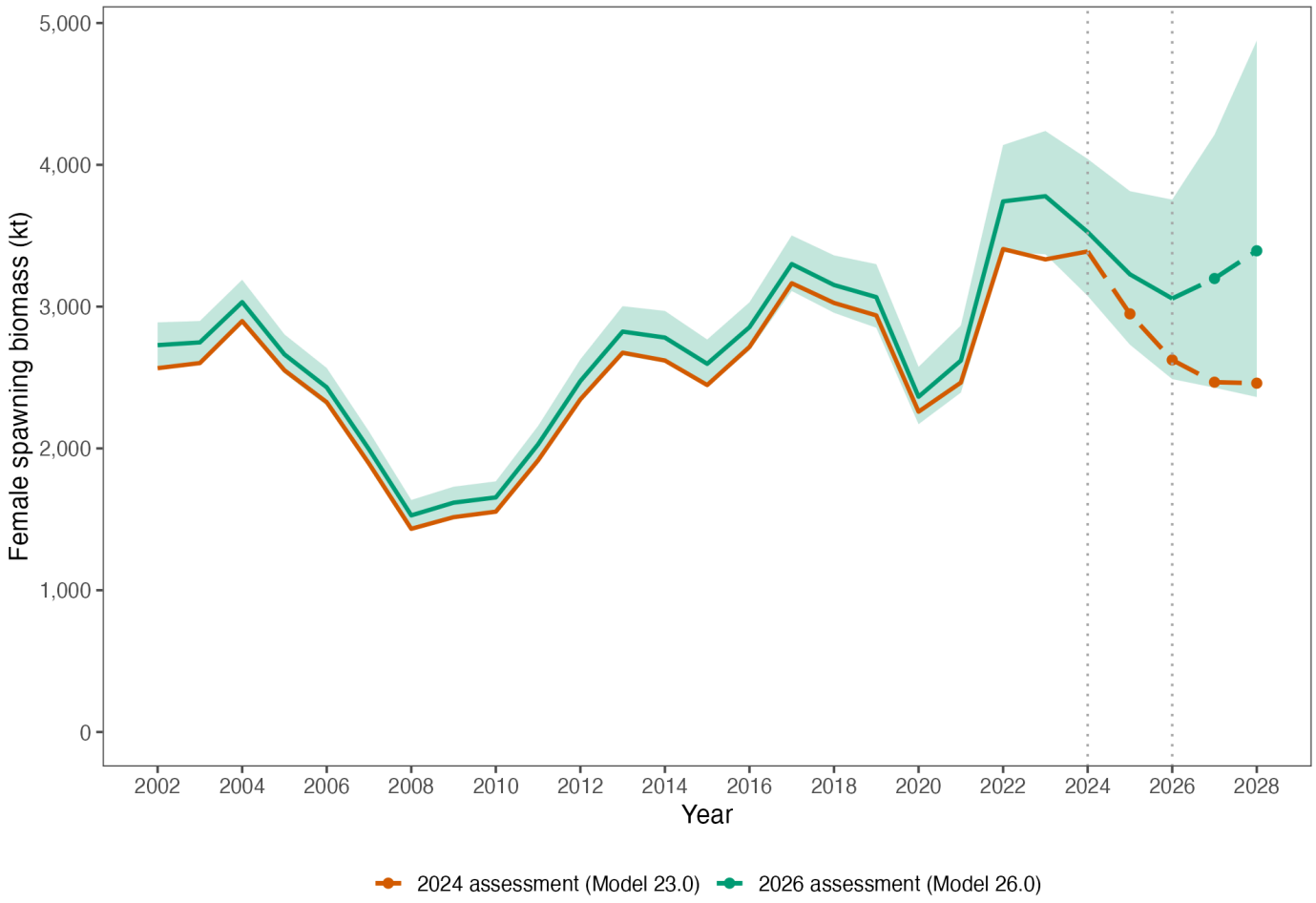



Figure 33: Female spawning biomass estimated and projected by the 2024 assessment (Model 23.0) and the 2026 assessment (Model 26.0), through 2028. Solid lines show estimates; dashed lines and points show Scenario 2 projections. The 2024 series reproduces published Tables 26 and 32, with planned catches of 1,350 kt in 2025–2026 followed by maximum ABC. The current projection uses 1,375 kt in 2027–2028. Shading shows approximate 95% log-scale delta-method intervals for current assessment-parameter uncertainty, conditional on fixed weights and mean future recruitment; independent future recruitment and weight-estimation uncertainty are excluded. No corresponding 2024 uncertainty band is shown. Dotted lines mark the assessment endpoints, 2024 and 2026.

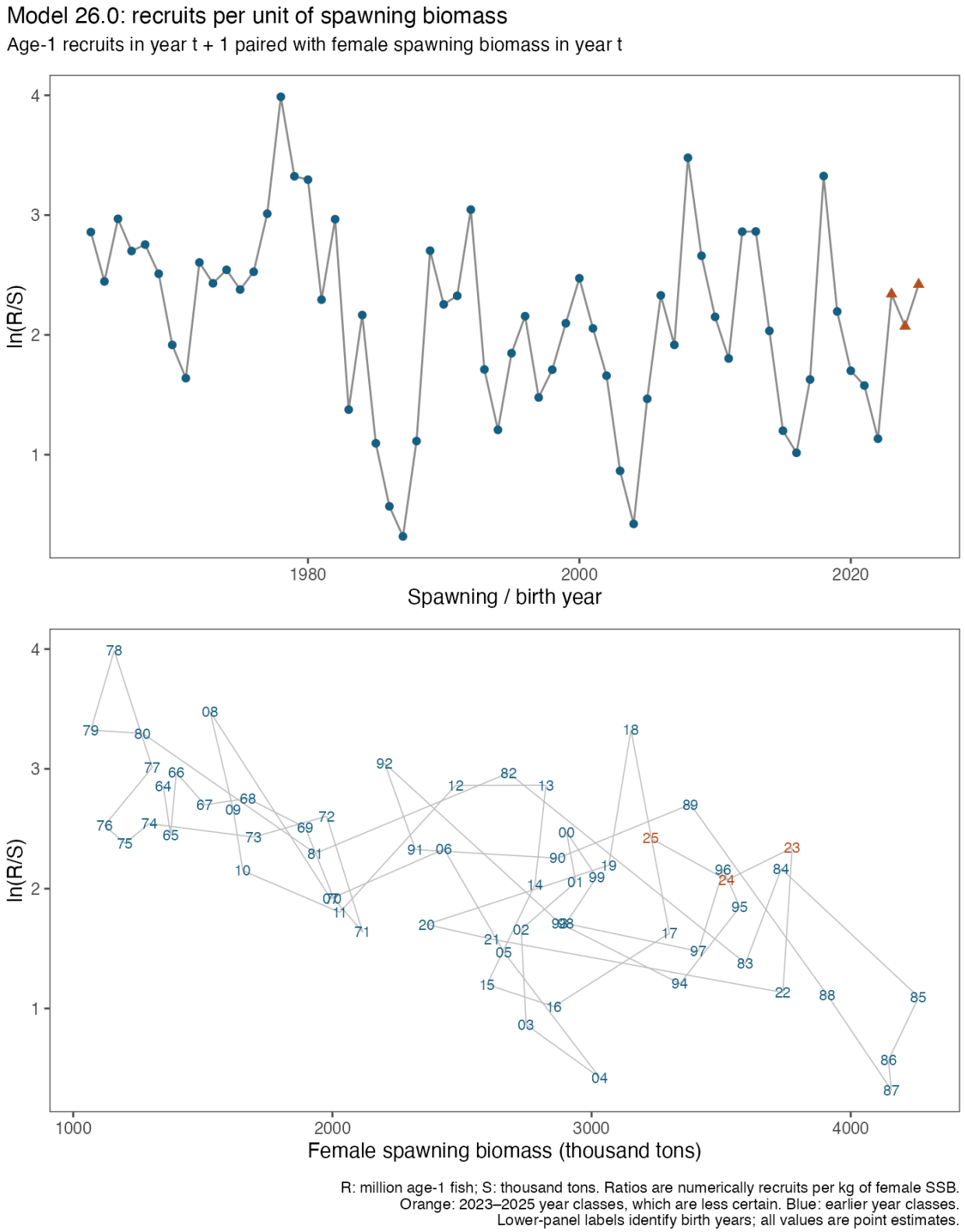



Figure 34: Model 26.0 log recruits per spawner, updating the two-panel diagnostic from the 2024 assessment. Age-1 recruits in year t+1 (million fish) are divided by female spawning biomass in year t (thousand tons), then natural-log transformed. The numerical ratio is recruits per kilogram of female spawning biomass. Upper panel: birth-year series, 1964–2025. Lower panel: the same ratios against spawning biomass, with two-digit birth-year labels and lines joining consecutive years. Orange identifies the less certain 2023–2025 year classes. The figure shows point estimates of the ratio.

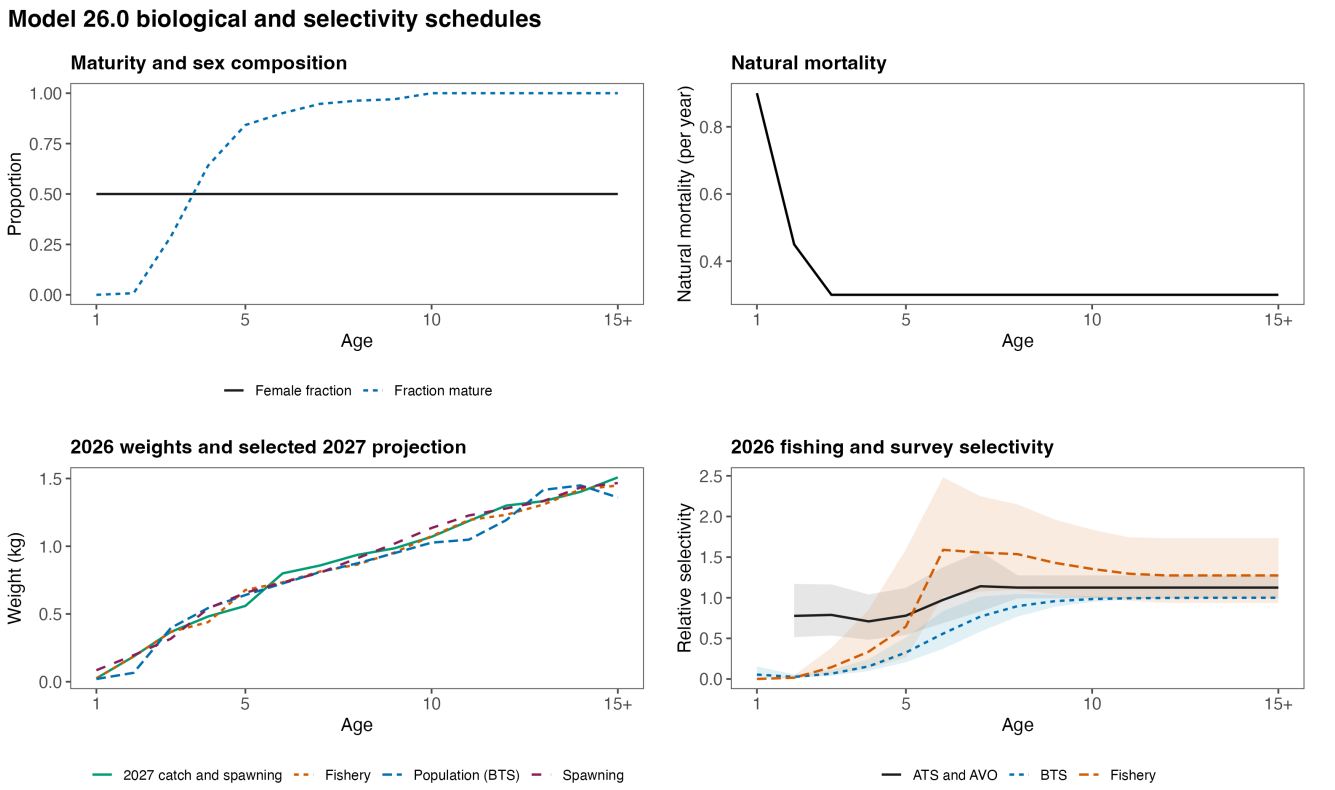



**Figure 35:** Model 26.0 biological and selectivity schedules. Maturity, female fraction and natural mortality are supplied inputs. The weight panel distinguishes the 2026 population (BTS), spawning and fishery schedules from the selected 2027 fishery forecast used for both catch and spawning in projections. The 2026 fishery estimates and 2027 forecast come from the standalone WtAgeRE fit omitting 2026 BTS weight observations; population and historical spawning weights retain their separate treatments. Fishing and survey selectivity curves have approximate 95% asymptotic ranges; ATS and AVO share their estimated curve. Supplied schedules have no fitted-parameter bands.

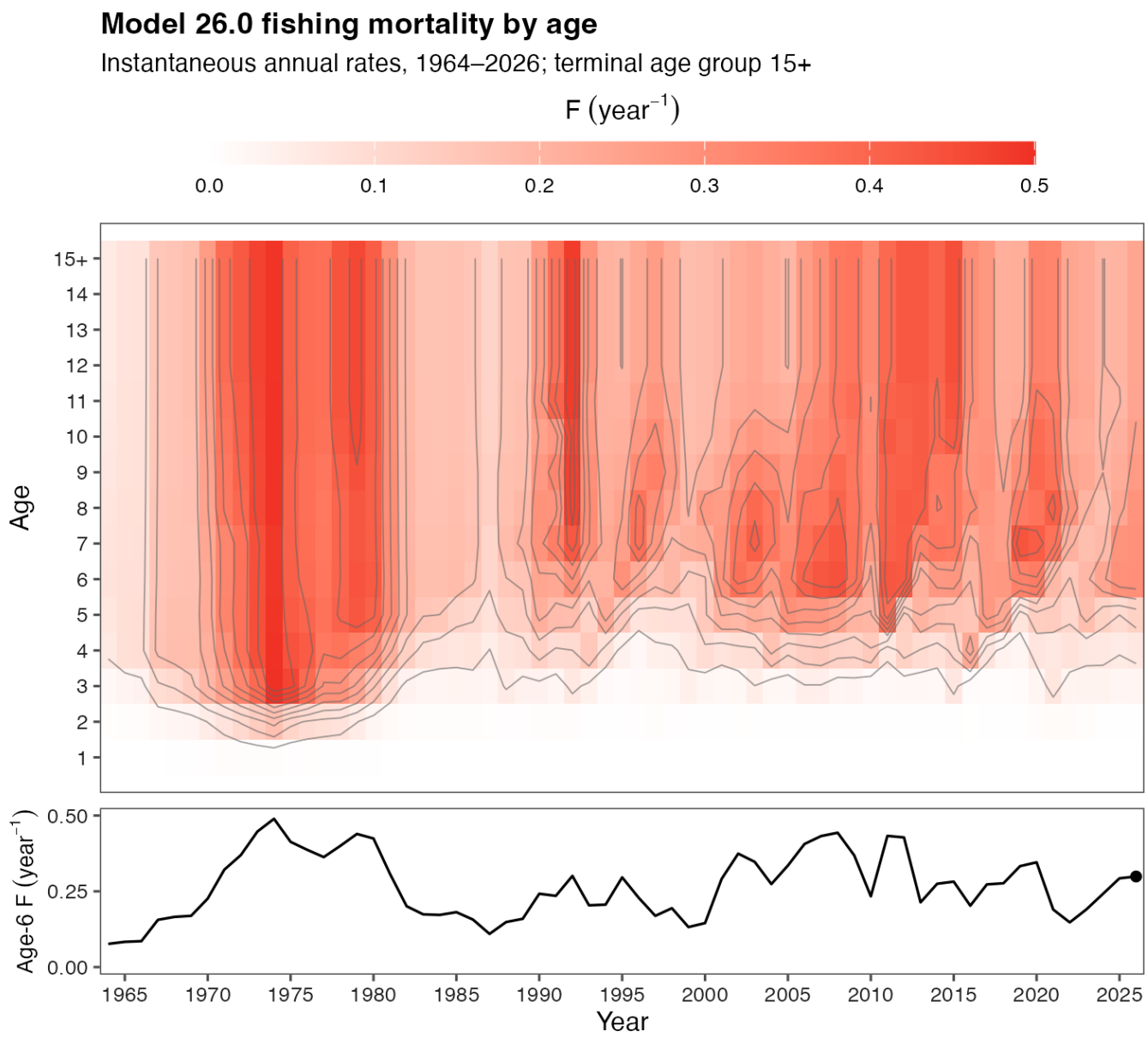



Figure 36: Model 26.0 instantaneous fishing mortality by age through 2026, using the verified fit. Tiles and contours show the same realized total fishing mortality rates; age 15+ remains the terminal group.

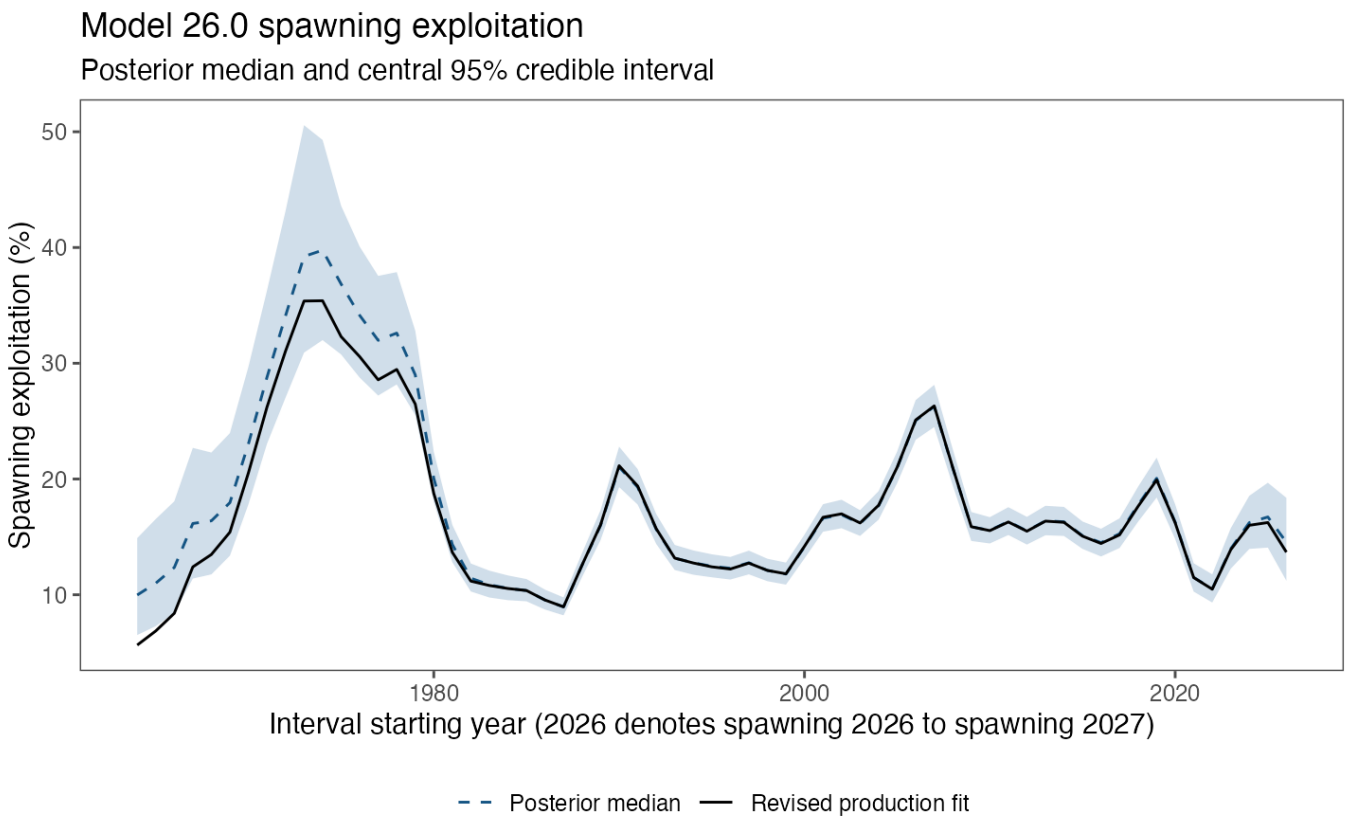



Figure 37: Model 26.0 spawning exploitation for spawning-to-spawning intervals starting in 1964–2026. Black lines show the revised production fit; dashed blue lines and shading show MCMC posterior medians and central 95% credible intervals from 5,000 existing draws. Each within-draw ratio compares next-spawning biomass with and without fishing from a shared spawning-date population. The terminal interval assumes 1.375 million t catch in 2027, within-draw mean 1978–2023 recruitment and mean 2023–2025 selectivity, and the revised deterministic fishery-weight schedule. Supplied biology is fixed. The terminal posterior median is 14.5333%, with interval 11.2075–18.3700%.

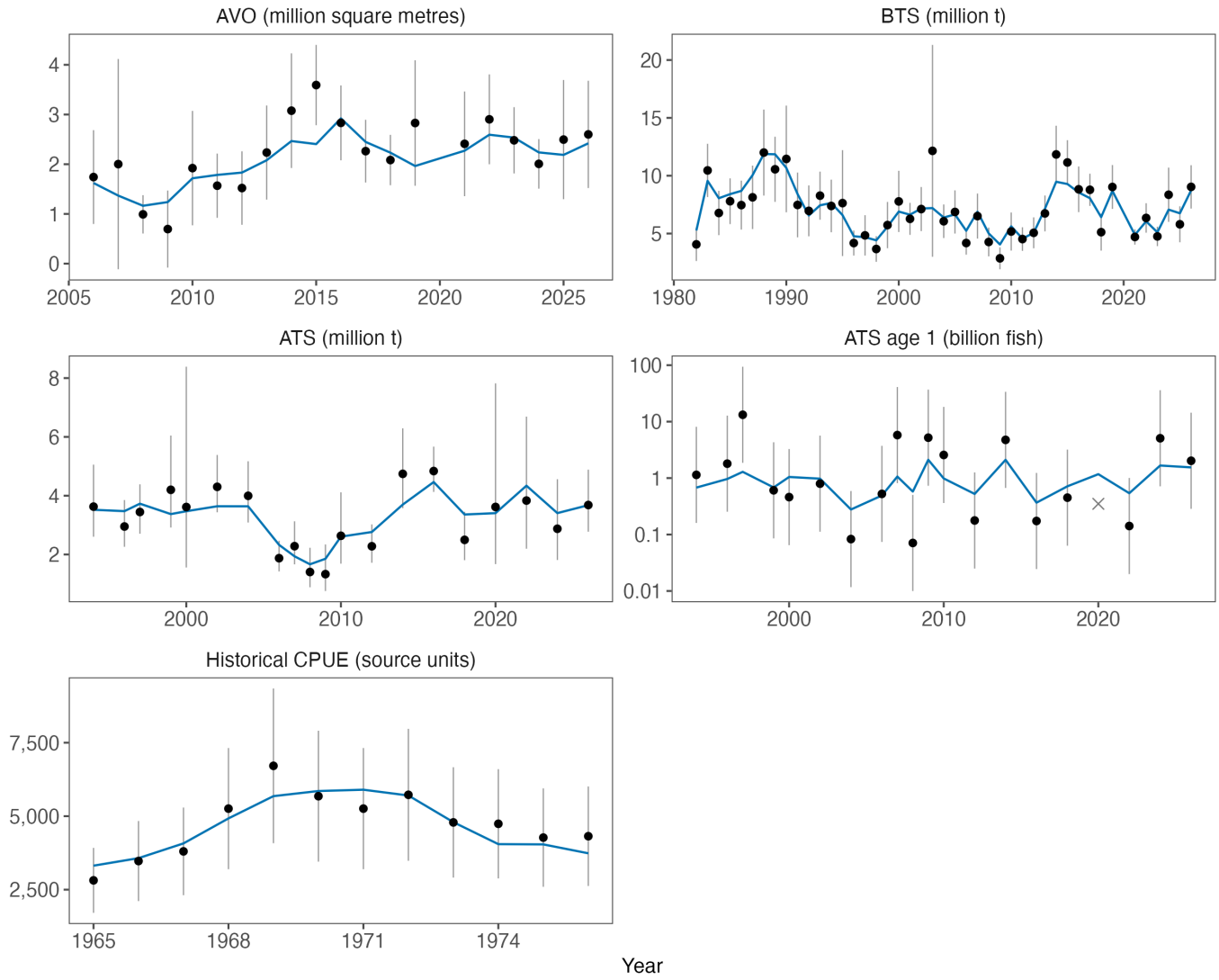



Figure 38: Observed indices and Model 26.0 fitted predictions. Black points show source observations, lines show fitted predictions, and bars show approximate 95% observation-error ranges. BTS uses its supplied covariance. The ATS age-1 panel uses a logarithmic scale in billions of fish; the retained 2020 age-1 placeholder is distinguished from the real 2020 USV biomass observation.

BTS errors account for the supplied covariance among survey years. Dashed lines: standard-normal 95% reference.
The gray 2020 ATS age-1 cross is an inherited placeholder, excluded from observed-index summaries.
The fitted 2020 USV biomass remains an observed ATS index.

Figure 39: Model 26.0 index residuals relative to the selected observation errors, with BTS accounting for supplied covariance among survey years. Dashed lines mark the standard-normal 95% reference range. The retained 2020 ATS age-1 placeholder is distinguished and excluded from observed-index summaries; the 2020 USV biomass remains an observation.

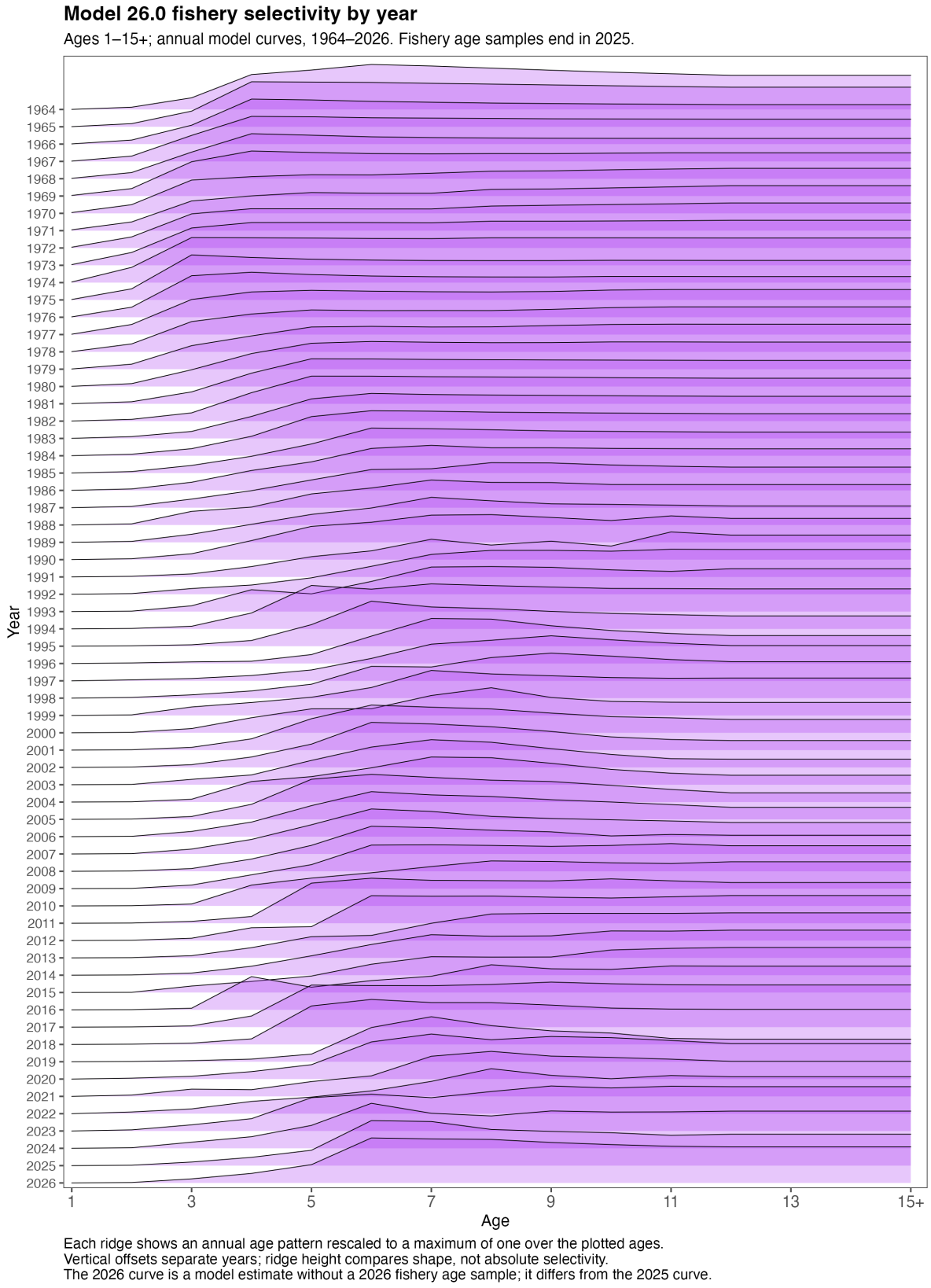



Figure 40: Annual fishery selectivity at ages 1–15+ from the Model 26.0 fit. Each ridge shows the estimated age pattern for that year, using the normalization recorded with the verified figure data.

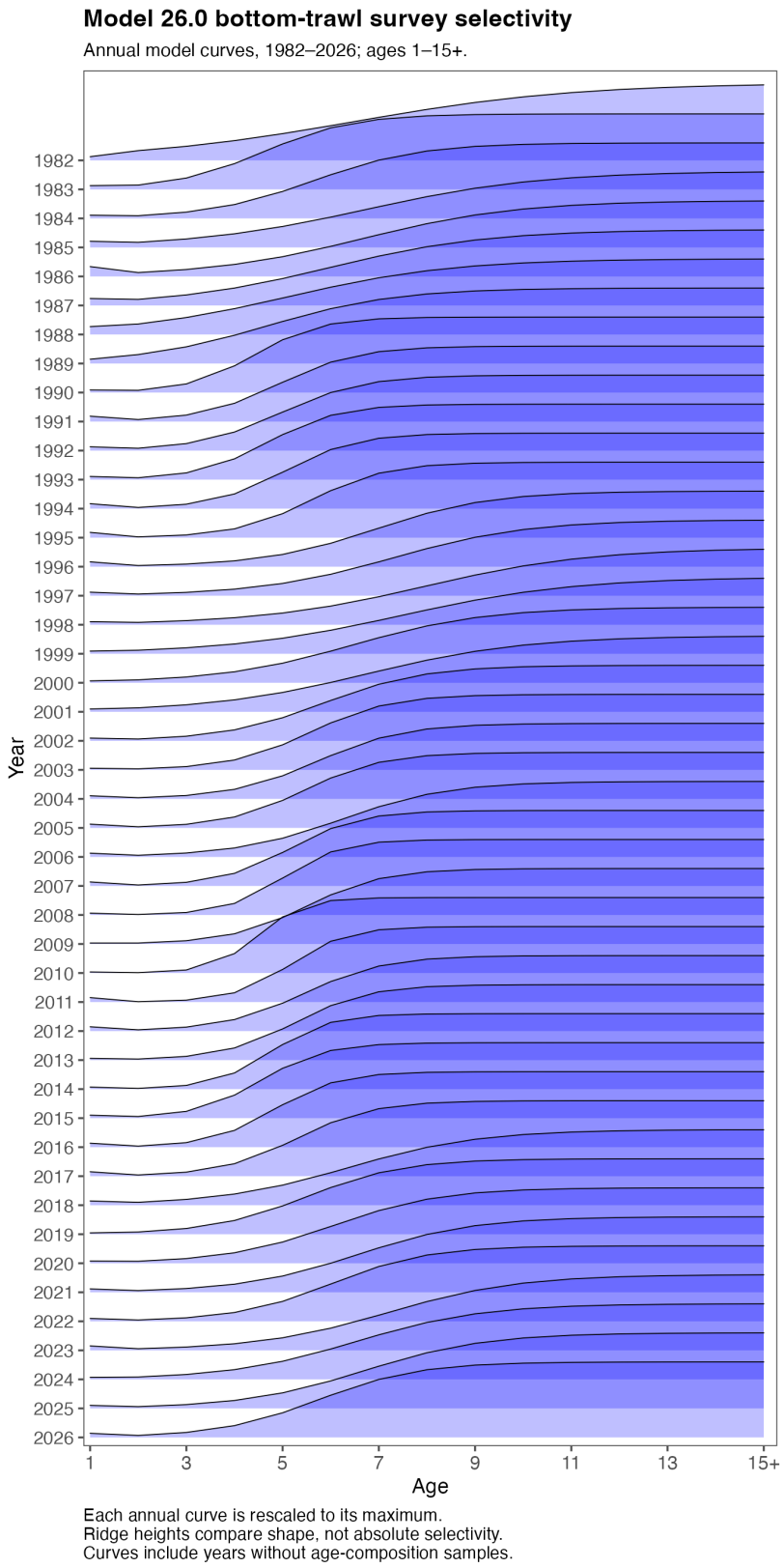



Figure 41: Annual bottom-trawl survey selectivity at ages 1–15+ from the Model 26.0 fit. Each ridge is divided by its maximum over the displayed ages. The taller, narrower display compares annual age-pattern shape, including years without an age-composition sample.

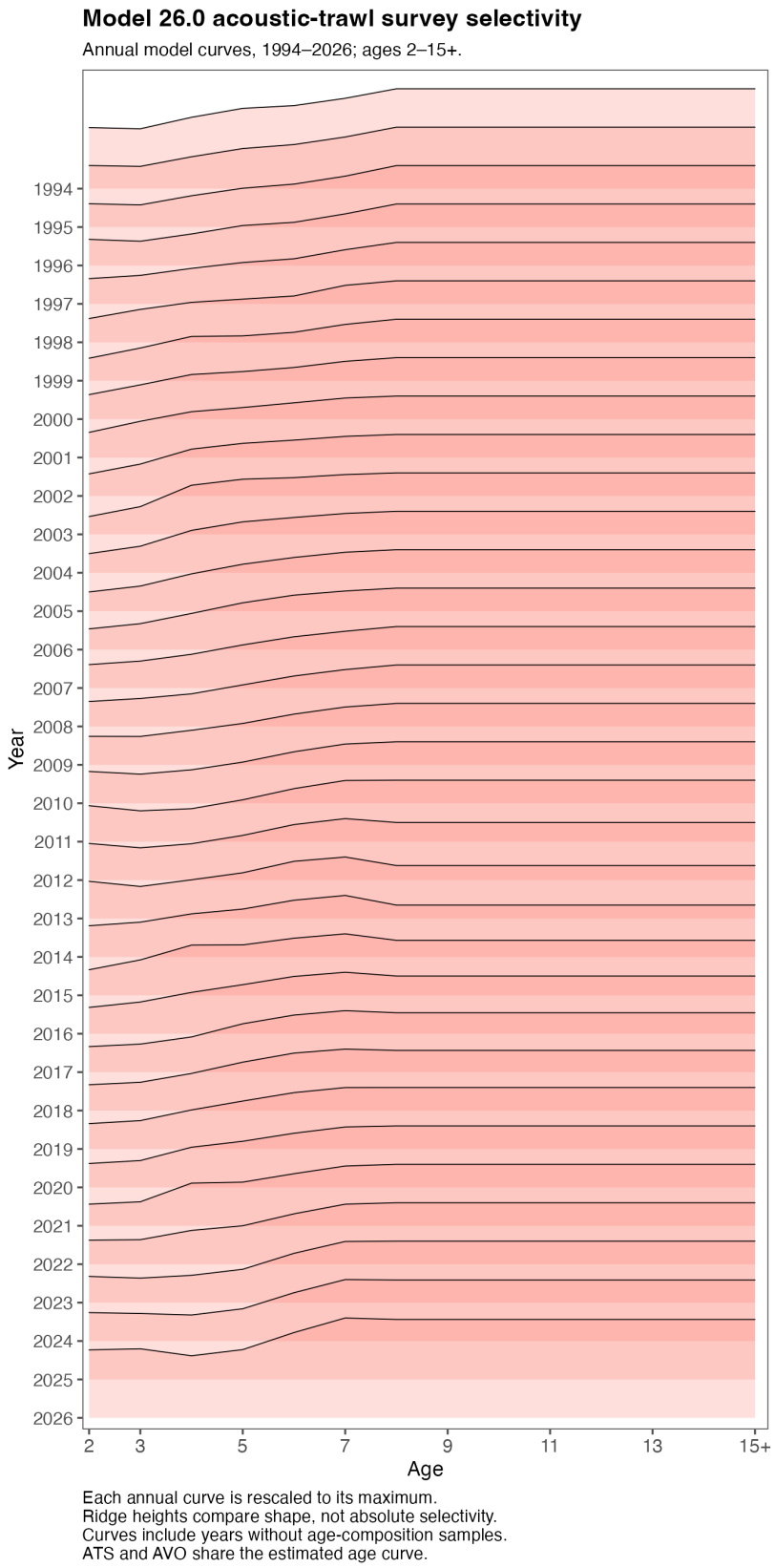



Figure 42: Annual acoustic-trawl survey selectivity at ages 2–15+ from the Model 26.0 fit. ATS and AVO share the estimated age curve; structural age 1 is excluded. The taller, narrower display includes model curves for years without an age-composition sample.

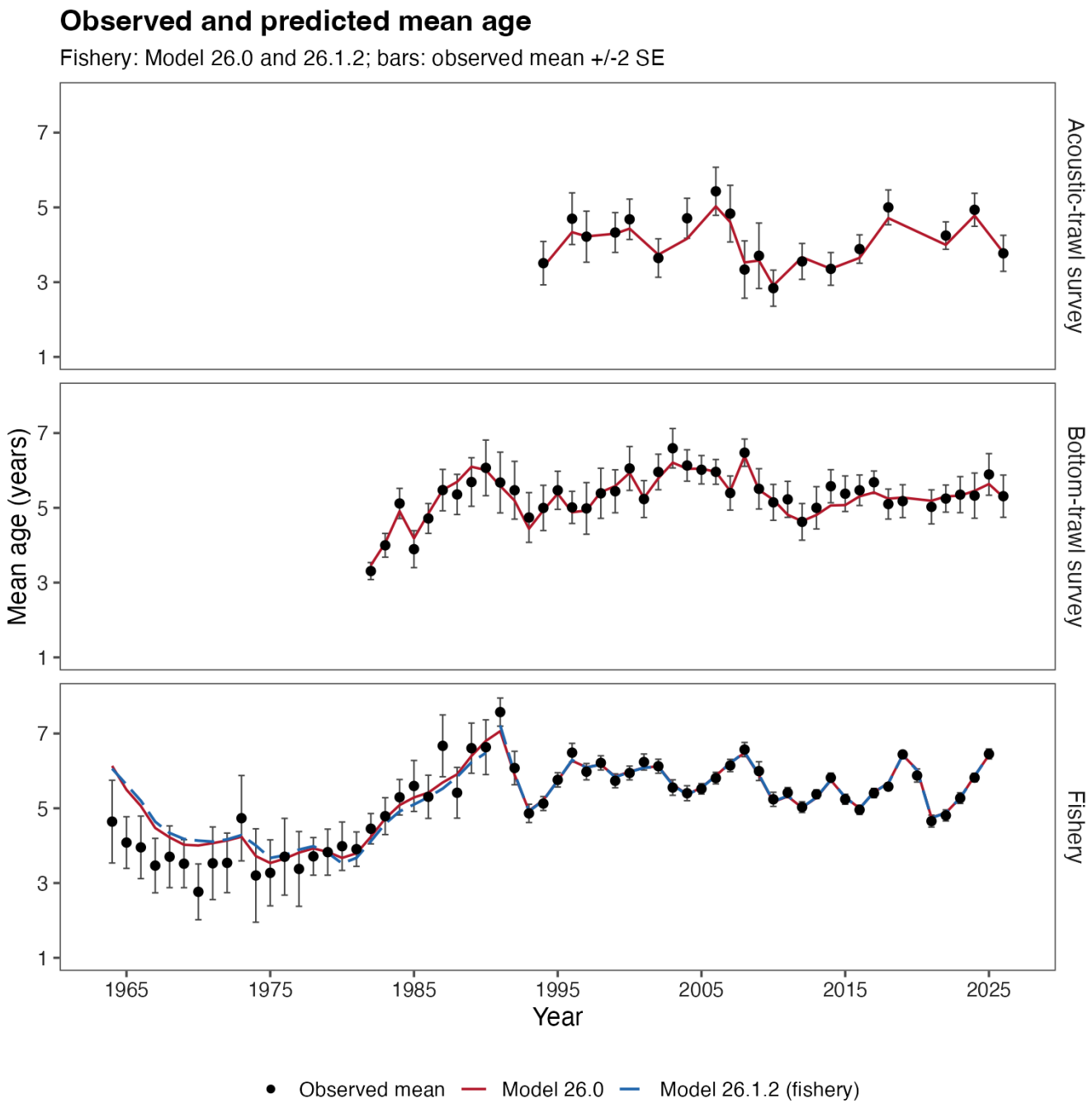



Figure 43: Observed grouped mean age for ATS, BTS and fishery with Model 26.0 predictions and the Model 26.1.2 fishery prediction. Bars are the observed mean ±2 SE, where SE = sqrt(sum[p_obs × (age score − observed mean)^2] / N), using the selected observed proportions and likelihood input N. Ages 1–14 and 15+ are scored 1–15; ATS excludes structural age 1 and the retained 2020 substitute. The same fishery observations appear once. The alternative prediction is evaluated at its fitted conditional selectivity-state modes. Bars show uncertainty in the observations.

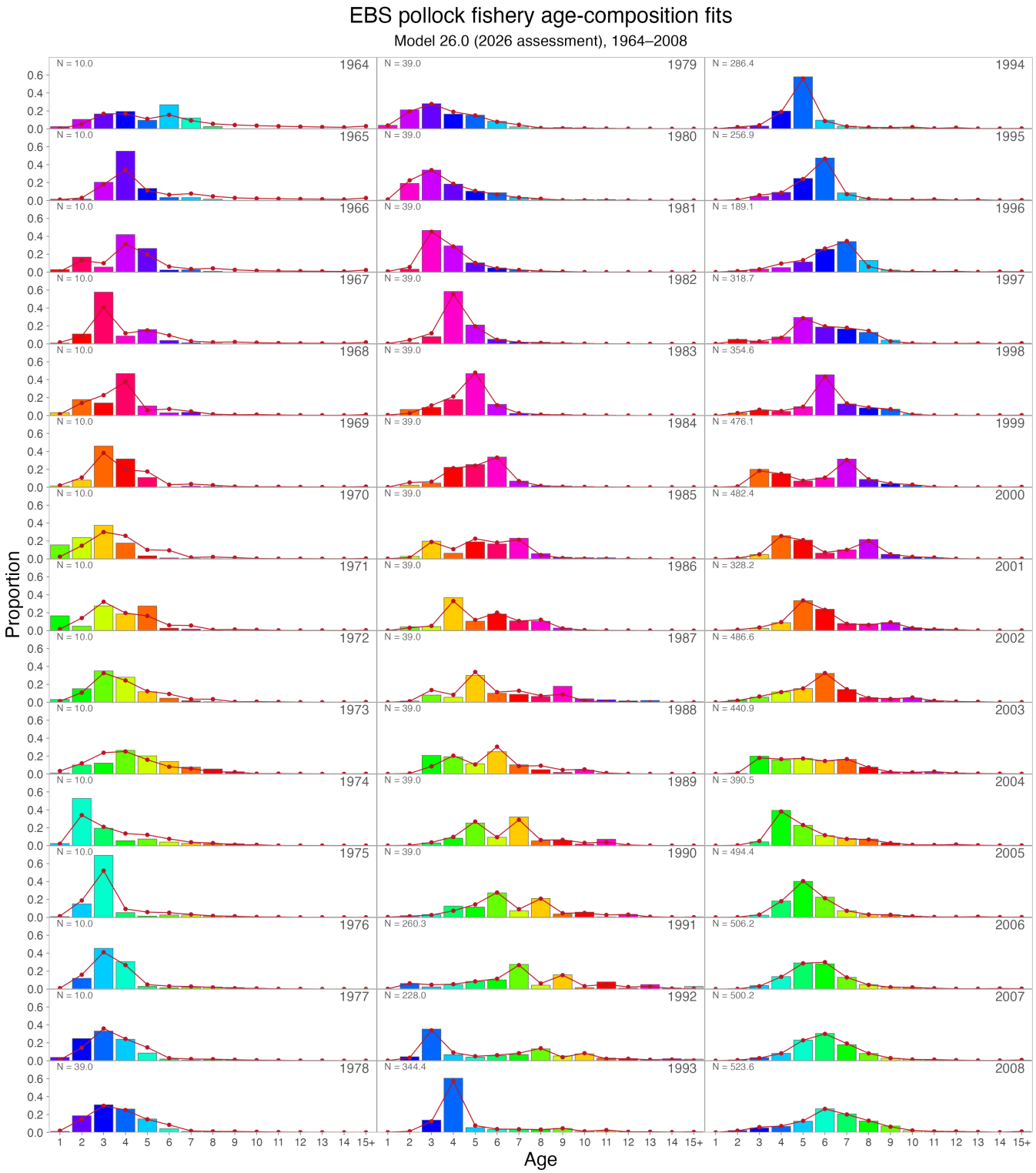



Figure 44: Fishery observed age proportions (cohort-colored bars) and Model 26.0 fitted proportions (points joined by lines), 1964– 2008. Annual panels are arranged down columns and then across, with a common proportion scale. Each panel shows its year and selected likelihood N. Age 15+ combines older cohorts.

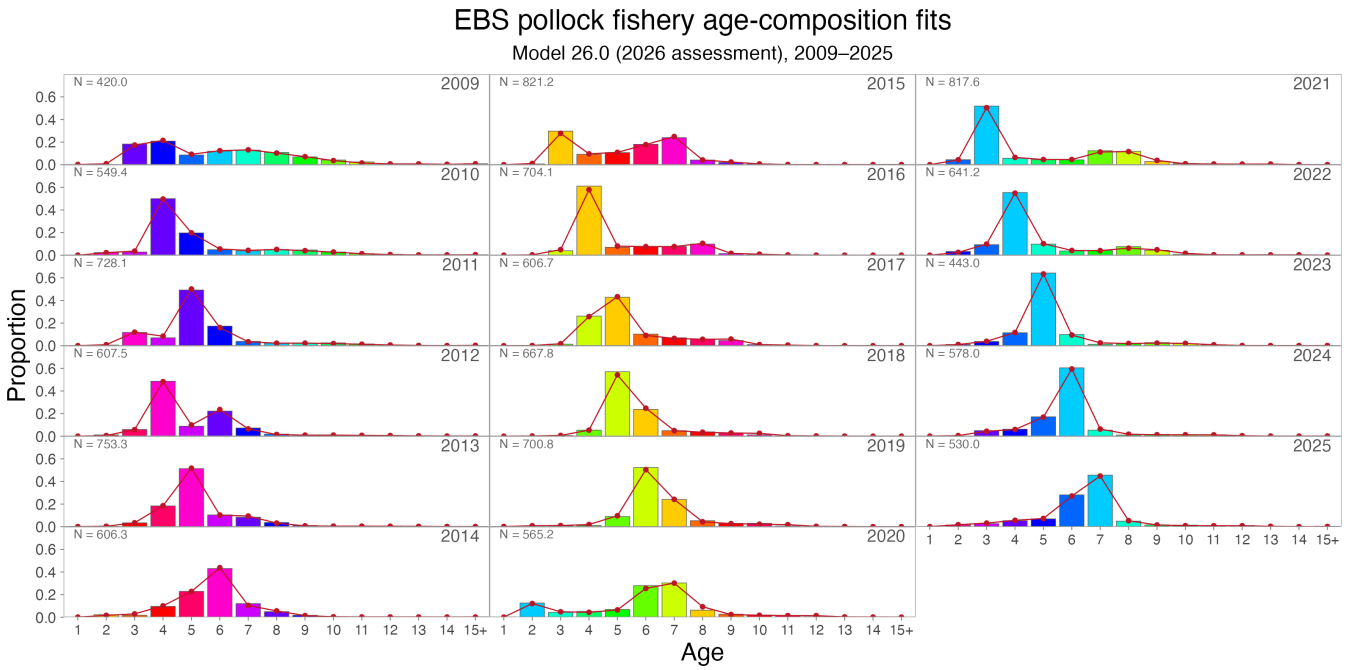



Figure 45: Fishery observed age proportions (cohort-colored bars) and Model 26.0 fitted proportions (points joined by lines), 2009– 2025. The proportion scale and cohort colors match the earlier-year figure. Each panel shows its year and selected likelihood N; age 15+ combines older cohorts.

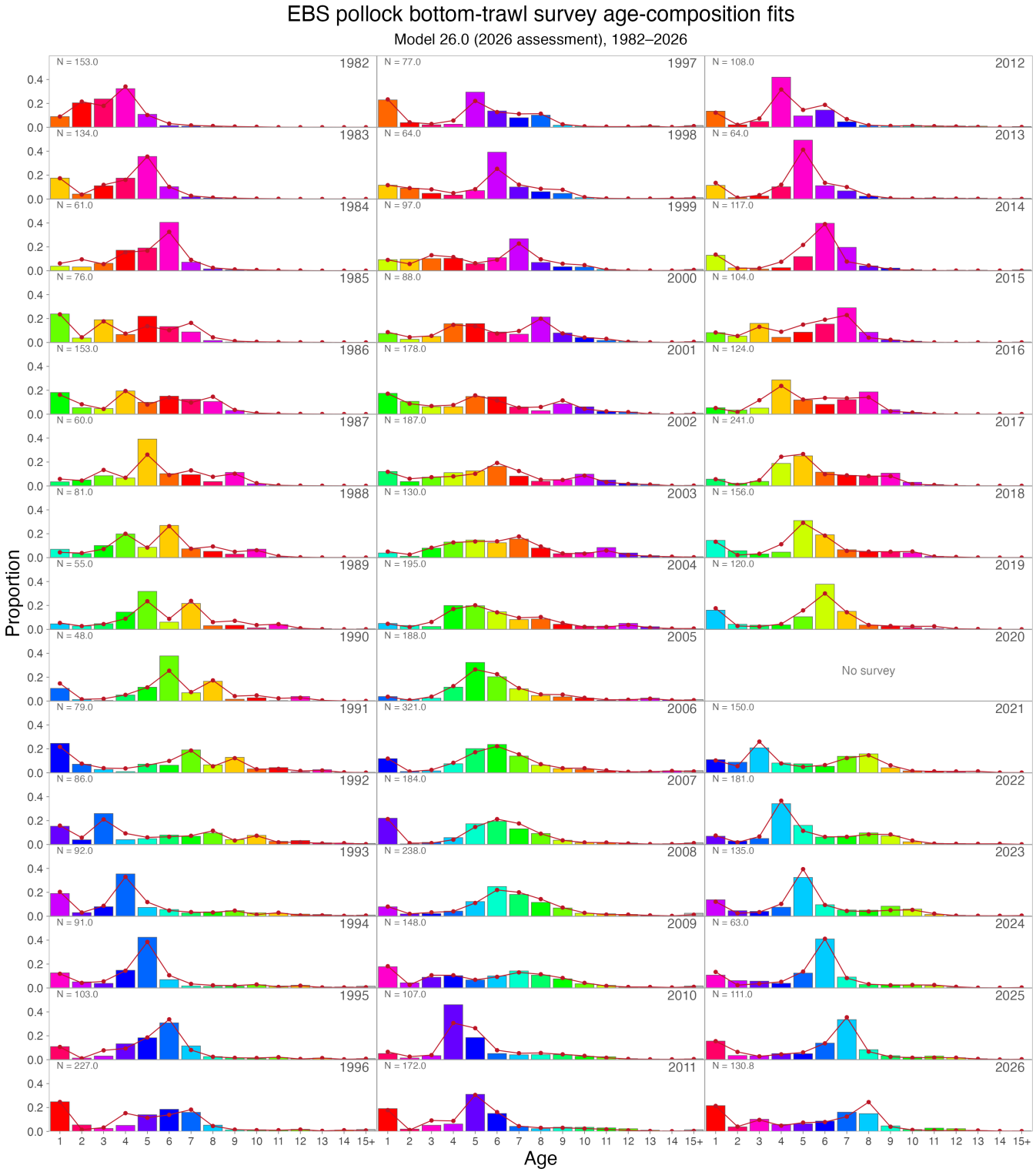



Figure 46: Bottom-trawl survey observed age proportions and Model 26.0 fitted proportions, 1982–2026. Annual panels share a proportion scale; the 2020 panel is explicitly labeled No survey. Sampled panels retain the selected likelihood Ns. Age 15+ combines older cohorts.

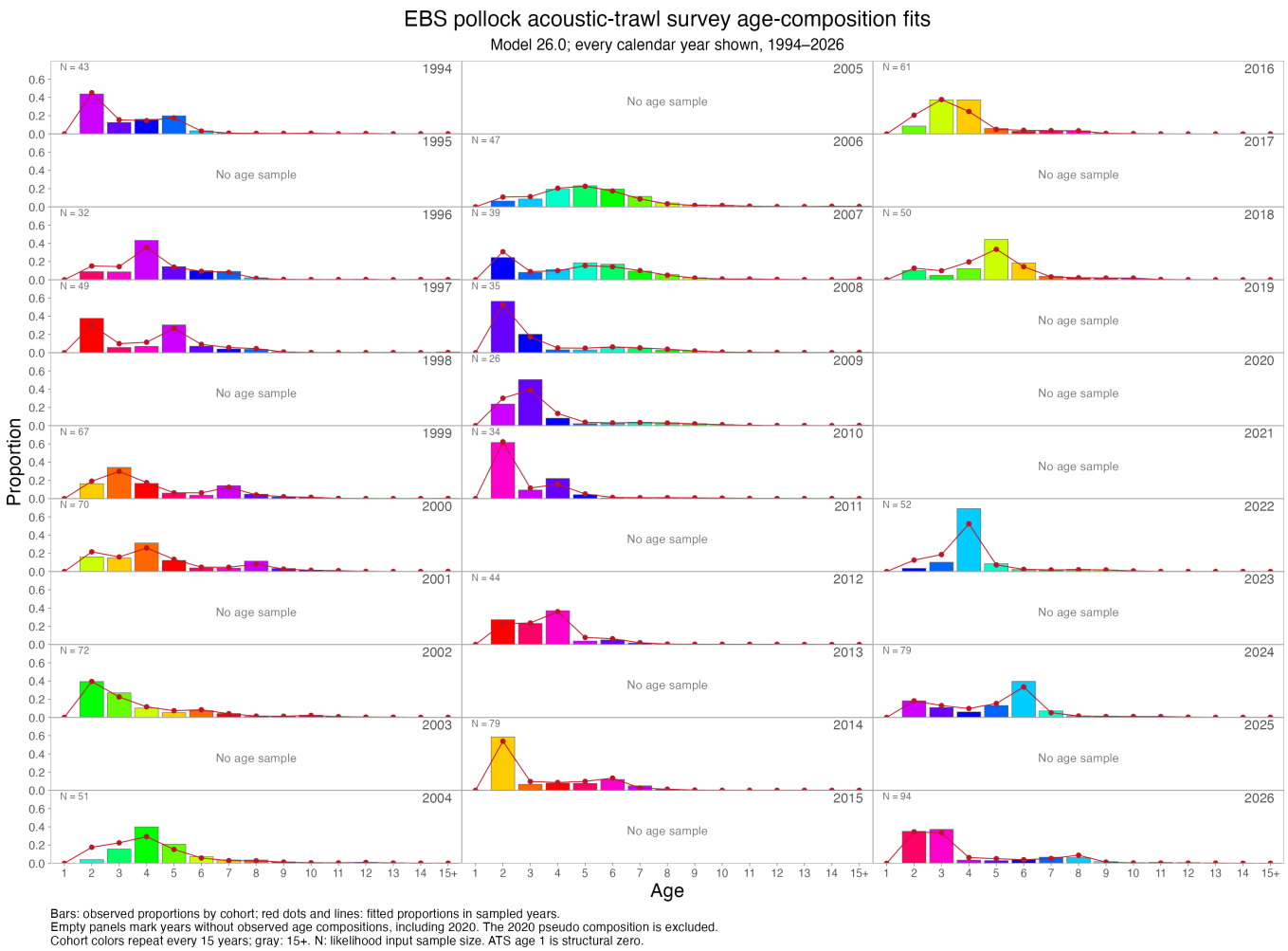
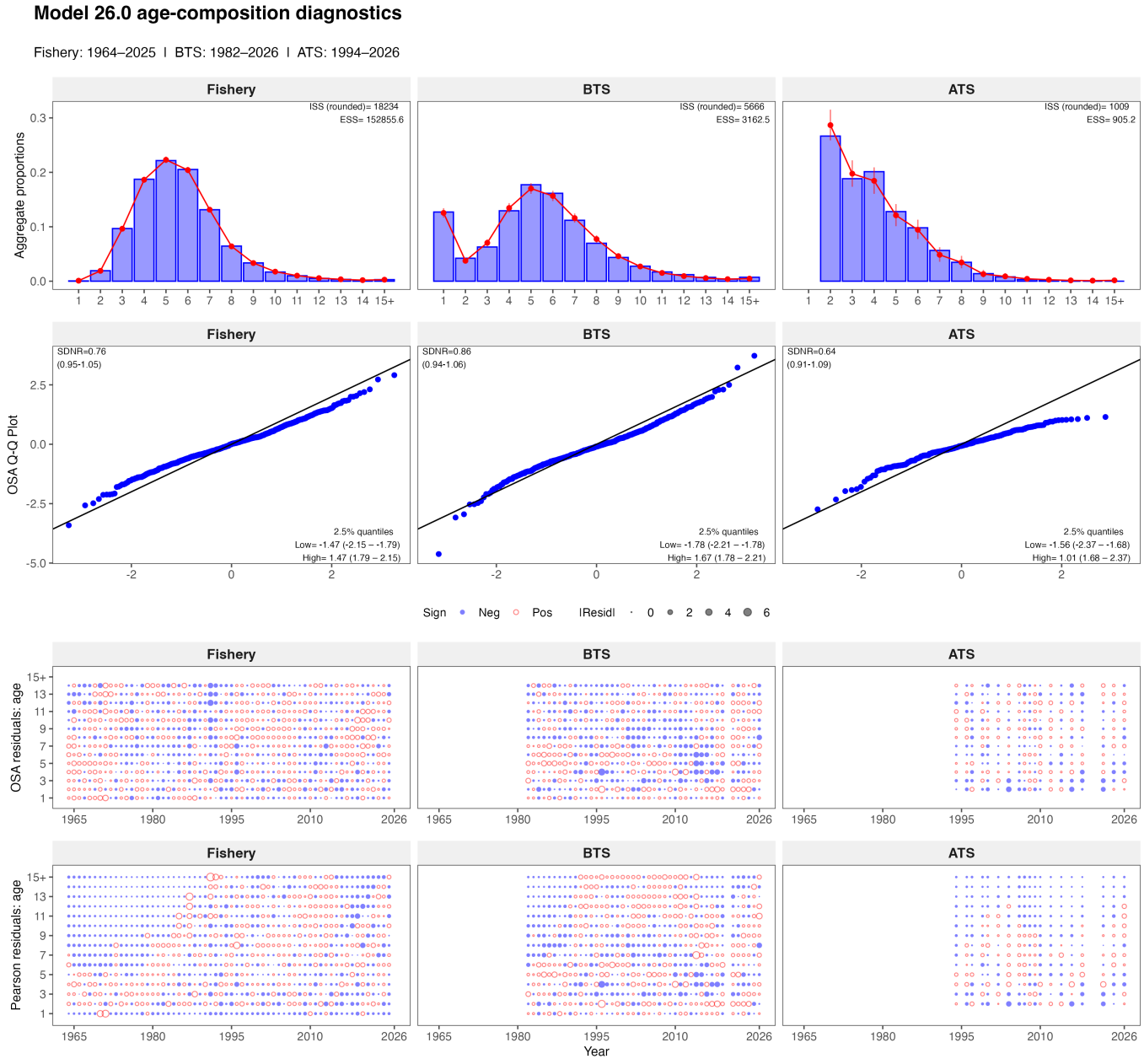



Figure 47: Acoustic-trawl survey observed and Model 26.0 fitted age proportions, shown on a complete 1994–2026 calendar. Empty panels mark every year without observed age-composition data, including 2020; the retained 2020 pseudo composition is excluded. Sampled panels retain their original observations, fitted proportions, likelihood Ns and common scale.




Blue filled circles: negative; red open circles: positive. Circle size represents absolute residual magnitude.
ATS uses ages 2–15+ and excludes the 2020 pseudo sample. The final age bin has no separate OSA residual.
Aggregate/Pearson displays use afscOSA rounded counts; native OSA values and assessment weights are preserved.

Figure 48: Standard afscOSA v0.0.1 diagnostics for Model 26.0, with fishery, BTS and ATS columns. Rows show aggregate observed and fitted compositions with 95% conditional observation ranges, normal-reference QQ plots with SDNR and tail summaries, OSA residual bubbles, and Pearson residual bubbles. Native OSAs and selected likelihood Ns are retained; ATS structural age 1 and the 2020 pseudo composition are excluded. Aggregate ranges condition on the fit; parameter and posterior predictive uncertainty are outside their scope.

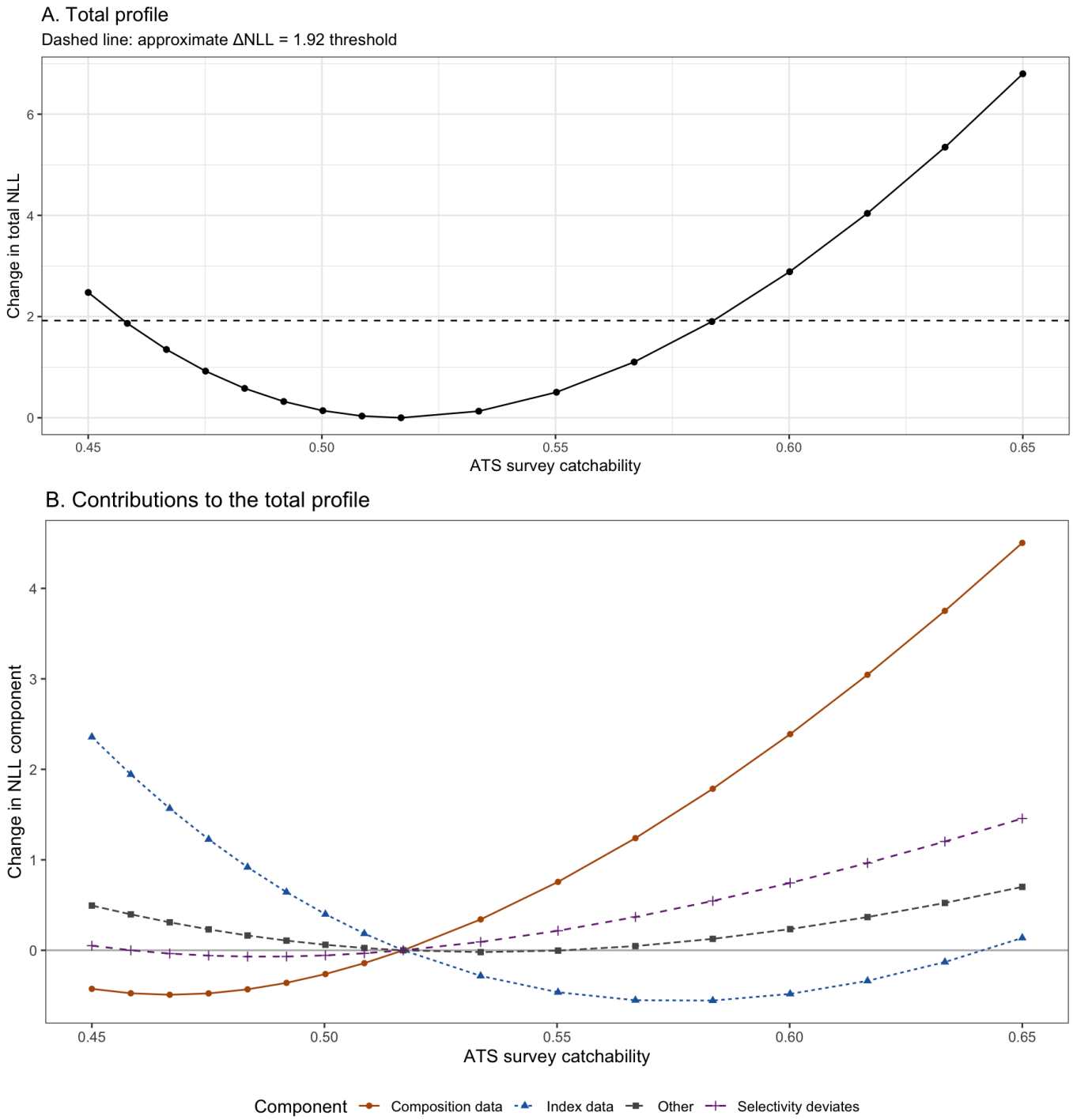



Figure 49: ATS catchability likelihood profile for Model 26.0. Seventeen fixed catchability values span 0.45–0.65, including the fitted value 0.51694. Panel A shows the change in total negative log likelihood; the dashed line marks 1.92. Panel B shows composition data, index data and selectivity-deviate contributions; Other pools the remaining components. All other active parameters were re-estimated at each fixed q.

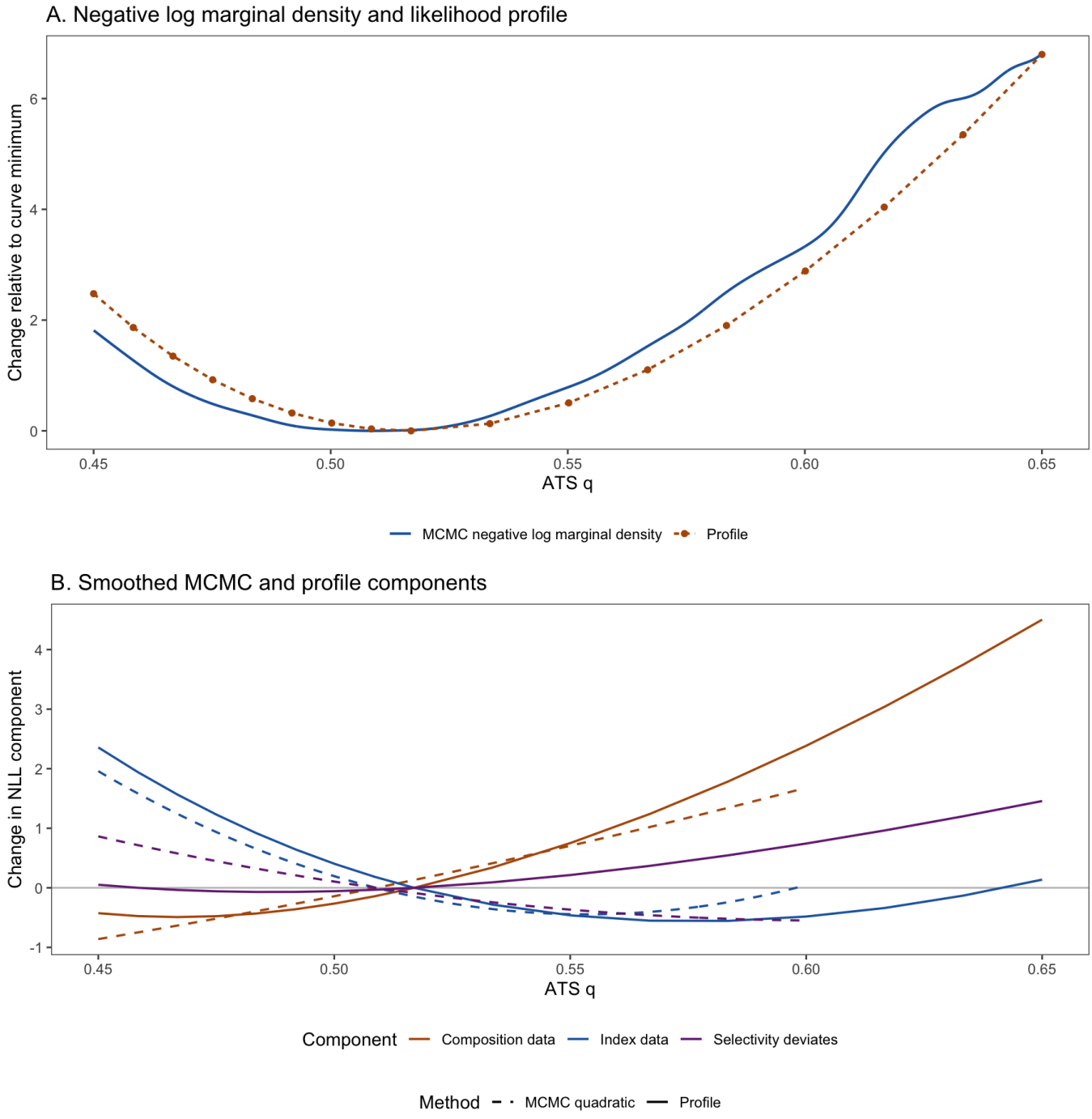



Figure 50: ATS-q negative log marginal density and likelihood profile for Model 26.0. Panel A compares the negative log of the MCMC marginal density, normalized to integrate to one, with the optimized profile NLL. Each curve is measured relative to its own minimum over ATS q. The marginal density uses a kernel estimate from all 5,000 saved draws; equal-grid probabilities also sum to one. Panel B pairs weighted quadratic regressions fitted directly to all 5,000 MCMC component values with profile NLL contributions for composition data, index data and selectivity deviates. The smoothers use individual draws and are displayed over the central 99% of sampled ATS q. MCMC curves are centered at the ATS-q marginal mode; profile components retain the profile minimum as their reference. These smoothers describe conditional mean native objective components; the negative log marginal density has no unique additive decomposition into those components. Other is omitted.

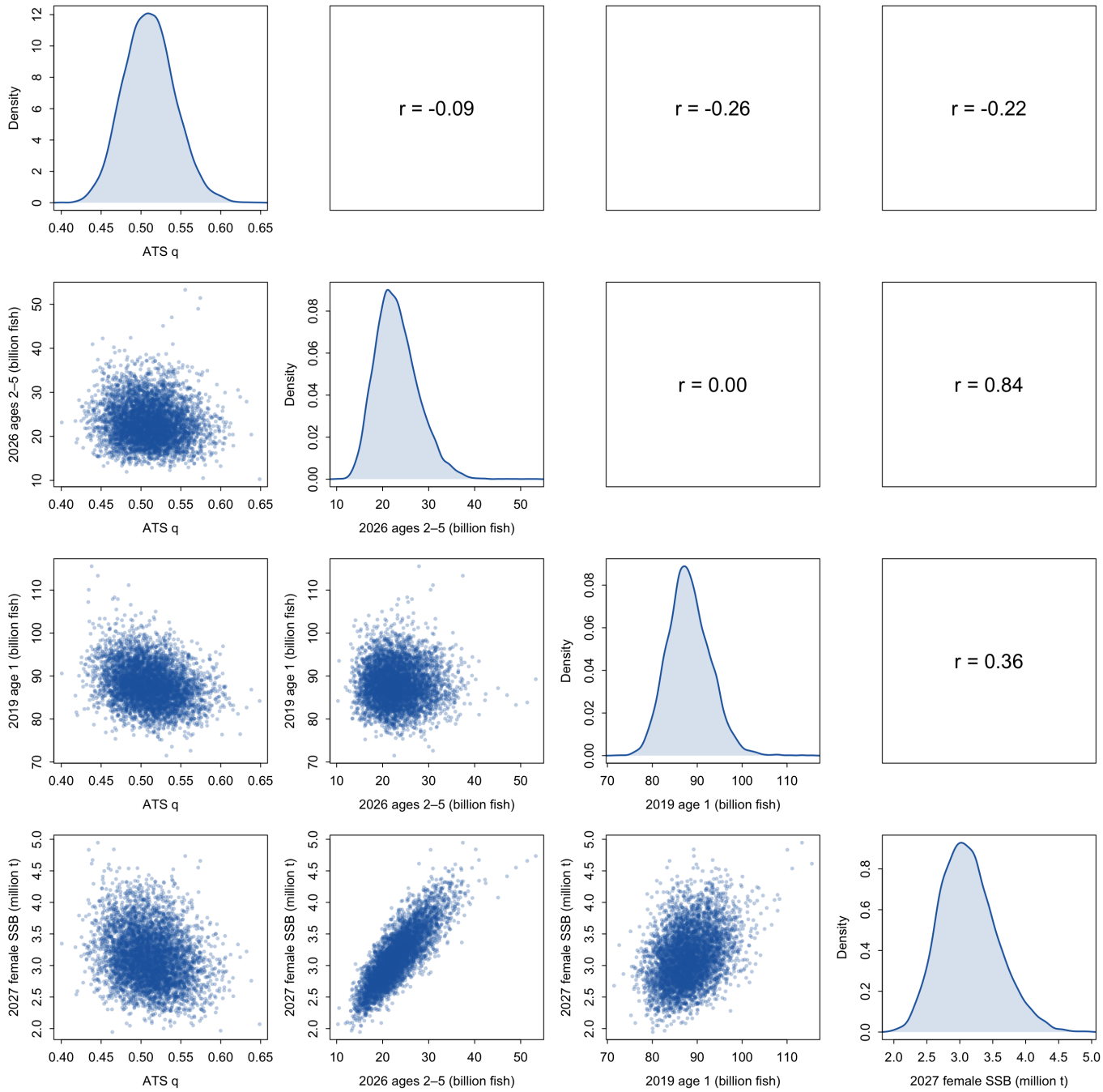



Figure 51: MCMC pairs plot of ATS catchability, 2026 abundance at ages 2–5, age-1 abundance in 2019 (the 2018 year class), and projected 2027 female spawning biomass for Model 26.0. Lower-triangle panels show paired values from all 5,000 saved draws with point alpha 0.3; upper-triangle panels give posterior Pearson correlations. Diagonal panels show marginal densities. Abundances are in billion fish and female SSB in million tons. The 2027 projection uses 1,375 kt catch, mean 2023–2025 fishery selectivity and within-draw mean 1978–2023 recruitment, with biological schedules fixed.

Figure 52: retrospective trajectories for Model 26.0 (left) and Model 26.1.2 with fixed process SDs (right). Black lines show each analysis's matched full 2026 reference; colors identify the nine peels ending in 2017–2025, and symbols mark their endpoints. Female spawning biomass is in million tons, age-1 recruitment is in billion fish, and maximum fishing mortality across ages is per year. Recruitment uses the saved age-1 calendar year rather than birth year. Panels show point estimates.

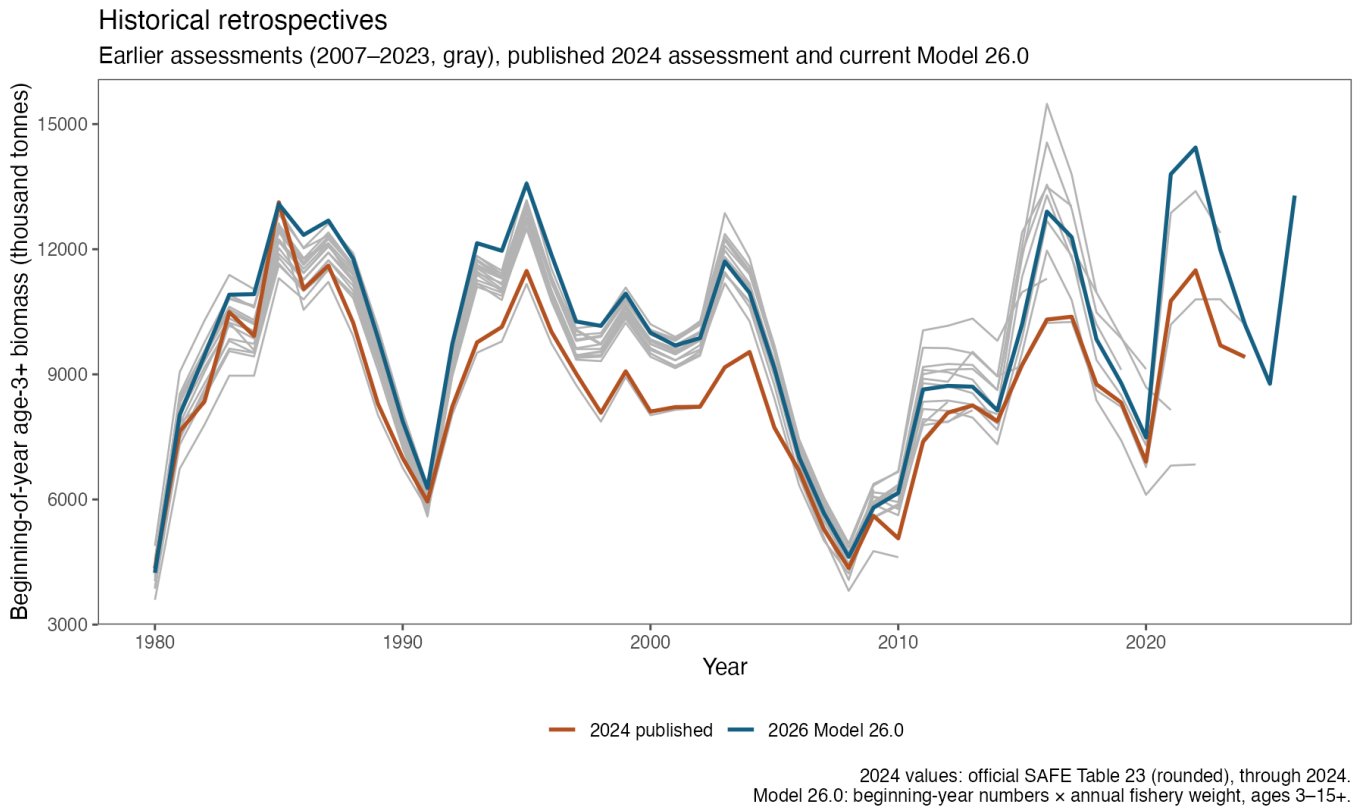



Figure 53: Historical retrospectives of beginning-of-year age-3+ EBS pollock biomass, updating Figure 63 of the 2024 SAFE. Gray lines show archived 2007–2023 assessments; orange shows the official 2024 SAFE Table 23 estimates through 2024; blue shows Model 26.0 estimates through 2026. Biomass sums beginning-of-year numbers at ages 3–15+ multiplied by the corresponding annual fishery weights at age, using the Model 26.0 fit. Historical series retain their original definitions and available endpoints, which may include next-year estimates. The 2024 tabulated values are rounded. No new assessment fit was produced in 2025. These are point estimates from successive assessments, encompassing changes in data, methods and biological inputs.

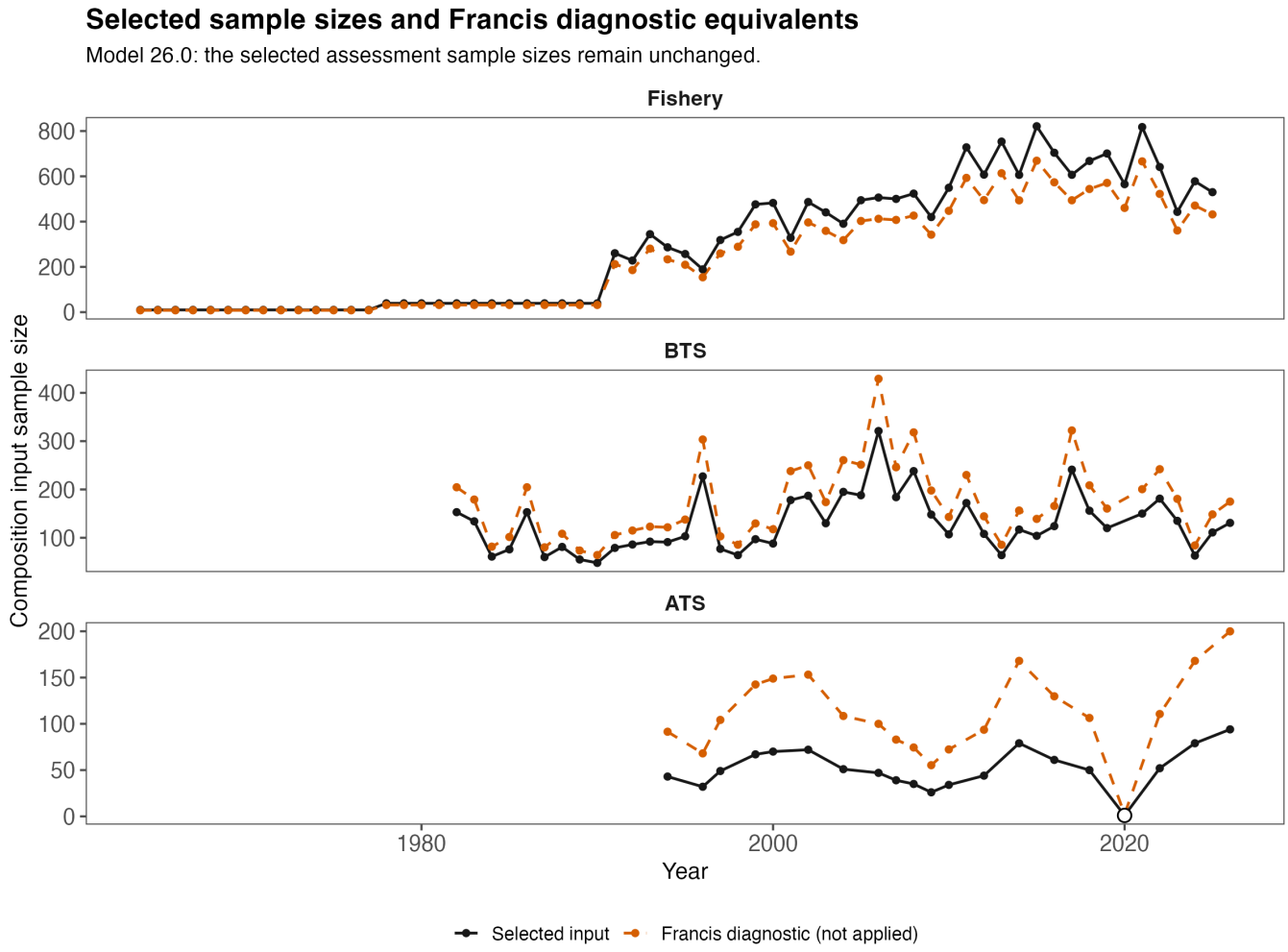



Figure 54: Selected annual composition sample sizes and their Francis diagnostic equivalents for the Model 26.0 fishery, BTS and ATS groups. This is a diagnostic comparison; no Francis reweighting is applied to the assessment inputs.

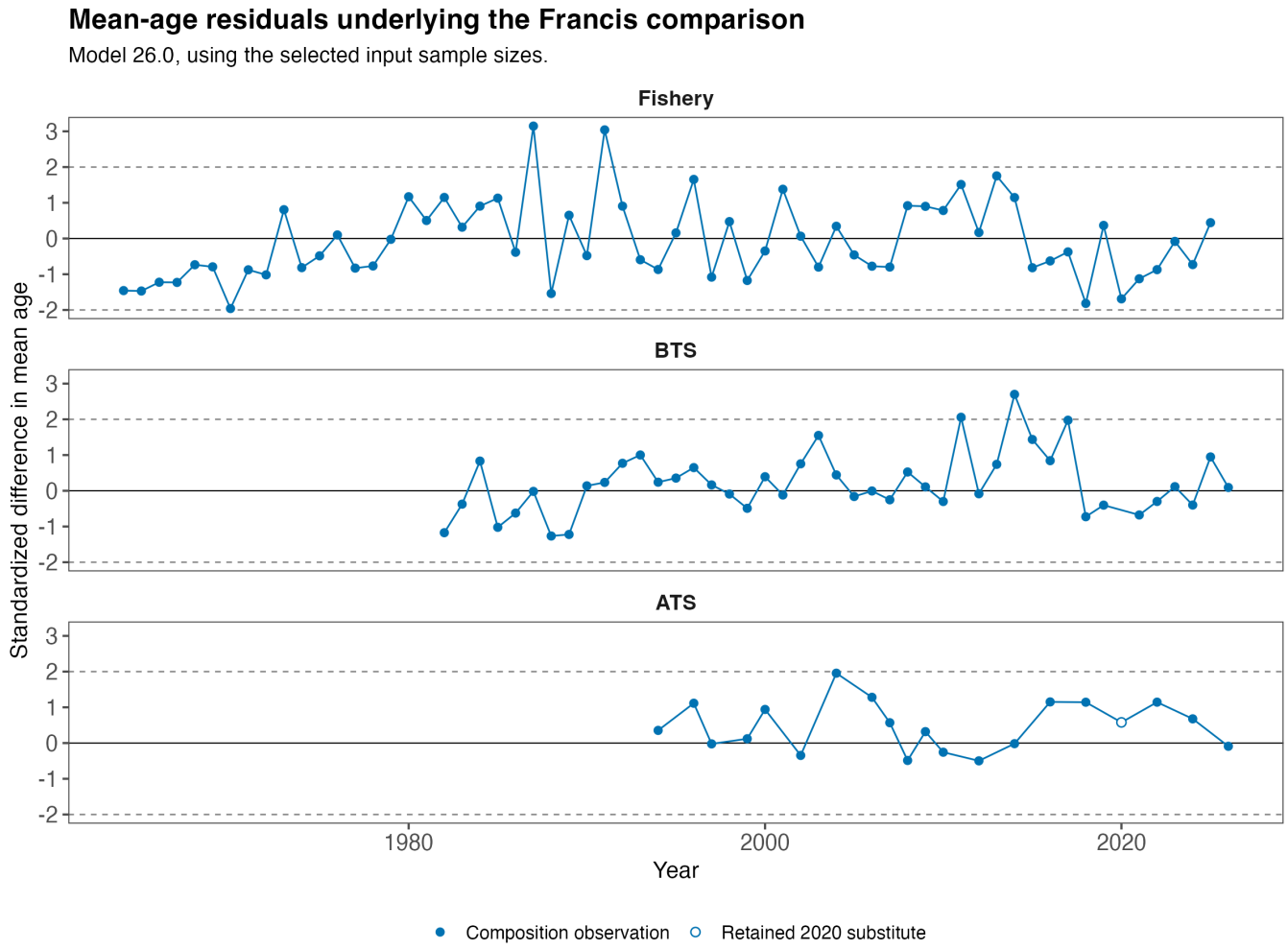



Figure 55: Standardized mean-age residuals used in the Francis diagnostic comparison for Model 26.0, with the selected input sample sizes. Positive values indicate older observed than predicted mean age. The original selected-data treatment and diagnostic formulas are retained.

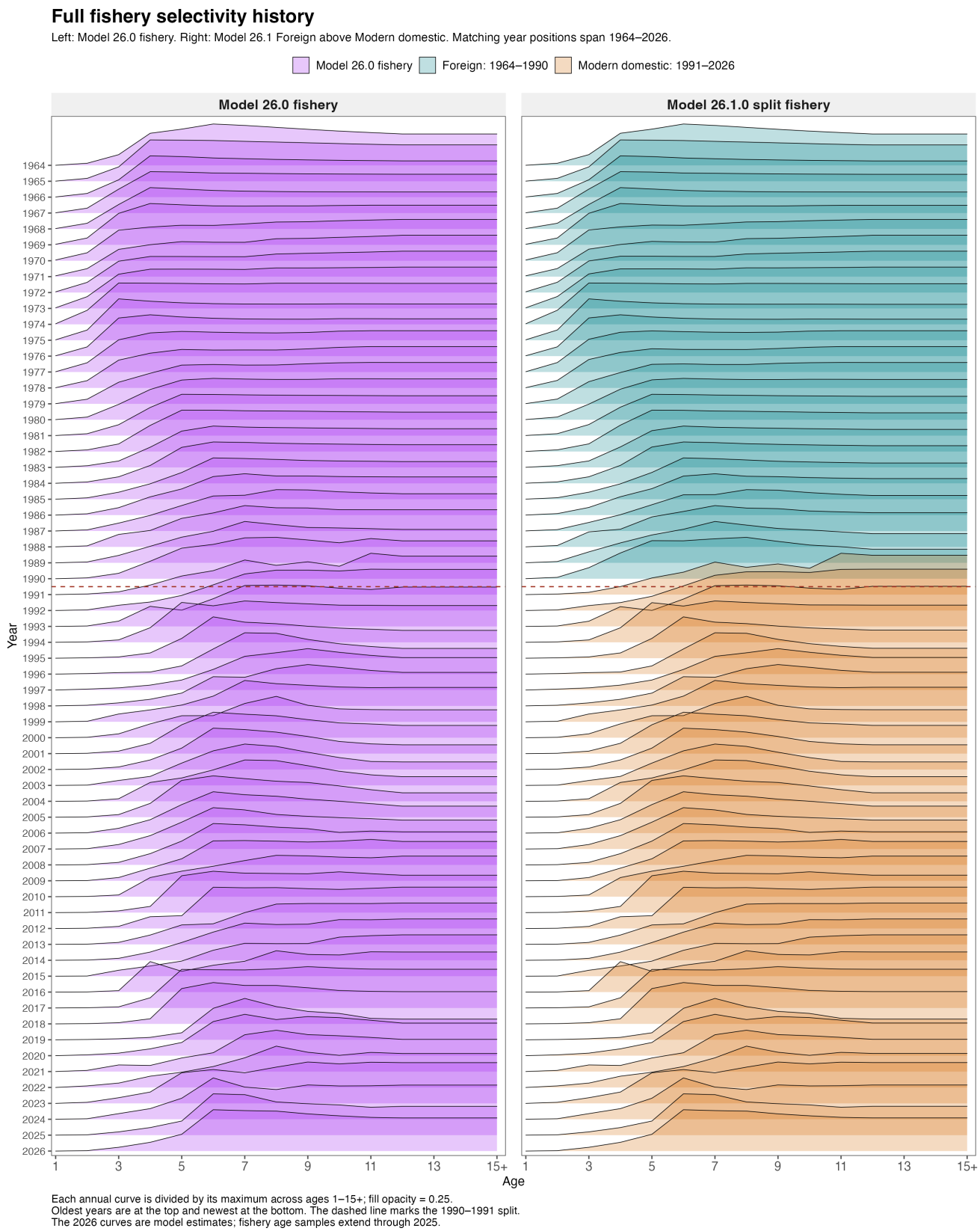



Figure 56: Annual fishery selectivity from Models 26.0 and 26.1.0. The split configuration separates Foreign and Modern domestic fisheries. Curves use the normalization recorded with the verified figure data.

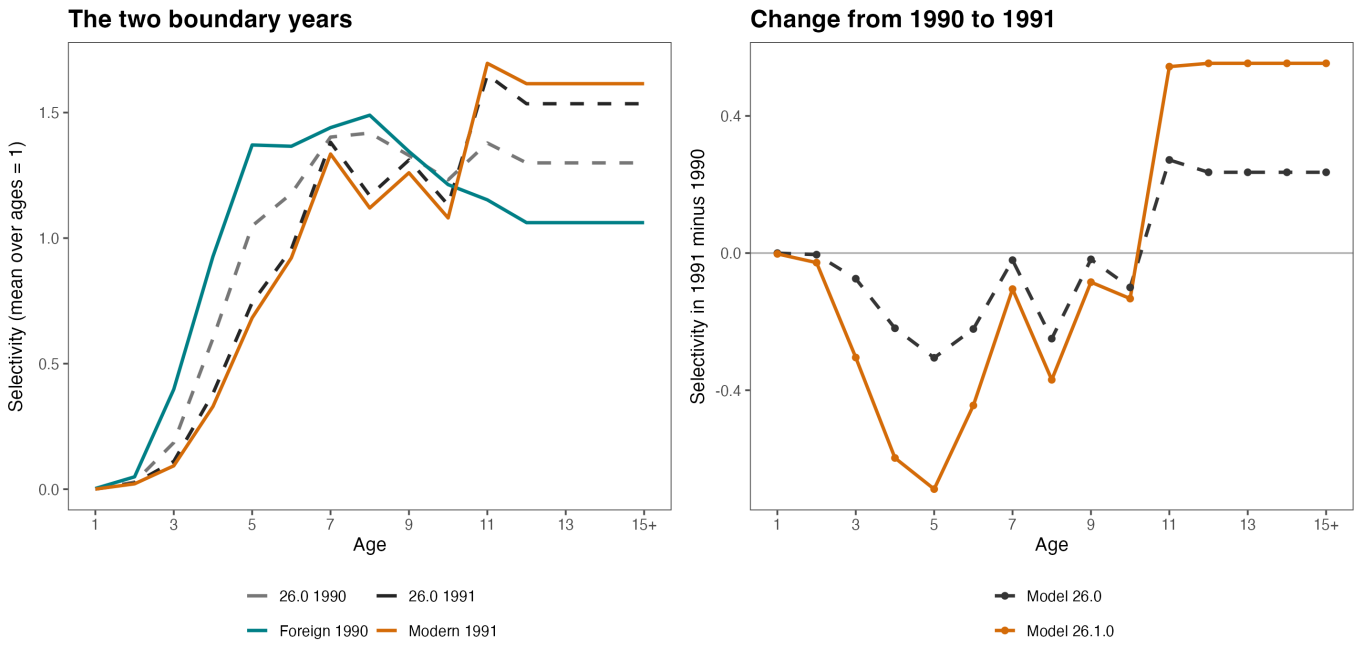



Figure 57: Fishery selectivity before and after the fleet split, with signed 1991-minus-1990 differences by age for Models 26.0 and 26.1.0. The figure preserves the existing boundary-year and normalization definitions.

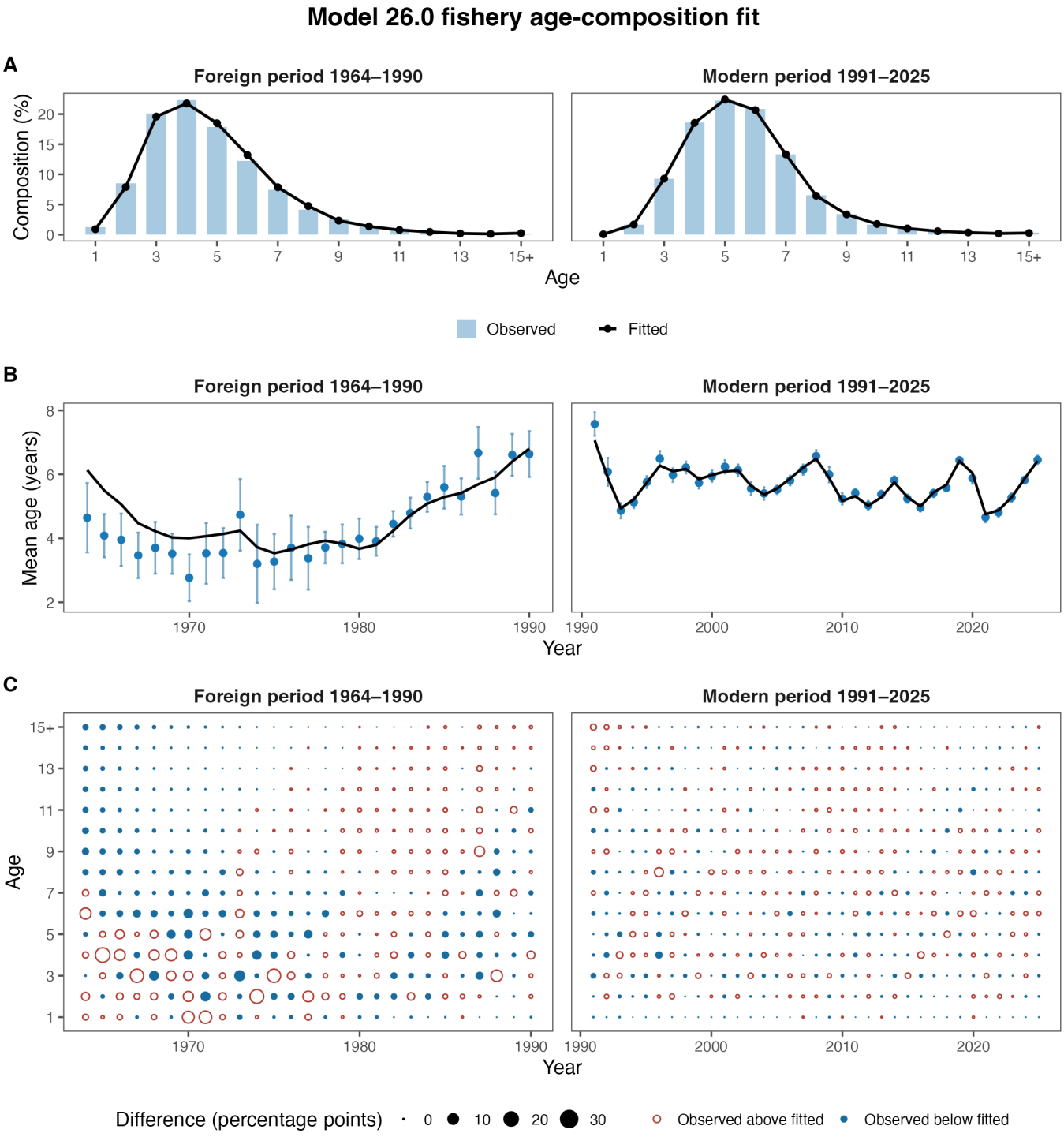



Figure 58: Model 26.0 fishery age-composition fits. We displayed the Foreign period (1964–1990, left) and Modern domestic period (1991–2025, right) separately for comparison with Model 26.1.2, although Model 26.0 fitted one fishery. Row A compared sample-weighted observed age proportions (bars) with fitted proportions (black lines). Row B compared observed annual mean ages and approximate 95% sampling intervals with fitted mean ages. Row C showed observed minus fitted proportions in percentage points; open red circles indicated observed proportions above the fit, and filled blue circles indicated proportions below it. Bubble areas represented difference magnitudes on a scale shared with the Model 26.1.2 figure. Mean ages treated the 15+ group as age 15.

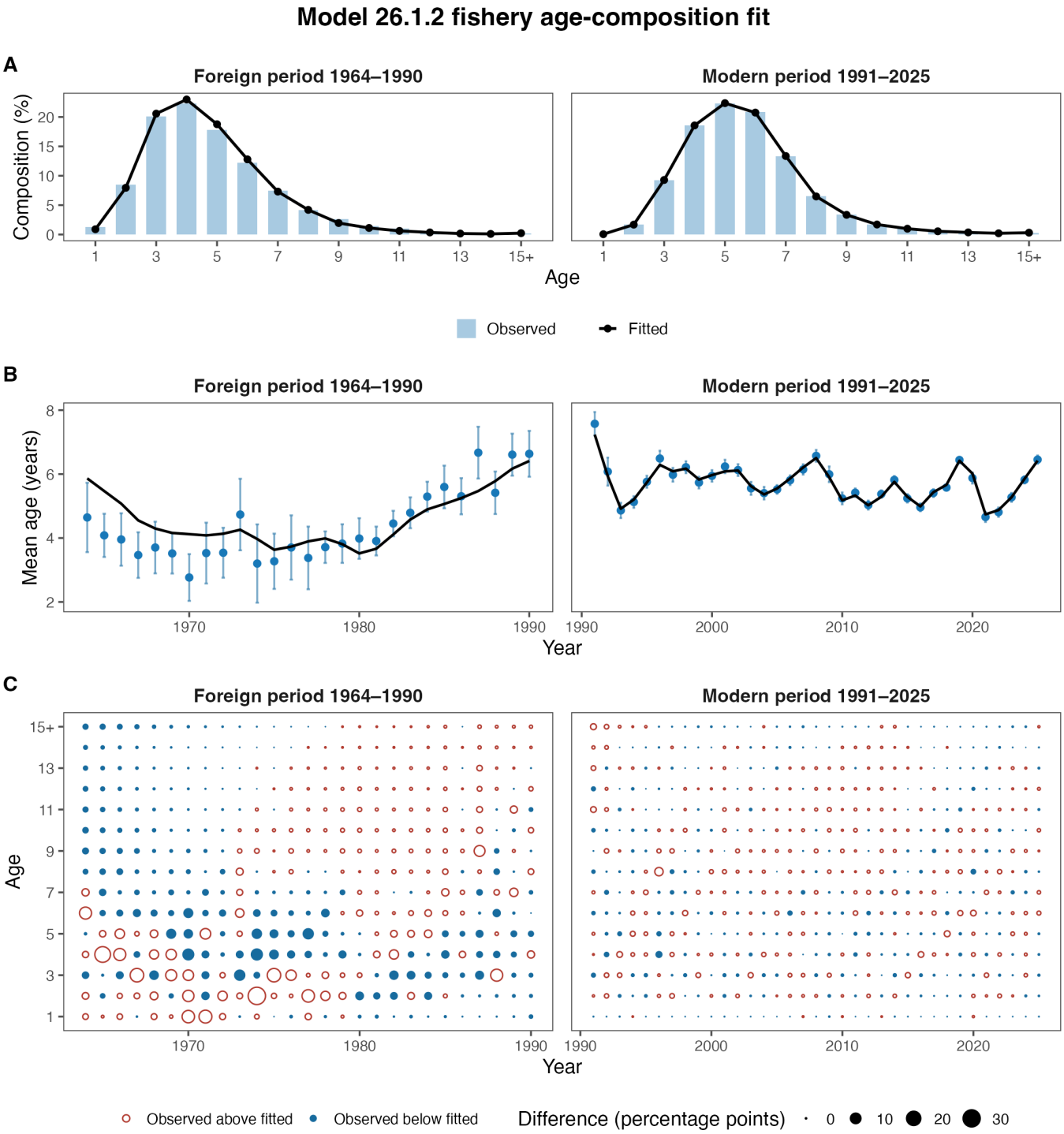



Figure 59: Model 26.1.2 fishery age-composition fits, using the fixed-process-SD conditional fit. Columns showed the Foreign and Modern domestic fisheries. Row A compared sample-weighted observed and fitted age proportions; row B compared observed annual mean ages and approximate 95% sampling intervals with fitted mean ages; row C showed observed minus fitted proportions in percentage points. The figure used the same observations, likelihood weights, age ranges and bubble scale as the Model 26.0 figure. The mean-age intervals described observation sampling uncertainty, and the fitted curves showed point estimates. Mean ages treated the 15+ group as age 15.

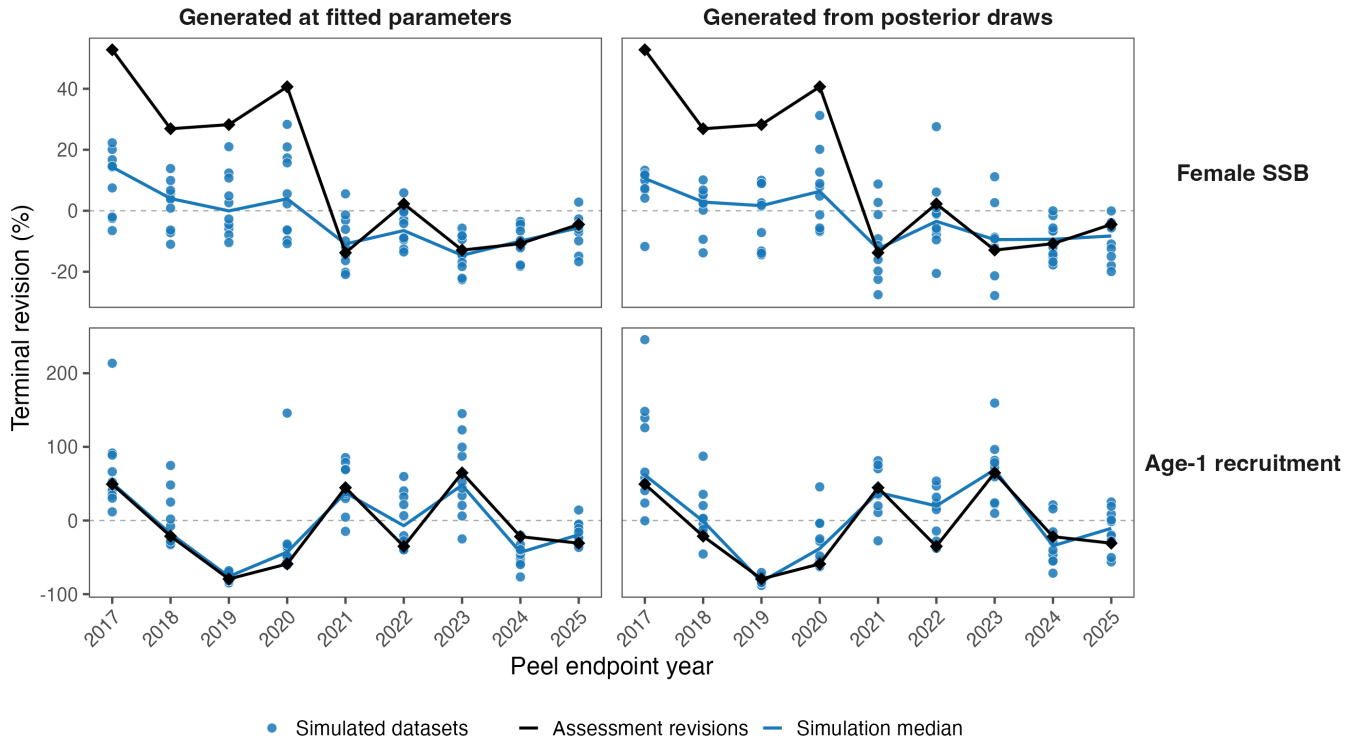



Figure 60: Conditional simulation-retrospective revisions for Model 26.0. Columns show ten datasets generated at the fitted parameters (left) and ten generated from regularized posterior parameter draws (right). Rows show female spawning biomass and age-1 recruitment. Blue points identify individual simulated datasets, blue lines join their medians, and black diamonds and lines show observed-data revisions. For a peel ending in year y, revision is 100 × [peel estimate in year y / its own full 2026 fit estimate for year y − 1]. Both columns use the same vertical scale within a quantity.

Figure 61: Native SparseNUTS pairs diagnostic for the six parameters with the lowest bulk effective sample size in the newly fitted regularized Model 26.0 posterior. Lower panels show all 5,000 retained draws in black, the saved regularized native MAP as a red dot, and the 95% bivariate Normal approximation from the matching full inverse augmented Hessian as a red ellipse. Upper panels report pooled posterior Pearson correlations. Diagonal panels show four retained chain traces, bulk ESS and R-hat. Parameters retain their native coordinate scales. SparseNUTS names the reference field mle; here it contains the regularized MAP, separate from the production penalized-ML fit.

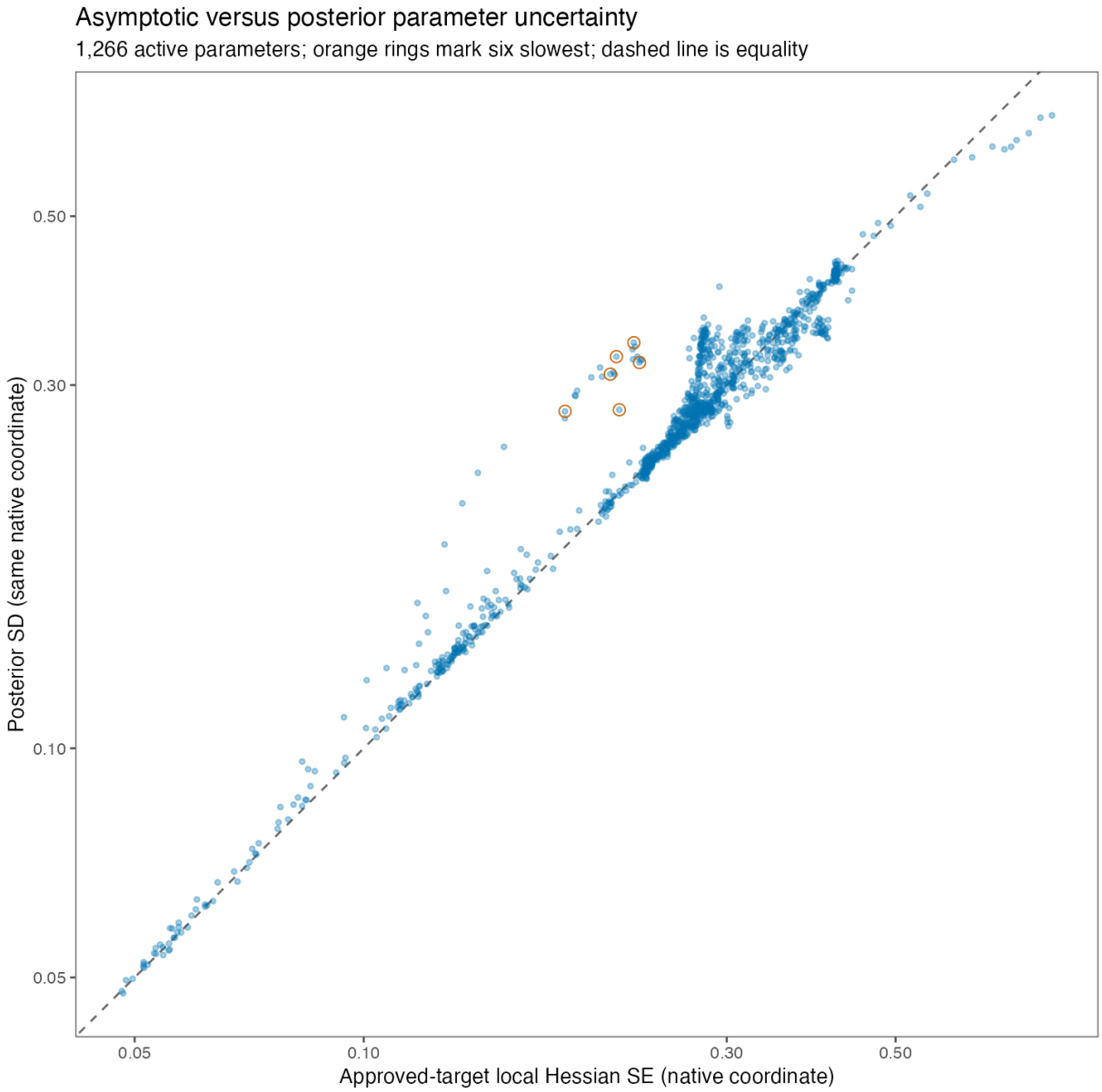



Figure 62: Matching native-coordinate local Hessian SE and posterior SD for all 1,266 active parameters under the same approved regularized target. Both axes use logarithmic scales; the dashed line denotes equality. Rings identify the six slowest parameters, with posterior-SD/local-SE ratios 1.28–1.52. Coordinate units differ among parameters. MCMC standard errors quantify simulation precision and are separate from the parameter uncertainties shown here.

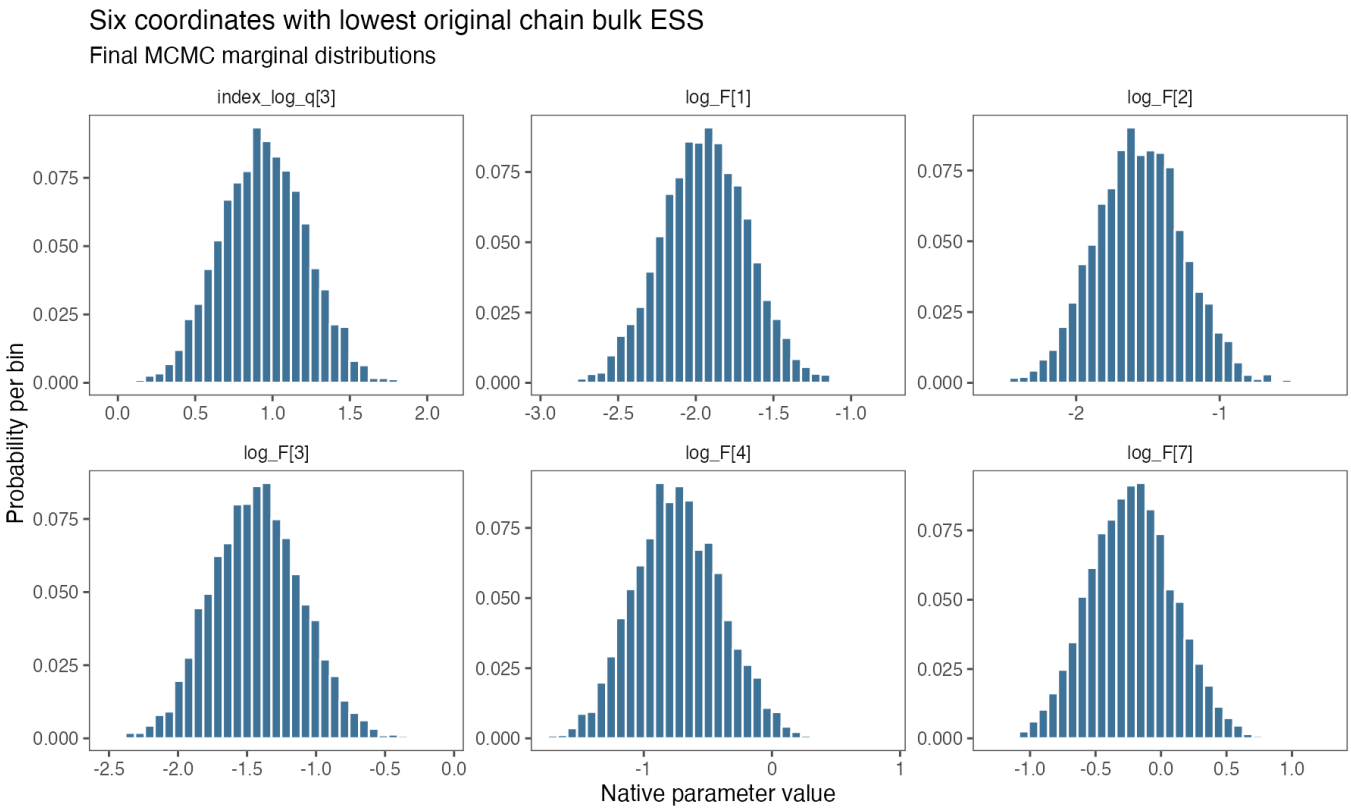



Figure 63: MCMC posterior histograms for the six native coordinates with the lowest bulk ESS in the original chains. All 5,000 saved parameter vectors contribute. Bar heights show probability per bin; the panels retain parameter units. The identity of these coordinates follows the original-chain diagnostic.

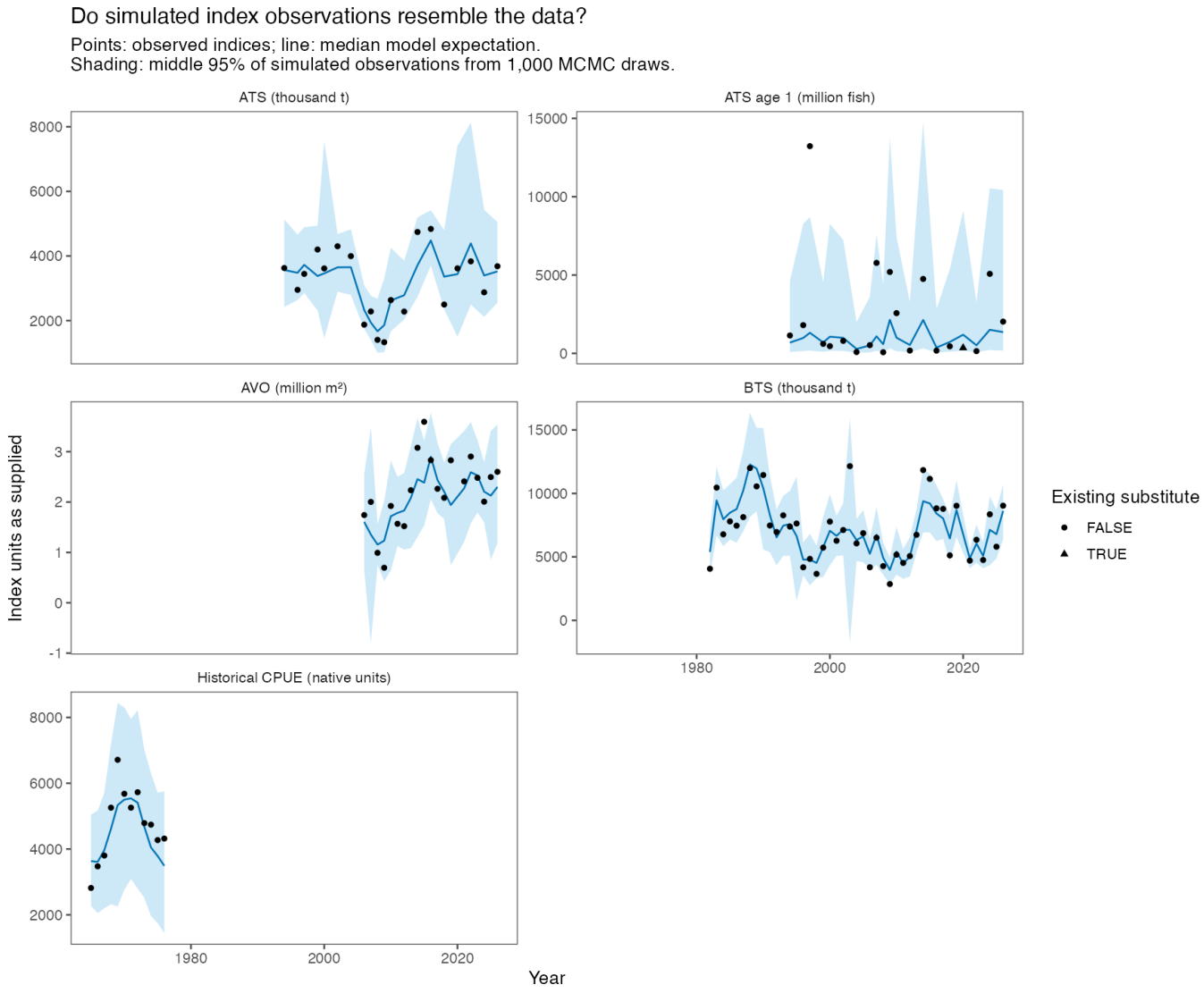



Figure 64: Index checks compare observed survey and historical catch-rate indices with simulated observations generated from 1,000 MCMC draws. Each draw supplies a model expectation; random observation errors are then added using the same error assumptions used to fit each index, including the relationships among BTS errors. Lines show median expectations and shading covers the middle 95% of simulated observations at each year. Circles are observed indices and triangles identify existing substitutes. Some simulated values are negative because the fitted Normal error distributions allow them. The BTS overall mismatch measure exceeds its observed value in 0.2% of simulations, indicating that the model rarely reproduces this feature of the BTS data. Units differ among panels. Biological inputs remain fixed.

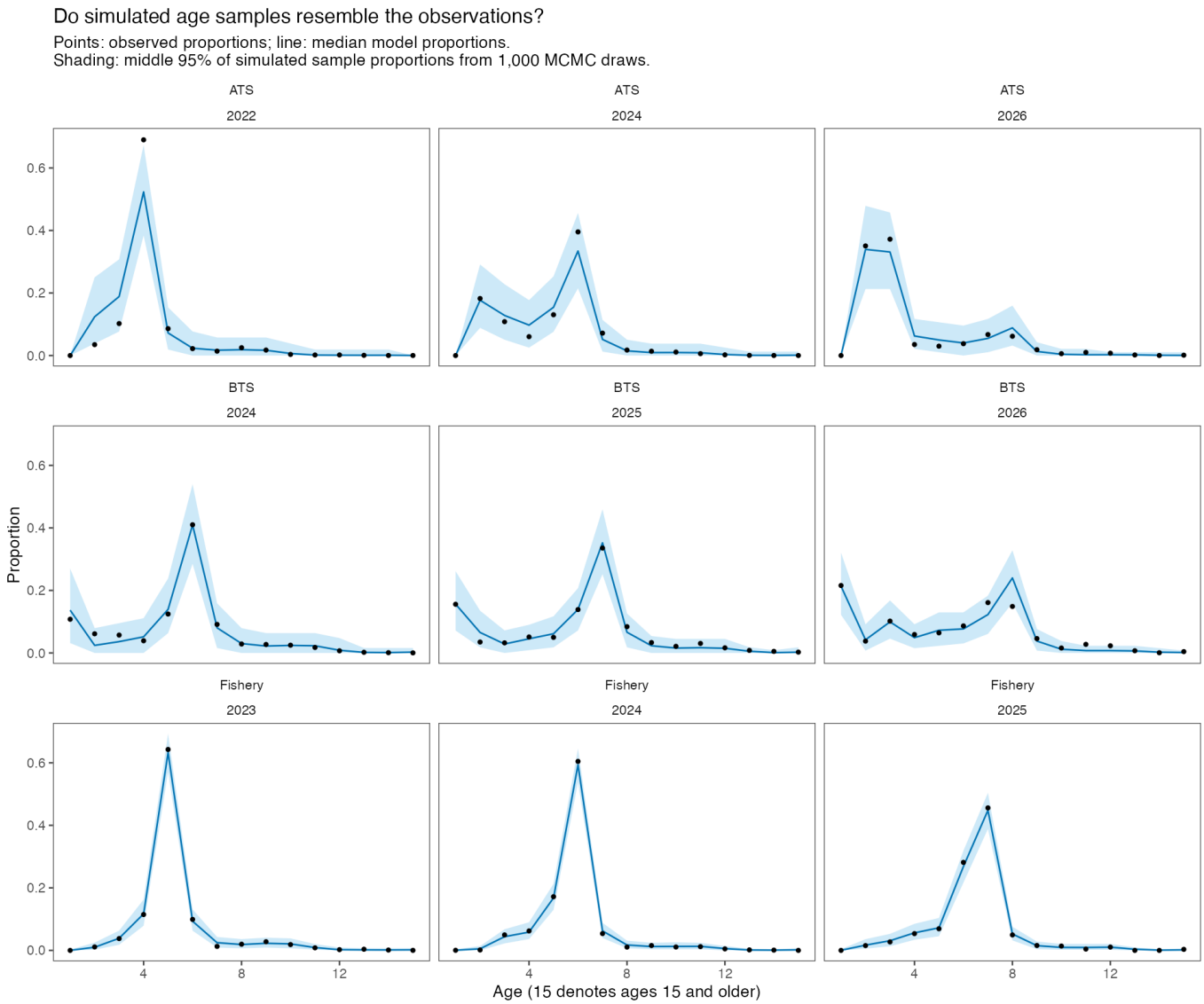



Figure 65: Age-composition checks compare the latest three observed samples from each data source with simulated samples generated from 1,000 MCMC draws. For each draw, fish are randomly assigned to age groups using the model-predicted age proportions and the sample size used in fitting, rounded to a whole number. Lines show median model proportions, shading covers the middle 95% of simulated sample proportions at each age, and points show observed proportions. The shading includes parameter uncertainty and the assumed sampling variation. This simulation approximates the age-composition fitting formula, which adds a small constant to age proportions when calculating fit. The 2020 ATS substitute is excluded. Age 15 includes all older fish. Age conversions, weights and other biology remain fixed.

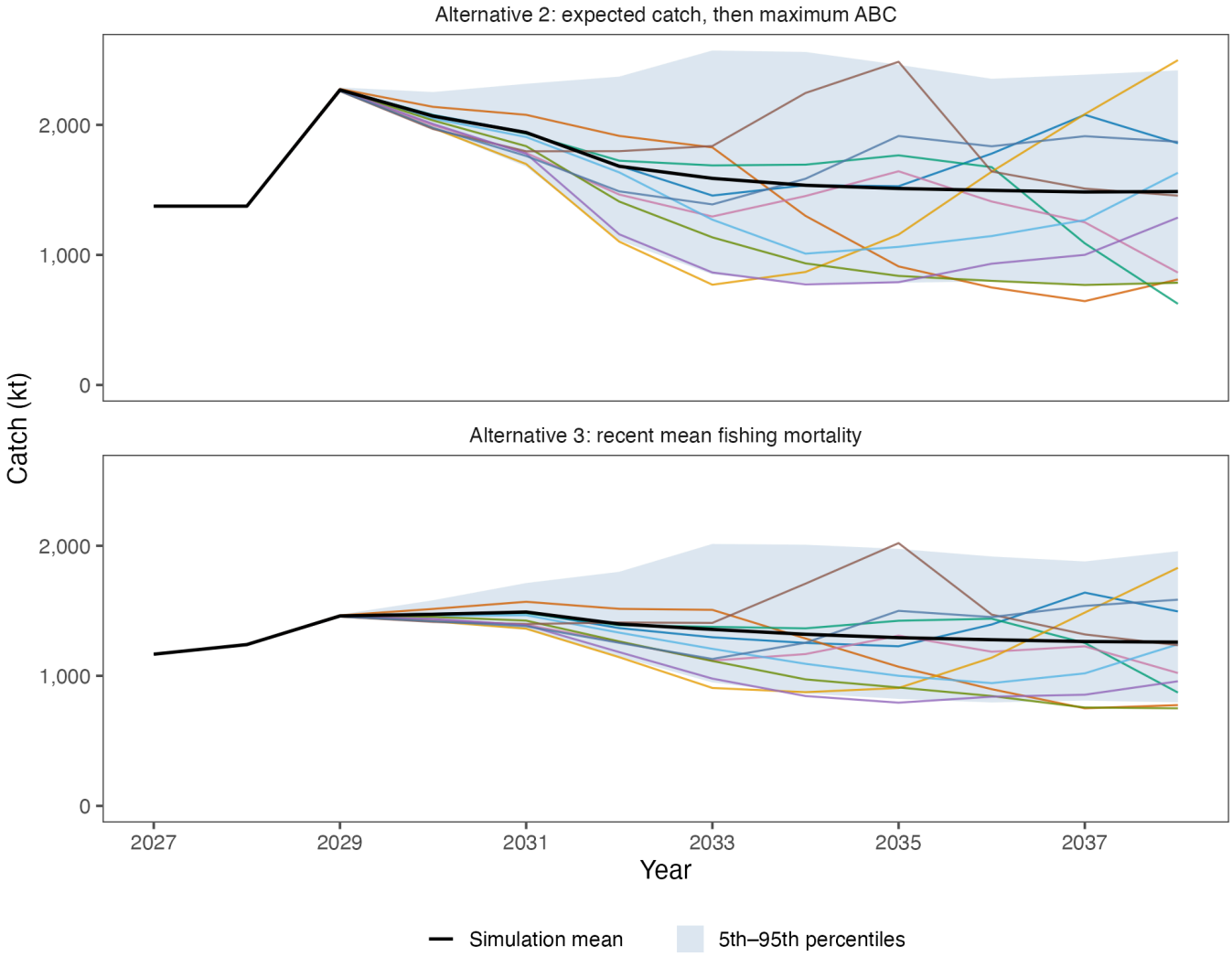



Figure 66: Catch projections for Model 26.0 under Alternative 2 (expected catch, top) and Alternative 3 (recent mean F, bottom). Black lines show simulation means, ribbons show the 5th–95th percentiles from 1,000 future-recruitment simulations, and thin colored lines show ten matched trajectories. Colors identify the same recruitment realization in both alternatives and the spawning-biomass figure. Alternative 2 fixes catch at 1,375 kt in 2027 and 2028, then applies the maximum-ABC rule; its paths coincide in the prescribed-catch years. The ribbons represent future recruitment variation; the advice tables retain historical-mean recruitment.

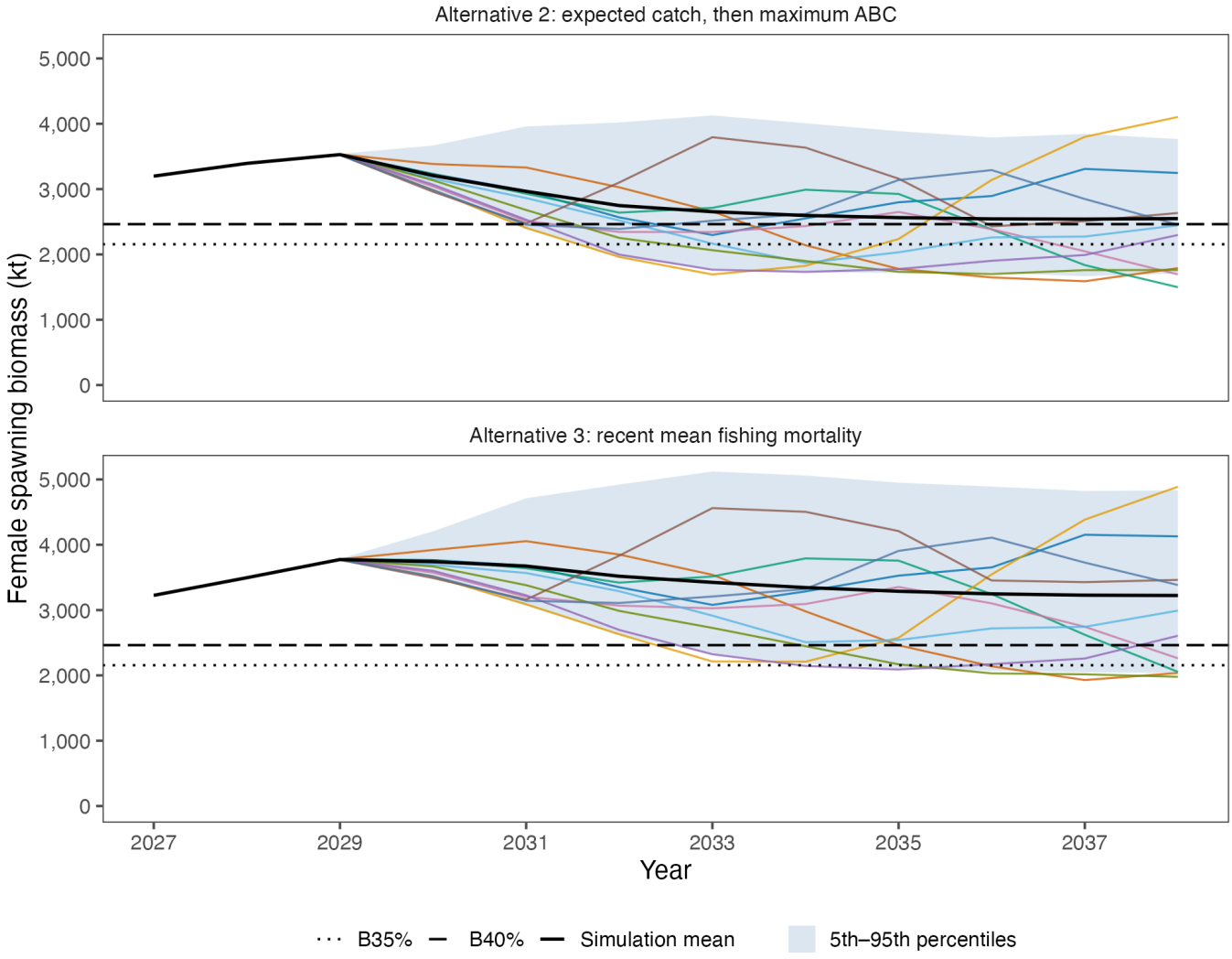



Figure 67: Female spawning-biomass projections for Model 26.0 under Alternative 2 (top) and Alternative 3 (bottom). Black lines show simulation means, ribbons show the 5th–95th percentiles from 1,000 future-recruitment simulations, and thin colored lines show the same ten matched trajectories as the catch figure. Future recruitment follows an inverse-Gaussian distribution fitted to the 1978–2023 age-1 entries. These ranges include future recruitment variation only. Dashed B40% and dotted B35% lines mark the primary projection reference points.

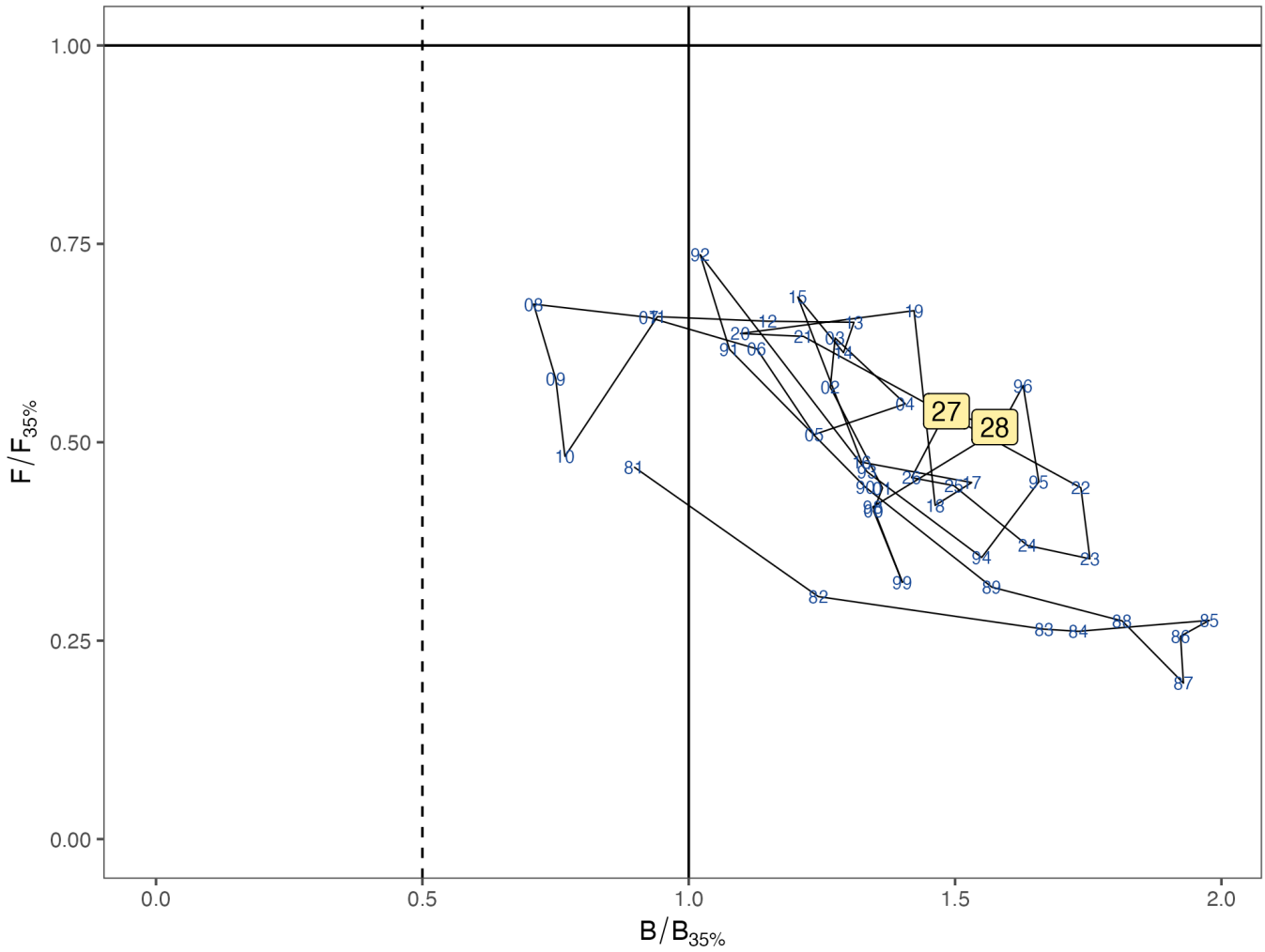



Figure 68: Historical trajectory of Model 26.0 female spawning biomass (B) and maximum fishing mortality at age (F), 1981–2026, with deterministic Alternative 2 projections for 2027–2028. Ratios use the current primary Tier 3 reference points, B35% = 2.155 million t and F35% = 0.658 per year, held constant across years. Lines connect consecutive years; two-digit labels identify years, and boxed labels with yellow shading identify projections. Solid lines mark ratios of one; the dashed vertical line marks half B35%. Projected catch is 1.375 million t in each year.

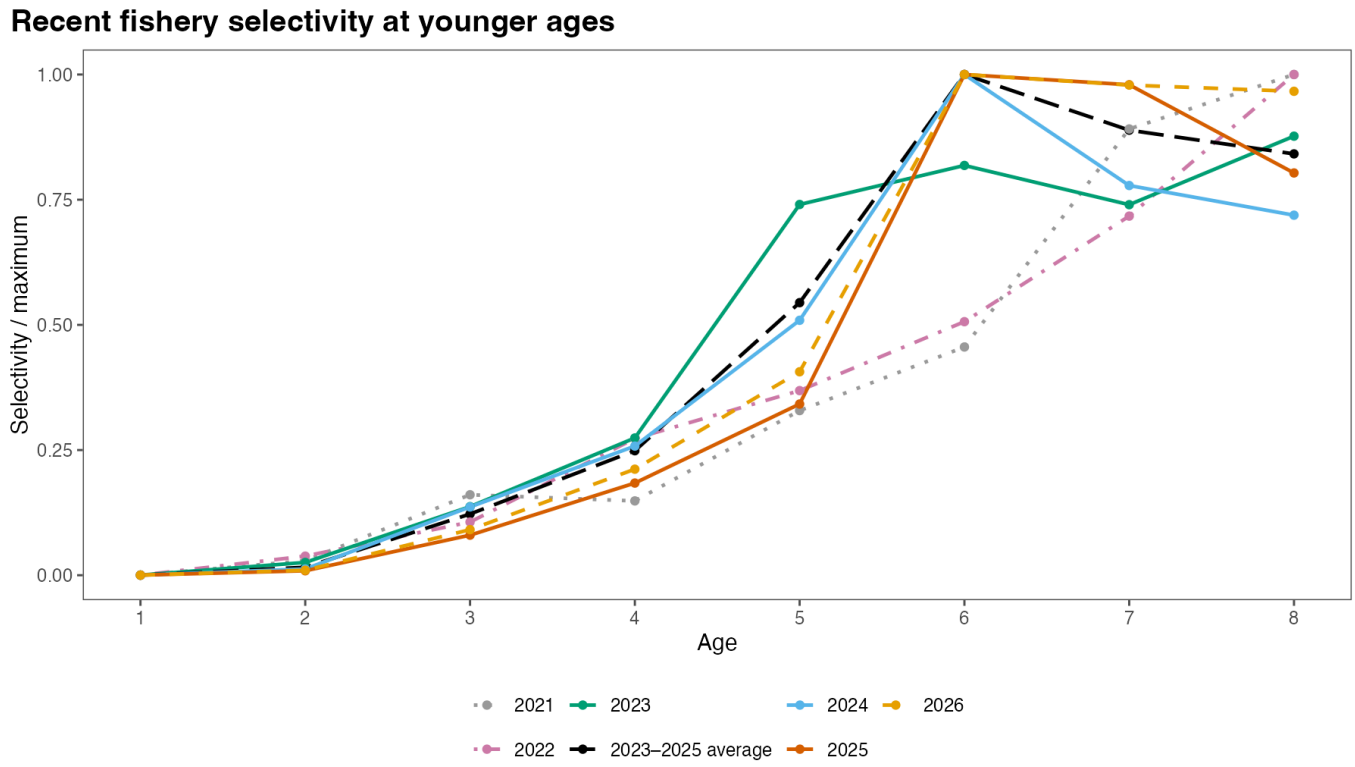



Figure 69: Recent normalized fishery selectivity curves from the Model 26.0 and the proposed raw-curve averaging reference. Half-maximum crossings and curve-shape summaries use the existing verified definitions.

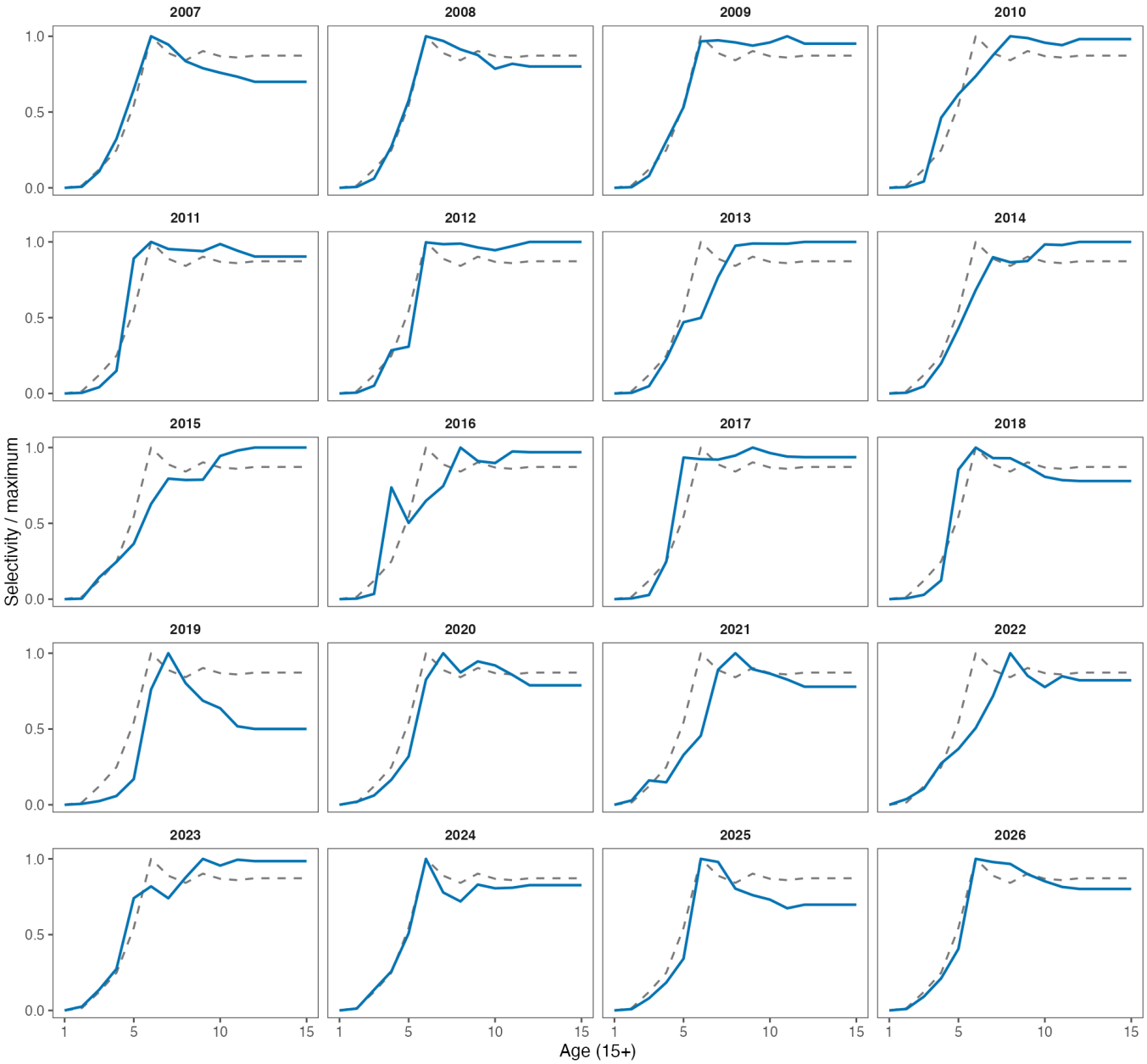



Figure 70: Annual Model 26.0 fishery selectivity curves for 2007–2026, each compared with the proposed 2023–2025 raw-curve average rescaled to maximum one. The inferred 2026 curve retains its distinction from directly informed earlier-year curves.

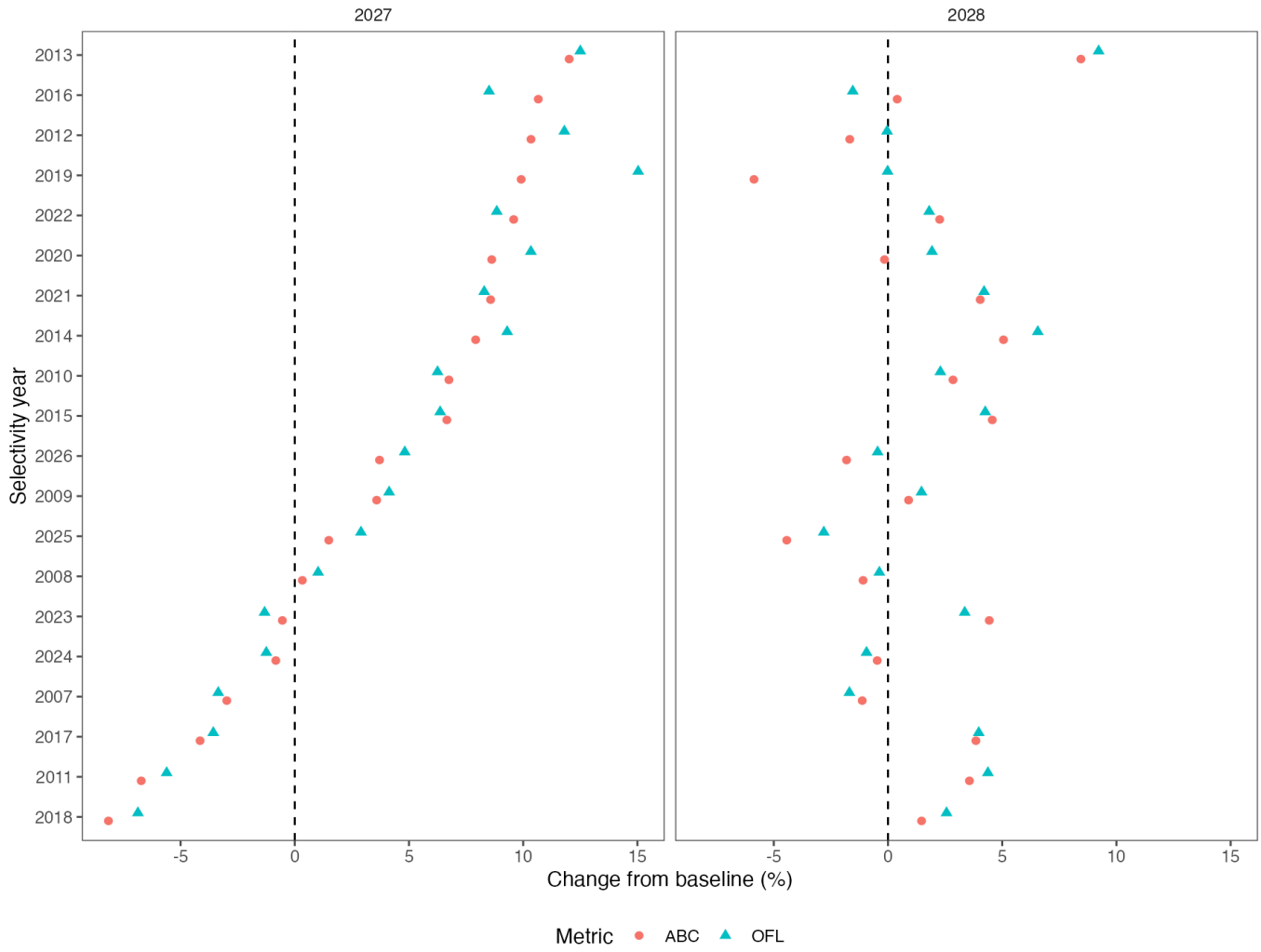



Figure 71: Changes in maximum ABC and OFL under the annual future-selectivity curves relative to the matching Model 26.0 baseline. Rows retain the recorded 2027 maximum-ABC ordering. These are projection selectivity sensitivities conditional on the same fitted source and projection design.

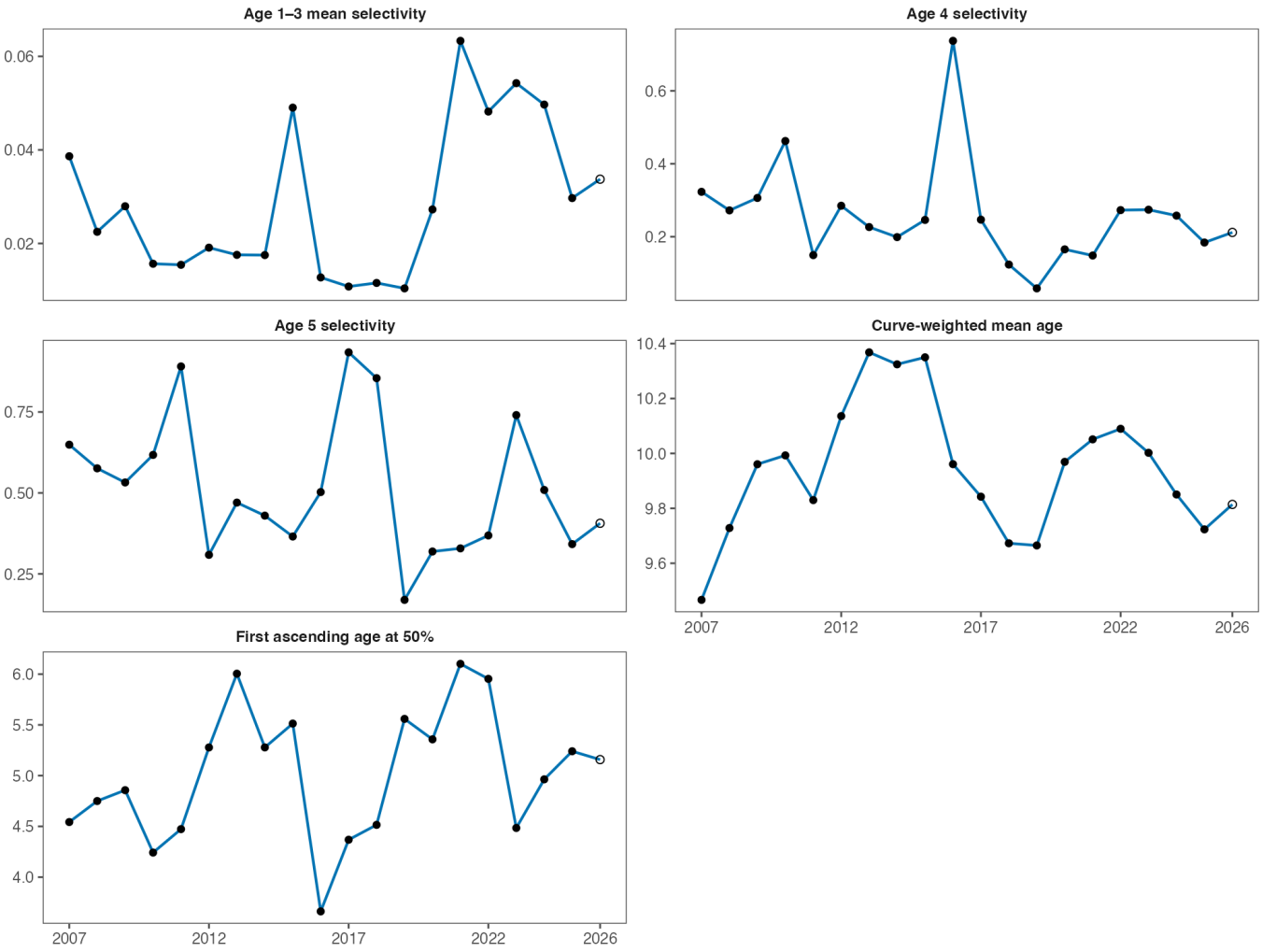



Figure 72: Annual younger-age selectivity measures for 2007–2026 from the Model 26.0. The inferred 2026 point is distinguished. Curve-weighted mean age is a shape measure under equal age weights, distinct from a population-weighted selected age.

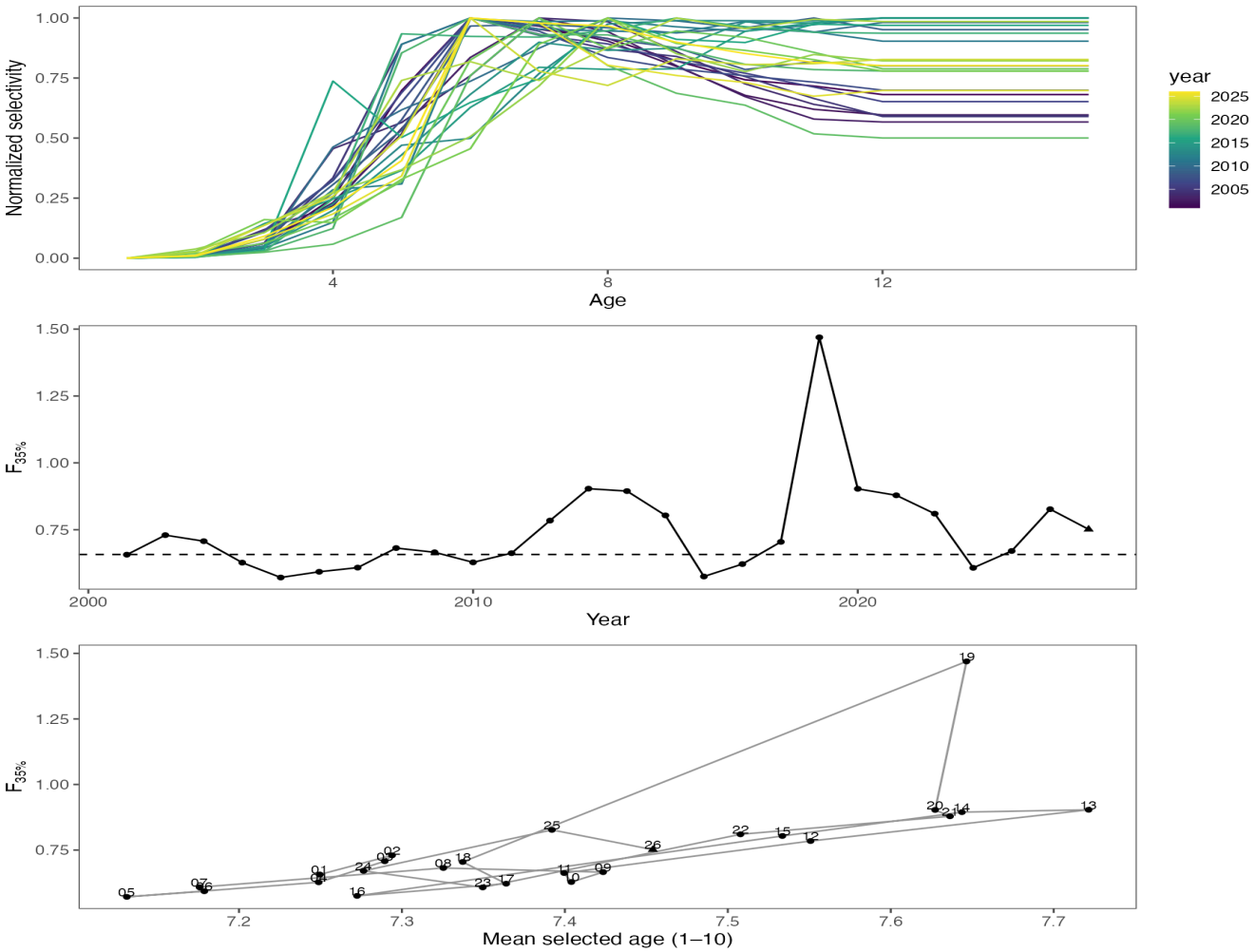



Figure 73: Annual Model 26.0 fishery selectivity and fully selected F35%, a proxy for FMSY, for 2001–2026. Mean selected age uses ages 1–10; SPR uses ages 1–14 and 15+, fixed mortality, maturity and the selected projection spawning weights, and quarter-year spawning timing. Curves are normalized to maximum one. The 2023–2025 mean-curve reference and inferred 2026 curve remain separately identified. The historical 2024 figure used FMSY, a different quantity.

# Appendices

## Appendix A
## SSC and Plan Team Requests and Responses

The Plan Team requests and assessment responses are summarized in Table A1.

Table A1 Responses to SSC and Plan Team Requests

| SSC and Plan Team requests | | | |
|---|---|---|---|
| ID | Recommendation | Response | Response status |
| PT-01 | Bring Model 26.0 forward for management use in November and present Model 23.0 for continuity and comparison during the transition to Rceattle. | Model 26.0 estimates, uncertainty, diagnostics, Tier 3 projections and Model 23.0 comparisons are presented. Regularized posterior summaries, predictive checks and observed and simulated-data retrospectives are included. Model 26.1.2 is a sensitivity comparison conditional on fixed Foreign and Modern process SDs of 0.1235 and 0.582. Risk categories are Level 1. | Assessment results and sensitivity comparisons presented |
| PT-02 | Evaluate alternative treatments of age-1 BTS selectivity, including the selectivity penalty and a time-invariant alternative, to assess their influence on the retrospective pattern. | The author proposes no additional age-1 BTS selectivity sensitivities for this assessment. Observed and simulated-data retrospective results are presented; these results do not separately identify the effect of an alternative age-1 selectivity treatment. | No additional age-1 selectivity tests proposed |
| PT-03 | Extend the Model 26.0 self-test with retrospective peels on simulated datasets generated from the fitted model, and examine retrospective patterns in time-varying selectivity estimates. | Two sets of ten simulated datasets use the fitted parameters and ten regularized posterior draws, respectively. Each dataset has a full-2026 fit and nine peels ending in 2017–2025. Observed spawning-biomass rho (+0.121) exceeds every simulated value in both sets. Posterior diagnostics and predictive checks are also presented. The simulations assess population revisions; they establish no single retrospective cause. | Paired simulation retrospectives and posterior predictive checks presented |
| PT-04 | Present retrospective results from Models 23.0 and 26.0 side by side in November. | Matched retrospective runs for Models 23.0 and 26.0 were completed and compared side by side. Both showed similar spawning-biomass retrospective patterns, with nine-peel Mohn's rho of +0.156 and +0.143, respectively (positive values indicate higher estimates in the shorter fits). See the side-by-side comparison and numerical summaries. | Completed; requested comparison provided |
| PT-05 | Further examine the influence of BTS index volatility and other potential data or ecosystem-related causes on the retrospective pattern. | The author proposes no additional BTS data-influence tests for this assessment. Model and cumulative-data comparisons, observed and simulated-data retrospectives, and posterior predictive checks are presented. The BTS mismatch measure is as large as observed in 0.2% of predictive simulations. These results identify model-data mismatch but establish no ecosystem cause. | Diagnostics presented; no additional BTS influence tests proposed |

| SSC and Plan Team requests | | | |
|---|---|---|---|
| ID | Recommendation | Response | Response status |
| PT-06 | Clearly document age-composition weighting in Model 26.0 and prioritize further evaluation of data weighting given patterns in the OSA diagnostics. | Age-composition information weights and likelihood multipliers are documented by data source and year. Model 26.0 and fixed-SD conditional Model 26.1.2 OSA and mean-age comparisons retain the selected sample sizes. Model 26.1.2 residuals condition on fitted selectivity states and exclude parameter uncertainty. Diagnostic equivalent weights are not applied as new inputs. Alternative-weight fits are outside the scope of this comparison. | Weighting and conditional diagnostics presented; alternative-weight fits remain separate |
| PT-07 | Continue development of material that could contribute to a future EBS pollock Ecosystem and Socioeconomic Profile, as resources allow. | A draft was developed in September and further work is included here following recommended format and guidance. | Draft has been developed and is included as Appendix B. |

*Note.* Responses to the September 2026 BSAI Groundfish Plan Team recommendations. Status distinguishes completed work, decisions about further tests and development of the ESP draft.

### Earlier SSC comments and response plan

The matrix below preserves the October 2025 SSC response plan as reproduced in the September 2026 development source. It records proposed actions at that stage; completion and outstanding work are identified in Section 1.4 and the response registers. Its RTMB and Model 25.0 entries are historical planning context rather than the software selection. The September 2026 Plan Team direction establishes Model 26.0 and Model 23.0 as the November main and comparison models; the RTMB and SPoRC work remains part of the development record.

The earlier SSC comments and response plan are reproduced in Table A2.

Table A2 Earlier SSC Comments and Response Plan

| SSC comment | Response |
|---|---|
| Use the 2024 base model as the source for an RTMB implementation, with routine data updates. | Use the accepted Model 23.0 (ADMB) configuration as the scientific source. First check the direct RTMB translation as Model 23.1, then use Model 25.0 (RTMB) for the assessment line begun in 2025. List each new data set and each model change separately. Save the model version, run files, diagnostics, projections, and management quantities used in the report. |
| Use the updated Groundfish Assessment Program estimates of survey biomass and age composition. | Add the sdmTMB biomass estimates and tinyVAST age compositions as revised survey data. Record when the estimates were received, the software version, model settings, and input files. Compare the old and new survey series and show whether the change affects the assessment results. |
| Use FT-NIRS age compositions for the 2025 fishery and surveys. | Document how the FT-NIRS ages were produced, screened, grouped, and assigned to fishery and survey data. Record sample sizes and aging-error matrices. Check the final model input against the source data before fitting the assessment. |
| Compare traditional microscope ages with FT-NIRS ages and show the effects on model results and management quantities. | Run the same assessment with traditional ages, FT-NIRS ages with one aging-error matrix, and FT-NIRS ages with fleet-specific aging-error matrices. Compare cohort patterns, fits to fishery and survey ages, residuals, recruitment, spawning biomass, reference points, ABC, and OFL. Save the full comparison in the assessment appendix and at a permanent web address. |
| Revisit the revised AVO index after the new uncertainty analysis has been reviewed. | After review of the final uncertainty method, compare the revised point estimates and uncertainty with the AVO series. Show their effect on model fit, stock estimates, and management quantities before proposing a change to the base model. |
| Develop a plan with the BSAI Groundfish Plan Team for moving beyond ADMB. | Use the model history and numbering plan above. Compare RTMB, Rceattle, and SPoRC with the same data and a common set of tests. Ask the Plan Team to agree on the tests before selecting the model software for future assessments. Keep the accepted assessment model as the control until a replacement gives stable estimates, sound uncertainty, acceptable retrospective results, and reproducible projections. |

| SSC comment | Response |
|---|---|
| Examine the effect of including Russian catches. | First confirm the source, area, years, units, species identification, and uncertainty of the Russian catch estimates. Then run clearly defined catch scenarios. Explain the assumptions about stock movement and population boundaries and show the effects on stock size, reference points, ABC, and OFL. |
| Explore time-varying natural mortality, selectivity, age-composition weighting, and cohort and temperature effects on weight-at-age. | Study each topic separately. State the scientific question before fitting a model. For each change, show model fit, parameter uncertainty, retrospective results, projections, and effects on management advice. Keep poorly estimated or unstable models as research results. |
| Continue work with CEATTLE estimates of natural mortality and consider Rceattle for time-varying mortality. | Test time-varying natural mortality as a sensitivity analysis. Check whether the model can separate mortality changes from recruitment, selectivity, and survey effects. The full set of assessment needs will guide the choice of model software. |
| Explain how the recruitment penalty affects the estimated 2018 year class and why the likelihood profile favored a larger recruitment estimate. | Prepare a likelihood profile for the 2018 recruitment deviation. Show how the recruitment penalty and each data source change across the profile. Explain which parts of the model favor a larger year class and how the estimate affects spawning biomass, reference points, ABC, and OFL. |
| Clearly identify the models proposed for the assessment, including model numbers. | Use Model 23.0 (ADMB) for the accepted 2024 source, Model 23.1 (RTMB) for the direct translation check, Model 25.0 (RTMB) for the line begun in 2025, and Model 26.0 (Rceattle) for the line begun in 2026. Use the same short names in the report, presentations, run folders, tables, and archive. For every model, list its purpose, parent model, data set, model changes, software version, and review status. State clearly which model is recommended for management advice. |
| Archive the October and December SAFE documents and presentations. | Archive the October 2025 material that exists and add a short note explaining that the funding lapse prevented the November and December SAFE package. For future reviews, save the Quarto source, rendered reports, presentations, model files, data list, software versions, and file checksums in one dated release. |

*Note.* Responses to the October 2025 SSC comments on EBS pollock. The SSC comments are summarized from pages 27 to 29 of the final October 2025 SSC report.

# Appendix B
# Ecosystem and Socioeconomic Profile of Eastern Bering Sea Walleye Pollock

This draft Ecosystem and Socioeconomic Profile (ESP) for EBS pollock brings together substantial work completed to date and provides a starting point for future ESP Team discussions. An AI agent helped compile and coordinate material from disparate information sources and research activities. Over the next year, the EBS pollock ESP Team will formalize its membership and begin meetings that engage a broader group of scientific experts to develop the profile. Please contact Jim Ianelli and Kalei Shotwell with questions or for further details about this document or the EBS pollock ESP.

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2. Responses to SSC and Plan Team Comments on ESPs in General
3. Responses to SSC and Plan Team Comments Specific to EBS Pollock
4. Introduction
5. Indicator Synthesis
6. Indicator Assessment
7. Data Gaps and Future Research Priorities
8. Conclusion
9. Literature Cited
10. Tables
15. Figures

## Executive Summary

The Ecosystem and Socioeconomic Profile (ESP) connects stock-specific environmental, biological, fishery, and community information with assessment interpretation and management questions. Its sequence of focus, synthesis, analysis, and communication provides a structured way to evaluate proposed indicators (Shotwell et al. 2023). This draft focuses initially on recruitment to age 1, then identifies processes affecting juvenile survival to age 3, adult growth and condition, survey interpretation, and fishery performance.

The ESP distinguishes two DSEM applications. The archived Rceattle pilot evaluates birth-year SST and age-1-year cold-pool proxies (Ianelli et al. 2026). A separate age-1 DSEM working paper analyzes 18 ecosystem series alongside an age-1 response estimated by the assessment. Regional age-0 energy density provides an additional measured-condition series (Page and Maselko 2026).

## Acceptable Biological Catch Considerations

## Candidate Predictive Indicators

Fall age-0 total energy, prey quality and abundance, and juvenile–predator overlap are priorities for testing. Field studies connect fall total energy with age-1 recruits per spawner, and laboratory experiments identify body size and energy reserves as determinants of survival during food deprivation (Heintz et al. 2013; Sogard and Olla 2000). Studies of pollock recruitment and spatial predation identify juvenile overlap with adult pollock and arrowtooth flounder as another important pathway (Mueter et al. 2006; Spencer et al. 2016).

The regional energy-density series contain 21 southeastern Bering Sea (SEBS) and 15 northern Bering Sea (NBS) annual means within 2002–2025, measured in kJ/g dry mass. The latest values, for 2025, are 19.23 and 20.84 kJ/g dry mass for SEBS and NBS, respectively. Both are lower than their 2024 observations; a 2026 energy-density observation is absent. Sampling metadata, observation uncertainty, and forecast evaluation remain prerequisites for a predictive role (Page and Maselko 2026).

## Contextual Indicators

Sea ice, seasonal temperature, cold-pool extent, plankton and euphausiids, age-0 abundance and condition, adult condition, weight-at-age, and predator overlap form the initial contextual set. The relevant biological question is whether each cohort encounters adequate prey, accumulates energy before winter, and survives subsequent predation. The poor 2007 cohort illustrates why a favorable temperature category alone can give an incomplete account: low summer production and poor age-0 condition occurred during a cold year (Gann et al. 2016).

The recorded 2026 physical and prey indicators show a mixed pattern relative to their own observed-period means. Sea ice and cold-pool extent are above those means; SST and bottom temperature are below. The open-water bloom indicator is lower, while chlorophyll a is higher. Spring small and large copepods and summer small copepods are below their respective means; summer large copepods are slightly above.

## Fishery Informed Indicators

Candidate indicators include age and size composition of catch, fishery weight-at-age, season and area of harvest, standardized fishery catch rates, catch and quota utilization, and salmon bycatch measures. Their interpretation must account for fishing choices, allocation and avoidance behavior, processing demand, and reporting coverage. These indicators may reveal operational changes or sampling differences that help interpret assessment inputs. Their proposed inclusion in this ESP requires a defined management question.

The draft supports evaluation of ecosystem and fishery concerns in the SAFE risk table. Any resulting change in risk classification or ABC must be documented by the assessment authors using the completed annual synthesis and assessment results. ABC and OFL remain unchanged by this contribution.

### Total Allowable Catch Considerations

Verified economic context covers BSAI catch, participation, ex-vessel and first-wholesale statistics through 2024, and U.S.-national trade through 2025 (Catch and Market Context). These scopes differ from the BS assessment stock and from each other. A speculative 2026 interpretation is that firmer product unit values could support gross returns, while operating costs and product mix could offset those gains unevenly among vessels, processors and communities. 2026 margins and community outcomes remain uncertain. This interpretation is qualitative.

### Responses to SSC and Plan Team Comments on ESPs in General

A summary of potential SSC/PT comments on ESPs in general includes: distinguish predictive from contextual indicators, evaluate competing mechanistic explanations, document DSEM diagnostics, and coordinate economic and community information with existing Council products. This pollock contribution addresses those questions by defining indicator roles, documenting the fitted pilot and its diagnostic limitation, and proposing an explicit socioeconomic inventory.

### Responses to SSC and Plan Team Comments Specific to EBS Pollock

December 2024 SSC, EBS Ecosystem Status Report (page 3): the SSC recommended pollock-focused DSEM work to reconcile contrasting larval, juvenile-condition, and adult signals, and to investigate changed recruitment relationships (North Pacific Fishery Management Council Scientific and Statistical Committee 2024).

Contribution to that work: The starter age-1 DSEM analysis appears in the 2026 EBS Ecosystem Status Report (Siddon 2026).

The assessment website records continuing work on contrasting early-life observations, cannibalism and biomass consumed, multispecies mortality, weight-at-age and condition, and stock distribution across U.S. and Russian waters (EBS Pollock Assessment Team 2026). Its retained earlier responses carry an explicit requirement for verification before use as final results. This draft translates those topics into proposed ESP tasks.

Pollock review topics and proposed ESP contributions are summarized in Table B1.

## Introduction

Pollock recruitment reflects survival through several life stages with different food requirements, habitats, and predator exposure. Large differences in egg or larval abundance can be modified by later feeding, growth, transport, and mortality. The first-year review by Duffy-Anderson et al. provides the organizing synthesis for these processes (Duffy- Anderson et al. 2016). The local recruitment review extends that structure through ages 1 and 2 to survival at age 3. In this appendix, age-1 recruitment is the focal response used by the DSEM pilot; survival to age 3 is a separate later- stage outcome that retains the identity of the same birth cohort.

## Justification

The ESP creates an evidence trail from a biological or socioeconomic question to an indicator and its proposed use. For EBS pollock, the immediate question is whether early-life information can improve interpretation of uncertain recruitment and future biomass. Additional questions concern adult productivity, changing spatial availability to surveys, fishery operations, and community exposure. The September 2026 document provides the starting framework for developing this appendix (Ianelli et al. 2026).

## Data

### Age-0 Energy Density

Annual mean energy density of age-0 walleye pollock is available for the northern and southeastern Bering Sea during 2002–2025, in kJ/g dry mass (Page and Maselko 2026). Johanna Page (Vollenweider) provided the data, with Jacek Maselko as a co-contact. The southeastern series contains 21 annual means and the northern series contains 15. Both regions lack observations for 2003, 2013 and 2020; the northern series also lacks 2008–2012 and 2014.

The source inventory and remaining data-assembly needs are summarized in Table B2.

## Indicator Synthesis

## Ecosystem Processes

### Spawning Supply and Transport

Spawning biomass and spawning location influence egg and early-larval supply. Smart et al. found associations between spawning biomass and early stages, with weaker correspondence after early larval development (Smart et al. 2012). Transport and the spatial arrangement of spawning also matter. The biophysical model of Petrik et al. reproduced warm–cold distribution differences most successfully through changes in spawning areas (Petrik et al. 2015). These results support separate hypotheses for cohort supply, transport, and access to nursery habitat.

### Feeding and Fall Age-0 Condition

Prey availability includes amount, nutritional quality, and spatial match with fish. Larval copepod concentrations and frontal habitat illustrate why a broad regional biomass index can differ from prey encountered by an individual fish (Smart et al. 2012; Brodeur et al. 2000). Diet studies show that prey use varies with fish size and that pollock and Pacific cod can partition prey (Buckley et al. 2016; Strasburger et al. 2014).

Siddon et al. describe a seasonal shift from larval growth and development toward lipid storage after transformation to the juvenile stage (Siddon et al. 2013). Heintz et al. found higher diet lipid content and age-0 energy density during cold conditions in their study; fall total energy was positively associated with age-1 recruits per spawner (Heintz et al. 2013).

Moss et al. also documented differences in age-0 energetic condition and food habits across thermal conditions (Moss et al. 2009). These findings motivate indicators of energy density, body mass, and total energy with explicit measurement definitions. The annual series measures energy density in kJ/g dry mass.

Eisner et al. evaluated large copepods as leading indicators of age-3 pollock abundance for the 2002–2015 year classes, excluding 2013 (Eisner et al. 2020). Their comparison reinforces the value of prey information while identifying a specific cohort range and recruitment age.

### Overwinter Survival

Fall condition provides the starting reserves for the first winter. Sogard and Olla's food-deprivation experiments show that greater size, higher initial condition, and colder water can extend survival (Sogard and Olla 2000). Field survival also involves winter feeding, movement, and predators. Laboratory results therefore support a mechanism and its direction, while annual field data are needed to quantify its effect on recruitment.

Sea ice, bloom timing, summer mixing, and temperature influence prey production and energy demand. Sigler et al. connect annual production of copepods, euphausiids, and juvenile pollock with seasonal environmental conditions (Sigler et al. 2016). Gann et al.'s analysis of 2007 identifies a plausible production limitation during a cold year (Gann et al. 2016). The ESP should evaluate these intermediate processes before interpreting a simple warm–cold contrast as a cohort forecast.

### Predation and Later Juvenile Survival

Adult pollock can increase juvenile mortality through cannibalism, while arrowtooth flounder and other predators create additional exposure. Mueter et al. identify both food-related and predation-related controls on pollock recruitment, including juvenile–adult overlap (Mueter et al. 2006). Spencer et al. model spatial predation by arrowtooth flounder and show how cold-pool extent can alter predator access to juvenile habitat (Spencer et al. 2016).

The timing of predation matters for the response. A summer overlap or cold-pool observation in year t+1 occurs after the beginning of the cohort's age-1 year. Its use to explain an age-1 entry state requires explicit justification or a revised response definition. Such observations may be more directly aligned with survival during age 1 and later ages.

Life stages, processes and priority observations are summarized in Table B3.

### Workshop Conceptual Models for Recruitment and Growth

The January 2025 pollock biology and ecology workshop developed preliminary conceptual models for Bering Sea recruitment and growth (Monnahan et al. 2026). Figure B1 organizes reproductive output, seasonal juvenile stages, prey quality, energetic condition, predation, and cold-pool extent. Figure B2 connects population weight-at-age with somatic growth, size-selective mortality, prey, competitors, fishing, and physical conditions. Together they support an ESP that examines recruitment and growth.

Recruitment pathways: spawning biomass and fecundity inform reproductive output. Seasonal surveys track the cohort from spring age 0 through age 2. Diet, krill, copepods, and prey lipid measurements inform forage quality and juvenile

energetic condition; predation indices and cold-pool extent inform later survival. The diagram highlights both candidate observations and intermediate biological states.

Growth pathways: the model separates growth of surviving fish from changes in population weight-at-age associated with size-selective mortality. Prey availability, density, genetic diversity, and spatial history enter the growth network alongside physical conditions and zooplankton. This distinction supports age- and season-specific weight and condition indicators with explicit survey or fishery sampling definitions.

The cohort sequence in Figure B3 summarizes the recruitment mechanisms developed in this draft and extends the survival question to age 3. The workshop diagrams provide a broader set of hypotheses for future indicator development.

### Socioeconomic Processes

Biological supply, access to fishing grounds, operating costs, product demand, and processing capacity can jointly affect realized harvest and its value. Those processes may alter where and when vessels fish, the size and age of fish landed, and the distribution of benefits. This component should therefore connect each indicator to a concrete question, such as whether lower quota utilization reflects reduced fish availability, market conditions, processing limits, or avoidance behavior.

The Economic SAFE provides source categories for catch, discards, prohibited species catch, production and value, participation, and effort (Alaska Fisheries Science Center 2026b). This draft proposes species- and sector-specific summaries where the source supports that detail. Community analysis requires clear distinctions among landing ports, vessel ownership, workforce residence, processor location, and CDQ beneficiaries. Each describes a different form of participation or exposure.

### Proposed Indicators

The mechanism table (Table B4) links a driver, a measurable process, and the intended response. Expected directions are hypotheses for the stated pathway. Several upstream physical effects depend on season, prey response, and spatial coverage, so their net sign can change across conditions.

### Indicator Assessment

### Indicator Selection

Selection starts with a mechanistic explanation, the population or socioeconomic response, and the date when the information becomes available. Candidates then require reliable annual production, adequate spatial and temporal coverage, quantified uncertainty, and additional information beyond the other indicators.

### Ecosystem Indicators

The initial set should remain small enough for scientific review: seasonal physical forcing; a documented prey index; fall age-0 size and energetic condition; juvenile–predator overlap or a reviewed mortality product; and adult growth and condition. Larval supply and transport add value when annual coverage and biological alignment support them. Indicator families should be assessed together so correlated temperature, ice, and prey signals receive an interpretation that accounts for shared processes.

The available age-0 energy-density series describe regional energetic condition (Figure B4). The recorded histories of the ecosystem inputs and age-1 response provide context for the DSEM analysis (Figure B5).

### Socioeconomic Indicators

Proposed socioeconomic indicator families, measures and management questions are summarized in Table B5.

## Catch and Market Context

The broader Bering Sea and Aleutian Islands (BSAI) pollock fishery reported 1.3163 million tons of total catch in 2024 (Table B6). Trawl catcher vessels retained 678.40 thousand tons, with an ex-vessel value of $213.43 million. These economic statistics cover the BSAI; their area and accounting definitions differ from the BS stock-assessment catch series. Published pollock-target vessel counts were 103 across harvesting sectors and 76 catcher vessels. The counts describe participation in the target fishery and include their source-specific gear definitions.

Published first-wholesale pollock production was 487.89 thousand tons of product weight and $1.4743 billion in 2024 (Table B7). The Economic SAFE excludes confidential components from some aggregates; the combined fillet quantities and derived values for 2021–2022 remain unavailable here. Product mix and market conditions both affect the observed unit values (Abelman et al. 2026).

Global Alaska pollock capture production was 3.573 million tons in 2024 (Table B8, panel A). The United States and Russian Federation accounted for 39.9% and 54.0%, respectively, and BSAI catch represented about 36.8% of the global total. U.S. Alaska-pollock domestic exports totaled 333 thousand tons of products valued at $1.00 billion in 2025, compared with 332 thousand tons and $0.946 billion in 2024 (panel B). These national trade statistics extend through 2025; global catch statistics extend through 2024. Their product weights, nominal export values and legacy import/destination baskets are defined in the table notes (FAO 2026; NOAA Fisheries 2026).

Among the three destination groups shown, selected European countries accounted for the largest export shares throughout 2020–2025 (Figure B6). Meat's share of export volume rose from 45.8% in 2020 to 53.3% in 2025, while its value share rose from 44.0% to 53.0% (Figure B7). The Russian Federation accounted for more than half of global catch in each year from 2020 to 2024 (Figure B8). First-wholesale surimi volume declined from 185.5 to 165.1 thousand tons between 2023 and 2024, while fillet volume rose from 142.9 to 158.6 thousand tons (Figure B9).

## Annual Indicator Status

### Age-0 Energy Density Through 2025

The latest five available observation years are 2021–2025 (Table B9). In 2025, the SEBS annual mean was 19.23 kJ/g dry mass, 0.64 lower than in 2024. The NBS annual mean was 20.84, 0.09 lower than in 2024 (Page and Maselko 2026). These are descriptive differences between annual means.

### Recorded Annual Indicator Series

The age-1 DSEM input spans 1964–2026 and contains 18 ecosystem series plus the assessment-derived age-1 response (Figure B5). Seventeen ecosystem series have recorded 2026 values; the energy-density series ends in 2025, and age-1 observations end in 2024.

Standardized departures below equal each recorded value minus that series' mean, divided by its sample standard deviation. Each baseline uses all available finite observations, including the latest recorded year, and retains the series'

gaps. These baselines are descriptive historical variability, rather than a common fixed climatology, sampling-error intervals or approved biological thresholds.

In 2026, recorded sea ice is +0.89 SD relative to its 1985–2026 baseline of 42 observations. Cold-pool extent is +1.12 SD, SST −0.39 SD and bottom temperature −0.79 SD relative to their separate 1982–2026 baselines of 44 available observations each. Sea ice and cold pool both exceed their recorded 2025 values. SST also increased from 2025 despite remaining below its observed-period mean, whereas bottom temperature fell.

The open-water bloom indicator is −1.54 SD and chlorophyll a is +0.77 SD relative to their separate 1998–2026 baselines of 29 observations each. Both declined from their recorded 2025 values. Spring small and large copepods are −0.45 and −0.66 SD using their separate 2000–2026 baselines of 25 observations; summer small and large copepods are −0.92 and +0.21 SD using their separate 2003–2026 baselines of 22 observations. The mixed physical and prey pattern supports contextual discussion rather than an overall favorable or adverse recruitment classification.

## Indicator Analysis

The September 2026 assessment development document presents a pilot DSEM analysis of SST and cold-pool pathways (Ianelli et al. 2026). Tables B10–B11 reproduce its fit comparisons and direct proxy coefficients as an example of analyses that could be provided in the ESP.

### Decision Support

A contextual indicator may inform the SAFE risk-table discussion when it identifies a documented condition relevant to recruitment, productivity, survey interpretation, or fishery operations. Its influence on a risk category requires an explanation of the evidence and uncertainty. A predictive indicator needs forecast evaluation and numerical checks. Incorporation into an assessment likelihood, projections, or a harvest rule requires a separate, reviewed implementation and evaluation of effects on biomass, uncertainty, ABC, and OFL.

### Socioeconomic Monitoring

Socioeconomic monitoring can build on the main assessment's discussion of fishery conditions, including fishing locations, catch rates, effort, size composition, and bycatch avoidance, alongside economic performance, participation, and community exposure. The catch-decision indicators and probabilities in Tables 55 and 56 provide a framework for comparing alternative catches in terms of fishing mortality, future spawning biomass, age composition, and relative fishing pressure. The fishing-mortality-based effort proxy describes relative pressure; observed vessel effort, catchability, operating costs, and revenues would provide the economic context for interpreting it. Future ESP drafts will pursue incorporating these fishery-condition observations and decision-table comparisons to connect projected stock outcomes with operational and socioeconomic consequences.

## Data Gaps and Future Research Priorities

### Ecosystem Priorities

One of our priorities is cohort-aligned predator overlap or mortality, with timing that matches the survival interval. Joint juvenile and predator distributions, diet and consumption estimates, and direct fall–spring cohort comparisons would strengthen interpretation.

Transport and nursery delivery, prey timing and lipid content, winter exposure, and annual observation-error estimates remain additional priorities. Climate relationships should be assessed for stability across recent warm conditions and historical regimes. Long-term climate projections in Mueter et al. and Spencer et al. describe conditional scenarios (Mueter et al. 2011; Spencer et al. 2016); their projected effects provide research context rather than annual 2026 indicator values.

### Socioeconomic Priorities

Confirm a compact pollock-specific indicator set with economic and social science contributors. Resolve sector and community definitions, annual update capacity, disclosure rules, cost comparability, and a common price base. Evaluate whether incomplete quota utilization or altered fishing patterns have identifiable operational, market, processing, or biological causes. Document the availability date for each product so the ESP distinguishes information usable during the specification process from information released later.

### Completion Priorities for the 2026 Appendix

Confirm the primary sources, units, seasonal and spatial definitions and observation uncertainty for the working annual series. Review the age-1 DAG and the proposed energy and predator pathways with the ESP and ecosystem teams. Preserve separate timing and scaling for the archived Rceattle pilot and standalone DSEM, and retain the AR1 forecast benchmark when evaluating any expanded model. Reconcile the historical pilot archive and its outstanding numerical checks before using the indicators to support an annual status or risk classification.

## Conclusion

The literature supports following cohorts through spawning and delivery, prey and fall age-0 energy, first-winter survival, and later predator exposure. The ESP also considers adult productivity, survey availability and socioeconomic conditions. The regional energy-density series supplies a measured-condition indicator through 2025.

Economic observations extend through 2024 for the BSAI fishery and 2025 for national trade; the 2026 interpretation is qualitative.

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## Tables

Table B1 Pollock Review Topics and Proposed ESP Contributions

| Review topic | Contribution in this draft | Completion needed |
|---|---|---|
| Contrasting larval, age-0 and adult signals | Follow cohorts through prey, condition and survival stages. | Verify source years, survey domains and cohort identity. |
| Cannibalism and biomass consumed | Distinguish overlap, diet occurrence, consumption and mortality. | Add reviewed consumption estimates with uncertainty. |
| Multispecies mortality | Treat age-specific predation as a candidate indicator. | Identify model version, predator coverage and sensitivity. |
| Weight-at-age and condition | Include growth and energetic context by age and season. | Update survey and fishery products and assess comparability. |
| U.S.–Russian distribution | Track survey domain and movement assumptions. | Use verified spatial products and explicit stock boundaries. |

*Note.* Pollock review topics and proposed ESP contributions. Topics are paraphrased from the assessment website; final responses require the original review record.

Table B2 Source Inventory for the EBS Pollock ESP

| Data group | Source and definition | Status in this contribution |
|---|---|---|
| Assessment states and survey inputs | EBS assessment; BTS, ATS and AVO products with age, season and spatial coverage. | Historical development material verified; align with the final 2026 SAFE run. |
| Temperature and cold pool | Seasonal SST; survey-derived cold-pool extent and temperatures. | Historical standardized pilot inputs archived; 2026 indicator update pending. |
| Prey and age-0 fish | BASIS/EcoFOCI and associated studies; size, biomass, diet and energetic measurements. | Literature reviewed; consolidate prey, size and sampling uncertainty. |
| Age-0 energy density | Page and Maselko annual NBS and SEBS means; 2002–2025; kJ/g dry mass. | Regional series received; season, weighting, uncertainty and cohort alignment require confirmation. |
| Predation | Diet observations, overlap products and multispecies model estimates. | Mechanisms synthesized; reviewed annual products required. |
| Catch and fishery behavior | Catch accounting, observer and fishery products; consistent season and area definitions. | Indicator specification proposed; extraction and quality review pending. |
| Economics and communities | Economic SAFE and approved species-level economic and community products. | Inventory proposed; latest complete years and disclosure rules require confirmation. |

*Note.* Source inventory for the EBS pollock contribution. Availability describes this draft's verified inputs and proposed data assembly.

Table B3 Life-Stage Synthesis for Eastern Bering Sea Pollock

| Stage and timing | Processes and evidence | Priority observations |
|---|---|---|
| Spawning and eggs, t | Spawner output, location and transport establish early supply. | Maturity, spawning timing, egg abundance and distribution. |
| Larvae, t | Prey encounter, growth and delivery to nursery habitat modify supply. | Local prey, larval size and growth, transport and habitat match. |
| Age 0, summer–fall t | Size and lipid storage determine energy available before winter. | Mass, energy density, total energy, diet quality and prey biomass. |
| First winter, t to t+1 | Reserves, demand, feeding and predation determine early survival. | Fall–spring cohort comparison and winter exposure. |
| Ages 1–2, t+1 to t+2 | Spatial overlap, cannibalism and other predation affect later survival. | Juvenile and predator distributions, diets and mortality. |
| Adults, later years | Growth, maturity, condition and distribution affect productivity and observation. | Weight-at-age, condition, maturity and comparable spatial indices. |

*Note.* Life-stage synthesis for eastern Bering Sea pollock. The same cohort is followed from birth year t to ages 1 and 3.

Table B4 Proposed Mechanisms and Responses

| Driver and mechanism | Expected relation and timing | Evidence and candidate measurement |
|---|---|---|
| Spawner output → early supply | Positive toward egg supply in birth year t; later survival modifies the effect. | Smart et al. 2012; SSB, eggs and spawning location. |
| Prey quality and match → fall energy | Positive toward age-0 energy in t. | Heintz 2013; Siddon 2013; prey lipid, biomass and fish–prey match. |
| Fall size and energy → winter survival | Positive toward age-1 survival in t+1. | Heintz 2013; Sogard and Olla 2000; mass, energy density and total energy. |
| Summer mixing → production and prey | Conditional through nutrient supply and prey production in t. | Gann 2016; wind mixing, stratification and nutrients. |
| Spawning and transport → habitat match | Spatially dependent effect before and during age 0. | Petrik 2015; Smart 2012; transport and nursery-delivery indices. |
| Predator overlap → juvenile mortality | Negative toward subsequent cohort survival in t+1 and t+2. | Mueter 2006; Spencer 2016; overlap, diet and mortality. |
| Cold pool → predator access | Hypothesized benefit through separation; depends on predator and stage. | Spencer 2016; common-domain cold pool and overlap. |
| Late-summer SST → energetic conditions | Hypothesized negative relation with age-1 recruitment after birth-year exposure. | Mueter 2011 and pilot; seasonal SST is a broad proxy. |

*Note.* Proposed mechanism table. Direction refers to the stated response; evidence supports hypotheses rather than a universal coefficient.

Table B5 Proposed Pollock Socioeconomic Indicators

| Indicator family | Proposed measure | Management question |
|---|---|---|
| Participation | Active vessels, processors and deliveries by sector and season. | Where is participation changing, and which operations face exposure? |
| Fishery performance | Retained catch, TAC utilization and standardized effort measures. | How do fishing conditions and operating choices affect realized harvest? |
| Value and product mix | Ex-vessel and first-wholesale value, prices and product shares. | How do product demand and prices affect value from the catch? |
| Costs | Fuel, labor or other comparable cost measures where available. | What evidence indicates changes in operating margins? |
| Community exposure | Pollock dependence, employment and participation under approved definitions. | Which communities and CDQ beneficiaries are exposed, and through which pathways? |
| Bycatch and access | Chinook and chum observations, defined rates, season and sector. | How do bycatch conditions and avoidance affect access and operations? |

*Note.* Proposed pollock socioeconomic indicators and the questions they address. All require defined indicators and assessment data.

Table B6 BSAI Pollock Catch, Participation and Ex-Vessel Economics, 2020–2024

| Measure | Unit | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|
| Total catch, all harvesting sectors | 1,000 t round weight | 1,370.4 | 1,378.1 | 1,108.7 | 1,314.5 | 1,316.3 |
| Retained catch, all harvesting sectors | 1,000 t round weight | 1,358.4 | 1,363.9 | 1,094.9 | 1,299.2 | 1,298.6 |
| Pollock-target vessels (all harvesting sectors) | Vessels | 117.0 | 116.0 | 114.0 | 105.0 | 103.0 |
| Retained catch, trawl catcher vessels | 1,000 t round weight | 725.0 | 718.2 | 570.0 | 678.2 | 678.4 |
| Ex-vessel value, trawl catcher vessels | US$ million (nominal) | 233.7 | 242.8 | 242.0 | 244.0 | 213.4 |
| Ex-vessel unit value, trawl catcher vessels | US$/lb (nominal) | 0.1 | 0.2 | 0.2 | 0.2 | 0.1 |
| Trawl catcher-vessel share of retained catch | % | 53.4% | 52.7% | 52.1% | 52.2% | 52.2% |
| Pollock-target catcher vessels | Vessels | 88.0 | 84.0 | 84.0 | 77.0 | 76.0 |

*Note.* BSAI denotes the Bering Sea and Aleutian Islands and differs from the BS-only assessment catch series. Total catch includes retained and discarded federal and state catch; retained catch and vessel counts cover federal TAC fisheries. The pollock target is assigned by vessel,

week, catching mode, NMFS area and gear. Published catcher-vessel counts describe the target-vessel group; only the catch and value rows explicitly marked trawl carry a separate trawl filter. Unit values and shares are derived from rounded published cells; catcher-vessel unit value combines the reported catcher-vessel catch and value across delivery modes. Monetary values are nominal. Values are rounded to one decimal place. Source: Abelman et al. (2026), Tables 4.1, 4.2, 4.9, 4.10 and 4.14 (printed pp. 34–35, 45–46, 48 and 59), using NMFS accounting and production reports and ADF&G COAR data compiled by AKFIN.

Table B7 BSAI Pollock First-Wholesale Products and Values, 2020–2024

| Measure | Unit | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|
| All products, published volume | 1,000 t product weight | 492.7 | 464.1 | 387.5 | 480.9 | 487.9 |
| All products, published value | US$ million (nominal) | 1,352.9 | 1,370.5 | 1,400.9 | 1,603.0 | 1,474.3 |
| All products, unit value | US$/lb (nominal) | 1.2 | 1.3 | 1.6 | 1.5 | 1.4 |
| At-sea share of published value | % | 60.4% | 61.7% | 61.6% | 55.6% | 57.9% |
| Fillets, volume | 1,000 t product weight | 149.4 | Unavailable | Unavailable | 142.9 | 158.6 |
| Fillets, share of published value | % | 37.4% | Unavailable | Unavailable | 40.2% | 40.6% |
| Fillets, unit value | US$/lb (nominal) | 1.5 | Unavailable | Unavailable | 2.0 | 1.7 |
| Surimi, volume | 1,000 t product weight | 171.8 | 187.8 | 154.8 | 185.5 | 165.1 |
| Surimi, share of published value | % | 35.2% | 42.8% | 37.8% | 36.0% | 32.2% |
| Surimi, unit value | US$/lb (nominal) | 1.3 | 1.4 | 1.6 | 1.4 | 1.3 |
| Roe, volume | 1,000 t product weight | 24.7 | 16.0 | 13.8 | 14.7 | 19.2 |
| Roe, share of published value | % | 8.2% | 6.5% | 7.1% | 6.0% | 7.5% |
| Roe, unit value | US$/lb (nominal) | 2.0 | 2.5 | 3.3 | 3.0 | 2.6 |
| At-sea minus shoreside all-product unit value | US$/lb (nominal) | 0.4 | 0.4 | 0.4 | 0.1 | 0.2 |

*Note.* BSAI denotes the Bering Sea and Aleutian Islands. Product weight differs from round-weight retained catch. Published aggregates can be incomplete because confidential components are excluded within a table (Economic SAFE §2.2.6). Fillets combine deep-skin and other fillets and exclude minced fish. Fillet volume, value share and unit value are unavailable for 2021–2022 because shoreside deep-skin components are confidential; unavailable cells are not zeros. Unit values and shares use rounded published values. The at-sea-minus-shoreside unit-value difference is unadjusted for product quality and reflects product mix. All dollar values are nominal. Values are rounded to one decimal place. Source: Abelman et al. (2026), Tables 4.16–4.18 (printed pp. 65, 70 and 74), using NMFS production reports and ADF&G COAR data compiled by AKFIN.

Table B8 Global Alaska Pollock Catch and U.S. Trade Panel A. Global Alaska Pollock Catch and Shares, 2020–2024

| Metric | Unit | 2020 | 2021 | 2022 | 2023 | 2024 |
|---|---|---|---|---|---|---|
| Global Alaska pollock catch | kt live weight | 3,544.3 | 3,484.2 | 3,380.4 | 3,543.5 | 3,573.5 |
| U.S. share of global catch | percent | 41.3% | 42.0% | 36.3% | 40.5% | 39.9% |
| Russian share of global catch | percent | 51.6% | 50.2% | 56.3% | 53.5% | 54.0% |
| BSAI share of global catch | percent | 38.7% | 39.6% | 32.8% | 37.1% | 36.8% |

Panel B. U.S. Alaska-Pollock Trade, 2020–2025

| Metric | Unit | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|
| Export volume | kt product weight | 323.5 | 311.4 | 274.1 | 326.3 | 332.5 | 333.1 |
| Export value | million nominal USD | 941.9 | 898.9 | 885.6 | 1,004.4 | 945.9 | 1,002.4 |
| Export unit value | USD/lb product weight | 1.3 | 1.3 | 1.5 | 1.4 | 1.3 | 1.4 |
| Import value (Alaska + unspecified pollock) | million nominal USD | 99.7 | 96.2 | 141.3 | 105.5 | 65.3 | 48.7 |
| Net exports (legacy basket) | million nominal USD | 842.2 | 802.6 | 744.4 | 899.0 | 880.5 | 953.6 |
| Japan export volume share | percent | 19.3% | 21.0% | 21.2% | 25.0% | 19.7% | 15.0% |
| Japan export value share | percent | 20.2% | 21.1% | 20.1% | 22.9% | 20.7% | 16.6% |
| China, Taiwan and Hong Kong (legacy aggregate) export volume share | percent | 9.7% | 6.8% | 11.6% | 7.7% | 13.3% | 10.6% |
| China, Taiwan and Hong Kong (legacy aggregate) export value share | percent | 6.5% | 5.3% | 7.5% | 5.2% | 8.5% | 6.7% |
| Selected European countries export volume share | percent | 37.0% | 32.7% | 31.4% | 32.1% | 34.2% | 37.1% |
| Selected European countries export value share | percent | 39.4% | 36.3% | 35.3% | 37.9% | 38.6% | 40.7% |
| Meat export volume share | percent | 45.8% | 41.1% | 42.7% | 40.6% | 47.5% | 53.3% |
| Meat export value share | percent | 44.0% | 40.8% | 42.5% | 43.4% | 48.0% | 53.0% |
| Surimi export volume share | percent | 47.0% | 53.8% | 52.2% | 54.3% | 46.3% | 40.4% |
| Surimi export value share | percent | 42.2% | 49.4% | 49.1% | 47.1% | 39.1% | 34.1% |
| Roe export volume share | percent | 7.3% | 5.1% | 5.1% | 5.2% | 6.2% | 6.3% |
| Roe export value share | percent | 13.8% | 9.8% | 8.4% | 9.5% | 13.0% | 13.0% |

*Note.* These observed historical statistics cover global catch and shares through 2024 and U.S. trade through 2025. Trade statistics remain revisable. Global catch is in thousand tons of live weight. Trade volumes are thousand tons of product weight. BSAI catch is a subset of U.S. catch and includes federal and state retained plus discarded catch from both the Bering Sea and Aleutian Islands. Trade covers U.S.-national Alaska-pollock products across their source fisheries. Export values are nominal U.S. dollars on a free-alongside-ship (FAS) export-value basis, distinct from wholesale revenue. Unit values reflect product mix as well as market conditions. Domestic exports exclude re-exports. The legacy import basket combines Alaska and unspecified (NSPF) pollock; legacy net exports subtract that broader basket from Alaska-pollock domestic exports. The legacy statistical aggregate combines mainland China, Taiwan and Hong Kong. The selected European aggregate comprises Austria, Belgium, Denmark, France, Germany, Greece, Ireland, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom. It is neither the European Union nor all Europe. Meat combines Alaska-pollock products other than surimi and roe, including frozen whole/headed-and-gutted fish, fillets and other meat. Product shares sum to 100% before rounding. Values are rounded to one decimal place from the source extract. Displayed net exports are rounded from the saved source totals rather than recalculated from separately rounded export and import values. Sources: NOAA Fisheries FOSS, retrieved 30 October 2026; FAO Global Capture Production release Capture_2026.1.0 (March 2026); Economic SAFE 2025, Table 4.1, p. 34, published 16 March 2026. Source links: NOAA foreign fishery trade data; FAO Global Capture Production; Economic SAFE.

Table B9 Annual Mean Age-0 Pollock Energy Density, 2021–2025

| Year | NBS (kJ/g dry mass) | SEBS (kJ/g dry mass) |
|---|---|---|
| 2021 | 21.5 | 20.1 |
| 2022 | 21.5 | 19.7 |
| 2023 | 21.3 | 19.8 |
| 2024 | 20.9 | 19.9 |
| 2025 | 20.8 | 19.2 |

*Note.* Annual mean age-0 pollock energy density for 2021–2025, in kJ/g dry mass. Regions remain separate; sampling uncertainty is pending. Values are rounded to one decimal place. Source: Page and Maselko (2026), DATA sheet.

Table B10 Archived DSEM Pilot Results

| Specification | Delta AIC | Residual variance reduction | Maximum gradient | Gradient below 1e-4 |
|---|---|---|---|---|
| Age-0 late-summer SST | 0.00 | 10.5% | 1.52e-04 | Requires refit |
| Cold pool + age-0 late-summer SST | 1.63 | 12.0% | 5.00e-05 | Pass |
| No-edge reference | 2.25 | 0.0% | 5.61e-05 | Pass |
| Cold pool | 4.10 | 0.3% | 5.69e-05 | Pass |

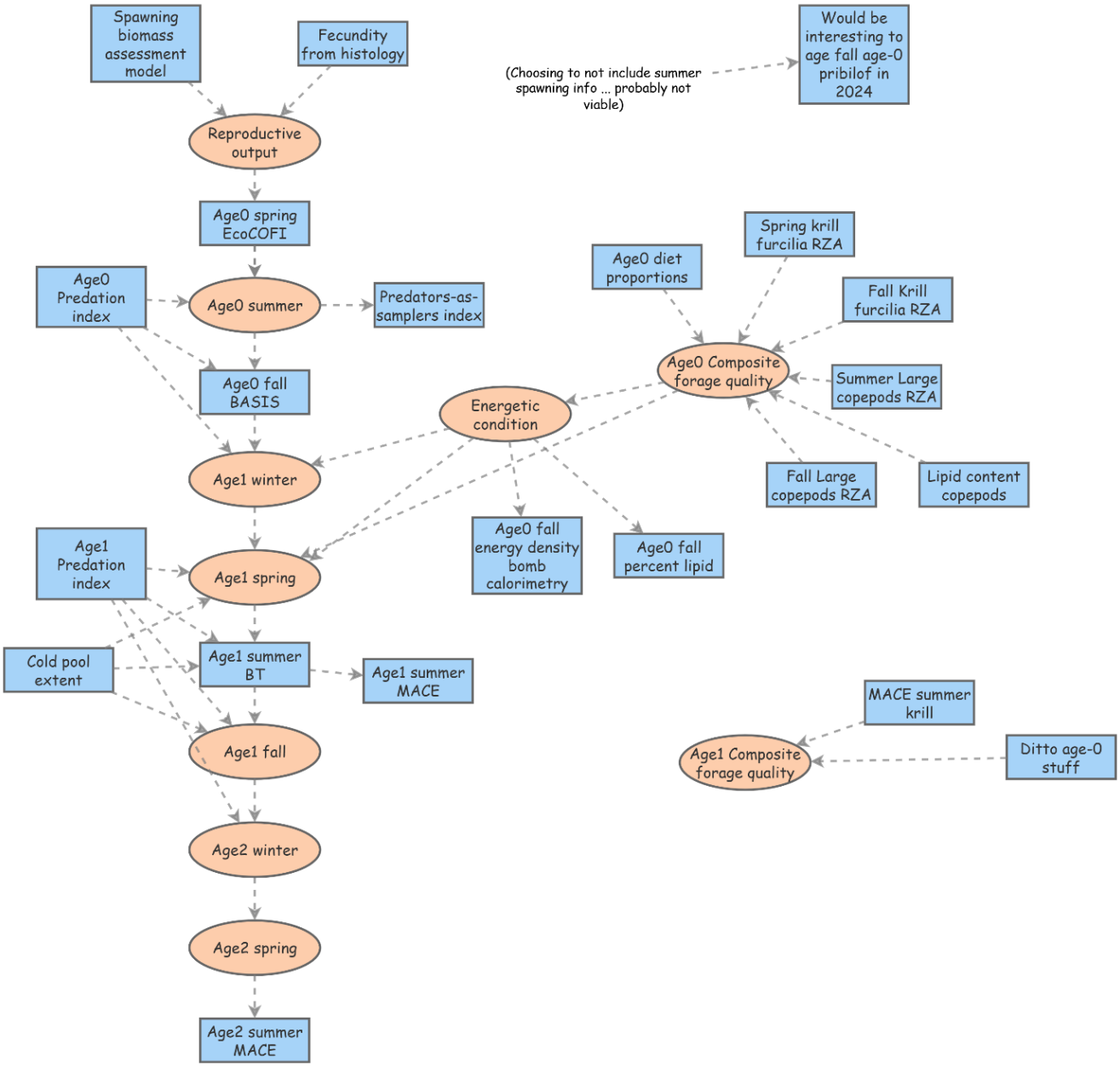
*Note.* Archived live DSEM pilot results. All Hessians are recorded as positive definite. The gradient column applies the strict threshold of 1e-4 independently of the saved OK status.

Table B11 Standardized Direct Proxy Coefficients From the Archived DSEM Pilot

| Specification | Pathway | Estimate | 95 percent interval |
|---|---|---|---|
| Cold pool | Cold pool | 0.041 | -0.170 to 0.251 |
| Age-0 late-summer SST | Age-0 SST | -0.222 | -0.426 to -0.017 |
| Cold pool + age-0 late-summer SST | Cold pool | -0.069 | -0.292 to 0.153 |
| Cold pool + age-0 late-summer SST | Age-0 SST | -0.254 | -0.484 to -0.024 |

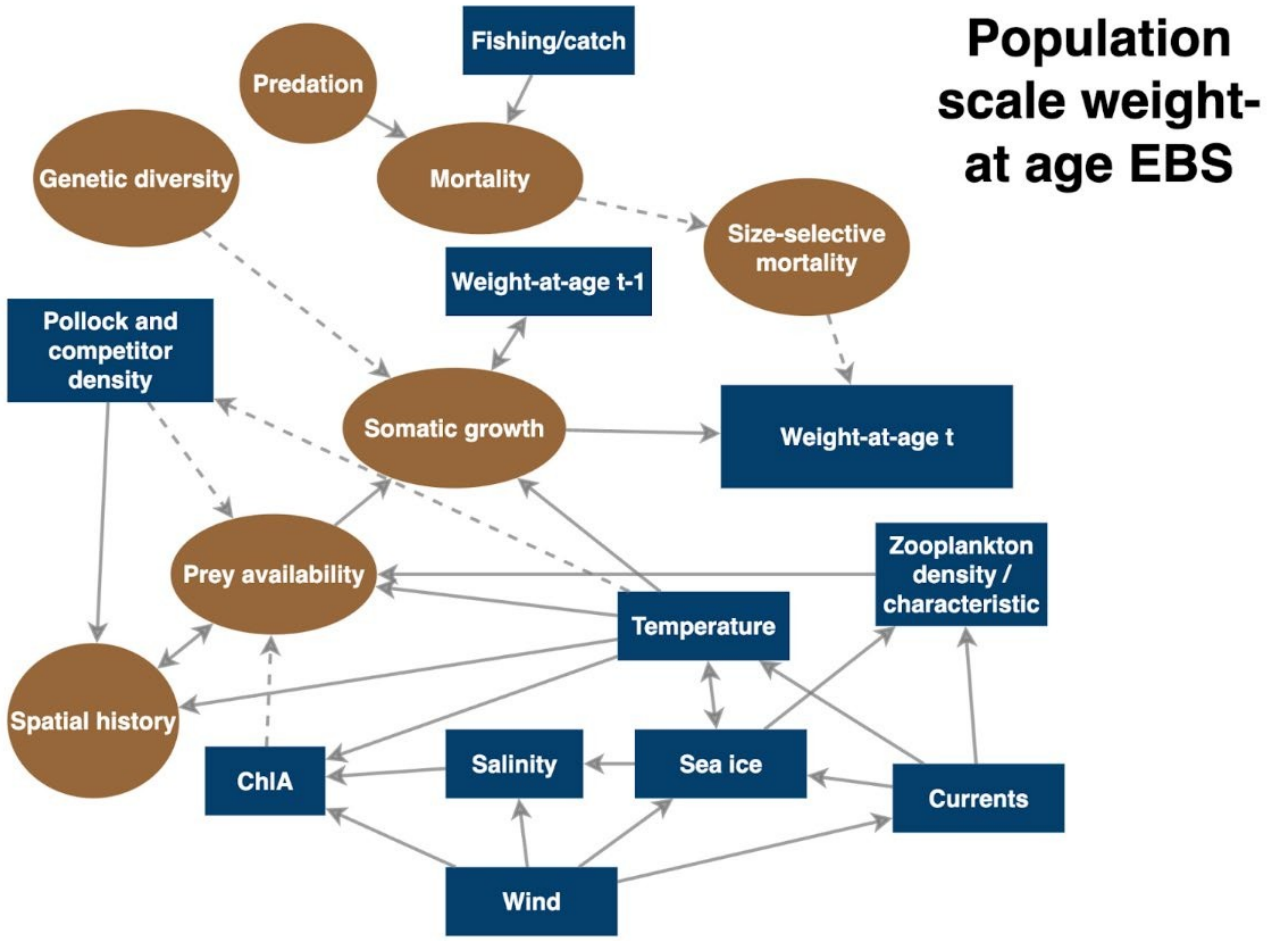
*Note.* Standardized direct proxy coefficients from the archived live DSEM pilot, with approximate 95 percent intervals. These fitted paths remain provisional research estimates.

## Figures



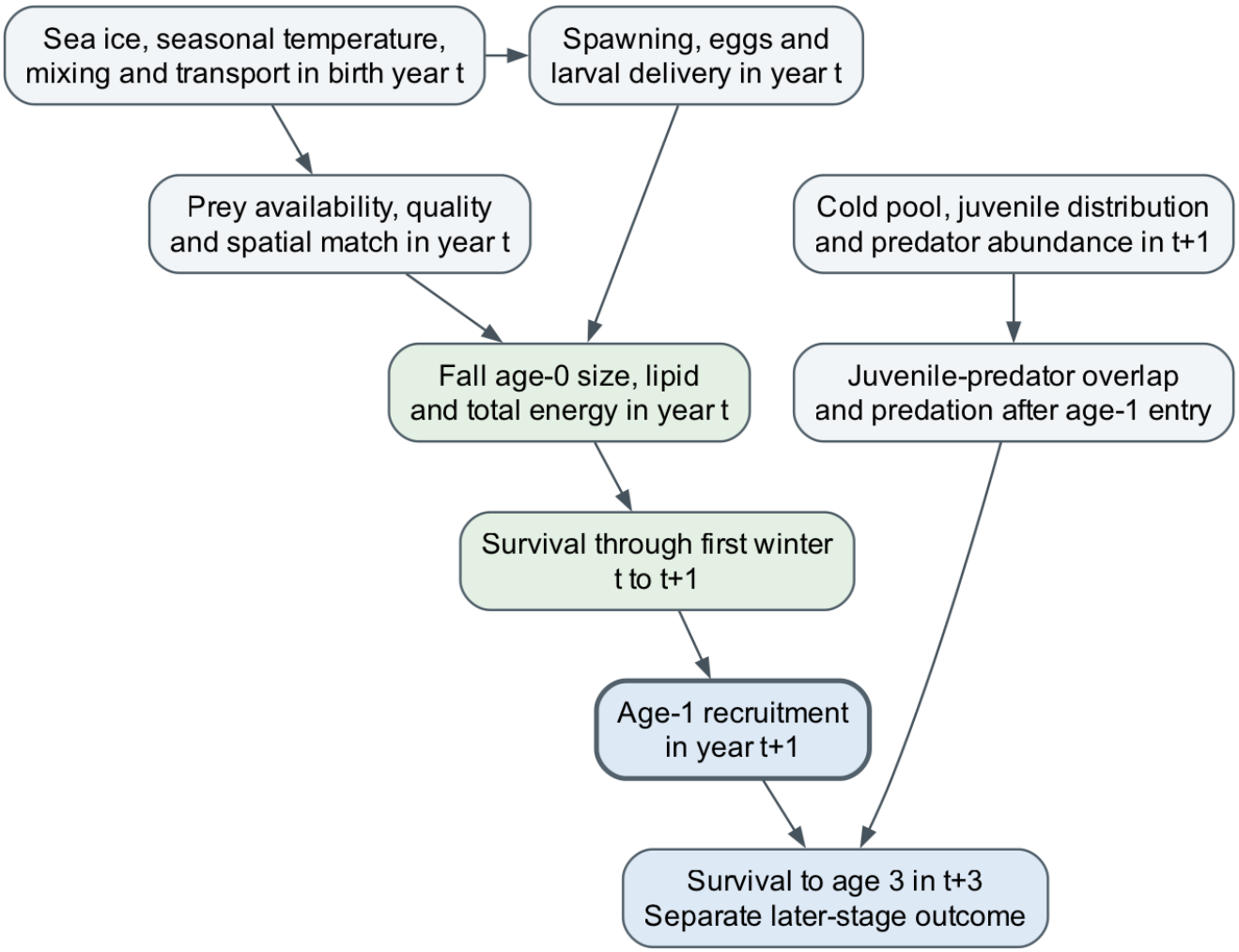

*Note.* Preliminary Bering Sea recruitment conceptual model reproduced from Monnahan et al. (2026), original Figure 1, printed page 19 (PDF page 27). Source: AFSC Processed Report 2026-09 (Monnahan et al. 2026).

Figure B1 Preliminary Bering Sea Recruitment Conceptual Model



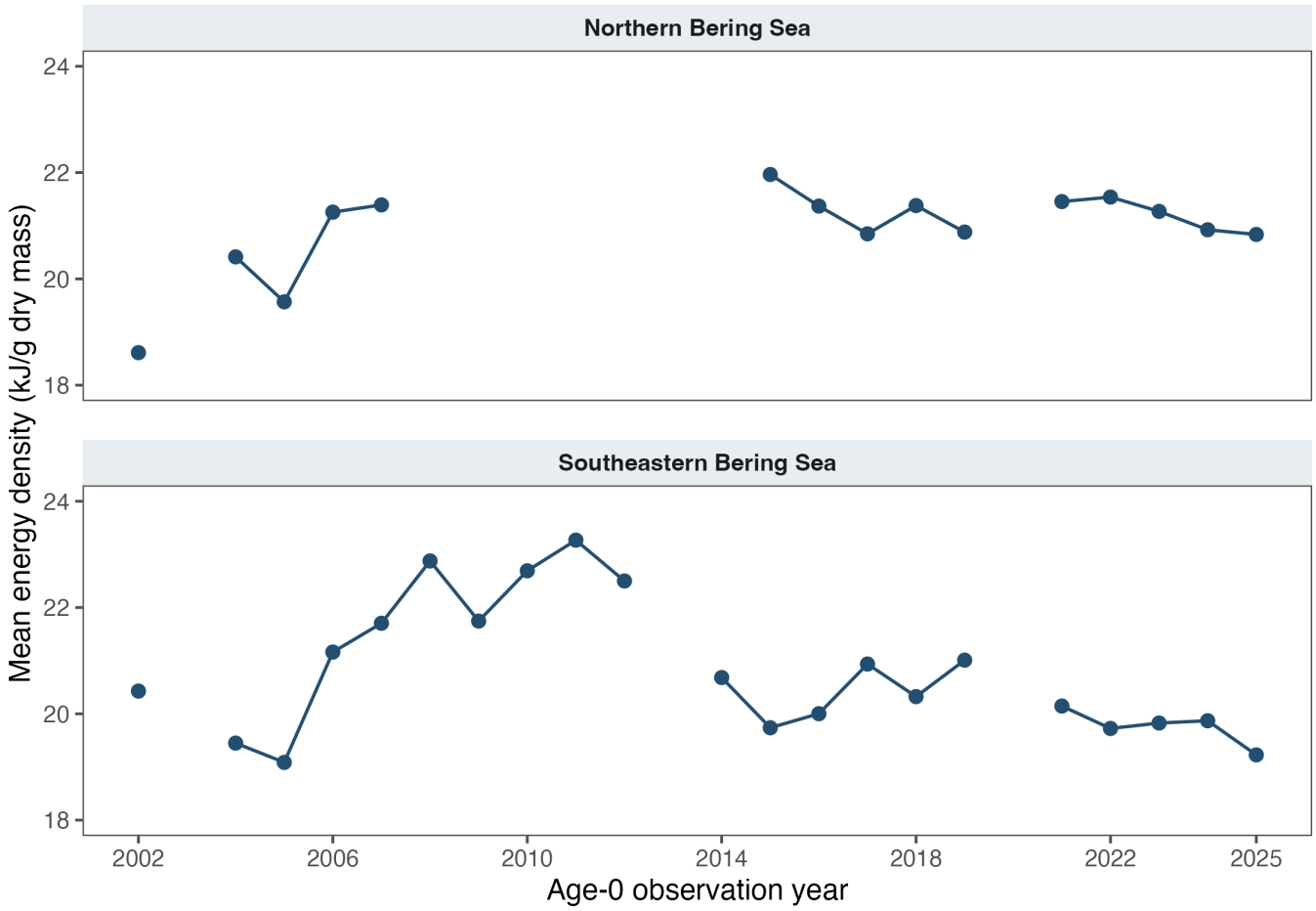

*Note.* Preliminary Bering Sea growth conceptual model reproduced from Monnahan et al. (2026), original Figure 2, printed page 20 (PDF page 28). Source: AFSC Processed Report 2026-09 (Monnahan et al. 2026).

Figure B2 Preliminary Bering Sea Growth Conceptual Model




*Note.* Conceptual cohort sequence from physical conditions and spawning through prey, fall energy, first-winter survival and age-1 recruitment, with subsequent predation affecting survival to age 3. Arrows represent hypotheses supported by the cited literature; they are separate from the fitted direct proxy paths.

Figure B3 Conceptual Cohort Sequence From Spawning Through Age-3 Survival




*Note.* Annual mean age-0 pollock energy density in the northern and southeastern Bering Seas, 2002–2025, in kJ/g dry mass. Points show regional means; lines connect consecutive observed years and break at missing years. Source: Page and Maselko (2026).

Figure B4 Annual Mean Age-0 Pollock Energy Density, 2002–2025

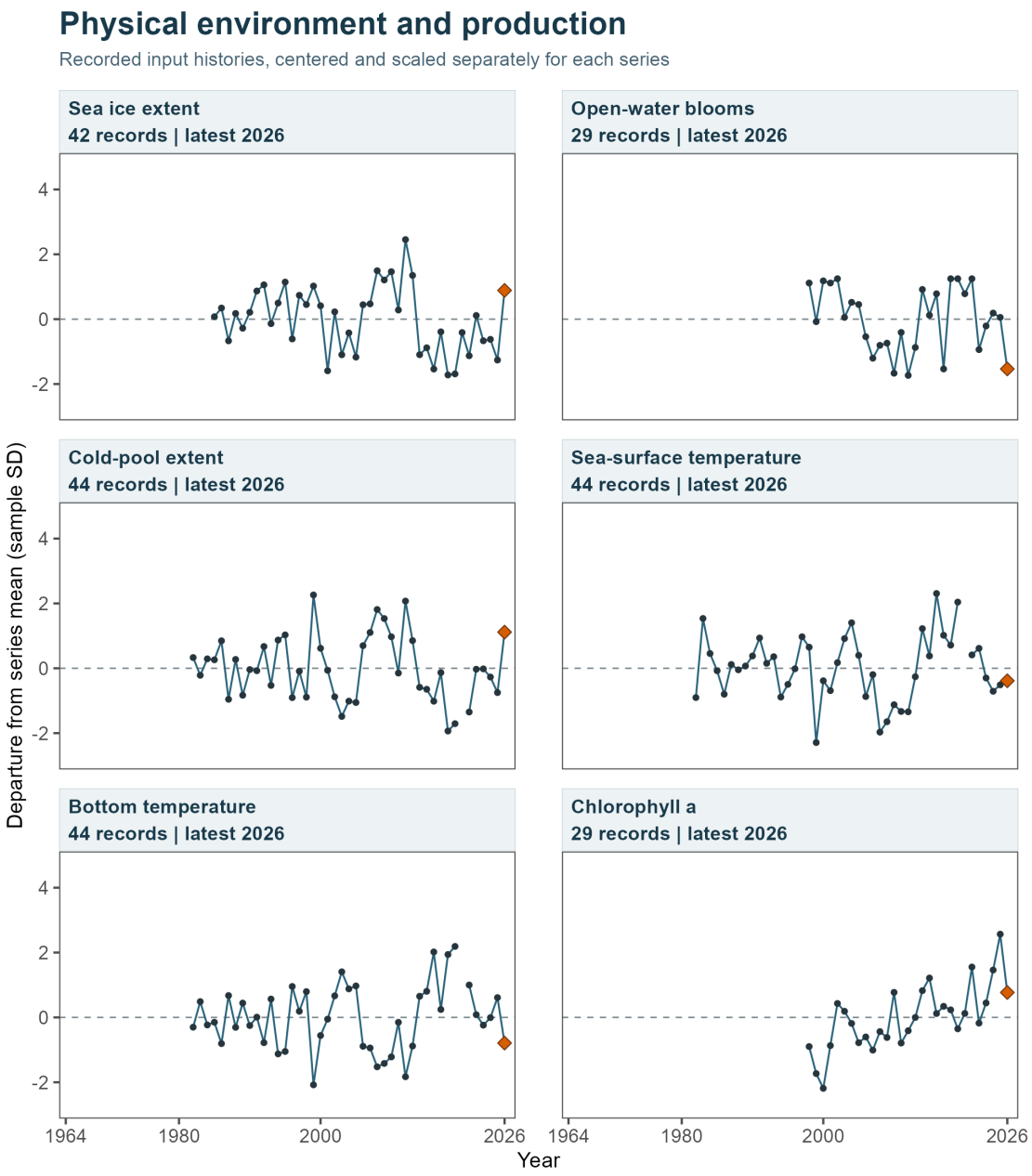

Physical environment and production
Recorded input histories, centered and scaled separately for each series
Sea ice extent
42 records | latest 2026
Open-water blooms
29 records | latest 2026
Cold-pool extent
44 records | latest 2026
Sea-surface temperature
44 records | latest 2026
Bottom temperature
44 records | latest 2026
Chlorophyll a
29 records | latest 2026
Departure from series mean (sample SD)
4
2
0
-2
1964
1980
2000
2026
Year
Means and sample SDs use each series' available records, including its latest year.
Lines join consecutive recorded years only; missing values remain blank. No model fits are shown.
Source: repository annual input matrix. Indicator metadata remain under review.

Figure B5 (continued)

**Zooplankton and prey**

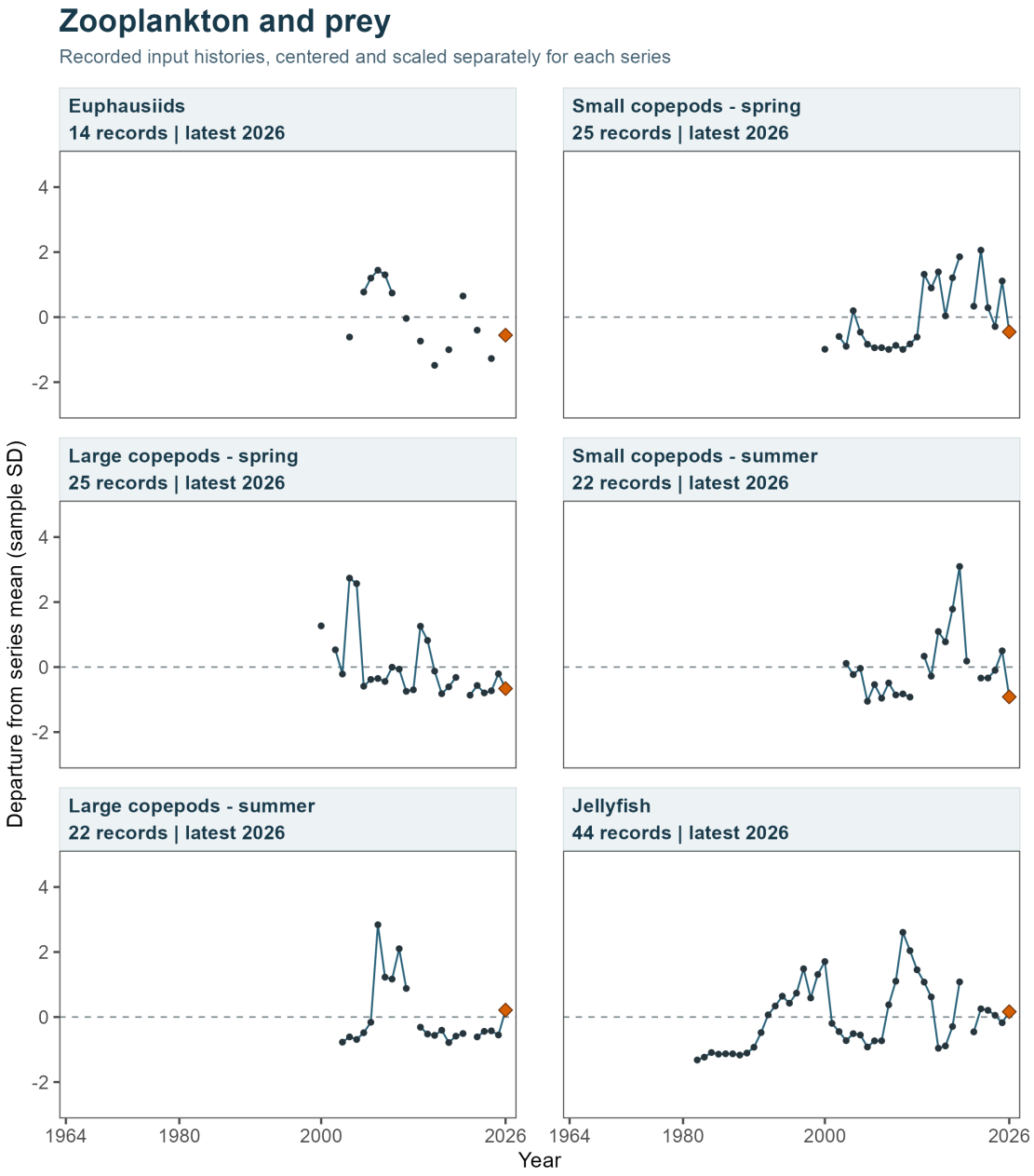
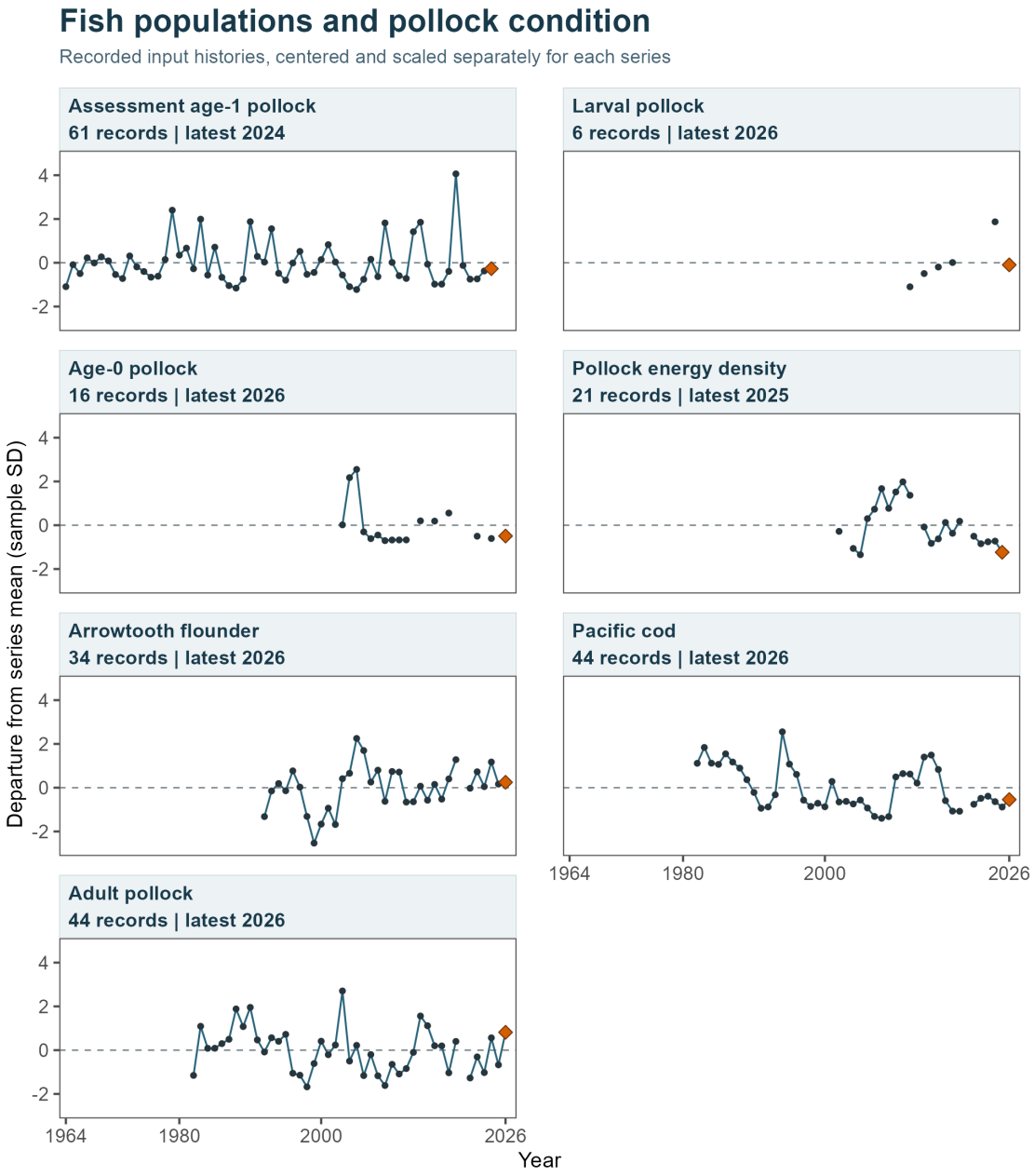


Figure B5 (continued)

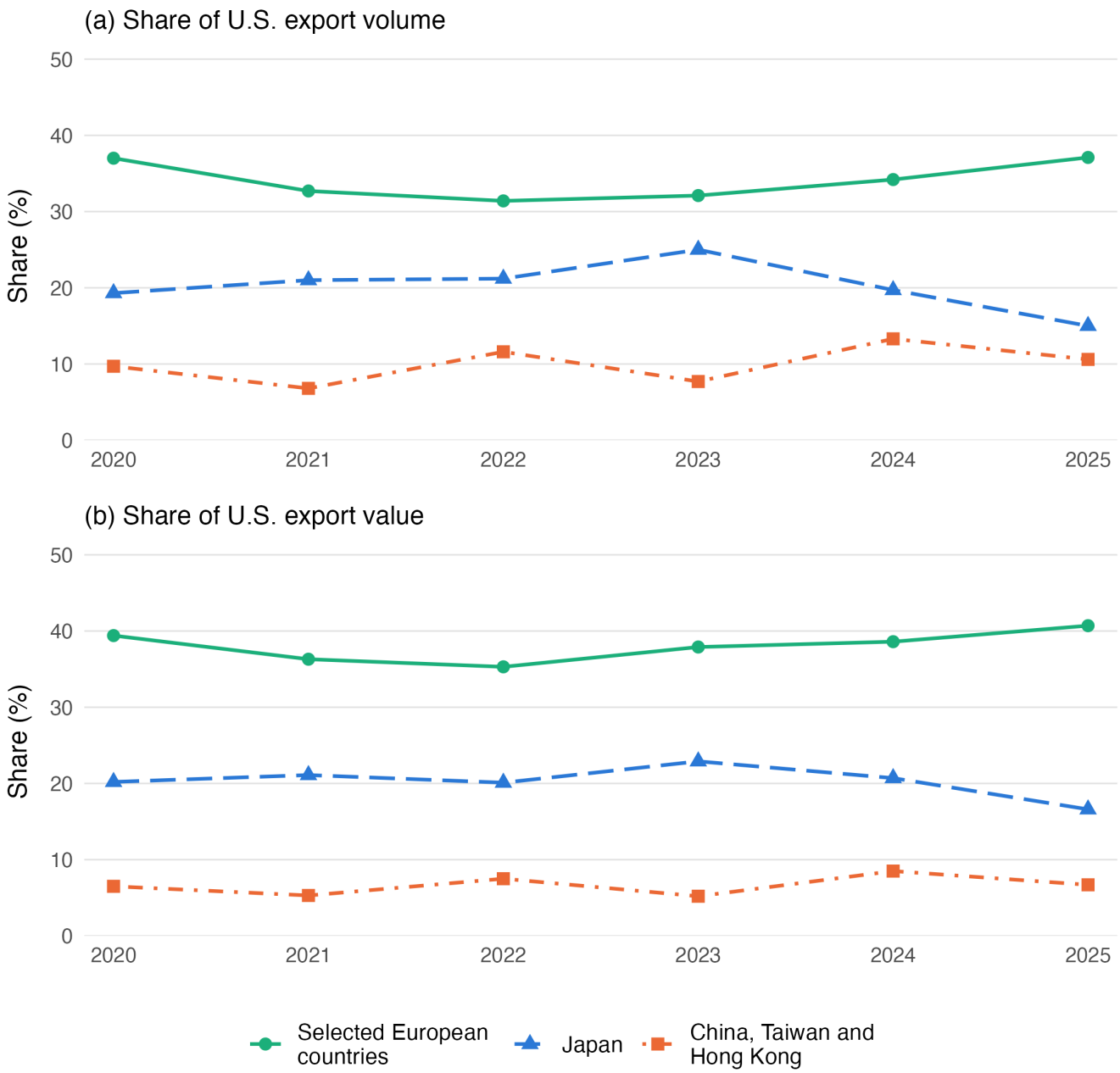
**Fish abundance, energy density and the age-1 response**




*Note.* Recorded annual input histories grouped for readability. Each value is centered on that series' available-record arithmetic mean and divided by its sample SD, including the latest recorded year. The zero line denotes that mean; gaps are retained without interpolation. Symbols identify recorded values and highlighted endpoints identify each latest record. The panels show 19 annual series.

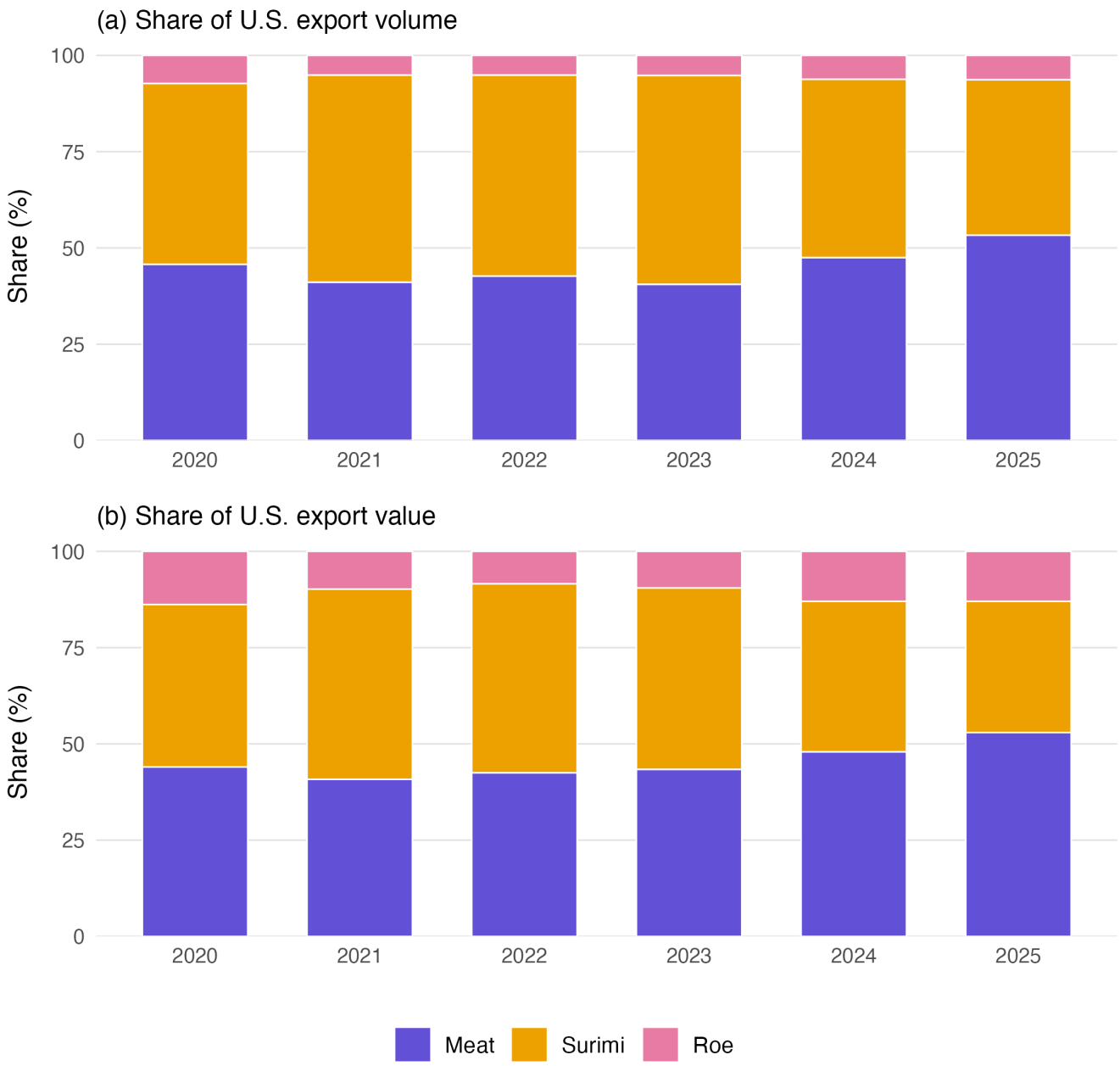
The energy record ends in 2025; the age-1 response ends in 2024. These descriptive histories contain no DSEM fitted states, forecast values or uncertainty bands. Primary-source metadata remain incomplete.

Figure B5 Recorded Indicator Histories in Standardized Units



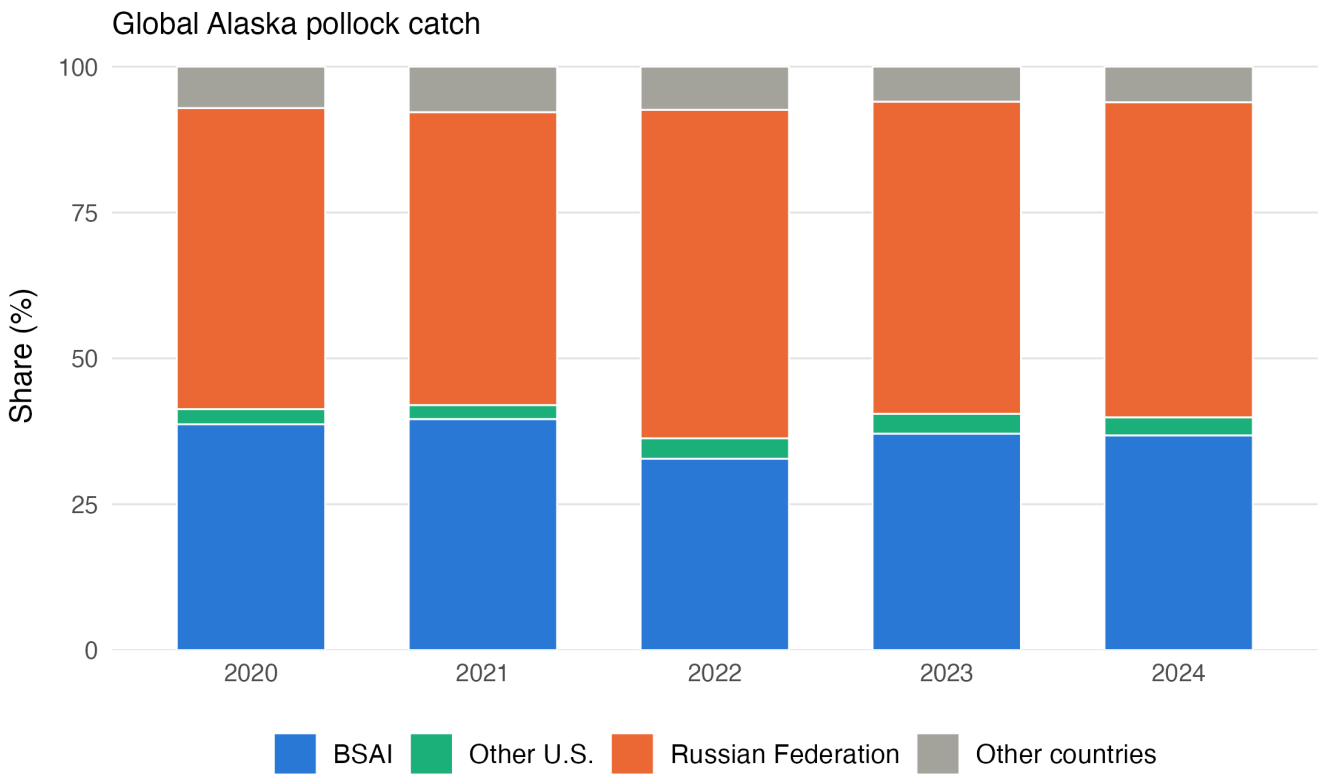

*Note.* Destination shares of U.S. Alaska pollock domestic exports by (a) product volume and (b) nominal FAS value. The three groups shown cover part of total exports. Selected European countries are Austria, Belgium, Denmark, France, Germany, Greece, Ireland, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland and the United Kingdom. China, Taiwan and Hong Kong form the legacy aggregate. Shares and definitions match Table B8, panel B. Source: (NOAA Fisheries 2026).

Figure B6 U.S. Alaska Pollock Export Destination Shares, 2020–2025



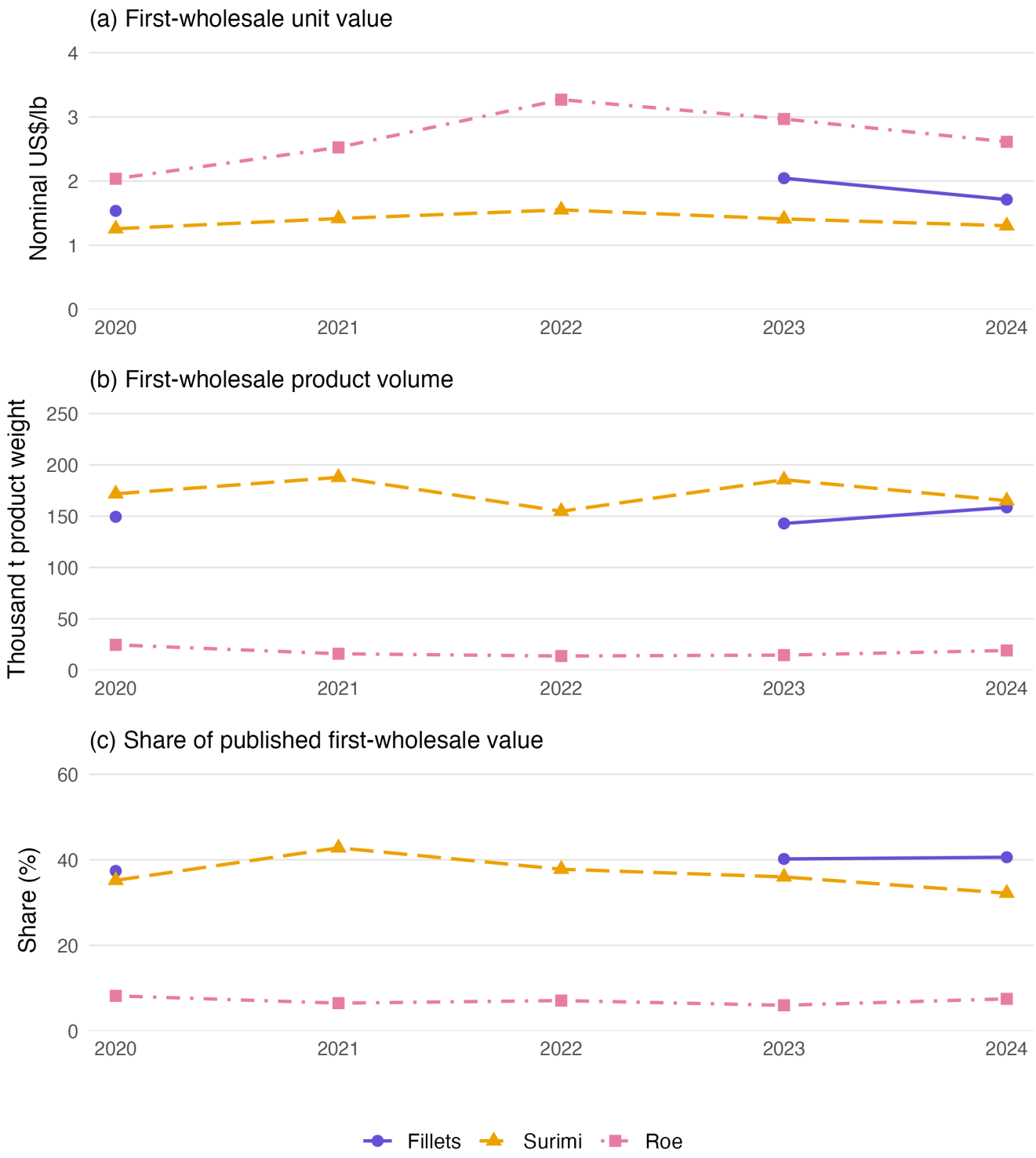

*Note.* Product shares of U.S. Alaska pollock domestic exports by (a) product volume and (b) nominal FAS value. Meat combines products other than surimi and roe, including frozen whole and headed-and-gutted fish and fillets. The one-decimal source shares are rescaled to sum to 100% within each year, as in the dashboard; the largest adjustment is 0.054 percentage points. Values and product definitions are in Table B8, panel B. Source: (NOAA Fisheries 2026).

Figure B7 U.S. Alaska Pollock Export Product Mix, 2020–2025




*Note.* Shares of global Alaska pollock capture production by live weight. Other U.S. is the U.S. share minus the BSAI share; other countries is the remainder after the United States and Russian Federation. BSAI includes Bering Sea and Aleutian Islands catch. FAO release Capture_2026.1.0 supplies the global totals; the Economic SAFE supplies BSAI catch. Definitions and totals are in Table B8, panel A. Sources: (FAO 2026); (Abelman et al. 2026).

Figure B8 Global Alaska Pollock Catch by Origin, 2020–2024




*Note.* BSAI pollock (a) nominal first-wholesale unit values, (b) production in thousand tons of product weight, and (c) shares of published first-wholesale value. Unit values equal total published value times the product value share, divided by product volume and converted to US$/lb; they reflect product mix as well as market conditions. Fillet observations for 2021–2022 are unavailable because confidential shoreside deep-skin components prevent their calculation. Gaps remain missing rather than zero. Other products account for the remaining value share. Values and definitions match Table B7. Source: (Abelman et al. 2026).

Figure B9 BSAI Pollock First-Wholesale Product Indicators, 2020–2024

# Appendix C
# Research and Other Noncommercial Pollock Removals

**Reported noncommercial catch**

Reported noncommercial pollock catches in the Bering Sea are summarized by collection in Table C1 and by year in Table C2. The AKR dataset covers 1982–2025 across the Gulf of Alaska, Bering Sea and Aleutian Islands; Bering Sea records begin in 1991, with broader multisurvey coverage from 2010. These are reported collection-year amounts, rather than verified complete EBS-stock totals. Missing years and collection cells mean no record in this extract. The extract ends in 2025.

The summaries use the reported Bering Sea FMP area, retaining stations from collections whose names mention other regions. Explicitly Bogoslof-named surveys are shown separately; other collections in reporting area 518 remain among the other reported Bering Sea catches. In 2024, the total was 216.280081 t, including 4.393640 t from the named Bogoslof survey and 211.886441 t from other reported Bering Sea collections. The corresponding 2025 reported total was 140.994428 t.

Experimental fishing permit (EFP) records account for 1,198.506090 t in 2011 and 771.516610 t in 2021. Their inclusion explains the large totals in those years. The sole 2020 record is a Bogoslof collection; other EBS research removals remain unverified for that year. Historical EFP inclusion in official catch accounting requires reconciliation. AKR metadata state that no 2025 EFP catches occurred outside CAS.

Weights are converted from kilograms to tons using the units corroborated by AKR dataset correspondence, dated 9 October and forwarded on 13 October 2026. Annual completeness of the AKR dataset received on 27 October remains unverified. The 2020–2023 totals agree, after rounding, with the accepted 2024 SAFE's research-catch table (Ianelli et al. 2024). Noncommercial removals are reported separately from assessment catch. The full annual summaries and source record retain the collection details and reporting qualifications.

**Exempted fishing and other removal categories**

The 2025 EFP 2025-01 permit and 2026 renewal allow up to **3,000 t of total groundfish**, rather than pollock alone, to be exempt from TAC during each permit's effective period; additional groundfish catch counts against TAC. This authorization ceiling is separate from observed removals. Actual pollock catch under these permits and its accounting treatment require permit-specific records.

EFP 2024-02 provides a footrope/gear exemption and otherwise retains applicable regulations. It supplies no basis for assuming a TAC exemption. The noncommercial dataset includes BS-labelled subsistence records in 1991 and 1994; complete recent stock-specific subsistence, recreational and other-FMP removals remain unverified.

**Tables**

Table C1 Reported Bering Sea Noncommercial Pollock Catch by Collection and Year

**Panel A. 1991–2009**

| Collection (agency) | 1991 | 1994 | 1997 | 1999 | 2001 | 2003 | 2005 | 2007 | 2009 |
|---|---|---|---|---|---|---|---|---|---|
| AFSC Annual Longline Survey (NMFS) | NA | NA | 1.0 | 2.1 | 4.0 | 3.1 | 1.1 | 0.9 | 1.4 |
| Subsistence Fishery (ADFG) | <0.1 | <0.1 | NA | NA | NA | NA | NA | NA | NA |

**Panel B. 2010–2017**

| Collection (agency) | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 |
|---|---|---|---|---|---|---|---|---|
| AFSC Annual Longline Survey (NMFS) | NA | 3.9 | NA | 2.3 | NA | 3.4 | NA | 1.6 |
| Aleutian Island Bottom Trawl Survey (NMFS) | 27.0 | NA | 4.5 | NA | 12.3 | NA | 2.6 | NA |
| Bering Sea Acoustic Survey (NMFS) | 49.6 | NA | NA | NA | NA | NA | NA | NA |
| Bering Sea Bottom Trawl Survey (NMFS) | 140.7 | NA | NA | NA | NA | NA | NA | NA |
| Bering Sea Slope Survey (NMFS) | 3.3 | NA | 7.7 | NA | NA | NA | 12.4 | NA |
| Bogoslof EIT Survey with Northern Extensions (NMFS) | NA | NA | 4.1 | NA | NA | NA | NA | NA |
| Eastern Bering Sea Bottom Trawl Survey (NMFS) | NA | 116.5 | 140.0 | 178.7 | 293.4 | 249.7 | 177.4 | 181.1 |
| Gulf of Alaska Bottom Trawl Survey (NMFS) | NA | <0.1 | NA | <0.1 | NA | <0.1 | NA | NA |
| IPHC Annual Longline Survey (IPHC) | 0.7 | 0.2 | 0.1 | 0.2 | 0.6 | 0.7 | 0.9 | 0.1 |
| Large-Mesh Trawl Survey (ADFG) | 3.7 | 3.2 | 2.4 | 2.2 | 1.4 | 2.1 | 4.6 | 5.0 |
| Northern Bering Sea Bottom Trawl Survey (NMFS) | 0.6 | NA | NA | NA | NA | NA | NA | 38.1 |
| Pollock EFP 11-01 (NMFS) | NA | 1,198.5 | 3.8 | NA | NA | NA | NA | NA |
| Pribilof Islands Crab Survey (ADFG) | NA | 0.1 | NA | NA | NA | NA | 0.3 | NA |
| St. Matthews Crab Survey (ADFG) | 0.2 | NA | NA | NA | NA | NA | NA | 0.2 |
| Summer EBS Survey with Russia (NMFS) | NA | NA | 45.2 | NA | NA | NA | NA | NA |

**Panel C. 2018–2025**

| Collection (agency) | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|---|
| AFSC Annual Longline Survey (NMFS) | NA | 1.3 | NA | 0.9 | NA | 0.7 | NA | NA |
| Aleutian Island Bottom Trawl Survey (NMFS) | 6.1 | NA | NA | NA | 9.7 | NA | 5.0 | NA |
| BSAI Trawl Salmon Excluder Device EFP 2018-03-02 (NMFS) | NA | NA | NA | 771.5 | NA | NA | NA | NA |
| Blue King Crab Pot Survey (ADFG) | <0.1 | NA | NA | NA | 0.1 | NA | NA | NA |
| Bogoslof Island Walleye Pollock Acoustic-Trawl Survey (NMFS) | 6.0 | NA | NA | NA | NA | NA | NA | NA |
| Eastern Bering Sea Bottom Trawl Survey (NMFS) | 111.7 | 188.4 | NA | 107.8 | 168.1 | 118.0 | 163.2 | 120.4 |
| Eastern Bering Sea Walleye Pollock Acoustic-Trawl Survey (NMFS) | 36.7 | NA | NA | NA | 48.3 | NA | NA | NA |
| Gulf of Alaska Bottom Trawl Survey (NMFS) | NA | <0.1 | NA | NA | NA | <0.1 | NA | NA |
| IPHC Annual Longline Survey (IPHC) | 0.1 | 0.1 | NA | <0.1 | 0.2 | NA | 0.1 | NA |
| Large-Mesh Trawl Survey (ADFG) | 1.8 | 1.8 | NA | 1.4 | 6.1 | 0.7 | 3.8 | 2.1 |
| Northern Bering Sea Bottom Trawl Survey (NMFS) | 15.0 | 38.6 | NA | 16.0 | 13.1 | NA | NA | 18.5 |
| Summer Acoustic-Trawl Survey of Walleye Pollock in the Eastern Bering Sea (NMFS) | NA | NA | NA | NA | NA | NA | 39.8 | NA |
| Winter Acoustic-Trawl Survey of Walleye Pollock Near Bogoslof Island and vicinity (NMFS) | NA | NA | 12.9 | NA | NA | NA | 4.4 | NA |

*Note.* Values are metric tons (source kilograms divided by 1,000), rounded to one decimal place. Source: received Noncommercial Fishery Catch (1).csv, 27 October 2026 received-date field. Reported records have FMP Area = BS and Species Group Code = PLCK. All reporting areas are retained, including 518. NA indicates missing records; 0.0 indicates a reported zero. Early panels show observed BS years only, and collection rows with no records in a panel are omitted from that panel. All collection-year cells and full source precision are retained in the accompanying CSV. Scope follows the FMP field, including collections whose names mention other regions. Positive amounts that round to 0.0 t are shown as <0.1; 0.0 denotes a source-recorded zero.

Table C2 Annual Reported Bering Sea Pollock Catch by Source Category

| Collection year | Reported BS total (t) | EFP (t) | Explicitly Bogoslof-named (t) | Other reported BS (t) | Source records |
|---|---|---|---|---|---|
| 1991 | <0.1 | NA | NA | <0.1 | 1 |
| 1992 | NA | NA | NA | NA | NA |
| 1993 | NA | NA | NA | NA | NA |
| 1994 | <0.1 | NA | NA | <0.1 | 1 |
| 1995 | NA | NA | NA | NA | NA |
| 1996 | NA | NA | NA | NA | NA |
| 1997 | 1.0 | NA | NA | 1.0 | 4 |
| 1998 | NA | NA | NA | NA | NA |
| 1999 | 2.1 | NA | NA | 2.1 | 4 |
| 2000 | NA | NA | NA | NA | NA |
| 2001 | 4.0 | NA | NA | 4.0 | 4 |
| 2002 | NA | NA | NA | NA | NA |
| 2003 | 3.1 | NA | NA | 3.1 | 4 |
| 2004 | NA | NA | NA | NA | NA |
| 2005 | 1.1 | NA | NA | 1.1 | 3 |
| 2006 | NA | NA | NA | NA | NA |
| 2007 | 0.9 | NA | NA | 0.9 | 4 |
| 2008 | NA | NA | NA | NA | NA |
| 2009 | 1.4 | NA | NA | 1.4 | 4 |
| 2010 | 226.0 | NA | NA | 226.0 | 65 |
| 2011 | 1,322.5 | 1,198.5 | NA | 124.0 | 30 |
| 2012 | 207.9 | 3.8 | 4.1 | 200.0 | 38 |
| 2013 | 183.4 | NA | NA | 183.4 | 56 |
| 2014 | 307.7 | NA | NA | 307.7 | 26 |
| 2015 | 255.9 | NA | NA | 255.9 | 30 |
| 2016 | 198.1 | NA | NA | 198.1 | 32 |
| 2017 | 226.1 | NA | NA | 226.1 | 29 |
| 2018 | 177.4 | NA | 6.0 | 171.4 | 38 |
| 2019 | 230.1 | NA | NA | 230.1 | 29 |
| 2020 | 12.9 | NA | 12.9 | NA | 1 |
| 2021 | 897.6 | 771.5 | NA | 126.1 | 34 |
| 2022 | 245.7 | NA | NA | 245.7 | 40 |
| 2023 | 119.4 | NA | NA | 119.4 | 19 |
| 2024 | 216.3 | NA | 4.4 | 211.9 | 33 |
| 2025 | 141.0 | NA | NA | 141.0 | 20 |
| 2026 | NA | NA | NA | NA | NA |

*Note.* Weights are metric tons, rounded to one decimal place. Categories are mutually exclusive source-label summaries. EFP comprises the two explicitly EFP-named collections; the Bogoslof column includes collections explicitly named for Bogoslof. Other reported BS includes all remaining BS-labelled records, including area-518 records from other collections. NA indicates missing records; 0.0 indicates a reported zero. The row count measures records. Totals cover the reported records; annual coverage and overlap with official catch accounting remain uncertain.

# Appendix D
# Detailed Current-Model Estimates

All estimated parameters and their asymptotic uncertainty are listed below (Table D1), followed by the largest parameter correlations (Table D2). These estimates use the selected model and fixed biological schedules. The full numerical products also retain natural-scale transformations and all shared parameter positions.

Table D1 Model 26.0 Estimated Parameters and Approximate 95% Confidence Intervals

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0001 | Recruitment level | R0 | 9.6728 | 0.1212 | 9.4352 | 9.9104 |
| P0002 | Recruitment deviation | 1964 | −1.0198 | 0.4681 | −1.9373 | −0.1023 |
| P0003 | Recruitment deviation | 1965 | 0.3917 | 0.2916 | −0.1798 | 0.9632 |
| P0004 | Recruitment deviation | 1966 | 0.0001 | 0.3593 | −0.7041 | 0.7042 |
| P0005 | Recruitment deviation | 1967 | 0.5387 | 0.2917 | −0.0330 | 1.1104 |
| P0006 | Recruitment deviation | 1968 | 0.3459 | 0.3165 | −0.2745 | 0.9662 |
| P0007 | Recruitment deviation | 1969 | 0.5038 | 0.2931 | −0.0705 | 1.0782 |
| P0008 | Recruitment deviation | 1970 | 0.3846 | 0.3052 | −0.2136 | 0.9828 |
| P0009 | Recruitment deviation | 1971 | −0.1528 | 0.3739 | −0.8856 | 0.5800 |
| P0010 | Recruitment deviation | 1972 | −0.3766 | 0.3851 | −1.1314 | 0.3782 |
| P0011 | Recruitment deviation | 1973 | 0.5213 | 0.2325 | 0.0655 | 0.9770 |
| P0012 | Recruitment deviation | 1974 | 0.1941 | 0.2371 | −0.2706 | 0.6589 |
| P0013 | Recruitment deviation | 1975 | 0.0354 | 0.2205 | −0.3967 | 0.4675 |
| P0014 | Recruitment deviation | 1976 | −0.2044 | 0.2104 | −0.6169 | 0.2081 |
| P0015 | Recruitment deviation | 1977 | −0.1243 | 0.1911 | −0.4989 | 0.2503 |
| P0016 | Recruitment deviation | 1978 | 0.5126 | 0.1555 | 0.2078 | 0.8175 |
| P0017 | Recruitment deviation | 1979 | 1.3692 | 0.1375 | 1.0997 | 1.6387 |
| P0018 | Recruitment deviation | 1980 | 0.6249 | 0.1492 | 0.3324 | 0.9174 |
| P0019 | Recruitment deviation | 1981 | 0.7685 | 0.1463 | 0.4817 | 1.0553 |
| P0020 | Recruitment deviation | 1982 | 0.1889 | 0.1614 | −0.1274 | 0.5052 |
| P0021 | Recruitment deviation | 1983 | 1.1863 | 0.1357 | 0.9204 | 1.4521 |
| P0022 | Recruitment deviation | 1984 | −0.1097 | 0.1634 | −0.4299 | 0.2104 |
| P0023 | Recruitment deviation | 1985 | 0.7176 | 0.1390 | 0.4450 | 0.9901 |
| P0024 | Recruitment deviation | 1986 | −0.2200 | 0.1558 | −0.5254 | 0.0853 |
| P0025 | Recruitment deviation | 1987 | −0.7740 | 0.1648 | −1.0971 | −0.4510 |
| P0026 | Recruitment deviation | 1988 | −1.0224 | 0.1658 | −1.3474 | −0.6974 |
| P0027 | Recruitment deviation | 1989 | −0.2878 | 0.1424 | −0.5669 | −0.0086 |
| P0028 | Recruitment deviation | 1990 | 1.1562 | 0.1262 | 0.9090 | 1.4035 |
| P0029 | Recruitment deviation | 1991 | 0.5442 | 0.1307 | 0.2881 | 0.8003 |
| P0030 | Recruitment deviation | 1992 | 0.4032 | 0.1325 | 0.1436 | 0.6628 |
| P0031 | Recruitment deviation | 1993 | 1.0698 | 0.1269 | 0.8211 | 1.3186 |
| P0032 | Recruitment deviation | 1994 | 0.0032 | 0.1356 | −0.2625 | 0.2690 |
| P0033 | Recruitment deviation | 1995 | −0.3517 | 0.1389 | −0.6239 | −0.0794 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0034 | Recruitment deviation | 1996 | 0.3553 | 0.1292 | 0.1020 | 0.6086 |
| P0035 | Recruitment deviation | 1997 | 0.6464 | 0.1269 | 0.3977 | 0.8950 |
| P0036 | Recruitment deviation | 1998 | −0.0602 | 0.1317 | −0.3183 | 0.1978 |
| P0037 | Recruitment deviation | 1999 | 0.0094 | 0.1306 | −0.2466 | 0.2654 |
| P0038 | Recruitment deviation | 2000 | 0.4371 | 0.1273 | 0.1876 | 0.6866 |
| P0039 | Recruitment deviation | 2001 | 0.7735 | 0.1252 | 0.5281 | 1.0189 |
| P0040 | Recruitment deviation | 2002 | 0.3678 | 0.1265 | 0.1199 | 0.6157 |
| P0041 | Recruitment deviation | 2003 | −0.1019 | 0.1286 | −0.3540 | 0.1502 |
| P0042 | Recruitment deviation | 2004 | −0.8894 | 0.1362 | −1.1564 | −0.6224 |
| P0043 | Recruitment deviation | 2005 | −1.2337 | 0.1415 | −1.5110 | −0.9563 |
| P0044 | Recruitment deviation | 2006 | −0.3192 | 0.1309 | −0.5758 | −0.0627 |
| P0045 | Recruitment deviation | 2007 | 0.4533 | 0.1271 | 0.2041 | 0.7025 |
| P0046 | Recruitment deviation | 2008 | −0.1584 | 0.1330 | −0.4192 | 0.1024 |
| P0047 | Recruitment deviation | 2009 | 1.1374 | 0.1249 | 0.8925 | 1.3823 |
| P0048 | Recruitment deviation | 2010 | 0.3763 | 0.1287 | 0.1240 | 0.6287 |
| P0049 | Recruitment deviation | 2011 | −0.1110 | 0.1328 | −0.3713 | 0.1493 |
| P0050 | Recruitment deviation | 2012 | −0.2546 | 0.1337 | −0.5166 | 0.0074 |
| P0051 | Recruitment deviation | 2013 | 1.0028 | 0.1249 | 0.7580 | 1.2475 |
| P0052 | Recruitment deviation | 2014 | 1.1366 | 0.1252 | 0.8912 | 1.3819 |
| P0053 | Recruitment deviation | 2015 | 0.2917 | 0.1313 | 0.0344 | 0.5490 |
| P0054 | Recruitment deviation | 2016 | −0.6103 | 0.1475 | −0.8995 | −0.3212 |
| P0055 | Recruitment deviation | 2017 | −0.6994 | 0.1519 | −0.9971 | −0.4018 |
| P0056 | Recruitment deviation | 2018 | 0.0578 | 0.1410 | −0.2187 | 0.3342 |
| P0057 | Recruitment deviation | 2019 | 1.7104 | 0.1302 | 1.4552 | 1.9656 |
| P0058 | Recruitment deviation | 2020 | 0.5518 | 0.1454 | 0.2667 | 0.8368 |
| P0059 | Recruitment deviation | 2021 | −0.2033 | 0.1786 | −0.5534 | 0.1467 |
| P0060 | Recruitment deviation | 2022 | −0.2247 | 0.2049 | −0.6263 | 0.1768 |
| P0061 | Recruitment deviation | 2023 | −0.3126 | 0.2286 | −0.7606 | 0.1353 |
| P0062 | Recruitment deviation | 2024 | 0.9050 | 0.2517 | 0.4118 | 1.3983 |
| P0063 | Recruitment deviation | 2025 | 0.5663 | 0.2920 | −0.0060 | 1.1387 |
| P0064 | Recruitment deviation | 2026 | 0.8281 | 0.5378 | −0.2260 | 1.8822 |
| P0065 | Initial-age deviation | Age1 | −0.3949 | 0.3509 | −1.0827 | 0.2928 |
| P0066 | Initial-age deviation | Age2 | −0.4683 | 0.3455 | −1.1455 | 0.2089 |
| P0067 | Initial-age deviation | Age3 | −1.2468 | 0.4532 | −2.1351 | −0.3586 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0068 | Initial-age deviation | Age4 | −1.4797 | 0.5231 | −2.5049 | −0.4545 |
| P0069 | Initial-age deviation | Age5 | −0.9725 | 0.4745 | −1.9026 | −0.0424 |
| P0070 | Initial-age deviation | Age6 | −1.1558 | 0.5508 | −2.2353 | −0.0762 |
| P0071 | Initial-age deviation | Age7 | −1.3132 | 0.6310 | −2.5499 | −0.0765 |
| P0072 | Initial-age deviation | Age8 | −1.2359 | 0.6710 | −2.5510 | 0.0791 |
| P0073 | Initial-age deviation | Age9 | −1.0702 | 0.6957 | −2.4338 | 0.2934 |
| P0074 | Initial-age deviation | Age10 | −0.9172 | 0.7220 | −2.3322 | 0.4979 |
| P0075 | Initial-age deviation | Age11 | −0.7788 | 0.7491 | −2.2470 | 0.6894 |
| P0076 | Initial-age deviation | Age12 | −0.6579 | 0.7760 | −2.1789 | 0.8631 |
| P0077 | Initial-age deviation | Age13 | −0.5491 | 0.8035 | −2.1239 | 1.0256 |
| P0078 | Initial-age deviation | Age14 | −1.0041 | 0.7100 | −2.3957 | 0.3874 |
| P0079 | Log fishing scale | Fishery; 1964 | −2.9386 | 0.2174 | −3.3647 | −2.5125 |
| P0080 | Log fishing scale | Fishery; 1965 | −2.7303 | 0.2116 | −3.1450 | −2.3155 |
| P0081 | Log fishing scale | Fishery; 1966 | −2.6737 | 0.2155 | −3.0960 | −2.2513 |
| P0082 | Log fishing scale | Fishery; 1967 | −2.0362 | 0.2270 | −2.4811 | −1.5913 |
| P0083 | Log fishing scale | Fishery; 1968 | −1.9723 | 0.2260 | −2.4153 | −1.5292 |
| P0084 | Log fishing scale | Fishery; 1969 | −1.9071 | 0.2317 | −2.3611 | −1.4530 |
| P0085 | Log fishing scale | Fishery; 1970 | −1.5364 | 0.2309 | −1.9890 | −1.0838 |
| P0086 | Log fishing scale | Fishery; 1971 | −1.1811 | 0.2325 | −1.6368 | −0.7254 |
| P0087 | Log fishing scale | Fishery; 1972 | −1.0495 | 0.2265 | −1.4935 | −0.6055 |
| P0088 | Log fishing scale | Fishery; 1973 | −0.9062 | 0.2295 | −1.3561 | −0.4563 |
| P0089 | Log fishing scale | Fishery; 1974 | −0.8209 | 0.2280 | −1.2679 | −0.3740 |
| P0090 | Log fishing scale | Fishery; 1975 | −0.9945 | 0.2055 | −1.3973 | −0.5916 |
| P0091 | Log fishing scale | Fishery; 1976 | −1.0737 | 0.2009 | −1.4673 | −0.6800 |
| P0092 | Log fishing scale | Fishery; 1977 | −1.1394 | 0.2153 | −1.5614 | −0.7174 |
| P0093 | Log fishing scale | Fishery; 1978 | −1.0271 | 0.2171 | −1.4525 | −0.6017 |
| P0094 | Log fishing scale | Fishery; 1979 | −0.9874 | 0.2107 | −1.4003 | −0.5745 |
| P0095 | Log fishing scale | Fishery; 1980 | −1.0814 | 0.1983 | −1.4701 | −0.6927 |
| P0096 | Log fishing scale | Fishery; 1981 | −1.4230 | 0.1935 | −1.8023 | −1.0438 |
| P0097 | Log fishing scale | Fishery; 1982 | −1.8679 | 0.1901 | −2.2404 | −1.4954 |
| P0098 | Log fishing scale | Fishery; 1983 | −2.0507 | 0.1833 | −2.4100 | −1.6914 |
| P0099 | Log fishing scale | Fishery; 1984 | −2.0663 | 0.1527 | −2.3655 | −1.7671 |
| P0100 | Log fishing scale | Fishery; 1985 | −2.0559 | 0.1408 | −2.3318 | −1.7801 |
| P0101 | Log fishing scale | Fishery; 1986 | −2.1239 | 0.1347 | −2.3878 | −1.8599 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0102 | Log fishing scale | Fishery; 1987 | −2.4130 | 0.1266 | −2.6611 | −2.1649 |
| P0103 | Log fishing scale | Fishery; 1988 | −2.0653 | 0.1178 | −2.2962 | −1.8344 |
| P0104 | Log fishing scale | Fishery; 1989 | −2.0069 | 0.1006 | −2.2040 | −1.8098 |
| P0105 | Log fishing scale | Fishery; 1990 | −1.5822 | 0.0944 | −1.7672 | −1.3972 |
| P0106 | Log fishing scale | Fishery; 1991 | −1.4032 | 0.0849 | −1.5696 | −1.2368 |
| P0107 | Log fishing scale | Fishery; 1992 | −1.1200 | 0.0920 | −1.3004 | −0.9396 |
| P0108 | Log fishing scale | Fishery; 1993 | −1.5672 | 0.1086 | −1.7800 | −1.3543 |
| P0109 | Log fishing scale | Fishery; 1994 | −1.8109 | 0.1176 | −2.0413 | −1.5804 |
| P0110 | Log fishing scale | Fishery; 1995 | −1.8155 | 0.1118 | −2.0347 | −1.5964 |
| P0111 | Log fishing scale | Fishery; 1996 | −1.6567 | 0.1114 | −1.8751 | −1.4383 |
| P0112 | Log fishing scale | Fishery; 1997 | −1.6452 | 0.1244 | −1.8890 | −1.4014 |
| P0113 | Log fishing scale | Fishery; 1998 | −1.8130 | 0.1174 | −2.0431 | −1.5830 |
| P0114 | Log fishing scale | Fishery; 1999 | −2.0245 | 0.1131 | −2.2463 | −1.8028 |
| P0115 | Log fishing scale | Fishery; 2000 | −1.9078 | 0.0941 | −2.0923 | −1.7232 |
| P0116 | Log fishing scale | Fishery; 2001 | −1.7581 | 0.0844 | −1.9236 | −1.5926 |
| P0117 | Log fishing scale | Fishery; 2002 | −1.5902 | 0.0776 | −1.7424 | −1.4381 |
| P0118 | Log fishing scale | Fishery; 2003 | −1.5098 | 0.0772 | −1.6611 | −1.3585 |
| P0119 | Log fishing scale | Fishery; 2004 | −1.5953 | 0.0838 | −1.7596 | −1.4311 |
| P0120 | Log fishing scale | Fishery; 2005 | −1.6359 | 0.0829 | −1.7985 | −1.4734 |
| P0121 | Log fishing scale | Fishery; 2006 | −1.4267 | 0.0820 | −1.5873 | −1.2660 |
| P0122 | Log fishing scale | Fishery; 2007 | −1.3555 | 0.0809 | −1.5141 | −1.1970 |
| P0123 | Log fishing scale | Fishery; 2008 | −1.2726 | 0.0841 | −1.4374 | −1.1078 |
| P0124 | Log fishing scale | Fishery; 2009 | −1.3188 | 0.0943 | −1.5036 | −1.1341 |
| P0125 | Log fishing scale | Fishery; 2010 | −1.5006 | 0.1034 | −1.7033 | −1.2978 |
| P0126 | Log fishing scale | Fishery; 2011 | −1.1973 | 0.1005 | −1.3944 | −1.0003 |
| P0127 | Log fishing scale | Fishery; 2012 | −1.2027 | 0.1268 | −1.4512 | −0.9542 |
| P0128 | Log fishing scale | Fishery; 2013 | −1.2580 | 0.1457 | −1.5435 | −0.9725 |
| P0129 | Log fishing scale | Fishery; 2014 | −1.3178 | 0.1579 | −1.6272 | −1.0085 |
| P0130 | Log fishing scale | Fishery; 2015 | −1.2383 | 0.1649 | −1.5615 | −0.9152 |
| P0131 | Log fishing scale | Fishery; 2016 | −1.5367 | 0.1456 | −1.8220 | −1.2513 |
| P0132 | Log fishing scale | Fishery; 2017 | −1.5614 | 0.1284 | −1.8130 | −1.3097 |
| P0133 | Log fishing scale | Fishery; 2018 | −1.7471 | 0.1197 | −1.9818 | −1.5124 |
| P0134 | Log fishing scale | Fishery; 2019 | −1.6372 | 0.1073 | −1.8475 | −1.4269 |
| P0135 | Log fishing scale | Fishery; 2020 | −1.3656 | 0.1186 | −1.5980 | −1.1331 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0136 | Log fishing scale | Fishery; 2021 | −1.4188 | 0.1220 | −1.6579 | −1.1798 |
| P0137 | Log fishing scale | Fishery; 2022 | −1.7708 | 0.1385 | −2.0422 | −1.4993 |
| P0138 | Log fishing scale | Fishery; 2023 | −1.8165 | 0.1494 | −2.1092 | −1.5238 |
| P0139 | Log fishing scale | Fishery; 2024 | −1.9059 | 0.1439 | −2.1878 | −1.6239 |
| P0140 | Log fishing scale | Fishery; 2025 | −1.8131 | 0.1338 | −2.0753 | −1.5508 |
| P0141 | Log fishing scale | Fishery; 2026 | −1.6691 | 0.1614 | −1.9853 | −1.3528 |
| P0142 | Log catchability | AVO | −8.2249 | 0.0711 | −8.3642 | −8.0855 |
| P0143 | Log catchability | ATS | −0.6599 | 0.0620 | −0.7814 | −0.5384 |
| P0144 | Log catchability | CPUE | −0.2481 | 0.1849 | −0.6104 | 0.1143 |
| P0145 | Selectivity increment | Fishery; Bin1; 1964 * | 1.2875 | 0.5969 | 0.1177 | 2.4573 |
| P0146 | Selectivity increment | Fishery; Bin2; 1964 * | 0.7997 | 0.4928 | −0.1663 | 1.7656 |
| P0147 | Selectivity increment | Fishery; Bin3; 1964 * | 0.4574 | 0.4383 | −0.4016 | 1.3164 |
| P0148 | Selectivity increment | Fishery; Bin4; 1964 * | 0.3677 | 0.3979 | −0.4122 | 1.1476 |
| P0149 | Selectivity increment | Fishery; Bin5; 1964 * | −0.1820 | 0.3965 | −0.9591 | 0.5952 |
| P0150 | Selectivity increment | Fishery; Bin6; 1964 * | −0.4579 | 0.2958 | −1.0377 | 0.1219 |
| P0151 | Selectivity increment | Fishery; Bin7; 1964 * | −0.5719 | 0.2747 | −1.1103 | −0.0335 |
| P0152 | Selectivity increment | Fishery; Bin8; 1964 * | −0.5861 | 0.2680 | −1.1114 | −0.0608 |
| P0153 | Selectivity increment | Fishery; Bin9; 1964 * | −0.4992 | 0.2677 | −1.0239 | 0.0255 |
| P0154 | Selectivity increment | Fishery; Bin10; 1964 * | −0.2953 | 0.2716 | −0.8277 | 0.2371 |
| P0155 | Selectivity increment | Fishery; Bin11; 1964 * | −0.2831 | 0.2799 | −0.8317 | 0.2654 |
| P0156 | Selectivity increment | Fishery; Bin12; 1964 * | −0.0367 | 0.2934 | −0.6117 | 0.5383 |
| P0157 | Selectivity increment | Fishery; Bin1; 1965 * | 0.3571 | 0.4151 | −0.4565 | 1.1708 |
| P0158 | Selectivity increment | Fishery; Bin2; 1965 * | 0.1619 | 0.4026 | −0.6272 | 0.9511 |
| P0159 | Selectivity increment | Fishery; Bin3; 1965 * | 0.1307 | 0.3944 | −0.6423 | 0.9037 |
| P0160 | Selectivity increment | Fishery; Bin4; 1965 * | 0.0902 | 0.3777 | −0.6500 | 0.8304 |
| P0161 | Selectivity increment | Fishery; Bin5; 1965 * | −0.0379 | 0.3792 | −0.7810 | 0.7053 |
| P0162 | Selectivity increment | Fishery; Bin6; 1965 * | −0.1784 | 0.3055 | −0.7771 | 0.4203 |
| P0163 | Selectivity increment | Fishery; Bin7; 1965 * | −0.1639 | 0.2839 | −0.7203 | 0.3926 |
| P0164 | Selectivity increment | Fishery; Bin8; 1965 * | −0.1318 | 0.2797 | −0.6799 | 0.4163 |
| P0165 | Selectivity increment | Fishery; Bin9; 1965 * | −0.0992 | 0.2797 | −0.6475 | 0.4491 |
| P0166 | Selectivity increment | Fishery; Bin10; 1965 * | −0.0684 | 0.2813 | −0.6198 | 0.4830 |
| P0167 | Selectivity increment | Fishery; Bin11; 1965 * | −0.0439 | 0.2832 | −0.5990 | 0.5111 |
| P0168 | Selectivity increment | Fishery; Bin12; 1965 * | −0.0165 | 0.2786 | −0.5626 | 0.5295 |
| P0169 | Selectivity increment | Fishery; Bin1; 1966 * | 0.3379 | 0.4159 | −0.4772 | 1.1530 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0170 | Selectivity increment | Fishery; Bin2; 1966 * | 0.1962 | 0.4019 | −0.5915 | 0.9838 |
| P0171 | Selectivity increment | Fishery; Bin3; 1966 * | 0.1357 | 0.3933 | −0.6351 | 0.9065 |
| P0172 | Selectivity increment | Fishery; Bin4; 1966 * | −0.0528 | 0.3090 | −0.6585 | 0.5528 |
| P0173 | Selectivity increment | Fishery; Bin5; 1966 * | −0.0653 | 0.2844 | −0.6227 | 0.4921 |
| P0174 | Selectivity increment | Fishery; Bin6; 1966 * | −0.0912 | 0.2786 | −0.6373 | 0.4549 |
| P0175 | Selectivity increment | Fishery; Bin7; 1966 * | −0.0904 | 0.2774 | −0.6341 | 0.4534 |
| P0176 | Selectivity increment | Fishery; Bin8; 1966 * | −0.0915 | 0.2773 | −0.6350 | 0.4519 |
| P0177 | Selectivity increment | Fishery; Bin9; 1966 * | −0.0850 | 0.2779 | −0.6298 | 0.4597 |
| P0178 | Selectivity increment | Fishery; Bin10; 1966 * | −0.0744 | 0.2795 | −0.6221 | 0.4733 |
| P0179 | Selectivity increment | Fishery; Bin11; 1966 * | −0.0638 | 0.2811 | −0.6149 | 0.4872 |
| P0180 | Selectivity increment | Fishery; Bin12; 1966 * | −0.0553 | 0.2749 | −0.5940 | 0.4834 |
| P0181 | Selectivity increment | Fishery; Bin1; 1967 * | 0.3044 | 0.4161 | −0.5112 | 1.1200 |
| P0182 | Selectivity increment | Fishery; Bin2; 1967 * | 0.1602 | 0.3988 | −0.6214 | 0.9418 |
| P0183 | Selectivity increment | Fishery; Bin3; 1967 * | 0.2062 | 0.3907 | −0.5595 | 0.9719 |
| P0184 | Selectivity increment | Fishery; Bin4; 1967 * | −0.1132 | 0.3085 | −0.7179 | 0.4915 |
| P0185 | Selectivity increment | Fishery; Bin5; 1967 * | −0.1023 | 0.2828 | −0.6566 | 0.4520 |
| P0186 | Selectivity increment | Fishery; Bin6; 1967 * | −0.0914 | 0.2781 | −0.6364 | 0.4536 |
| P0187 | Selectivity increment | Fishery; Bin7; 1967 * | −0.0794 | 0.2773 | −0.6229 | 0.4641 |
| P0188 | Selectivity increment | Fishery; Bin8; 1967 * | −0.0677 | 0.2772 | −0.6110 | 0.4757 |
| P0189 | Selectivity increment | Fishery; Bin9; 1967 * | −0.0629 | 0.2779 | −0.6075 | 0.4818 |
| P0190 | Selectivity increment | Fishery; Bin10; 1967 * | −0.0573 | 0.2794 | −0.6049 | 0.4904 |
| P0191 | Selectivity increment | Fishery; Bin11; 1967 * | −0.0511 | 0.2811 | −0.6020 | 0.4998 |
| P0192 | Selectivity increment | Fishery; Bin12; 1967 * | −0.0455 | 0.2747 | −0.5838 | 0.4928 |
| P0193 | Selectivity increment | Fishery; Bin1; 1968 * | 0.3127 | 0.4163 | −0.5033 | 1.1286 |
| P0194 | Selectivity increment | Fishery; Bin2; 1968 * | 0.1419 | 0.3978 | −0.6377 | 0.9215 |
| P0195 | Selectivity increment | Fishery; Bin3; 1968 * | 0.0120 | 0.3855 | −0.7436 | 0.7675 |
| P0196 | Selectivity increment | Fishery; Bin4; 1968 * | −0.0108 | 0.3097 | −0.6179 | 0.5963 |
| P0197 | Selectivity increment | Fishery; Bin5; 1968 * | −0.0366 | 0.2850 | −0.5952 | 0.5220 |
| P0198 | Selectivity increment | Fishery; Bin6; 1968 * | −0.0514 | 0.2793 | −0.5988 | 0.4960 |
| P0199 | Selectivity increment | Fishery; Bin7; 1968 * | −0.0611 | 0.2781 | −0.6061 | 0.4839 |
| P0200 | Selectivity increment | Fishery; Bin8; 1968 * | −0.0644 | 0.2790 | −0.6113 | 0.4824 |
| P0201 | Selectivity increment | Fishery; Bin9; 1968 * | −0.0626 | 0.2800 | −0.6114 | 0.4863 |
| P0202 | Selectivity increment | Fishery; Bin10; 1968 * | −0.0612 | 0.2825 | −0.6149 | 0.4925 |
| P0203 | Selectivity increment | Fishery; Bin11; 1968 * | −0.0596 | 0.2834 | −0.6150 | 0.4958 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0204 | Selectivity increment | Fishery; Bin12; 1968 * | −0.0589 | 0.2764 | −0.6007 | 0.4829 |
| P0205 | Selectivity increment | Fishery; Bin1; 1969 * | 0.2851 | 0.4165 | −0.5312 | 1.1014 |
| P0206 | Selectivity increment | Fishery; Bin2; 1969 * | 0.0924 | 0.3968 | −0.6852 | 0.8701 |
| P0207 | Selectivity increment | Fishery; Bin3; 1969 * | 0.1579 | 0.3857 | −0.5981 | 0.9139 |
| P0208 | Selectivity increment | Fishery; Bin4; 1969 * | −0.0960 | 0.3101 | −0.7037 | 0.5117 |
| P0209 | Selectivity increment | Fishery; Bin5; 1969 * | −0.0947 | 0.2880 | −0.6590 | 0.4697 |
| P0210 | Selectivity increment | Fishery; Bin6; 1969 * | −0.0822 | 0.2892 | −0.6490 | 0.4846 |
| P0211 | Selectivity increment | Fishery; Bin7; 1969 * | −0.0710 | 0.3110 | −0.6805 | 0.5385 |
| P0212 | Selectivity increment | Fishery; Bin8; 1969 * | −0.0543 | 0.3363 | −0.7134 | 0.6047 |
| P0213 | Selectivity increment | Fishery; Bin9; 1969 * | −0.0484 | 0.3372 | −0.7094 | 0.6125 |
| P0214 | Selectivity increment | Fishery; Bin10; 1969 * | −0.0360 | 0.4001 | −0.8201 | 0.7481 |
| P0215 | Selectivity increment | Fishery; Bin11; 1969 * | −0.0272 | 0.3244 | −0.6629 | 0.6085 |
| P0216 | Selectivity increment | Fishery; Bin12; 1969 * | −0.0257 | 0.2858 | −0.5858 | 0.5345 |
| P0217 | Selectivity increment | Fishery; Bin1; 1970 * | 0.3011 | 0.4170 | −0.5161 | 1.1183 |
| P0218 | Selectivity increment | Fishery; Bin2; 1970 * | 0.1488 | 0.3965 | −0.6283 | 0.9259 |
| P0219 | Selectivity increment | Fishery; Bin3; 1970 * | −0.0316 | 0.3840 | −0.7843 | 0.7211 |
| P0220 | Selectivity increment | Fishery; Bin4; 1970 * | −0.2072 | 0.3635 | −0.9196 | 0.5052 |
| P0221 | Selectivity increment | Fishery; Bin5; 1970 * | −0.1218 | 0.3161 | −0.7413 | 0.4976 |
| P0222 | Selectivity increment | Fishery; Bin6; 1970 * | −0.1011 | 0.3166 | −0.7217 | 0.5194 |
| P0223 | Selectivity increment | Fishery; Bin7; 1970 * | −0.0529 | 0.3685 | −0.7751 | 0.6693 |
| P0224 | Selectivity increment | Fishery; Bin8; 1970 * | −0.0096 | 0.3991 | −0.7920 | 0.7727 |
| P0225 | Selectivity increment | Fishery; Bin9; 1970 * | −0.0009 | 0.3998 | −0.7845 | 0.7828 |
| P0226 | Selectivity increment | Fishery; Bin10; 1970 * | 0.0133 | 0.4076 | −0.7855 | 0.8122 |
| P0227 | Selectivity increment | Fishery; Bin11; 1970 * | 0.0230 | 0.4049 | −0.7706 | 0.8167 |
| P0228 | Selectivity increment | Fishery; Bin12; 1970 * | 0.0389 | 0.3034 | −0.5557 | 0.6335 |
| P0229 | Selectivity increment | Fishery; Bin1; 1971 * | 0.0965 | 0.4172 | −0.7211 | 0.9142 |
| P0230 | Selectivity increment | Fishery; Bin2; 1971 * | 0.0278 | 0.3955 | −0.7474 | 0.8031 |
| P0231 | Selectivity increment | Fishery; Bin3; 1971 * | −0.0921 | 0.3808 | −0.8385 | 0.6544 |
| P0232 | Selectivity increment | Fishery; Bin4; 1971 * | −0.0308 | 0.3714 | −0.7587 | 0.6972 |
| P0233 | Selectivity increment | Fishery; Bin5; 1971 * | 0.0085 | 0.3196 | −0.6179 | 0.6350 |
| P0234 | Selectivity increment | Fishery; Bin6; 1971 * | −0.0046 | 0.3151 | −0.6222 | 0.6129 |
| P0235 | Selectivity increment | Fishery; Bin7; 1971 * | −0.0485 | 0.3697 | −0.7730 | 0.6761 |
| P0236 | Selectivity increment | Fishery; Bin8; 1971 * | 0.0021 | 0.4013 | −0.7845 | 0.7887 |
| P0237 | Selectivity increment | Fishery; Bin9; 1971 * | 0.0038 | 0.4016 | −0.7833 | 0.7910 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0238 | Selectivity increment | Fishery; Bin10; 1971 * | 0.0043 | 0.4080 | −0.7953 | 0.8040 |
| P0239 | Selectivity increment | Fishery; Bin11; 1971 * | 0.0104 | 0.4051 | −0.7835 | 0.8044 |
| P0240 | Selectivity increment | Fishery; Bin12; 1971 * | 0.0224 | 0.3007 | −0.5671 | 0.6118 |
| P0241 | Selectivity increment | Fishery; Bin1; 1972 * | −0.1540 | 0.4171 | −0.9715 | 0.6635 |
| P0242 | Selectivity increment | Fishery; Bin2; 1972 * | 0.1905 | 0.3966 | −0.5868 | 0.9678 |
| P0243 | Selectivity increment | Fishery; Bin3; 1972 * | 0.0918 | 0.3791 | −0.6513 | 0.8349 |
| P0244 | Selectivity increment | Fishery; Bin4; 1972 * | 0.0742 | 0.3605 | −0.6324 | 0.7808 |
| P0245 | Selectivity increment | Fishery; Bin5; 1972 * | −0.0186 | 0.3073 | −0.6208 | 0.5836 |
| P0246 | Selectivity increment | Fishery; Bin6; 1972 * | −0.0055 | 0.2955 | −0.5847 | 0.5738 |
| P0247 | Selectivity increment | Fishery; Bin7; 1972 * | −0.0047 | 0.3201 | −0.6322 | 0.6227 |
| P0248 | Selectivity increment | Fishery; Bin8; 1972 * | −0.0323 | 0.4015 | −0.8192 | 0.7546 |
| P0249 | Selectivity increment | Fishery; Bin9; 1972 * | −0.0229 | 0.4016 | −0.8100 | 0.7641 |
| P0250 | Selectivity increment | Fishery; Bin10; 1972 * | −0.0315 | 0.4079 | −0.8311 | 0.7680 |
| P0251 | Selectivity increment | Fishery; Bin11; 1972 * | −0.0396 | 0.4049 | −0.8332 | 0.7540 |
| P0252 | Selectivity increment | Fishery; Bin12; 1972 * | −0.0473 | 0.3006 | −0.6364 | 0.5419 |
| P0253 | Selectivity increment | Fishery; Bin1; 1973 * | −0.1854 | 0.4165 | −1.0018 | 0.6309 |
| P0254 | Selectivity increment | Fishery; Bin2; 1973 * | 0.1006 | 0.3966 | −0.6767 | 0.8779 |
| P0255 | Selectivity increment | Fishery; Bin3; 1973 * | 0.0425 | 0.3793 | −0.7010 | 0.7860 |
| P0256 | Selectivity increment | Fishery; Bin4; 1973 * | 0.0386 | 0.3526 | −0.6525 | 0.7298 |
| P0257 | Selectivity increment | Fishery; Bin5; 1973 * | 0.0410 | 0.3027 | −0.5522 | 0.6342 |
| P0258 | Selectivity increment | Fishery; Bin6; 1973 * | 0.0434 | 0.2924 | −0.5296 | 0.6165 |
| P0259 | Selectivity increment | Fishery; Bin7; 1973 * | 0.0391 | 0.3170 | −0.5822 | 0.6604 |
| P0260 | Selectivity increment | Fishery; Bin8; 1973 * | 0.0045 | 0.3969 | −0.7735 | 0.7825 |
| P0261 | Selectivity increment | Fishery; Bin9; 1973 * | −0.0153 | 0.3987 | −0.7967 | 0.7660 |
| P0262 | Selectivity increment | Fishery; Bin10; 1973 * | −0.0262 | 0.4063 | −0.8225 | 0.7701 |
| P0263 | Selectivity increment | Fishery; Bin11; 1973 * | −0.0364 | 0.4038 | −0.8279 | 0.7550 |
| P0264 | Selectivity increment | Fishery; Bin12; 1973 * | −0.0463 | 0.3011 | −0.6364 | 0.5438 |
| P0265 | Selectivity increment | Fishery; Bin1; 1974 * | −0.1610 | 0.4155 | −0.9753 | 0.6534 |
| P0266 | Selectivity increment | Fishery; Bin2; 1974 * | 0.1384 | 0.3933 | −0.6324 | 0.9092 |
| P0267 | Selectivity increment | Fishery; Bin3; 1974 * | 0.1528 | 0.3689 | −0.5702 | 0.8759 |
| P0268 | Selectivity increment | Fishery; Bin4; 1974 * | 0.0118 | 0.3473 | −0.6690 | 0.6926 |
| P0269 | Selectivity increment | Fishery; Bin5; 1974 * | 0.0042 | 0.3027 | −0.5891 | 0.5975 |
| P0270 | Selectivity increment | Fishery; Bin6; 1974 * | −0.0021 | 0.2923 | −0.5750 | 0.5709 |
| P0271 | Selectivity increment | Fishery; Bin7; 1974 * | −0.0010 | 0.3158 | −0.6200 | 0.6180 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0272 | Selectivity increment | Fishery; Bin8; 1974 * | −0.0203 | 0.3456 | −0.6978 | 0.6571 |
| P0273 | Selectivity increment | Fishery; Bin9; 1974 * | −0.0200 | 0.3422 | −0.6907 | 0.6507 |
| P0274 | Selectivity increment | Fishery; Bin10; 1974 * | −0.0257 | 0.4021 | −0.8137 | 0.7624 |
| P0275 | Selectivity increment | Fishery; Bin11; 1974 * | −0.0326 | 0.4002 | −0.8171 | 0.7518 |
| P0276 | Selectivity increment | Fishery; Bin12; 1974 * | −0.0446 | 0.3017 | −0.6360 | 0.5468 |
| P0277 | Selectivity increment | Fishery; Bin1; 1975 * | −0.1574 | 0.4148 | −0.9705 | 0.6557 |
| P0278 | Selectivity increment | Fishery; Bin2; 1975 * | −0.1858 | 0.3879 | −0.9460 | 0.5745 |
| P0279 | Selectivity increment | Fishery; Bin3; 1975 * | 0.1323 | 0.3078 | −0.4710 | 0.7357 |
| P0280 | Selectivity increment | Fishery; Bin4; 1975 * | 0.0781 | 0.2842 | −0.4789 | 0.6351 |
| P0281 | Selectivity increment | Fishery; Bin5; 1975 * | 0.0430 | 0.2808 | −0.5074 | 0.5933 |
| P0282 | Selectivity increment | Fishery; Bin6; 1975 * | 0.0290 | 0.2859 | −0.5313 | 0.5893 |
| P0283 | Selectivity increment | Fishery; Bin7; 1975 * | 0.0231 | 0.3077 | −0.5801 | 0.6262 |
| P0284 | Selectivity increment | Fishery; Bin8; 1975 * | 0.0028 | 0.3377 | −0.6589 | 0.6646 |
| P0285 | Selectivity increment | Fishery; Bin9; 1975 * | 0.0035 | 0.3448 | −0.6722 | 0.6792 |
| P0286 | Selectivity increment | Fishery; Bin10; 1975 * | 0.0104 | 0.3139 | −0.6049 | 0.6257 |
| P0287 | Selectivity increment | Fishery; Bin11; 1975 * | 0.0105 | 0.2910 | −0.5598 | 0.5807 |
| P0288 | Selectivity increment | Fishery; Bin12; 1975 * | 0.0104 | 0.2783 | −0.5351 | 0.5558 |
| P0289 | Selectivity increment | Fishery; Bin1; 1976 * | −0.1591 | 0.4145 | −0.9714 | 0.6533 |
| P0290 | Selectivity increment | Fishery; Bin2; 1976 * | −0.0964 | 0.3884 | −0.8577 | 0.6650 |
| P0291 | Selectivity increment | Fishery; Bin3; 1976 * | −0.0776 | 0.3491 | −0.7618 | 0.6066 |
| P0292 | Selectivity increment | Fishery; Bin4; 1976 * | 0.0621 | 0.3017 | −0.5292 | 0.6534 |
| P0293 | Selectivity increment | Fishery; Bin5; 1976 * | 0.0496 | 0.2832 | −0.5055 | 0.6048 |
| P0294 | Selectivity increment | Fishery; Bin6; 1976 * | 0.0379 | 0.2801 | −0.5111 | 0.5870 |
| P0295 | Selectivity increment | Fishery; Bin7; 1976 * | 0.0310 | 0.2819 | −0.5214 | 0.5835 |
| P0296 | Selectivity increment | Fishery; Bin8; 1976 * | 0.0269 | 0.2950 | −0.5513 | 0.6051 |
| P0297 | Selectivity increment | Fishery; Bin9; 1976 * | 0.0252 | 0.3416 | −0.6443 | 0.6947 |
| P0298 | Selectivity increment | Fishery; Bin10; 1976 * | 0.0334 | 0.3214 | −0.5965 | 0.6633 |
| P0299 | Selectivity increment | Fishery; Bin11; 1976 * | 0.0339 | 0.2922 | −0.5388 | 0.6066 |
| P0300 | Selectivity increment | Fishery; Bin12; 1976 * | 0.0329 | 0.2785 | −0.5129 | 0.5787 |
| P0301 | Selectivity increment | Fishery; Bin1; 1977 * | −0.1490 | 0.4140 | −0.9604 | 0.6624 |
| P0302 | Selectivity increment | Fishery; Bin2; 1977 * | −0.0115 | 0.3875 | −0.7709 | 0.7479 |
| P0303 | Selectivity increment | Fishery; Bin3; 1977 * | −0.1853 | 0.3694 | −0.9094 | 0.5387 |
| P0304 | Selectivity increment | Fishery; Bin4; 1977 * | −0.0719 | 0.3558 | −0.7692 | 0.6254 |
| P0305 | Selectivity increment | Fishery; Bin5; 1977 * | 0.0195 | 0.3047 | −0.5777 | 0.6167 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0306 | Selectivity increment | Fishery; Bin6; 1977 * | 0.0338 | 0.2861 | −0.5269 | 0.5945 |
| P0307 | Selectivity increment | Fishery; Bin7; 1977 * | 0.0379 | 0.2880 | −0.5266 | 0.6024 |
| P0308 | Selectivity increment | Fishery; Bin8; 1977 * | 0.0395 | 0.3108 | −0.5697 | 0.6487 |
| P0309 | Selectivity increment | Fishery; Bin9; 1977 * | 0.0514 | 0.3946 | −0.7220 | 0.8247 |
| P0310 | Selectivity increment | Fishery; Bin10; 1977 * | 0.0697 | 0.3962 | −0.7069 | 0.8463 |
| P0311 | Selectivity increment | Fishery; Bin11; 1977 * | 0.0816 | 0.3251 | −0.5556 | 0.7189 |
| P0312 | Selectivity increment | Fishery; Bin12; 1977 * | 0.0844 | 0.2885 | −0.4811 | 0.6498 |
| P0313 | Selectivity increment | Fishery; Bin1; 1978 * | −0.1935 | 0.4131 | −1.0032 | 0.6162 |
| P0314 | Selectivity increment | Fishery; Bin2; 1978 * | −0.1580 | 0.3803 | −0.9035 | 0.5874 |
| P0315 | Selectivity increment | Fishery; Bin3; 1978 * | −0.0660 | 0.3645 | −0.7803 | 0.6483 |
| P0316 | Selectivity increment | Fishery; Bin4; 1978 * | −0.0381 | 0.3640 | −0.7515 | 0.6753 |
| P0317 | Selectivity increment | Fishery; Bin5; 1978 * | 0.0269 | 0.3143 | −0.5891 | 0.6429 |
| P0318 | Selectivity increment | Fishery; Bin6; 1978 * | 0.0307 | 0.3083 | −0.5736 | 0.6349 |
| P0319 | Selectivity increment | Fishery; Bin7; 1978 * | 0.0457 | 0.3154 | −0.5725 | 0.6639 |
| P0320 | Selectivity increment | Fishery; Bin8; 1978 * | 0.0496 | 0.3259 | −0.5891 | 0.6883 |
| P0321 | Selectivity increment | Fishery; Bin9; 1978 * | 0.0673 | 0.3989 | −0.7144 | 0.8491 |
| P0322 | Selectivity increment | Fishery; Bin10; 1978 * | 0.0741 | 0.3989 | −0.7076 | 0.8559 |
| P0323 | Selectivity increment | Fishery; Bin11; 1978 * | 0.0807 | 0.3341 | −0.5741 | 0.7354 |
| P0324 | Selectivity increment | Fishery; Bin12; 1978 * | 0.0808 | 0.2890 | −0.4856 | 0.6472 |
| P0325 | Selectivity increment | Fishery; Bin1; 1979 * | −0.1527 | 0.4072 | −0.9509 | 0.6455 |
| P0326 | Selectivity increment | Fishery; Bin2; 1979 * | −0.3807 | 0.3608 | −1.0879 | 0.3265 |
| P0327 | Selectivity increment | Fishery; Bin3; 1979 * | −0.1744 | 0.3505 | −0.8614 | 0.5126 |
| P0328 | Selectivity increment | Fishery; Bin4; 1979 * | −0.0494 | 0.3568 | −0.7488 | 0.6499 |
| P0329 | Selectivity increment | Fishery; Bin5; 1979 * | 0.0853 | 0.3491 | −0.5990 | 0.7695 |
| P0330 | Selectivity increment | Fishery; Bin6; 1979 * | 0.1164 | 0.3292 | −0.5289 | 0.7617 |
| P0331 | Selectivity increment | Fishery; Bin7; 1979 * | 0.0994 | 0.3399 | −0.5669 | 0.7656 |
| P0332 | Selectivity increment | Fishery; Bin8; 1979 * | 0.1032 | 0.3687 | −0.6194 | 0.8257 |
| P0333 | Selectivity increment | Fishery; Bin9; 1979 * | 0.1059 | 0.4008 | −0.6796 | 0.8914 |
| P0334 | Selectivity increment | Fishery; Bin10; 1979 * | 0.0896 | 0.3983 | −0.6910 | 0.8703 |
| P0335 | Selectivity increment | Fishery; Bin11; 1979 * | 0.0792 | 0.3286 | −0.5649 | 0.7232 |
| P0336 | Selectivity increment | Fishery; Bin12; 1979 * | 0.0784 | 0.2878 | −0.4856 | 0.6424 |
| P0337 | Selectivity increment | Fishery; Bin1; 1980 * | −0.1661 | 0.4085 | −0.9667 | 0.6344 |
| P0338 | Selectivity increment | Fishery; Bin2; 1980 * | −0.4732 | 0.3535 | −1.1660 | 0.2197 |
| P0339 | Selectivity increment | Fishery; Bin3; 1980 * | −0.2547 | 0.3436 | −0.9282 | 0.4188 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0340 | Selectivity increment | Fishery; Bin4; 1980 * | 0.0750 | 0.3564 | −0.6234 | 0.7734 |
| P0341 | Selectivity increment | Fishery; Bin5; 1980 * | 0.1095 | 0.3585 | −0.5930 | 0.8121 |
| P0342 | Selectivity increment | Fishery; Bin6; 1980 * | 0.1351 | 0.3138 | −0.4800 | 0.7502 |
| P0343 | Selectivity increment | Fishery; Bin7; 1980 * | 0.1332 | 0.3106 | −0.4757 | 0.7420 |
| P0344 | Selectivity increment | Fishery; Bin8; 1980 * | 0.1202 | 0.3686 | −0.6023 | 0.8427 |
| P0345 | Selectivity increment | Fishery; Bin9; 1980 * | 0.0907 | 0.3988 | −0.6909 | 0.8722 |
| P0346 | Selectivity increment | Fishery; Bin10; 1980 * | 0.0826 | 0.3937 | −0.6891 | 0.8542 |
| P0347 | Selectivity increment | Fishery; Bin11; 1980 * | 0.0751 | 0.3175 | −0.5472 | 0.6975 |
| P0348 | Selectivity increment | Fishery; Bin12; 1980 * | 0.0726 | 0.2867 | −0.4893 | 0.6345 |
| P0349 | Selectivity increment | Fishery; Bin1; 1981 * | −0.1370 | 0.4123 | −0.9450 | 0.6710 |
| P0350 | Selectivity increment | Fishery; Bin2; 1981 * | −0.2530 | 0.3700 | −0.9782 | 0.4722 |
| P0351 | Selectivity increment | Fishery; Bin3; 1981 * | −0.2579 | 0.3394 | −0.9230 | 0.4072 |
| P0352 | Selectivity increment | Fishery; Bin4; 1981 * | 0.0107 | 0.3493 | −0.6739 | 0.6953 |
| P0353 | Selectivity increment | Fishery; Bin5; 1981 * | 0.1230 | 0.3471 | −0.5573 | 0.8033 |
| P0354 | Selectivity increment | Fishery; Bin6; 1981 * | 0.0783 | 0.3012 | −0.5121 | 0.6686 |
| P0355 | Selectivity increment | Fishery; Bin7; 1981 * | 0.0847 | 0.2895 | −0.4827 | 0.6522 |
| P0356 | Selectivity increment | Fishery; Bin8; 1981 * | 0.0882 | 0.3089 | −0.5172 | 0.6936 |
| P0357 | Selectivity increment | Fishery; Bin9; 1981 * | 0.0812 | 0.3921 | −0.6872 | 0.8497 |
| P0358 | Selectivity increment | Fishery; Bin10; 1981 * | 0.0626 | 0.3238 | −0.5721 | 0.6973 |
| P0359 | Selectivity increment | Fishery; Bin11; 1981 * | 0.0595 | 0.2952 | −0.5191 | 0.6382 |
| P0360 | Selectivity increment | Fishery; Bin12; 1981 * | 0.0596 | 0.2817 | −0.4925 | 0.6117 |
| P0361 | Selectivity increment | Fishery; Bin1; 1982 * | −0.1040 | 0.4154 | −0.9182 | 0.7102 |
| P0362 | Selectivity increment | Fishery; Bin2; 1982 * | −0.0931 | 0.3823 | −0.8423 | 0.6562 |
| P0363 | Selectivity increment | Fishery; Bin3; 1982 * | −0.3008 | 0.3536 | −0.9937 | 0.3922 |
| P0364 | Selectivity increment | Fishery; Bin4; 1982 * | −0.0092 | 0.3403 | −0.6762 | 0.6579 |
| P0365 | Selectivity increment | Fishery; Bin5; 1982 * | 0.0660 | 0.3012 | −0.5244 | 0.6565 |
| P0366 | Selectivity increment | Fishery; Bin6; 1982 * | 0.0694 | 0.2805 | −0.4804 | 0.6193 |
| P0367 | Selectivity increment | Fishery; Bin7; 1982 * | 0.0670 | 0.2772 | −0.4763 | 0.6104 |
| P0368 | Selectivity increment | Fishery; Bin8; 1982 * | 0.0685 | 0.2771 | −0.4746 | 0.6116 |
| P0369 | Selectivity increment | Fishery; Bin9; 1982 * | 0.0662 | 0.2794 | −0.4813 | 0.6137 |
| P0370 | Selectivity increment | Fishery; Bin10; 1982 * | 0.0600 | 0.2812 | −0.4911 | 0.6111 |
| P0371 | Selectivity increment | Fishery; Bin11; 1982 * | 0.0560 | 0.2836 | −0.5000 | 0.6119 |
| P0372 | Selectivity increment | Fishery; Bin12; 1982 * | 0.0538 | 0.2777 | −0.4905 | 0.5981 |
| P0373 | Selectivity increment | Fishery; Bin1; 1983 * | −0.1223 | 0.4165 | −0.9386 | 0.6940 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0374 | Selectivity increment | Fishery; Bin2; 1983 * | 0.1032 | 0.3892 | −0.6597 | 0.8660 |
| P0375 | Selectivity increment | Fishery; Bin3; 1983 * | −0.0970 | 0.3645 | −0.8113 | 0.6174 |
| P0376 | Selectivity increment | Fishery; Bin4; 1983 * | −0.1890 | 0.3452 | −0.8656 | 0.4876 |
| P0377 | Selectivity increment | Fishery; Bin5; 1983 * | −0.0624 | 0.3207 | −0.6909 | 0.5661 |
| P0378 | Selectivity increment | Fishery; Bin6; 1983 * | 0.0676 | 0.2926 | −0.5059 | 0.6410 |
| P0379 | Selectivity increment | Fishery; Bin7; 1983 * | 0.0552 | 0.2786 | −0.4908 | 0.6012 |
| P0380 | Selectivity increment | Fishery; Bin8; 1983 * | 0.0486 | 0.2766 | −0.4935 | 0.5906 |
| P0381 | Selectivity increment | Fishery; Bin9; 1983 * | 0.0482 | 0.2775 | −0.4956 | 0.5921 |
| P0382 | Selectivity increment | Fishery; Bin10; 1983 * | 0.0497 | 0.2797 | −0.4985 | 0.5978 |
| P0383 | Selectivity increment | Fishery; Bin11; 1983 * | 0.0494 | 0.2820 | −0.5033 | 0.6021 |
| P0384 | Selectivity increment | Fishery; Bin12; 1983 * | 0.0488 | 0.2760 | −0.4922 | 0.5898 |
| P0385 | Selectivity increment | Fishery; Bin1; 1984 * | −0.1099 | 0.4185 | −0.9301 | 0.7102 |
| P0386 | Selectivity increment | Fishery; Bin2; 1984 * | −0.1521 | 0.3905 | −0.9174 | 0.6132 |
| P0387 | Selectivity increment | Fishery; Bin3; 1984 * | 0.0489 | 0.3700 | −0.6763 | 0.7742 |
| P0388 | Selectivity increment | Fishery; Bin4; 1984 * | −0.0874 | 0.3505 | −0.7743 | 0.5996 |
| P0389 | Selectivity increment | Fishery; Bin5; 1984 * | 0.0264 | 0.3403 | −0.6405 | 0.6933 |
| P0390 | Selectivity increment | Fishery; Bin6; 1984 * | 0.0367 | 0.2951 | −0.5417 | 0.6151 |
| P0391 | Selectivity increment | Fishery; Bin7; 1984 * | 0.0512 | 0.2785 | −0.4946 | 0.5970 |
| P0392 | Selectivity increment | Fishery; Bin8; 1984 * | 0.0411 | 0.2762 | −0.5002 | 0.5824 |
| P0393 | Selectivity increment | Fishery; Bin9; 1984 * | 0.0380 | 0.2771 | −0.5051 | 0.5811 |
| P0394 | Selectivity increment | Fishery; Bin10; 1984 * | 0.0360 | 0.2792 | −0.5113 | 0.5833 |
| P0395 | Selectivity increment | Fishery; Bin11; 1984 * | 0.0358 | 0.2814 | −0.5158 | 0.5874 |
| P0396 | Selectivity increment | Fishery; Bin12; 1984 * | 0.0354 | 0.2754 | −0.5043 | 0.5751 |
| P0397 | Selectivity increment | Fishery; Bin1; 1985 * | −0.1272 | 0.4190 | −0.9484 | 0.6940 |
| P0398 | Selectivity increment | Fishery; Bin2; 1985 * | 0.0402 | 0.3949 | −0.7338 | 0.8141 |
| P0399 | Selectivity increment | Fishery; Bin3; 1985 * | 0.1307 | 0.3677 | −0.5900 | 0.8513 |
| P0400 | Selectivity increment | Fishery; Bin4; 1985 * | −0.0902 | 0.3563 | −0.7885 | 0.6081 |
| P0401 | Selectivity increment | Fishery; Bin5; 1985 * | −0.2409 | 0.3470 | −0.9210 | 0.4392 |
| P0402 | Selectivity increment | Fishery; Bin6; 1985 * | 0.0564 | 0.2947 | −0.5213 | 0.6340 |
| P0403 | Selectivity increment | Fishery; Bin7; 1985 * | 0.0519 | 0.2754 | −0.4879 | 0.5917 |
| P0404 | Selectivity increment | Fishery; Bin8; 1985 * | 0.0500 | 0.2751 | −0.4892 | 0.5891 |
| P0405 | Selectivity increment | Fishery; Bin9; 1985 * | 0.0345 | 0.2765 | −0.5074 | 0.5765 |
| P0406 | Selectivity increment | Fishery; Bin10; 1985 * | 0.0325 | 0.2789 | −0.5141 | 0.5790 |
| P0407 | Selectivity increment | Fishery; Bin11; 1985 * | 0.0318 | 0.2811 | −0.5191 | 0.5827 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0408 | Selectivity increment | Fishery; Bin12; 1985 * | 0.0304 | 0.2749 | −0.5084 | 0.5693 |
| P0409 | Selectivity increment | Fishery; Bin1; 1986 * | −0.1344 | 0.4196 | −0.9569 | 0.6881 |
| P0410 | Selectivity increment | Fishery; Bin2; 1986 * | −0.0383 | 0.3972 | −0.8168 | 0.7401 |
| P0411 | Selectivity increment | Fishery; Bin3; 1986 * | 0.0606 | 0.3691 | −0.6628 | 0.7840 |
| P0412 | Selectivity increment | Fishery; Bin4; 1986 * | 0.1677 | 0.3543 | −0.5267 | 0.8622 |
| P0413 | Selectivity increment | Fishery; Bin5; 1986 * | −0.0207 | 0.3550 | −0.7164 | 0.6751 |
| P0414 | Selectivity increment | Fishery; Bin6; 1986 * | −0.0709 | 0.3319 | −0.7214 | 0.5796 |
| P0415 | Selectivity increment | Fishery; Bin7; 1986 * | 0.0143 | 0.2940 | −0.5620 | 0.5905 |
| P0416 | Selectivity increment | Fishery; Bin8; 1986 * | −0.0173 | 0.2782 | −0.5626 | 0.5279 |
| P0417 | Selectivity increment | Fishery; Bin9; 1986 * | 0.0089 | 0.2789 | −0.5378 | 0.5555 |
| P0418 | Selectivity increment | Fishery; Bin10; 1986 * | 0.0072 | 0.2788 | −0.5393 | 0.5537 |
| P0419 | Selectivity increment | Fishery; Bin11; 1986 * | 0.0093 | 0.2807 | −0.5408 | 0.5594 |
| P0420 | Selectivity increment | Fishery; Bin12; 1986 * | 0.0136 | 0.2745 | −0.5244 | 0.5517 |
| P0421 | Selectivity increment | Fishery; Bin1; 1987 * | −0.1333 | 0.4198 | −0.9561 | 0.6895 |
| P0422 | Selectivity increment | Fishery; Bin2; 1987 * | −0.0846 | 0.4008 | −0.8702 | 0.7009 |
| P0423 | Selectivity increment | Fishery; Bin3; 1987 * | 0.1086 | 0.3713 | −0.6192 | 0.8364 |
| P0424 | Selectivity increment | Fishery; Bin4; 1987 * | −0.1061 | 0.3582 | −0.8082 | 0.5959 |
| P0425 | Selectivity increment | Fishery; Bin5; 1987 * | 0.0218 | 0.3538 | −0.6717 | 0.7153 |
| P0426 | Selectivity increment | Fishery; Bin6; 1987 * | −0.0462 | 0.3557 | −0.7433 | 0.6509 |
| P0427 | Selectivity increment | Fishery; Bin7; 1987 * | −0.0955 | 0.3359 | −0.7538 | 0.5628 |
| P0428 | Selectivity increment | Fishery; Bin8; 1987 * | 0.1044 | 0.2962 | −0.4762 | 0.6850 |
| P0429 | Selectivity increment | Fishery; Bin9; 1987 * | 0.0984 | 0.2793 | −0.4491 | 0.6459 |
| P0430 | Selectivity increment | Fishery; Bin10; 1987 * | 0.0660 | 0.2781 | −0.4791 | 0.6112 |
| P0431 | Selectivity increment | Fishery; Bin11; 1987 * | 0.0418 | 0.2801 | −0.5072 | 0.5908 |
| P0432 | Selectivity increment | Fishery; Bin12; 1987 * | 0.0248 | 0.2743 | −0.5127 | 0.5624 |
| P0433 | Selectivity increment | Fishery; Bin1; 1988 * | −0.1650 | 0.4198 | −0.9878 | 0.6577 |
| P0434 | Selectivity increment | Fishery; Bin2; 1988 * | −0.1229 | 0.4025 | −0.9118 | 0.6660 |
| P0435 | Selectivity increment | Fishery; Bin3; 1988 * | 0.4321 | 0.3737 | −0.3003 | 1.1645 |
| P0436 | Selectivity increment | Fishery; Bin4; 1988 * | 0.0242 | 0.3611 | −0.6835 | 0.7320 |
| P0437 | Selectivity increment | Fishery; Bin5; 1988 * | 0.0879 | 0.3542 | −0.6062 | 0.7820 |
| P0438 | Selectivity increment | Fishery; Bin6; 1988 * | −0.0565 | 0.3542 | −0.7508 | 0.6377 |
| P0439 | Selectivity increment | Fishery; Bin7; 1988 * | 0.1187 | 0.3368 | −0.5414 | 0.7788 |
| P0440 | Selectivity increment | Fishery; Bin8; 1988 * | −0.0812 | 0.2935 | −0.6565 | 0.4941 |
| P0441 | Selectivity increment | Fishery; Bin9; 1988 * | −0.0785 | 0.2772 | −0.6218 | 0.4648 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0442 | Selectivity increment | Fishery; Bin10; 1988 * | −0.0796 | 0.2776 | −0.6236 | 0.4644 |
| P0443 | Selectivity increment | Fishery; Bin11; 1988 * | −0.0497 | 0.2829 | −0.6042 | 0.5048 |
| P0444 | Selectivity increment | Fishery; Bin12; 1988 * | −0.0295 | 0.2753 | −0.5690 | 0.5100 |
| P0445 | Selectivity increment | Fishery; Bin1; 1989 * | −0.1307 | 0.4196 | −0.9531 | 0.6916 |
| P0446 | Selectivity increment | Fishery; Bin2; 1989 * | −0.0816 | 0.4021 | −0.8696 | 0.7065 |
| P0447 | Selectivity increment | Fishery; Bin3; 1989 * | −0.3948 | 0.3821 | −1.1437 | 0.3540 |
| P0448 | Selectivity increment | Fishery; Bin4; 1989 * | 0.1355 | 0.3658 | −0.5814 | 0.8525 |
| P0449 | Selectivity increment | Fishery; Bin5; 1989 * | 0.0220 | 0.3542 | −0.6723 | 0.7163 |
| P0450 | Selectivity increment | Fishery; Bin6; 1989 * | 0.0546 | 0.3531 | −0.6374 | 0.7466 |
| P0451 | Selectivity increment | Fishery; Bin7; 1989 * | 0.1293 | 0.2978 | −0.4543 | 0.7130 |
| P0452 | Selectivity increment | Fishery; Bin8; 1989 * | 0.1041 | 0.2772 | −0.4392 | 0.6473 |
| P0453 | Selectivity increment | Fishery; Bin9; 1989 * | 0.0309 | 0.2754 | −0.5088 | 0.5706 |
| P0454 | Selectivity increment | Fishery; Bin10; 1989 * | 0.0641 | 0.2773 | −0.4795 | 0.6076 |
| P0455 | Selectivity increment | Fishery; Bin11; 1989 * | 0.0445 | 0.2815 | −0.5072 | 0.5962 |
| P0456 | Selectivity increment | Fishery; Bin12; 1989 * | 0.0221 | 0.2747 | −0.5163 | 0.5605 |
| P0457 | Selectivity increment | Fishery; Bin1; 1990 * | −0.1246 | 0.4192 | −0.9462 | 0.6971 |
| P0458 | Selectivity increment | Fishery; Bin2; 1990 * | −0.0556 | 0.3998 | −0.8392 | 0.7279 |
| P0459 | Selectivity increment | Fishery; Bin3; 1990 * | −0.3454 | 0.3826 | −1.0953 | 0.4046 |
| P0460 | Selectivity increment | Fishery; Bin4; 1990 * | 0.0258 | 0.3704 | −0.7002 | 0.7518 |
| P0461 | Selectivity increment | Fishery; Bin5; 1990 * | 0.1451 | 0.3528 | −0.5464 | 0.8366 |
| P0462 | Selectivity increment | Fishery; Bin6; 1990 * | 0.0542 | 0.3493 | −0.6303 | 0.7387 |
| P0463 | Selectivity increment | Fishery; Bin7; 1990 * | −0.0416 | 0.3272 | −0.6830 | 0.5997 |
| P0464 | Selectivity increment | Fishery; Bin8; 1990 * | 0.0504 | 0.2894 | −0.5169 | 0.6176 |
| P0465 | Selectivity increment | Fishery; Bin9; 1990 * | 0.0621 | 0.2804 | −0.4875 | 0.6116 |
| P0466 | Selectivity increment | Fishery; Bin10; 1990 * | 0.0002 | 0.2925 | −0.5731 | 0.5736 |
| P0467 | Selectivity increment | Fishery; Bin11; 1990 * | 0.1328 | 0.2978 | −0.4508 | 0.7164 |
| P0468 | Selectivity increment | Fishery; Bin12; 1990 * | 0.0966 | 0.2719 | −0.4364 | 0.6295 |
| P0469 | Selectivity increment | Fishery; Bin1; 1991 * | −0.0571 | 0.4181 | −0.8765 | 0.7624 |
| P0470 | Selectivity increment | Fishery; Bin2; 1991 * | −0.0210 | 0.3915 | −0.7883 | 0.7462 |
| P0471 | Selectivity increment | Fishery; Bin3; 1991 * | −0.3596 | 0.3748 | −1.0942 | 0.3750 |
| P0472 | Selectivity increment | Fishery; Bin4; 1991 * | −0.2984 | 0.3583 | −1.0006 | 0.4038 |
| P0473 | Selectivity increment | Fishery; Bin5; 1991 * | −0.1882 | 0.3484 | −0.8710 | 0.4946 |
| P0474 | Selectivity increment | Fishery; Bin6; 1991 * | −0.0522 | 0.3230 | −0.6853 | 0.5809 |
| P0475 | Selectivity increment | Fishery; Bin7; 1991 * | 0.1413 | 0.3233 | −0.4924 | 0.7750 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0476 | Selectivity increment | Fishery; Bin8; 1991 * | −0.0373 | 0.2924 | −0.6104 | 0.5359 |
| P0477 | Selectivity increment | Fishery; Bin9; 1991 * | 0.1423 | 0.2774 | −0.4014 | 0.6859 |
| P0478 | Selectivity increment | Fishery; Bin10; 1991 * | 0.0714 | 0.2959 | −0.5085 | 0.6513 |
| P0479 | Selectivity increment | Fishery; Bin11; 1991 * | 0.3360 | 0.2985 | −0.2491 | 0.9211 |
| P0480 | Selectivity increment | Fishery; Bin12; 1991 * | 0.3228 | 0.2650 | −0.1966 | 0.8421 |
| P0481 | Selectivity increment | Fishery; Bin1; 1992 * | −0.1291 | 0.4153 | −0.9430 | 0.6848 |
| P0482 | Selectivity increment | Fishery; Bin2; 1992 * | −0.0575 | 0.3344 | −0.7129 | 0.5978 |
| P0483 | Selectivity increment | Fishery; Bin3; 1992 * | 0.5222 | 0.3061 | −0.0779 | 1.1222 |
| P0484 | Selectivity increment | Fishery; Bin4; 1992 * | −0.2344 | 0.3156 | −0.8530 | 0.3842 |
| P0485 | Selectivity increment | Fishery; Bin5; 1992 * | −0.3277 | 0.3180 | −0.9509 | 0.2955 |
| P0486 | Selectivity increment | Fishery; Bin6; 1992 * | −0.0479 | 0.3127 | −0.6607 | 0.5649 |
| P0487 | Selectivity increment | Fishery; Bin7; 1992 * | −0.0629 | 0.2927 | −0.6365 | 0.5108 |
| P0488 | Selectivity increment | Fishery; Bin8; 1992 * | 0.2020 | 0.2872 | −0.3608 | 0.7648 |
| P0489 | Selectivity increment | Fishery; Bin9; 1992 * | 0.0885 | 0.2802 | −0.4607 | 0.6377 |
| P0490 | Selectivity increment | Fishery; Bin10; 1992 * | 0.2149 | 0.2878 | −0.3491 | 0.7790 |
| P0491 | Selectivity increment | Fishery; Bin11; 1992 * | −0.1178 | 0.2882 | −0.6827 | 0.4471 |
| P0492 | Selectivity increment | Fishery; Bin12; 1992 * | −0.0503 | 0.2607 | −0.5612 | 0.4607 |
| P0493 | Selectivity increment | Fishery; Bin1; 1993 * | −0.1187 | 0.4147 | −0.9315 | 0.6940 |
| P0494 | Selectivity increment | Fishery; Bin2; 1993 * | −0.4719 | 0.3522 | −1.1621 | 0.2184 |
| P0495 | Selectivity increment | Fishery; Bin3; 1993 * | −0.0290 | 0.2736 | −0.5654 | 0.5073 |
| P0496 | Selectivity increment | Fishery; Bin4; 1993 * | 0.8219 | 0.2800 | 0.2731 | 1.3708 |
| P0497 | Selectivity increment | Fishery; Bin5; 1993 * | 0.0377 | 0.3059 | −0.5619 | 0.6373 |
| P0498 | Selectivity increment | Fishery; Bin6; 1993 * | 0.0368 | 0.3239 | −0.5980 | 0.6717 |
| P0499 | Selectivity increment | Fishery; Bin7; 1993 * | 0.0738 | 0.3239 | −0.5609 | 0.7086 |
| P0500 | Selectivity increment | Fishery; Bin8; 1993 * | −0.0135 | 0.2905 | −0.5830 | 0.5559 |
| P0501 | Selectivity increment | Fishery; Bin9; 1993 * | −0.0312 | 0.2806 | −0.5812 | 0.5188 |
| P0502 | Selectivity increment | Fishery; Bin10; 1993 * | −0.0706 | 0.2792 | −0.6178 | 0.4766 |
| P0503 | Selectivity increment | Fishery; Bin11; 1993 * | −0.1520 | 0.3012 | −0.7423 | 0.4384 |
| P0504 | Selectivity increment | Fishery; Bin12; 1993 * | −0.0833 | 0.2738 | −0.6200 | 0.4534 |
| P0505 | Selectivity increment | Fishery; Bin1; 1994 * | −0.0143 | 0.4166 | −0.8309 | 0.8023 |
| P0506 | Selectivity increment | Fishery; Bin2; 1994 * | 0.0152 | 0.3582 | −0.6868 | 0.7172 |
| P0507 | Selectivity increment | Fishery; Bin3; 1994 * | −0.7774 | 0.3087 | −1.3825 | −0.1722 |
| P0508 | Selectivity increment | Fishery; Bin4; 1994 * | −0.3070 | 0.2656 | −0.8275 | 0.2136 |
| P0509 | Selectivity increment | Fishery; Bin5; 1994 * | 0.9075 | 0.2652 | 0.3877 | 1.4273 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0510 | Selectivity increment | Fishery; Bin6; 1994 * | 0.2876 | 0.3068 | −0.3137 | 0.8890 |
| P0511 | Selectivity increment | Fishery; Bin7; 1994 * | 0.0218 | 0.3155 | −0.5966 | 0.6402 |
| P0512 | Selectivity increment | Fishery; Bin8; 1994 * | −0.0271 | 0.2831 | −0.5820 | 0.5277 |
| P0513 | Selectivity increment | Fishery; Bin9; 1994 * | −0.0449 | 0.2690 | −0.5721 | 0.4824 |
| P0514 | Selectivity increment | Fishery; Bin10; 1994 * | −0.0192 | 0.2727 | −0.5538 | 0.5153 |
| P0515 | Selectivity increment | Fishery; Bin11; 1994 * | 0.0142 | 0.2870 | −0.5483 | 0.5767 |
| P0516 | Selectivity increment | Fishery; Bin12; 1994 * | −0.0564 | 0.2782 | −0.6018 | 0.4889 |
| P0517 | Selectivity increment | Fishery; Bin1; 1995 * | 0.0094 | 0.4170 | −0.8078 | 0.8267 |
| P0518 | Selectivity increment | Fishery; Bin2; 1995 * | 0.1923 | 0.3679 | −0.5288 | 0.9134 |
| P0519 | Selectivity increment | Fishery; Bin3; 1995 * | −0.2946 | 0.3221 | −0.9259 | 0.3368 |
| P0520 | Selectivity increment | Fishery; Bin4; 1995 * | −0.7007 | 0.2923 | −1.2736 | −0.1279 |
| P0521 | Selectivity increment | Fishery; Bin5; 1995 * | −0.3795 | 0.2662 | −0.9013 | 0.1423 |
| P0522 | Selectivity increment | Fishery; Bin6; 1995 * | 0.4455 | 0.2661 | −0.0760 | 0.9671 |
| P0523 | Selectivity increment | Fishery; Bin7; 1995 * | 0.1832 | 0.2782 | −0.3621 | 0.7285 |
| P0524 | Selectivity increment | Fishery; Bin8; 1995 * | 0.1784 | 0.2707 | −0.3522 | 0.7090 |
| P0525 | Selectivity increment | Fishery; Bin9; 1995 * | 0.1368 | 0.2714 | −0.3951 | 0.6687 |
| P0526 | Selectivity increment | Fishery; Bin10; 1995 * | 0.1109 | 0.2724 | −0.4231 | 0.6449 |
| P0527 | Selectivity increment | Fishery; Bin11; 1995 * | 0.0758 | 0.2748 | −0.4628 | 0.6143 |
| P0528 | Selectivity increment | Fishery; Bin12; 1995 * | 0.0424 | 0.2681 | −0.4831 | 0.5680 |
| P0529 | Selectivity increment | Fishery; Bin1; 1996 * | −0.0097 | 0.4173 | −0.8277 | 0.8083 |
| P0530 | Selectivity increment | Fishery; Bin2; 1996 * | 0.5075 | 0.3722 | −0.2219 | 1.2369 |
| P0531 | Selectivity increment | Fishery; Bin3; 1996 * | 0.2794 | 0.3348 | −0.3768 | 0.9357 |
| P0532 | Selectivity increment | Fishery; Bin4; 1996 * | −0.7854 | 0.3054 | −1.3840 | −0.1867 |
| P0533 | Selectivity increment | Fishery; Bin5; 1996 * | −0.7408 | 0.2866 | −1.3026 | −0.1790 |
| P0534 | Selectivity increment | Fishery; Bin6; 1996 * | −0.3569 | 0.2674 | −0.8811 | 0.1672 |
| P0535 | Selectivity increment | Fishery; Bin7; 1996 * | 0.2781 | 0.2585 | −0.2285 | 0.7847 |
| P0536 | Selectivity increment | Fishery; Bin8; 1996 * | 0.3081 | 0.2639 | −0.2091 | 0.8253 |
| P0537 | Selectivity increment | Fishery; Bin9; 1996 * | 0.2215 | 0.2701 | −0.3080 | 0.7510 |
| P0538 | Selectivity increment | Fishery; Bin10; 1996 * | 0.1476 | 0.2737 | −0.3889 | 0.6841 |
| P0539 | Selectivity increment | Fishery; Bin11; 1996 * | 0.0906 | 0.2752 | −0.4489 | 0.6301 |
| P0540 | Selectivity increment | Fishery; Bin12; 1996 * | 0.0601 | 0.2678 | −0.4648 | 0.5849 |
| P0541 | Selectivity increment | Fishery; Bin1; 1997 * | −0.1094 | 0.4177 | −0.9282 | 0.7093 |
| P0542 | Selectivity increment | Fishery; Bin2; 1997 * | 0.3323 | 0.3622 | −0.3776 | 1.0422 |
| P0543 | Selectivity increment | Fishery; Bin3; 1997 * | 0.1555 | 0.3403 | −0.5114 | 0.8224 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0544 | Selectivity increment | Fishery; Bin4; 1997 * | 0.6559 | 0.3059 | 0.0563 | 1.2556 |
| P0545 | Selectivity increment | Fishery; Bin5; 1997 * | −0.0012 | 0.2806 | −0.5511 | 0.5488 |
| P0546 | Selectivity increment | Fishery; Bin6; 1997 * | −0.3984 | 0.2751 | −0.9376 | 0.1409 |
| P0547 | Selectivity increment | Fishery; Bin7; 1997 * | −0.4036 | 0.2694 | −0.9315 | 0.1243 |
| P0548 | Selectivity increment | Fishery; Bin8; 1997 * | −0.2889 | 0.2735 | −0.8249 | 0.2471 |
| P0549 | Selectivity increment | Fishery; Bin9; 1997 * | −0.0227 | 0.2787 | −0.5690 | 0.5235 |
| P0550 | Selectivity increment | Fishery; Bin10; 1997 * | 0.0186 | 0.2746 | −0.5197 | 0.5569 |
| P0551 | Selectivity increment | Fishery; Bin11; 1997 * | 0.0284 | 0.2758 | −0.5121 | 0.5689 |
| P0552 | Selectivity increment | Fishery; Bin12; 1997 * | 0.0335 | 0.2671 | −0.4900 | 0.5570 |
| P0553 | Selectivity increment | Fishery; Bin1; 1998 * | −0.0772 | 0.4183 | −0.8971 | 0.7427 |
| P0554 | Selectivity increment | Fishery; Bin2; 1998 * | −0.3764 | 0.3344 | −1.0318 | 0.2791 |
| P0555 | Selectivity increment | Fishery; Bin3; 1998 * | 0.2778 | 0.3214 | −0.3521 | 0.9077 |
| P0556 | Selectivity increment | Fishery; Bin4; 1998 * | 0.2308 | 0.3062 | −0.3693 | 0.8309 |
| P0557 | Selectivity increment | Fishery; Bin5; 1998 * | 0.1763 | 0.2739 | −0.3605 | 0.7132 |
| P0558 | Selectivity increment | Fishery; Bin6; 1998 * | 0.2773 | 0.2583 | −0.2290 | 0.7837 |
| P0559 | Selectivity increment | Fishery; Bin7; 1998 * | −0.2341 | 0.2704 | −0.7642 | 0.2960 |
| P0560 | Selectivity increment | Fishery; Bin8; 1998 * | −0.0717 | 0.2888 | −0.6378 | 0.4944 |
| P0561 | Selectivity increment | Fishery; Bin9; 1998 * | −0.0713 | 0.2812 | −0.6223 | 0.4798 |
| P0562 | Selectivity increment | Fishery; Bin10; 1998 * | −0.0478 | 0.2738 | −0.5845 | 0.4889 |
| P0563 | Selectivity increment | Fishery; Bin11; 1998 * | −0.0464 | 0.2758 | −0.5870 | 0.4942 |
| P0564 | Selectivity increment | Fishery; Bin12; 1998 * | −0.0374 | 0.2688 | −0.5643 | 0.4895 |
| P0565 | Selectivity increment | Fishery; Bin1; 1999 * | −0.1247 | 0.4186 | −0.9452 | 0.6958 |
| P0566 | Selectivity increment | Fishery; Bin2; 1999 * | −0.3584 | 0.3509 | −1.0462 | 0.3294 |
| P0567 | Selectivity increment | Fishery; Bin3; 1999 * | 0.8134 | 0.2823 | 0.2601 | 1.3666 |
| P0568 | Selectivity increment | Fishery; Bin4; 1999 * | 0.4525 | 0.2907 | −0.1172 | 1.0221 |
| P0569 | Selectivity increment | Fishery; Bin5; 1999 * | 0.1329 | 0.2861 | −0.4279 | 0.6938 |
| P0570 | Selectivity increment | Fishery; Bin6; 1999 * | −0.2625 | 0.2614 | −0.7749 | 0.2499 |
| P0571 | Selectivity increment | Fishery; Bin7; 1999 * | 0.2338 | 0.2526 | −0.2613 | 0.7289 |
| P0572 | Selectivity increment | Fishery; Bin8; 1999 * | −0.1112 | 0.2746 | −0.6493 | 0.4269 |
| P0573 | Selectivity increment | Fishery; Bin9; 1999 * | −0.2599 | 0.2671 | −0.7833 | 0.2636 |
| P0574 | Selectivity increment | Fishery; Bin10; 1999 * | −0.2373 | 0.2691 | −0.7648 | 0.2901 |
| P0575 | Selectivity increment | Fishery; Bin11; 1999 * | −0.1704 | 0.2922 | −0.7431 | 0.4023 |
| P0576 | Selectivity increment | Fishery; Bin12; 1999 * | −0.1081 | 0.2841 | −0.6649 | 0.4487 |
| P0577 | Selectivity increment | Fishery; Bin1; 2000 * | −0.0558 | 0.4185 | −0.8761 | 0.7645 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0578 | Selectivity increment | Fishery; Bin2; 2000 * | −0.1443 | 0.3623 | −0.8544 | 0.5658 |
| P0579 | Selectivity increment | Fishery; Bin3; 2000 * | −0.5590 | 0.2797 | −1.1072 | −0.0108 |
| P0580 | Selectivity increment | Fishery; Bin4; 2000 * | 0.2946 | 0.2566 | −0.2083 | 0.7976 |
| P0581 | Selectivity increment | Fishery; Bin5; 2000 * | 0.4186 | 0.2725 | −0.1155 | 0.9528 |
| P0582 | Selectivity increment | Fishery; Bin6; 2000 * | −0.0055 | 0.2800 | −0.5544 | 0.5433 |
| P0583 | Selectivity increment | Fishery; Bin7; 2000 * | −0.0436 | 0.2605 | −0.5542 | 0.4669 |
| P0584 | Selectivity increment | Fishery; Bin8; 2000 * | 0.2266 | 0.2481 | −0.2597 | 0.7129 |
| P0585 | Selectivity increment | Fishery; Bin9; 2000 * | 0.0273 | 0.2558 | −0.4741 | 0.5287 |
| P0586 | Selectivity increment | Fishery; Bin10; 2000 * | −0.0436 | 0.2647 | −0.5624 | 0.4753 |
| P0587 | Selectivity increment | Fishery; Bin11; 2000 * | −0.0517 | 0.2881 | −0.6163 | 0.5129 |
| P0588 | Selectivity increment | Fishery; Bin12; 2000 * | −0.0637 | 0.2858 | −0.6239 | 0.4965 |
| P0589 | Selectivity increment | Fishery; Bin1; 2001 * | −0.0146 | 0.4182 | −0.8342 | 0.8051 |
| P0590 | Selectivity increment | Fishery; Bin2; 2001 * | −0.2128 | 0.3639 | −0.9260 | 0.5005 |
| P0591 | Selectivity increment | Fishery; Bin3; 2001 * | −0.4501 | 0.3107 | −1.0591 | 0.1588 |
| P0592 | Selectivity increment | Fishery; Bin4; 2001 * | −0.3103 | 0.2707 | −0.8409 | 0.2202 |
| P0593 | Selectivity increment | Fishery; Bin5; 2001 * | 0.2544 | 0.2536 | −0.2428 | 0.7515 |
| P0594 | Selectivity increment | Fishery; Bin6; 2001 * | 0.6106 | 0.2699 | 0.0815 | 1.1397 |
| P0595 | Selectivity increment | Fishery; Bin7; 2001 * | 0.1237 | 0.2651 | −0.3958 | 0.6432 |
| P0596 | Selectivity increment | Fishery; Bin8; 2001 * | −0.1064 | 0.2514 | −0.5990 | 0.3863 |
| P0597 | Selectivity increment | Fishery; Bin9; 2001 * | 0.0296 | 0.2532 | −0.4666 | 0.5259 |
| P0598 | Selectivity increment | Fishery; Bin10; 2001 * | 0.0467 | 0.2626 | −0.4679 | 0.5614 |
| P0599 | Selectivity increment | Fishery; Bin11; 2001 * | 0.0368 | 0.2697 | −0.4918 | 0.5654 |
| P0600 | Selectivity increment | Fishery; Bin12; 2001 * | −0.0076 | 0.2700 | −0.5368 | 0.5215 |
| P0601 | Selectivity increment | Fishery; Bin1; 2002 * | −0.0584 | 0.4185 | −0.8785 | 0.7618 |
| P0602 | Selectivity increment | Fishery; Bin2; 2002 * | 0.1591 | 0.3560 | −0.5386 | 0.8567 |
| P0603 | Selectivity increment | Fishery; Bin3; 2002 * | 0.0693 | 0.3069 | −0.5323 | 0.6709 |
| P0604 | Selectivity increment | Fishery; Bin4; 2002 * | −0.0089 | 0.2773 | −0.5524 | 0.5345 |
| P0605 | Selectivity increment | Fishery; Bin5; 2002 * | −0.2313 | 0.2558 | −0.7326 | 0.2699 |
| P0606 | Selectivity increment | Fishery; Bin6; 2002 * | 0.0666 | 0.2451 | −0.4137 | 0.5470 |
| P0607 | Selectivity increment | Fishery; Bin7; 2002 * | 0.0798 | 0.2495 | −0.4092 | 0.5689 |
| P0608 | Selectivity increment | Fishery; Bin8; 2002 * | 0.0548 | 0.2547 | −0.4443 | 0.5539 |
| P0609 | Selectivity increment | Fishery; Bin9; 2002 * | 0.0383 | 0.2538 | −0.4592 | 0.5357 |
| P0610 | Selectivity increment | Fishery; Bin10; 2002 * | −0.0247 | 0.2576 | −0.5297 | 0.4802 |
| P0611 | Selectivity increment | Fishery; Bin11; 2002 * | −0.0785 | 0.2672 | −0.6022 | 0.4452 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0612 | Selectivity increment | Fishery; Bin12; 2002 * | −0.0660 | 0.2655 | −0.5864 | 0.4544 |
| P0613 | Selectivity increment | Fishery; Bin1; 2003 * | −0.0709 | 0.4187 | −0.8916 | 0.7499 |
| P0614 | Selectivity increment | Fishery; Bin2; 2003 * | −0.3090 | 0.3568 | −1.0082 | 0.3902 |
| P0615 | Selectivity increment | Fishery; Bin3; 2003 * | 0.6699 | 0.2727 | 0.1354 | 1.2045 |
| P0616 | Selectivity increment | Fishery; Bin4; 2003 * | −0.0671 | 0.2628 | −0.5821 | 0.4478 |
| P0617 | Selectivity increment | Fishery; Bin5; 2003 * | 0.0369 | 0.2581 | −0.4690 | 0.5428 |
| P0618 | Selectivity increment | Fishery; Bin6; 2003 * | −0.1845 | 0.2516 | −0.6775 | 0.3085 |
| P0619 | Selectivity increment | Fishery; Bin7; 2003 * | 0.0299 | 0.2463 | −0.4528 | 0.5126 |
| P0620 | Selectivity increment | Fishery; Bin8; 2003 * | 0.0377 | 0.2530 | −0.4581 | 0.5336 |
| P0621 | Selectivity increment | Fishery; Bin9; 2003 * | −0.0010 | 0.2590 | −0.5085 | 0.5066 |
| P0622 | Selectivity increment | Fishery; Bin10; 2003 * | −0.0130 | 0.2594 | −0.5213 | 0.4954 |
| P0623 | Selectivity increment | Fishery; Bin11; 2003 * | −0.0731 | 0.2643 | −0.5912 | 0.4449 |
| P0624 | Selectivity increment | Fishery; Bin12; 2003 * | −0.0559 | 0.2639 | −0.5732 | 0.4614 |
| P0625 | Selectivity increment | Fishery; Bin1; 2004 * | −0.0277 | 0.4190 | −0.8489 | 0.7936 |
| P0626 | Selectivity increment | Fishery; Bin2; 2004 * | −0.1140 | 0.3769 | −0.8527 | 0.6247 |
| P0627 | Selectivity increment | Fishery; Bin3; 2004 * | −0.6762 | 0.2837 | −1.2323 | −0.1201 |
| P0628 | Selectivity increment | Fishery; Bin4; 2004 * | 0.7169 | 0.2537 | 0.2196 | 1.2142 |
| P0629 | Selectivity increment | Fishery; Bin5; 2004 * | 0.0216 | 0.2568 | −0.4817 | 0.5249 |
| P0630 | Selectivity increment | Fishery; Bin6; 2004 * | −0.1190 | 0.2671 | −0.6425 | 0.4044 |
| P0631 | Selectivity increment | Fishery; Bin7; 2004 * | −0.0233 | 0.2567 | −0.5265 | 0.4799 |
| P0632 | Selectivity increment | Fishery; Bin8; 2004 * | 0.0195 | 0.2529 | −0.4762 | 0.5151 |
| P0633 | Selectivity increment | Fishery; Bin9; 2004 * | 0.0478 | 0.2620 | −0.4656 | 0.5613 |
| P0634 | Selectivity increment | Fishery; Bin10; 2004 * | 0.0528 | 0.2673 | −0.4710 | 0.5766 |
| P0635 | Selectivity increment | Fishery; Bin11; 2004 * | 0.0774 | 0.2682 | −0.4484 | 0.6031 |
| P0636 | Selectivity increment | Fishery; Bin12; 2004 * | 0.0243 | 0.2639 | −0.4929 | 0.5415 |
| P0637 | Selectivity increment | Fishery; Bin1; 2005 * | −0.0551 | 0.4191 | −0.8765 | 0.7662 |
| P0638 | Selectivity increment | Fishery; Bin2; 2005 * | 0.1003 | 0.3875 | −0.6593 | 0.8598 |
| P0639 | Selectivity increment | Fishery; Bin3; 2005 * | −0.0105 | 0.3079 | −0.6139 | 0.5929 |
| P0640 | Selectivity increment | Fishery; Bin4; 2005 * | −0.3443 | 0.2523 | −0.8388 | 0.1502 |
| P0641 | Selectivity increment | Fishery; Bin5; 2005 * | 0.4211 | 0.2497 | −0.0682 | 0.9104 |
| P0642 | Selectivity increment | Fishery; Bin6; 2005 * | 0.2411 | 0.2598 | −0.2682 | 0.7503 |
| P0643 | Selectivity increment | Fishery; Bin7; 2005 * | −0.1040 | 0.2576 | −0.6089 | 0.4009 |
| P0644 | Selectivity increment | Fishery; Bin8; 2005 * | −0.1580 | 0.2557 | −0.6591 | 0.3431 |
| P0645 | Selectivity increment | Fishery; Bin9; 2005 * | −0.0584 | 0.2607 | −0.5694 | 0.4527 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0646 | Selectivity increment | Fishery; Bin10; 2005 * | 0.0069 | 0.2686 | −0.5195 | 0.5333 |
| P0647 | Selectivity increment | Fishery; Bin11; 2005 * | 0.0034 | 0.2723 | −0.5304 | 0.5371 |
| P0648 | Selectivity increment | Fishery; Bin12; 2005 * | −0.0424 | 0.2634 | −0.5586 | 0.4738 |
| P0649 | Selectivity increment | Fishery; Bin1; 2006 * | −0.0831 | 0.4190 | −0.9043 | 0.7381 |
| P0650 | Selectivity increment | Fishery; Bin2; 2006 * | 0.0868 | 0.3896 | −0.6768 | 0.8504 |
| P0651 | Selectivity increment | Fishery; Bin3; 2006 * | 0.5507 | 0.3137 | −0.0642 | 1.1655 |
| P0652 | Selectivity increment | Fishery; Bin4; 2006 * | −0.0815 | 0.2585 | −0.5881 | 0.4251 |
| P0653 | Selectivity increment | Fishery; Bin5; 2006 * | −0.3062 | 0.2441 | −0.7846 | 0.1723 |
| P0654 | Selectivity increment | Fishery; Bin6; 2006 * | −0.0503 | 0.2432 | −0.5269 | 0.4264 |
| P0655 | Selectivity increment | Fishery; Bin7; 2006 * | −0.0551 | 0.2479 | −0.5410 | 0.4308 |
| P0656 | Selectivity increment | Fishery; Bin8; 2006 * | −0.0252 | 0.2559 | −0.5268 | 0.4763 |
| P0657 | Selectivity increment | Fishery; Bin9; 2006 * | −0.0731 | 0.2605 | −0.5836 | 0.4375 |
| P0658 | Selectivity increment | Fishery; Bin10; 2006 * | −0.0319 | 0.2663 | −0.5540 | 0.4901 |
| P0659 | Selectivity increment | Fishery; Bin11; 2006 * | 0.0189 | 0.2723 | −0.5148 | 0.5526 |
| P0660 | Selectivity increment | Fishery; Bin12; 2006 * | 0.0499 | 0.2638 | −0.4671 | 0.5670 |
| P0661 | Selectivity increment | Fishery; Bin1; 2007 * | −0.0442 | 0.4186 | −0.8647 | 0.7763 |
| P0662 | Selectivity increment | Fishery; Bin2; 2007 * | −0.0142 | 0.3863 | −0.7714 | 0.7429 |
| P0663 | Selectivity increment | Fishery; Bin3; 2007 * | −0.0473 | 0.3156 | −0.6659 | 0.5713 |
| P0664 | Selectivity increment | Fishery; Bin4; 2007 * | 0.0167 | 0.2701 | −0.5127 | 0.5461 |
| P0665 | Selectivity increment | Fishery; Bin5; 2007 * | −0.0412 | 0.2468 | −0.5249 | 0.4425 |
| P0666 | Selectivity increment | Fishery; Bin6; 2007 * | 0.0189 | 0.2405 | −0.4525 | 0.4904 |
| P0667 | Selectivity increment | Fishery; Bin7; 2007 * | 0.0371 | 0.2432 | −0.4395 | 0.5138 |
| P0668 | Selectivity increment | Fishery; Bin8; 2007 * | −0.0466 | 0.2513 | −0.5391 | 0.4459 |
| P0669 | Selectivity increment | Fishery; Bin9; 2007 * | −0.0207 | 0.2599 | −0.5301 | 0.4886 |
| P0670 | Selectivity increment | Fishery; Bin10; 2007 * | 0.0048 | 0.2650 | −0.5145 | 0.5241 |
| P0671 | Selectivity increment | Fishery; Bin11; 2007 * | 0.0473 | 0.2699 | −0.4817 | 0.5762 |
| P0672 | Selectivity increment | Fishery; Bin12; 2007 * | 0.0894 | 0.2630 | −0.4261 | 0.6048 |
| P0673 | Selectivity increment | Fishery; Bin1; 2008 * | −0.0079 | 0.4177 | −0.8266 | 0.8108 |
| P0674 | Selectivity increment | Fishery; Bin2; 2008 * | −0.1140 | 0.3624 | −0.8243 | 0.5962 |
| P0675 | Selectivity increment | Fishery; Bin3; 2008 * | −0.5247 | 0.3047 | −1.1219 | 0.0724 |
| P0676 | Selectivity increment | Fishery; Bin4; 2008 * | −0.1226 | 0.2806 | −0.6725 | 0.4274 |
| P0677 | Selectivity increment | Fishery; Bin5; 2008 * | −0.0705 | 0.2571 | −0.5745 | 0.4335 |
| P0678 | Selectivity increment | Fishery; Bin6; 2008 * | 0.0489 | 0.2404 | −0.4224 | 0.5201 |
| P0679 | Selectivity increment | Fishery; Bin7; 2008 * | 0.0738 | 0.2405 | −0.3975 | 0.5452 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0680 | Selectivity increment | Fishery; Bin8; 2008 * | 0.1378 | 0.2459 | −0.3442 | 0.6198 |
| P0681 | Selectivity increment | Fishery; Bin9; 2008 * | 0.1536 | 0.2555 | −0.3472 | 0.6544 |
| P0682 | Selectivity increment | Fishery; Bin10; 2008 * | 0.0828 | 0.2751 | −0.4564 | 0.6220 |
| P0683 | Selectivity increment | Fishery; Bin11; 2008 * | 0.1590 | 0.2830 | −0.3956 | 0.7137 |
| P0684 | Selectivity increment | Fishery; Bin12; 2008 * | 0.1838 | 0.2618 | −0.3292 | 0.6969 |
| P0685 | Selectivity increment | Fishery; Bin1; 2009 * | −0.0405 | 0.4164 | −0.8567 | 0.7757 |
| P0686 | Selectivity increment | Fishery; Bin2; 2009 * | −0.3417 | 0.3505 | −1.0287 | 0.3453 |
| P0687 | Selectivity increment | Fishery; Bin3; 2009 * | 0.1995 | 0.2750 | −0.3395 | 0.7385 |
| P0688 | Selectivity increment | Fishery; Bin4; 2009 * | 0.0605 | 0.2709 | −0.4704 | 0.5914 |
| P0689 | Selectivity increment | Fishery; Bin5; 2009 * | −0.1362 | 0.2743 | −0.6738 | 0.4015 |
| P0690 | Selectivity increment | Fishery; Bin6; 2009 * | −0.0916 | 0.2610 | −0.6032 | 0.4199 |
| P0691 | Selectivity increment | Fishery; Bin7; 2009 * | −0.0530 | 0.2484 | −0.5399 | 0.4339 |
| P0692 | Selectivity increment | Fishery; Bin8; 2009 * | −0.0082 | 0.2472 | −0.4927 | 0.4763 |
| P0693 | Selectivity increment | Fishery; Bin9; 2009 * | 0.0108 | 0.2611 | −0.5009 | 0.5224 |
| P0694 | Selectivity increment | Fishery; Bin10; 2009 * | 0.1422 | 0.2987 | −0.4433 | 0.7277 |
| P0695 | Selectivity increment | Fishery; Bin11; 2009 * | 0.1431 | 0.2927 | −0.4305 | 0.7167 |
| P0696 | Selectivity increment | Fishery; Bin12; 2009 * | 0.1152 | 0.2631 | −0.4006 | 0.6309 |
| P0697 | Selectivity increment | Fishery; Bin1; 2010 * | 0.0380 | 0.4155 | −0.7764 | 0.8524 |
| P0698 | Selectivity increment | Fishery; Bin2; 2010 * | 0.0713 | 0.3478 | −0.6103 | 0.7529 |
| P0699 | Selectivity increment | Fishery; Bin3; 2010 * | −0.6053 | 0.2855 | −1.1648 | −0.0458 |
| P0700 | Selectivity increment | Fishery; Bin4; 2010 * | 0.4401 | 0.2487 | −0.0475 | 0.9276 |
| P0701 | Selectivity increment | Fishery; Bin5; 2010 * | 0.1763 | 0.2702 | −0.3533 | 0.7059 |
| P0702 | Selectivity increment | Fishery; Bin6; 2010 * | −0.2436 | 0.2851 | −0.8024 | 0.3152 |
| P0703 | Selectivity increment | Fishery; Bin7; 2010 * | −0.0793 | 0.2864 | −0.6407 | 0.4821 |
| P0704 | Selectivity increment | Fishery; Bin8; 2010 * | 0.0697 | 0.2604 | −0.4407 | 0.5800 |
| P0705 | Selectivity increment | Fishery; Bin9; 2010 * | 0.0800 | 0.2622 | −0.4339 | 0.5940 |
| P0706 | Selectivity increment | Fishery; Bin10; 2010 * | 0.0262 | 0.2937 | −0.5494 | 0.6018 |
| P0707 | Selectivity increment | Fishery; Bin11; 2010 * | −0.0324 | 0.2989 | −0.6181 | 0.5533 |
| P0708 | Selectivity increment | Fishery; Bin12; 2010 * | 0.0589 | 0.2752 | −0.4806 | 0.5984 |
| P0709 | Selectivity increment | Fishery; Bin1; 2011 * | 0.0295 | 0.4160 | −0.7858 | 0.8448 |
| P0710 | Selectivity increment | Fishery; Bin2; 2011 * | −0.0180 | 0.3414 | −0.6872 | 0.6511 |
| P0711 | Selectivity increment | Fishery; Bin3; 2011 * | 0.0445 | 0.2837 | −0.5116 | 0.6005 |
| P0712 | Selectivity increment | Fishery; Bin4; 2011 * | −1.0769 | 0.2551 | −1.5769 | −0.5770 |
| P0713 | Selectivity increment | Fishery; Bin5; 2011 * | 0.4190 | 0.2469 | −0.0649 | 0.9030 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0714 | Selectivity increment | Fishery; Bin6; 2011 * | 0.3585 | 0.2725 | −0.1757 | 0.8927 |
| P0715 | Selectivity increment | Fishery; Bin7; 2011 * | 0.1392 | 0.2841 | −0.4176 | 0.6961 |
| P0716 | Selectivity increment | Fishery; Bin8; 2011 * | −0.0029 | 0.2677 | −0.5275 | 0.5217 |
| P0717 | Selectivity increment | Fishery; Bin9; 2011 * | 0.0017 | 0.2713 | −0.5299 | 0.5334 |
| P0718 | Selectivity increment | Fishery; Bin10; 2011 * | 0.0820 | 0.2732 | −0.4534 | 0.6174 |
| P0719 | Selectivity increment | Fishery; Bin11; 2011 * | 0.0533 | 0.2812 | −0.4978 | 0.6044 |
| P0720 | Selectivity increment | Fishery; Bin12; 2011 * | −0.0299 | 0.2765 | −0.5719 | 0.5121 |
| P0721 | Selectivity increment | Fishery; Bin1; 2012 * | −0.0139 | 0.4163 | −0.8298 | 0.8021 |
| P0722 | Selectivity increment | Fishery; Bin2; 2012 * | 0.2336 | 0.3613 | −0.4745 | 0.9417 |
| P0723 | Selectivity increment | Fishery; Bin3; 2012 * | 0.1890 | 0.2724 | −0.3450 | 0.7229 |
| P0724 | Selectivity increment | Fishery; Bin4; 2012 * | 0.6247 | 0.2561 | 0.1228 | 1.1266 |
| P0725 | Selectivity increment | Fishery; Bin5; 2012 * | −1.0798 | 0.2559 | −1.5813 | −0.5782 |
| P0726 | Selectivity increment | Fishery; Bin6; 2012 * | −0.0236 | 0.2490 | −0.5115 | 0.4644 |
| P0727 | Selectivity increment | Fishery; Bin7; 2012 * | 0.0121 | 0.2637 | −0.5047 | 0.5289 |
| P0728 | Selectivity increment | Fishery; Bin8; 2012 * | 0.0237 | 0.2771 | −0.5194 | 0.5668 |
| P0729 | Selectivity increment | Fishery; Bin9; 2012 * | 0.0051 | 0.2797 | −0.5432 | 0.5533 |
| P0730 | Selectivity increment | Fishery; Bin10; 2012 * | −0.0627 | 0.2976 | −0.6459 | 0.5205 |
| P0731 | Selectivity increment | Fishery; Bin11; 2012 * | 0.0107 | 0.3307 | −0.6375 | 0.6590 |
| P0732 | Selectivity increment | Fishery; Bin12; 2012 * | 0.0811 | 0.2825 | −0.4726 | 0.6347 |
| P0733 | Selectivity increment | Fishery; Bin1; 2013 * | 0.0205 | 0.4165 | −0.7958 | 0.8368 |
| P0734 | Selectivity increment | Fishery; Bin2; 2013 * | −0.1915 | 0.3668 | −0.9104 | 0.5275 |
| P0735 | Selectivity increment | Fishery; Bin3; 2013 * | 0.0270 | 0.2910 | −0.5434 | 0.5974 |
| P0736 | Selectivity increment | Fishery; Bin4; 2013 * | −0.1384 | 0.2482 | −0.6249 | 0.3481 |
| P0737 | Selectivity increment | Fishery; Bin5; 2013 * | 0.5116 | 0.2562 | 0.0094 | 1.0138 |
| P0738 | Selectivity increment | Fishery; Bin6; 2013 * | −0.6014 | 0.2588 | −1.1087 | −0.0941 |
| P0739 | Selectivity increment | Fishery; Bin7; 2013 * | −0.1571 | 0.2700 | −0.6863 | 0.3720 |
| P0740 | Selectivity increment | Fishery; Bin8; 2013 * | 0.0768 | 0.3021 | −0.5153 | 0.6688 |
| P0741 | Selectivity increment | Fishery; Bin9; 2013 * | 0.1177 | 0.2916 | −0.4538 | 0.6892 |
| P0742 | Selectivity increment | Fishery; Bin10; 2013 * | 0.1362 | 0.2948 | −0.4415 | 0.7139 |
| P0743 | Selectivity increment | Fishery; Bin11; 2013 * | 0.1077 | 0.3419 | −0.5625 | 0.7778 |
| P0744 | Selectivity increment | Fishery; Bin12; 2013 * | 0.0911 | 0.2877 | −0.4728 | 0.6550 |
| P0745 | Selectivity increment | Fishery; Bin1; 2014 * | −0.0208 | 0.4159 | −0.8359 | 0.7943 |
| P0746 | Selectivity increment | Fishery; Bin2; 2014 * | 0.1368 | 0.3573 | −0.5635 | 0.8372 |
| P0747 | Selectivity increment | Fishery; Bin3; 2014 * | −0.0265 | 0.3022 | −0.6189 | 0.5659 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0748 | Selectivity increment | Fishery; Bin4; 2014 * | −0.1390 | 0.2622 | −0.6529 | 0.3749 |
| P0749 | Selectivity increment | Fishery; Bin5; 2014 * | −0.0989 | 0.2478 | −0.5845 | 0.3868 |
| P0750 | Selectivity increment | Fishery; Bin6; 2014 * | 0.3036 | 0.2553 | −0.1967 | 0.8038 |
| P0751 | Selectivity increment | Fishery; Bin7; 2014 * | 0.1480 | 0.2667 | −0.3747 | 0.6708 |
| P0752 | Selectivity increment | Fishery; Bin8; 2014 * | −0.1281 | 0.2924 | −0.7011 | 0.4449 |
| P0753 | Selectivity increment | Fishery; Bin9; 2014 * | −0.1334 | 0.3287 | −0.7777 | 0.5110 |
| P0754 | Selectivity increment | Fishery; Bin10; 2014 * | −0.0146 | 0.3099 | −0.6219 | 0.5927 |
| P0755 | Selectivity increment | Fishery; Bin11; 2014 * | −0.0181 | 0.3208 | −0.6467 | 0.6106 |
| P0756 | Selectivity increment | Fishery; Bin12; 2014 * | −0.0092 | 0.2918 | −0.5810 | 0.5627 |
| P0757 | Selectivity increment | Fishery; Bin1; 2015 * | −0.0049 | 0.4173 | −0.8228 | 0.8129 |
| P0758 | Selectivity increment | Fishery; Bin2; 2015 * | −0.4664 | 0.3403 | −1.1335 | 0.2006 |
| P0759 | Selectivity increment | Fishery; Bin3; 2015 * | 1.0864 | 0.2825 | 0.5327 | 1.6401 |
| P0760 | Selectivity increment | Fishery; Bin4; 2015 * | 0.1881 | 0.2654 | −0.3321 | 0.7083 |
| P0761 | Selectivity increment | Fishery; Bin5; 2015 * | −0.1869 | 0.2562 | −0.6890 | 0.3151 |
| P0762 | Selectivity increment | Fishery; Bin6; 2015 * | −0.1074 | 0.2478 | −0.5930 | 0.3783 |
| P0763 | Selectivity increment | Fishery; Bin7; 2015 * | −0.1479 | 0.2510 | −0.6399 | 0.3441 |
| P0764 | Selectivity increment | Fishery; Bin8; 2015 * | −0.1207 | 0.2715 | −0.6527 | 0.4113 |
| P0765 | Selectivity increment | Fishery; Bin9; 2015 * | −0.1278 | 0.3310 | −0.7765 | 0.5209 |
| P0766 | Selectivity increment | Fishery; Bin10; 2015 * | −0.0646 | 0.3500 | −0.7505 | 0.6213 |
| P0767 | Selectivity increment | Fishery; Bin11; 2015 * | −0.0234 | 0.3684 | −0.7454 | 0.6986 |
| P0768 | Selectivity increment | Fishery; Bin12; 2015 * | −0.0245 | 0.2928 | −0.5983 | 0.5493 |
| P0769 | Selectivity increment | Fishery; Bin1; 2016 * | 0.0709 | 0.4184 | −0.7491 | 0.8909 |
| P0770 | Selectivity increment | Fishery; Bin2; 2016 * | 0.0205 | 0.3679 | −0.7006 | 0.7417 |
| P0771 | Selectivity increment | Fishery; Bin3; 2016 * | −1.4595 | 0.2654 | −1.9797 | −0.9393 |
| P0772 | Selectivity increment | Fishery; Bin4; 2016 * | 1.0623 | 0.2498 | 0.5726 | 1.5519 |
| P0773 | Selectivity increment | Fishery; Bin5; 2016 * | 0.2840 | 0.2593 | −0.2243 | 0.7922 |
| P0774 | Selectivity increment | Fishery; Bin6; 2016 * | −0.0027 | 0.2623 | −0.5167 | 0.5114 |
| P0775 | Selectivity increment | Fishery; Bin7; 2016 * | −0.0982 | 0.2597 | −0.6071 | 0.4107 |
| P0776 | Selectivity increment | Fishery; Bin8; 2016 * | 0.2058 | 0.2624 | −0.3085 | 0.7201 |
| P0777 | Selectivity increment | Fishery; Bin9; 2016 * | 0.1100 | 0.3019 | −0.4817 | 0.7018 |
| P0778 | Selectivity increment | Fishery; Bin10; 2016 * | −0.0858 | 0.3388 | −0.7498 | 0.5782 |
| P0779 | Selectivity increment | Fishery; Bin11; 2016 * | −0.0413 | 0.3728 | −0.7720 | 0.6894 |
| P0780 | Selectivity increment | Fishery; Bin12; 2016 * | −0.0660 | 0.2958 | −0.6458 | 0.5138 |
| P0781 | Selectivity increment | Fishery; Bin1; 2017 * | 0.0527 | 0.4179 | −0.7664 | 0.8719 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0782 | Selectivity increment | Fishery; Bin2; 2017 * | 0.2806 | 0.3829 | −0.4698 | 1.0311 |
| P0783 | Selectivity increment | Fishery; Bin3; 2017 * | −0.2525 | 0.3120 | −0.8641 | 0.3590 |
| P0784 | Selectivity increment | Fishery; Bin4; 2017 * | −1.1177 | 0.2428 | −1.5936 | −0.6418 |
| P0785 | Selectivity increment | Fishery; Bin5; 2017 * | 0.5963 | 0.2497 | 0.1068 | 1.0857 |
| P0786 | Selectivity increment | Fishery; Bin6; 2017 * | 0.3303 | 0.2619 | −0.1831 | 0.8436 |
| P0787 | Selectivity increment | Fishery; Bin7; 2017 * | 0.1867 | 0.2691 | −0.3407 | 0.7141 |
| P0788 | Selectivity increment | Fishery; Bin8; 2017 * | −0.0775 | 0.2733 | −0.6132 | 0.4583 |
| P0789 | Selectivity increment | Fishery; Bin9; 2017 * | 0.0698 | 0.2694 | −0.4582 | 0.5978 |
| P0790 | Selectivity increment | Fishery; Bin10; 2017 * | 0.0476 | 0.2886 | −0.5180 | 0.6133 |
| P0791 | Selectivity increment | Fishery; Bin11; 2017 * | −0.0586 | 0.3038 | −0.6541 | 0.5369 |
| P0792 | Selectivity increment | Fishery; Bin12; 2017 * | −0.0577 | 0.2800 | −0.6065 | 0.4910 |
| P0793 | Selectivity increment | Fishery; Bin1; 2018 * | 0.0269 | 0.4176 | −0.7916 | 0.8454 |
| P0794 | Selectivity increment | Fishery; Bin2; 2018 * | 0.3297 | 0.3844 | −0.4237 | 1.0831 |
| P0795 | Selectivity increment | Fishery; Bin3; 2018 * | 0.1383 | 0.3454 | −0.5386 | 0.8153 |
| P0796 | Selectivity increment | Fishery; Bin4; 2018 * | −0.5920 | 0.2685 | −1.1182 | −0.0658 |
| P0797 | Selectivity increment | Fishery; Bin5; 2018 * | 0.0098 | 0.2416 | −0.4637 | 0.4833 |
| P0798 | Selectivity increment | Fishery; Bin6; 2018 * | 0.1787 | 0.2486 | −0.3085 | 0.6660 |
| P0799 | Selectivity increment | Fishery; Bin7; 2018 * | 0.1098 | 0.2599 | −0.3996 | 0.6191 |
| P0800 | Selectivity increment | Fishery; Bin8; 2018 * | 0.0802 | 0.2732 | −0.4553 | 0.6156 |
| P0801 | Selectivity increment | Fishery; Bin9; 2018 * | −0.0363 | 0.2658 | −0.5572 | 0.4847 |
| P0802 | Selectivity increment | Fishery; Bin10; 2018 * | −0.0785 | 0.2671 | −0.6021 | 0.4451 |
| P0803 | Selectivity increment | Fishery; Bin11; 2018 * | −0.0815 | 0.2758 | −0.6221 | 0.4591 |
| P0804 | Selectivity increment | Fishery; Bin12; 2018 * | −0.0851 | 0.2740 | −0.6221 | 0.4518 |
| P0805 | Selectivity increment | Fishery; Bin1; 2019 * | 0.0466 | 0.4171 | −0.7710 | 0.8641 |
| P0806 | Selectivity increment | Fishery; Bin2; 2019 * | 0.4753 | 0.3768 | −0.2631 | 1.2138 |
| P0807 | Selectivity increment | Fishery; Bin3; 2019 * | 0.2133 | 0.3498 | −0.4724 | 0.8989 |
| P0808 | Selectivity increment | Fishery; Bin4; 2019 * | −0.3828 | 0.3044 | −0.9794 | 0.2138 |
| P0809 | Selectivity increment | Fishery; Bin5; 2019 * | −1.2476 | 0.2535 | −1.7444 | −0.7507 |
| P0810 | Selectivity increment | Fishery; Bin6; 2019 * | 0.0949 | 0.2440 | −0.3833 | 0.5732 |
| P0811 | Selectivity increment | Fishery; Bin7; 2019 * | 0.4408 | 0.2500 | −0.0491 | 0.9307 |
| P0812 | Selectivity increment | Fishery; Bin8; 2019 * | 0.2212 | 0.2554 | −0.2793 | 0.7217 |
| P0813 | Selectivity increment | Fishery; Bin9; 2019 * | 0.1280 | 0.2577 | −0.3771 | 0.6332 |
| P0814 | Selectivity increment | Fishery; Bin10; 2019 * | 0.1319 | 0.2624 | −0.3825 | 0.6463 |
| P0815 | Selectivity increment | Fishery; Bin11; 2019 * | −0.0475 | 0.2710 | −0.5786 | 0.4836 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0816 | Selectivity increment | Fishery; Bin12; 2019 * | −0.0742 | 0.2705 | −0.6044 | 0.4561 |
| P0817 | Selectivity increment | Fishery; Bin1; 2020 * | −0.0191 | 0.4206 | −0.8434 | 0.8052 |
| P0818 | Selectivity increment | Fishery; Bin2; 2020 * | 0.6177 | 0.3313 | −0.0316 | 1.2671 |
| P0819 | Selectivity increment | Fishery; Bin3; 2020 * | 0.4207 | 0.3273 | −0.2208 | 1.0622 |
| P0820 | Selectivity increment | Fishery; Bin4; 2020 * | 0.5413 | 0.3105 | −0.0673 | 1.1498 |
| P0821 | Selectivity increment | Fishery; Bin5; 2020 * | 0.1307 | 0.2757 | −0.4096 | 0.6709 |
| P0822 | Selectivity increment | Fishery; Bin6; 2020 * | −0.4191 | 0.2437 | −0.8968 | 0.0586 |
| P0823 | Selectivity increment | Fishery; Bin7; 2020 * | −0.5005 | 0.2439 | −0.9786 | −0.0224 |
| P0824 | Selectivity increment | Fishery; Bin8; 2020 * | −0.4159 | 0.2603 | −0.9260 | 0.0942 |
| P0825 | Selectivity increment | Fishery; Bin9; 2020 * | −0.1799 | 0.2700 | −0.7091 | 0.3493 |
| P0826 | Selectivity increment | Fishery; Bin10; 2020 * | −0.1332 | 0.2641 | −0.6509 | 0.3844 |
| P0827 | Selectivity increment | Fishery; Bin11; 2020 * | 0.0035 | 0.2686 | −0.5231 | 0.5300 |
| P0828 | Selectivity increment | Fishery; Bin12; 2020 * | −0.0460 | 0.2683 | −0.5718 | 0.4798 |
| P0829 | Selectivity increment | Fishery; Bin1; 2021 * | −0.0184 | 0.4212 | −0.8439 | 0.8071 |
| P0830 | Selectivity increment | Fishery; Bin2; 2021 * | 0.3251 | 0.2749 | −0.2138 | 0.8639 |
| P0831 | Selectivity increment | Fishery; Bin3; 2021 * | 0.9147 | 0.2699 | 0.3857 | 1.4437 |
| P0832 | Selectivity increment | Fishery; Bin4; 2021 * | −0.1532 | 0.2838 | −0.7093 | 0.4030 |
| P0833 | Selectivity increment | Fishery; Bin5; 2021 * | −0.0152 | 0.2871 | −0.5779 | 0.5474 |
| P0834 | Selectivity increment | Fishery; Bin6; 2021 * | −0.6384 | 0.2690 | −1.1655 | −0.1113 |
| P0835 | Selectivity increment | Fishery; Bin7; 2021 * | −0.1600 | 0.2500 | −0.6500 | 0.3299 |
| P0836 | Selectivity increment | Fishery; Bin8; 2021 * | 0.0909 | 0.2552 | −0.4093 | 0.5910 |
| P0837 | Selectivity increment | Fishery; Bin9; 2021 * | −0.1014 | 0.2714 | −0.6334 | 0.4305 |
| P0838 | Selectivity increment | Fishery; Bin10; 2021 * | −0.1057 | 0.2667 | −0.6285 | 0.4170 |
| P0839 | Selectivity increment | Fishery; Bin11; 2021 * | −0.0815 | 0.2704 | −0.6114 | 0.4484 |
| P0840 | Selectivity increment | Fishery; Bin12; 2021 * | −0.0567 | 0.2670 | −0.5800 | 0.4666 |
| P0841 | Selectivity increment | Fishery; Bin1; 2022 * | 0.0125 | 0.4227 | −0.8160 | 0.8411 |
| P0842 | Selectivity increment | Fishery; Bin2; 2022 * | 0.2365 | 0.3079 | −0.3670 | 0.8399 |
| P0843 | Selectivity increment | Fishery; Bin3; 2022 * | −0.4506 | 0.2537 | −0.9478 | 0.0466 |
| P0844 | Selectivity increment | Fishery; Bin4; 2022 * | 0.5723 | 0.2483 | 0.0856 | 1.0591 |
| P0845 | Selectivity increment | Fishery; Bin5; 2022 * | 0.0776 | 0.2801 | −0.4714 | 0.6266 |
| P0846 | Selectivity increment | Fishery; Bin6; 2022 * | 0.0682 | 0.2955 | −0.5109 | 0.6473 |
| P0847 | Selectivity increment | Fishery; Bin7; 2022 * | −0.2542 | 0.2823 | −0.8076 | 0.2991 |
| P0848 | Selectivity increment | Fishery; Bin8; 2022 * | −0.0369 | 0.2568 | −0.5401 | 0.4664 |
| P0849 | Selectivity increment | Fishery; Bin9; 2022 * | −0.0857 | 0.2572 | −0.5898 | 0.4184 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0850 | Selectivity increment | Fishery; Bin10; 2022 * | −0.1450 | 0.2819 | −0.6976 | 0.4076 |
| P0851 | Selectivity increment | Fishery; Bin11; 2022 * | −0.0111 | 0.2962 | −0.5915 | 0.5694 |
| P0852 | Selectivity increment | Fishery; Bin12; 2022 * | 0.0163 | 0.2718 | −0.5165 | 0.5491 |
| P0853 | Selectivity increment | Fishery; Bin1; 2023 * | 0.0335 | 0.4240 | −0.7975 | 0.8646 |
| P0854 | Selectivity increment | Fishery; Bin2; 2023 * | −0.5536 | 0.3485 | −1.2367 | 0.1295 |
| P0855 | Selectivity increment | Fishery; Bin3; 2023 * | 0.1024 | 0.3004 | −0.4864 | 0.6911 |
| P0856 | Selectivity increment | Fishery; Bin4; 2023 * | −0.1481 | 0.2631 | −0.6638 | 0.3677 |
| P0857 | Selectivity increment | Fishery; Bin5; 2023 * | 0.5445 | 0.2616 | 0.0319 | 1.0572 |
| P0858 | Selectivity increment | Fishery; Bin6; 2023 * | 0.3275 | 0.2881 | −0.2372 | 0.8923 |
| P0859 | Selectivity increment | Fishery; Bin7; 2023 * | −0.1214 | 0.2972 | −0.7039 | 0.4612 |
| P0860 | Selectivity increment | Fishery; Bin8; 2023 * | −0.2840 | 0.3093 | −0.8901 | 0.3222 |
| P0861 | Selectivity increment | Fishery; Bin9; 2023 * | 0.0082 | 0.2801 | −0.5408 | 0.5572 |
| P0862 | Selectivity increment | Fishery; Bin10; 2023 * | 0.0548 | 0.2938 | −0.5211 | 0.6307 |
| P0863 | Selectivity increment | Fishery; Bin11; 2023 * | 0.0069 | 0.3066 | −0.5941 | 0.6079 |
| P0864 | Selectivity increment | Fishery; Bin12; 2023 * | 0.0292 | 0.2763 | −0.5124 | 0.5707 |
| P0865 | Selectivity increment | Fishery; Bin1; 2024 * | 0.0203 | 0.4247 | −0.8122 | 0.8528 |
| P0866 | Selectivity increment | Fishery; Bin2; 2024 * | −0.5495 | 0.3664 | −1.2676 | 0.1685 |
| P0867 | Selectivity increment | Fishery; Bin3; 2024 * | 0.1606 | 0.3170 | −0.4607 | 0.7819 |
| P0868 | Selectivity increment | Fishery; Bin4; 2024 * | 0.1045 | 0.2938 | −0.4713 | 0.6804 |
| P0869 | Selectivity increment | Fishery; Bin5; 2024 * | −0.2087 | 0.2602 | −0.7186 | 0.3012 |
| P0870 | Selectivity increment | Fishery; Bin6; 2024 * | 0.3659 | 0.2595 | −0.1427 | 0.8745 |
| P0871 | Selectivity increment | Fishery; Bin7; 2024 * | 0.2160 | 0.2755 | −0.3239 | 0.7559 |
| P0872 | Selectivity increment | Fishery; Bin8; 2024 * | −0.0331 | 0.3171 | −0.6545 | 0.5884 |
| P0873 | Selectivity increment | Fishery; Bin9; 2024 * | −0.0209 | 0.2958 | −0.6007 | 0.5589 |
| P0874 | Selectivity increment | Fishery; Bin10; 2024 * | −0.0052 | 0.2983 | −0.5898 | 0.5795 |
| P0875 | Selectivity increment | Fishery; Bin11; 2024 * | −0.0403 | 0.3419 | −0.7104 | 0.6298 |
| P0876 | Selectivity increment | Fishery; Bin12; 2024 * | −0.0097 | 0.2891 | −0.5763 | 0.5569 |
| P0877 | Selectivity increment | Fishery; Bin1; 2025 * | 0.0465 | 0.4277 | −0.7917 | 0.8847 |
| P0878 | Selectivity increment | Fishery; Bin2; 2025 * | −0.1805 | 0.3689 | −0.9034 | 0.5425 |
| P0879 | Selectivity increment | Fishery; Bin3; 2025 * | −0.3735 | 0.3337 | −1.0275 | 0.2805 |
| P0880 | Selectivity increment | Fishery; Bin4; 2025 * | −0.1775 | 0.3158 | −0.7964 | 0.4414 |
| P0881 | Selectivity increment | Fishery; Bin5; 2025 * | −0.2383 | 0.2955 | −0.8174 | 0.3409 |
| P0882 | Selectivity increment | Fishery; Bin6; 2025 * | 0.1600 | 0.2522 | −0.3343 | 0.6542 |
| P0883 | Selectivity increment | Fishery; Bin7; 2025 * | 0.3901 | 0.2527 | −0.1051 | 0.8854 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0884 | Selectivity increment | Fishery; Bin8; 2025 * | 0.2710 | 0.2754 | −0.2687 | 0.8107 |
| P0885 | Selectivity increment | Fishery; Bin9; 2025 * | 0.0724 | 0.2890 | −0.4940 | 0.6388 |
| P0886 | Selectivity increment | Fishery; Bin10; 2025 * | 0.0628 | 0.2921 | −0.5098 | 0.6354 |
| P0887 | Selectivity increment | Fishery; Bin11; 2025 * | −0.0233 | 0.3339 | −0.6777 | 0.6312 |
| P0888 | Selectivity increment | Fishery; Bin12; 2025 * | −0.0097 | 0.2915 | −0.5811 | 0.5617 |
| P0889 | Selectivity increment | Fishery; Bin1; 2026 * | 0.0057 | 0.4326 | −0.8422 | 0.8537 |
| P0890 | Selectivity increment | Fishery; Bin2; 2026 * | −0.0131 | 0.4291 | −0.8541 | 0.8278 |
| P0891 | Selectivity increment | Fishery; Bin3; 2026 * | 0.0024 | 0.4306 | −0.8416 | 0.8464 |
| P0892 | Selectivity increment | Fishery; Bin4; 2026 * | 0.0136 | 0.4330 | −0.8351 | 0.8622 |
| P0893 | Selectivity increment | Fishery; Bin5; 2026 * | 0.0458 | 0.4381 | −0.8128 | 0.9045 |
| P0894 | Selectivity increment | Fishery; Bin6; 2026 * | −0.1266 | 0.3089 | −0.7320 | 0.4787 |
| P0895 | Selectivity increment | Fishery; Bin7; 2026 * | −0.1276 | 0.2822 | −0.6807 | 0.4256 |
| P0896 | Selectivity increment | Fishery; Bin8; 2026 * | 0.0580 | 0.2800 | −0.4908 | 0.6068 |
| P0897 | Selectivity increment | Fishery; Bin9; 2026 * | 0.0408 | 0.2804 | −0.5087 | 0.5903 |
| P0898 | Selectivity increment | Fishery; Bin10; 2026 * | 0.0264 | 0.2850 | −0.5322 | 0.5850 |
| P0899 | Selectivity increment | Fishery; Bin11; 2026 * | 0.0629 | 0.3040 | −0.5330 | 0.6587 |
| P0900 | Selectivity increment | Fishery; Bin12; 2026 * | 0.0118 | 0.2926 | −0.5617 | 0.5853 |
| P0901 | Selectivity increment | AVO; Bin2; 1994 * | 0.4268 | 0.3221 | −0.2045 | 1.0581 |
| P0902 | Selectivity increment | AVO; Bin3; 1994 * | 0.2014 | 0.3578 | −0.4998 | 0.9027 |
| P0903 | Selectivity increment | AVO; Bin4; 1994 * | 0.0307 | 0.3479 | −0.6511 | 0.7125 |
| P0904 | Selectivity increment | AVO; Bin5; 1994 * | 0.1390 | 0.3399 | −0.5273 | 0.8052 |
| P0905 | Selectivity increment | AVO; Bin6; 1994 * | 0.1843 | 0.3570 | −0.5154 | 0.8839 |
| P0906 | Selectivity increment | AVO; Bin7; 1994 * | −0.2044 | 0.3577 | −0.9055 | 0.4967 |
| P0907 | Selectivity increment | AVO; Bin8; 1994 * | 0.0739 | 0.3182 | −0.5498 | 0.6976 |
| P0908 | Selectivity increment | AVO; Bin2; 1995 * | 0.2193 | 0.2842 | −0.3376 | 0.7763 |
| P0909 | Selectivity increment | AVO; Bin3; 1995 * | 0.2252 | 0.2938 | −0.3507 | 0.8010 |
| P0910 | Selectivity increment | AVO; Bin4; 1995 * | 0.1966 | 0.2925 | −0.3766 | 0.7699 |
| P0911 | Selectivity increment | AVO; Bin5; 1995 * | 0.1891 | 0.2912 | −0.3817 | 0.7599 |
| P0912 | Selectivity increment | AVO; Bin6; 1995 * | 0.2052 | 0.2922 | −0.3676 | 0.7779 |
| P0913 | Selectivity increment | AVO; Bin7; 1995 * | 0.2084 | 0.2921 | −0.3640 | 0.7808 |
| P0914 | Selectivity increment | AVO; Bin8; 1995 * | 0.2116 | 0.2641 | −0.3061 | 0.7293 |
| P0915 | Selectivity increment | AVO; Bin2; 1996 * | 0.2181 | 0.2742 | −0.3193 | 0.7556 |
| P0916 | Selectivity increment | AVO; Bin3; 1996 * | 0.2133 | 0.2742 | −0.3240 | 0.7507 |
| P0917 | Selectivity increment | AVO; Bin4; 1996 * | 0.2070 | 0.2725 | −0.3271 | 0.7412 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0918 | Selectivity increment | AVO; Bin5; 1996 * | 0.1986 | 0.2723 | −0.3351 | 0.7323 |
| P0919 | Selectivity increment | AVO; Bin6; 1996 * | 0.2015 | 0.2730 | −0.3335 | 0.7364 |
| P0920 | Selectivity increment | AVO; Bin7; 1996 * | 0.2067 | 0.2721 | −0.3266 | 0.7400 |
| P0921 | Selectivity increment | AVO; Bin8; 1996 * | 0.2151 | 0.2472 | −0.2694 | 0.6997 |
| P0922 | Selectivity increment | AVO; Bin2; 1997 * | 0.2620 | 0.2631 | −0.2535 | 0.7776 |
| P0923 | Selectivity increment | AVO; Bin3; 1997 * | 0.2517 | 0.2607 | −0.2594 | 0.7627 |
| P0924 | Selectivity increment | AVO; Bin4; 1997 * | 0.2206 | 0.2594 | −0.2877 | 0.7289 |
| P0925 | Selectivity increment | AVO; Bin5; 1997 * | 0.2321 | 0.2585 | −0.2745 | 0.7387 |
| P0926 | Selectivity increment | AVO; Bin6; 1997 * | 0.2202 | 0.2597 | −0.2888 | 0.7292 |
| P0927 | Selectivity increment | AVO; Bin7; 1997 * | 0.2254 | 0.2590 | −0.2822 | 0.7331 |
| P0928 | Selectivity increment | AVO; Bin8; 1997 * | 0.2178 | 0.2340 | −0.2408 | 0.6764 |
| P0929 | Selectivity increment | AVO; Bin2; 1998 * | 0.2201 | 0.2506 | −0.2710 | 0.7112 |
| P0930 | Selectivity increment | AVO; Bin3; 1998 * | 0.2988 | 0.2509 | −0.1930 | 0.7905 |
| P0931 | Selectivity increment | AVO; Bin4; 1998 * | 0.2887 | 0.2506 | −0.2025 | 0.7799 |
| P0932 | Selectivity increment | AVO; Bin5; 1998 * | 0.2513 | 0.2507 | −0.2401 | 0.7427 |
| P0933 | Selectivity increment | AVO; Bin6; 1998 * | 0.2567 | 0.2509 | −0.2351 | 0.7485 |
| P0934 | Selectivity increment | AVO; Bin7; 1998 * | 0.2612 | 0.2497 | −0.2283 | 0.7507 |
| P0935 | Selectivity increment | AVO; Bin8; 1998 * | 0.2331 | 0.2239 | −0.2057 | 0.6719 |
| P0936 | Selectivity increment | AVO; Bin2; 1999 * | 0.2312 | 0.2468 | −0.2526 | 0.7149 |
| P0937 | Selectivity increment | AVO; Bin3; 1999 * | 0.3187 | 0.2435 | −0.1584 | 0.7959 |
| P0938 | Selectivity increment | AVO; Bin4; 1999 * | 0.3135 | 0.2427 | −0.1622 | 0.7891 |
| P0939 | Selectivity increment | AVO; Bin5; 1999 * | 0.2774 | 0.2435 | −0.1999 | 0.7547 |
| P0940 | Selectivity increment | AVO; Bin6; 1999 * | 0.2693 | 0.2437 | −0.2083 | 0.7470 |
| P0941 | Selectivity increment | AVO; Bin7; 1999 * | 0.2849 | 0.2420 | −0.1894 | 0.7593 |
| P0942 | Selectivity increment | AVO; Bin8; 1999 * | 0.2551 | 0.2166 | −0.1694 | 0.6796 |
| P0943 | Selectivity increment | AVO; Bin2; 2000 * | 0.2540 | 0.2396 | −0.2156 | 0.7236 |
| P0944 | Selectivity increment | AVO; Bin3; 2000 * | 0.2719 | 0.2393 | −0.1972 | 0.7410 |
| P0945 | Selectivity increment | AVO; Bin4; 2000 * | 0.3283 | 0.2384 | −0.1390 | 0.7955 |
| P0946 | Selectivity increment | AVO; Bin5; 2000 * | 0.2944 | 0.2392 | −0.1744 | 0.7632 |
| P0947 | Selectivity increment | AVO; Bin6; 2000 * | 0.2994 | 0.2398 | −0.1707 | 0.7694 |
| P0948 | Selectivity increment | AVO; Bin7; 2000 * | 0.2678 | 0.2390 | −0.2007 | 0.7363 |
| P0949 | Selectivity increment | AVO; Bin8; 2000 * | 0.2747 | 0.2126 | −0.1419 | 0.6913 |
| P0950 | Selectivity increment | AVO; Bin2; 2001 * | 0.3133 | 0.2345 | −0.1463 | 0.7729 |
| P0951 | Selectivity increment | AVO; Bin3; 2001 * | 0.3023 | 0.2373 | −0.1628 | 0.7673 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0952 | Selectivity increment | AVO; Bin4; 2001 * | 0.2844 | 0.2370 | −0.1800 | 0.7489 |
| P0953 | Selectivity increment | AVO; Bin5; 2001 * | 0.3131 | 0.2372 | −0.1517 | 0.7780 |
| P0954 | Selectivity increment | AVO; Bin6; 2001 * | 0.3172 | 0.2374 | −0.1480 | 0.7824 |
| P0955 | Selectivity increment | AVO; Bin7; 2001 * | 0.2960 | 0.2370 | −0.1685 | 0.7605 |
| P0956 | Selectivity increment | AVO; Bin8; 2001 * | 0.2805 | 0.2102 | −0.1314 | 0.6925 |
| P0957 | Selectivity increment | AVO; Bin2; 2002 * | 0.3008 | 0.2355 | −0.1608 | 0.7625 |
| P0958 | Selectivity increment | AVO; Bin3; 2002 * | 0.3180 | 0.2342 | −0.1411 | 0.7771 |
| P0959 | Selectivity increment | AVO; Bin4; 2002 * | 0.3056 | 0.2342 | −0.1533 | 0.7645 |
| P0960 | Selectivity increment | AVO; Bin5; 2002 * | 0.3188 | 0.2346 | −0.1410 | 0.7786 |
| P0961 | Selectivity increment | AVO; Bin6; 2002 * | 0.3244 | 0.2345 | −0.1353 | 0.7840 |
| P0962 | Selectivity increment | AVO; Bin7; 2002 * | 0.3085 | 0.2345 | −0.1510 | 0.7680 |
| P0963 | Selectivity increment | AVO; Bin8; 2002 * | 0.2908 | 0.2078 | −0.1165 | 0.6980 |
| P0964 | Selectivity increment | AVO; Bin2; 2003 * | 0.2507 | 0.2309 | −0.2019 | 0.7033 |
| P0965 | Selectivity increment | AVO; Bin3; 2003 * | 0.2393 | 0.2357 | −0.2227 | 0.7013 |
| P0966 | Selectivity increment | AVO; Bin4; 2003 * | 0.3117 | 0.2351 | −0.1490 | 0.7725 |
| P0967 | Selectivity increment | AVO; Bin5; 2003 * | 0.3291 | 0.2351 | −0.1317 | 0.7898 |
| P0968 | Selectivity increment | AVO; Bin6; 2003 * | 0.3118 | 0.2353 | −0.1493 | 0.7729 |
| P0969 | Selectivity increment | AVO; Bin7; 2003 * | 0.3008 | 0.2349 | −0.1597 | 0.7612 |
| P0970 | Selectivity increment | AVO; Bin8; 2003 * | 0.3001 | 0.2080 | −0.1075 | 0.7077 |
| P0971 | Selectivity increment | AVO; Bin2; 2004 * | 0.2241 | 0.2342 | −0.2349 | 0.6830 |
| P0972 | Selectivity increment | AVO; Bin3; 2004 * | 0.2365 | 0.2351 | −0.2244 | 0.6973 |
| P0973 | Selectivity increment | AVO; Bin4; 2004 * | 0.3237 | 0.2336 | −0.1342 | 0.7816 |
| P0974 | Selectivity increment | AVO; Bin5; 2004 * | 0.3248 | 0.2340 | −0.1338 | 0.7835 |
| P0975 | Selectivity increment | AVO; Bin6; 2004 * | 0.3064 | 0.2345 | −0.1533 | 0.7661 |
| P0976 | Selectivity increment | AVO; Bin7; 2004 * | 0.2994 | 0.2344 | −0.1600 | 0.7588 |
| P0977 | Selectivity increment | AVO; Bin8; 2004 * | 0.2971 | 0.2078 | −0.1101 | 0.7043 |
| P0978 | Selectivity increment | AVO; Bin2; 2005 * | 0.3122 | 0.2340 | −0.1465 | 0.7708 |
| P0979 | Selectivity increment | AVO; Bin3; 2005 * | 0.2707 | 0.2370 | −0.1939 | 0.7353 |
| P0980 | Selectivity increment | AVO; Bin4; 2005 * | 0.2088 | 0.2361 | −0.2539 | 0.6714 |
| P0981 | Selectivity increment | AVO; Bin5; 2005 * | 0.2422 | 0.2355 | −0.2193 | 0.7037 |
| P0982 | Selectivity increment | AVO; Bin6; 2005 * | 0.2737 | 0.2355 | −0.1878 | 0.7352 |
| P0983 | Selectivity increment | AVO; Bin7; 2005 * | 0.2837 | 0.2354 | −0.1776 | 0.7450 |
| P0984 | Selectivity increment | AVO; Bin8; 2005 * | 0.2886 | 0.2091 | −0.1212 | 0.6985 |
| P0985 | Selectivity increment | AVO; Bin2; 2006 * | 0.2699 | 0.2375 | −0.1956 | 0.7355 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P0986 | Selectivity increment | AVO; Bin3; 2006 * | 0.2465 | 0.2377 | −0.2194 | 0.7124 |
| P0987 | Selectivity increment | AVO; Bin4; 2006 * | 0.2044 | 0.2360 | −0.2583 | 0.6670 |
| P0988 | Selectivity increment | AVO; Bin5; 2006 * | 0.2238 | 0.2353 | −0.2374 | 0.6850 |
| P0989 | Selectivity increment | AVO; Bin6; 2006 * | 0.2520 | 0.2353 | −0.2091 | 0.7131 |
| P0990 | Selectivity increment | AVO; Bin7; 2006 * | 0.2653 | 0.2354 | −0.1961 | 0.7267 |
| P0991 | Selectivity increment | AVO; Bin8; 2006 * | 0.2690 | 0.2100 | −0.1426 | 0.6806 |
| P0992 | Selectivity increment | AVO; Bin2; 2007 * | 0.2692 | 0.2368 | −0.1949 | 0.7332 |
| P0993 | Selectivity increment | AVO; Bin3; 2007 * | 0.2569 | 0.2368 | −0.2071 | 0.7210 |
| P0994 | Selectivity increment | AVO; Bin4; 2007 * | 0.2281 | 0.2359 | −0.2343 | 0.6904 |
| P0995 | Selectivity increment | AVO; Bin5; 2007 * | 0.2419 | 0.2351 | −0.2189 | 0.7027 |
| P0996 | Selectivity increment | AVO; Bin6; 2007 * | 0.2516 | 0.2350 | −0.2090 | 0.7122 |
| P0997 | Selectivity increment | AVO; Bin7; 2007 * | 0.2401 | 0.2350 | −0.2206 | 0.7008 |
| P0998 | Selectivity increment | AVO; Bin8; 2007 * | 0.2437 | 0.2092 | −0.1664 | 0.6538 |
| P0999 | Selectivity increment | AVO; Bin2; 2008 * | 0.2695 | 0.2362 | −0.1933 | 0.7324 |
| P1000 | Selectivity increment | AVO; Bin3; 2008 * | 0.2401 | 0.2375 | −0.2253 | 0.7055 |
| P1001 | Selectivity increment | AVO; Bin4; 2008 * | 0.1914 | 0.2376 | −0.2743 | 0.6571 |
| P1002 | Selectivity increment | AVO; Bin5; 2008 * | 0.1803 | 0.2377 | −0.2856 | 0.6462 |
| P1003 | Selectivity increment | AVO; Bin6; 2008 * | 0.1896 | 0.2375 | −0.2758 | 0.6551 |
| P1004 | Selectivity increment | AVO; Bin7; 2008 * | 0.2055 | 0.2371 | −0.2592 | 0.6702 |
| P1005 | Selectivity increment | AVO; Bin8; 2008 * | 0.2248 | 0.2109 | −0.1886 | 0.6382 |
| P1006 | Selectivity increment | AVO; Bin2; 2009 * | 0.2256 | 0.2352 | −0.2354 | 0.6867 |
| P1007 | Selectivity increment | AVO; Bin3; 2009 * | 0.2164 | 0.2381 | −0.2503 | 0.6832 |
| P1008 | Selectivity increment | AVO; Bin4; 2009 * | 0.1855 | 0.2380 | −0.2810 | 0.6519 |
| P1009 | Selectivity increment | AVO; Bin5; 2009 * | 0.1817 | 0.2386 | −0.2860 | 0.6494 |
| P1010 | Selectivity increment | AVO; Bin6; 2009 * | 0.1921 | 0.2389 | −0.2761 | 0.6602 |
| P1011 | Selectivity increment | AVO; Bin7; 2009 * | 0.2021 | 0.2383 | −0.2650 | 0.6692 |
| P1012 | Selectivity increment | AVO; Bin8; 2009 * | 0.2012 | 0.2114 | −0.2132 | 0.6156 |
| P1013 | Selectivity increment | AVO; Bin2; 2010 * | 0.2367 | 0.2353 | −0.2245 | 0.6979 |
| P1014 | Selectivity increment | AVO; Bin3; 2010 * | 0.1889 | 0.2390 | −0.2795 | 0.6574 |
| P1015 | Selectivity increment | AVO; Bin4; 2010 * | 0.2100 | 0.2375 | −0.2554 | 0.6754 |
| P1016 | Selectivity increment | AVO; Bin5; 2010 * | 0.1807 | 0.2382 | −0.2861 | 0.6476 |
| P1017 | Selectivity increment | AVO; Bin6; 2010 * | 0.1855 | 0.2386 | −0.2821 | 0.6531 |
| P1018 | Selectivity increment | AVO; Bin7; 2010 * | 0.1926 | 0.2382 | −0.2742 | 0.6595 |
| P1019 | Selectivity increment | AVO; Bin8; 2010 * | 0.1823 | 0.2112 | −0.2316 | 0.5962 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1020 | Selectivity increment | AVO; Bin2; 2011 * | 0.2198 | 0.2349 | −0.2407 | 0.6802 |
| P1021 | Selectivity increment | AVO; Bin3; 2011 * | 0.1828 | 0.2393 | −0.2863 | 0.6519 |
| P1022 | Selectivity increment | AVO; Bin4; 2011 * | 0.1560 | 0.2379 | −0.3103 | 0.6223 |
| P1023 | Selectivity increment | AVO; Bin5; 2011 * | 0.1683 | 0.2383 | −0.2987 | 0.6354 |
| P1024 | Selectivity increment | AVO; Bin6; 2011 * | 0.1801 | 0.2385 | −0.2872 | 0.6475 |
| P1025 | Selectivity increment | AVO; Bin7; 2011 * | 0.1860 | 0.2383 | −0.2810 | 0.6531 |
| P1026 | Selectivity increment | AVO; Bin8; 2011 * | 0.1717 | 0.2116 | −0.2430 | 0.5863 |
| P1027 | Selectivity increment | AVO; Bin2; 2012 * | 0.2201 | 0.2381 | −0.2465 | 0.6868 |
| P1028 | Selectivity increment | AVO; Bin3; 2012 * | 0.1874 | 0.2380 | −0.2791 | 0.6539 |
| P1029 | Selectivity increment | AVO; Bin4; 2012 * | 0.1557 | 0.2357 | −0.3062 | 0.6177 |
| P1030 | Selectivity increment | AVO; Bin5; 2012 * | 0.1715 | 0.2374 | −0.2937 | 0.6367 |
| P1031 | Selectivity increment | AVO; Bin6; 2012 * | 0.1797 | 0.2375 | −0.2857 | 0.6451 |
| P1032 | Selectivity increment | AVO; Bin7; 2012 * | 0.1825 | 0.2377 | −0.2834 | 0.6484 |
| P1033 | Selectivity increment | AVO; Bin8; 2012 * | 0.1626 | 0.2112 | −0.2513 | 0.5765 |
| P1034 | Selectivity increment | AVO; Bin2; 2013 * | 0.2090 | 0.2344 | −0.2504 | 0.6683 |
| P1035 | Selectivity increment | AVO; Bin3; 2013 * | 0.2204 | 0.2364 | −0.2429 | 0.6837 |
| P1036 | Selectivity increment | AVO; Bin4; 2013 * | 0.2464 | 0.2350 | −0.2142 | 0.7070 |
| P1037 | Selectivity increment | AVO; Bin5; 2013 * | 0.2349 | 0.2351 | −0.2258 | 0.6957 |
| P1038 | Selectivity increment | AVO; Bin6; 2013 * | 0.2267 | 0.2351 | −0.2342 | 0.6875 |
| P1039 | Selectivity increment | AVO; Bin7; 2013 * | 0.2029 | 0.2354 | −0.2585 | 0.6643 |
| P1040 | Selectivity increment | AVO; Bin8; 2013 * | 0.1606 | 0.2081 | −0.2473 | 0.5686 |
| P1041 | Selectivity increment | AVO; Bin2; 2014 * | 0.2387 | 0.2348 | −0.2216 | 0.6989 |
| P1042 | Selectivity increment | AVO; Bin3; 2014 * | 0.2282 | 0.2328 | −0.2281 | 0.6845 |
| P1043 | Selectivity increment | AVO; Bin4; 2014 * | 0.2623 | 0.2318 | −0.1920 | 0.7167 |
| P1044 | Selectivity increment | AVO; Bin5; 2014 * | 0.2438 | 0.2321 | −0.2111 | 0.6986 |
| P1045 | Selectivity increment | AVO; Bin6; 2014 * | 0.2492 | 0.2316 | −0.2048 | 0.7032 |
| P1046 | Selectivity increment | AVO; Bin7; 2014 * | 0.2315 | 0.2330 | −0.2251 | 0.6881 |
| P1047 | Selectivity increment | AVO; Bin8; 2014 * | 0.1816 | 0.2054 | −0.2211 | 0.5842 |
| P1048 | Selectivity increment | AVO; Bin2; 2015 * | 0.1440 | 0.2281 | −0.3030 | 0.5910 |
| P1049 | Selectivity increment | AVO; Bin3; 2015 * | 0.2648 | 0.2323 | −0.1904 | 0.7200 |
| P1050 | Selectivity increment | AVO; Bin4; 2015 * | 0.2820 | 0.2321 | −0.1729 | 0.7370 |
| P1051 | Selectivity increment | AVO; Bin5; 2015 * | 0.2515 | 0.2330 | −0.2053 | 0.7082 |
| P1052 | Selectivity increment | AVO; Bin6; 2015 * | 0.2218 | 0.2332 | −0.2352 | 0.6788 |
| P1053 | Selectivity increment | AVO; Bin7; 2015 * | 0.2269 | 0.2336 | −0.2309 | 0.6847 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1054 | Selectivity increment | AVO; Bin8; 2015 * | 0.2145 | 0.2058 | −0.1888 | 0.6178 |
| P1055 | Selectivity increment | AVO; Bin2; 2016 * | 0.1171 | 0.2337 | −0.3409 | 0.5752 |
| P1056 | Selectivity increment | AVO; Bin3; 2016 * | 0.2099 | 0.2333 | −0.2474 | 0.6671 |
| P1057 | Selectivity increment | AVO; Bin4; 2016 * | 0.2846 | 0.2324 | −0.1708 | 0.7401 |
| P1058 | Selectivity increment | AVO; Bin5; 2016 * | 0.2301 | 0.2344 | −0.2294 | 0.6896 |
| P1059 | Selectivity increment | AVO; Bin6; 2016 * | 0.2050 | 0.2348 | −0.2552 | 0.6652 |
| P1060 | Selectivity increment | AVO; Bin7; 2016 * | 0.2006 | 0.2352 | −0.2604 | 0.6616 |
| P1061 | Selectivity increment | AVO; Bin8; 2016 * | 0.2355 | 0.2086 | −0.1734 | 0.6444 |
| P1062 | Selectivity increment | AVO; Bin2; 2017 * | 0.2189 | 0.2350 | −0.2416 | 0.6795 |
| P1063 | Selectivity increment | AVO; Bin3; 2017 * | 0.1560 | 0.2388 | −0.3121 | 0.6240 |
| P1064 | Selectivity increment | AVO; Bin4; 2017 * | 0.1020 | 0.2369 | −0.3624 | 0.5664 |
| P1065 | Selectivity increment | AVO; Bin5; 2017 * | 0.1924 | 0.2378 | −0.2738 | 0.6585 |
| P1066 | Selectivity increment | AVO; Bin6; 2017 * | 0.2094 | 0.2389 | −0.2588 | 0.6777 |
| P1067 | Selectivity increment | AVO; Bin7; 2017 * | 0.2076 | 0.2398 | −0.2625 | 0.6776 |
| P1068 | Selectivity increment | AVO; Bin8; 2017 * | 0.2362 | 0.2134 | −0.1821 | 0.6545 |
| P1069 | Selectivity increment | AVO; Bin2; 2018 * | 0.1941 | 0.2406 | −0.2776 | 0.6657 |
| P1070 | Selectivity increment | AVO; Bin3; 2018 * | 0.1519 | 0.2425 | −0.3234 | 0.6271 |
| P1071 | Selectivity increment | AVO; Bin4; 2018 * | 0.1254 | 0.2409 | −0.3467 | 0.5975 |
| P1072 | Selectivity increment | AVO; Bin5; 2018 * | 0.1977 | 0.2397 | −0.2722 | 0.6675 |
| P1073 | Selectivity increment | AVO; Bin6; 2018 * | 0.2034 | 0.2414 | −0.2698 | 0.6766 |
| P1074 | Selectivity increment | AVO; Bin7; 2018 * | 0.2057 | 0.2430 | −0.2707 | 0.6820 |
| P1075 | Selectivity increment | AVO; Bin8; 2018 * | 0.2230 | 0.2169 | −0.2022 | 0.6481 |
| P1076 | Selectivity increment | AVO; Bin2; 2019 * | 0.2102 | 0.2403 | −0.2607 | 0.6811 |
| P1077 | Selectivity increment | AVO; Bin3; 2019 * | 0.2093 | 0.2441 | −0.2692 | 0.6878 |
| P1078 | Selectivity increment | AVO; Bin4; 2019 * | 0.2316 | 0.2446 | −0.2477 | 0.7110 |
| P1079 | Selectivity increment | AVO; Bin5; 2019 * | 0.2064 | 0.2446 | −0.2730 | 0.6857 |
| P1080 | Selectivity increment | AVO; Bin6; 2019 * | 0.2099 | 0.2448 | −0.2698 | 0.6897 |
| P1081 | Selectivity increment | AVO; Bin7; 2019 * | 0.2061 | 0.2448 | −0.2737 | 0.6859 |
| P1082 | Selectivity increment | AVO; Bin8; 2019 * | 0.2140 | 0.2181 | −0.2135 | 0.6416 |
| P1083 | Selectivity increment | AVO; Bin2; 2020 * | 0.1981 | 0.2447 | −0.2815 | 0.6777 |
| P1084 | Selectivity increment | AVO; Bin3; 2020 * | 0.2065 | 0.2472 | −0.2781 | 0.6910 |
| P1085 | Selectivity increment | AVO; Bin4; 2020 * | 0.2306 | 0.2472 | −0.2538 | 0.7151 |
| P1086 | Selectivity increment | AVO; Bin5; 2020 * | 0.1997 | 0.2474 | −0.2853 | 0.6847 |
| P1087 | Selectivity increment | AVO; Bin6; 2020 * | 0.1935 | 0.2475 | −0.2916 | 0.6787 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1088 | Selectivity increment | AVO; Bin7; 2020 * | 0.2048 | 0.2475 | −0.2802 | 0.6898 |
| P1089 | Selectivity increment | AVO; Bin8; 2020 * | 0.2187 | 0.2214 | −0.2152 | 0.6527 |
| P1090 | Selectivity increment | AVO; Bin2; 2021 * | 0.1941 | 0.2490 | −0.2939 | 0.6820 |
| P1091 | Selectivity increment | AVO; Bin3; 2021 * | 0.1939 | 0.2508 | −0.2977 | 0.6855 |
| P1092 | Selectivity increment | AVO; Bin4; 2021 * | 0.2316 | 0.2500 | −0.2584 | 0.7216 |
| P1093 | Selectivity increment | AVO; Bin5; 2021 * | 0.1960 | 0.2505 | −0.2951 | 0.6871 |
| P1094 | Selectivity increment | AVO; Bin6; 2021 * | 0.1941 | 0.2507 | −0.2972 | 0.6854 |
| P1095 | Selectivity increment | AVO; Bin7; 2021 * | 0.2074 | 0.2505 | −0.2837 | 0.6985 |
| P1096 | Selectivity increment | AVO; Bin8; 2021 * | 0.2175 | 0.2256 | −0.2246 | 0.6596 |
| P1097 | Selectivity increment | AVO; Bin2; 2022 * | 0.1766 | 0.2549 | −0.3230 | 0.6761 |
| P1098 | Selectivity increment | AVO; Bin3; 2022 * | 0.1694 | 0.2554 | −0.3312 | 0.6701 |
| P1099 | Selectivity increment | AVO; Bin4; 2022 * | 0.2445 | 0.2534 | −0.2521 | 0.7411 |
| P1100 | Selectivity increment | AVO; Bin5; 2022 * | 0.1840 | 0.2548 | −0.3154 | 0.6834 |
| P1101 | Selectivity increment | AVO; Bin6; 2022 * | 0.1910 | 0.2550 | −0.3088 | 0.6908 |
| P1102 | Selectivity increment | AVO; Bin7; 2022 * | 0.2070 | 0.2551 | −0.2930 | 0.7069 |
| P1103 | Selectivity increment | AVO; Bin8; 2022 * | 0.2136 | 0.2312 | −0.2396 | 0.6667 |
| P1104 | Selectivity increment | AVO; Bin2; 2023 * | 0.2450 | 0.2591 | −0.2628 | 0.7527 |
| P1105 | Selectivity increment | AVO; Bin3; 2023 * | 0.2141 | 0.2631 | −0.3016 | 0.7298 |
| P1106 | Selectivity increment | AVO; Bin4; 2023 * | 0.0901 | 0.2626 | −0.4245 | 0.6048 |
| P1107 | Selectivity increment | AVO; Bin5; 2023 * | 0.1396 | 0.2622 | −0.3742 | 0.6535 |
| P1108 | Selectivity increment | AVO; Bin6; 2023 * | 0.1864 | 0.2622 | −0.3274 | 0.7002 |
| P1109 | Selectivity increment | AVO; Bin7; 2023 * | 0.2082 | 0.2629 | −0.3070 | 0.7234 |
| P1110 | Selectivity increment | AVO; Bin8; 2023 * | 0.2057 | 0.2401 | −0.2648 | 0.6762 |
| P1111 | Selectivity increment | AVO; Bin2; 2024 * | 0.2261 | 0.2708 | −0.3046 | 0.7569 |
| P1112 | Selectivity increment | AVO; Bin3; 2024 * | 0.1919 | 0.2739 | −0.3449 | 0.7287 |
| P1113 | Selectivity increment | AVO; Bin4; 2024 * | 0.0979 | 0.2727 | −0.4366 | 0.6324 |
| P1114 | Selectivity increment | AVO; Bin5; 2024 * | 0.1236 | 0.2711 | −0.4077 | 0.6549 |
| P1115 | Selectivity increment | AVO; Bin6; 2024 * | 0.1798 | 0.2705 | −0.3504 | 0.7100 |
| P1116 | Selectivity increment | AVO; Bin7; 2024 * | 0.2037 | 0.2727 | −0.3309 | 0.7382 |
| P1117 | Selectivity increment | AVO; Bin8; 2024 * | 0.1924 | 0.2513 | −0.3001 | 0.6849 |
| P1118 | Selectivity increment | AVO; Bin2; 2025 * | 0.2192 | 0.2804 | −0.3304 | 0.7688 |
| P1119 | Selectivity increment | AVO; Bin3; 2025 * | 0.2323 | 0.2862 | −0.3287 | 0.7934 |
| P1120 | Selectivity increment | AVO; Bin4; 2025 * | 0.1656 | 0.2871 | −0.3970 | 0.7283 |
| P1121 | Selectivity increment | AVO; Bin5; 2025 * | 0.1703 | 0.2867 | −0.3916 | 0.7321 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1122 | Selectivity increment | AVO; Bin6; 2025 * | 0.1740 | 0.2869 | −0.3883 | 0.7362 |
| P1123 | Selectivity increment | AVO; Bin7; 2025 * | 0.1884 | 0.2866 | −0.3734 | 0.7501 |
| P1124 | Selectivity increment | AVO; Bin8; 2025 * | 0.1812 | 0.2649 | −0.3380 | 0.7004 |
| P1125 | Selectivity increment | AVO; Bin2; 2026 * | 0.2093 | 0.3050 | −0.3886 | 0.8072 |
| P1126 | Selectivity increment | AVO; Bin3; 2026 * | 0.2386 | 0.3046 | −0.3585 | 0.8356 |
| P1127 | Selectivity increment | AVO; Bin4; 2026 * | 0.1551 | 0.3058 | −0.4443 | 0.7544 |
| P1128 | Selectivity increment | AVO; Bin5; 2026 * | 0.1576 | 0.3063 | −0.4428 | 0.7581 |
| P1129 | Selectivity increment | AVO; Bin6; 2026 * | 0.1761 | 0.3067 | −0.4251 | 0.7772 |
| P1130 | Selectivity increment | AVO; Bin7; 2026 * | 0.1927 | 0.3064 | −0.4078 | 0.7932 |
| P1131 | Selectivity increment | AVO; Bin8; 2026 * | 0.1815 | 0.2852 | −0.3775 | 0.7406 |
| P1132 | Log logistic slope | Asc; BTS | −0.8376 | 0.1962 | −1.2221 | −0.4531 |
| P1133 | Logistic midpoint | Asc; BTS | 6.9805 | 0.6409 | 5.7244 | 8.2367 |
| P1134 | BTS age-1 log selectivity | Age-1 component; BTS | −3.0479 | 0.3067 | −3.6490 | −2.4469 |
| P1135 | BTS slope deviation | Asc; BTS; 1983 | 0.9087 | 0.2274 | 0.4630 | 1.3543 |
| P1136 | BTS slope deviation | Asc; BTS; 1984 | 0.7382 | 0.2286 | 0.2901 | 1.1863 |
| P1137 | BTS slope deviation | Asc; BTS; 1985 | 0.2475 | 0.2461 | −0.2348 | 0.7297 |
| P1138 | BTS slope deviation | Asc; BTS; 1986 | 0.3667 | 0.2364 | −0.0965 | 0.8300 |
| P1139 | BTS slope deviation | Asc; BTS; 1987 | 0.3474 | 0.2949 | −0.2306 | 0.9254 |
| P1140 | BTS slope deviation | Asc; BTS; 1988 | 0.3002 | 0.3348 | −0.3559 | 0.9564 |
| P1141 | BTS slope deviation | Asc; BTS; 1989 | 0.5436 | 0.3985 | −0.2374 | 1.3245 |
| P1142 | BTS slope deviation | Asc; BTS; 1990 | 1.2073 | 0.3076 | 0.6043 | 1.8102 |
| P1143 | BTS slope deviation | Asc; BTS; 1991 | 1.0413 | 0.2581 | 0.5355 | 1.5471 |
| P1144 | BTS slope deviation | Asc; BTS; 1992 | 0.9879 | 0.2549 | 0.4883 | 1.4875 |
| P1145 | BTS slope deviation | Asc; BTS; 1993 | 1.1449 | 0.2594 | 0.6365 | 1.6532 |
| P1146 | BTS slope deviation | Asc; BTS; 1994 | 1.1404 | 0.2336 | 0.6827 | 1.5982 |
| P1147 | BTS slope deviation | Asc; BTS; 1995 | 1.0736 | 0.2332 | 0.6165 | 1.5307 |
| P1148 | BTS slope deviation | Asc; BTS; 1996 | 0.6664 | 0.2162 | 0.2428 | 1.0901 |
| P1149 | BTS slope deviation | Asc; BTS; 1997 | 0.5019 | 0.2354 | 0.0406 | 0.9632 |
| P1150 | BTS slope deviation | Asc; BTS; 1998 | 0.2949 | 0.2285 | −0.1530 | 0.7427 |
| P1151 | BTS slope deviation | Asc; BTS; 1999 | 0.2464 | 0.2318 | −0.2079 | 0.7008 |
| P1152 | BTS slope deviation | Asc; BTS; 2000 | 0.5104 | 0.2470 | 0.0264 | 0.9944 |
| P1153 | BTS slope deviation | Asc; BTS; 2001 | 0.3417 | 0.2233 | −0.0959 | 0.7793 |
| P1154 | BTS slope deviation | Asc; BTS; 2002 | 0.7946 | 0.2240 | 0.3556 | 1.2337 |
| P1155 | BTS slope deviation | Asc; BTS; 2003 | 1.0199 | 0.2354 | 0.5585 | 1.4812 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1156 | BTS slope deviation | Asc; BTS; 2004 | 0.9485 | 0.2234 | 0.5106 | 1.3864 |
| P1157 | BTS slope deviation | Asc; BTS; 2005 | 1.0394 | 0.2342 | 0.5803 | 1.4985 |
| P1158 | BTS slope deviation | Asc; BTS; 2006 | 0.7304 | 0.2221 | 0.2950 | 1.1657 |
| P1159 | BTS slope deviation | Asc; BTS; 2007 | 1.1604 | 0.2459 | 0.6785 | 1.6424 |
| P1160 | BTS slope deviation | Asc; BTS; 2008 | 1.3498 | 0.2211 | 0.9164 | 1.7832 |
| P1161 | BTS slope deviation | Asc; BTS; 2009 | 1.0541 | 0.2284 | 0.6065 | 1.5016 |
| P1162 | BTS slope deviation | Asc; BTS; 2010 | 1.5895 | 0.2493 | 1.1008 | 2.0781 |
| P1163 | BTS slope deviation | Asc; BTS; 2011 | 1.3653 | 0.2144 | 0.9450 | 1.7856 |
| P1164 | BTS slope deviation | Asc; BTS; 2012 | 1.0054 | 0.2373 | 0.5403 | 1.4705 |
| P1165 | BTS slope deviation | Asc; BTS; 2013 | 1.1035 | 0.2417 | 0.6297 | 1.5773 |
| P1166 | BTS slope deviation | Asc; BTS; 2014 | 1.3500 | 0.2227 | 0.9136 | 1.7864 |
| P1167 | BTS slope deviation | Asc; BTS; 2015 | 1.2379 | 0.2284 | 0.7902 | 1.6856 |
| P1168 | BTS slope deviation | Asc; BTS; 2016 | 1.2450 | 0.2421 | 0.7705 | 1.7195 |
| P1169 | BTS slope deviation | Asc; BTS; 2017 | 1.0643 | 0.2207 | 0.6317 | 1.4968 |
| P1170 | BTS slope deviation | Asc; BTS; 2018 | 0.5311 | 0.2347 | 0.0711 | 0.9911 |
| P1171 | BTS slope deviation | Asc; BTS; 2019 | 0.8316 | 0.2327 | 0.3755 | 1.2876 |
| P1172 | BTS slope deviation | Asc; BTS; 2020 | 0.7195 | 0.2985 | 0.1345 | 1.3046 |
| P1173 | BTS slope deviation | Asc; BTS; 2021 | 0.6483 | 0.2169 | 0.2232 | 1.0734 |
| P1174 | BTS slope deviation | Asc; BTS; 2022 | 0.8388 | 0.2317 | 0.3848 | 1.2929 |
| P1175 | BTS slope deviation | Asc; BTS; 2023 | 0.5333 | 0.2269 | 0.0887 | 0.9779 |
| P1176 | BTS slope deviation | Asc; BTS; 2024 | 0.5612 | 0.2527 | 0.0660 | 1.0564 |
| P1177 | BTS slope deviation | Asc; BTS; 2025 | 0.6074 | 0.2290 | 0.1586 | 1.0562 |
| P1178 | BTS slope deviation | Asc; BTS; 2026 | 0.8008 | 0.2609 | 0.2895 | 1.3121 |
| P1179 | BTS midpoint deviation | Asc; BTS; 1983 | −0.3111 | 0.0866 | −0.4808 | −0.1414 |
| P1180 | BTS midpoint deviation | Asc; BTS; 1984 | −0.1267 | 0.0895 | −0.3021 | 0.0488 |
| P1181 | BTS midpoint deviation | Asc; BTS; 1985 | 0.0383 | 0.0942 | −0.1463 | 0.2229 |
| P1182 | BTS midpoint deviation | Asc; BTS; 1986 | 0.0274 | 0.0975 | −0.1637 | 0.2186 |
| P1183 | BTS midpoint deviation | Asc; BTS; 1987 | −0.0750 | 0.1102 | −0.2909 | 0.1409 |
| P1184 | BTS midpoint deviation | Asc; BTS; 1988 | −0.2158 | 0.1232 | −0.4573 | 0.0258 |
| P1185 | BTS midpoint deviation | Asc; BTS; 1989 | −0.2963 | 0.1204 | −0.5324 | −0.0603 |
| P1186 | BTS midpoint deviation | Asc; BTS; 1990 | −0.3506 | 0.1113 | −0.5688 | −0.1323 |
| P1187 | BTS midpoint deviation | Asc; BTS; 1991 | −0.2509 | 0.1111 | −0.4688 | −0.0331 |
| P1188 | BTS midpoint deviation | Asc; BTS; 1992 | −0.2448 | 0.1085 | −0.4575 | −0.0322 |
| P1189 | BTS midpoint deviation | Asc; BTS; 1993 | −0.2921 | 0.1020 | −0.4920 | −0.0922 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1190 | BTS midpoint deviation | Asc; BTS; 1994 | −0.2291 | 0.0979 | −0.4210 | −0.0373 |
| P1191 | BTS midpoint deviation | Asc; BTS; 1995 | −0.1343 | 0.1012 | −0.3326 | 0.0640 |
| P1192 | BTS midpoint deviation | Asc; BTS; 1996 | 0.0653 | 0.0961 | −0.1230 | 0.2536 |
| P1193 | BTS midpoint deviation | Asc; BTS; 1997 | 0.1110 | 0.1035 | −0.0917 | 0.3138 |
| P1194 | BTS midpoint deviation | Asc; BTS; 1998 | 0.1916 | 0.1005 | −0.0054 | 0.3885 |
| P1195 | BTS midpoint deviation | Asc; BTS; 1999 | 0.1329 | 0.1076 | −0.0780 | 0.3439 |
| P1196 | BTS midpoint deviation | Asc; BTS; 2000 | −0.0032 | 0.1111 | −0.2210 | 0.2145 |
| P1197 | BTS midpoint deviation | Asc; BTS; 2001 | 0.0394 | 0.1009 | −0.1583 | 0.2372 |
| P1198 | BTS midpoint deviation | Asc; BTS; 2002 | −0.0939 | 0.1011 | −0.2920 | 0.1042 |
| P1199 | BTS midpoint deviation | Asc; BTS; 2003 | −0.1374 | 0.1034 | −0.3402 | 0.0653 |
| P1200 | BTS midpoint deviation | Asc; BTS; 2004 | −0.1130 | 0.0971 | −0.3033 | 0.0773 |
| P1201 | BTS midpoint deviation | Asc; BTS; 2005 | −0.1589 | 0.0977 | −0.3504 | 0.0326 |
| P1202 | BTS midpoint deviation | Asc; BTS; 2006 | −0.0355 | 0.0960 | −0.2237 | 0.1527 |
| P1203 | BTS midpoint deviation | Asc; BTS; 2007 | −0.2089 | 0.0993 | −0.4035 | −0.0143 |
| P1204 | BTS midpoint deviation | Asc; BTS; 2008 | −0.2335 | 0.0961 | −0.4219 | −0.0451 |
| P1205 | BTS midpoint deviation | Asc; BTS; 2009 | −0.1517 | 0.0988 | −0.3454 | 0.0420 |
| P1206 | BTS midpoint deviation | Asc; BTS; 2010 | −0.3339 | 0.0978 | −0.5256 | −0.1421 |
| P1207 | BTS midpoint deviation | Asc; BTS; 2011 | −0.2094 | 0.0940 | −0.3936 | −0.0251 |
| P1208 | BTS midpoint deviation | Asc; BTS; 2012 | −0.1593 | 0.0986 | −0.3526 | 0.0339 |
| P1209 | BTS midpoint deviation | Asc; BTS; 2013 | −0.1905 | 0.0946 | −0.3760 | −0.0050 |
| P1210 | BTS midpoint deviation | Asc; BTS; 2014 | −0.2790 | 0.0959 | −0.4670 | −0.0911 |
| P1211 | BTS midpoint deviation | Asc; BTS; 2015 | −0.3231 | 0.0976 | −0.5144 | −0.1318 |
| P1212 | BTS midpoint deviation | Asc; BTS; 2016 | −0.2689 | 0.0984 | −0.4617 | −0.0762 |
| P1213 | BTS midpoint deviation | Asc; BTS; 2017 | −0.1865 | 0.0941 | −0.3709 | −0.0020 |
| P1214 | BTS midpoint deviation | Asc; BTS; 2018 | −0.0140 | 0.0995 | −0.2090 | 0.1811 |
| P1215 | BTS midpoint deviation | Asc; BTS; 2019 | −0.1490 | 0.0969 | −0.3390 | 0.0410 |
| P1216 | BTS midpoint deviation | Asc; BTS; 2020 | −0.0624 | 0.1177 | −0.2930 | 0.1683 |
| P1217 | BTS midpoint deviation | Asc; BTS; 2021 | 0.0125 | 0.0977 | −0.1790 | 0.2041 |
| P1218 | BTS midpoint deviation | Asc; BTS; 2022 | −0.0671 | 0.0992 | −0.2617 | 0.1274 |
| P1219 | BTS midpoint deviation | Asc; BTS; 2023 | 0.0978 | 0.0985 | −0.0952 | 0.2909 |
| P1220 | BTS midpoint deviation | Asc; BTS; 2024 | 0.0068 | 0.1052 | −0.1993 | 0.2130 |
| P1221 | BTS midpoint deviation | Asc; BTS; 2025 | 0.0317 | 0.1021 | −0.1684 | 0.2319 |
| P1222 | BTS midpoint deviation | Asc; BTS; 2026 | −0.1099 | 0.1145 | −0.3344 | 0.1146 |
| P1223 | BTS age-1 deviation | Age-1 component; BTS; 1983 | −0.0145 | 0.1177 | −0.2452 | 0.2162 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1224 | BTS age-1 deviation | Age-1 component; BTS; 1984 | 0.0371 | 0.1648 | −0.2858 | 0.3600 |
| P1225 | BTS age-1 deviation | Age-1 component; BTS; 1985 | −0.1923 | 0.1416 | −0.4698 | 0.0851 |
| P1226 | BTS age-1 deviation | Age-1 component; BTS; 1986 | −0.4013 | 0.1448 | −0.6851 | −0.1174 |
| P1227 | BTS age-1 deviation | Age-1 component; BTS; 1987 | −0.2452 | 0.1911 | −0.6198 | 0.1294 |
| P1228 | BTS age-1 deviation | Age-1 component; BTS; 1988 | −0.2987 | 0.1852 | −0.6616 | 0.0643 |
| P1229 | BTS age-1 deviation | Age-1 component; BTS; 1989 | −0.0598 | 0.1804 | −0.4134 | 0.2939 |
| P1230 | BTS age-1 deviation | Age-1 component; BTS; 1990 | 0.1030 | 0.1485 | −0.1881 | 0.3941 |
| P1231 | BTS age-1 deviation | Age-1 component; BTS; 1991 | −0.1431 | 0.1368 | −0.4113 | 0.1250 |
| P1232 | BTS age-1 deviation | Age-1 component; BTS; 1992 | −0.0165 | 0.1370 | −0.2849 | 0.2520 |
| P1233 | BTS age-1 deviation | Age-1 component; BTS; 1993 | 0.0411 | 0.1297 | −0.2131 | 0.2953 |
| P1234 | BTS age-1 deviation | Age-1 component; BTS; 1994 | −0.1102 | 0.1451 | −0.3946 | 0.1741 |
| P1235 | BTS age-1 deviation | Age-1 component; BTS; 1995 | −0.1370 | 0.1474 | −0.4259 | 0.1519 |
| P1236 | BTS age-1 deviation | Age-1 component; BTS; 1996 | −0.1046 | 0.1174 | −0.3346 | 0.1254 |
| P1237 | BTS age-1 deviation | Age-1 component; BTS; 1997 | 0.0048 | 0.1318 | −0.2534 | 0.2631 |
| P1238 | BTS age-1 deviation | Age-1 component; BTS; 1998 | 0.0560 | 0.1447 | −0.2275 | 0.3395 |
| P1239 | BTS age-1 deviation | Age-1 component; BTS; 1999 | 0.0971 | 0.1390 | −0.1753 | 0.3695 |
| P1240 | BTS age-1 deviation | Age-1 component; BTS; 2000 | 0.1927 | 0.1380 | −0.0778 | 0.4631 |
| P1241 | BTS age-1 deviation | Age-1 component; BTS; 2001 | 0.0776 | 0.1172 | −0.1521 | 0.3073 |
| P1242 | BTS age-1 deviation | Age-1 component; BTS; 2002 | 0.0837 | 0.1210 | −0.1534 | 0.3209 |
| P1243 | BTS age-1 deviation | Age-1 component; BTS; 2003 | 0.2335 | 0.1390 | −0.0389 | 0.5059 |
| P1244 | BTS age-1 deviation | Age-1 component; BTS; 2004 | 0.0473 | 0.1386 | −0.2242 | 0.3189 |
| P1245 | BTS age-1 deviation | Age-1 component; BTS; 2005 | −0.0147 | 0.1483 | −0.3054 | 0.2759 |
| P1246 | BTS age-1 deviation | Age-1 component; BTS; 2006 | −0.0132 | 0.1166 | −0.2417 | 0.2153 |
| P1247 | BTS age-1 deviation | Age-1 component; BTS; 2007 | −0.0358 | 0.1179 | −0.2668 | 0.1952 |
| P1248 | BTS age-1 deviation | Age-1 component; BTS; 2008 | 0.2024 | 0.1181 | −0.0291 | 0.4340 |
| P1249 | BTS age-1 deviation | Age-1 component; BTS; 2009 | 0.3586 | 0.1115 | 0.1400 | 0.5773 |
| P1250 | BTS age-1 deviation | Age-1 component; BTS; 2010 | 0.3397 | 0.1320 | 0.0809 | 0.5984 |
| P1251 | BTS age-1 deviation | Age-1 component; BTS; 2011 | −0.0718 | 0.1226 | −0.3121 | 0.1685 |
| P1252 | BTS age-1 deviation | Age-1 component; BTS; 2012 | −0.0599 | 0.1366 | −0.3276 | 0.2078 |
| P1253 | BTS age-1 deviation | Age-1 component; BTS; 2013 | 0.2090 | 0.1341 | −0.0539 | 0.4719 |
| P1254 | BTS age-1 deviation | Age-1 component; BTS; 2014 | 0.1700 | 0.1236 | −0.0723 | 0.4123 |
| P1255 | BTS age-1 deviation | Age-1 component; BTS; 2015 | 0.0604 | 0.1400 | −0.2139 | 0.3347 |
| P1256 | BTS age-1 deviation | Age-1 component; BTS; 2016 | −0.0342 | 0.1521 | −0.3322 | 0.2639 |
| P1257 | BTS age-1 deviation | Age-1 component; BTS; 2017 | −0.0710 | 0.1368 | −0.3392 | 0.1972 |

| Complete parameter estimates and uncertainty | | | | | | |
|---|---|---|---|---|---|---|
| ID | Group | Parameter context | Estimate | SE | 95% lower | 95% upper |
| P1258 | BTS age-1 deviation | Age-1 component; BTS; 2018 | −0.0545 | 0.1279 | −0.3053 | 0.1962 |
| P1259 | BTS age-1 deviation | Age-1 component; BTS; 2019 | 0.2778 | 0.1171 | 0.0483 | 0.5074 |
| P1260 | BTS age-1 deviation | Age-1 component; BTS; 2020 | 0.1488 | 0.2105 | −0.2638 | 0.5614 |
| P1261 | BTS age-1 deviation | Age-1 component; BTS; 2021 | 0.0198 | 0.1341 | −0.2431 | 0.2826 |
| P1262 | BTS age-1 deviation | Age-1 component; BTS; 2022 | 0.0699 | 0.1373 | −0.1992 | 0.3390 |
| P1263 | BTS age-1 deviation | Age-1 component; BTS; 2023 | −0.0696 | 0.1428 | −0.3494 | 0.2102 |
| P1264 | BTS age-1 deviation | Age-1 component; BTS; 2024 | 0.1744 | 0.1502 | −0.1201 | 0.4689 |
| P1265 | BTS age-1 deviation | Age-1 component; BTS; 2025 | 0.0580 | 0.1478 | −0.2317 | 0.3477 |
| P1266 | BTS age-1 deviation | Age-1 component; BTS; 2026 | −0.0438 | 0.2085 | −0.4524 | 0.3649 |
| * A parameter is shared across additional positions. The full shared-position mappings are supplied in the numerical parameter table. | | | | | | |
| The BTS age-1 component reuses native coordinates named Descending; it is a separate first-age component rather than a descending limb. Its baseline is log selectivity, and its annual deviations multiply that baseline log value. Midpoint deviations likewise multiply the baseline midpoint. The selectivity table combines these parameters and their uncertainty. | | | | | | |

*Note.* All estimated parameters and approximate 95% confidence intervals in Model 26.0, on their estimation scales. The parameter ID links to the complete numerical data and shared-parameter mappings. Recruitment and selectivity deviations are expressed on their fitted deviation or increment scales; the biological and selectivity schedules show their realized effects. The full precision data also supply natural-scale transformations where applicable.

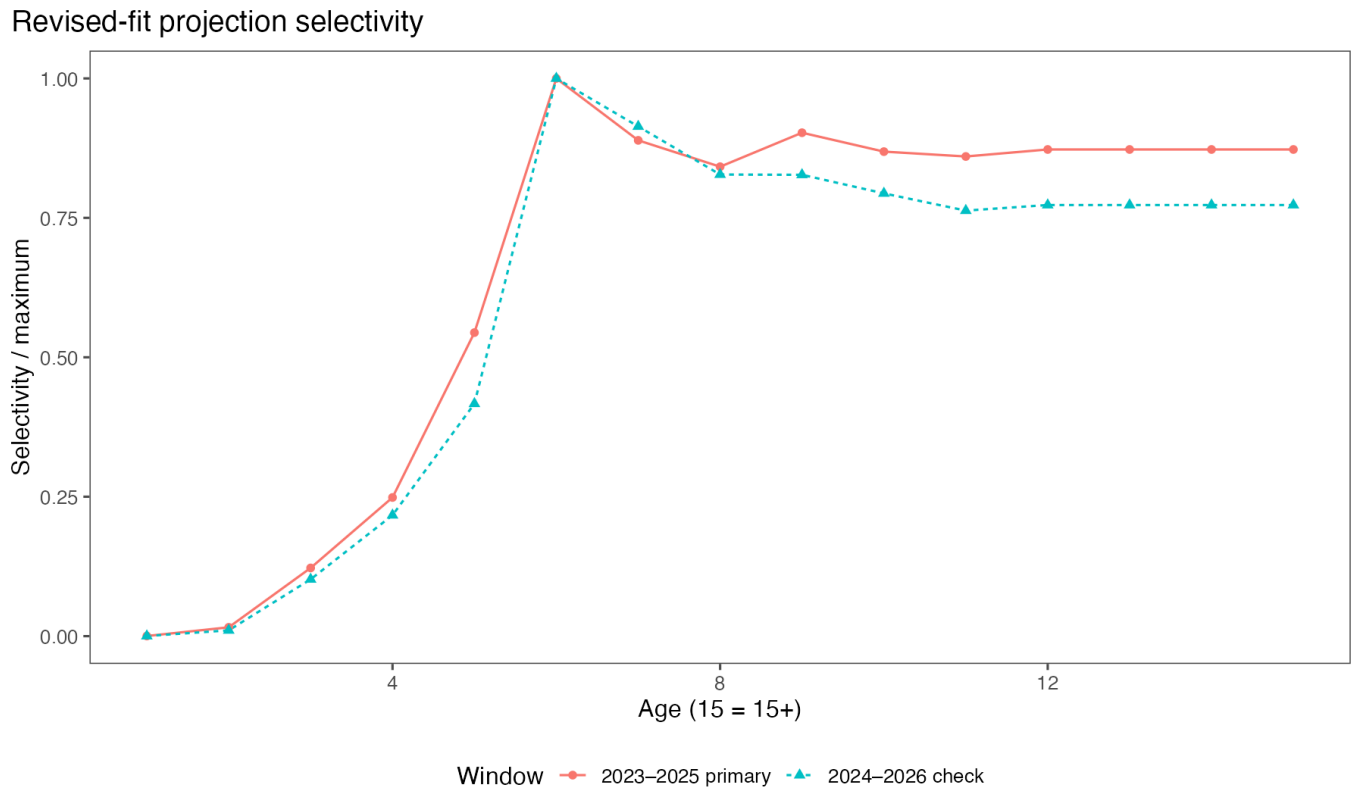
Table D2 Largest Pairwise Correlations Among Model 26.0 Parameters

| Strong parameter correlations | | |
|---|---|---|
| Parameter 1 | Parameter 2 | Correlation |
| P0001: Recruitment level: EBS Pollock; R0 | P0039: Recruitment deviation: EBS Pollock; 2001 | −0.96375 |
| P0001: Recruitment level: EBS Pollock; R0 | P0051: Recruitment deviation: EBS Pollock; 2013 | −0.96370 |
| P0001: Recruitment level: EBS Pollock; R0 | P0047: Recruitment deviation: EBS Pollock; 2009 | −0.96198 |
| P0001: Recruitment level: EBS Pollock; R0 | P0052: Recruitment deviation: EBS Pollock; 2014 | −0.96036 |
| P0001: Recruitment level: EBS Pollock; R0 | P0028: Recruitment deviation: EBS Pollock; 1990 | −0.95676 |
| P1133: Logistic midpoint: Ascending; BTS; Sex combined | P1207: BTS midpoint deviation: Ascending; BTS; Sex combined; 2011 | −0.95670 |
| P1133: Logistic midpoint: Ascending; BTS; Sex combined | P1213: BTS midpoint deviation: Ascending; BTS; Sex combined; 2017 | −0.95529 |
| P0001: Recruitment level: EBS Pollock; R0 | P0040: Recruitment deviation: EBS Pollock; 2002 | −0.95229 |
| P0001: Recruitment level: EBS Pollock; R0 | P0035: Recruitment deviation: EBS Pollock; 1997 | −0.95002 |
| P0001: Recruitment level: EBS Pollock; R0 | P0045: Recruitment deviation: EBS Pollock; 2007 | −0.94802 |

*Note.* Ten largest absolute pairwise correlations among Model 26.0 estimated parameters. IDs link to the complete parameter table; correlations close to plus or minus one identify parameter combinations that change together. These are parameter correlations rather than correlations between survey observations.

# Appendix E
# Proposed Projection Input Curves

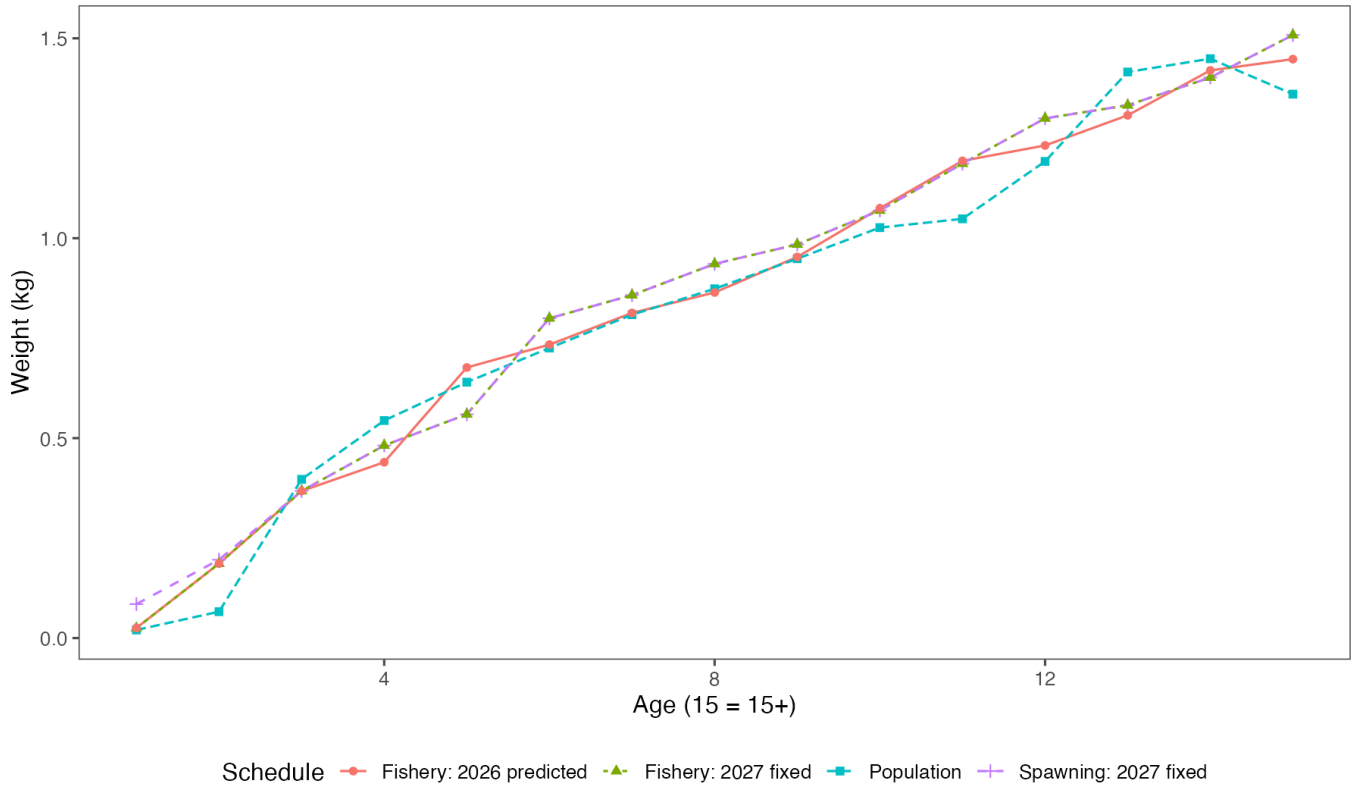
The proposed three-year fishery-selectivity average is compared with the window including the inferred 2026 curve (Figure E1). The technical appendix records their conditional projection effects.




*Note.* Proposed three-year fishery selectivity and the alternative window check. The two curves are distinguished by colour, line type and point shape.

Figure E1  Proposed Fishery Selectivity and Alternative Averaging Window

The projection uses the fixed 2027 fishery forecast for both catch and spawning weights, alongside the population-weight schedule (Figure E2).




Note. Projection weight schedules: the fixed 2027 fishery forecast supplies catch and spawning weights at ages 3–15+; population weights retain the supplied schedule.

Figure E2 Weight Schedules for the Proposed Projection

# Appendix F
# Model Equations

**Scope.** This appendix carries forward the structure and model components of the 2024 *Appendix EBS Pollock Model Description* (SAFE pp. 172–182). The equations describe the Model 23.0 ADMB comparison. The proposed Model 26.0 configuration and its weight schedules are described in Section 5 and Section 4.1.3. The historical Tier 1 material documents an available calculation; the December 2024 SSC adopted Tier 3a.

**Dynamics**

The statistical age-structured model combines the catch equation with annual survival and an accumulating terminal age group (Fournier and Archibald 1982; Hilborn and Walters 1992; Schnute and Richards 1995; McAllister and Ianelli 1997). Let $t$ index years, $a=1,\ldots,A$ index ages, and $A=15$ denote ages 15 and older. $N_{t,a}$ is abundance at the start of the year, $F_{t,a}$ and $M_{t,a}$ are annual instantaneous mortality rates, and $Z_{t,a}$ is their sum. Catch at age and catch biomass are

$$\begin{aligned} Z_{t,a} &= F_{t,a} + M_{t,a}, \\ C_{t,a} &= N_{t,a}\frac{F_{t,a}}{Z_{t,a}}\left(1-e^{-Z_{t,a}}\right), \\ Y_t &= \sum_{a=1}^{A} C_{t,a}\, w^{f}_{t,a},\; p^{f}_{t,a} = \frac{C_{t,a}}{\sum_{b} C_{t,b}}. \end{aligned} \tag{2}$$

Here $w^{f}_{t,a}$ is fishery weight-at-age. Abundance in millions of fish multiplied by kg per fish gives biomass in kt. The population advances as

$$\begin{aligned} N_{t+1,1} &= R_{t+1}, \\ N_{t+1,a+1} &= N_{t,a}\, e^{-Z_{t,a}},\; 1 \le a < A, \\ N_{t+1,A} &= N_{t,A-1}\, e^{-Z_{t,A-1}} + N_{t,A}\, e^{-Z_{t,A}}. \end{aligned} \tag{3}$$

The middle line applies to entry from each younger age; the final line adds surviving members of the plus group. This corrects the age subscript in 2024 Eq. 14.2, which printed $a-1$ on the right side while advancing $a$ to $a+1$ on the left.

Fishing mortality separates annual intensity from selectivity at age:

$$F_{t,a} = s^{f}_{t,a}\mu_F\, e^{\epsilon_{t,a}}. \tag{4}$$

The assessment estimates selectivity coefficients with penalties on changes over time and on curvature across ages (Butterworth et al. 2003). In a general penalty notation, $\Delta_t \log s_{t,a}$ controls annual changes and $\Delta_a^2 \log s_{t,a}$ controls smoothness across ages. Their weights and allowed change years come from the run controls.

The bottom-trawl survey (BTS) uses logistic selectivity above age one and a separate age-one term. The implemented form is

$$\begin{aligned} s^{b}_{t,a} &= \left\{1+\exp\left[-k_t\left(a-a_{50,t}\right)\right]\right\}^{-1},\; a>1, \\ \log s^{b}_{t,1} &= b_1 \exp\left(d_t\right), \\ k_t &= k\exp\left(u_t\right),\; a_{50,t} = \acute{a}_{50}\exp\left(v_t\right). \end{aligned} \tag{5}$$

This follows compute_selectivity() and the age-one assignment in Get_Selectivity(). It replaces the ambiguous slope/intercept and age-one expressions printed in 2024. The 2024 description reported a process variance of 0.04 for an earlier random-walk formulation. The retained controls instead select the penalty on changes in log-selectivity at the

specified ages, plus the separate age-one deviation penalty. Its coefficient is 8 for the squared first differences of $d_t$. The actual controls and source identify which parameter or log-selectivity penalty is applied.

Acoustic-trawl survey (ATS) selectivity is flexible across ages; the optional separate age-one index leaves ages 2–15+ in its composition likelihood. Constant survey catchability and time-varying selectivity represent separate model components.

**Recruitment**

Recruitment is abundance at age one. Female spawning biomass accounts for survival from the start of the year to spawning:

$$B_t=\sum_{a=1}^{A} N_{t,a}\exp(-\tau\, Z_{t,a})\,\phi_a\, w^s_{t,a}, \tau=0.25 \,. (6)$$

Here $w^s_{t,a}$ is spawning weight and $\phi_a$ includes both maturity and the female fraction. The assessment reads a maturity schedule and applies the female fraction internally; the one-sex SPM handoff reads maturity and halves total abundance. Applying the female fraction once preserves the same biomass definition. The spawning time is April 1 (month 4).

Annual recruitment deviations accompany a stock–recruit function, $R_t=f(B_{t-1})\exp(\varepsilon_t)$, with log-scale deviation penalties and the implementation's bias adjustment. The Ricker option selected by SrType=1 is

$$f(B)=\frac{B}{\psi_0}\exp\left[a_R\left(1-\frac{B}{B_0}\right)\right], B_0=R_0\psi_0, a_R=\log\left(\frac{4\,h_{ADMB}}{1-h_{ADMB}}\right). (7)$$

$\psi_0$ is unfished spawning biomass per recruit. The final expression is the source code's transformation of its parameter named steepness; it is retained as $h_{ADMB}$ to identify that convention. The 2024 printed Ricker equation misplaced $R_0$ in the biomass ratio. The expression here follows SRecruit() and Get_Bzero() in the preserved template.

The alternative Beverton–Holt relationship (Francis 1992) is

$$f_{BH}(B)=\frac{B}{a_{BH}+\beta_{BH}B}, a_{BH}=\psi_0\frac{1-h}{4h}, \beta_{BH}=\frac{5h-1}{4hR_0}. (8)$$

For this alternative, $h$ is the fraction of unfished recruitment at 20% of unfished spawning biomass. The 2024 controls fix $\sigma_R=1$ and specify a symmetric beta prior with shape parameters 14.93209877 for the selected stock–recruit parameter. The earlier Beverton–Holt example in the 2024 appendix used a different prior; it remains an alternative rather than the selected 2024 setting.

**Diagnostics**

The historical replay diagnostic retains fitted recruitment anomalies while adjusting recruitment for the change in spawning biomass under a hypothetical history with fishing set to zero:

$$R_t'=\hat{R}_t\frac{f(B'_{t-1})}{f(B_{t-1})}. (9)$$

Retrospective analyses repeat estimation after removing recent years and compare estimates for common years (Parma 1993; Ianelli and Fournier 1998). Their fit controls and terminal years accompany each saved run.

The effective number of ages contributing to spawning biomass is the exponential of Shannon entropy (Jost 2006):

$$D_t=\exp\left(-\sum_a p^B_{t,a}\log p^B_{t,a}\right), p^B_{t,a}=\frac{N_{t,a}e^{-\tau Z_{t,a}}\phi_a w^s_{t,a}}{B_t}. (10)$$

The mean-age diagnostic uses observed and predicted composition means and the predicted variance at age (Francis 2011):

$$\bar{a}_t=\sum_a a\, p_{t,a},\ \hat{\bar{a}}_t=\sum_a a\, \hat{p}_{t,a},$$
$$V_t=\sum_a (a-\hat{\bar{a}}_t)^2\, \hat{p}_{t,a}, \qquad (11)$$
$$z_t=\frac{\bar{a}_t-\hat{\bar{a}}_t}{\sqrt{V_t/n_t}},\ \nu=\{Var_t(z_t)\}^{-1}.$$

The implied adjusted sample size is $\nu\, n_t$. This diagnostic supports evaluation of weighting; changing assessment inputs requires a recorded decision. The expression for $V_t$ replaces the malformed $\varkappa_t$ formula in the printed 2024 appendix.

**Parameter estimation**

The objective adds negative log likelihoods, prior densities and selectivity penalties. Phase-based estimation introduces parameters in stages. For age compositions, the multinomial contribution, up to a data-only constant, is

$$\ell_{age}=-\sum_t n_t \sum_a p_{t,a}\log(p_{t,a}+c),\ p_t^{read}=\hat{p}_t^{true}\, E. \qquad (12)$$

$c$ is the small numerical constant in the implementation. When enabled, $E_{a,b}$ is the probability that true age $a$ is read as age $b$. The selected historical controls have use_age_err=0. The original appendix also described a robust composition option (Fournier et al. 1990), with variance inflation and a small outlier component. In the retained 2024 controls its activation phase is 1350; the six-phase completed comparison uses the multinomial form. This distinction prevents the general software description from implying that the robust option was active.

For survey $s$ at within-year time $\tau_s$, expected abundance and biomass at age are

$$\hat{N}^s_{t,a}=q_s\, s^s_{t,a}\, N_{t,a}\, e^{-\tau_s Z_{t,a}},\ \hat{I}^s_t=\sum_a \hat{N}^s_{t,a}\, w^s_{t,a}. \qquad (13)$$

An abundance index omits the weight multiplier. The original description used midyear survey timing; actual run inputs specify the timing used in each comparison. The AVO index shares ATS selectivity and its documented weight basis, while retaining a separate catchability coefficient.

The independent lognormal-index and multivariate BTS contributions have the forms

$$\ell_s=\frac{1}{2}\sum_t \frac{[\log I^{obs}_{s,t}-\log \hat{I}_{s,t}]^2}{\sigma^2_{s,t}},\ \ell_{BTS}=\frac{1}{2} r^T \Sigma^{-1} r,\ r_t=\log I^{obs}_t-\log \hat{I}_t. \qquad (14)$$

Fixed-variance normalization terms are constant for these comparisons. The normal-index option described in 2024 instead uses residuals on the original scale. The BTS covariance provides $\Sigma$ for the multivariate lognormal fit. Catch biomass enters the implemented objective as $\lambda_C \sum_t [\log(Y^{obs}_t+c)-\log(\hat{Y}_t+c)]^2$, with $\lambda_C=200$ in the retained controls. This equals $1/(2\sigma_C^2)$ for $\sigma_C=0.05$ and removes the extra factor of two in the printed 2024 likelihood expression. Priors and penalties retain their actual source-code weights.

**Uncertainty in mean body mass**

The 2024 appendix described estimating future weights within the assessment to propagate uncertainty into derived quantities (Ianelli and Williamson 2001). Its earlier option used $w_a^{future} \sim N(\bar{w}_a, \sigma^2_{w,a})$. The retained implementation fits a mean growth curve plus cohort and year effects. For starting modeled age $a_0$, its mean curve is

$$w_a = a_w \left\{ L_1 + (L_2 - L_1) \frac{1 - K^{a - a_0}}{1 - K^{A-1}} \right\}^3. \quad (15)$$

At the final estimation phase, the recurrence in Est_Fixed_Effects_wts() is

$$\begin{aligned} \hat{w}_{t,a_0} &= w_{a_0} \exp(\sigma_c^2/2 + \sigma_c c_t), \\ \hat{w}_{t,a} &= \hat{w}_{t-1,a-1} + (w_a - w_{a-1}) \exp(\sigma_y^2/2 + \sigma_y y_t), a > a_0. \end{aligned} \quad (16)$$

The initial year starts at $w_a$; dataset-specific scale vectors map the latent weights to observation series. This notation corrects the increment index in 2024 and explicitly retains the implementation's sigma scaling and final-phase offsets. The cohort and year deviations receive standard-normal penalties. The four-term weight objective is

$$\ell_w = \frac{1}{2} \sum_{h,t,a} \frac{(w_{h,t,a}^{obs} - w_a)^2}{\sigma_{h,t,a}^2} + \frac{1}{2} \sum_{h,t,a} \frac{(w_{h,t,a}^{obs} - \hat{w}_{h,t,a})^2}{\sigma_{h,t,a}^2} + \frac{1}{2} \sum_t c_t^2 + \frac{1}{2} \sum_t y_t^2. \quad (17)$$

The 2026 numerical baseline allocates all four terms and retains their sum.

**Tier 1 projections: historical calculation**

The 2024 description retained a Tier 1 calculation based on fishable biomass, the harmonic-mean harvest rate at MSY, and a spawning-biomass adjustment:

$$\begin{aligned} ABC_t &= B_{GM,t}^f u_{HM} \zeta_t, \\ B_{GM,t}^f &= \exp\{\log \hat{B}_t^f - \sigma^2_{\log B^f}/2\}, \\ u_{HM} &= \exp\{\log u_{MSY} - \sigma^2_{\log u}/2\}, \\ \zeta_t &= \begin{cases} (B_t/B_{MSY} - 0.05)/0.95, & B_t < B_{MSY}, \\ 1, & B_t \geq B_{MSY}. \end{cases} \end{aligned} \quad (18)$$

The geometric and harmonic adjustments assume the documented lognormal approximations. This historical formula describes the earlier calculation over its applicable biomass range; final advice uses the adopted tier and the complete corresponding harvest-control rule.

**Tier 3 projections**

For fixed mortality, maturity, weights and selectivity, equilibrium spawning biomass per recruit at fishing intensity $F$ is

$$\begin{aligned} l_1(F) &= 1, l_a(F) = l_{a-1}(F) e^{-M_{a-1} - F s_{a-1}}, 1 < a < A, \\ l_A(F) &= \frac{l_{A-1}(F) e^{-M_{A-1} - F s_{A-1}}}{1 - e^{-M_A - F s_A}}, \\ \psi(F) &= \sum_a l_a(F) \phi_a w_a^s e^{-\tau(M_a + F s_a)}, \\ B_{100\%} &= R\psi(0), B_{x\%} = \frac{x}{100} B_{100\%}, \frac{\psi(F_{x\%})}{\psi(0)} = \frac{x}{100}. \end{aligned} \quad (19)$$

Tier 3a uses $F_{40\%}$ for maximum ABC and $F_{35\%}$ for OFL, with the adopted biomass-dependent rule applied when needed. Projection catch follows Equation 2. The 2024 SPM calculation used one sex category split equally into females and males, fixed biological schedules and selectivity, and stochastic recruitment generated from the historical arithmetic and harmonic means with an inverse-Gaussian distribution. Its 1,000 simulations retained recruitment process variability; uncertainty in fitted abundance, mean recruitment, selectivity and weights was omitted. The saved seed is 123 in the historical SPM source.

The recovered 2024 handoff uses five preceding selectivity years (2019–2023), April spawning, the terminal spawning weights and fitted future fishery weights. SPM averages age-one recruitment over 1978–2023; the assessment's $B_{100\%}$ calculation includes 2024. These different recruitment means produce different reference-point values. The selected scenario fixes catches at 1,350 kt in both 2025 and 2026. Its archived simulation means provide the historical forecast comparison.

# Appendix G
# FT-NIRS Age Prediction for EBS Pollock

Authored by: Derek Chamberlin, Esther Goldstein, Beth Matta, and Jason Conner

FT-NIRS Team: John Brogan, Jason Conner, Derek Chamberlin, Esther Goldstein, Brenna Hsieh, Beth Matta, Jon Short

FT-NIRS predicts age from otolith spectra and associated specimen information, using traditional microscopic ages for calibration and validation. The workflow proceeds from age prediction through tester-age conversion to the assessment likelihood (Table G1). The 2024 and 2025 fishery age compositions use their respective independent tester-age keys applied to released FT-NIRS predictions. The 2025 source is model a00-d3. The year-specific keys also convert biological weight inputs (Table G2). Both composition series use the assessment's identity ageing-error assumption after their selected age treatment.

Year-specific conditional keys redistribute source abundance and biomass among tester ages. Mean weights are then calculated as converted biomass divided by converted abundance, with terminal ages pooled using those totals. The partial 2024 fishery diagnostic retains unsupported source-age-13 mass (Table G3). Selected assessment weights fill that row using pooled age offsets and assign a 100% CV to the target-age-15+ weight observation. The 2025 conversion uses 668 independent tester pairs and fills the unsupported source-age-13 row with the midpoint of the age-12 and age-14 conditional probabilities. The same key supplies the 2025 age composition and weight observations (Table G4).

BTS 2025 and 2026 use separate tester-age keys. The 2026 key retains 307 pairs after excluding 22 flagged pairs, with exact TMA ages 1–18 converted before ages 15–18 are pooled into 15+. Total abundance and biomass are conserved. Converted weight SEs propagate the saved joint abundance–weight covariance (Table G5). The fitted weight model retains its existing marginal observation likelihood, and empirical-key uncertainty is excluded. Numerical checks distinguish fitted use from the retained full-pair conversion diagnostics (Table G6).

Provisional 2026 ATS ages use the BTS age–length key (ALK) applied to ATS population length-frequency estimates, followed by conversion to the TMA basis. FT-NIRS ages from the 2026 ATS samples will be available in 2027. Unsupported lengths remain unallocated. The resulting weights also supply the 2026 AVO predictions.

**Fourier transform near-infrared spectroscopy (FT-NIRS) Overview**

Fourier transform near-infrared spectroscopy (FT-NIRS) has emerged as an effective secondary age prediction tool and has successfully been implemented in production ageing settings (Matta and Helser, 2024), with FT-NIRS predicted ages recently being used to produce stock assessment age composition data (Dick et al., 2025; Ianelli, 2025). FT-NIRS is a non-destructive technique that interrogates a sample using light in the near-infrared region of the light spectrum (typically 4,000 to 12,000 $cm^{-1}$).

Scanning with FT-NIRS produces a spectral signature of the sample based on the absorption of near infrared light by different chemical bonds, such as C-H (aliphatic), C-H (aromatic), C-O (carboxyl), O-H (hydroxyl), and N-H (amine and amide) (Murray and Williams, 1987). For otoliths, these spectra capture changes in the organic protein matrix (otolin) relative to the inorganic calcium carbonate structure. This relationship correlates strongly with fish growth and age (Passerotti et al., 2022) and has been used to develop models, calibrated with traditional microscopic ages (TMA) and FT-NIR spectra, to predict the age of unknown samples based on their spectral signatures with an estimated time-cost saving of 23-40% (Adamack et al., 2025). FT-NIRS has been applied to estimate age for a multitude of fish species from Australia (Wedding et al., 2014), the Mediterranean Sea (Basilone et al., 2024), the eastern Bering Sea (Benson et al., 2023), and the Atlantic Ocean (Adamack et al., 2025).

**Traditional Microscopic Ageing (TMA) QA/QC**

The FT-NIRS models rely on traditional microscopic ageing (TMA) to calibrate the model, thus the accuracy of the age predictions is directly impacted by the accuracy of the calibration ages (Chamberlin et al., 2026). The AFSC Age and Growth Program (AGP) has a long history of ageing walleye pollock by microscopic examination of otolith growth patterns (Matta et al., 2026) and has employed a rigorous quality control program since 1983 (Kimura and Anderl, 2005). In TMA of pollock, presumed annual marks ("annuli") are enumerated from the otolith surface in younger fish or from burned or toasted otolith cross sections in older fish (Matta and Kimura, 2012; Matta et al., 2026). The accuracy of the

AGP's TMA for pollock has been independently corroborated using a variety of methods, including radiometric ageing, edge and marginal increment analysis, year-class tracking, and captive-rearing studies (Kastelle and Kimura, 2006; Kimura et al., 2006; Kimura et al., 2007; Lai, 1985).

To test FT-NIRS model prediction skill, random 20% subsamples of pollock otoliths from each collection source (fishery, bottom-trawl survey, and hydroacoustic midwater trawl) were subjected to TMA. Spectral outliers (i.e., those that predicted fish lengths outside an established threshold) were also selected for TMA; see FT-NIRS QA/QC below for more details. Each otolith selected for TMA was independently aged by two experienced readers. The TMA subsamples were evaluated for inter-reader precision and bias. All discrepancies between the two readers were resolved. Any TMA age-length outliers were reexamined, and re-aged or recorded as collection errors when appropriate (Kimura and Anderl, 2005).

**FT-NIRS QA/QC**

In addition to the TMA QA/QC, the FT-NIR spectra underwent a three-step QA/QC process to identify poor quality spectra due to data collection errors, instrument issues, or otolith irregularities such as crystallization, contamination, or fragmentation (Goldstein et al., 2024). Spectrometer operators conducted qualitative QA/QC concurrent with data collection to identify missing data and unusual spectra using length predictions from a species- and region-specific partial least-squares regression (PLSR) model in comparison to observed lengths. Real-time length predictions were used to identify transcription errors and provide assurance of proper instrument function.

Following data collection, rigorous QA/QC was conducted to identify outliers using principal component analysis (PCA) and PLSR models developed from a multi-year collection of spectral data from each data source (AFSC groundfish survey, AFSC hydroacoustic survey, fishery data) and projections (PCA) and predictions (PLSR) of new data. PCA scores by scan date were used to visually identify instrument drift. Gross spectral data outliers were identified from PCA using Hotelling's $T^2$ and Q residuals (Pomerantsev and Rodionova, 2014) and PLSR outlier analyses identified discrepancies between spectra and specimen length by combining Hotelling's $T^2$ and Q residuals (full X-distance) and Y-distance (observed vs. predicted; Rodionova and Pomerantsev, 2020). Analyses used a threshold of 1% to identify outliers. Outliers were then aged using traditional microscopic methods. Once aged, outlier spectra were excluded from the final dataset if age predictions from the MMCNN models were also outside the MMCNN 95% Monte Carlo dropout prediction intervals discussed below.

**FT-NIRS Model**

Age was predicted from FT-NIR spectra using a multimodal convolutional neural network model (MMCNN). MMCNNs identify non-linear relationships in spectral data, incorporate additional inputs such as fish length or capture location, and learn relevant spectral features during training. Classical chemometric approaches such as PLS require the user to select key wavenumbers (Chen and Wang, 2019). The MMCNN used here is an updated version of the MMCNN presented in Benson et al. (2023) and Chamberlin et al. (2026). The MMCNN includes one convolutional layer and six dense layers to extract information from spectral and associated fish length data to predict age.

The first layer of the MMCNN is the spectral input data which is fed to a 1D convolutional layer, which acts as an automated feature extractor to identify patterns in the spectra. This output is then processed with a Rectified Linear Unit (ReLU) activation function to introduce nonlinearity into the network necessary for approximating continuous functions (Brownlee, 2019; Géron, 2019). The data are then flattened into a one-dimensional array and passed through a dense layer (all dense layers use Exponential Linear Unit (ELU) activation). This reduces the dimensionality of the spectral input and balances its contribution relative to the metadata. The metadata input is fed to a dense layer, then concatenated with the spectral input layer. The output is passed through a fully connected dense layer with ELU activation to model the interactions between spectral data and metadata. To mitigate overfitting, 5 dropout layers follow each dense layer to randomly ignore a subset of neurons and prevent codependency between nodes (Srivastava et al., 2014). Crucially, these dropout layers remain active during the inference phase, a technique known as Monte Carlo (MC) dropout, allowing the model to perform multiple forward passes per sample to generate a distribution of age estimates and calculate Monte Carlo prediction intervals. Finally, a fully connected dense layer with a single node and a batch normalization layer are used to produce a continuous age estimate, allowing for loss estimation via a Mean Absolute Error (MAE) function. The model's hyperparameters (the number of filters in the convolutional layer, the number of nodes in the concatenated layer,

learning rate, batch size and the dropout rate) were optimized using hyperband optimization (Li et al., 2018) with early stopping.

The walleye pollock model is trained on spectra and TMA ages from 2014-2025 using an "n−1" annual model updating approach. Under this scheme, spectral and age data from all fleets up to the year prior to the target prediction year are included (for example, predicting 2025 fishery ages uses data from 2024 and earlier). For MMCNN training, input data are first split into development (80%) and testing (20%) pools using a randomly shuffled split that is stratified on fleet and year. Samples receive a normalized weighting score based on 2 coefficients: *decay* is an integer [1, 2, 3] that defines the rate of exponential decay for samples based on year, and *alpha* [0, 0.1, 0.2, ..., 1.0] defines whether sample year or sample age frequency has more influence in weighting the sample, with *alpha* = 1 turning decay off. This allows for testing of whether model performance improves when more recent calibration data or rare age classes are weighted more heavily. For hyperparameter optimization, the development data are further split into training (90%) and validation (10%) datasets to inform early stopping, using the previously defined sample weights. For each scenario a series of 10 trainings, with random seeds, are run to estimate the median number of epochs needed to train the model but avoid overfitting, prior to a final training run based on the median number of epochs.

In total, 31 training models were produced with a grid search for the 2 sample weighting coefficients, and age predictions were generated for bottom trawl survey, hydroacoustic survey and fishery fleets. The final candidate model for each fleet was selected using four statistical metrics: Lin's concordance correlation coefficient (Lin's CCC), the mean Winkler score, $R^2$, and the slope of the regression between FT-NIRS predicted ages and corresponding TMA ages. Lin's CCC measures how closely the observed data aligns with a 45° line of perfect agreement, while the mean Winkler score evaluates average performance based on the width of the Monte Carlo prediction intervals and the frequency of samples falling outside them. The four metrics capture different aspects of model accuracy, and visual assessments complemented them. Ultimately, the best-performing model was identified by synthesizing these visual and quantitative criteria. Age predictions were then finalized by removing any samples that were both flagged as a spectral outlier and an outlier from the Monte Carlo prediction intervals. This finalized dataset was then transmitted to end users and coded as FT-NIRS predicted ages.

**Age conversion and assessment use**

The assessment uses year-specific independent tester TMA conversions for the 2024 and 2025 fishery compositions, matching annual weight conversions and separate survey-specific age conversions (Table G1). These treatments serve different purposes: predicting age from spectra, allocating abundance and biomass among ages, and specifying the assessment likelihood are separate steps. Current input and fit records establish their use; the retained partial-conversion diagnostics describe their original support and uncertainty limits.

The 2026 BTS conditional tester-age key excludes calibration pairs marked nir_outlier = Y, removing 22 of 329 complete pairs and retaining 307. It targets independent test_age conditional on FT-NIRS final_nir_age; each FT-NIRS row is normalized by its retained paired count. Conversion uses exact TMA ages 1–18 before pooling ages 15–18 into 15+. Screening applies to calibration pairs, preserving the released abundance vector. All unflagged complete pairs are eligible, including checks outside the systematic 20% sample.

Screening changed the exact mean converted age from 5.318978 to 5.316448 years, with a largest age-proportion change of approximately 0.124 percentage points and unchanged total abundance. The same 307-pair key supplies BTS weight means and their joint-covariance delta-method SEs, provisional ATS age allocations and biomass-derived weights, and the 2026 AVO prediction weights. The standalone RE-WAA refit and assessment fits using these inputs are complete S10. The 2025 BTS key is unchanged. The original 329-pair products remain historical evidence. Empirical-key uncertainty is excluded; the fitted weight model retains its existing marginal observation likelihood rather than using the full converted covariance as a new likelihood.

The contribution describes **31 MMCNN training candidates**, distinguished by sample-weight settings and fleet-specific selection. PCA and PLSR provide spectral QA/QC; PLSR also provides length predictions. The a00-d3 and a03-d1 labels on the saved 600-fish comparison identify prediction sets, separately from the assessment's cumulative 0.x input sequence.

Table G1 FT-NIRS Use and Transformations by Data Source and Year

| Gear or survey | Collection year | Input or model version | Age basis and year-specific key | Application and verified status |
|---|---|---|---|---|
| Commercial fishery, pelagic trawl (NORPAC gear 2) | 2024 | Saved sampler_2024_2025_20261016T160737Z; current fishery CAA | Released FT-NIRS ages; distinct 2024 key recovered from the earlier release: 2,985 released records and 693 independent tester pairs. Specimen/age-field hashes match the original-release verification record. | The ordinary assessment uses the independent tester TMA composition key, with the source-age-13 row completed from pooled 2024 source-age offsets at ages 9–12 (34 pairs) and likelihood N=578. The selected weight input uses the same key and a 100% CV for the target-age-15+ observation. The retained partial weight diagnostic describes its historical supported-only scope. The accepted marginal Model 26.1.2 fit uses the same year-specific composition and weight inputs. S2, S3, S6, S10. |
| Commercial fishery, pelagic trawl (NORPAC gear 2) | 2025 | FT-NIRS model a00-d3; annual catch-at-age expansion | Independent tester TMA ages; year-specific conditional key from 668 complete pairs. The unsupported source-age-13 row is the midpoint of the age-12 and age-14 rows. | Converted proportions use ages 1–14 and 15+ with selected likelihood N = 530 and identity ageing error. Total numbers are preserved. The matching weight observations use the same key. Key and interpolation uncertainty are additional to sampling uncertainty. S1, S2, S3, S10. |
| Commercial fishery, pelagic trawl | 2025 | Matching annual weight observations | Same independent tester-age key and source-age-13 midpoint. Source age 1 has zero abundance. | The key redistributes abundance and biomass in each of 1,000 saved bootstrap draws; weight ratios and SDs are calculated from the converted quantities. These observations supply the selected fitted weights. Target ages 13 and 14 have no converted abundance. Empirical-key and interpolation uncertainty are excluded. S3, S10. |
| Bottom-trawl survey (BTS), EBS/NBS | 2025 | Revised tinyVAST_abundance.csv / tinyVAST_props.csv; saved bts_2026_candidate lineage | FT-NIRS age estimates; bts_2025 independent tester key, dated 2025-09-24, 390 pairs. | Converted abundance and age proportions are used in the working assessment. Exact TMA ages 1–18 are converted before ages 15–18 are pooled into 15+. This key transfer supplies no additional abundance SE or empirical-key uncertainty. The earlier candidate exclusion is retained only in the historical staging record. S1, S4, S6, S10. |

| Gear or survey | Collection year | Input or model version | Age basis and year-specific key | Application and verified status |
|---|---|---|---|---|
| Bottom-trawl survey (BTS), EBS/NBS | 2026 | Revised spatial estimates; screened 29 October rerun input | Independent test_age conditional key from the 2026-09-24 release: 329 complete pairs before screening, 307 after excluding 22 nir_outlier = Y pairs. | Screened key applied to working CAA and WAA inputs. Exact TMA ages 1–18 are converted before 15+ pooling; source abundance, sample-size choices and 2025 key are retained. Original full-pair products are retained unchanged. S1, S4, S10. |
| Bottom-trawl survey, Indivero/VAST weight products | 2025 and 2026, converted separately | Saved VAST joint N/W covariance; 29 October screened-key preparation and standalone RE-WAA refit | The 2025 full tester key is retained; 2026 uses the screened 307-pair key with the matching VAST abundance and weight estimates. | Ratios and delta-method SEs use abundance–weight and cross-age/year covariance under the same method. Nine 2026 observation means and nine SEs change; 2025 observations remain unchanged. The standalone RE-WAA numerical checks pass. Empirical-key uncertainty is excluded. S3, S5, S10. |
| BTS historical age-comparison material | 2014–2018 | FTNIR Compare.qmd uses historical fishery/survey paired releases | Historical comparison data span 2014–2018. The new contribution describes training data spanning 2014–2025 with annual n−1 updating. | These series use different data updates. The historical comparison files alone do not establish the precise training set/model ID used for each 2025/2026 release; do not transfer old keys to new years. S7, supplied contribution above. |
| Acoustic-trawl survey (ATS), acoustic expansion with trawl biological samples | 2024 | ATS_2024_age comp ALK.xlsx, length totals × age-length proportions | The preparation record quotes the supplied 2025-09-08 email as TMA ages sent to MACE. The workbook has a 2024 age-length pivot and matrix multiplication of expanded numbers/biomass. | This local workbook verifies a length-to-age step, not an automatic FT-NIRS designation. A separate ATS age-age conversion and exact predictive model/key lineage are unverified here. S3, S8. |
| Acoustic-trawl survey (ATS) | 2026 | ATS/ stock_assessment_output/ *202604.csv; 29 October provisional rerun products | Released BTS FT-NIRS length key followed by the screened 307-pair BTS conditional tester-age key; source FT-NIRS allocations and unsupported lengths are preserved. | Abundance and biomass are redistributed with the same key; pooled ratios supply ATS and 2026 AVO prediction weights. Provisional estimates are used in the completed working fits and await direct ATS age-composition data in 2027. Transfer assumptions and uncertainty scope remain unchanged. S9, S10. |

*Note.* FT-NIRS use and transformations by collection and year. Current fitted applications are distinguished from retained historical preparation diagnostics. Unfiltered counts describe the original complete released tester pairs; the 2026 screened input excludes the 22 flagged complete pairs.

The original keys contain 693 and 668 pairs for fishery 2024 and 2025, and 390 and 329 for BTS 2025 and 2026 before screening. The current 2026 BTS key retains 307 pairs. The fishery gear label follows the saved NORPAC records: gear 2 is PELAGIC in both 2024 and 2025. The contribution's fishery, bottom-trawl survey and hydroacoustic categories retain their own age basis and sampling conditions.

**Predictive model and assessment input differences**

The MMCNN combines spectral and fish-length information, MC dropout prediction intervals, hyperband optimization and 31 sample-weight settings. Its annual n−1 updating rule uses information available before the prediction year; for

example, a 2025 prediction uses data through 2024. The selected fleet-by-year model checksum, coefficient setting and exact mapping to the saved 600-fish prediction files remain to be supplied.

The full tester-key release uses final_nir_age as predictor and independent test_age as target, excluding resolved final_age as the calibration target. It contains 3,000 fishery-2025 records, 1,638 BTS-2025 records and 1,586 BTS-2026 records; 2,992, 1,619 and 1,582 have released FT-NIRS ages. Their complete-pair counts are 668, 390 and 329, respectively; the screened 2026 BTS application retains 307 S1.

For the 2025 fishery, model a00-d3 FT-NIRS ages are converted to independent tester ages using the 668 complete paired observations. The conditional probability of tester age $j$ given FT-NIRS age $i$ is $K_{ij}=n_{ij}/\sum_j n_{ij}$. Converted numbers are $N_j^{TMA}=\sum_i N_i^{FTNIR}K_{ij}$. Empirical rows retain the released paired observations, including flagged pairs. The unsupported source-age-13 row is $K_{13,j}=(K_{12,j}+K_{14,j})/2$, giving probabilities of one third at tester age 10, one sixth at age 11 and one half at age 15. The age-12 row has six pairs and the age-14 row one pair, so this interpolation has limited direct support. Source and tester ages in the release are exact ages 2–15; the assessment terminal group contains ages 15 and older. The selected likelihood sample size is 530. The key is treated as fixed; empirical-key and interpolation uncertainty are excluded.

The 2024 fishery composition uses its 693-pair independent tester key. The unsupported source-age-13 row uses pooled TMA-minus-FT-NIRS offsets from the same year's source ages 9–12 (34 pairs). Empirical rows remain unchanged; the selected likelihood input is 578. Both recent years are converted before fitting and use a 15-by-15 identity ageing-error matrix, equivalent to the ADMB setting with ageing error switched off. Key and tail-allocation uncertainty are additional to the sampling precision represented by the selected likelihood weights.

Weight conversion applies the same conditional key to source numbers and biomass, then divides converted biomass by converted numbers. Terminal-age weights use pooled totals. The partial empirical 2024 diagnostic leaves unsupported source-age-13 mass unallocated. The selected weight treatment fills that row using pooled offsets from source ages 9–12 and assigns a 100% CV to the target-age-15+ observation. Those fixed approximations add no empirical calibration pairs; their uncertainty differs from a sampling SE. The 2025 weight observations use the same source-age-13 midpoint as the age composition. Their zero-abundance source-age-1 padding contributes no converted numbers or biomass. The midpoint is specific to 2025; source age 14 also lacks pairs in the 2024 key.

The older staging record S6 describes a 2024 endpoint in pm_26_candidate.dat: fishery CAA at 0.4, provisional BTS WAA/biomass at 0.5/0.6, ATS at 0.7, and separate AVO-only 0.8/0.9 components. These are historical input-sequence labels from the incremental-model plan. Current fitted use follows the documented working input and run records, not those earlier staging exclusions.

**Source and validation limits**

The partial 2024 empirical diagnostic retains its unsupported source-age-13 mass. Selected assessment weights use the pooled-offset rule and 100% CV described above. The 2024 and 2025 compositions use their respective tester-age keys and selected support rules. Weight SEs propagate the available joint abundance–weight covariance, while the fitted model retains its marginal observation likelihood. Empirical-key uncertainty remains excluded.

The assessment author identified a00-d3 as the selected 2025 fishery model. The exact training-run checksum, coefficient setting and complete collection-by-year release record remain to be documented. The 31 predictive candidates and the assessment's cumulative input stages identify different choices. The earlier mixed-basis BTS workbook remains unavailable; the separately identified current workbook is excluded from assessment assembly. Its source basis must be established before using it in a comparison.

Provisional ATS products retain their transferred BTS age–length and tester-age keys, unsupported-length limits and separate sampling conditions. Direct ATS age and weight validation remains open, with age-composition data expected in 2027. Archived full-pair calibration products remain available alongside the screened-key input records. Prior fit, projection and posterior records retain their original dates and input identities. Final model and recommendation selection remain separate review decisions.

**Sources and reproducibility**

The paired-age releases, preparation products and current run records retain their source identities. Detailed file locations, checksums and the original preparation narrative are preserved in the technical source record.

**S1 — Paired tester-age release**: the full year-specific tester-key workbook and tidy release, with independent tester ages as the calibration target.

**S2 — Fishery observations**: catch-expanded annual numbers based on released FT-NIRS ages, with year-specific tester-age conversions: 693 pairs and N=578 for 2024; 668 pairs and N=530 for 2025.

**S3 — Weight conversion**: year-specific fishery conversion keys, partial 2024 diagnostics and independent fishery arithmetic checks.

**S4 — Survey age compositions**: revised BTS abundance and proportion estimates, with their source age assignments.

**S5 — Survey weights and covariance**: matching model-based abundance and weight estimates, joint covariance and delta-method validation.

**S6 — Historical staging**: the earlier preparation sequence, retained separately from current selected input and fit records.

**S7 — Earlier age comparison**: fishery and survey comparisons for 2014–2018, distinct from later training and prediction releases.

**S8 — ATS length-to-age calculation**: the 2024 age–length workbook and expanded number/biomass calculations.

**S9 — ATS survey products**: regional survey length and biomass products used for the provisional BTS-key transfer.

**S10 — Current screened-key inputs and fits**: the 307-pair calibration, completed standalone RE-WAA refit and current assessment run records. The original 329-pair products retain their historical identity.

**Year-specific weight-conversion diagnostics**

The retained full-pair conversion record separates three completed year-specific conversions from the partial 2024 fishery diagnostic (conversion-status table). Fishery 2025 and BTS 2025–2026 use independent tester-age keys; positive unsupported source-age-13 mass remains unallocated in fishery 2024.

Mean weights generally increase with age, with the largest source-versus- converted differences concentrated at older ages (2024 fishery weight table). The partial fishery 2024 values at TMA ages 13 and 14 are both 1.2878 kg, while its target 15+ weight remains missing. Its prepared older-age SEs are large relative to the means; the partial converted SDs condition on the fixed key and supported source ages.

The 2025 fishery conversion raises the 15+ mean from 0.8102 to 1.2676 kg; target ages 13 and 14 retain missing weights because converted abundance is zero (fishery weight comparison). Thus the older-age differences include changes in support as well as in ratios.

BTS 2025 converted weights are generally below the source weights from age 5 upward; in 2026, weights at ages 3–8 remain close to or above their source values, while older-age weights are lower (BTS weight comparison; Figure G1). The 15+ mean changes from 1.7646 to 1.5474 kg in 2025 and from 1.8390 to 1.3665 kg in 2026. These are conversion contrasts that vary by age and year. The smaller conditional SEs in several converted cells retain the stated fixed-key uncertainty scope.

Independent fishery arithmetic, preservation of earlier fishery values and survey covariance/derivative checks establish computational consistency (conversion checks). The partial 2024 diagnostic leaves unsupported source-age-13 mass unallocated. Current selected weights use the separate pooled-offset tail treatment and 100% CV, and the screened-key standalone RE-WAA and assessment fits are complete. These distinctions preserve the historical diagnostic's support limits and the current fit's fixed-key uncertainty scope.

**Adjustments to FT-NIRs for assessment application**

The following tables retain the full-pair calibration used to evaluate conversion arithmetic, including the partial 2024 fishery results. Current assessment inputs use the separately documented selected tail treatment and screened 2026 key. The table values preserve their original calibration basis.

Complete conversion report, including Table 3 for fishery 2024–2025

**Diagnostic basis and current use**

The retained diagnostics convert fishery **2025** and BTS **2025–2026** abundance and biomass with their year-specific independent tester-age keys. Fishery **2024** has a distinct 693-pair key, but its empirical partial conversion leaves positive source-age-13 mass unallocated. The selected assessment weight treatment fills that row using pooled offsets from source ages 9–12 and assigns a 100% CV to the target-age-15+ weight observation. The current standalone RE-WAA refit and assessment fits are complete; the historical partial diagnostic remains separate. The BTS tables below retain the original full-pair calibration, while the current 2026 fit uses the screened 307-pair key.

Table G2 Conversion Status by Data Source

| Stream | Conversion key | Current action |
|---|---|---|
| Fishery 2024 | Distinct 2024 key: 693 pairs | Partial diagnostic; source age 13 unallocated |
| Fishery 2025 | 2025 fishery tester-age key: 668 pairs | Converted with two approved probability rules |
| BTS 2025 | 2025 BTS tester-age key: 390 pairs | Converted with joint model uncertainty |
| BTS 2026 | 2026 BTS tester-age key: 329 pairs | Converted with joint model uncertainty |

*Note.* Conversion status by new data source.

**Conversion and approved approximations**

Each key gives the distribution of TMA tester age conditional on FTNIR age. Abundance and biomass are redistributed together, using the same conditional probabilities. Converted mean weight is converted biomass divided by converted abundance. Within a source FTNIR bin, this calculation assigns the same source mean weight to every target age.

For fishery 2025, Jim approved two rules for unsupported source ages:

> **Age 1:** a dummy identity row assigns FTNIR age 1 to TMA age 1. All saved source age-1 catch estimates are zero, so this row retains zero catch.
>
> **Age 13:** each target-age probability is the arithmetic average of its probabilities in the source age-12 and age-14 rows. The resulting row sums to one. This approximation uses the existing age-12 and age-14 pairs.

Converted target-age 13 and 14 abundances are zero in the 2025 fishery key. Their weights and SEs remain paired missing observations. The historical data continue to inform these ages in the future wtAgeRE fit. All pre-2025 fishery mean/SE decimal values are preserved exactly.

Exact BTS target ages 1–18 are retained during conversion. Target ages 15–18 are then pooled into 15+, conserving total abundance and biomass. The 2024 fishery key also retains exact target ages through 18; the 2025 fishery key has target ages through 15. The same terminal pooling convention applies. Age selection for wtAgeRE occurs after conversion; its input bins are 3–15+.

**Fishery 2024 results and remaining gap**

The recovered 2024 release contains **2,985 released FTNIR ages and 693 complete independent tester pairs**. The target is test_age; the resolved final_age is a separate field. Canonical specimen and age-field hashes match the independent original-workbook verification record. The original workbook itself remains outside this local snapshot. The empirical key uses all complete released pairs, including additional QA/QC selections; their representativeness remains an assumption.

Source ages **1–12** have empirical probability rows. Source ages **13–15** lack pairs; only age 13 has positive mass in the saved sampler draws. Its unallocated observed-point mass is **0.0323% of abundance** and **0.0817% of biomass**.

Allocated plus unallocated totals balance across all 1,001 saved rows to relative errors below 6.7e-16. The diagnostic retains that mass separately and preserves the original totals.

The table and figure report the **partial, fixed-key bootstrap mean and SD** from the supported source rows. Every converted 2024 bin remains conditional on the unresolved source-age-13 allocation. Target TMA age 15+ receives zero allocated abundance, so its mean and SD remain missing. Source FTNIR age 13 and target TMA age 13 are distinct: supported source ages can populate target ages 13 and 14.

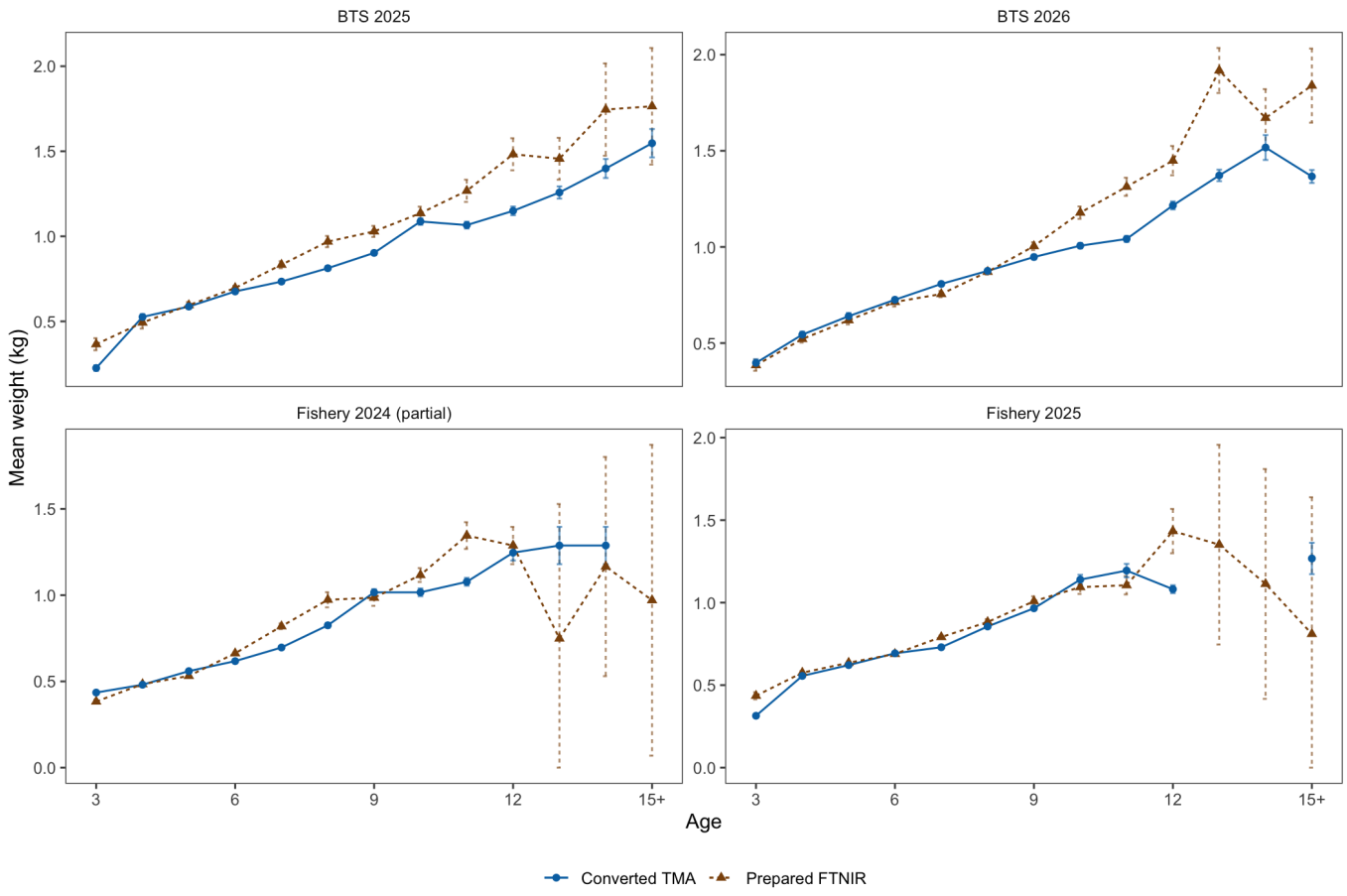
The 2025 midpoint rule stays specific to its approved key. In 2024, source age 14 also lacks calibration pairs; applying the age-12/14 midpoint rule would require additional empirical support. The 2024 key supports source age 1 directly; its positive source catch is retained.

Table G3 Fishery 2024 Weights and Partial Supported-Only Conversion

| Age | Prepared mean | Prepared SE | Partial TMA mean | Partial TMA SD |
|---|---|---|---|---|
| 3 | 0.3832 | 0.0093 | 0.4353 | 0.0053 |
| 4 | 0.4841 | 0.0067 | 0.4807 | 0.0033 |
| 5 | 0.5317 | 0.0035 | 0.5596 | 0.0021 |
| 6 | 0.6628 | 0.0040 | 0.6175 | 0.0023 |
| 7 | 0.8195 | 0.0166 | 0.6965 | 0.0060 |
| 8 | 0.9733 | 0.0439 | 0.8252 | 0.0138 |
| 9 | 0.9850 | 0.0469 | 1.0162 | 0.0182 |
| 10 | 1.1167 | 0.0401 | 1.0163 | 0.0227 |
| 11 | 1.3455 | 0.0774 | 1.0774 | 0.0227 |
| 12 | 1.2878 | 0.1082 | 1.2464 | 0.0455 |
| 13 | 0.7485 | 0.7800 | 1.2878 | 0.1082 |
| 14 | 1.1658 | 0.6356 | 1.2878 | 0.1082 |
| 15+ | 0.9705 | 0.9011 | Missing | Missing |

*Note.* Fishery 2024 weights in kilograms: preserved prepared values and partial supported-only conversion. SDs are conditional fixed-key diagnostics; all converted bins remain incomplete.

**Weight and uncertainty comparisons**




*Note.* Mean weights before and after conversion, with source SEs and converted bootstrap SDs or survey SEs. Fishery 2024 shows a partial supported-only diagnostic; source-age-13 mass remains unallocated.

Figure G1 Mean Weights Before and After Conversion

Converted fishery weights and bootstrap SDs are recalculated from the first 1,000 paired sampler bootstrap draws after transforming each draw's abundance and biomass. The final observed-data point row is excluded from those summaries. The previously selected legacy convention is retained: a target age absent in a bootstrap draw has zero weight for that draw. The SD uses the sample denominator. Known limitations of the legacy sampler's resampling selectors remain in the source record.

BTS uncertainty uses the saved joint VAST parameter covariance and conditional random-effect covariance. It includes dependence between source abundance and weight, among ages through shared parameters, and between 2025 and 2026. The ratio's numerator and denominator both contribute to the delta-method SE. The calculation evaluates the saved VAST model at its recorded parameter estimates.

**All conversion keys are treated as fixed.** The reported SEs exclude uncertainty in the empirical keys, the dummy row, and the interpolated row.

Table G4 Fishery 2024–2025 Weights and Uncertainty

| Year | Age | Original mean | Original SE | TMA mean | TMA SE / SD |
|---|---|---|---|---|---|
| 2024 | 3 | 0.3832 | 0.0093 | 0.4353 | 0.0053 |
| 2024 | 4 | 0.4841 | 0.0067 | 0.4807 | 0.0033 |
| 2024 | 5 | 0.5317 | 0.0035 | 0.5596 | 0.0021 |
| 2024 | 6 | 0.6628 | 0.0040 | 0.6175 | 0.0023 |
| 2024 | 7 | 0.8195 | 0.0166 | 0.6965 | 0.0060 |
| 2024 | 8 | 0.9733 | 0.0439 | 0.8252 | 0.0138 |
| 2024 | 9 | 0.9850 | 0.0469 | 1.0162 | 0.0182 |
| 2024 | 10 | 1.1167 | 0.0401 | 1.0163 | 0.0227 |
| 2024 | 11 | 1.3455 | 0.0774 | 1.0774 | 0.0227 |
| 2024 | 12 | 1.2878 | 0.1082 | 1.2464 | 0.0455 |
| 2024 | 13 | 0.7485 | 0.7800 | 1.2878 | 0.1082 |
| 2024 | 14 | 1.1658 | 0.6356 | 1.2878 | 0.1082 |
| 2024 | 15+ | 0.9705 | 0.9011 | Missing | Missing |
| 2025 | 3 | 0.4354 | 0.0217 | 0.3143 | 0.0115 |
| 2025 | 4 | 0.5753 | 0.0102 | 0.5559 | 0.0085 |
| 2025 | 5 | 0.6356 | 0.0063 | 0.6214 | 0.0066 |
| 2025 | 6 | 0.6891 | 0.0050 | 0.6929 | 0.0027 |
| 2025 | 7 | 0.7913 | 0.0086 | 0.7295 | 0.0029 |
| 2025 | 8 | 0.8824 | 0.0175 | 0.8555 | 0.0073 |
| 2025 | 9 | 1.0084 | 0.0296 | 0.9657 | 0.0115 |
| 2025 | 10 | 1.0939 | 0.0411 | 1.1395 | 0.0292 |
| 2025 | 11 | 1.1054 | 0.0557 | 1.1943 | 0.0406 |
| 2025 | 12 | 1.4336 | 0.1344 | 1.0815 | 0.0241 |
| 2025 | 13 | 1.3513 | 0.6052 | Missing | Missing |
| 2025 | 14 | 1.1133 | 0.6965 | Missing | Missing |
| 2025 | 15+ | 0.8102 | 0.8284 | 1.2676 | 0.0950 |

*Note.* Fishery 2024–2025 weights and uncertainty in kilograms. The 2024 TMA results are a partial supported-only conversion; source-age-13 mass remains unallocated. Missing target ages have no converted abundance.

2024: partial bootstrap mean and fixed-key SD, conditional on the supported source ages; target age 15+ remains missing. 2025: converted mean and bootstrap SE.

Table G5 BTS Mean Weights and Standard Errors After Joint Covariance Propagation

| Year | Age | Original mean | Original SE | TMA mean | TMA SE |
|---|---|---|---|---|---|
| 2025 | 3 | 0.3655 | 0.0357 | 0.2255 | 0.0161 |
| 2025 | 4 | 0.4928 | 0.0352 | 0.5260 | 0.0166 |
| 2025 | 5 | 0.5951 | 0.0186 | 0.5871 | 0.0134 |
| 2025 | 6 | 0.6964 | 0.0114 | 0.6758 | 0.0098 |
| 2025 | 7 | 0.8329 | 0.0206 | 0.7340 | 0.0100 |
| 2025 | 8 | 0.9690 | 0.0331 | 0.8125 | 0.0125 |
| 2025 | 9 | 1.0288 | 0.0325 | 0.9026 | 0.0131 |
| 2025 | 10 | 1.1360 | 0.0383 | 1.0872 | 0.0197 |
| 2025 | 11 | 1.2675 | 0.0653 | 1.0659 | 0.0200 |
| 2025 | 12 | 1.4825 | 0.0943 | 1.1496 | 0.0257 |
| 2025 | 13 | 1.4562 | 0.1232 | 1.2577 | 0.0358 |
| 2025 | 14 | 1.7453 | 0.2714 | 1.3987 | 0.0558 |
| 2025 | 15+ | 1.7646 | 0.3427 | 1.5474 | 0.0834 |
| 2026 | 3 | 0.3850 | 0.0291 | 0.3971 | 0.0190 |
| 2026 | 4 | 0.5211 | 0.0184 | 0.5440 | 0.0155 |
| 2026 | 5 | 0.6188 | 0.0218 | 0.6400 | 0.0165 |
| 2026 | 6 | 0.7139 | 0.0237 | 0.7254 | 0.0109 |
| 2026 | 7 | 0.7551 | 0.0162 | 0.8076 | 0.0101 |
| 2026 | 8 | 0.8707 | 0.0144 | 0.8755 | 0.0101 |
| 2026 | 9 | 1.0041 | 0.0203 | 0.9476 | 0.0125 |
| 2026 | 10 | 1.1782 | 0.0322 | 1.0064 | 0.0132 |
| 2026 | 11 | 1.3129 | 0.0472 | 1.0415 | 0.0156 |
| 2026 | 12 | 1.4490 | 0.0762 | 1.2159 | 0.0209 |
| 2026 | 13 | 1.9177 | 0.1171 | 1.3719 | 0.0298 |
| 2026 | 14 | 1.6711 | 0.1496 | 1.5171 | 0.0648 |
| 2026 | 15+ | 1.8390 | 0.1928 | 1.3665 | 0.0336 |

*Note.* BTS mean weights and SEs in kilograms after propagating the joint model covariance.

**Validation and source record**

Table G6 Validation of the Transformations

| Check | Result |
|---|---|
| Fishery independent recalculation | PASS |
| Pre-2025 fishery decimal preservation | Pass |
| Survey source SE reconstruction: largest relative difference | 1.22e-15 |
| Ratio derivative check: largest scaled difference | 1.68e-11 |
| Analytic versus numerical converted SE: largest relative difference | 3.15e-11 |
| Survey random-effect gradient: maximum | 4.36e-12 |
| Saved survey fixed Hessian positive definite | Yes |
| 2024 specimen/age-field identity and partial arithmetic | Pass: 22 checks; positive source-age-13 support unresolved |
| Input adapter | Selected inputs used in completed working fits; historical partial-conversion support limits retained |
| wtAgeRE optimization | Screened-key standalone RE-WAA fit complete; earlier partial-conversion branch remains a separate diagnostic |

*Note.* Validation of the transformations. Numerical checks retain the full-pair conversion basis; current fitted inputs use the selected 2024 tail treatment and screened 2026 key.

The source records retain matching abundance and weight estimates from the same saved survey fit, along with input/output hashes and fixed-key approximation rules. The empirical key is treated as fixed. Current fitting records and the historical conversion diagnostics are identified separately in the technical source record.

The historical 2024 partial-conversion diagnostics preserve their original inputs and supported-age limits. The current selected tail allocation, screened BTS key and completed fits have separate source and run identities. Earlier preparation states and the original verification wording remain archived in the technical source record.

This is an internal methods record. It uses a compact Quarto report and TAF-compatible source, transformation, output, and provenance stages. Assessment adoption is documented in the separate input and run records.

Historical partial-diagnostic fishery input: converted 2025, preserved 2024

Converted BTS input

Fishery approximation key

Conversion manifest

BTS uncertainty validation

2024 empirical key

2024 partial bootstrap summaries

2024 exact and pooled point candidates

2024 key support and unallocated source mass

2024 validation

Complete numbered conversion report · Editable analytical Quarto source · Subsection source manifest

**Provisional ATS age compositions**

In 2026, ATS age compositions were estimated by borrowing the BTS age–length keys and applying them to ATS abundance-at-length data. Following usual practice, age data for the 2026 ATS length samples will be applied in 2027.

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# Appendix H
# BTS Spatiotemporal Biomass Index and Abundance at Age, Including the Northern Bering Sea

This appendix describes the spatiotemporal modeling approach used to estimate the bottom-trawl survey (BTS) biomass index and abundance at age for the combined eastern and northern Bering Sea (EBS and NBS). This approach was adopted in the 2024 assessment for both abundance at age and survey indices and follows the approach used by the AFSC Groundfish Assessment Program (GAP) in previous assessments. The sections below describe its application to the 2026 survey data, including model specifications and uncertainty estimates.

**Scope and relationship to the earlier VAST appendix**

The earlier VAST appendix describes a different software generation and mixed 1982–2023/2024 coverage. It is retained as a historical source. The description uses the settings recorded in the saved production fit objects. The saved production fit objects identify **sdmTMB 1.0.0** for the biomass index and **tinyVAST 1.6.3** for abundance at age. Model-based integration across EBS and NBS extends the approach used to address pollock distribution shifts (O'Leary et al. 2020). The sdmTMB framework is described by Anderson et al. (2022).

Both output series contain **44 years, 1982–2019 and 2021–2026**. The survey was absent in 2020. The index fit retains 2020 as an additional latent time step representing a model prediction. The age fit retains annual time levels across 1982–2026, while the abundance and composition data files omit 2020. NBS sampling coverage varies among years within the prediction domain.

**Biomass index**

The pollock index code reads density-dependence-corrected station biomass density from the VAST-named input file. The file name is historical; the saved model class is sdmTMB. Input density is converted from kg/ha to kg/km² by multiplying by 100. The saved fit uses a Poisson-link delta-gamma observation model with separate year effects and spatial and spatiotemporal fields. Both components have spatial fields and annual AR(1) spatiotemporal structure. The retained call enables geometric anisotropy and includes a spatially varying coefficient for the environmental covariate.

The companion pollock code defines that covariate as standardized cold-pool area, the area of bottom water at or below 2°C (Rohan et al. 2022). The saved object contains its numerical covariate and the formula; the complete annual cold-pool release manifest remains unavailable. The exact environmental release and production script revision require confirmation.

Predicted density is integrated over a fixed regional grid. For grid-cell area ($A_{r,j}$) in km² and predicted biomass density ($d_{t,j}$) in kg/km², the regional biomass index in tons is

$$I_{r,t} = \frac{1}{1000} \sum_{j \in r} A_{r,j} \hat{d}_{t,j}. \quad (20)$$

The biomass-index estimates include separate EBS, NBS and combined **Both** products. The combined 2026 point estimate is **9,034,017.809 t**; the separate EBS and NBS point estimates sum to that value within numerical precision. These survey estimates remain subject to assessment input selection and acceptance.

**Abundance at age**

The saved tinyVAST fit uses numerical density at station and age, with age-by-year fixed effects in both delta components. All 15 labelled age classes have age-specific spatial variance terms and a common estimated lag-1 temporal coefficient in the spatiotemporal specification. The saved control has anisotropy disabled. This model uses age-by-year effects with the environmental terms excluded.

The companion pollock code multiplies CPUE_num by 100, converting fish/ha to fish/km². Its comment describing this as a conversion of kg is inconsistent with the numerical input; the calculation is an abundance calculation. Numerical density is integrated over the combined EBS–NBS grid and scaled to billions of fish:

$$\hat{N}_{r,t,a}[\,billion\ f\ i\ sh] = \frac{1}{10^9} \sum_{j \in r} A_{r,j}\, n_{t,a,j}\, p_{r,t,a} = \frac{\hat{N}_{r,t,a}}{\sum_b \hat{N}_{r,t,b}}. \quad (21)$$

The abundance-at-age matrix contains 15 rows and 44 year columns. Its row order agrees with the companion age_1 through age_15 composition columns: normalizing every matrix column reproduces the composition file to a maximum absolute difference of (8.4^{-16}). The 2026 combined total is **16.064690 billion fish**. The companion composition file supplies the age labels for the matrix and must remain attached to it. Displayed ages are 1–14 and 15+; the terminal category represents fish aged 15 and older. The original source column names are retained in the data. The age export supplies only the combined EBS–NBS estimates (terminal-age interpretation).

These expanded compositions use the age assignments and age–length-key preparation in their station-level inputs. Age assignment and FT-NIRS-to-traditional-age conversion precede the tinyVAST expansion. The exact release, assignment rules for 2025–2026, and any age-age correction used in this production input remain part of the source reconciliation.

**Verified settings and prediction-code evidence**

The retained index fit uses 250 interior mesh knots, anisotropy and a spatially varying environmental coefficient; the age fit uses 50 interior knots with anisotropy disabled and a separate spatial variance structure by age (Table H1). Prediction-grid and integration settings are identified separately as companion-code evidence. The local code checkout is 2a9e6f26062d34ee0375f8eac9462e51273e65a3. Its default species selections are Pacific cod for the index and yellowfin sole for age compositions. The saved pollock fit objects supply the production evidence for this appendix; the generic default selections remain code context.

Table H1 Delivered 2026 Pollock Fit and Companion Prediction Settings

| Setting | Biomass-index fit | Abundance-at-age fit |
|---|---|---|
| Platform and retained version | sdmTMB 1.0.0 | tinyVAST 1.6.3 |
| Response | Biomass density | Numerical density by age |
| Observation family | Poisson-link delta-gamma | Poisson-link delta-gamma |
| Fixed effects | Year in both components | Age × year in both components |
| Spatial mesh | 250 interior knots; 307 total vertices | 50 interior knots; 106 total vertices |
| Spatiotemporal structure | AR(1) in both components | Shared lag-1 coefficient; age-specific variance structure |
| Environmental coefficient | Spatially varying coefficient for env | None retained |
| Anisotropy | Enabled in saved call | Disabled in saved control |
| Observation years | 1982–2019, 2021–2026 | 1982–2019, 2021–2026 |
| Saved input rows | 17,236 | 265,350 station–age rows |
| Reported output regions | EBS, NBS, Both | Both |
| Prediction grid in companion code | 5 nmi; 1,578 EBS + 672 NBS cells | Same combined 2,250-cell grid |
| Integration bias/SE settings in companion code | get_index: bias_correct=TRUE | apply.epsilon=TRUE; bias.correct=FALSE; getsd=FALSE |

*Note.* Retained settings in the 2026 pollock fit objects, with prediction-grid and integration settings identified separately as companion-code evidence. The complete production prediction record remains unavailable.

The code grid has 492,989.921 km² in the EBS and 198,866.805 km² in the NBS, summing to 691,856.726 km². Those grid files are hashed in the source record. The complete production prediction grid and expansion invocation remain unavailable. This draft therefore distinguishes companion-code settings from verified saved-fit settings. The historical 2-nmi grid and 750-knot index describe the earlier VAST workflow.

**Uncertainty, diagnostics and outstanding review**

The index code requests bias-corrected integration. It also contains a 500-simulation covariance workflow. The assessment selects its covariance file and uncertainty treatment separately from those integration settings. The age

expansion code sets bias.correct=FALSE and getsd=FALSE despite apply.epsilon=TRUE. Its age output therefore excludes epsilon bias correction and a full estimated age-composition covariance. Assessment weighting and the selected design-based covariance remain separate decisions.

The saved index object reports a positive definite Hessian and maximum absolute retained gradient **0.0138148**. The saved age object reports optimizer return code 0, a positive definite Hessian and maximum retained gradient **0.000205084**. These recorded diagnostics require provider review for numerical acceptance. Both gradients exceed the workspace's (10^{-4}) assessment-model convergence guardrail; their suitability for these survey fits requires provider review. This appendix uses the saved survey fits by the provider.

The saved index data have **17,236 rows**, while the adjacent raw index export has **17,237**; the counts differ in 2026 (350 versus 351). The saved age fit has **265,350 rows**, versus **258,555** in the adjacent age-density export. Historical differences occur in 1982, 1985, 1988, 1991, 1994, 2001, 2005, 2006 and 2018; 2026 has 5,250 fitted rows versus 5,265 exported rows. These differences prevent a claim that the saved objects exactly reproduce the neighboring raw files. Their causes, the intended final input versions and the fit-to-export chain require confirmation from the survey analysts.

Before finalizing this appendix, confirm the production script revision, prediction grid, annual environmental release, full R/TMB/dependency versions, age-assignment lineage, treatment of the input mismatches, and the accepted covariance and diagnostic assessment. The latest linked general writeup and terms-of-reference table remain unavailable through the connected source. Their agreement with this draft requires author review. Historical VAST dependency versions retain their original dates.

**Source and reproducibility record**

The local provenance bundle retains saved-object metadata and checks, source file hashes, output consistency checks, and the read-only extraction scripts. It also preserves the original historical VAST text and historical PDF. The original fit objects and station-level inputs remain in their owning data directory. The report consumes saved metadata and aggregate outputs.

Current-method source code: index workflow and age-composition workflow. Author-documents awaiting access: general methods writeup and terms-of-reference settings.

# Appendix I
# Assessment Development and Supporting Record

The September 2026 assessment development document (Ianelli et al. 2026), published 3 September, presents the starting model comparisons and exploratory appendices using data through 2024. Corrected through-2024 results are archived with that document; the main assessment uses observations through 2026.

The Plan Team excerpt provides the seven recommendation summaries (Table A1). Its final report citation and any subsequent SSC direction require verification before release.

The source guidelines are *Alaska Groundfish Stock Assessment Guidelines*, July 2024, in a file named *Stock Assessment Guidelines Tier 3.docx*. Their publication procedures inform eventual release preparation. The development copy remains local under the user's release timing.

**Early–late fishery sensitivity summaries**

Model 26.1.2 evaluates sensitivity to separating the early and late fisheries. Model 26.0 supplies the assessment recommendations.

| Model 26.1.2 Early-late fishery split comparison | | | | |
|---|---|---|---|---|
| Quantity | As estimated or specified in 2024 for: | | Model 26.1.2 sensitivity for: | |
| | 2025 | 2026 | 2027 | 2028 |
| Natural mortality (annual rate, ages 3+) | 0.300 | 0.300 | 0.300 | 0.300 |
| Tier and sub-tier | 3a | 3a | 3a | 3a |
| Projected total biomass (ages 3+, t) | 8,526,000 | 8,075,000 | 11,392,000 | 12,488,000 |
| Projected female spawning biomass (t) | 3,118,000 | 3,342,000 | 3,223,000 | 3,432,000 |
| B100% (t) | 5,902,000 | 5,902,000 | 6,220,000 | 6,220,000 |
| B40% (t) | 2,361,000 | 2,361,000 | 2,488,000 | 2,488,000 |
| B35% (t) | 2,066,000 | 2,066,000 | 2,177,000 | 2,177,000 |
| F_OFL (annual rate) | 0.513 | 0.513 | 0.666 | 0.666 |
| Maximum F_ABC (annual rate) | 0.394 | 0.394 | 0.524 | 0.524 |
| Recommended F_ABC (annual rate) | 0.394 | 0.394 | Not selected | Not selected |
| OFL (t) | 2,957,000 | 2,496,000 | 2,269,000 | 2,399,000 |
| Maximum ABC (t) | 2,417,000 | 2,036,000 | 1,855,000 | 1,958,000 |
| Recommended ABC (t) | 2,417,000 | 2,036,000 | Not selected | Not selected |
| Historical source: December 2024 SAFE, Executive Summary, Tier 3 version, p. 4; B40% is printed in section 7.2, p. 35. | | | | |
| 2027–2028 projections assume catch of 1,375,000 t in each year, mean 2023–2025 selectivity and fixed biological inputs. Total biomass covers ages 3+ at the start of the year. F_OFL and maximum F_ABC are the fully selected F35% and F40% rates. Biomass and catch are rounded to the nearest 1,000 t. | | | | |
| Model 26.1.2 is a sensitivity analysis. Recruitment is fixed at the fitted 1978–2023 mean; these projections show point estimates. Model 26.0 supplies the assessment recommendation. | | | | |

| Model 26.1.2 Early-late fishery split stock-status summary. The 2025–2026 columns report status for 2025 and 2026; historical determinations are from the 2024 assessment. | | | | |
|---|---|---|---|---|
| Status | As determined in 2024 for: | | As determined this year for: | |
| | 2023 | 2024 | 2025 | 2026 |
| Overfishing | No | n/a | No | n/a |
| Overfished | n/a | No | n/a | No |
| Approaching overfished | n/a | No | n/a | No |
| Historical determinations are from the December 2024 EBS pollock SAFE, Executive Summary, p. 4. | | | | |
| Assessment conclusions are separate from final NMFS determinations. Updated 2025 catch (1,326,215.61 t) is below its 2,957,000-t OFL. For Model 26.1.2, 2026 spawning biomass (3.065 million t) and Alternative 7 biomass in 2028 (3.179 million t) both exceed B35% (2.177 million t), supporting the two 2026 "No" entries. Section 9 describes the status tests. Each two-year column pair reports overfishing for the earlier catch year and stock status for the later assessment year; cells outside those year-specific determinations are marked n/a. NMFS issues the final status determinations. | | | | |

**Assessment archive record**

The assessment archive links model source and executable versions, prepared inputs, run controls, saved fits, diagnostics, retrospectives, projections and report products. Source manifests document provenance and access restrictions. Scientific acceptance is recorded separately from numerical diagnostics and report preparation.

**Annual data-update register**

The annual register identifies the input streams, observation years, revisions and weight-at-age treatments used in the assessment (Table I1).

Table I1 Annual Data Additions, Revisions and Weight-at-Age Assumptions

| Annual data and weight-at-age register | | |
|---|---|---|
| Data source | Year | Update or weight-at-age treatment |
| Fishery catch | 2024 | Historical catch revision check |
| Fishery catch | 2025 | Observed annual update |
| Fishery catch | 2026 | Year-to-date plus annual completion forecast |
| BTS index | 2025 | Observed annual update |
| BTS index | 2026 | Observed annual update |
| ATS index | 2026 | Observed annual update |
| AVO index | 2025 | Observed annual update |
| AVO index | 2026 | Observed annual update |
| BTS ages | 2025 | Observed annual update |
| BTS ages | 2026 | Observed annual update |
| ATS ages | 2026 | Observed annual update |
| Fishery ages | 2024 | Observed annual update |
| Fishery ages | 2025 | Observed annual update |
| Fishery weight-at-age | 2024 | Observed inputs plus forecast 2026 |
| Fishery weight-at-age | 2025 | Observed inputs plus forecast 2026 |
| Fishery weight-at-age | 2026 | Forecast schedule |
| Spawning weight-at-age | 2024 | A-season schedule correction |
| Spawning weight-at-age | 2025 | A-season schedule correction |
| Spawning weight-at-age | 2026 | A-season schedule correction |
| BTS weight-at-age | 2025 | Survey product with recorded fills |
| BTS weight-at-age | 2026 | Survey product with recorded fills |
| ATS weight-at-age | 2025 | Assumed carry of 2024 WAA |
| ATS weight-at-age | 2026 | Carry or provisional survey product |

*Note.* Annual data additions, revisions and weight-at-age assumptions relative to the 2024 assessment. Listed updates are incorporated in the Model 23.x and 26.x fits. Cumulative input-update fits retain their stated stepwise history.

# Appendix J
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